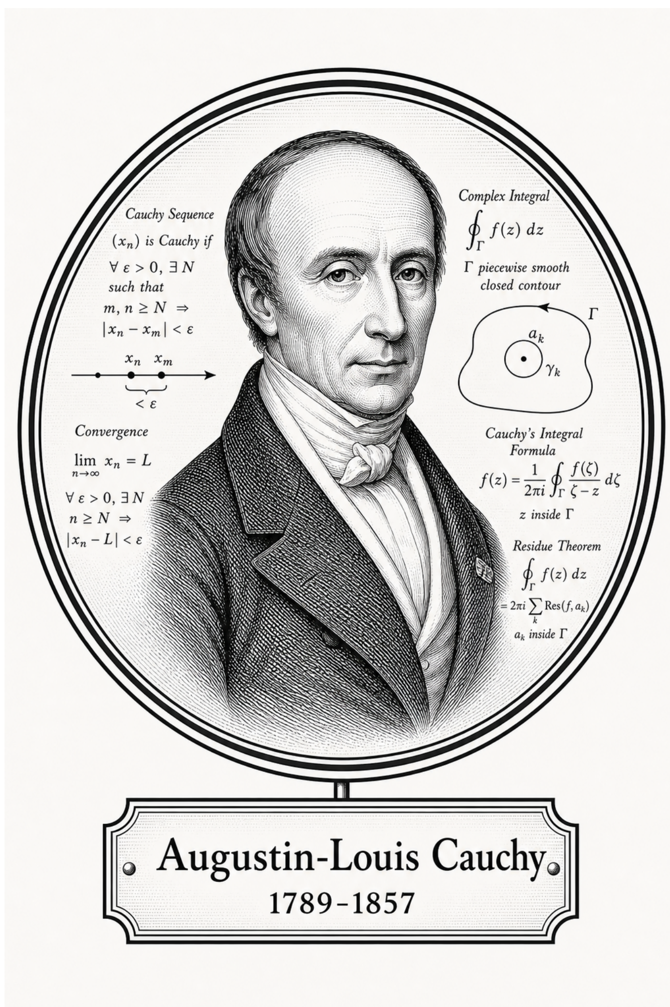
FROM CANTOR TO ITO

Classical Analysis



Augustin-Louis Cauchy

1789-1857

Self-Study Notes

William Sollers

June 30, 2026

For every reader learning to make mathematics their own.

Contents

Part I

Classical Analysis

Chapter 1

Real Analysis

SERIES
From Cantor to Ito

VOLUME III
Classical Analysis

BOOK I
Bounds, Sequences, and Series

CHAPTER 1
Real Analysis

1.1 Foundational Analysis Properties

1.1.1 Foundational Analysis Properties

Foundational Analysis Properties — Quick Reference

Concept	Goal	Proof mechanism / logical form
Well-definedness	Consistency	$x \sim y \Rightarrow F(x) = F(y)$
Existence	Feasibility	$\exists x\ P(x)$
Uniqueness	Singularity	$P(x_1) \wedge P(x_2) \Rightarrow x_1 = x_2$
Linearity	Decomposition	$L(f+g) = L(f) + L(g)$ and $L(cf) = cL(f)$
Uniformity	One choice works globally	$\forall \varepsilon > 0, \exists \delta > 0, \forall x, y,\ Q(\varepsilon, \delta, x, y)$
Equivalence	Interchangeable descriptions	$A \Rightarrow B$ and $B \Rightarrow A$
Characterisation	Usable test	$P(x) \iff C(x)$
Estimate	Control	$A \leq B$, $\lvert A \rvert < \varepsilon$, or $d(A, B) < \varepsilon$
Boundedness	Two-sided control	$\exists l, u \in S\ \forall x \in A\ (l \leq x \leq u)$
Upper/lower control	One-sided domination	$\forall x \in A\ (x \leq u)$ or $\forall x \in A\ (l \leq x)$
Extremality	Realized optimality	$m \in A \wedge \mathrm{Bound}(m, A)$
Least/greatest character	Optimal bound	$\mathrm{Bound}(b, A) \wedge \forall c\, (\mathrm{Bound}(c, A) \Rightarrow b \preceq c)$
Monotonicity	Order preservation	Start with $x \leq y$ and prove $f(x) \leq f(y)$, or the reverse
Closure	Staying inside a class	$P(x) \wedge P(y) \Rightarrow P(x \star y)$
Approximation	Arbitrary precision	$\forall \varepsilon > 0\ \exists x\ d(x, a) < \varepsilon$

Many proofs in analysis are not isolated inventions. They prove recurring structural properties: a definition is independent of choices, an object exists, there is only one possible object, an operation respects addition and scalars, a single parameter works everywhere, or two definitions describe the same phenomenon. Naming the property first tells us what the proof must do.

Well-Definedness

Well-definedness appears whenever an object is defined using choices: representatives of equivalence classes, selected sequences, selected partitions, or selected approximations. The goal is to prove that different allowed choices give the same output.

$$x \sim x' \Rightarrow F(x) = F(x').$$

Remark 1.1.1 (Proof sketch).

1. Identify the choice built into the definition.
2. Take two allowed choices, usually x and x' with $x \sim x'$.
3. Compute or compare the two proposed outputs.

4. Show the outputs agree in the codomain.

Remark 1.1.2 (Crucial logic). The output must be independent of the chosen representative. If changing the representative changes the output, the proposed definition is not a function on the quotient or chosen domain.

Existence

Existence proves that at least one object satisfying a property is available:

$$\exists x\ P(x).$$

In analysis the object might be a limit, a supremum, a point in an interval, a subsequence, a bound, an index N, a radius δ, or a partition.

Remark 1.1.3 (Proof sketch).

1. Determine every requirement in the property P.
2. Produce a candidate explicitly, or invoke a theorem that guarantees one.
3. Verify the candidate belongs to the required domain.
4. Verify the candidate satisfies every part of P.

Remark 1.1.4 (Constructive and nonconstructive existence). A constructive proof names the object directly. A nonconstructive proof uses a theorem such as completeness, the intermediate value theorem, or Bolzano–Weierstrass. In both cases, the final burden is the same: the object must satisfy the stated property.

Uniqueness

Uniqueness proves that at most one object can satisfy a property:

$$P(x_1) \wedge P(x_2) \Rightarrow x_1 = x_2.$$

It does not by itself prove existence. It only proves that two successful candidates cannot be distinct.

Remark 1.1.5 (Proof sketch).

1. Assume x_1 and x_2 both satisfy P.
2. Use the defining property of x_1 and the defining property of x_2.
3. Derive two-sided comparison, zero distance, or identical defining data.
4. Conclude $x_1 = x_2$.

Remark 1.1.6 (Common analysis forms). For suprema, show $s \leq t$ and $t \leq s$. For limits, show $|L - M| < \varepsilon$ for every $\varepsilon > 0$, hence $L = M$. In metric spaces, show $d(x_1, x_2) = 0$.

Linearity

Linearity proves that an operation decomposes over addition and scalar multiplication:

$$L(f+g) = L(f) + L(g), \qquad L(cf) = cL(f).$$

The goal is not merely algebraic neatness. Linearity says that a complicated input can be analysed by splitting it into simpler pieces.

Remark 1.1.7 (Proof sketch).

1. Take arbitrary admissible inputs f, g and scalar c.
2. Expand $L(f+g)$ using the definition of L and the definition of addition.
3. Rearrange the expression until $L(f) + L(g)$ appears.
4. Repeat with $L(cf)$ and show it equals $cL(f)$.

Remark 1.1.8 (Typical analysis examples). Limits, derivatives, integrals, and many summation operators have linearity theorems. Each proof follows the same shape: expand the left side, split the expression, and identify the two pieces.

Uniformity

Uniformity means that one choice works for all relevant points at once. The prototype is not just uniform continuity, but any statement where a witness is chosen before the point is allowed to vary:

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y,\ Q(\varepsilon, \delta, x, y).$$

The key phrase is "one choice." The same N, δ, bound, or partition must work globally over the specified domain.

Remark 1.1.9 (Proof sketch).

1. Fix the tolerance or target requirement.
2. Choose a single witness that depends only on the allowed global data.
3. Let the point, pair of points, function, or index vary arbitrarily.
4. Verify that the same witness works in every permitted case.

Remark 1.1.10 (Contrast with pointwise statements). Pointwise statements allow the witness to depend on the point. Uniform statements do not. The order of quantifiers is the whole issue:

$$\forall x\, \exists \delta \quad \text{is pointwise, while} \quad \exists \delta\, \forall x \quad \text{is uniform.}$$

Equivalence

Equivalence proves that two statements are logically interchangeable:

$$A \iff B.$$

Such theorems are common because analysis often has several descriptions of the same phenomenon: epsilon-delta, sequential, order-theoretic, topological, or integral.

Remark 1.1.11 (Proof sketch).

1. Prove $A \Rightarrow B$.
2. Prove $B \Rightarrow A$.
3. Keep the assumptions for the two directions separate.
4. After both directions are proved, either description may be used.

Remark 1.1.12 (Typical analysis examples). Sequential and epsilon-delta definitions of continuity are equivalent. Riemann and Darboux integrability are equivalent under the appropriate hypotheses. A set is closed exactly when it contains the limits of all convergent sequences from the set.

Characterisation

A characterisation is an equivalence whose purpose is to turn a definition into a usable test:

$$P(x) \iff C(x).$$

The theorem does not merely say two statements agree. It supplies a criterion that can be used in later proofs.

Remark 1.1.13 (Proof sketch).

1. Identify the original definition.
2. Identify the proposed criterion.
3. Prove definition $\Rightarrow$ criterion.
4. Prove criterion $\Rightarrow$ definition.

Remark 1.1.14 (Typical analysis examples). The epsilon characterisation of the supremum turns least-upper-bound language into an approximation rule. Cauchy criteria turn convergence questions into internal distance estimates. Sequential criteria turn topological conditions into statements about sequences.

Estimates and Bounds

An estimate proves control:

$$A \leq B, \qquad |A| < \varepsilon, \qquad d(A, B) < \varepsilon.$$

The aim is not to compute the exact value. The aim is to put the target quantity inside a range that is strong enough for the theorem.

Remark 1.1.15 (Proof sketch).

1. Identify the quantity that must be controlled.
2. Replace it by a larger or simpler quantity using valid inequalities.
3. Use hypotheses to replace variable terms by constants or small errors.
4. End with the required bound.

Remark 1.1.16 (Typical analysis examples). Showing a sequence is bounded, proving an error term tends to zero, bounding a Riemann-sum error, or proving $|f(x) - L| < \varepsilon$ are all estimate problems. The proof is a chain of control.

Boundedness

Boundedness proves two-sided control. In an ordered setting, the usual form is that a set or sequence is bounded above and bounded below:

$$\exists l, u \in S \ \forall x \in A \ (l \leq x \leq u).$$

Remark 1.1.17 (Proof sketch).

1. Prove bounded above by finding a candidate upper bound.
2. Prove bounded below by finding a candidate lower bound.
3. Verify both candidates work for every element under consideration.
4. Conclude that the object is bounded.

Upper and Lower Control

Upper and lower control prove one-sided domination. The proof focuses on a candidate bound and checks that every element lies on the required side of it:

$$\forall x \in A \ (x \leq u), \qquad \forall x \in A \ (l \leq x).$$

Remark 1.1.18 (Proof sketch).

1. Name the candidate upper or lower bound.
2. Let an arbitrary element of the set be given.
3. Use the hypotheses or definition of the set to compare that element with the candidate.
4. Since the element was arbitrary, conclude the one-sided bound.

Extremality

Extremality proves that an optimal value is actually realized by the object being studied. A maximum is not merely an upper bound; it is an upper bound that belongs to the set. A minimum is the corresponding realized lower bound.

$$\mathrm{Maximum}(m, A) \iff (m \in A) \wedge \mathrm{UpperBound}(m, A),$$

$$\mathrm{Minimum}(m, A) \iff (m \in A) \wedge \mathrm{LowerBound}(m, A).$$

Remark 1.1.19 (Proof sketch).

1. Prove the candidate belongs to the set.
2. Prove the candidate is the appropriate one-sided bound.
3. Combine membership with the bound condition.
4. Conclude maximum or minimum.

Least and Greatest Character

Least and greatest character proves optimality among competitors. For a supremum, the candidate must first be an upper bound, and then it must be no larger than every other upper bound. Infimum proofs reverse the direction.

$$\text{Supremum}(s, A) \iff \text{UpperBound}(s, A) \wedge \forall u \in S \left(\text{UpperBound}(u, A) \Rightarrow s \leq u\right),$$

$$\text{Infimum}(i, A) \iff \text{LowerBound}(i, A) \wedge \forall l \in S \left(\text{LowerBound}(l, A) \Rightarrow l \leq i\right).$$

Remark 1.1.20 (Proof sketch).

1. Prove the candidate is a bound of the required kind.
2. Let an arbitrary competing bound be given.
3. Compare the candidate with that competing bound.
4. Conclude the candidate is the least upper bound or greatest lower bound.

Monotonicity

Monotonicity proves order preservation or order reversal:

$$x \leq y \Rightarrow f(x) \leq f(y), \qquad x \leq y \Rightarrow f(y) \leq f(x).$$

The proof begins with an arbitrary ordered pair and carries that order through the definitions.

Remark 1.1.21 (Proof sketch).

1. Take arbitrary points x and y with $x \leq y$.
2. Expand the definitions of the function or operation.
3. Use order rules and hypotheses to compare the outputs.
4. Conclude that the order is preserved or reversed.

Closure

Closure proves that a class is stable under an operation. One starts with objects already known to belong to the class, combines them, and then verifies that the result still satisfies the defining property.

$$P(x) \wedge P(y) \Rightarrow P(x \star y).$$

For closure under scalar operations, the same pattern becomes

$$P(x) \Rightarrow P(c \star x)$$

for every admissible scalar or parameter c.

Remark 1.1.22 (Proof sketch).

1. Take arbitrary objects from the class.
2. Form the object produced by the operation.
3. Check every condition in the definition of the class.
4. Conclude the result remains inside the class.

Approximation

Approximation proves arbitrary precision. The proof usually begins with a tolerance and constructs an object close enough to a target:

$$\forall \varepsilon > 0 \; \exists x \; d(x, a) < \varepsilon.$$

Remark 1.1.23 (Proof sketch).

1. Fix an arbitrary tolerance $\varepsilon > 0$.
2. Choose or construct an approximating object.
3. Estimate the error between the approximation and the target.
4. Show the error is below ε.

A Classification Pass

Before proving a theorem, classify its job:

- Does the statement depend on choices? Check well-definedness.
- Does it ask for an object? Prove existence.
- Does it say only one object can work? Prove uniqueness.
- Does it say an operation splits across sums and scalars? Prove linearity.
- Does one witness have to work for every point? Prove uniformity.
- Does it say two descriptions agree? Prove equivalence.
- Does it give a test for a definition? Prove a characterisation.
- Does it ask for smallness or control? Prove an estimate.
- Does it ask for two-sided order control? Prove boundedness.
- Does it ask for one-sided order control? Prove an upper or lower bound.
- Does it ask for a realized best element? Prove extremality.
- Does it ask for the best bound among all bounds? Prove least or greatest character.
- Does it preserve or reverse order? Prove monotonicity.
- Does it stay inside a class after an operation? Prove closure.
- Does it ask for arbitrary precision? Prove approximation.

Remark 1.1.24 (The point). The classification does not solve the proof, but it tells us what a completed proof must contain. For example, an existence proof must exhibit or obtain an object, while a uniqueness proof must compare two candidates. An equivalence proof must have two directions, while a uniformity proof must choose one witness before the arbitrary point is fixed.

1.1.2 Manipulating Inequalities and the Logic of Bounding

Inequality Toolkit — Quick Reference

Tool	Statement	Typical use
Triangle inequality	$\lvert a+b\rvert \leq \lvert a\rvert + \lvert b\rvert$	Split a distance at an intermediate poin
Distance form	$\lvert a-b\rvert \leq \lvert a-c\rvert + \lvert c-b\rvert$	Introduce pivot c to separate two estima
Reverse triangle ineq.	$\big\lvert \lvert a\rvert - \lvert b\rvert \big\rvert \leq \lvert a-b\rvert$	Bound magnitudes using closeness
Absolute value interval	$\lvert x\rvert \leq y \iff -y \leq x \leq y$	Convert $\lvert\cdot\rvert$ bound to two-sided inequali
Boundedness replacement	$\lvert b_n\rvert \leq M$ (constant): replace $\lvert b_n\rvert$ by M	Eliminate variable factors in products
ε-budget split	$\lvert A\rvert < \frac{\varepsilon}{2}$ and $\lvert B\rvert < \frac{\varepsilon}{2} \Rightarrow \lvert A+B\rvert < \varepsilon$	Combine two independent estimates
$+1$ trick	Replace $\lvert L\rvert$ by $\lvert L\rvert+1$ in denominators	Avoid division by zero when L may vani

The Fundamental Asymmetry Between Equality and Inequality

Equality is bilateral: $a = b$ licenses substitution in either direction and carries no information about magnitude. An inequality is directional: $a \leq b$ locates a relative to b on the line, says nothing about the gap, and does not support backwards substitution.

This asymmetry is the source of a strategic shift. In an equality argument the goal is to show that two expressions are the same object. In a bounding argument the goal is to show that one expression is *controlled* by another. Control is inherently one-sided: a ceiling above $|f(x)|$ is useful regardless of how far below it $|f(x)|$ actually falls.

Remark 1.1.25 (Consequence for proof strategy). The question in analysis is rarely "what is $|a_n - L|$?" It is "is $|a_n - L|$ smaller than ε?" A valid crude bound that answers the second question is more useful than an exact answer to the first. Deliberate over-estimation — absent from algebra — is a required habit.

Remark 1.1.26 (Common error). Treating an inequality as if it were an equality by reversing direction. Given $a \leq b$, negation gives $-a \geq -b$, not $-a \leq -b$. Multiplication by a negative constant reverses direction. Failing to track sign changes when multiplying or negating is the most frequent algebraic error in ε-δ computations.

The Basic Rules of Inequality Arithmetic

Proposition 1.1.27 (Order arithmetic). *Let $a, b, c, d \in \mathbb{R}$.*

(i) If $a \leq b$ and $c \leq d$, then $a + c \leq b + d$.

(ii) If $a \leq b$ and $c > 0$, then $ac \leq bc$.

(iii) If $a \leq b$ and $c < 0$, then $ac \geq bc$.

(iv) $|x| \leq y$ (with $y \geq 0$) if and only if $-y \leq x \leq y$.

Go to proof.

Remark.

Remark 1.1.28 (Fully quantified form). (i) $\forall a, b, c, d \in \mathbb{R},\ (a \le b) \wedge (c \le d) \Rightarrow a + c \le b + d$. (iv) $\forall x \in \mathbb{R},\ \forall y \ge 0,\ |x| \le y \ \Leftrightarrow\ -y \le x \le y$.

Remark 1.1.29 (Inequalities add but do not subtract). Rule (i) licenses adding inequalities in the same direction. There is no subtraction rule: $a \le b$ and $c \le d$ yields no conclusion about $a - c$ versus $b - d$ in general. The direction of $a - c$ versus $b - d$ depends on the relative sizes of $(b - a)$ and $(d - c)$, which are not controlled.

Remark 1.1.30 (Consequence for absolute values). Rule (iv) is the bridge between $|\cdot|$ bounds and two-sided linear inequalities. It is invoked silently every time an ε-N argument converts $|a_n - L| < \varepsilon$ into positional information about a_n.

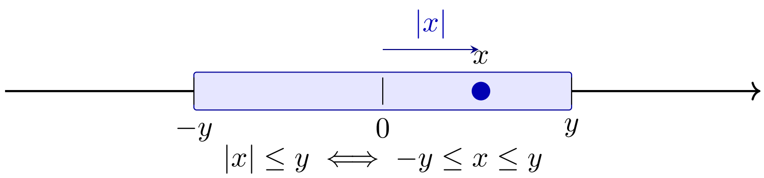


Figure 1.1: The absolute value characterisation as a symmetric interval. The shaded region is $[-y, y]$; the point x lies inside with $|x| \le y$. The equivalence $|x| \le y \Leftrightarrow -y \le x \le y$ (Proposition 1.1.27(iv)) converts an absolute value bound into a two-sided linear constraint.

The Triangle Inequality and the Pivot Move

Theorem (Triangle Inequality)

For all $a, b \in \mathbb{R},\ |a+b| \le |a|+|b|$. Equivalently, for all $a, b, c \in \mathbb{R},\ |a-b| \le |a-c|+|c-b|$.

Remark 1.1.31 (Fully quantified form). $\forall a, b \in \mathbb{R},\ |a + b| \le |a| + |b|$.

Remark 1.1.32 (The pivot move). The distance form $|a-b| \le |a-c| + |c-b|$ is the operative version in analysis. The element c is a *pivot*: an intermediate point that decomposes a hard distance into two pieces, each controllable by hypothesis. In the proof that $a_n + b_n \to L + M$, the pivot $c = L + b_n$ gives

$$|(a_n + b_n) - (L + M)| = |(a_n - L) + (b_n - M)| \le |a_n - L| + |b_n - M|,$$

and the two pieces are controlled by the individual convergences.

Remark 1.1.33 (Reverse triangle inequality). $\big||a| - |b|\big| \le |a - b|$. Proof: apply the standard inequality to $a = (a - b) + b$ to get $|a| - |b| \le |a - b|$; symmetry gives the other side. Consequence: $|\cdot| : \mathbb{R} \to \mathbb{R}$ is Lipschitz with constant 1, so closeness of inputs implies closeness of magnitudes.

Remark 1.1.34 (Common error). Writing $|a| + |b| \le |a + b|$. The triangle inequality always produces an *upper* bound on a sum, never a lower bound. The reverse $(|1| + |-1| = 2 > |1 + (-1)| = 0)$ is a one-line counterexample.

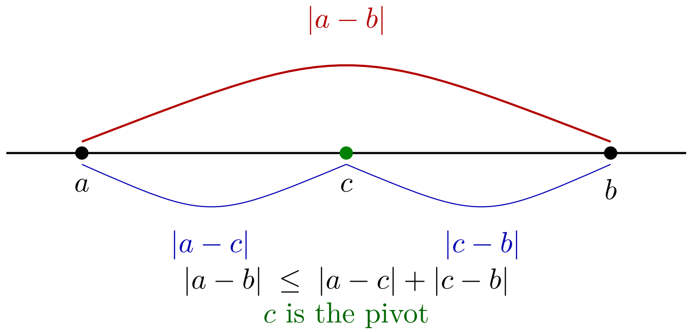


Figure 1.2: The triangle inequality in distance form. The direct distance $|a-b|$ (red) is bounded above by the sum of indirect distances $|a-c|$ and $|c-b|$ (blue) through pivot c. In analysis, c is chosen to split $|a-b|$ into two pieces each controlled by a separate hypothesis.

Bounding Is Not Solving

In algebra, a computation terminates when the exact value is found. In analysis, a bounding argument terminates when the expression is shown to fall within an acceptable range. These are structurally different goals.

Remark 1.1.35 (Deliberate crudeness is valid). To bound $|x^2-9x+1|$ on $[-1,5]$:

$$|x^2-9x+1| \le |x|^2 + 9|x| + 1 \le 25 + 45 + 1 = 71.$$

The actual supremum is far smaller than 71. That is irrelevant. The bound is valid, explicit, and was obtained cheaply. The test for a bound is not "is this sharp?" but "is this true and sufficient for the argument's purpose?"

Remark 1.1.36 (Consequence for proof reading). A reader who checks that each inequality is valid can trust the conclusion without caring whether the bounds are optimal. Optimality is irrelevant to logical correctness.

The ε-Budget Split

The prototype of all two-piece bounding arguments:

$$|A+B| \le |A| + |B| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

The total tolerance ε is a budget allocated among the summands. With k summands, allocate ε/k each; with summands of unequal difficulty, split unevenly provided the shares sum to at most ε.

Remark 1.1.37 (The four-step procedure). Standard structure of an ε-N or ε-δ proof:

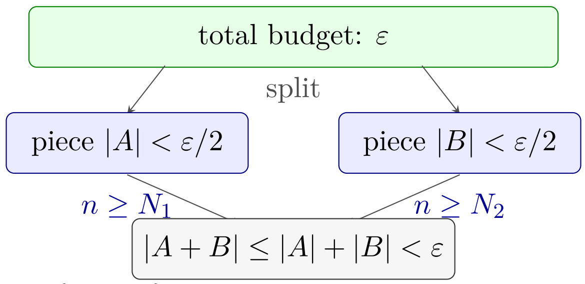
1. Apply the triangle inequality: express $|f-L|$ as a sum $|A_1| + \cdots + |A_k|$.
2. Assign budgets $\varepsilon_i > 0$ with $\sum_i \varepsilon_i = \varepsilon$.

3. For each piece, find N_i (or δ_i) so that $|A_i| < \varepsilon_i$ for all $n \geq N_i$ (or $|x - a| < \delta_i$).

4. Set $N = \max_i N_i$ (or $\delta = \min_i \delta_i$) to enforce all constraints simultaneously.

Step 4 reflects the logical structure of the conclusion: all conditions must hold at once, so the threshold is the most restrictive one.

Remark 1.1.38 (Common error). Taking $N = N_1$ and ignoring N_2. For $n \geq N_1$ the first piece is controlled; the second is not. Both pieces must be within budget simultaneously, which requires $n \geq \max\{N_1, N_2\}$.



take $N = \max\{N_1, N_2\}$ to enforce both constraints simultaneously

Figure 1.3: The ε-budget split. The total tolerance ε is distributed as shares $\varepsilon/2$ between two pieces. Each piece is controlled independently for $n \geq N_i$. Taking $N = \max\{N_1, N_2\}$ enforces both bounds simultaneously, giving $|A+B| < \varepsilon$ for all $n \geq N$. With k pieces, assign ε/k each and take $N = \max_i N_i$.

The Boundedness Replacement

Proposition 1.1.39 (Convergent sequences are bounded)**.** *If $(b_n) \to b$, then there exists $M > 0$ such that $|b_n| \leq M$ for all $n \in \mathbb{N}$. Go to proof.*

Remark.

Remark 1.1.40 (Fully quantified form). $\forall (b_n) \subset \mathbb{R},\ \lim_{n\to\infty} b_n = b \ \Rightarrow\ \exists M > 0,\ \forall n \in \mathbb{N},\ |b_n| \leq M$.

Remark 1.1.41 (The replacement move). Given $|b_n| \leq M$:

$$|b_n| \cdot |a_n - L| \leq M \cdot |a_n - L|.$$

The variable factor is gone. Choose N so that $|a_n - L| < \varepsilon/M$ for $n \geq N$; the whole product is then less than ε. The constant M need not be tight.

Remark 1.1.42 (Common error). Choosing $|a_n - L| < \varepsilon/|b_n|$. This is circular: N would depend on n, since $|b_n|$ changes with n. The threshold N must be a fixed number, independent of n. Replacing $|b_n|$ by the constant M breaks the circularity.

The Work-Backwards Principle

ε-N proofs are *stated* forwards but *discovered* backwards. The proof asserts: "Let $\varepsilon > 0$. Choose $N = \lceil C/\varepsilon \rceil$." The value N was found by scratchwork: "the expression simplifies to C/n; this is less than ε when $n > C/\varepsilon$; so set $N = \lceil C/\varepsilon \rceil$."

The scratchwork is a private computation that produces the *witness*. The proof states the witness and verifies it, working forwards.

Scratchwork (private):	Work backwards from $\lvert a_n - L \rvert < \varepsilon$ to find what N must satisfy.
Proof (public):	State N explicitly. Verify $n \geq N \Rightarrow \lvert a_n - L \rvert < \varepsilon$.

Remark 1.1.43 (Common error). Allowing scratchwork to appear in the proof: "we need $n > C/\varepsilon$, so choose n this large." The proof must assert a fixed N and verify the conclusion holds. A threshold stated as "n large enough" is an incomplete scratchwork note, not a proof.

Remark 1.1.44 (Connection to the geometric phase). The work-backwards process is the analytic counterpart of the geometric phase in limit proofs. The geometric phase identifies structure and chooses a pivot; the backwards computation determines the explicit threshold. The proof states both and verifies the chain forwards.

Chaining Inequalities

A standard proof move builds a chain $|f(x)| \leq A(x) \leq B \leq \varepsilon$, where each step uses a different tool: the triangle inequality, a domain bound, and the choice of δ.

Remark 1.1.45 (Replacing by a larger quantity). A quantity on the right-hand side of $\leq$ may be freely replaced by anything larger, since this only weakens the inequality. If $|x| \leq |x-1|+1$ and $|x-1| < \delta$, then $|x| < \delta + 1$. The replacement of $|x-1|$ by δ is valid because $|x-1|$ appears on the right. The same replacement on the *left* would be invalid.

Remark 1.1.46 (Common error). Including a step where the inequality points the wrong direction, or where a step holds only generically rather than pointwise. A chain is only as strong as its weakest link; every $\leq$ must be unconditionally valid.

The +1 Trick for Denominators

When $\varepsilon/|L|$ appears in an estimate and L may be zero, replace $|L|$ by $|L| + 1 \geq 1$:

$$\frac{\varepsilon}{|L| + 1} \leq \varepsilon.$$

Remark 1.1.47 (Fully quantified form). $\forall L \in \mathbb{R},\ |L| + 1 \geq 1 > 0$. Hence $\varepsilon/(|L| + 1)$ is well-defined and positive for all $\varepsilon > 0$ and all $L \in \mathbb{R}$.

Remark 1.1.48 (Generalisation). The +1 is an instance of a wider principle: any constant intended for a denominator must be guaranteed strictly positive. If the constant is a given quantity C that might be zero, replace it by $C+1$. The choice of 1 is conventional; any fixed positive number serves the same purpose.

Remark 1.1.49 (Common error). Using $|L|$ as a denominator without first verifying $L \neq 0$. The product limit proof $a_n b_n \to LM$ is the canonical location for this error: the bound $\varepsilon/(2|L|)$ fails when $L = 0$, and the $L = 0$ case requires separate treatment or the use of $|L| + 1$.

Summary: The Bounding Mindset

Algebra	**Analysis**
Find the exact value of x	Show $|x - L| < \varepsilon$
Equality: both sides are the same object	Inequality: one side controls the other
Work backwards and forwards symmetrically	Discover witness backwards; verify forwards
Exact answer is the goal	Any valid bound is sufficient
Fractions must be simplified	Crude over-estimates are acceptable
Multiply both sides freely	Track sign changes; direction is load-bearing

Remark 1.1.50 (The structure of a bounding proof). A bounding proof is a *chain of control*: inequalities connecting the target expression to ε, where each link uses one of the toolkit tools above. The proof is complete when the chain reaches ε. Every link must be unconditionally valid; every variable factor must be replaced by a constant before the chain can close.

1.1.3 Completeness as a Constructive Engine

Completeness Construction Toolkit — Quick Reference

Tool	**What it supplies**
ε-characterisation of sup	$\forall \varepsilon > 0,\ \exists x \in S,\ x > \sup S - \varepsilon$
Inductive selection	Monotone $(x_n) \subset S$ with $x_n \to \sup S$
Monotone Approximation	Such a sequence exists for any nonempty bounded S
LUB $\Leftrightarrow$ MCT	The two completeness statements are equivalent

The ε-Characterisation of Suprema and Infima

Proposition 1.1.51 (ε-characterisation of sup)**.** *Let $S \subset \mathbb{R}$ be nonempty and bounded above, and let $s = \sup S$. Then for every $\varepsilon > 0$ there exists $x \in S$ with $x > s - \varepsilon$.* *Go to proof.*

Remark.

Remark 1.1.52 (Fully quantified form). $\forall \varepsilon > 0,\ \exists x \in S,\ s - \varepsilon < x \leq s$. Equivalently: any $M < s$ is not an upper bound for S.

Remark 1.1.53 (Proof strategy). If no such x existed for some $\varepsilon_0 > 0$, then $s - \varepsilon_0$ would be an upper bound for S smaller than s, contradicting $s = \sup S$.

Remark 1.1.54 (Operative role). The ε-characterisation is the *existential engine* of constructive completeness arguments. At each inductive step it is invoked with a specific ε value, discharging the universal quantifier and yielding a concrete existence claim. Existential instantiation then names a witness — call it x_{n+1} — and the rest of the argument uses only the single property $x_{n+1} > s - \varepsilon$. The witness is underspecified; the argument makes no commitment about which element of S is chosen, and no further properties of x_{n+1} are used.

Remark 1.1.55 (Symmetric form for inf). For $t = \inf S$: for every $\varepsilon > 0$ there exists $y \in S$ with $y < t + \varepsilon$. The proof is symmetric: any $M > t$ is not a lower bound.

Inductive Selection at Bounds

Theorem (Inductive Selection at $\sup$)

Let $S \subset \mathbb{R}$ be nonempty and bounded above, and let $s = \sup S$. Then there exists a strictly increasing sequence $(x_n)_{n=1}^{\infty} \subset S$ with $x_n \to s$.

Remark 1.1.56 (Fully quantified form). $\exists (x_n) \subset S$: $\forall n,\ x_n < x_{n+1};\ \ \forall n,\ s - \frac{1}{n} < x_n \le s$; $\lim_{n\to\infty} x_n = s$.

Remark 1.1.57 (Why no algebraic formula is possible in general). The set S may have large gaps; any fixed formula might land outside S entirely. The construction must be inductive, selecting from S at each stage using the ε-characterisation rather than evaluating a closed expression.

Remark 1.1.58 (Proof strategy — the max device). The inductive step must simultaneously satisfy two lower bounds: $x_{n+1} > x_n$ (monotonicity) and $x_{n+1} > s - \frac{1}{n+1}$ (proximity). A single invocation of the existential can impose only one threshold. Setting

$$M_n := \max\Bigl(x_n,\ s - \tfrac{1}{n+1}\Bigr)$$

collapses both requirements into one: $x_{n+1} > M_n$ delivers both $x_{n+1} > x_n$ and $x_{n+1} > s - \frac{1}{n+1}$ simultaneously. Since $M_n < s$ (both entries are strictly below s whenever $s \notin S$, or the constant sequence $x_n = s$ handles the case $s \in S$), the ε-characterisation with $\varepsilon = s - M_n > 0$ supplies the required witness.

Base case. Set $\varepsilon = 1$. The ε-characterisation gives $x_1 \in S$ with $x_1 > s - 1$; since $x_1 \in S$, also $x_1 \le s$.

Inductive step. Given x_n with $s - \frac{1}{n} < x_n \le s$, form M_n as above and instantiate the ε-characterisation with $\varepsilon = s - M_n > 0$ to obtain $x_{n+1} \in S$ with $x_{n+1} > M_n$. Then:

$$\begin{aligned} x_{n+1} > M_n \ge x_n \quad &\Rightarrow \quad x_{n+1} > x_n, \\ x_{n+1} > M_n \ge s - \tfrac{1}{n+1} \quad &\Rightarrow \quad s - \tfrac{1}{n+1} < x_{n+1} \le s. \end{aligned}$$

Convergence. $0 \le s - x_n < \frac{1}{n}$; the squeeze theorem and the Archimedean Property give $x_n \to s$.

Remark 1.1.59 (What the construction requires). Three ingredients: (i) completeness of $\mathbb{R}$ ($\sup S$ exists); (ii) the ε-characterisation (Proposition 1.1.51), supplying the witness at each step; (iii) the Archimedean Property (Theorem 3.1.4), forcing $|s - x_n| \le \frac{1}{n} \to 0$. The internal structure of S — its cardinality, openness, density — plays no role.

Remark 1.1.60 (Common error). Instantiating the existential with only the proximity threshold $s - \frac{1}{n+1}$ and ignoring x_n. The resulting witness satisfies proximity but not monotonicity: one might obtain $x_1 = 0.9$, $x_2 = 0.4$, $x_3 = 0.85$, each close to s but not increasing. The max is not a convenience; it is the minimal device that encodes both requirements into one threshold.

Remark 1.1.61 (Infimum case). Replace max by min, $s - \frac{1}{n}$ by $t + \frac{1}{n}$, and reverse all strict inequalities. The resulting sequence $(y_n) \subset S$ is strictly decreasing with $y_n \to t = \inf S$.

Monotone Approximation of Bounds

Proposition 1.1.62 (Monotone Approximation of Bounds)**.** *Let $S \subset \mathbb{R}$ be nonempty and bounded. Then there exist monotone sequences $(x_n), (y_n) \subset S$ with $x_n \to \sup S$ and $y_n \to \inf S$.* *Go to proof.*

Remark.

Remark 1.1.63 (Fully quantified form). $\forall S \neq \emptyset$ bounded: $\exists (x_n) \subset S$, $x_n \uparrow \sup S$; $\exists (y_n) \subset S$, $y_n \downarrow \inf S$.

Remark 1.1.64 (Proof strategy). The strictly increasing sequence from Section 1.1.3 is already monotone and converges to $\sup S$. Taking $X_n = \max\{x_1, \ldots, x_n\}$ makes it explicitly non-decreasing (useful when the inductive step does not guarantee strict increase). The infimum case is symmetric.

Remark 1.1.65 (Consequence). Every supremum is the limit of a sequence from the set. The sequential characterisation of sup and inf — that they are realised as limits of internal sequences — follows from this construction, making Proposition 1.1.62 the canonical proof of that characterisation.

Remark 1.1.66 (Forward connections). The inductive selection pattern reappears in:

- **Monotone Convergence Theorem** — every increasing bounded sequence converges to the supremum of its range; this construction exhibits that the supremum is always realised as such a limit.
- **Nested Interval Property** — the left and right endpoints of nested intervals form monotone sequences converging to a common point by exactly this mechanism.
- **Existence of square roots** — the standard proof sets $s = \sup\{x \in \mathbb{R} : x^2 < 2\}$ and extracts an approximating sequence from that set by inductive selection.

LUB $\iff$ MCT: The Completeness Equivalence

Definition (Least Upper Bound Property)

$\mathbb{R}$ has the **least upper bound property** if every nonempty set $A \subset \mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Theorem (LUB $\iff$ MCT)

The following are equivalent over the Archimedean ordered field axioms:

(i) $\mathbb{R}$ has the least upper bound property (LUB).

(ii) Every bounded monotone sequence of real numbers converges (MCT).

Remark 1.1.67 (Fully quantified form). (i) $\forall A \subset \mathbb{R},\ (A \neq \emptyset \wedge \exists M,\ \forall a \in A,\ a \leq M) \Rightarrow \exists \sup A \in \mathbb{R}$.

(ii) $\forall (x_n) \subset \mathbb{R},\ (\forall n,\ x_n \leq x_{n+1}) \wedge (\exists M,\ \forall n,\ x_n \leq M) \Rightarrow \exists L \in \mathbb{R},\ \lim_{n\to\infty} x_n = L$.

Remark 1.1.68 (Proof strategy — LUB $\Rightarrow$ MCT). Let (x_n) be increasing and bounded. The image set $A = \{x_n : n \in \mathbb{N}\}$ is bounded above, so LUB gives $\alpha = \sup A$. The ε-characterisation produces N with $x_N > \alpha - \varepsilon$. Monotonicity promotes this to a tail statement: $\alpha - \varepsilon < x_n \leq \alpha$ for all $n \geq N$. Hence $x_n \to \alpha$.

The quantifier structure is:

$$\alpha = \sup A \ \Rightarrow\ (\forall \varepsilon > 0)\ \exists N\ (x_N > \alpha - \varepsilon) \xrightarrow{\text{monotone}} (\forall \varepsilon > 0)\ \exists N\ (\forall n \geq N)\ (|x_n - \alpha| < \varepsilon).$$

Remark 1.1.69 (Proof strategy — MCT $\Rightarrow$ LUB). Given $A \neq \emptyset$ and bounded above by u_1, pick $l_1 \in A$. Build a bisection bracket: at each stage, let $m_n = (l_n + u_n)/2$; if m_n is an upper bound for A set $u_{n+1} = m_n$, otherwise set $l_{n+1} = m_n$ (and pick a witnessing $a \in A$ with $a > l_{n+1}$). The sequences (l_n) and (u_n) are monotone and bounded, so MCT gives limits $l_n \to s$ and $u_n \to s$ (since $u_n - l_n = (u_1 - l_1)/2^n \to 0$). The invariants — "$u_n$ is an upper bound for A" and "l_n is not" — force $s = \sup A$.

The construction maintains:

- a *universal* invariant: $(\forall a \in A)(a \leq u_n)$,
- a *failure* invariant: $(\exists a \in A)(a > l_n)$.

These two invariants, together with $u_n - l_n \to 0$, pin s as the least upper bound.

Remark 1.1.70 (Why this matters structurally). The chain LUB $\Leftrightarrow$ MCT $\Leftrightarrow$ NIP $\Leftrightarrow$ BW $\Leftrightarrow$ CC (Cauchy Criterion) is the backbone of the equivalence cluster at the heart of real analysis. All five statements assert that $\mathbb{R}$ has no gaps, each in its own language. The inductive selection technique — building sequences from sets using the ε-characterisation — is the constructive face of all five equivalences.

Remark 1.1.71 (Downstream reuse). The quantifier pattern

$$(\forall \varepsilon > 0)(\exists \text{witness}) \ \Rightarrow\ (\exists \text{sequence with controlled tail})$$

reappears in Bolzano–Weierstrass (extracting convergent subsequences), $\limsup/\liminf$ (tail suprema as a decreasing sequence), compactness (finite subcovers as finite witnesses), and approximation theorems throughout analysis.

1.1.4 Walking a Sequence with a Predicate

Predicate-Walking Toolkit — Quick Reference

Theorem	Predicate $P(n)$	Tool
Monotone Subseq.	n is a peak of (x_n)	Dichotomy on peak count
Bolzano–Weierstrass	interval has ∞ many terms	Bisection + pigeonhole
Arzelà–Ascoli	converges at $d_1, \ldots, d_m$	Diagonal extraction

The Infinitely-Often / Eventually Dichotomy

Definition (Infinitely Often / Eventually)

Let (x_n) be a sequence and P a predicate on $\mathbb{N}$. P holds **infinitely often** if $\{n \in \mathbb{N} : P(n)\}$ is infinite; P holds **eventually** if there exists $N \in \mathbb{N}$ such that $P(n)$ holds for all $n \geq N$.

Remark 1.1.72 (Fully quantified form). P infinitely often: $\forall N \in \mathbb{N},\ \exists n \geq N,\ P(n)$.
P eventually: $\exists N \in \mathbb{N},\ \forall n \geq N,\ P(n)$.

Proposition 1.1.73 (The dichotomy)**.** *For any predicate P on $\mathbb{N}$, exactly one of the following holds:*

(i) P holds infinitely often.

(ii) $\neg P$ holds eventually.

Go to proof.

Remark.

Remark 1.1.74 (Fully quantified form). $\big(\forall N,\ \exists n \geq N,\ P(n)\big)\ \veebar\ \big(\exists N,\ \forall n \geq N,\ \neg P(n)\big)$. The two cases are exhaustive because $\mathbb{N}$ is infinite: if P fails for all but finitely many n, those indices have a maximum N, and $\neg P(n)$ holds for all $n > N$.

Remark 1.1.75 (Relation to Infinite Pigeonhole). The index set $\mathbb{N}$ is partitioned into $\{n : P(n)\}$ and $\{n : \neg P(n)\}$; an infinite set cannot have both parts finite. The "infinitely often" case provides an inexhaustible reservoir of indices from which to select subsequence terms inductively.

The Abstract Extraction Template

Remark 1.1.76 (Four-step pattern). Every predicate-walking argument follows the same skeleton:

1. *Define the predicate.* Identify the property $P(n)$ to propagate: peak, infinite-occupancy, pointwise convergence, etc.

2. *Apply the dichotomy.* Either P holds infinitely often, or $\neg P$ holds eventually. Handle each case.

3. *Extract the subsequence.* In the "infinitely often" case, enumerate indices satisfying P in increasing order: $n_1 < n_2 < \cdots$. In the "eventually" case, work in the $\neg P$-tail and build inductively, using $\neg P$ as the fuel.

4. *Verify the property.* Show (x_{n_k}) has the desired property, using the predicate and any additional hypotheses (boundedness, MCT, NIP, equicontinuity).

The critical structural requirement: at each inductive stage, one must find $n_{k+1} > n_k$ satisfying the relevant condition. The "infinitely often" guarantee ensures the reservoir is never exhausted.

The Peak Argument: Monotone Subsequence Theorem

Definition (Peak Index)

An index $m \in \mathbb{N}$ is a **peak** of (x_n) if $x_m \geq x_n$ for all $n \geq m$.

Remark 1.1.77 (Fully quantified form). m is a peak: $\forall n \in \mathbb{N},\ n \geq m \Rightarrow x_m \geq x_n$.

Theorem (Monotone Subsequence)

Every sequence (x_n) in $\mathbb{R}$ has a monotone subsequence.

Remark 1.1.78 (Fully quantified form). $\forall (x_n) \subset \mathbb{R},\ \exists (n_k)$ strictly increasing, (x_{n_k}) is monotone

Remark 1.1.79 (Proof strategy — peak dichotomy). Apply the dichotomy to $P(m) =$ "m is a peak."

Case 1: infinitely many peaks. Let $n_1 < n_2 < \cdots$ enumerate the peaks. Since n_k is a peak and $n_{k+1} > n_k$: $x_{n_k} \geq x_{n_{k+1}}$. The subsequence is decreasing.

Case 2: finitely many peaks. Let N be an index beyond all peaks. No $n \geq N$ is a peak, so for each such n there exists $m > n$ with $x_m > x_n$. Set $n_1 = N$. Since n_1 is not a peak, $\exists n_2 > n_1$ with $x_{n_2} > x_{n_1}$. Inductively, this produces a strictly increasing subsequence.

The no-peak condition $\neg P(n_k)$ is the fuel for each inductive step: it is precisely the statement "there exists a later term exceeding x_{n_k}," which supplies n_{k+1}.

Remark 1.1.80 (Consequence: Bolzano–Weierstrass). Every bounded sequence has a bounded monotone subsequence. By MCT, that subsequence converges. Bolzano–Weierstrass follows immediately.

Remark 1.1.81 (Common error). Assuming Case 2 produces an increasing subsequence "by default." It does, but only because $\neg P(n)$ supplies a strictly larger term. If the predicate were defined differently, $\neg P$ might provide no useful structure. The conclusion depends on the specific content of $\neg P$, not the abstract case structure.

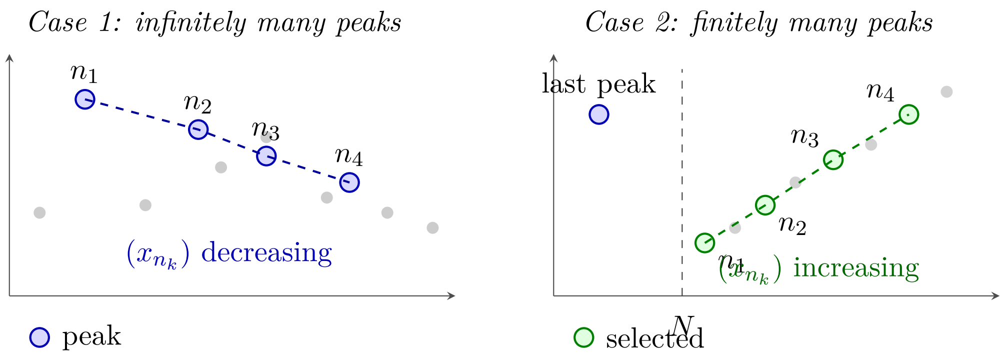


Figure 1.4: The two cases of the Monotone Subsequence Theorem (Section 1.1.4). *Left:* Infinitely many peaks (blue) yield a decreasing subsequence (dashed blue). *Right:* After the last peak, every $n \geq N$ is a non-peak, so a later term always exceeds the current one; this fuels the inductive construction of an increasing subsequence (dashed green).

The Infinite-Occupancy Predicate: Bolzano–Weierstrass via Bisection

Proposition 1.1.82 (Bolzano–Weierstrass via bisection)**.** *Every bounded sequence in* $\mathbb{R}$ *has a convergent subsequence. Go to proof.*

Remark.

Remark 1.1.83 (Proof strategy — infinite-occupancy predicate)*.* The predicate at each stage: $P(I) =$"interval I contains infinitely many terms of (x_n)." Since $(x_n) \subseteq [a_0, b_0]$ for some bounded interval, $P([a_0, b_0])$ holds.

Inductive step. Given $I_k = [a_k, b_k]$ with $P(I_k)$, bisect at $m_k = (a_k + b_k)/2$. The two halves cover I_k; at least one contains infinitely many terms. Set I_{k+1} to be that half. Pick $x_{n_{k+1}} \in I_{k+1}$ with $n_{k+1} > n_k$. The predicate $P(I_{k+1})$ holds by construction.

Convergence. The intervals have $|I_k| = (b_0 - a_0)/2^k \to 0$. By NIP there exists a unique $L \in \bigcap I_k$. Since $a_k \leq x_{n_k} \leq b_k$ and both endpoints converge to L, the squeeze theorem gives $x_{n_k} \to L$.

Remark 1.1.84 (Comparison with the peak proof)*.* The peak proof is combinatorial: it operates on the values of the sequence directly and produces a monotone subsequence without requiring boundedness. Convergence follows from MCT.

The bisection proof is geometric: it operates on intervals containing the sequence's range and produces a subsequence squeezed to a limit directly. Boundedness is required from the outset to produce the initial interval.

Both proofs use the same dichotomy, but in the peak proof it is applied to an index predicate; in the bisection proof, to an interval predicate.

Diagonal Extraction: Arzelà–Ascoli

Remark 1.1.85 (Diagonal extraction pattern). The diagonal argument generalises predicate-walking to sequences of functions. Let (f_n) be bounded and equicontinuous in $C^0([a,b],\mathbb{R})$, and let $D = \{d_1, d_2, \ldots\} = \mathbb{Q} \cap [a,b]$.

Stage 1. $(f_n(d_1))$ is bounded; extract by BW a subsequence $(f_{1,k})$ converging at d_1.
Stage 2. $(f_{1,k}(d_2))$ is bounded; extract a sub-subsequence $(f_{2,k})$ converging at d_2. As a subsequence of $(f_{1,k})$, it still converges at d_1.
Stage m. Extract $(f_{m,k})$ as a subsequence of $(f_{m-1,k})$ converging at d_m. By induction, $(f_{m,k})$ converges at $d_1, \ldots, d_m$.
Diagonal step. Define $g_m = f_{m,m}$. For any fixed j and $m \geq j$, the term $g_m = f_{m,m}$ is a term of $(f_{j,k})$ with index $k = m \geq j$. Hence $g_m(d_j) \to y_j$. The sequence (g_m) converges pointwise on all of D.

Equicontinuity then propagates pointwise convergence on D to uniform convergence on $[a,b]$ via an $\varepsilon/3$ argument.

Remark 1.1.86 (Why the diagonal works). At stage m, the predicate "converges at $d_1, \ldots, d_m$" holds for all terms of $(f_{m,k})$. The diagonal term $g_m = f_{m,m}$ is a tail term of every earlier row $f_{j,k}$ for $j \leq m$, so it satisfies the predicate for all $j \leq m$ simultaneously. Sending $m \to \infty$ gives convergence at every d_j.

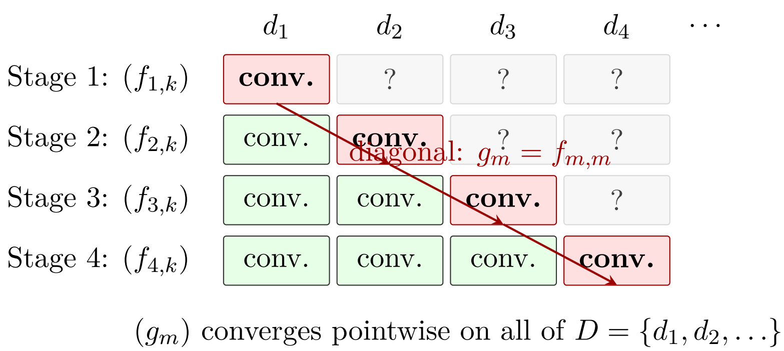


Figure 1.5: Diagonal extraction in the Arzelà–Ascoli argument. Each row $(f_{m,k})$ converges at $d_1, \ldots, d_m$ (green) and is a subsequence of the row above. The diagonal $g_m = f_{m,m}$ (red) is a tail of every row, so it converges at every d_j. Equicontinuity then extends pointwise convergence on the dense set D to uniform convergence on $[a,b]$.

Limitations of the Technique

Remark 1.1.87 (The predicate must be decidable at each stage). At each inductive step, one must determine which branch of the dichotomy applies. If the predicate requires knowledge of the limit — a quantity not yet known — the construction cannot proceed. Predicate-walking requires a finitary, combinatorial, or topological condition on the terms already seen.

Remark 1.1.88 (Multiple simultaneous predicates may exhaust the reservoir). The conjunction $P \wedge Q$ can hold only finitely often even when P and Q each hold infinitely often. The diagonal argument avoids this by handling one predicate at a time, composing via the diagonal.

Remark 1.1.89 (Subsequences, not sequences). Predicate-walking yields existence of a subsequence with the desired property, not a conclusion about the full sequence. The Monotone Subsequence Theorem does not assert (x_n) is monotone, or that the extracted subsequence is unique.

Remark 1.1.90 (Convergence requires a separate argument). The extracted subsequence has structural properties but not yet a limit. Converting structure into convergence requires a separate theorem: monotone subsequence → MCT (requires boundedness); bisection subsequence → NIP and squeeze; pointwise convergence on D → equicontinuity and $\varepsilon/3$.

Remark 1.1.91 (Quantitative control is absent). Predicate-walking proofs are pure existence arguments. They specify neither the rate of convergence nor which indices are selected. Explicit rates or constructive witnesses require a different approach.

Remark 1.1.92 (Countability is essential for the diagonal). The Arzelà–Ascoli diagonal requires a countable dense D so that points can be enumerated and handled one at a time. In non-separable metric spaces (those without a countable dense subset) the argument breaks down.

The Unifying Principle

Remark 1.1.93 (Infinite Pigeonhole as the common core). Every predicate-walking argument rests on the fact that an infinite set cannot be partitioned into two finite parts. Applied to $\mathbb{N}$: if P holds only finitely often, the P-indices have a maximum, and $\neg P$ holds from that point on. The method reduces a complex existence question to a sequence of binary choices, each justified by the Infinite Pigeonhole Principle. The construction is inductive, automatic, and never runs dry, provided the predicate is well-chosen and the reservoir is genuinely infinite at each stage.

Remark 1.1.94 (Dependency summary).

1. Monotone Subsequence Theorem — peak dichotomy alone; no completeness.
2. Bolzano–Weierstrass via MCT — Monotone Subseq. + MCT (requires completeness via AoC).
3. Bolzano–Weierstrass via bisection — NIP + squeeze (requires completeness via NIP $\equiv$ AoC).
4. Arzelà–Ascoli — BW at each diagonal stage + equicontinuity.

The Monotone Subsequence Theorem is the only result in the cluster requiring no completeness. All others depend on completeness via MCT or NIP.

1.1.5 Interval Bisection and Nested Intervals

Bisection and NIP Toolkit — Quick Reference

Label	Statement	Proof method
NIP	$\bigcap_{n=1}^{\infty} I_n \neq \emptyset$ for nested closed bounded intervals	Supremum of left endpoints
IVT	f continuous, $f(a) < L < f(b) \Rightarrow \exists\, c \in (a,b),\ f(c) = L$	Bisection + NIP
UNC	$\mathbb{R}$ is uncountable	Nested interval diagonalisation

The Nested Interval Property

Theorem (Nested Interval Property)

For each $n \in \mathbb{N}$, let $I_n = [a_n, b_n]$ be a closed bounded interval with $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$. Then $\displaystyle\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Remark 1.1.95 (Fully quantified form). $\forall n \in \mathbb{N},\ I_n = [a_n, b_n] \neq \emptyset,\ [a_{n+1}, b_{n+1}] \subseteq [a_n, b_n] \implies \exists\, x \in \mathbb{R},\ \forall n,\ x \in I_n$.

Remark 1.1.96 (Proof strategy). Let $A = \{a_n : n \in \mathbb{N}\}$. Nestedness implies every b_n is an upper bound for A. By AoC, $x = \sup A$ exists. Then $a_n \leq x \leq b_n$ for all n, so $x \in I_n$ for all n.

Remark 1.1.97 (Closedness is essential). The open intervals $(0, 1/n)$ are nested but $\bigcap (0, 1/n) = \emptyset$. Closedness ensures the endpoints can serve as witnesses in the AoC argument.

Remark 1.1.98 (Boundedness is essential). The unbounded closed intervals $[n, \infty)$ are nested but their intersection is empty. Both hypotheses are necessary.

Remark 1.1.99 (Singleton intersection). If additionally $b_n - a_n \to 0$, the intersection is a singleton $\{x\}$, since $a_n \leq x \leq b_n$ and the squeeze theorem forces $a_n \to x$ and $b_n \to x$. This strengthened form is used in most applications.

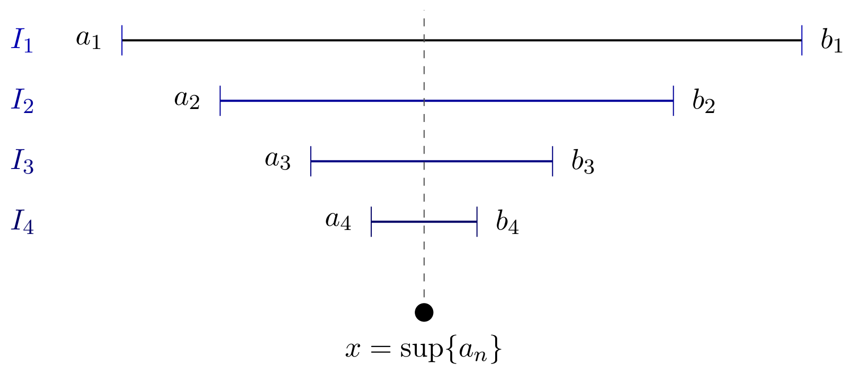


Figure 1.6: The Nested Interval Property (Section 1.1.5). Left endpoints a_n increase; right endpoints b_n decrease; each b_n is an upper bound for $\{a_n\}$. By the Axiom of Completeness, $x = \sup\{a_n\}$ exists and satisfies $a_n \leq x \leq b_n$ for all n, so $x \in \bigcap I_n$.

Remark 1.1.100 (Equivalence with completeness). NIP is equivalent to AoC over the Archimedean ordered field axioms. In particular, NIP fails over $\mathbb{Q}$: the intervals $[1, \sqrt{2}\,] \cap \mathbb{Q}$ shrink to a gap that $\mathbb{Q}$ cannot fill.

The Bisection Technique

Definition (Bisection Sequence)

Let $I_0 = [a_0, b_0]$ carry a property P. The **bisection sequence** (I_n) is defined inductively: given $I_n = [a_n, b_n]$, let $m_n = (a_n + b_n)/2$ and set

$$I_{n+1} = \begin{cases} [a_n, m_n] & \text{if } [a_n, m_n] \text{ inherits } P, \\ [m_n, b_n] & \text{otherwise.} \end{cases}$$

Remark 1.1.101 (Fully quantified form). $I_0 = [a_0, b_0];\ \forall n,\ I_{n+1} \subseteq I_n;\ |I_n| = (b_0 - a_0)/2^n$.

Remark 1.1.102 (Length control). Bisection gives automatic length control: $|I_n| = (b_0 - a_0)/2^n \to 0$. No separate argument for the length condition of NIP is required.

Remark 1.1.103 (Abstract template). Every bisection proof has the same five-step shape:

1. *Initialize.* Identify P and starting interval I_0.
2. *Bisect.* Check which half inherits P.
3. *Extract.* Apply NIP to obtain $x \in \bigcap I_n$; note $a_n \to x$ and $b_n \to x$.
4. *Conclude.* Pass P through the limit, using continuity, MCT, or squeeze as appropriate.
5. *Verify hypotheses.* Confirm intervals are closed and nested; length control is free from bisection.

Remark 1.1.104 (Connection to predicate-walking). The bisection technique is a special case of predicate-walking: the predicate $P(I)$ = "interval I inherits the relevant property" is propagated at each stage by bisecting and keeping the half that satisfies P. The infinite-pigeonhole dichotomy appears in the Bolzano–Weierstrass proof (Proposition 1.1.82) but not in the IVT proof, where neither half is discarded based on cardinality — instead the sign-test determines the choice deterministically.

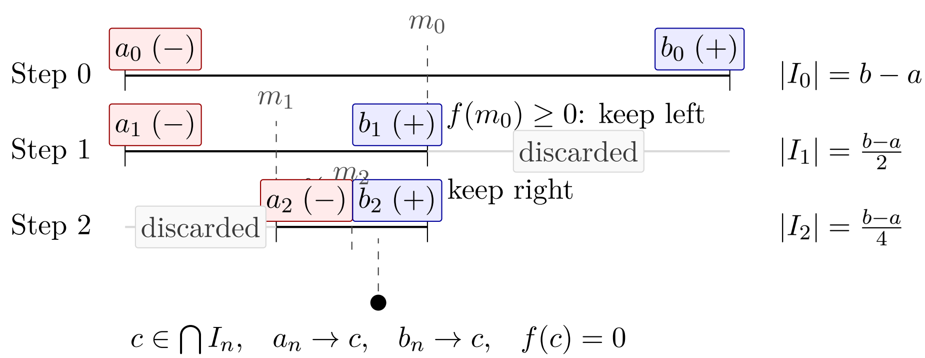


Figure 1.7: Three steps of the bisection proof of the Intermediate Value Theorem (Proposition 1.1.105) for $f(a) < 0 < f(b)$. At each step the half preserving a sign change is retained and the other discarded; lengths halve as $(b - a)/2^n \to 0$. By NIP, $c \in \bigcap I_n$ exists uniquely; continuity forces $f(c) = 0$.

The Intermediate Value Theorem via Bisection

Proposition 1.1.105 (Intermediate Value Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. If $L \in \mathbb{R}$ satisfies $f(a) < L < f(b)$ or $f(a) > L > f(b)$, then there exists $c \in (a, b)$ with $f(c) = L$. Go to proof.*

Remark.

Remark 1.1.106 (Proof strategy — bisection approach). Reduce to $L = 0$, $f(a) < 0 < f(b)$ (replace f by $f - L$).

Property to propagate. $f(a_n) < 0$ and $f(b_n) \geq 0$ — a sign change across the interval.
Branching rule. Let $m_n = (a_n + b_n)/2$: if $f(m_n) \geq 0$ set $[a_{n+1}, b_{n+1}] = [a_n, m_n]$; otherwise set $[a_{n+1}, b_{n+1}] = [m_n, b_n]$. The sign-change property is preserved in both cases.
Closing argument. NIP yields $c \in \bigcap I_n$ with $a_n \to c$ and $b_n \to c$. Since $f(a_n) < 0$ for all n: continuity gives $f(c) \leq 0$. Since $f(b_n) \geq 0$ for all n: continuity gives $f(c) \geq 0$. Therefore $f(c) = 0$.

Remark 1.1.107 (Two completeness paths). IVT admits two proofs from completeness:

1. *AoC path.* Set $K = \{x \in [a, b] : f(x) \leq 0\}$, $c = \sup K$; rule out $f(c) > 0$ and $f(c) < 0$ using the LUB property.

2. *NIP path.* Bisection as above.

The two paths are equivalent over the Archimedean ordered field axioms.

Remark 1.1.108 (Applied interpretation). The bisection construction is directly algorithmic: it locates a root to any desired precision in finitely many steps. After n bisections the root is known to within $(b - a)/2^n$. This is the bisection method of numerical analysis.

Uncountability of $\mathbb{R}$ via Nested Intervals

Proposition 1.1.109. $\mathbb{R}$ *is uncountable. Go to proof.*

Remark.

Remark 1.1.110 (Proof strategy — nested interval diagonalisation)*.* Suppose for contradiction $\mathbb{R} = \{x_1, x_2, \ldots\}$. Build nested intervals (I_n) inductively with $I_{n+1} \subseteq I_n$ and $x_n \notin I_{n+1}$: at each step I_n contains two disjoint closed subintervals; x_{n+1} lies in at most one, so choose the other.

By NIP, $\bigcap I_n \neq \emptyset$: pick $y \in \bigcap I_n$. Then $y \neq x_n$ for every n (since $y \in I_n$ but $x_n \notin I_n$), contradicting the assumption that $\{x_n\}$ enumerates $\mathbb{R}$.

Remark 1.1.111 (Relation to Cantor's diagonal argument)*.* This is the geometric form of Cantor's diagonalisation: instead of altering a decimal digit, the nth real is excluded from the nth nested interval. The witness y is the interval analogue of the "diagonal" real.

Summary: When to Use Each Technique

Remark 1.1.112 (Bisection vs. general nested intervals)*.*

Feature	**Bisection**
How built	Halving at midpoints
Length control	Automatic: $(b_0 - a_0)/2^n$
Typical use	Property propagation via sig
Canonical examples	IVT, BW
Guarantor	NIP $\equiv$ AoC

Remark 1.1.113 (Completeness is indispensable)*.* Neither technique works over $\mathbb{Q}$. The bisection procedure for $f(x) = x^2 - 2$ on $[1, 2]$ runs mechanically over $\mathbb{Q}$, but the limiting point $\sqrt{2}$ is not in $\mathbb{Q}$. Every bisection proof is, at bottom, an appeal to the completeness of $\mathbb{R}$.

Remark 1.1.114 (Cross-section of the chapter)*.* The four sections of this chapter are not independent: each builds on the last. Inequality bounding (§1.1.2) requires only the ordered field axioms. Completeness as a constructive engine (§1.1.3) requires AoC and the Archimedean Property. Predicate-walking (§1.1.4) requires completeness via MCT or NIP, depending on the variant. Bisection and NIP (§1.1.5) is the geometric synthesis: it encapsulates completeness in a form that drives the IVT, BW, and uncountability proofs simultaneously, connecting back to both the inductive-selection constructions of §1.1.3 and the predicate-walking patterns of §1.1.4.

1.1.6 Subsequence Partition via Residue Classes

Residue Partition Toolkit — Quick Reference

Tool	**Statement**	**Use**
k-Periodicity Criterion	$(a_n) \to L \iff \forall r,\ (a_{kn+r}) \to L$	Prove convergence by analysing k c
Forward direction	Subseqs. of convergent seqs. converge to L	Reduce to subsequence analysis
Divergence corollary	Two classes $\to$ different limits $\Rightarrow (a_n)$ diverges	Detect divergence by comparing res
Index translation	$m = kn + r \Rightarrow n \geq N_r$ once $m \geq k \max_r N_r$	Core of the backward direction

The Residue-Class Partition

For any sequence (a_n) and any integer $k \geq 2$, the division algorithm partitions $\mathbb{N}$ into k disjoint residue classes:

$$\mathbb{N} = \bigsqcup_{r=0}^{k-1} \{kn + r : n \geq 0\}.$$

Each class defines a subsequence: $(a_{kn+r})_{n\geq 0}$ selects every kth term of (a_n) beginning at position r. These k subsequences partition (a_n) exactly — every term appears in precisely one class, order within each class is preserved, and their union recovers (a_n).

Remark 1.1.115 (Fully quantified form of the partition). $\forall n \in \mathbb{N},\ \exists!\, r \in \{0, \ldots, k-1\},\ \exists!\, q \in \mathbb{N},\ n = kq + r$. Hence $\{(a_{kn+r})_{n\geq 0} : r = 0, \ldots, k-1\}$ is a partition of (a_n) into k subsequences.

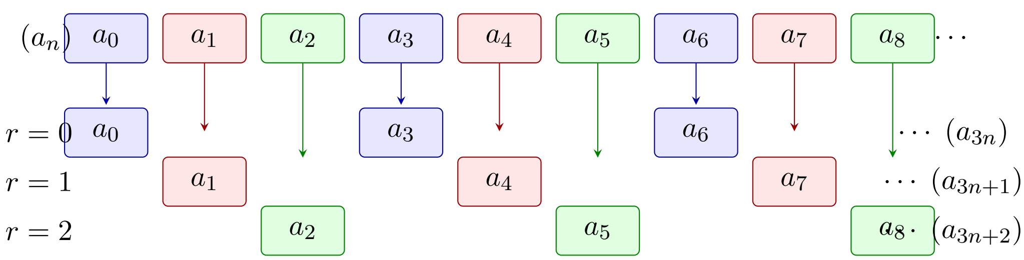


Figure 1.8: The residue-class partition of (a_n) with modulus $k = 3$. Each term is coloured by residue class: blue ($r = 0$), red ($r = 1$), green ($r = 2$). The partition is exact; the k-Periodicity Criterion (Section 1.1.6) states that $(a_n) \to L$ if and only if each coloured subsequence converges to the same L.

The k-Periodicity Convergence Criterion

Theorem (k-Periodicity Convergence Criterion)

Let (a_n) be a sequence, $k \geq 2$ an integer, and $L \in \mathbb{R}$. The following are equivalent:

(i) $(a_n) \to L$.

(ii) For every $r \in \{0, 1, \ldots, k-1\}$, the subsequence $(a_{kn+r}) \to L$.

Remark 1.1.116 (Fully quantified form). $(i) \Leftrightarrow (ii)$, where:
(i): $\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall m \geq N,\ |a_m - L| < \varepsilon$.
(ii): $\forall r \in \{0, \ldots, k-1\},\ \forall \varepsilon > 0,\ \exists N_r \in \mathbb{N},\ \forall n \geq N_r,\ |a_{kn+r} - L| < \varepsilon$.

Remark 1.1.117 (Proof strategy). **Forward direction** $(i) \Rightarrow (ii)$. Each (a_{kn+r}) is a subsequence of (a_n). Subsequences of a convergent sequence converge to the same limit (Proposition 1.1.39).

Backward direction $(ii) \Rightarrow (i)$. The operative direction. Given $\varepsilon > 0$, for each residue r let N_r satisfy $|a_{kn+r} - L| < \varepsilon$ for all $n \geq N_r$. Set $N^* = k \cdot \max_r N_r$. For any $m \geq N^*$, write $m = kn + r$ via the division algorithm. Then:

$$n = \frac{m-r}{k} \geq \frac{N^* - (k-1)}{k} \geq \frac{k \max_r N_r - (k-1)}{k} \geq N_r,$$

so $|a_m - L| = |a_{kn+r} - L| < \varepsilon$.

Remark 1.1.118 (The index translation). The key step converts a global index m to a local index n within its residue class via $m = kn + r$. The factor k in the definition of N^* absorbs the offset $r \leq k-1$ from every class simultaneously, ensuring that $n \geq N_r$ holds for whichever class m lands in.

The Divergence Corollary

Proposition 1.1.119 (Residue divergence criterion). *If there exist residues $r, s \in \{0, \ldots, k-1\}$ such that $(a_{kn+r}) \to L$ and $(a_{kn+s}) \to M$ with $L \neq M$, then (a_n) diverges. Go to proof.*

Remark.

Remark 1.1.120 (Fully quantified form). $(a_{kn+r}) \to L \ \wedge\ (a_{kn+s}) \to M \ \wedge\ L \neq M \ \Rightarrow$ (a_n) diverges.

Remark 1.1.121 (Proof strategy). Contrapositively: if $(a_n) \to L$, then every subsequence converges to L (forward direction of Section 1.1.6). If two residue-class subsequences converge to different limits, (a_n) cannot converge.

Remark 1.1.122 (Canonical example). The sequence $((-1)^n)$ has $a_{2n} = 1 \to 1$ and $a_{2n+1} = -1 \to -1$. Since $1 \neq -1$, the sequence diverges — established immediately without any ε-N computation.

The Alternating Series Test via Even/Odd Partition

The $k = 2$ case of Section 1.1.6 drives the cleanest proof of the Alternating Series Test.

Theorem (Alternating Series Test)

Let (a_n) be a positive decreasing sequence with $a_n \to 0$. Then the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges.

Remark 1.1.123 (Fully quantified form). $(\forall n,\ a_n > 0) \ \wedge\ (\forall n,\ a_{n+1} \leq a_n) \ \wedge\ (a_n \to 0) \ \Rightarrow$ $\exists L,\ \sum_{n=1}^{\infty} (-1)^{n+1} a_n = L$.

Remark 1.1.124 (Proof strategy — even/odd partition). Let s_n denote the nth partial sum. Apply Section 1.1.6 with $k = 2$: it suffices to show $(s_{2k}) \to L$ and $(s_{2k+1}) \to L$ for the same L.

Even subsequence (s_{2k}): decreasing and bounded. Group consecutive pairs:

$$s_{2k} = \sum_{j=1}^{k} (-a_{2j-1} + a_{2j}).$$

Since (a_n) is decreasing, each summand $-a_{2j-1} + a_{2j} \le 0$, so (s_{2k}) is decreasing. Also:

$$s_{2k} = -a_1 + \sum_{j=1}^{k-1} (a_{2j} - a_{2j+1}) + a_{2k} \ge -a_1,$$

since each term in the sum is non-negative. Thus (s_{2k}) is decreasing and bounded below by $-a_1$. By MCT, $s_{2k} \to L$ for some $L \ge -a_1$.

Odd subsequence (s_{2k+1}): linked to the even. Observe $s_{2k+1} = s_{2k} + a_{2k+1}$. Since $s_{2k} \to L$ and $a_{2k+1} \to 0$, the Algebraic Limit Theorem gives $s_{2k+1} \to L$.

Applying the criterion. Both residue-class subsequences converge to the same L. By Section 1.1.6 with $k = 2$, $(s_n) \to L$.

Remark 1.1.125 (Why the partition is the right tool). The even and odd partial sums have structurally different characters: the even sums are monotone (enabling MCT), the odd sums are then controlled via the algebraic relationship $s_{2k+1} = s_{2k} + a_{2k+1}$. No single uniform argument handles both. The partition separates the two cases into independent analyses, then the criterion reassembles the conclusion.

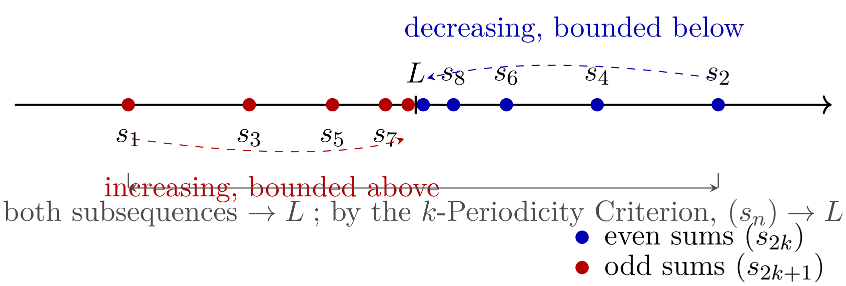


Figure 1.9: Even and odd partial sums of an alternating series $\sum(-1)^{n+1}a_n$ with (a_n) positive, decreasing, and tending to zero. Even sums (s_{2k}) (blue) decrease and are bounded below; by MCT they converge to L. Odd sums (s_{2k+1}) (red) satisfy $s_{2k+1} = s_{2k} + a_{2k+1} \to L$. Since both residue classes (mod $k = 2$) converge to L, the k-Periodicity Criterion (Section 1.1.6) gives $(s_n) \to L$.

Remark 1.1.126 (Alternative proofs). The Alternating Series Test admits three proofs from different completeness avatars:

1. *Cauchy criterion* — show (s_n) is eventually ε-steady directly.

2. *NIP* — the nested intervals $[s_{2k}, s_{2k-1}]$ shrink to a point by the Nested Interval Property.

3. *MCT + residue partition* — as above.

The partition proof is the most transparent structurally: the monotone behaviour and the limit identity are isolated into the two residue classes rather than handled simultaneously.

Higher k and Choosing the Right Period

For sequences with period $k > 2$, the criterion applies directly. The sequence $a_n = \cos(2\pi n/3)$ takes values $1, -\frac{1}{2}, -\frac{1}{2}, 1, -\frac{1}{2}, -\frac{1}{2}, \ldots$, cycling with period 3. Partitioning modulo 3:

$$\begin{aligned} a_{3n} &= 1 \to 1, \\ a_{3n+1} &= -\tfrac{1}{2} \to -\tfrac{1}{2}, \\ a_{3n+2} &= -\tfrac{1}{2} \to -\tfrac{1}{2}. \end{aligned}$$

The three limits differ ($1 \neq -\frac{1}{2}$), so the sequence diverges by Proposition 1.1.119.

Remark 1.1.127 (Choosing k to match the period)*.* The right choice of k is the natural period of the sequence's structure. Using $k = 2$ for a period-3 sequence produces residue classes that are themselves period-3 subsequences; convergence of each class then requires a further partition with $k = 3$. Choosing k to match the period compresses the argument to a single application of the criterion.

Remark 1.1.128 (Applying the criterion to prove convergence)*.* To prove $(a_n) \to L$ via the criterion, one must:

1. Select k matching the period of the sequence's behaviour.

2. Prove $(a_{kn+r}) \to L$ for each $r \in \{0, \ldots, k-1\}$, using whatever tool fits that residue class.

3. Invoke Section 1.1.6 to conclude $(a_n) \to L$.

The method does not produce L — it confirms a candidate. Identifying L requires separate work (direct computation, MCT applied to a class, etc.) before the criterion is invoked.

Contrast with Predicate-Walking

Remark 1.1.129 (Dual logical directions)*.* The residue partition technique and predicate-walking (§1.1.4) operate in opposite logical directions.

Feature	**Predicate-walking**	**Residue partition**
Flow	Whole $\to$ one part	All parts $\to$ whole
Goal	Property of extracted subseq.	Convergence of full (a_n)
Coverage	One "good" subsequence	All k residue classes
Divergence use	Two classes, different limits	Same: two classes, different limits
Completeness role	Via MCT or NIP	Via MCT or algebraic limit theorem

Predicate-walking extracts what is needed; the residue partition reassembles from all pieces. The divergence criterion is available in both frameworks: two subsequences with different limits immediately imply divergence.

Limitations

Remark 1.1.130 (The same-limit condition is non-negotiable). The backward direction of Section 1.1.6 requires all k residue-class subsequences to converge to the *same* L. There is no weakening: a convergent sequence has all its subsequences converging to the identical limit, so any two classes converging to different limits already certify divergence. The technique cannot produce convergence from mixed limits.

Remark 1.1.131 (The limit must be known in advance). The criterion confirms a candidate limit L; it does not produce L from scratch. In practice, L is identified by direct computation on one residue class (e.g., the even partial sums converge to L by MCT), then the criterion assembles global convergence.

Remark 1.1.132 (Exact periodicity is required). The technique applies when the sequence's qualitative behaviour repeats with exact period k in the index. For sequences whose oscillation decays without exact periodicity, the residue classes are not individually tractable, and a direct ε-N argument or a squeeze is more appropriate.

Remark 1.1.133 (Common error). Applying the criterion with k smaller than the true period, then concluding convergence after verifying only the even residue class. If the odd class is not verified, the conclusion does not follow. Every residue class must be checked.

Remark 1.1.134 (Forward connections). The residue partition technique reappears in:

- **Cesàro means** — the Cesàro average of a sequence (a_n) can be analysed by partitioning the partial sums of the average into residue classes.
- **Fourier series** — the convergence of partial sums of a Fourier series is often established by analysing even and odd partial sums (or partial sums along specific subsequences of the approximation order).
- **Recursively defined sequences** — when a recurrence has period k, the residue-class subsequences each satisfy a simpler recurrence, enabling separate analysis.

1.2 Asymptotic Notation

Asymptotic Notation — Quick Reference

Concept/Statement	Meaning/Role
Little-o at a point	One function is negligible relative to another near a point.
Little-o at infinity	Negligibility along an unbounded input range.
Increment little-o	The local error notation $r(h) = o(h)$ as $h \to 0$.
Quotient criterion	Little-o is detected by a quotient tending to 0.
Sum rule	Negligible errors add.
Bounded factor rule	Bounded multipliers preserve negligible errors.

Little-o Notation

Asymptotic notation separates the main term of an expression from the part that becomes negligible in a limiting process. The notation $f = o(g)$ means that f is smaller than every fixed multiple of g near the limiting point. It does not say that f is zero. It says that f is negligible relative to the chosen scale g.

Definition 1.2.1 (Little-o at a Point)**.** Let $a \in \mathbb{R}$, let f and g be real-valued functions defined on a punctured neighbourhood of a, and suppose $g(x) \neq 0$ for all x sufficiently close to a with $x \neq a$. We write

$$f(x) = o(g(x)) \qquad (x \to a)$$

if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x)| \leq \varepsilon |g(x)|.$$

Remark (Standard quantified statement)*.*

$$f(x) = o(g(x)) \ (x \to a) \Longleftrightarrow \forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \\ \big(0 < |x - a| < \delta \Rightarrow |f(x)| \leq \varepsilon |g(x)|\big).$$

Remark (Interpretation)*.* The scale g is the unit of comparison. The assertion $f = o(g)$ says that no matter how small a tolerance multiple εg is chosen, f eventually fits inside that multiple.

Definition 1.2.2 (Little-o at Infinity)**.** Let f and g be real-valued functions defined on some interval (M, ∞), and suppose $g(x) \neq 0$ for all sufficiently large x. We write

$$f(x) = o(g(x)) \qquad (x \to \infty)$$

if and only if for every $\varepsilon > 0$ there exists $R > 0$ such that

$$x > R \quad \Longrightarrow \quad |f(x)| \leq \varepsilon |g(x)|.$$

Remark (Standard quantified statement)*.*

$$f(x) = o(g(x)) \ (x \to \infty) \Longleftrightarrow \forall \varepsilon > 0 \ \exists R > 0 \ \forall x \\ \big(x > R \Rightarrow |f(x)| \leq \varepsilon |g(x)|\big).$$

Remark (Interpretation). The point version and the infinity version have the same logical shape. Only the neighbourhood changes: small punctured intervals around a are replaced by tails (R, ∞).

Definition 1.2.3 (Increment Little-o)**.** Let r be a real-valued function defined for all sufficiently small nonzero h. We write

$$r(h) = o(h) \qquad (h \to 0)$$

if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |h| < \delta \quad \Longrightarrow \quad |r(h)| \le \varepsilon |h|.$$

Remark (Standard quantified statement).

$$r(h) = o(h)\ (h \to 0) \Longleftrightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall h\ \big(0 < |h| < \delta \Rightarrow |r(h)| \le \varepsilon |h|\big).$$

Remark (Interpretation). This is the form used in first-order approximation. A remainder $r(h)$ is smaller than the increment h itself, so the quotient $r(h)/h$ tends to zero.

Remark (Dependencies).

- Little-o at a Point

Proposition 1.2.4 (Little-o Quotient Characterisation)**.** *Let $a \in \mathbb{R}$, and let f and g be real-valued functions defined on a punctured neighbourhood of a, with $g(x) \neq 0$ for x sufficiently close to a and $x \neq a$. Then*

$$f(x) = o(g(x))\ (x \to a) \quad \Longleftrightarrow \quad \frac{f(x)}{g(x)} \to 0\ (x \to a).$$

Go to proof.

Remark (Standard quantified statement).

$$[\forall \varepsilon > 0\ \exists \delta > 0\ 0 < |x - a| < \delta \Rightarrow |f(x)| \le \varepsilon |g(x)|] \Longleftrightarrow \frac{f(x)}{g(x)} \to 0.$$

Remark (Interpretation). The quotient form is often the fastest test for little-o. The epsilon form is often the safest proof form because it avoids manipulating a quotient until the nonvanishing of the scale has been checked.

Remark (Dependencies).

- Little-o at a Point

Proposition 1.2.5 (Sum Rule for Little-o)**.** *If $f_1(x) = o(g(x))$ and $f_2(x) = o(g(x))$ as $x \to a$, then*

$$f_1(x) + f_2(x) = o(g(x)) \qquad (x \to a).$$

Go to proof.

Remark (Standard quantified statement).

$$f_1 = o(g) \wedge f_2 = o(g) \implies f_1 + f_2 = o(g).$$

Remark (Interpretation). Negligible errors can be added. The proof is the usual epsilon-budget split: make each error smaller than half the final tolerance.

Remark (Dependencies).

- Little-o at a Point

Proposition 1.2.6 (Bounded Factor Rule for Little-o). *If $f(x) = o(g(x))$ as $x \to a$ and m is bounded near a, then*

$$m(x)f(x) = o(g(x)) \qquad (x \to a).$$

Go to proof.

Remark (Standard quantified statement).

$$f = o(g) \wedge \exists M > 0\ \exists \eta > 0\ \forall x \left(0 < |x - a| < \eta \Rightarrow |m(x)| \le M\right) \implies mf = o(g).$$

Remark (Interpretation). A bounded multiplier cannot turn a negligible error into a main-order term. It only changes the constant in the epsilon estimate.

Remark (Dependencies).

- Little-o at a Point

1.3 Inequality

Inequality — Quick Reference

Concept/Statement	**Meaning/Role**
Add both sides	Adding the same term preserves strict inequalities.
Nonstrict add both sides	Addition also preserves nonstrict inequalities.
Add inequalities	Pairwise strict inequalities add.
Mixed addition	Strict and nonstrict hypotheses combine predictably.
Multiply by positive	Positive factors preserve order.
Multiply by negative	Negative factors reverse order.
Squeeze theorem	Double bounding by one value forces equality.
Reciprocal order	Reciprocals reverse positive inequalities.
Square monotonicity	Squaring preserves order on nonnegative inputs.
AM–GM	Arithmetic mean dominates geometric mean.
Bernoulli inequality	Powers of $1 + x$ dominate the tangent line.

Inequality

Theorem 1.3.1 (Addition Preserves Strict Inequality)**.** *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \;\Rightarrow\; a + c < b + c.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a < b \;\Rightarrow\; a + c < b + c.$$

Remark (Dependencies)*.*
Ordered field, ordered field axioms, additive cancellation.

Theorem 1.3.2 (Addition Preserves Nonstrict Inequality)**.** *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \;\Rightarrow\; a + c \leq b + c.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \;\Rightarrow\; a + c \leq b + c.$$

Remark (Dependencies)*.*
Ordered field, addition preserves strict inequality.

Theorem 1.3.3 (Addition of Strict Inequalities)**.** *For all* $a, b, c, d \in \mathbb{R}$*,*

$$a < b \;\wedge\; c < d \;\Rightarrow\; a + c < b + d.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c, d \in \mathbb{R}, \quad a < b \;\wedge\; c < d \;\Rightarrow\; a + c < b + d.$$

Remark (Dependencies)*.*
addition preserves strict inequality, transitivity of strict inequality.

Theorem 1.3.4 (Addition of Nonstrict Inequalities)**.** *For all* $a, b, c, d \in \mathbb{R}$*,*

$$a \leq b \;\wedge\; c \leq d \;\Rightarrow\; a + c \leq b + d.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c, d \in \mathbb{R}, \quad a \leq b \;\wedge\; c \leq d \;\Rightarrow\; a + c \leq b + d.$$

Remark (Dependencies)*.*
addition preserves nonstrict inequality.

Theorem 1.3.5 (Mixed Addition of Inequalities)**.** *For all* $a, b, c, d \in \mathbb{R}$*,*

$$a \leq b \ \wedge \ c < d \ \Rightarrow \ a + c < b + d.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c, d \in \mathbb{R}, \quad a \leq b \ \wedge \ c < d \ \Rightarrow \ a + c < b + d.$$

Remark (Dependencies)*.*

addition preserves nonstrict inequality, addition preserves strict inequality, mixed transitivity.

Theorem 1.3.6 (Multiplication by a Positive Scalar Preserves Strict Inequality)**.** *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \ \wedge \ 0 < c \ \Rightarrow \ ac < bc.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a < b \ \wedge \ 0 < c \ \Rightarrow \ ac < bc.$$

Remark (Dependencies)*.*

Ordered field, multiply positive, positive cone.

Theorem 1.3.7 (Multiplication by a Negative Scalar Reverses Strict Inequality)**.** *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \ \wedge \ c < 0 \ \Rightarrow \ ac > bc.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a < b \ \wedge \ c < 0 \ \Rightarrow \ ac > bc.$$

Remark (Dependencies)*.*

Ordered field, multiply negative, positive cone.

Theorem 1.3.8 (Multiplication by a Positive Scalar Preserves Nonstrict Inequality)**.** *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \ \wedge \ 0 < c \ \Rightarrow \ ac \leq bc.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \ \wedge \ 0 < c \ \Rightarrow \ ac \leq bc.$$

Remark (Dependencies)*.*

multiplication by positive preserves strict inequality.

Theorem 1.3.9 (Multiplication by a Nonneg Scalar Preserves Nonstrict Inequality)**.** *For all $a, b, c \in \mathbb{R}$,*

$$a \le b \ \wedge \ 0 \le c \ \Rightarrow \ ac \le bc.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a \le b \ \wedge \ 0 \le c \ \Rightarrow \ ac \le bc.$$

Remark (Interpretation)*.* This extends the positive-scalar result to include $c = 0$, where both sides become 0 and the inequality holds trivially.

Remark (Dependencies)*.*

multiplication by positive preserves nonstrict inequality, ordered field.

Theorem (Squeeze)

Theorem 1.3.10 (Squeeze)**.** *For all $a, b, c \in \mathbb{R}$,*

$$a \le b \le c \ \wedge \ a = c \ \Rightarrow \ b = a.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a \le b \le c \ \wedge \ a = c \ \Rightarrow \ b = a.$$

Remark (Interpretation)*.* The squeeze principle forces equality when a value is simultaneously bounded above and below by equal quantities. The continuous analogue (if $f(x) \le g(x) \le h(x)$ and $f(x) \to L$ and $h(x) \to L$ then $g(x) \to L$) is the central tool for computing limits. The algebraic version stated here is its foundation.

Remark (Dependencies)*.*

Ordered field, additive cancellation.

Theorem 1.3.11 (Transitivity of Strict Inequality)**.** *For all $a, b, c \in \mathbb{R}$,*

$$a < b \ \wedge \ b < c \ \Rightarrow \ a < c.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a < b \ \wedge \ b < c \ \Rightarrow \ a < c.$$

Remark (Dependencies)*.*

Ordered field, positive cone.

Theorem 1.3.12 (Mixed Transitivity)**.** *For all $a, b, c \in \mathbb{R}$,*

$$a \le b \ \wedge \ b < c \ \Rightarrow \ a < c.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \wedge b < c \Rightarrow a < c.$$

Remark (Dependencies).
Ordered field, transitivity of strict inequality.

Theorem 1.3.13 (Reciprocal Reverses Strict Inequality for Positives)**.** *For all $a, b \in \mathbb{R}$,*

$$0 < a < b \Rightarrow 0 < \frac{1}{b} < \frac{1}{a}.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad 0 < a < b \Rightarrow 0 < \tfrac{1}{b} < \tfrac{1}{a}.$$

Remark (Dependencies).
Ordered field, inverse positive, reciprocal order for positive elements, multiplication by positive preserves strict inequality.

Theorem 1.3.14 (Reciprocal Equivalence for Positives)**.** *For all $a, b \in \mathbb{R}$ with $a > 0$ and $b > 0$,*

$$a < b \iff \frac{1}{b} < \frac{1}{a}.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R},\ a > 0 \wedge b > 0 \Rightarrow \big(a < b \iff \tfrac{1}{b} < \tfrac{1}{a}\big).$$

Remark (Dependencies).
reciprocal reverses strict inequality for positives, reciprocal order for positive elements.

Theorem 1.3.15 (Square Is Strictly Monotone on Nonneg Reals)**.** *For all $a, b \in \mathbb{R}$,*

$$0 \leq a < b \Rightarrow a^2 < b^2.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad 0 \leq a < b \Rightarrow a^2 < b^2.$$

Remark (Dependencies).
Ordered field, addition of strict inequalities, multiplication by positive preserves strict inequality, squares are nonnegative.

Theorem 1.3.16 (Square Root Is Monotone on Nonneg Reals). *For all $a, b \in \mathbb{R}$,*

$$0 \le a \le b \Rightarrow \sqrt{a} \le \sqrt{b}.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad 0 \le a \le b \Rightarrow \sqrt{a} \le \sqrt{b}.$$

Remark (Audit note). This theorem depends on `def:square-root`, which does not yet appear in the repository label inventory. That definition must be created and registered before this result can be fully formalised.

Remark (Dependencies).

`def:square-root` (not yet registered — see audit note above), square is strictly monotone on nonneg reals.

Theorem 1.3.17 (AM-GM Inequality for Two Terms). *For all $a, b \ge 0$,*

$$\sqrt{ab} \le \frac{a+b}{2}.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \ a \ge 0 \ \wedge \ b \ge 0 \Rightarrow \sqrt{ab} \le \frac{a+b}{2}.$$

Remark (Interpretation). The arithmetic–geometric mean inequality for two terms is the simplest case of the AM-GM family. The standard proof proceeds by squaring both sides (using square monotonicity) and reducing to $(\sqrt{a} - \sqrt{b})^2 \ge 0$, which follows from squares being nonnegative.

Remark (Dependencies).

squares are nonnegative, square is strictly monotone on nonneg reals, `def:square-root` (not yet registered — see audit note in square root monotonicity).

Theorem (Bernoulli's Inequality)

Theorem 1.3.18 (Bernoulli's Inequality). *For all $x \in \mathbb{R}$ with $x \ge -1$ and all $n \in \mathbb{N}$,*

$$(1+x)^n \ge 1 + nx.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R}, \ \forall n \in \mathbb{N}, \ x \ge -1 \Rightarrow (1+x)^n \ge 1 + nx.$$

Remark (Interpretation). Bernoulli's inequality is proved by induction and is a key estimate in analysis: it provides a linear lower bound for exponential growth and is used to establish the divergence of the harmonic series, bounds on compound interest, and properties of the exponential function.

Remark (Dependencies).

Ordered field, multiplication by nonneg preserves nonstrict inequality, addition preserves nonstrict inequality.

1.4 Modulus

Modulus — Quick Reference

Concept/Statement	**Meaning/Role**
Absolute value	Magnitude stripped of sign.
Nonnegativity	Modulus is always nonnegative.
Zero characterisation	Only 0 has modulus 0.
Self or negation	$\lvert a\rvert$ is either a or $-a$.
Symmetry	Negation does not change magnitude.
Product rule	Modulus is multiplicative.
Quotient rule	Division respects modulus when the denominator is nonzero.
Bounds by modulus	a lies between $-\lvert a\rvert$ and $\lvert a\rvert$.
Nonstrict modulus inequality	$\lvert a\rvert \le r$ is equivalent to a two-sided bound.
Strict modulus inequality	$\lvert a\rvert < r$ is equivalent to a strict two-sided bound.
Reverse triangle inequality	Magnitudes vary no faster than distance.
Finite triangle inequality	The modulus of a finite sum is bounded by summed moduli.

Modulus

Definition (Absolute Value)

Definition 1.4.1 (Absolute Value)**.** Let $a \in \mathbb{R}$. The *absolute value* (or *modulus*) of a is

$$|a| \;:=\; \begin{cases} a & \text{if } a \ge 0, \\ -a & \text{if } a < 0. \end{cases}$$

Equivalently, $|a| := \max(a, -a)$.

Remark (Standard quantified statement).

$$|a| \;=\; \max(a,\, -a) \;:=\; \begin{cases} a & \text{if } a \ge 0, \\ -a & \text{if } a < 0. \end{cases}$$

Remark (Interpretation). The absolute value measures magnitude independent of sign. On the real line it equals the distance from a to 0: $|a| = d(a, 0)$ under the standard metric. The two-branch definition partitions $\mathbb{R}$ at 0 and reflects negative values to positive ones.

Remark (Dependencies).
Ordered field, $\mathbb{R}$ is an ordered field.

Theorem 1.4.2 (Absolute Value Is Nonnegative)**.** *For all* $a \in \mathbb{R}$*,*

$$|a| \geq 0.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{R}, \quad |a| \geq 0.$$

Remark (Dependencies).
Absolute value, squares are nonnegative, ordered field.

Theorem 1.4.3 (Absolute Value Zero Iff Zero)**.** *For all* $a \in \mathbb{R}$*,*

$$|a| = 0 \iff a = 0.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{R}, \quad |a| = 0 \iff a = 0.$$

Remark (Dependencies).
Absolute value, absolute value is nonnegative, ordered field.

Theorem 1.4.4 (Absolute Value Equals Self or Negation)**.** *For all* $a \in \mathbb{R}$*, either* $|a| = a$ *or* $|a| = -a$*.*

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{R}, \quad |a| = a \ \lor \ |a| = -a.$$

Remark (Dependencies).
Absolute value.

Theorem 1.4.5 (Absolute Value Is Symmetric)**.** *For all* $a \in \mathbb{R}$*,*

$$|-a| = |a|.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{R}, \quad |-a| = |a|.$$

Remark (Dependencies).

Absolute value, double negation.

Theorem 1.4.6 (Absolute Value of a Product)**.** *For all $a, b \in \mathbb{R}$,*

$$|ab| = |a|\,|b|.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b \in \mathbb{R}, \quad |ab| = |a|\,|b|.$$

Remark (Dependencies).

Absolute value, negation of product, product of negatives, squares are nonnegative.

Theorem 1.4.7 (Absolute Value of a Quotient)**.** *For all $a \in \mathbb{R}$ and $b \in \mathbb{R}$ with $b \neq 0$,*

$$\left|\frac{a}{b}\right| = \frac{|a|}{|b|}.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{R},\ \forall b \in \mathbb{R} \setminus \{0\}, \quad \left|\frac{a}{b}\right| = \frac{|a|}{|b|}.$$

Remark (Dependencies).

Absolute value, absolute value of a product, absolute value zero iff zero, uniqueness of multiplicative inverse.

Theorem 1.4.8 (Bound by Modulus)**.** *For all $a \in \mathbb{R}$,*

$$-|a| \leq a \leq |a|.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{R}, \quad -|a| \leq a \leq |a|.$$

Remark (Dependencies).

Absolute value, absolute value is nonnegative, ordered field.

Theorem (Nonstrict Modulus Inequality)

Theorem 1.4.9 (Nonstrict Modulus Inequality)**.** *For all $a \in \mathbb{R}$ and $r \geq 0$,*

$$|a| \leq r \iff -r \leq a \leq r.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R},\ \forall r \in \mathbb{R},\ r \geq 0 \ \Rightarrow\ \big(|a| \leq r \iff -r \leq a \leq r\big).$$

Remark (Interpretation). This biconditional is the primary tool for replacing a modulus inequality with a pair of simultaneous linear inequalities. It underlies the ε-δ manipulations throughout analysis: the condition $|x - a| \leq \varepsilon$ is equivalent to $a - \varepsilon \leq x \leq a + \varepsilon$.

Remark (Dependencies).
Absolute value, bound by modulus, ordered field.

Theorem 1.4.10 (Strict Modulus Inequality). *For all $a \in \mathbb{R}$ and $r > 0$,*

$$|a| < r \iff -r < a < r.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R},\ \forall r \in \mathbb{R},\ r > 0 \ \Rightarrow\ \big(|a| < r \iff -r < a < r\big).$$

Remark (Dependencies).
Absolute value, nonstrict modulus inequality, ordered field.

Theorem (Triangle Inequality)

Theorem (Triangle Inequality). *For all $a, b \in \mathbb{R}$,*

$$|a + b| \leq |a| + |b|.$$

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad |a + b| \leq |a| + |b|.$$

Remark (Interpretation). The triangle inequality is the foundational estimate of analysis. Every convergence argument, continuity proof, and metric-space result that requires bounding a sum reduces, at some step, to this inequality. The name reflects its geometric meaning: the length of one side of a triangle does not exceed the sum of the other two.

Theorem 1.4.11 (Reverse Triangle Inequality). *For all $a, b \in \mathbb{R}$,*

$$\big|\,|a| - |b|\,\big| \leq |a - b|.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad \big|\,|a| - |b|\,\big| \leq |a - b|.$$

Remark (Interpretation). The reverse triangle inequality provides a lower bound on $|a-b|$ in terms of the difference of moduli. It is used to bound the modulus of a difference from below, particularly when proving that a function is not uniformly continuous or that a sequence does not converge.

Remark (Dependencies).

Absolute value, triangle inequality, absolute value is symmetric.

Theorem 1.4.12 (Generalised Triangle Inequality). *For all $n \in \mathbb{N}$ and $a_1, \ldots, a_n \in \mathbb{R}$,*

$$\left|\sum_{k=1}^{n} a_k\right| \leq \sum_{k=1}^{n} |a_k|.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N},\ \forall a_1, \ldots, a_n \in \mathbb{R}, \quad \left|\sum_{k=1}^{n} a_k\right| \leq \sum_{k=1}^{n} |a_k|.$$

Remark (Dependencies).

Absolute value, triangle inequality.

Chapter 2

Structure of the Real Line

2.1 Real Line One Dimensional

The Line as Order Plus Distance

The real numbers carry two compatible structures at once: a linear order $<$ and a notion of distance $d(x, y) = |x - y|$. Together these make $\mathbb{R}$ a one-dimensional space — a line. This chapter develops only the metric topology of $\mathbb{R}$, not topology in general spaces. Every open set is built from intervals, and compactness is introduced only on $\mathbb{R}$, with Heine–Borel as the capstone.

The intuition is geometric: points are positions on a line, order is "left of," and distance is the length of the segment between two points. The caution to keep throughout is that $\mathbb{R}$ is special. Order and metric agree here in a way that fails on a general space, so definitions are stated in the form that survives to metric spaces and manifolds, with $\mathbb{R}$ as the first instance rather than the only one.

2.2 Order Distance Length

Distance on the Line

The absolute value of a difference turns $\mathbb{R}$ into a metric space. Length of an interval is the distance between its endpoints. These are the quantitative tools underneath every later definition of nearness.

Definition (Distance on the Real Line)

Definition 2.2.1 (Distance on the Real Line)**.** The *distance* on $\mathbb{R}$ is the map $d : \mathbb{R}\times\mathbb{R} \to \mathbb{R}$ given by

$$d(x, y) = |x - y|.$$

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}\ (d(x, y) = |x - y|).$$

Remark (Interpretation). The distance between two points is the length of the segment joining them. It is the metric that makes "close" precise and is the basis for balls, neighborhoods, and open sets.

Remark (Dependencies).

- Triangle Inequality

Definition (Length of a Bounded Interval)

Definition 2.2.2 (Length of a Bounded Interval)**.** Let $a, b \in \mathbb{R}$ with $a \le b$. The *length* of any bounded interval with endpoints a and b is

$$\ell = b - a.$$

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ \big(a \le b \Longrightarrow \ell([a, b]) = b - a = d(a, b)\big).$$

Remark (Interpretation). Length depends only on the endpoints, not on whether they are included. It is the distance $d(a, b)$ and measures the size of the basic pieces of the line.

Remark (Dependencies).

- Distance on the Real Line

Theorem (The Distance Function Is a Metric)

Theorem 2.2.3 (The Distance Function Is a Metric)**.** *The map $d(x, y) = |x-y|$ satisfies, for all $x, y, z \in \mathbb{R}$: (i) $d(x, y) \ge 0$ with $d(x, y) = 0$ iff $x = y$; (ii) $d(x, y) = d(y, x)$; and (iii) $d(x, z) \le d(x, y) + d(y, z)$. Hence $(\mathbb{R}, d)$ is a metric space. Go to proof.*

Remark (Standard quantified statement).

$$\forall x, y, z \in \mathbb{R}\ \big(d(x, y) \ge 0 \land (d(x, y) = 0 \Leftrightarrow x = y) \land d(x, y) = d(y, x) \land d(x, z) \le d(x, y)+d(y, z)\big).$$

Remark (Interpretation). The three metric axioms are exactly the properties of $|\cdot|$: nonnegativity with the zero-distance characterization, symmetry, and the triangle inequality. This is the precise sense in which $\mathbb{R}$ is a one-dimensional metric space.

Remark (Dependencies).

- Distance on the Real Line
- Triangle Inequality

2.3 Absolute Value as Distance

Absolute Value Is Distance to Zero

The metric d specializes to absolute value when one point is the origin. This is the geometric reading of $|a|$: how far a sits from 0 on the line.

Proposition (Absolute Value Is Distance to the Origin)

Proposition 2.3.1 (Absolute Value Is Distance to the Origin)**.** *For every* $a \in \mathbb{R}$, $|a| = d(a, 0)$. *Go to proof.*

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R}\ (|a| = d(a, 0)).$$

Remark (Interpretation). Absolute value is not a separate idea from distance; it is the distance to the origin. The two-sided bound $|x| \leq r \Leftrightarrow -r \leq x \leq r$ is then exactly the statement that x lies within distance r of 0, i.e. in the interval $[-r, r]$.

Remark (Dependencies).

- Distance on the Real Line

2.4 Intervals as Subsets

Intervals as the Basic Pieces

The interval shapes were introduced in §??. Here they are recast as subsets of the metric line and supplemented with boundedness, which is the size condition compactness will need.

Definition (Bounded Subset of the Real Line)

Definition 2.4.1 (Bounded Subset of the Real Line)**.** A set $E \subseteq \mathbb{R}$ is *bounded* exactly when there exists $M \geq 0$ such that $|x| \leq M$ for all $x \in E$; equivalently, E is contained in some closed interval $[-M, M]$.

Remark (Standard quantified statement).

$$E \text{ is bounded} \iff \exists M \geq 0\ \forall x \in E\ (|x| \leq M).$$

Remark (Negated quantified statement).

$$E \text{ is unbounded} \iff \forall M \geq 0\ \exists x \in E\ (|x| > M).$$

Remark (Interpretation). Boundedness says E fits inside a single closed interval — it does not run off to infinity. Together with closedness it is one half of the Heine–Borel characterization of compact subsets of $\mathbb{R}$.

Remark (Dependencies).

- Closed Interval
- Triangle Inequality

Set Operations on Intervals

Intervals as Sets

Once intervals are regarded as subsets of $\mathbb{R}$, the usual set operations apply to them. This is the first place where intervals behave like the building blocks of topology rather than merely as pieces of notation on the number line. Intersections often remain intervals, while unions, complements, and differences may split into several interval pieces.

Definition (Union of Intervals)

Definition 2.4.2 (Union of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. Their *union* is the set

$$I \cup J = \{x \in \mathbb{R} : x \in I \text{ or } x \in J\}.$$

Remark (Dependencies).

- Open Interval
- Closed Interval
- Union

Definition (Intersection of Intervals)

Definition 2.4.3 (Intersection of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. Their *intersection* is the set

$$I \cap J = \{x \in \mathbb{R} : x \in I \text{ and } x \in J\}.$$

Remark (Dependencies).

- Open Interval
- Closed Interval
- Intersection

Definition (Complement of an Interval)

Definition 2.4.4 (Complement of an Interval)**.** Let $I \subseteq \mathbb{R}$ be an interval. Its *complement* in $\mathbb{R}$ is the set

$$I^c = \mathbb{R} \setminus I = \{x \in \mathbb{R} : x \notin I\}.$$

Remark (Dependencies).

- Open Interval
- Closed Interval
- Complement

Definition (Difference of Intervals)

Definition 2.4.5 (Difference of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. The *difference* of I by J is the set

$$I \setminus J = \{x \in \mathbb{R} : x \in I \text{ and } x \notin J\} = I \cap J^c.$$

Remark (Predicate readings)*.* For any $x \in \mathbb{R}$,

$$x \in I \cup J \iff (x \in I) \vee (x \in J),$$

$$x \in I \cap J \iff (x \in I) \wedge (x \in J),$$

$$x \in I^c \iff x \notin I, \qquad x \in I \setminus J \iff (x \in I) \wedge (x \notin J).$$

These are set-theoretic operations; no arithmetic operation is being performed on the endpoints.

Remark (Dependencies)*.*

- Open Interval
- Closed Interval
- Left-Half-Open Interval
- Right-Half-Open Interval
- Union of Intervals
- Intersection of Intervals
- Complement of an Interval

Remark (Intersections of intervals)*.* The intersection of two intervals is either empty or another interval. For example, if $a \leq b$ and $c \leq d$, then

$$[a, b] \cap [c, d] = \begin{cases} [\max\{a, c\}, \min\{b, d\}], & \max\{a, c\} \leq \min\{b, d\}, \\ \varnothing, & \max\{a, c\} > \min\{b, d\}. \end{cases}$$

Endpoint inclusion follows the endpoint rules of the original intervals.

Remark (Unions of intervals)*.* The union of two intervals need not be an interval:

$$(0, 1) \cup (2, 3)$$

has a gap. If two intervals overlap or touch without a missing endpoint, their union is again an interval, for instance

$$(0, 2] \cup [2, 5) = (0, 5).$$

The missing or included endpoints determine whether the joined set is truly one interval.

Remark (Complements of intervals). The complement of a bounded interval usually splits into two rays:

$$(a, b)^c = (-\infty, a] \cup [b, \infty), \qquad [a, b]^c = (-\infty, a) \cup (b, \infty).$$

Complements reverse endpoint inclusion: an endpoint excluded from the interval is included in the complement, and an endpoint included in the interval is excluded from the complement.

Remark (Differences of intervals). A difference is an intersection with a complement, so it can also split:

$$(0, 5) \setminus [2, 3] = (0, 2) \cup (3, 5).$$

If the removed interval only cuts off one side, the result may remain a single interval:

$$(0, 5) \setminus (3, \infty) = (0, 3].$$

The bracket at 3 appears because $3 \notin (3, \infty)$, so it is not removed.

Remark (Topological motivation). Open sets in $\mathbb{R}$ are built from unions of open intervals. Closed sets are understood through complements of open sets. Boundaries arise when a point is not cleanly inside a set or its complement. Thus interval set operations are the grammar behind the metric topology of the real line.

2.5 Neighborhoods and Balls

Balls Are Open Intervals

On the line, the ball of radius r about a point is an open interval centered at that point. Neighborhoods are sets that contain such a ball. These are the local objects from which open sets are assembled.

Definition (Open Ball on the Real Line)

Definition 2.5.1 (Open Ball on the Real Line)**.** Let $x \in \mathbb{R}$ and $r > 0$. The *open ball* of radius r about x is

$$B_r(x) = \{\, y \in \mathbb{R} : |y - x| < r \,\} = (x - r,\, x + r).$$

Remark (Standard quantified statement).

$$B_r(x) = \{\, y \in \mathbb{R} : d(y, x) < r \,\} = (x - r, x + r).$$

Remark (Interpretation). A ball on the line is just a symmetric open interval: all points strictly within distance r of x. Writing balls as intervals keeps the chapter inside the metric topology of $\mathbb{R}$.

Remark (Dependencies).

- Distance on the Real Line

- Open Interval

Definition (Neighborhood of a Point)

Definition 2.5.2 (Neighborhood of a Point)**.** A set $N \subseteq \mathbb{R}$ is a *neighborhood* of $x \in \mathbb{R}$ exactly when there exists $r > 0$ with $B_r(x) \subseteq N$.

Remark (Standard quantified statement)*.*

$$N \text{ is a neighborhood of } x \iff \exists r > 0\ (B_r(x) \subseteq N).$$

Remark (Interpretation)*.* A neighborhood of x is any set with room to spare around x — it contains a whole ball, not just x itself. Neighborhoods are the language of "near x" used to define interior points and limits.

Remark (Dependencies)*.*

- Open Ball on the Real Line

2.6 Open Sets

Sets Built from Balls

An open set is one that contains a ball around each of its points. On the line these are exactly the unions of open intervals. The closure properties recorded here are the metric-topology core.

Definition (Open Set in $\mathbb{R}$)

Definition 2.6.1 (Open Set in $\mathbb{R}$)**.** A set $U \subseteq \mathbb{R}$ is *open* exactly when

$$\forall x \in U\ \exists r > 0\ (B_r(x) \subseteq U).$$

Remark (Standard quantified statement)*.*

$$U \text{ is open} \iff \forall x \in U\ \exists r > 0\ (B_r(x) \subseteq U).$$

Remark (Negated quantified statement)*.*

$$U \text{ is not open} \iff \exists x \in U\ \forall r > 0\ (B_r(x) \not\subseteq U).$$

Remark (Failure modes)*.* Openness fails exactly when some point of U has no ball contained in U — a boundary point trapped inside U, every ball around which pokes outside.

Remark (Interpretation)*.* An open set has interior room around each of its points: a small step in either direction remains inside the set. The empty set and $\mathbb{R}$ are open, and open intervals are the model examples.

Remark (Dependencies)*.*

- Open Ball on the Real Line

Theorem (Open Intervals Are Open)

Theorem 2.6.2 (Open Intervals Are Open)**.** *Every open interval (a, b), every open ray (a, ∞) and $(-\infty, b)$, and $\mathbb{R}$ itself are open sets. Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ (a < b \Longrightarrow (a, b) \text{ is open}),$$
$$\forall a \in \mathbb{R}\ \big((a, \infty) \text{ is open} \wedge (-\infty, a) \text{ is open}\big), \qquad \mathbb{R} \text{ is open}.$$

Remark (Interpretation). The name "open interval" is justified: each such interval satisfies the open-set condition, with the radius at x taken as the distance to the nearer endpoint.

Remark (Dependencies).

- Open Set in $\mathbb{R}$
- Open Interval

Theorem (Unions and Finite Intersections of Open Sets)

Theorem 2.6.3 (Unions and Finite Intersections of Open Sets)**.** *An arbitrary union of open subsets of $\mathbb{R}$ is open, and a finite intersection of open subsets of $\mathbb{R}$ is open. Go to proof.*

Remark (Standard quantified statement).

$$\Big(\forall i \in I\ U_i \text{ open}\Big) \Longrightarrow \Big(\bigcup_{i \in I} U_i\Big) \text{ open}; \qquad \Big(\bigwedge_{k=1}^{n} U_k \text{ open}\Big) \Longrightarrow \Big(\bigcap_{k=1}^{n} U_k\Big) \text{ open}.$$

Remark (Interpretation). Openness is preserved by arbitrary unions but only finite intersections. The infinite intersection $\bigcap_n(-1/n, 1/n) = \{0\}$ shows why "finite" cannot be dropped: a shrinking family of open intervals can close down onto a single point.

Remark (Dependencies).

- Open Set in $\mathbb{R}$

2.7 Closed Sets

Complements of Open Sets

Closed sets are the complements of open sets. They can equivalently be described as the sets that contain all of their limit points, the description used when reasoning about convergence.

Definition (Closed Set in $\mathbb{R}$)

Definition 2.7.1 (Closed Set in $\mathbb{R}$)**.** A set $F \subseteq \mathbb{R}$ is *closed* exactly when its complement $\mathbb{R} \setminus F$ is open.

Remark (Standard quantified statement).

$$F \text{ is closed} \iff (\mathbb{R} \setminus F) \text{ is open.}$$

Remark (Interpretation). Closed and open are complementary, not opposite: a set can be both (e.g. $\emptyset$, $\mathbb{R}$) or neither (e.g. $[0, 1)$). Closed intervals $[a, b]$ are closed, as are finite sets and $\mathbb{R}$ itself.

Remark (Dependencies).

- Open Set in $\mathbb{R}$

Theorem (Closed Sets Contain Their Limit Points)

Theorem 2.7.2 (Closed Sets Contain Their Limit Points)**.** *A set $F \subseteq \mathbb{R}$ is closed if and only if it contains every one of its limit points. Go to proof.*

Remark (Standard quantified statement).

$$F \text{ closed} \iff \forall x \in \mathbb{R} \; \big(x \text{ is a limit point of } F \Rightarrow x \in F\big).$$

Remark (Interpretation). This is the convergence-flavored description of closedness: a closed set cannot be approached from outside without already containing the approached point. It is the form used when showing limits of sequences in F stay in F.

Remark (Dependencies).

- Closed Set in $\mathbb{R}$
- Limit Point

2.8 Interior Exterior Boundary

Inside, Outside, and Edge

Relative to a set E, each point of the line is interior to E, exterior to E, or on its boundary. These three regions partition $\mathbb{R}$ and refine the open/closed distinction.

Definition (Interior Point)

Definition 2.8.1 (Interior Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an interior point of E exactly when

$$\exists r > 0 \; (B_r(x) \subseteq E).$$

Remark (Standard quantified statement).

$$x \text{ is an interior point of } E \iff \exists r > 0\ (B_r(x) \subseteq E).$$

Remark (Interpretation). An interior point of E sits inside E with room to spare: a whole ball around it lies in E.

Remark (Dependencies).

- Open Ball on the Real Line

Definition (Exterior Point)

Definition 2.8.2 (Exterior Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an exterior point of E exactly when

$$\exists r > 0\ (B_r(x) \subseteq \mathbb{R} \setminus E).$$

Remark (Standard quantified statement).

$$x \text{ is an exterior point of } E \iff \exists r > 0\ (B_r(x) \subseteq \mathbb{R} \setminus E).$$

Remark (Interpretation). An exterior point of E is interior to the complement: a whole ball around it misses E.

Remark (Dependencies).

- Open Ball on the Real Line

Definition (Boundary Point)

Definition 2.8.3 (Boundary Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a boundary point of E exactly when

$$\forall r > 0\ (B_r(x) \cap E \neq \emptyset\ \wedge\ B_r(x) \cap (\mathbb{R} \setminus E) \neq \emptyset).$$

Remark (Standard quantified statement).

$$x \text{ is a boundary point of } E \iff \forall r > 0\ (B_r(x) \cap E \neq \emptyset\ \wedge\ B_r(x) \cap (\mathbb{R} \setminus E) \neq \emptyset).$$

Remark (Interpretation). A boundary point of E is approached from both sides: every ball around it meets both E and its complement.

Remark (Dependencies).

- Open Ball on the Real Line

Definition (Interior of a Set)

Definition 2.8.4 (Interior of a Set)**.** The *interior* of $E \subseteq \mathbb{R}$, written $\operatorname{int}(E)$, is the set of all interior points of E.

Remark (Standard quantified statement).

$$\operatorname{int}(E) = \{\, x \in \mathbb{R} : \exists r > 0\ (B_r(x) \subseteq E)\,\}.$$

Remark (Interpretation). The interior is the largest open set inside E; E is open exactly when $E = \operatorname{int}(E)$. The boundary is what E and its complement share, and a set is closed exactly when it contains its boundary.

Remark (Dependencies).

- Interior Point

2.9 Limit and Isolated Points

Approachable and Detached Points

A limit point of E is one that E crowds arbitrarily closely, even with the point itself removed. An isolated point of E belongs to E but has a ball meeting E only at itself. These notions drive the convergence description of closed sets.

Definition (Limit Point)

Definition 2.9.1 (Limit Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a *limit point* of E exactly when

$$\forall r > 0\ \exists y \in E\ \big(0 < |y - x| < r\big).$$

Remark (Standard quantified statement).

$$x \text{ is a limit point of } E \iff \forall r > 0\ \exists y \in E\ (0 < |y - x| < r).$$

Remark (Negated quantified statement).

$$x \text{ is not a limit point of } E \iff \exists r > 0\ \forall y \in E\ (y = x \vee |y - x| \geq r).$$

Remark (Interpretation). Every punctured ball around a limit point meets E: the set accumulates at x. The strict inequality $0 < |y - x|$ excludes x itself, so x need not belong to E to be a limit point. This is the exact condition under which a sequence in E can converge to x.

Remark (Dependencies).

- Open Ball on the Real Line

Definition (Isolated Point)

Definition 2.9.2 (Isolated Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in E$. Then x is an *isolated point* of E exactly when

$$\exists r > 0\ \forall y \in E\ \big(|y - x| < r \Rightarrow y = x\big).$$

Remark (Standard quantified statement).

$$x \text{ is an isolated point of } E \iff x \in E \wedge \exists r > 0\ \forall y \in E\ \big(|y - x| < r \Rightarrow y = x\big).$$

Remark (Interpretation). An isolated point sits in E but stands apart: some ball around it contains no other point of E. A point of E is either isolated or a limit point of E, never both.

Remark (Dependencies).

- Open Ball on the Real Line

2.10 Unions and Intersections

Closure Laws for Open and Closed Sets

The open closure laws of §2.6 dualize to closed sets by complementation. Collecting them gives the bookkeeping rules used throughout the rest of the development.

Theorem (Closure Laws for Closed Sets)

Theorem 2.10.1 (Closure Laws for Closed Sets)**.** *An arbitrary intersection of closed subsets of* $\mathbb{R}$ *is closed, and a finite union of closed subsets of* $\mathbb{R}$ *is closed. Go to proof.*

Remark (Standard quantified statement).

$$\Big(\forall i \in I\ F_i \text{ closed}\Big) \Longrightarrow \Big(\bigcap_{i\in I} F_i\Big) \text{ closed;} \qquad \Big(\bigwedge_{k=1}^{n} F_k \text{ closed}\Big) \Longrightarrow \Big(\bigcup_{k=1}^{n} F_k\Big) \text{ closed.}$$

Remark (Interpretation). The roles of union and intersection swap relative to open sets: closed sets are stable under arbitrary intersection but only finite union. The asymmetry mirrors the open case under De Morgan's laws.

Remark (Dependencies).

- Closed Set in $\mathbb{R}$
- Unions and Finite Intersections of Open Sets

2.11 Covers and Subcovers

Covering a Set with Open Pieces

A cover of E is a family of sets whose union contains E; an open cover uses open sets. A subcover keeps enough of the family to still cover, and a finite subcover keeps only finitely many. These are the exact ingredients of compactness, defined so as to survive beyond the line.

Definition (Open Cover)

Definition 2.11.1 (Open Cover)**.** Let $E \subseteq \mathbb{R}$. An *open cover* of E is a family $\{U_i\}_{i\in I}$ of open subsets of $\mathbb{R}$ with

$$E \subseteq \bigcup_{i\in I} U_i.$$

Remark (Standard quantified statement).

$$\{U_i\}_{i\in I} \text{ is an open cover of } E \iff \big(\forall i \in I\ U_i \text{ open}\big) \wedge E \subseteq \textstyle\bigcup_{i\in I} U_i.$$

Remark (Interpretation). An open cover is a way of surrounding every point of E with an open piece of the family. The index set I may be infinite; the question compactness asks is whether finitely many pieces already suffice.

Remark (Dependencies).

- Open Set in $\mathbb{R}$

Definition (Subcover and Finite Subcover)

Definition 2.11.2 (Subcover and Finite Subcover)**.** Let $\{U_i\}_{i\in I}$ be an open cover of E. A *subcover* is a subfamily $\{U_i\}_{i\in J}$ with $J \subseteq I$ that still covers E. It is a *finite subcover* when J is finite.

Remark (Standard quantified statement).

$$\{U_i\}_{i\in J} \text{ is a finite subcover of } E \iff \big(J \subseteq I \wedge J \text{ is finite} \wedge E \subseteq \textstyle\bigcup_{i\in J} U_i\big).$$

Remark (Interpretation). A subcover discards some pieces while still covering E; a finite subcover discards all but finitely many. The existence of a finite subcover for *every* open cover is the defining property of compactness.

Remark (Dependencies).

- Open Cover

2.12 Compactness on R

The Open-Cover Definition

Compactness is defined by the open-cover property, deliberately and not as "closed and bounded." The open-cover formulation is the one that survives to general metric spaces and to manifolds such as the torus, where "bounded" has no intrinsic meaning. On $\mathbb{R}$

the two descriptions coincide, and that coincidence is the Heine–Borel theorem of the next section, proved rather than assumed.

Definition (Compact Set in $\mathbb{R}$)

Definition 2.12.1 (Compact Set in $\mathbb{R}$)**.** A set $K \subseteq \mathbb{R}$ is *compact* exactly when every open cover of K admits a finite subcover:

$$\forall \{U_i\}_{i\in I} \text{ open cover of } K\ \exists \text{finite } J \subseteq I\ \Big(K \subseteq \bigcup_{i\in J} U_i\Big).$$

Remark (Standard quantified statement)*.*

$$K \text{ compact} \iff \forall \{U_i\}_{i\in I}\Big(\big(\forall i \in I\ U_i \text{ open}\big) \wedge K \subseteq \textstyle\bigcup_{i\in I} U_i \Rightarrow \exists \text{finite } J \subseteq I\ K \subseteq \bigcup_{i\in J} U_i\Big).$$

Remark (Negated quantified statement)*.*

$$K \text{ not compact} \iff \exists \{U_i\}_{i\in I}\Big(\big(\forall i \in I\ U_i \text{ open}\big) \wedge K \subseteq \textstyle\bigcup_{i\in I} U_i \wedge \forall \text{finite } J \subseteq I\ K \not\subseteq \bigcup_{i\in J} U_i\Big).$$

Remark (Failure modes)*.* Compactness fails exactly when some open cover has no finite subcover. Two standard witnesses: $(0, 1]$ fails via the cover $\{(1/n, 2)\}_n$ (not closed), and $[0, \infty)$ fails via $\{(-1, n)\}_n$ (not bounded).

Remark (Interpretation)*.* Compactness is a finiteness property phrased entirely in open sets: no matter how a set is covered by open pieces, finitely many already do the job. This is what later guarantees, on a compact domain, that continuous functions attain extrema and are uniformly continuous — the facts Analysis I relies on. Crucially, this definition does not mention order or boundedness, so it transfers unchanged to the torus.

Remark (Dependencies)*.*

- Open Cover
- Subcover and Finite Subcover

Theorem (Compact Subsets of $\mathbb{R}$ Are Closed and Bounded)

Theorem 2.12.2 (Compact Subsets of $\mathbb{R}$ Are Closed and Bounded)**.** *If $K \subseteq \mathbb{R}$ is compact, then K is closed and bounded.* *Go to proof.*

Remark (Standard quantified statement)*.*

$$K \text{ compact} \Longrightarrow (K \text{ closed} \wedge K \text{ bounded}).$$

Remark (Interpretation)*.* This is the easy half of Heine–Borel and holds the same way in every metric space. Boundedness comes from covering K by growing balls about a fixed point; closedness comes from separating an outside point from K by disjoint balls. The converse — closed and bounded implies compact — is special to $\mathbb{R}^n$ and is the next theorem.

Remark (Dependencies).

- Compact Set in $\mathbb{R}$
- Closed Set in $\mathbb{R}$
- Bounded Subset of the Real Line

2.13 Heine Borel

The Capstone of the Real Line

Heine–Borel closes the chapter by proving the converse direction: on $\mathbb{R}$, closed and bounded is enough for compactness. The keystone is that every closed bounded interval $[a, b]$ is compact; the general statement follows because a closed subset of a compact set is compact.

Theorem (Closed Bounded Intervals Are Compact)

Theorem 2.13.1 (Closed Bounded Intervals Are Compact)**.** *Every closed bounded interval $[a, b] \subseteq \mathbb{R}$ is compact. Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R} \left(a \leq b \Longrightarrow [a, b] \text{ is compact}\right).$$

Remark (Interpretation). This is the hard half. The standard proof uses completeness directly: let S be the set of $x \in [a, b]$ such that $[a, x]$ has a finite subcover, and show $\sup S = b$ is attained. Bisection (the nested-interval method) gives the same result. Either route shows completeness is what makes the line compact in the large.

Remark (Dependencies).

- Compact Set in $\mathbb{R}$
- Closed Interval
- Least-Upper-Bound Property

Theorem (Heine–Borel Theorem for $\mathbb{R}$)

Theorem 2.13.2 (Heine–Borel Theorem for $\mathbb{R}$)**.** *A set $K \subseteq \mathbb{R}$ is compact if and only if K is closed and bounded. Go to proof.*

Remark (Standard quantified statement).

$$K \text{ compact} \Longleftrightarrow (K \text{ closed} \wedge K \text{ bounded}).$$

Remark (Interpretation). Heine–Borel is the bridge into Analysis I: it certifies that the closed bounded sets — the ones on which the Extreme Value and uniform-continuity theorems live — are exactly the compact ones. It is stated here as the equivalence *on* $\mathbb{R}$; the open-cover side is the definition that later carries to the torus, while "closed and bounded" does not.

Remark (Exposition). With Heine–Borel in hand the chain is complete: number construction $\to$ embedded number systems $\to$ the real line as an ordered metric space $\to$ Analysis I. Completeness, introduced as the last rung of the embedding ladder, is exactly the property that makes the line compact in the large, and compactness is the hypothesis Analysis I leans on.

Remark (Dependencies).

- Compact Subsets of $\mathbb{R}$ Are Closed and Bounded
- Closed Bounded Intervals Are Compact

Chapter 3

Bounds

SERIES
From Cantor to Ito

VOLUME III
Classical Analysis

BOOK I
Bounds, Sequences, and Series

CHAPTER 3
Bounds

3.1 Completeness Axiom

Axiom of Completeness

Remark (Why completeness is needed). The ordered field axioms alone do not prevent "holes" in the number line. The rationals $\mathbb{Q}$ satisfy all field and order axioms, yet fail completeness. Consider the set

$$S = \{x \in \mathbb{Q} : x^2 < 2\}.$$

This set is nonempty (e.g. $1 \in S$) and bounded above in $\mathbb{Q}$ (e.g. 2 is an upper bound). Yet $\sup S$ does not exist as a rational number: the candidate $\sqrt{2}$ is irrational. The set S has no least upper bound in $\mathbb{Q}$ — a hole sits exactly where $\sqrt{2}$ should be.

The Completeness Axiom asserts that $\mathbb{R}$ has no such holes: every nonempty bounded-above set has a supremum *inside* $\mathbb{R}$. This single axiom is what distinguishes $\mathbb{R}$ from $\mathbb{Q}$, and it underlies every major theorem in analysis.

Definition (Least Upper Bound Property)

Definition 3.1.1 (Least Upper Bound Property)**.** Let $(S, \leq)$ be an ordered set. The set S has the *Least Upper Bound Property* if and only if every nonempty subset of S that is bounded above in S has a supremum in S.

Remark (Standard quantified statement).

Let $(S, \leq)$ be an ordered set.

S has the Least Upper Bound Property

$$\iff \forall A \subseteq S \; \Big[\big(A \neq \varnothing \land \exists u \in S \; \forall x \in A \; (x \leq u)\big) \to \exists s \in S \; \operatorname{Supremum}(s, A)\Big].$$

Remark (Definition predicate reading).

$$\mathrm{LeastUpperBoundProperty}(S) := \forall A \subseteq S \left[\big(A \neq \varnothing \wedge \exists u \in S \; \forall x \in A \; (x \leq u) \big) \to \exists s \in S \;\; \mathrm{Supremum}(s, A) \right]$$

Remark (Negated quantified statement).

Let $(S, \leq)$ be an ordered set.
S does not have the Least Upper Bound Property
$$\iff \exists A \subseteq S \left[A \neq \varnothing \wedge \exists u \in S \; \forall x \in A \; (x \leq u) \wedge \forall s \in S \; \neg\, \mathrm{Supremum}(s, A) \right].$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{LeastUpperBoundProperty}(S) \iff \exists A \subseteq S \left[A \neq \varnothing \wedge \exists u \in S \; \forall x \in A \; (x \leq u) \wedge \forall s \in S \; \neg\, \mathrm{Supremum}(s, A) \right.$$

Remark (Failure modes). The property fails when there is a nonempty subset of the ambient ordered set that has at least one upper bound but has no least upper bound inside that same ambient set.

Remark (Failure mode decomposition).

$$\neg\, \mathrm{LeastUpperBoundProperty}(S) = \underbrace{\exists A \subseteq S \; (A \neq \varnothing)}_{\text{nonempty witness set}} \wedge \underbrace{\exists u \in S \; \forall x \in A \; (x \leq u)}_{\text{bounded above in } S} \wedge \underbrace{\forall s \in S \; \neg\, \mathrm{Supremum}(s,}_{\text{no supremum in } S}$$

Remark (Interpretation). The Least Upper Bound Property says that upper-bounded nonempty subsets do not point toward missing ceiling elements. Whenever the order supplies at least one upper bound, it also supplies a smallest upper bound in the same ambient set. Failure therefore indicates a hole in the ordered structure rather than a defect in the subset alone.

Remark (Dependencies).

- Ordered Set
- Subset
- Upper Bound
- Supremum

Axiom (Completeness of $\mathbb{R}$)

Axiom 3.1.2 (Completeness of $\mathbb{R}$)**.** The ordered field $\mathbb{R}$ has the Least Upper Bound Property.

Remark (Standard quantified statement).

$$\mathrm{LeastUpperBoundProperty}(\mathbb{R}).$$

Equivalently,

$$\forall A \subseteq \mathbb{R} \left[\big(A \neq \varnothing \wedge \exists u \in \mathbb{R} \; \forall x \in A \; (x \leq u) \big) \to \exists s \in \mathbb{R} \;\; \mathrm{Supremum}(s, A) \right].$$

Remark (Axiom predicate reading).

$$\mathrm{LeastUpperBoundProperty}(\mathbb{R}).$$

Remark (Negated quantified statement).

$$\begin{aligned} &\neg\,\mathrm{LeastUpperBoundProperty}(\mathbb{R}) \\ &\iff \exists A \subseteq \mathbb{R}\,\Big[A \neq \varnothing \wedge \exists u \in \mathbb{R}\ \forall x \in A\ (x \le u) \wedge \forall s \in \mathbb{R}\ \neg\,\mathrm{Supremum}(s, A)\Big]. \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{LeastUpperBoundProperty}(\mathbb{R}) \iff \exists A \subseteq \mathbb{R}\,\Big[A \neq \varnothing \wedge \exists u \in \mathbb{R}\ \forall x \in A\ (x \le u) \wedge \forall s \in \mathbb{R}\ \neg\,\mathrm{Supremum}(s, A$$

Remark (Failure modes). Completeness would fail if some nonempty subset of $\mathbb{R}$ were bounded above in $\mathbb{R}$ but had no supremum in $\mathbb{R}$.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{LeastUpperBoundProperty}(\mathbb{R}) = \underbrace{\exists A \subseteq \mathbb{R}\ (A \neq \varnothing)}_{\text{nonempty witness set}} \wedge \underbrace{\exists u \in \mathbb{R}\ \forall x \in A\ (x \le u)}_{\text{bounded above in } \mathbb{R}} \wedge \underbrace{\forall s \in \mathbb{R}\ \neg\,\mathrm{Supremum}(s,}_{\text{missing real supremum}}$$

Remark (Interpretation). The completeness axiom asserts that the real line has no order gaps of the least-upper-bound kind. Nonempty sets of real numbers that are bounded above always determine a real number sitting exactly at the least possible ceiling. In Volume II, $\mathbb{R}$ is constructed as the complete ordered field arising from Dedekind cuts; in this volume, completeness is used as the standing least-upper-bound principle for real analysis. This is the structural property that distinguishes $\mathbb{R}$ from ordered fields such as $\mathbb{Q}$.

Remark (Dependencies).

- The Real Numbers
- Ordered Field
- Least Upper Bound Property

3.1.1 Nested Intervals

Nested Interval Property

Theorem (Nested Interval Property)

Theorem 3.1.3 (Nested Interval Property)**.** *Let (I_n) be a sequence of nonempty closed intervals in $\mathbb{R}$. If $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$, then*

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (a_n) \subseteq \mathbb{R}\ \forall (b_n) \subseteq \mathbb{R} \\
&\Big[\big(\forall n \in \mathbb{N}\ (a_n \le b_n)\big) \land \big(\forall n \in \mathbb{N}\ ([a_{n+1}, b_{n+1}] \subseteq [a_n, b_n])\big) \\
&\qquad \to \exists x \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x \in [a_n, b_n])\Big].
\end{aligned}$$

Remark (Predicate reading).

$$\text{NestedClosedIntervals}(I_n) \implies \text{NonemptyIntersection}\left(\bigcap_{n=1}^{\infty} I_n\right).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\exists (a_n) \subseteq \mathbb{R}\ \exists (b_n) \subseteq \mathbb{R} \\
&\Big[\big(\forall n \in \mathbb{N}\ (a_n \le b_n)\big) \land \big(\forall n \in \mathbb{N}\ ([a_{n+1}, b_{n+1}] \subseteq [a_n, b_n])\big) \\
&\qquad \land \forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x \notin [a_n, b_n])\Big].
\end{aligned}$$

Remark (Negation predicate reading).

$$\text{NestedClosedIntervals}(I_n) \land \neg\, \text{NonemptyIntersection}\left(\bigcap_{n=1}^{\infty} I_n\right).$$

Remark (Failure modes). The theorem fails when a nested sequence of nonempty closed real intervals has no common real point.

Remark (Failure mode decomposition).

$$\underbrace{\text{NestedClosedIntervals}(I_n)}_{\text{closed intervals remain nested}} \land \underbrace{\forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x \notin I_n)}_{\text{no real point lies in every interval}}.$$

Remark (Interpretation). Nested closed intervals cannot keep shrinking or relocating in a way that loses every real point. Nesting keeps the intervals aligned, nonemptiness prevents trivial collapse at each stage, and completeness supplies a real point that survives all stages at once.

Remark (Dependencies).

Axiom 3.1.2.

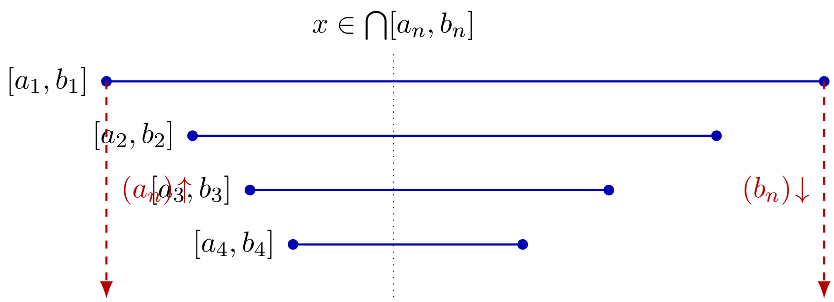


Figure 3.1: Nested intervals $[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$ pinch toward a common point. The sequence of left endpoints is increasing and bounded above; its supremum x lies in every interval.

3.1.2 Archimedean Property

Archimedean Property

Remark (Why this requires proof)*.* The Archimedean Property states that the natural numbers are unbounded in $\mathbb{R}$. That conclusion is not a formal consequence of the ordered-field axioms alone. There exist ordered fields satisfying all ordered-field axioms in which the natural numbers are bounded above, so the statement must be justified from stronger structure.

The needed structure is completeness. If $\mathbb{N}$ were bounded above in $\mathbb{R}$, then the Least Upper Bound Property would produce a supremum $\alpha \in \mathbb{R}$ for $\mathbb{N}$. But then $\alpha - 1$ cannot be an upper bound, so there exists $n \in \mathbb{N}$ with $n > \alpha - 1$. Hence $n + 1 > \alpha$, contradicting the fact that α is an upper bound of $\mathbb{N}$.

Go to proof.

Theorem (Archimedean Property)

Theorem 3.1.4 (Archimedean Property)**.** *For every real number x, there exists a natural number n such that $n > x$.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (n > x).$$

Remark (Predicate reading)*.*

$$\mathrm{ArchimedeanProperty}(\mathbb{R}) \iff \forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (n > x).$$

Remark (Negated quantified statement)*.*

$$\exists x \in \mathbb{R}\ \forall n \in \mathbb{N}\ (n \le x).$$

Remark (Negation predicate reading)*.*

$$\neg\, \mathrm{ArchimedeanProperty}(\mathbb{R}) \iff \exists x \in \mathbb{R}\ \forall n \in \mathbb{N}\ (n \le x).$$

Remark (Failure modes). The theorem fails exactly when the natural numbers are bounded above by some real number.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{ArchimedeanProperty}(\mathbb{R}) = \underbrace{\exists x \in \mathbb{R}}_{\text{real upper bound}} \quad \underbrace{\forall n \in \mathbb{N}\ (n \le x)}_{\text{all natural numbers remain below it}} .$$

Remark (Interpretation). The Archimedean Property says that the natural numbers do not stop below any real ceiling. No real number is so large that every natural number remains below it. In this chapter, the result is a consequence of completeness: a bounded copy of $\mathbb{N}$ would have a supremum, and that supremum is contradicted by shifting one natural number upward.

Remark (Dependencies).

Axiom 3.1.2, Definition 3.2.9.

Corollary (Archimedean Reciprocal Corollary)

Corollary 3.1.5 (Archimedean Reciprocal Corollary)**.** *For every $x > 0$ and every $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $nx > y$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}\ \big(x > 0 \to \exists n \in \mathbb{N}\ (nx > y)\big).$$

Remark (Predicate reading).

$$\mathrm{ArchimedeanReciprocal}(\mathbb{R}) \iff \forall x, y \in \mathbb{R}\ \big(x > 0 \to \exists n \in \mathbb{N}\ (nx > y)\big).$$

Remark (Negated quantified statement).

$$\exists x, y \in \mathbb{R}\ \big(x > 0 \land \forall n \in \mathbb{N}\ (nx \le y)\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{ArchimedeanReciprocal}(\mathbb{R}) \iff \exists x, y \in \mathbb{R}\ \big(x > 0 \land \forall n \in \mathbb{N}\ (nx \le y)\big).$$

Remark (Failure modes). The corollary fails when a positive real step size has all of its natural multiples bounded above by some real threshold.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{ArchimedeanReciprocal}(\mathbb{R}) = \underbrace{\exists x \in \mathbb{R}\ (x > 0)}_{\text{positive step size}} \land \underbrace{\exists y \in \mathbb{R}\ \forall n \in \mathbb{N}\ (nx \le y)}_{\text{all natural multiples stay bounded}} .$$

Remark (Interpretation). Any fixed positive real step eventually passes any real threshold when repeated enough times. The statement is the Archimedean Property applied after rescaling by the positive number x: growth by units becomes growth by steps of size x.

Remark (Dependencies).

Theorem 3.1.4.

3.1.3 Density

Density

Definition (Dense Subset)

Definition 3.1.6 (Dense Subset)**.** Let $D \subseteq S \subseteq \mathbb{R}$. The set D is *dense in* S if and only if for every $x \in S$ and every $\varepsilon > 0$ there exists $d \in D$ such that $|x - d| < \varepsilon$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } D \subseteq S \subseteq \mathbb{R}.\\ &D \text{ is dense in } S \iff \forall x \in S\ \forall \varepsilon > 0\ \exists d \in D\ \big(|x - d| < \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\operatorname{DenseSubset}(D, S) :\equiv \forall x \in S\ \forall \varepsilon > 0\ \exists d \in D\ \big(|x - d| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } D \subseteq S \subseteq \mathbb{R}.\\ &D \text{ is not dense in } S \iff \exists x \in S\ \exists \varepsilon_0 > 0\ \forall d \in D\ \big(|x - d| \geq \varepsilon_0\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg \operatorname{DenseSubset}(D, S) \iff \exists x \in S\ \exists \varepsilon_0 > 0\ \forall d \in D\ \big(|x - d| \geq \varepsilon_0\big).$$

Remark (Failure modes). Density fails when some point of S has a positive-radius neighborhood that contains no point of D.

Remark (Failure mode decomposition).

$$\neg \operatorname{DenseSubset}(D, S) = \underbrace{\exists x \in S}_{\text{point not approximated}} \quad \underbrace{\exists \varepsilon_0 > 0}_{\text{positive exclusion radius}} \quad \underbrace{\forall d \in D\ \big(|x - d| \geq \varepsilon_0\big)}_{\text{no point of } D \text{ enters the neighborhood}} .$$

Remark (Interpretation). Density is an approximation property inside S. It does not say that every point of S already belongs to D; it says that points of D occur arbitrarily close to every point of S. Thus D can be a proper subset of S while still leaving no positive-width gap inside S.

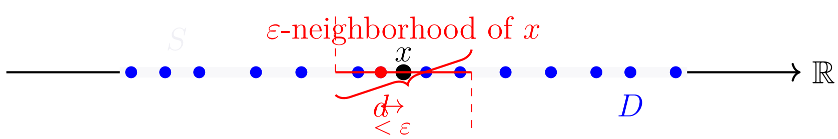


Figure. Illustration of a Dense Subset $D \subseteq S$. Given any $x \in S$ (black dot) and any $\varepsilon > 0$ (red interval width), there exists $d \in D$ (red-filled blue dot) such that $|x - d| < \varepsilon$.

Definition (Order Dense Subset)

Definition 3.1.7 (Order Dense Subset)**.** Let X be a linearly ordered set and let $A \subseteq X$. The subset A is *order dense in* X if and only if for every $a, b \in X$ with $a < b$ there exists $c \in A$ such that $a < c < b$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } X \text{ be a linearly ordered set and } A \subseteq X.\\ &A \text{ is order dense in } X \iff \forall a, b \in X \ \big(a < b \to \exists c \in A \ (a < c < b)\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\operatorname{OrderDenseSubset}(A, X) :\!= \forall a, b \in X \ \big(a < b \to \exists c \in A \ (a < c < b)\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } X \text{ be a linearly ordered set and } A \subseteq X.\\ &A \text{ is not order dense in } X \iff \exists a, b \in X \ \big(a < b \wedge \forall c \in A \ (c \leq a \vee b \leq c)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg \operatorname{OrderDenseSubset}(A, X) \iff \exists a, b \in X \ \big(a < b \wedge \forall c \in A \ (c \leq a \vee b \leq c)\big).$$

Remark (Failure modes). Order density fails when there is a nonempty open interval of X that contains no point of A.

Remark (Failure mode decomposition).

$$\neg \operatorname{OrderDenseSubset}(A, X) = \underbrace{\exists a, b \in X \ (a < b)}_{\text{nonempty interval}} \wedge \underbrace{\forall c \in A \ (c \leq a \vee b \leq c)}_{\text{no point of } A \text{ lies between } a \text{ and } b}.$$

Remark (Interpretation). Order density is an interval-penetration property. It ignores metric distance and asks only whether every ordered gap contains a point of the subset. A subset can therefore be order dense in X precisely when no nonempty open interval of X is missed entirely by that subset.

Remark (Dependencies).

- Total Order
- Subset

Theorem (Density of the Rational Numbers in $\mathbb{R}$)

Theorem 3.1.8 (Density of the Rational Numbers in $\mathbb{R}$)**.** *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ such that $a < q < b$.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R} \ \big(a < b \to \exists q \in \mathbb{Q} \ (a < q < b)\big).$$

Remark (Predicate reading).

$$\mathrm{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \iff \forall a, b \in \mathbb{R}\ \big(a < b \to \exists q \in \mathbb{Q}\ (a < q < b)\big).$$

Remark (Negated quantified statement).

$$\exists a, b \in \mathbb{R}\ \big(a < b \land \forall q \in \mathbb{Q}\ (q \leq a \lor b \leq q)\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \iff \exists a, b \in \mathbb{R}\ \big(a < b \land \forall q \in \mathbb{Q}\ (q \leq a \lor b \leq q)\big).$$

Remark (Failure modes). The theorem would fail precisely if some nonempty real interval contained no rational number.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) = \underbrace{\exists a, b \in \mathbb{R}\ (a < b)}_{\text{nonempty real interval}} \land \underbrace{\forall q \in \mathbb{Q}\ (q \leq a \lor b \leq q)}_{\text{no rational lies inside}}.$$

Remark (Interpretation). Rational numbers penetrate every real interval. The theorem does not say that every real number is rational; it says that rational numbers are never separated from a real interval by a positive order gap. This is the interval form of rational density in the real line.

Remark (Dependencies).
Definition 3.1.7, Theorem 3.1.4.

Theorem (Density of the Irrational Numbers in $\mathbb{R}$)

Theorem 3.1.9 (Density of the Irrational Numbers in $\mathbb{R}$)**.** *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $s \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < s < b$. Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ \big(a < b \to \exists s \in \mathbb{R} \setminus \mathbb{Q}\ (a < s < b)\big).$$

Remark (Predicate reading).

$$\mathrm{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}) \iff \forall a, b \in \mathbb{R}\ \big(a < b \to \exists s \in \mathbb{R} \setminus \mathbb{Q}\ (a < s < b)\big).$$

Remark (Negated quantified statement).

$$\exists a, b \in \mathbb{R}\ \big(a < b \land \forall s \in \mathbb{R} \setminus \mathbb{Q}\ (s \leq a \lor b \leq s)\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}) \iff \exists a, b \in \mathbb{R}\ \big(a < b \land \forall s \in \mathbb{R} \setminus \mathbb{Q}\ (s \leq a \lor b \leq s)\big).$$

Remark (Failure modes). The theorem would fail precisely if some nonempty real interval contained no irrational number.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{OrderDenseSubset}(\mathbb{R}\setminus\mathbb{Q},\mathbb{R}) = \underbrace{\exists a,b\in\mathbb{R}\ (a<b)}_{\text{nonempty real interval}} \ \wedge\ \underbrace{\forall s\in\mathbb{R}\setminus\mathbb{Q}\ (s\le a\vee b\le s)}_{\text{no irrational lies inside}}.$$

Remark (Interpretation). Irrational numbers also penetrate every real interval. One way to see the result is to insert a rational point into a translated and rescaled interval, then shift it by an irrational amount. The outcome is that the irrational numbers, like the rational numbers, leave no interval gap in $\mathbb{R}$.

Remark (Dependencies).

Definition 3.1.7, Theorem 3.1.8.

Corollary (Rational and Irrational Points in Every Open Interval)

Corollary 3.1.10 (Rational and Irrational Points in Every Open Interval)**.** *Every open interval in $\mathbb{R}$ contains both a rational number and an irrational number. Go to proof.*

Remark (Standard quantified statement).

$$\forall a,b\in\mathbb{R}\ \Big[a<b\to\big(\exists q\in\mathbb{Q}\ (a<q<b)\wedge\exists s\in\mathbb{R}\setminus\mathbb{Q}\ (a<s<b)\big)\Big].$$

Remark (Predicate reading).

$$\mathrm{OrderDenseSubset}(\mathbb{Q},\mathbb{R})\wedge\mathrm{OrderDenseSubset}(\mathbb{R}\setminus\mathbb{Q},\mathbb{R}).$$

Remark (Negated quantified statement).

$$\exists a,b\in\mathbb{R}\ \Big[a<b\wedge\big(\forall q\in\mathbb{Q}\ (q\le a\vee b\le q)\vee\forall s\in\mathbb{R}\setminus\mathbb{Q}\ (s\le a\vee b\le s)\big)\Big].$$

Remark (Negation predicate reading).

$$\neg\Big[\mathrm{OrderDenseSubset}(\mathbb{Q},\mathbb{R})\wedge\mathrm{OrderDenseSubset}(\mathbb{R}\setminus\mathbb{Q},\mathbb{R})\Big].$$

Remark (Failure modes). The corollary fails if a nonempty real interval misses all rational points or misses all irrational points.

Remark (Failure mode decomposition).

$$\exists a,b\in\mathbb{R}\ (a<b)\ \wedge\ \Big[\underbrace{\forall q\in\mathbb{Q}\ (q\le a\vee b\le q)}_{\text{no rational point}}\ \vee\ \underbrace{\forall s\in\mathbb{R}\setminus\mathbb{Q}\ (s\le a\vee b\le s)}_{\text{no irrational point}}\Big].$$

Remark (Interpretation). The corollary combines the two density theorems. Every real interval, no matter how small, contains representatives of both arithmetic types. Thus neither the rational numbers nor the irrational numbers occupy isolated regions of the real line.

Remark (Dependencies).

Theorem 3.1.8, Theorem 3.1.9.

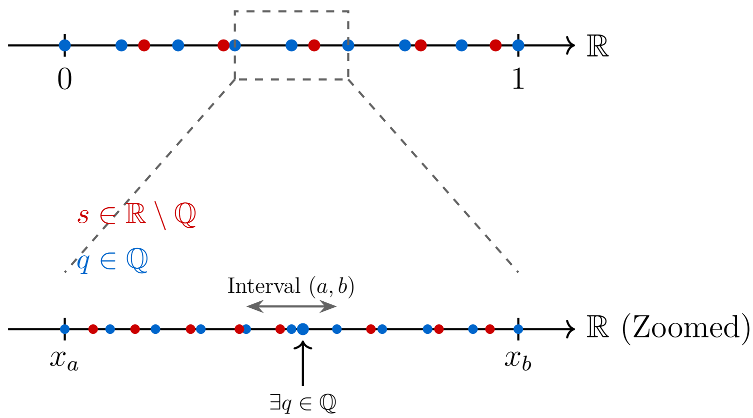


Figure 3.2: Visualization of the Density of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$. Zooming into any interval reveals that both sets are always present.

Remark (Zoom invariance and structural consequences).
Density in $\mathbb{R}$ has a scale-invariant consequence: no matter how small a nonempty open interval becomes, it still contains both rational and irrational points. The interleaving of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ is therefore not a large-scale phenomenon. It persists at every interval scale.

This has three important consequences. First, the continuum does not become locally discrete under magnification. For every nonempty open interval (a, b), both

$$\mathbb{Q} \cap (a, b) \qquad \text{and} \qquad (\mathbb{R} \setminus \mathbb{Q}) \cap (a, b)$$

remain nonempty, and in fact infinite. Density is therefore stable under repeated restriction to smaller intervals.

Second, this behavior sharply distinguishes the real line from discrete numerical models. Floating-point systems form a discrete set of representable values, so repeated magnification eventually reaches a scale at which adjacent representable numbers leave no further point strictly between them. At that stage, interval penetration fails in the discrete model even though it continues to hold in $\mathbb{R}$.

Third, the Archimedean Property supplies the scale mechanism behind this behavior. Given any positive scale ε, there exists $n \in \mathbb{N}$ with

$$\frac{1}{n} < \varepsilon.$$

This makes it possible to construct rational approximations at arbitrarily small interval scales, which is one of the basic engines behind density arguments.

The resulting contrast is structural. Smooth mathematical models are built in a setting where interval penetration persists at every scale, whereas finite computational models eventually encounter a smallest representable separation. The transition from continuum behavior to discrete behavior occurs exactly where nonempty intervals in the numerical model cease to contain further
representable points.

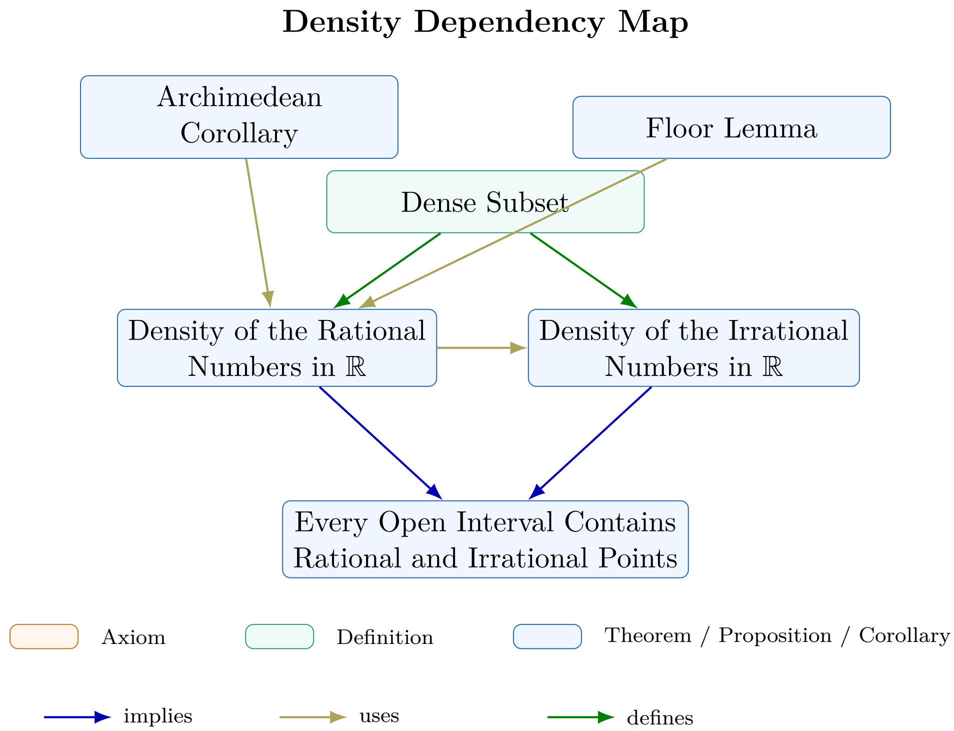


Figure 3.3: Dependency map for the density subchapter. The Floor Lemma and the Archimedean Corollary support the proof of the density of the rational numbers in $\mathbb{R}$. The irrational density theorem is then derived from the rational density result, and the final corollary combines both density theorems into a single interval statement. The definition records the local vocabulary of the subchapter.

3.1.4 Completeness Summary

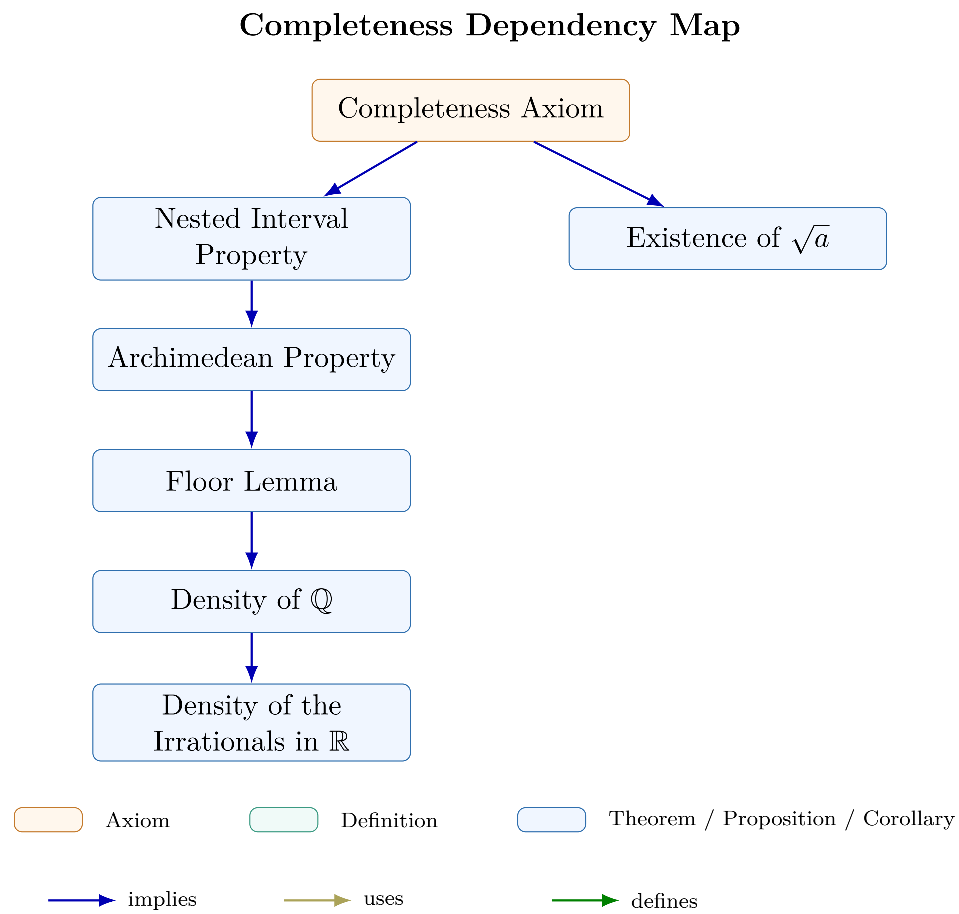


Figure 3.4: Dependency map for the completeness subchapter. The Completeness Axiom yields the Nested Interval Property and the existence of $\sqrt{a}$; the Nested Interval Property then supports the Archimedean Property, the Floor Lemma, and the density results for $\mathbb{Q}$ and for the irrational numbers.

Remark (Connections forward). Completeness appears in every major limit theorem in the notes. The Monotone Convergence Theorem (convergence chapter) constructs the limit of a bounded monotone sequence as $\sup\{a_n : n \in \mathbb{N}\}$; existence of this supremum requires completeness. The Bolzano–Weierstrass Theorem applies the Nested Interval Property directly to extract a convergent subsequence from any bounded sequence. The Cauchy Criterion identifies convergent sequences purely in terms of their own spacing, without naming the limit in advance; its proof constructs the limit as a supremum and again requires completeness. In metric spaces (Distance chapter), completeness in the sense of Cauchy sequences is the natural generalization of the same idea to abstract settings. Every ε-δ proof that locates a limit or extremum is drawing on this chapter's infrastructure.

3.2 Suprema and Infima

3.2.1 Bounds and Extremal Values

Remark (Predicate vs. Existence: Structural Template). Each bound-type concept (maximum, minimum, upper bound, supremum, etc.) is defined by a predicate relative to an ambient ordered set S. Given $A \subseteq S$, let $P_A(x)$ denote such a predicate.

Predicate form. The statement "x is a P-object of A" means that $P_A(x)$ holds for a specific identified element $x \in S$. To prove this, one exhibits the element and verifies each condition in $P_A(x)$.

Existence form. The statement "A has a P-object" means that

$$\exists x \in S \; P_A(x).$$

Once established, one may fix such an x and reason with the fact that $P_A(x)$ holds.

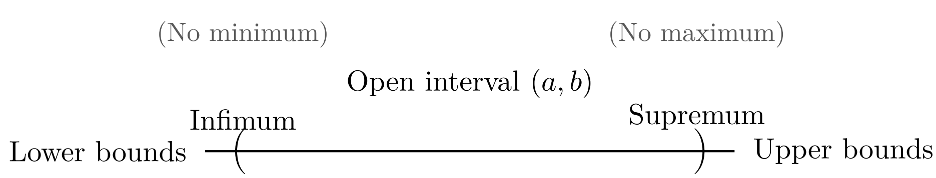
Notational glosses.

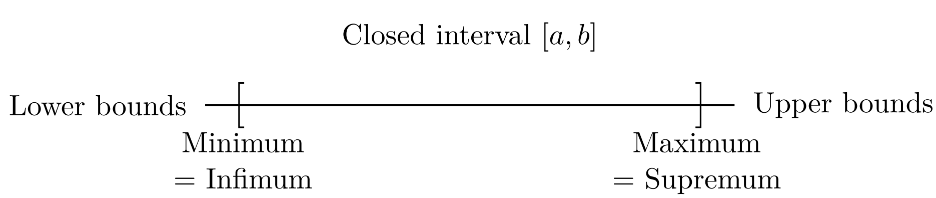
- "x is a P-object of A" abbreviates $P_A(x)$.
- "A has a P-object" abbreviates $\exists x \in S \; P_A(x)$.

Proof strategy summary.

Goal	**Form**	**Strategy**
$P_A(x)$	Predicate	Exhibit x and verify $P_A(x)$ directly.
$\exists x \in S \; P_A(x)$	Existence	Construct or identify a candidate and verify $P_A(x)$.
Hypothesis: $\exists x \in S \; P_A(x)$	—	Fix such an x and use the fact that $P_A(x)$ holds.

Note. The hypothesis row is not a proof goal. It records the standard move of fixing an existential witness once existence has been established.





Bounds, Extrema, Infimum, and Supremum for Subsets of S

Concept	Predicate $P_A(x)$	Predicate Form	Existence Form
Bound	$\big[\forall a \in A\ (a \le x)\big] \vee \big[\forall a \in A\ (x \le a)\big]$	x is a bound for A	A has a bound
Upper bound	$\forall a \in A\ (a \le x)$	x is an upper bound of A	A is bounded above
Lower bound	$\forall a \in A\ (x \le a)$	x is a lower bound of A	A is bounded below
Bounded	$\mathrm{BoundedAbove}(A) \wedge \mathrm{BoundedBelow}(A)$	–	A is bounded
Maximum	$(x \in A) \wedge \mathrm{UpperBound}(x, A)$	x is a maximum of A	A has a maximum
Minimum	$(x \in A) \wedge \mathrm{LowerBound}(x, A)$	x is a minimum of A	A has a minimum
Supremum	$\mathrm{UpperBound}(x, A) \wedge \forall u \in S\ (\mathrm{UpperBound}(u, A) \to x \le u)$	x [redacted] the supremum of A	A has a supremum
Infimum	$\mathrm{LowerBound}(x, A) \wedge \forall \ell \in S\ (\mathrm{LowerBound}(\ell, A) \to \ell \le x)$	x [redacted]s the infimum of A	A has an infimum

Remark (Reading the summary table)*.* The predicate column gives the condition that must hold for a specific identified element $x \in S$. The existence column records the corresponding existential statement. The rows are grouped by logical complexity: bounds use a single universal condition, extrema add membership, and suprema and infima add the least or greatest comparison against all competing bounds.

Bounds and Extremal Elements

Definition (Bound)

Definition 3.2.1 (Bound)**.** Let $(S, \le)$ be an ordered set, let $A \subseteq S$, and let $b \in S$. The element b is a *bound* for A if and only if b is a uniform one-sided comparison point for all elements of A:

$$\mathrm{Bound}(b, A) \iff \big[\forall x \in A\ (x \le b)\big] \vee \big[\forall x \in A\ (b \le x)\big].$$

Remark (Orientation). A bound always has a direction. The first alternative bounds A from above, and the second alternative bounds A from below. Thus "bound" is the common template, while upper and lower bounds are the two oriented cases.

Remark (Dependencies).

- Ordered Set
- Subset

Definition (Upper Bound)

Definition 3.2.2 (Upper Bound)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$, and let $u \in S$. The element u is an *upper bound* of A if and only if u is a bound for A from above, meaning $x \leq u$ for every $x \in A$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } u \in S.\\ &u \text{ is an upper bound of } A \iff \forall x \in A\ (x \leq u).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{UpperBound}(u, A) := \forall x \in A\ (x \leq u).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } u \in S.\\ &u \text{ is not an upper bound of } A \iff \exists x \in A\ (u < x).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{UpperBound}(u, A) \iff \exists x \in A\ (u < x).$$

Remark (Failure modes). The attempt to declare u an upper bound collapses precisely when some element of A lies strictly above u, providing a concrete counterexample to the domination condition.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{UpperBound}(u, A) = \underbrace{\exists x \in A\ (u < x)}_{\text{witness in } A \text{ exceeding } u}.$$

Remark (Interpretation). An upper bound caps a subset of an ordered structure from above, so every element of the subset fits beneath the chosen guard u. The negation highlights the standard obstruction: locating even one element of the set beyond u demolishes the claim. Geometrically, u places a horizontal ceiling over A on the number line, and the only failure mode consists of punching through that ceiling with a higher point of A.

Remark (Dependencies).

- Ordered Set
- Subset
- Bound

Definition (Bounded Above)

Definition 3.2.3 (Bounded Above)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$. The set A is *bounded above* in S if and only if there exists $u \in S$ such that u is an upper bound of A.

Remark (Existence reading).

$$A \text{ is bounded above in } S \iff \exists u \in S \ \text{UpperBound}(u, A).$$

Remark (Dependencies).

- Ordered Set
- Subset
- Upper Bound

Definition (Lower Bound)

Definition 3.2.4 (Lower Bound)**.** Let $(S, \leq)$ be an ordered set, $A \subseteq S$, and $l \in S$. The element l is a *lower bound* of A if and only if l is a bound for A from below, meaning every $x \in A$ satisfies $l \leq x$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } l \in S.\\ &l \text{ is a lower bound of } A \iff \forall x \in A \ (l \leq x).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{LowerBound}(l, A) :\equiv \forall x \in A \ (l \leq x).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } l \in S.\\ &\neg\big[l \text{ is a lower bound of } A\big] \iff \exists x \in A \ (x < l).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg \text{LowerBound}(l, A) \iff \exists x \in A \ (x < l).$$

Remark (Failure modes). Failure occurs precisely when some element of A lies strictly below l, so l cannot bound A from below.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{LowerBound}(l, A) = \underbrace{\exists x \in A \text{ with } x < l}_{\text{witness below } l}.$$

Remark (Interpretation). A lower bound clamps the set A from below: starting at l and moving upward along the order one encounters all of A, but no point of A sits beneath l. Failure means that at least one element of A pokes under l, revealing that l was chosen too high to constrain the set from below.

Remark (Dependencies).

- Ordered Set
- Subset
- Bound

Definition (Bounded Below)

Definition 3.2.5 (Bounded Below)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$. The set A is *bounded below* in S if and only if there exists $l \in S$ such that l is a lower bound of A.

Remark (Existence reading).

$$A \text{ is bounded below in } S \iff \exists l \in S\ \ \mathrm{LowerBound}(l, A).$$

Remark (Dependencies).

- Ordered Set
- Subset
- Lower Bound

Definition (Bounded)

Definition 3.2.6 (Bounded)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$. The set A is *bounded* in S if and only if A is bounded above in S and bounded below in S.

Remark (Two-sided existence reading).

$$A \text{ is bounded in } S \iff \big(\exists u \in S\ \ \mathrm{UpperBound}(u, A)\big) \wedge \big(\exists l \in S\ \ \mathrm{LowerBound}(l, A)\big).$$

Remark (Dependencies).

- Ordered Set
- Subset

- Bounded Above
- Bounded Below

Definition (Minimum)

Definition 3.2.7 (Minimum)**.** Let $(S, \leq)$ be an ordered set, $A \subseteq S$, and $m \in S$. The element m is the *minimum* of A if and only if $m \in A$ and m is a lower bound of A.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } m \in S.\\ &m \text{ is the minimum of } A \iff \big(m \in A \wedge \forall x \in A\ (m \leq x)\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\operatorname{Minimum}(m, A) :\!= (m \in A) \wedge \forall x \in A\ (m \leq x).$$

$$\operatorname{Minimum}(m, A) \iff (m \in A) \wedge \operatorname{LowerBound}(m, A).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } m \in S.\\ &\neg\big(m \text{ is the minimum of } A\big) \iff \big(m \notin A \vee \exists x \in A\ (m \nleq x)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg \operatorname{Minimum}(m, A) \iff \big(m \notin A \vee \exists x \in A\ (m \nleq x)\big).$$

Remark (Failure modes). Failure occurs either because m is not a member of A, or because some element of A lies strictly below m, violating the global lower-bound condition.

Remark (Failure mode decomposition).

$$\neg \operatorname{Minimum}(m, A) = \underbrace{m \notin A}_{\text{not even a candidate}} \vee \underbrace{\exists x \in A\ (m \nleq x)}_{\text{candidate but not below all elements}}.$$

Remark (Interpretation). A minimum point both belongs to the set and sits at its absolute bottom: every other element rests above it in the ambient order. The definition splits into the membership requirement, ensuring the bound is realized, and the universal inequality that promotes the element from a mere lower bound to a genuine extremal. Failure arises if the candidate lies outside the set or if another element dips below it; geometrically, the set either has no realized floor at that point or its supposed floor is undercut elsewhere.

Remark (Dependencies).

- Ordered Set
- Subset

- Bound
- Lower Bound

Definition (Maximum)

Definition 3.2.8 (Maximum)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$. An element $M \in S$ is the maximum of A if and only if $M \in A$ and M is an upper bound of A.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (S, \leq) \text{ be an ordered set and } A \subseteq S.\\ &M \text{ is the maximum of } A \iff (M \in A) \wedge \forall x \in A\ (x \leq M).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Maximum}(M, A) :\equiv (M \in A) \wedge \forall x \in A\ (x \leq M).$$

$$\mathrm{Maximum}(M, A) \iff (M \in A) \wedge \mathrm{UpperBound}(M, A).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (S, \leq) \text{ be an ordered set and } A \subseteq S.\\ &\neg\big(M \text{ is the maximum of } A\big) \iff (M \notin A) \vee \exists x \in A\ (M < x).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Maximum}(M, A) \iff (M \notin A) \vee \exists x \in A\ (M < x).$$

Remark (Failure modes). The property fails either because the candidate value is missing from A or because some element of A sits strictly above it, violating the domination requirement.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{Maximum}(M, A) = \underbrace{(M \notin A)}_{\text{membership failure}} \vee \underbrace{\big(\exists x \in A\ (M < x)\big)}_{\text{domination failure}}.$$

Remark (Interpretation). A maximum is an element of the set that simultaneously belongs to the set and dominates it under the ambient order. The domination clause guarantees that no point of A lies above the maximum, so the element captures the extreme right edge of the subset. Failure occurs either when the supposed extremal is not actually part of the set or when another member protrudes farther to the right, showing the candidate was not truly maximal. In geometric terms, the maximum is the endpoint where the subset comes to rest when viewed along the ordering axis.

Remark (Dependencies).

- Ordered Set

- Subset
- Bound
- Upper Bound

Definition (Supremum)

Definition 3.2.9 (Supremum)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$, and let $s \in S$. The element s is the supremum of A if and only if s is an upper bound of A and every upper bound u of A satisfies $s \leq u$.

Remark (Standard quantified statement).

$$\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } s \in S.$$
$$s \text{ is the supremum of } A \iff \Big[(\forall x \in A)(x \leq s) \wedge \forall u \in S\,\big((\forall x \in A)(x \leq u) \to s \leq u\big)\Big].$$

Remark (Definition predicate reading).

$$\mathrm{Supremum}(s, A) :\equiv \mathrm{UpperBound}(s, A) \wedge \forall u \in S\,\big(\mathrm{UpperBound}(u, A) \to s \leq u\big).$$

Remark (Negated quantified statement).

$$\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } s \in S.$$
$$\neg\big(s \text{ is the supremum of } A\big) \iff \Big[\exists x \in A\,(s < x) \vee \exists u \in S\,\big((\forall x \in A)(x \leq u) \wedge u < s\big)\Big].$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Supremum}(s, A) \iff \big(\exists x \in A\,(s < x)\big) \vee \big(\exists u \in S\,(\mathrm{UpperBound}(u, A) \wedge u < s)\big).$$

Remark (Failure modes). The definition fails in exactly two ways: either s is not an upper bound of A, or there exists a strictly smaller upper bound u of A, showing that s is not the least among all upper bounds.

Remark (Failure mode decomposition).

$$\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } s \in S.$$
$$\neg\,\mathrm{Supremum}(s, A) \iff \underbrace{\exists x \in A\,(s < x)}_{\text{fails to be an upper bound}} \vee \underbrace{\exists u \in S\,\big(\mathrm{UpperBound}(u, A) \wedge u < s\big)}_{\text{upper bound lies below } s}.$$

Remark (Interpretation). The supremum of A is the smallest ambient element that dominates the whole set. It exists precisely when s bounds A from above and no stricter bound sits below it. Failure occurs either because A pokes above s or because another upper bound nestles strictly beneath s, contradicting minimality. Geometrically, the supremum is the leftmost point in the order that caps the set, so every other cap must lie to its right.

Remark (Dependencies).

- Ordered Set
- Subset
- Upper Bound

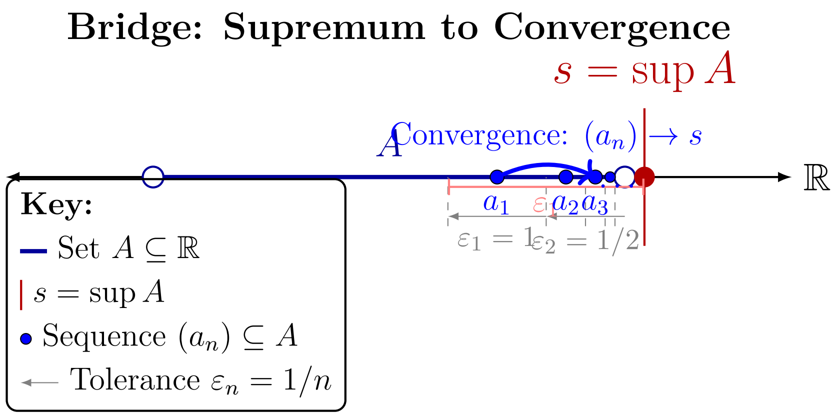


Figure 3.5: Illustrating the constructive connection between the static existence of a supremum (s) and the dynamic process of convergence. The ε-characterization acts as a sequence generator. By selecting decreasing tolerances $\varepsilon_n = 1/n$, we are guaranteed to find corresponding elements $a_n \in A$ that satisfy $s - \frac{1}{n} < a_n \leq s$. The sequence (a_n), contained strictly within A, is forced by the 'squeeze' of the tolerances to converge to s as $n \to \infty$. This construction provides the required sequence $(a_n) \subseteq A$ such that $\lim_{n\to\infty} a_n = \sup A$.

Definition (Infimum)

Definition 3.2.10 (Infimum)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$, and let $i \in S$. The element i is the infimum of A if and only if i is a lower bound of A and every lower bound l of A satisfies $l \leq i$.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } i \in S. \\
&i \text{ is the infimum of } A \\
&\iff \Big[\forall x \in A\ (i \leq x)\Big] \wedge \forall l \in S\ \Big(\big[\forall x \in A\ (l \leq x)\big] \Rightarrow l \leq i\Big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Infimum}(i, A) :\equiv \mathrm{LowerBound}(i, A) \wedge \forall l \in S\ \big(\mathrm{LowerBound}(l, A) \Rightarrow l \leq i\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } i \in S. \\
&i \text{ is not the infimum of } A \\
&\iff \Big[\exists x \in A\ (i \not\leq x)\Big] \vee \exists l \in S\ \Big(\big[\forall x \in A\ (l \leq x)\big] \wedge l \not\leq i\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{Infimum}(i, A) \iff \neg\,\text{LowerBound}(i, A) \lor \exists l \in S\ (\text{LowerBound}(l, A) \land l \not\leq i).$$

Remark (Failure modes). The definition fails either because i is not a lower bound of A, or because some other lower bound l is not beneath i, preventing i from being the greatest among all lower bounds.

Remark (Failure mode decomposition).

$$\neg\,\text{Infimum}(i, A) = \underbrace{\neg\,\text{LowerBound}(i, A)}_{\text{does not bound } A \text{ below}} \lor \underbrace{\exists l \in S\ (\text{LowerBound}(l, A) \land l \not\leq i)}_{\text{competing lower bound not below } i}.$$

Remark (Interpretation). The infimum of A is the greatest lower bound: it still lies below every element of A, yet no other lower bound stands above it. If i fails to be an infimum, either it already misses some element of A or there exists a better lower bound lying strictly higher in the order. Geometrically, in a number line picture, the infimum is the highest point lying to the left of A that is still a universal left barrier for the set.

Remark (Dependencies).

- Lower Bound
- Least Upper Bound Property

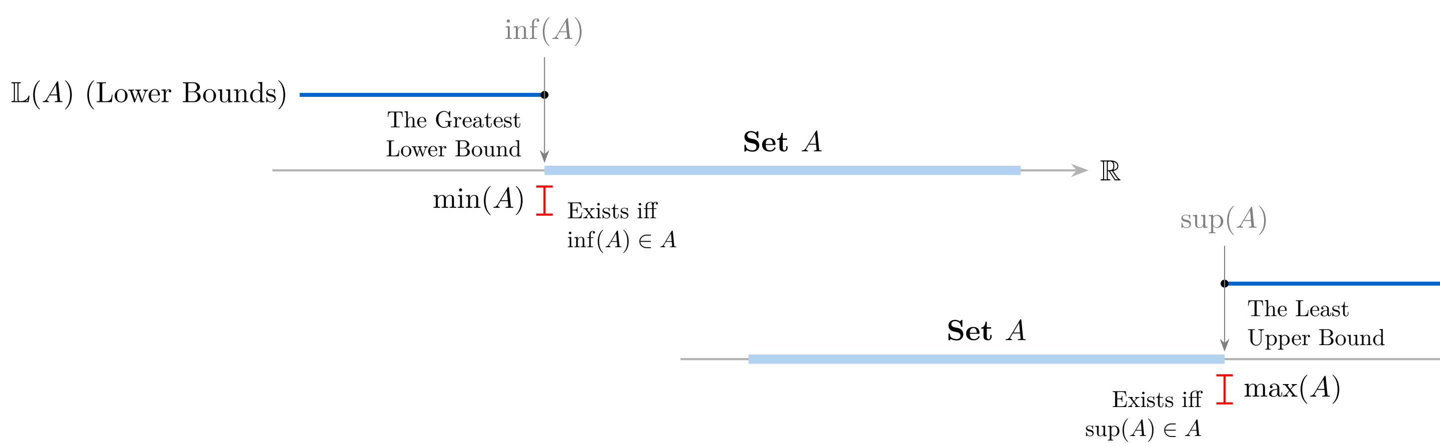


Figure 3.6: Separated geometric relationships for lower and upper bounds.

Contrast: Maximum vs. Supremum.

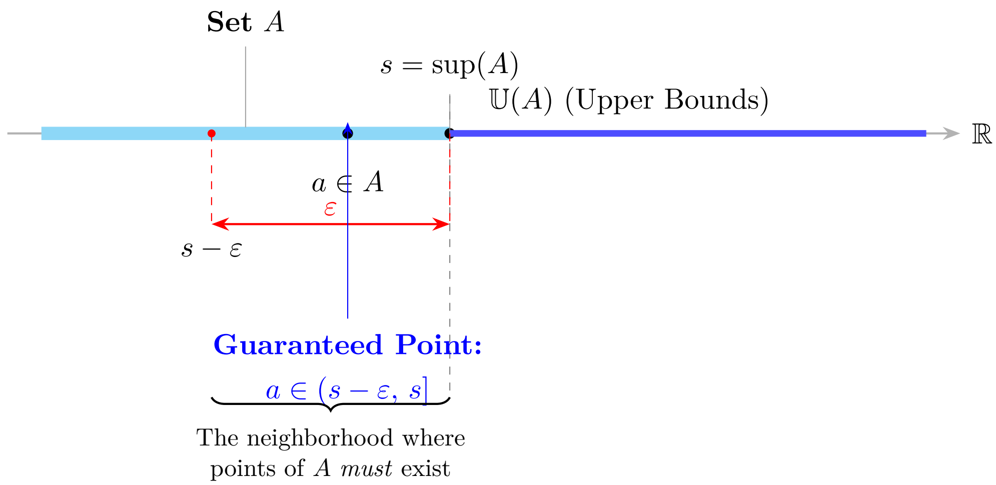


Figure 3.7: Refined visualization of the ε-characterization. Labels occupy staggered vertical lanes with leader lines to eliminate all horizontal collision, while the dashed spine preserves the geometric correspondence between the real line and the probe below it.

Property	**Maximum** $m = \max A$	**Supremum** $s = \sup A$
Must belong to A?	Yes. $m \in A$ is required.	No. s may or may not belong to A.
Must be an upper bound?	Yes. $\forall x \in A\ (x \leq m)$.	Yes. $\forall x \in A\ (x \leq s)$.
Must be the least upper bound?	Not stated separately, but membership together with the upper-bound condition forces it.	Yes. This is required explicitly.
Existence guaranteed?	Only when the set attains its supremum.	For nonempty sets bounded above in $\mathbb{R}$, yes, by the Completeness Axiom.

Remark (Maximum versus supremum). A maximum is an element of the set, whereas a supremum is an extremal upper bound that may lie outside the set. The two coincide exactly when the supremum belongs to the set.

Theorem (Least Upper Bound Property Implies Existence of Suprema)

Theorem 3.2.11 (Least Upper Bound Property Implies Existence of Suprema)**.** *Let S be an ordered set with the least upper bound property. If $A \subseteq S$ is nonempty and bounded above in S, then A has a supremum in S.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } S \text{ be an ordered set with the least upper bound property.}$$
$$\forall A \subseteq S\ \big[A \neq \varnothing \wedge \exists u \in S\ \forall x \in A\ (x \leq u) \implies \exists s \in S\ (s = \sup_S A)\big].$$

Remark (Predicate reading).

$$\text{LeastUpperBoundProperty}(S) \implies \forall A \subseteq S\ \big[A \neq \varnothing \wedge \text{BoundedAbove}_S(A) \implies \exists s \in S\ \text{Supremum}_S(s, A)\big].$$

Remark (Negated quantified statement).

$$\text{LeastUpperBoundProperty}(S) \quad \text{and} \quad \exists A \subseteq S : A \neq \varnothing,\ A \text{ bounded above in } S, \text{ and } \sup_S A \text{ does not exist}$$

is impossible.

Remark (Negation predicate reading).

$$\text{LeastUpperBoundProperty}(S) \wedge \exists A \subseteq S\ \big[A \neq \varnothing \wedge \text{BoundedAbove}_S(A) \wedge \neg \exists s \in S\ \text{Supremum}_S(s, A)\big]$$

is a contradictory configuration.

Remark (Failure modes). The theorem could fail only if an admissible subset were nonempty and bounded above while still lacking a least upper bound in S. That is precisely the situation ruled out by the least upper bound property.

Remark (Failure mode decomposition).

$$\text{LeastUpperBoundProperty}(S) = \underbrace{\forall A \subseteq S\ \big[A \neq \varnothing \wedge \text{BoundedAbove}_S(A) \implies \exists s \in S\ \text{Supremum}_S(s, A)\big]}_{\text{every admissible subset has a least upper bound in } S}.$$

Remark (Interpretation). This theorem is the operational use of the least upper bound property. Once the ambient ordered set has that property, every nonempty bounded-above subset receives the least ceiling that the definition promises.

Remark (Dependencies).

Least Upper Bound Property; Upper Bound; Supremum.

Proposition (Upper Bound Property of Supremum)

Proposition 3.2.12 (Upper Bound Property of Supremum). *Let $A \subseteq \mathbb{R}$ and let $s = \sup A$. Then $x \leq s$ for every $x \in A$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } s = \sup A.$$
$$\forall x \in A\ (x \leq s).$$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \implies \text{UpperBound}(s, A).$$

Remark (Negated quantified statement).

$$s = \sup A \quad \text{and} \quad \exists x \in A\ (s < x)$$

is impossible.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s, A) \wedge \neg\, \mathrm{UpperBound}(s, A)$$

is a contradictory configuration.

Remark (Failure modes). The proposition could fail only if a point of A sat strictly above its claimed supremum. Such a point would show that the claimed supremum is not even an upper bound.

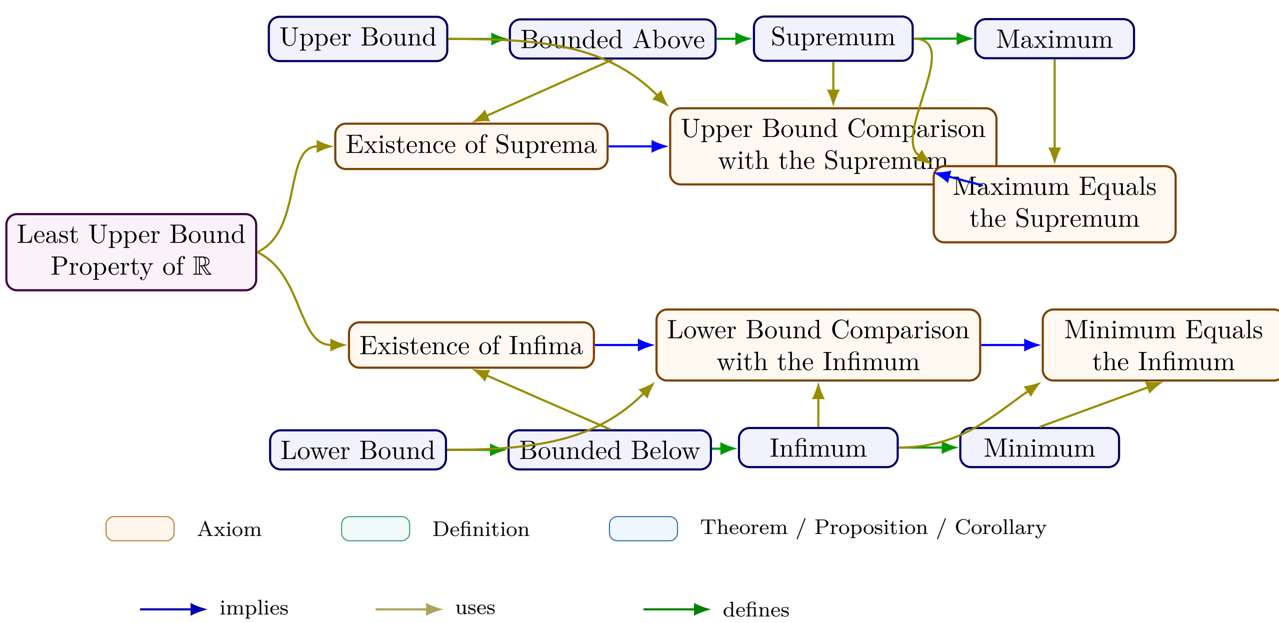
Remark (Failure mode decomposition).

$$\neg\, \mathrm{UpperBound}(s, A) \iff \underbrace{\exists x \in A\ (s < x)}_{\text{element above the claimed supremum}}.$$

Remark (Interpretation). The supremum has two jobs: it must bound the set from above, and it must be least among all such bounds. This proposition isolates the first job.

Remark (Dependencies).

Upper Bound; Supremum.



Consequences

Remark (Logical structure). Bounds, extrema, and least/greatest bounds form a hierarchy of increasing logical complexity. Upper and lower bounds are defined by a universal quantifier alone. Maximum and minimum add a membership condition: the extremum must belong to the set. Supremum and infimum relax membership but add a minimality (or maximality) condition: the bound must be the *least* (or *greatest*) among all bounds. Maximum implies supremum — when $\max A$ exists, it equals $\sup A$ (Proposition 3.3.3 and the proof at maximum-implies-supremum) — but the converse fails: $(0, 1)$ has $\sup = 1$ but no maximum.

Remark (Connections forward). The ε-characterization is the workhorse tool of the bounding chapter. Every proof in the Bound Algebra section invokes it to verify that a candidate value is the least upper bound, not merely an upper bound. The Monotone Convergence Theorem (convergence chapter) uses the same mechanism: the limit of a bounded monotone sequence is identified as $\sup\{a_n\}$, and the ε-characterization produces the required N.

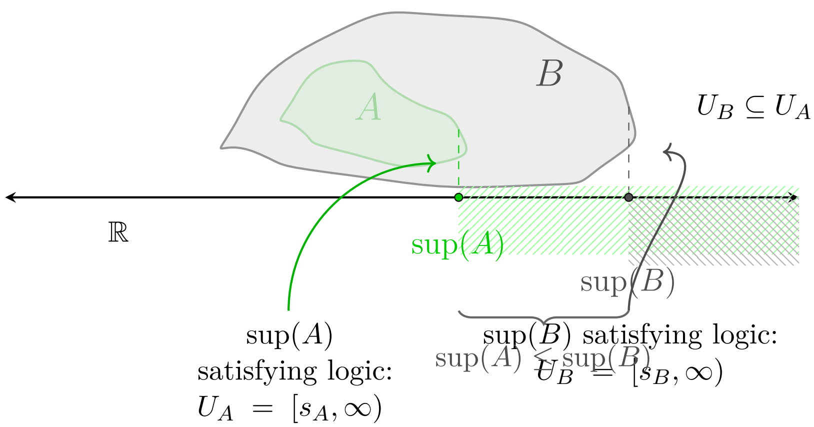


Figure 3.8: The Monotonicity of Supremum Under Set Inclusion. This visualization illustrates the arithmetic consequence of set inclusion, $A \subseteq B \subset \mathbb{R}$. The topology of the sets is represented by nested blobs: A (green) is contained within B (gray). Vertically dashed projections link the supremal boundary of each set down to the shared real axis, defining the unique points $s_A = \sup(A)$ and $s_B = \sup(B)$. The figure demonstrates that because the territory of A is restricted by B, its maximum guard point s_A cannot exceed s_B. Finally, the shaded regions to the right illustrate the territories of upper bounds (U_A and U_B), clearly showing that U_B is a proper subset of U_A. Any value guarding the larger set B necessarily satisfies the guard condition for the smaller set A.

3.2.2 Epsilon Characterization

ε-Characterization of Supremum

Definition (Epsilon Characterization of Supremum)

Definition 3.2.13 (Epsilon Characterization of Supremum). Let $A \subseteq \mathbb{R}$ and let $s \in \mathbb{R}$. The element s satisfies the *epsilon characterization of the supremum* of A if and only if s is an upper bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $s - \varepsilon < a$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } s \in \mathbb{R}.\\ &s \text{ satisfies the epsilon characterization of } \sup A\\ &\iff \Big[\forall x \in A\ (x \le s)\Big] \wedge \Big[\forall \varepsilon > 0\ \exists a \in A\ (s - \varepsilon < a)\Big].\end{aligned}$$

Remark (Definition predicate reading).

$$\text{EpsilonSupremum}(s, A) :\equiv \text{UpperBound}(s, A) \wedge \forall \varepsilon > 0\ \exists a \in A\ (s - \varepsilon < a).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } s \in \mathbb{R}.\\ &s \text{ does not satisfy the epsilon characterization of } \sup A\\ &\iff \Big[\exists x \in A\ (s < x)\Big] \vee \Big[\exists \varepsilon_0 > 0\ \forall a \in A\ (a \le s - \varepsilon_0)\Big].\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{EpsilonSupremum}(s, A) \iff \Big[\exists x \in A\ (s < x)\Big] \vee \Big[\exists \varepsilon_0 > 0\ \forall a \in A\ (a \le s - \varepsilon_0)\Big].$$

Remark (Failure modes). The characterization fails either because s is not an upper bound of A, or because A remains at least some positive distance below s.

Remark (Failure mode decomposition).

$$\neg\,\text{EpsilonSupremum}(s, A) = \underbrace{\exists x \in A\ (s < x)}_{\text{upper-bound failure}} \vee \underbrace{\exists \varepsilon_0 > 0\ \forall a \in A\ (a \le s - \varepsilon_0)}_{\text{positive gap below } s}.$$

Remark (Interpretation). This condition says that s caps the set from above while points of A approach s from below at every positive scale. The upper-bound clause prevents A from crossing above s, and the epsilon clause prevents s from floating strictly above the set. Geometrically, s is a ceiling that the set presses against arbitrarily closely.

Remark (Dependencies).

Definition 3.2.2.

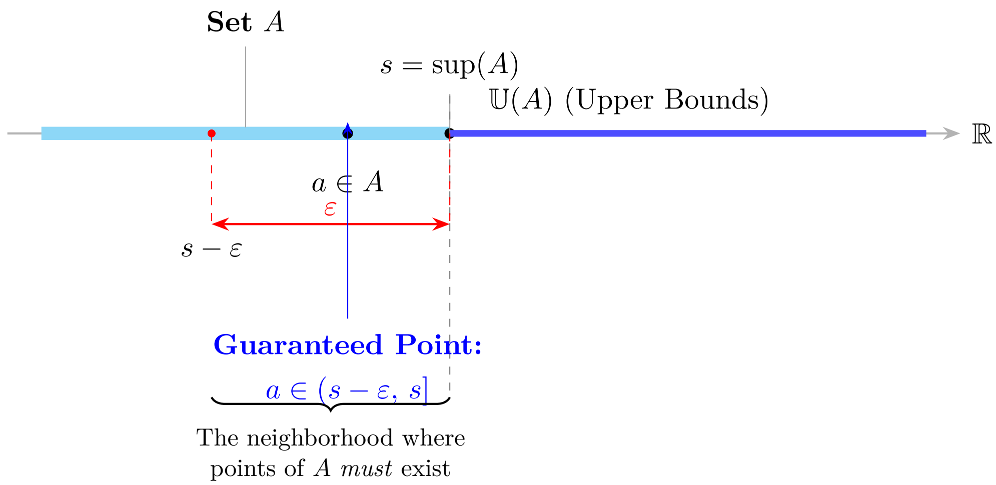


Figure 3.9: Refined visualization of the ε-characterization. Labels occupy staggered vertical lanes with leader lines to eliminate all horizontal collision, while the dashed spine preserves the geometric correspondence between the real line and the probe below it.

Definition (Epsilon Characterization of Infimum)

Definition 3.2.14 (Epsilon Characterization of Infimum)**.** Let $A \subseteq \mathbb{R}$ and let $t \in \mathbb{R}$. The element t satisfies the *epsilon characterization of the infimum* of A if and only if t is a lower bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $a < t + \varepsilon$.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } t \in \mathbb{R}.\\ &t \text{ satisfies the epsilon characterization of } \inf A\\ &\iff \Big[\forall x \in A\ (t \le x)\Big] \land \Big[\forall \varepsilon > 0\ \exists a \in A\ (a < t + \varepsilon)\Big].\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{EpsilonInfimum}(t, A) :\!= \mathrm{LowerBound}(t, A) \land \forall \varepsilon > 0\ \exists a \in A\ (a < t + \varepsilon).$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } t \in \mathbb{R}.\\ &t \text{ does not satisfy the epsilon characterization of } \inf A\\ &\iff \Big[\exists x \in A\ (x < t)\Big] \lor \Big[\exists \varepsilon_0 > 0\ \forall a \in A\ (t + \varepsilon_0 \le a)\Big].\end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{EpsilonInfimum}(t, A) \iff \Big[\exists x \in A\ (x < t)\Big] \lor \Big[\exists \varepsilon_0 > 0\ \forall a \in A\ (t + \varepsilon_0 \le a)\Big].$$

Remark (Failure modes)*.* The characterization fails either because t is not a lower bound of A, or because A remains at least some positive distance above t.

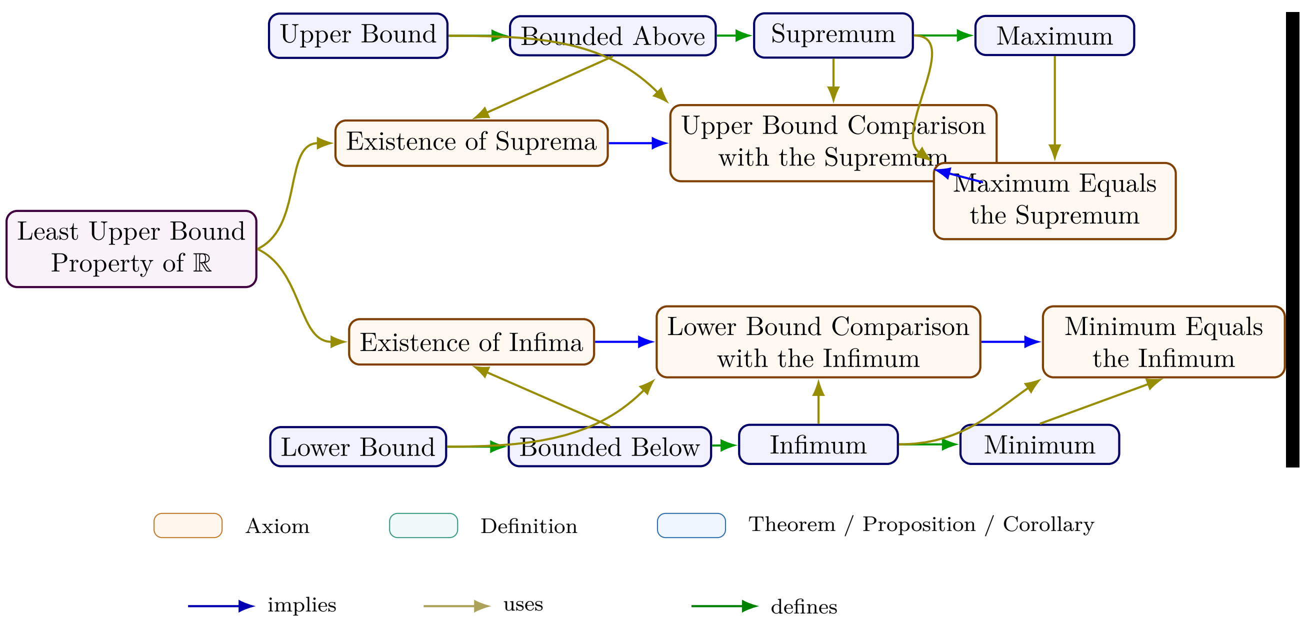
Remark (Failure mode decomposition).

$$\neg\,\mathrm{EpsilonInfimum}(t, A) = \underbrace{\exists x \in A\ (x < t)}_{\text{lower-bound failure}} \vee \underbrace{\exists \varepsilon_0 > 0\ \forall a \in A\ (t + \varepsilon_0 \le a)}_{\text{positive gap above } t}.$$

Remark (Interpretation). This condition says that t sits below the set while points of A approach t from above at every positive scale. The lower-bound clause prevents A from crossing below t, and the epsilon clause prevents t from sitting strictly below the set with a positive gap. Geometrically, t is a floor that the set presses against arbitrarily closely from above.

Remark (Dependencies).
Definition 3.2.4.

3.2.3 Extremal Bounds Summary



Remark (Equivalent formulations).
Over the ordered-field axioms, completeness is equivalent to each of the following:

- Every nonempty set bounded below has an infimum.
- Every Cauchy sequence in $\mathbb{R}$ converges.
- *(Nested Interval Property)* Every nested sequence of nonempty closed bounded intervals has nonempty intersection.

3.3 Algebra of Bounds

Algebraic Properties of Supremum and Infimum

Proposition (Uniqueness of the Maximum)

Proposition 3.3.1 (Uniqueness of the Maximum)**.** *Let $A \subseteq \mathbb{R}$. If m_1 and m_2 are both maximum elements of A, then $m_1 = m_2$.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } m_1, m_2 \in A.\\ &\big[m_1 \text{ is a maximum of } A \ \wedge \ m_2 \text{ is a maximum of } A\big] \implies m_1 = m_2.\end{aligned}$$

Remark (Predicate reading)*.*

$$\mathrm{Maximum}(m_1, A) \wedge \mathrm{Maximum}(m_2, A) \implies m_1 = m_2.$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } m_1, m_2 \in A.\\ &m_1 \text{ and } m_2 \text{ are both maximum elements of } A \text{ and } m_1 \neq m_2\end{aligned}$$

is impossible.

Remark (Negation predicate reading)*.*

$$\mathrm{Maximum}(m_1, A) \wedge \mathrm{Maximum}(m_2, A) \wedge m_1 \neq m_2$$

is a contradictory configuration.

Remark (Failure modes)*.* The uniqueness claim could fail only if two distinct elements of A each dominated all elements of A. In the real order this cannot occur because each maximum must be less than or equal to the other.

Remark (Failure mode decomposition)*.*

$$\begin{aligned}&\mathrm{Maximum}(m_1, A) \wedge \mathrm{Maximum}(m_2, A)\\ &\implies \underbrace{m_1 \leq m_2}_{m_2 \text{ dominates } A} \ \wedge \ \underbrace{m_2 \leq m_1}_{m_1 \text{ dominates } A} \implies m_1 = m_2.\end{aligned}$$

Remark (Interpretation)*.* A maximum is not merely a high point of the set; it is the greatest point of the set. If two elements both claim that role, each must sit above the other, and antisymmetry of the real order forces them to be the same number.

Remark (Dependencies)*.*

Maximum.

Proposition (Uniqueness of the Supremum)

Proposition 3.3.2 (Uniqueness of the Supremum)**.** *Let $A \subseteq \mathbb{R}$. If s_1 and s_2 are both suprema of A, then $s_1 = s_2$.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\text{Let } A \subseteq \mathbb{R} \text{ and } s_1, s_2 \in \mathbb{R}.$$
$$\big[s_1 = \sup A \ \wedge\ s_2 = \sup A\big] \implies s_1 = s_2.$$

Remark (Predicate reading)*.*

$$\text{Supremum}(s_1, A) \wedge \text{Supremum}(s_2, A) \implies s_1 = s_2.$$

Remark (Negated quantified statement)*.*

$$\text{Let } A \subseteq \mathbb{R} \text{ and } s_1, s_2 \in \mathbb{R}.$$
$$s_1 = \sup A, \quad s_2 = \sup A, \quad \text{and} \quad s_1 \neq s_2$$

is impossible.

Remark (Negation predicate reading)*.*

$$\text{Supremum}(s_1, A) \wedge \text{Supremum}(s_2, A) \wedge s_1 \neq s_2$$

is a contradictory configuration.

Remark (Failure modes)*.* The uniqueness claim could fail only if two different real numbers were both least upper bounds of the same set. Since each supremum is less than or equal to every upper bound, each must be less than or equal to the other.

Remark (Failure mode decomposition)*.*

$$\begin{aligned}&\text{Supremum}(s_1, A) \wedge \text{Supremum}(s_2, A)\\ &\implies \underbrace{s_1 \le s_2}_{s_2 \text{ is an upper bound and } s_1 \text{ is least}} \wedge \underbrace{s_2 \le s_1}_{s_1 \text{ is an upper bound and } s_2 \text{ is least}} \implies s_1 = s_2.\end{aligned}$$

Remark (Interpretation)*.* The supremum is the least possible ceiling over a set. Two ceilings can both be least only if neither one is actually above the other, so the order on $\mathbb{R}$ identifies them as the same number.

Remark (Dependencies)*.*

Supremum.

Proposition (Maximum Implies Supremum)

Proposition 3.3.3 (Maximum Implies Supremum)**.** *Let $A \subseteq \mathbb{R}$ and let $m \in A$. If m is a maximum of A, then $m = \sup A$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } m \in A.\\&m \text{ is a maximum of } A \implies m = \sup A.\end{aligned}$$

Remark (Predicate reading).

$$\text{Maximum}(m, A) \implies \text{Supremum}(m, A).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } m \in A.\\&m \text{ is a maximum of } A \text{ and } m \neq \sup A\end{aligned}$$

is impossible.

Remark (Negation predicate reading).

$$\text{Maximum}(m, A) \wedge \neg\,\text{Supremum}(m, A)$$

is a contradictory configuration.

Remark (Failure modes). The implication could fail only if the greatest element of A were not the least upper bound of A. But a maximum already bounds the set from above, and every upper bound must lie at least as high as that maximum.

Remark (Failure mode decomposition).

$$\begin{aligned}\text{Maximum}(m, A) \implies &\underbrace{\forall x \in A\ (x \leq m)}_{m \text{ is an upper bound}}\\&\wedge \underbrace{\forall u \in \mathbb{R}\left[(\forall x \in A\ (x \leq u)) \implies m \leq u\right]}_{m \text{ is least among upper bounds}}.\end{aligned}$$

Remark (Interpretation). A maximum belongs to the set and sits above every element of the set. That makes it an upper bound, and no smaller upper bound can exist because it would have to sit below the maximum while still bounding the maximum itself.

Remark (Dependencies).

Maximum; Supremum.

Proposition (Supremum in the Set Is the Maximum)

Proposition 3.3.4 (Supremum in the Set Is the Maximum)**.** *Let $A \subseteq \mathbb{R}$ and let $s = \sup A$. If $s \in A$, then s is a maximum of A.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } s \in \mathbb{R}.\\&\left[s = \sup A\ \wedge\ s \in A\right] \implies s \text{ is a maximum of } A.\end{aligned}$$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge s \in A \Longrightarrow \text{Maximum}(s, A).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } s \in \mathbb{R}.\\ &s = \sup A, \quad s \in A, \quad \text{and } s \text{ is not a maximum of } A\end{aligned}$$

is impossible.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge s \in A \wedge \neg \text{Maximum}(s, A)$$

is a contradictory configuration.

Remark (Failure modes). The implication could fail only if the least upper bound belonged to the set but did not dominate every element of the set. That cannot happen because every supremum is already an upper bound.

Remark (Failure mode decomposition).

$$\begin{aligned}\text{Supremum}(s, A) \wedge s \in A \Longrightarrow &\underbrace{\forall x \in A\ (x \leq s)}_{s \text{ is an upper bound}}\\ &\wedge \quad \underbrace{s \in A}_{s \text{ belongs to the set}} \quad \Longrightarrow \text{Maximum}(s, A).\end{aligned}$$

Remark (Interpretation). A supremum is usually only a least ceiling. When that ceiling is actually a member of the set, it becomes the greatest element of the set, so the extremal value is attained rather than merely approached.

Remark (Dependencies).

Supremum; Maximum.

Lemma (Subset Inclusion Preserves Upper Bounds)

Lemma 3.3.5 (Subset Inclusion Preserves Upper Bounds)**.** *Let $A \subseteq B \subseteq \mathbb{R}$. If u is an upper bound of B, then u is an upper bound of A.*

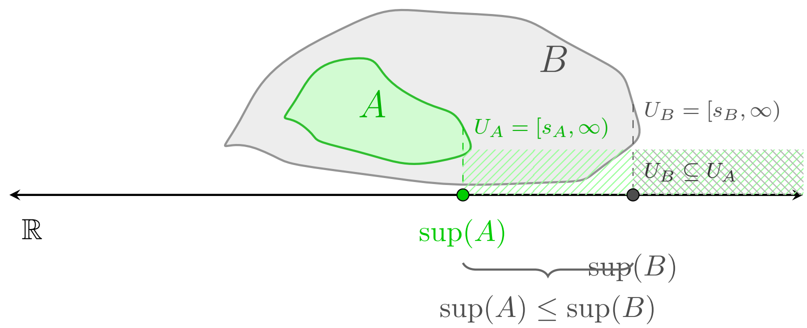
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq B \subseteq \mathbb{R} \text{ and } u \in \mathbb{R}.\\ &\big[\forall y \in B\ (y \leq u)\big] \Longrightarrow \big[\forall x \in A\ (x \leq u)\big].\end{aligned}$$

Remark (Predicate reading).

$$A \subseteq B \wedge \text{UpperBound}(u, B) \Longrightarrow \text{UpperBound}(u, A).$$



H]

Figure 3.10: The Monotonicity of Supremum Under Set Inclusion. This visualization illustrates the arithmetic consequence of set inclusion, $A \subseteq B \subset \mathbb{R}$. The topology of the sets is represented by nested blobs where A (green) is contained within B (gray). Vertically dashed projections link the supremal boundary of each set down to the shared real axis, defining the unique points $s_A = \sup(A)$ and $s_B = \sup(B)$. The figure demonstrates that because the territory of A is restricted by B, its maximum guard point s_A cannot exceed s_B. Finally, the shaded regions illustrate the territories of upper bounds (U_A and U_B), clearly showing that U_B is a proper subset of U_A. Any value guarding the larger set B necessarily satisfies the guard condition for the smaller set A.

Remark (Negated quantified statement).

$$\begin{gathered} A \subseteq B, \quad u \text{ is an upper bound of } B, \\ \text{and } u \text{ is not an upper bound of } A \end{gathered}$$

is impossible.

Remark (Negation predicate reading).

$$A \subseteq B \wedge \mathrm{UpperBound}(u, B) \wedge \neg\, \mathrm{UpperBound}(u, A)$$

is a contradictory configuration.

Remark (Failure modes). The implication could fail only if some $x \in A$ exceeded u. Since $A \subseteq B$, that same x would belong to B, contradicting that u bounds B from above.

Remark (Failure mode decomposition).

$$\begin{aligned} \neg\, \mathrm{UpperBound}(u, A) &\iff \underbrace{\exists x \in A\ (u < x)}_{\text{witness in } A} \\ &\implies \underbrace{\exists x \in B\ (u < x)}_{A \subseteq B} \iff \neg\, \mathrm{UpperBound}(u, B). \end{aligned}$$

Remark (Interpretation). A bound for a larger set automatically bounds every smaller set inside it. Inclusion can only remove possible counterexamples, so it cannot destroy an existing upper bound.

Remark (Dependencies).

Upper Bound.

Theorem (Supremum Is Monotone Under Inclusion)

Theorem 3.3.6 (Supremum Is Monotone Under Inclusion)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty sets that are bounded above. If $A \subseteq B$, then $\sup A \le \sup B$.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\text{Let } A, B \subseteq \mathbb{R} \text{ be nonempty and bounded above.}$$
$$A \subseteq B \implies \sup A \le \sup B.$$

Remark (Predicate reading)*.*

$$A \subseteq B \wedge \mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \implies s_A \le s_B.$$

Remark (Negated quantified statement)*.*

$$A \subseteq B, \quad \sup B < \sup A$$

is impossible when A and B are nonempty and bounded above.

Remark (Negation predicate reading)*.*

$$A \subseteq B \wedge \mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \wedge s_B < s_A$$

is a contradictory configuration.

Remark (Failure modes)*.* The monotonicity statement could fail only if the larger set had a strictly smaller least upper bound. But every upper bound for B is also an upper bound for A, so the least upper bound of A cannot sit above $\sup B$.

Remark (Failure mode decomposition)*.*

$$\begin{aligned} A \subseteq B &\implies \underbrace{\mathrm{UpperBound}(\sup B, A)}_{\sup B \text{ bounds the smaller set}} \\ &\implies \underbrace{\sup A \le \sup B}_{\sup A \text{ is least among upper bounds of } A}. \end{aligned}$$

Remark (Interpretation)*.* Making a set larger can only push its least ceiling upward or leave it unchanged. It cannot lower the supremum, because the larger set contains all of the old points that already had to be bounded.

Remark (Dependencies)*.*

Upper Bound; Supremum; Subset Inclusion Preserves Upper Bounds.

Proposition (Upper Bound Comparison with the Supremum)

Proposition 3.3.7 (Upper Bound Comparison with the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. For $u \in \mathbb{R}$, the number u is an upper bound of A if and only if $s \le u$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded above, and let } s = \sup A.\\ &\forall u \in \mathbb{R}\ \big[u \text{ is an upper bound of } A \iff s \le u\big].\end{aligned}$$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \implies \forall u \in \mathbb{R}\ \big[\text{UpperBound}(u, A) \iff s \le u\big].$$

Remark (Negated quantified statement).

$$\exists u \in \mathbb{R}\ \big[(u \text{ is an upper bound of } A \text{ and } u < s) \vee (s \le u \text{ and } u \text{ is not an upper bound of } A)\big]$$

is impossible when $s = \sup A$.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \exists u \in \mathbb{R}\ \big[(\text{UpperBound}(u, A) \wedge u < s) \vee (s \le u \wedge \neg\,\text{UpperBound}(u, A))\big]$$

is a contradictory configuration.

Remark (Failure modes). There are two apparent ways the equivalence could fail. Either an upper bound lies below the supremum, contradicting leastness, or a number above the supremum fails to bound the set, contradicting that the supremum already bounds the set.

Remark (Failure mode decomposition).

$$\begin{aligned}\text{UpperBound}(u, A) &\implies s \le u,\\ s \le u &\implies \underbrace{\forall x \in A\ (x \le s)}_{s \text{ bounds } A} \implies \underbrace{\forall x \in A\ (x \le u)}_{u \text{ also bounds } A}.\end{aligned}$$

Remark (Interpretation). Once the supremum is known, all upper bounds are exactly the real numbers at or above it. The supremum is therefore the threshold where upper bounds begin.

Remark (Dependencies).

Upper Bound; Supremum.

Proposition (Every Element Lies Below the Supremum)

Proposition 3.3.8 (Every Element Lies Below the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. Then $x \le s$ for every $x \in A$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded above, and let } s = \sup A.\\ &\forall x \in A\ (x \le s).\end{aligned}$$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \implies \forall x \in A\ (x \le s).$$

Remark (Negated quantified statement).

$$s = \sup A \quad \text{and} \quad \exists x \in A\ (s < x)$$

is impossible.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \exists x \in A\ (s < x)$$

is a contradictory configuration.

Remark (Failure modes). The statement could fail only if some element of A sat strictly above $\sup A$. That would mean $\sup A$ is not even an upper bound of A.

Remark (Failure mode decomposition).

$$\neg\big[\forall x \in A\ (x \le s)\big] \iff \underbrace{\exists x \in A\ (s < x)}_{\text{element above the alleged supremum}} \iff \neg\,\text{UpperBound}(s, A).$$

Remark (Interpretation). The supremum is a ceiling for the set. It may or may not be a point of the set, but every actual element of the set must lie at or below it.

Remark (Dependencies).

Upper Bound; Supremum.

Theorem (Infimum Is Less Than or Equal to the Supremum)

Theorem 3.3.9 (Infimum Is Less Than or Equal to the Supremum)**.** *Let* $A \subseteq \mathbb{R}$ *be nonempty, bounded below, and bounded above. Then* $\inf A \le \sup A$*.*

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be nonempty, bounded below, and bounded above.
$\inf A \le \sup A$.

Remark (Predicate reading).

$$\text{Infimum}(t, A) \wedge \text{Supremum}(s, A) \Longrightarrow t \le s.$$

Remark (Negated quantified statement).

$$\inf A > \sup A$$

is impossible for a nonempty set that has both bounds.

Remark (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge \text{Supremum}(s, A) \wedge s < t$$

is a contradictory configuration.

Remark (Failure modes). The inequality could fail only if the greatest lower bound were strictly above the least upper bound. Any element of the nonempty set must lie between those two bounding values, which rules out this order.

Remark (Failure mode decomposition).

$$\text{Choose } a \in A.$$
$$\underbrace{\inf A \leq a}_{\inf A \text{ is a lower bound threshold}} \quad \text{and} \quad \underbrace{a \leq \sup A}_{\sup A \text{ is an upper bound threshold}} \implies \inf A \leq \sup A.$$

Remark (Interpretation). A nonempty bounded set cannot have its floor above its ceiling. Any point of the set witnesses the bridge from the infimum up to the supremum.

Remark (Dependencies).

Infimum; Supremum.

Proposition (Order Comparison of Suprema)

Proposition 3.3.10 (Order Comparison of Suprema)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If for every $x \in A$ there exists $y \in B$ such that $x \leq y$, then* $\sup A \leq \sup B$*.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A, B \subseteq \mathbb{R} \text{ be nonempty and bounded above.}$$
$$\big[\forall x \in A\ \exists y \in B\ (x \leq y)\big] \implies \sup A \leq \sup B.$$

Remark (Predicate reading).

$$\big[\forall x \in A\ \exists y \in B\ (x \leq y)\big] \wedge \mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \implies s_A \leq s_B.$$

Remark (Negated quantified statement).

$$\big[\forall x \in A\ \exists y \in B\ (x \leq y)\big] \quad \text{and} \quad \sup B < \sup A$$

is impossible.

Remark (Negation predicate reading).

$$\big[\forall x \in A\ \exists y \in B\ (x \leq y)\big] \wedge \mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \wedge s_B < s_A$$

is a contradictory configuration.

Remark (Failure modes). The comparison could fail only if the supremum of A rose above the supremum of B. But every element of A is dominated by some element of B, and every element of B lies below $\sup B$, so $\sup B$ is an upper bound for A.

Remark (Failure mode decomposition).

$$\begin{aligned} \forall x \in A\ \exists y \in B\ (x \leq y) &\implies \underbrace{\forall x \in A\ (x \leq \sup B)}_{\sup B \text{ bounds } A} \\ &\implies \underbrace{\sup A \leq \sup B}_{\sup A \text{ is least among upper bounds of } A}. \end{aligned}$$

Remark (Interpretation). This criterion compares two sets without requiring inclusion. If every point of A can be matched below some point of B, then B reaches at least as high as A in the supremum sense.

Remark (Dependencies).

Supremum; Every Element Lies Below the Supremum; Upper Bound Comparison with the Supremum.

Theorem (Translation Invariance of the Supremum)

Theorem 3.3.11 (Translation Invariance of the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $c \in \mathbb{R}$. Then*

$$\sup(A+c) = \sup A + c, \qquad A+c = \{a+c : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded above, and let } c \in \mathbb{R}.\\ &\sup\{a+c : a \in A\} = \sup A + c.\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{Supremum}(s, A) \implies \mathrm{Supremum}(s+c, A+c).$$

Remark (Negated quantified statement).

$$\sup(A+c) \neq \sup A + c$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s, A) \wedge \neg\,\mathrm{Supremum}(s+c, A+c)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if translation either failed to preserve upper bounds or failed to preserve leastness. Adding the same real number to every comparison preserves the order, so neither failure mode can occur.

Remark (Failure mode decomposition).

$$\begin{gathered}\forall a \in A\ (a \le s) \iff \forall a \in A\ (a+c \le s+c),\\ \forall u\,[\mathrm{UpperBound}(u, A) \implies s \le u] \iff \forall v\,[\mathrm{UpperBound}(v, A+c) \implies s+c \le v].\end{gathered}$$

Remark (Interpretation). Translation shifts the entire set and all of its ceilings by the same amount. The least ceiling therefore shifts by exactly that amount.

Remark (Dependencies).

Upper Bound; Supremum.

Theorem (Scalar Multiplication by a Positive Scalar)

Theorem 3.3.12 (Scalar Multiplication by a Positive Scalar)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k > 0$. Then*

$$\sup(kA) = k \sup A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded above, and let } k > 0.\\ &\sup\{ka : a \in A\} = k \sup A.\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{Supremum}(s, A) \wedge k > 0 \Longrightarrow \mathrm{Supremum}(ks, kA).$$

Remark (Negated quantified statement).

$$k > 0, \quad \sup(kA) \neq k \sup A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s, A) \wedge k > 0 \wedge \neg\,\mathrm{Supremum}(ks, kA)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if multiplying by a positive scalar failed to preserve upper bounds or failed to preserve leastness. Positive multiplication preserves the order, so both properties scale exactly.

Remark (Failure mode decomposition).

$$\begin{gathered}\forall a \in A\ (a \leq s) \Longleftrightarrow \forall a \in A\ (ka \leq ks),\\ \mathrm{UpperBound}(u, A) \Longleftrightarrow \mathrm{UpperBound}(ku, kA).\end{gathered}$$

Remark (Interpretation). Positive scaling stretches the set without reversing order. The least ceiling stretches by the same positive factor.

Remark (Dependencies).

Upper Bound; Supremum.

Theorem (Scalar Multiplication by a Negative Scalar)

Theorem 3.3.13 (Scalar Multiplication by a Negative Scalar)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k < 0$. Then*

$$\sup(kA) = k \inf A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{gathered}\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded below, and let } k < 0. \\ \sup\{ka : a \in A\} = k \inf A.\end{gathered}$$

Remark (Predicate reading)*.*

$$\text{Infimum}(t, A) \wedge k < 0 \Longrightarrow \text{Supremum}(kt, kA).$$

Remark (Negated quantified statement)*.*

$$k < 0, \quad \sup(kA) \neq k \inf A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading)*.*

$$\text{Infimum}(t, A) \wedge k < 0 \wedge \neg\,\text{Supremum}(kt, kA)$$

is a contradictory configuration.

Remark (Failure modes)*.* The formula could fail only if negative multiplication did not reverse lower bounds into upper bounds, or if the extremal value lost its leastness after reversal. Multiplication by a negative scalar reverses order exactly, so the infimum of A becomes the supremum of kA.

Remark (Failure mode decomposition)*.*

$$\begin{gathered}\forall a \in A\ (t \leq a) \Longleftrightarrow \forall a \in A\ (ka \leq kt), \\ \text{LowerBound}(v, A) \Longleftrightarrow \text{UpperBound}(kv, kA) \qquad (k < 0).\end{gathered}$$

Remark (Interpretation)*.* Negative scaling reflects the set and changes floors into ceilings. The lowest point threshold of A becomes the highest point threshold of the reflected and scaled set.

Remark (Dependencies)*.*

Lower Bound; Upper Bound; Infimum; Supremum.

Theorem (Negation Exchanges Supremum and Infimum)

Theorem 3.3.14 (Negation Exchanges Supremum and Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below. Then $-A = \{-a : a \in A\}$ is bounded above and*

$$\sup(-A) = -\inf A.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded below.}\\&\sup\{-a : a \in A\} = -\inf A.\end{aligned}$$

Remark (Predicate reading).

$$\text{Infimum}(t, A) \Longrightarrow \text{Supremum}(-t, -A).$$

Remark (Negated quantified statement).

$$\sup(-A) \neq -\inf A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge \neg\,\text{Supremum}(-t, -A)$$

is a contradictory configuration.

Remark (Failure modes). The theorem could fail only if negation did not convert lower bounds into upper bounds or did not preserve the extremal threshold after reversing order. The map $x \mapsto -x$ reverses all inequalities, so the exchange is exact.

Remark (Failure mode decomposition).

$$\begin{aligned}\forall a \in A\ (t \leq a) &\Longleftrightarrow \forall a \in A\ (-a \leq -t),\\ \text{LowerBound}(v, A) &\Longleftrightarrow \text{UpperBound}(-v, -A).\end{aligned}$$

Remark (Interpretation). Negation reflects the real line through the origin. Floors become ceilings, so the infimum of the original set becomes the negative of the supremum of the reflected set.

Remark (Dependencies).

Lower Bound; Upper Bound; Infimum; Supremum.

Theorem (Supremum of a Sum Set)

Theorem 3.3.15 (Supremum of a Sum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup(A + B) = \sup A + \sup B, \qquad A + B = \{a + b : a \in A,\ b \in B\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A, B \subseteq \mathbb{R} \text{ be nonempty and bounded above.}\\&\sup\{a + b : a \in A,\ b \in B\} = \sup A + \sup B.\end{aligned}$$

Remark (Predicate reading).

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \implies \text{Supremum}(s_A + s_B, A + B).$$

Remark (Negated quantified statement).

$$\sup(A + B) \neq \sup A + \sup B$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \wedge \neg\, \text{Supremum}(s_A + s_B, A + B)$$

is a contradictory configuration.

Remark (Failure modes). The equality could fail only if the sum of the two suprema did not bound the sum set, or if it were not the least such bound. The first part follows from adding the two upper-bound inequalities; the second follows by approximating each supremum from within its set.

Remark (Failure mode decomposition).

$$\begin{aligned} a \leq \sup A, \quad b \leq \sup B &\implies a + b \leq \sup A + \sup B, \\ \varepsilon > 0 &\implies \exists a \in A,\ b \in B : \sup A + \sup B - \varepsilon < a + b. \end{aligned}$$

Remark (Interpretation). The highest limiting value of all pairwise sums is obtained by combining values that approach the separate ceilings of A and B.

Remark (Dependencies).

Supremum; Epsilon Characterization of Supremum.

Theorem (Supremum of a Difference Set)

Theorem 3.3.16 (Supremum of a Difference Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded above and B bounded below. Then*

$$\sup(A - B) = \sup A - \inf B, \qquad A - B = \{a - b : a \in A,\ b \in B\}.$$

Go to proof.

Remark (Standard quantified statement).

Let $A, B \subseteq \mathbb{R}$ be nonempty, A bounded above, and B bounded below.
$\sup\{a - b : a \in A,\ b \in B\} = \sup A - \inf B.$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, B) \implies \text{Supremum}(s - t, A - B).$$

Remark (Negated quantified statement).

$$\sup(A - B) \neq \sup A - \inf B$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, B) \wedge \neg\,\text{Supremum}(s - t, A - B)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if subtracting elements of B did not convert the lower extremal behavior of B into an upper contribution. Since $a - b = a + (-b)$, this is the sum theorem combined with the negation exchange.

Remark (Failure mode decomposition).

$$A - B = A + (-B), \qquad \sup(-B) = -\inf B, \qquad \sup(A + (-B)) = \sup A + \sup(-B).$$

Remark (Interpretation). The largest possible difference comes from making the first term as large as possible and the subtracted term as small as possible.

Remark (Dependencies).

Supremum; Infimum; Negation Exchanges Supremum and Infimum; Supremum of a Sum Set.

Theorem (Supremum of a Dilation)

Theorem 3.3.17 (Supremum of a Dilation). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and let $\lambda \in \mathbb{R}$. Then*

$$\sup(\lambda A) = \max\{\lambda \sup A, \lambda \inf A\}, \qquad \lambda A = \{\lambda a : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and let $\lambda \in \mathbb{R}$.
$\sup\{\lambda a : a \in A\} = \max\{\lambda \sup A, \lambda \inf A\}$.

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, A) \Longrightarrow \text{Supremum}(\max\{\lambda s, \lambda t\}, \lambda A).$$

Remark (Negated quantified statement).

$$\sup(\lambda A) \neq \max\{\lambda \sup A, \lambda \inf A\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\operatorname{Supremum}(s, A) \wedge \operatorname{Infimum}(t, A) \wedge \neg \operatorname{Supremum}(\max\{\lambda s, \lambda t\}, \lambda A)$$

is a contradictory configuration.

Remark (Failure modes). The formula separates into the cases $\lambda > 0$, $\lambda = 0$, and $\lambda < 0$. Positive dilation scales the supremum, zero dilation collapses the set to $\{0\}$, and negative dilation turns the infimum into the supremum.

Remark (Failure mode decomposition).

$$\sup(\lambda A) = \begin{cases} \lambda \sup A, & \lambda > 0, \\ 0, & \lambda = 0, \\ \lambda \inf A, & \lambda < 0. \end{cases}$$

Remark (Interpretation). A dilation either preserves the order, collapses all points to zero, or reverses the order. The displayed maximum packages those three cases into one formula.

Remark (Dependencies).

Supremum; Infimum; Maximum; Scalar Multiplication by a Positive Scalar; Scalar Multiplication by a Negative Scalar.

Theorem (Supremum of the Absolute-Value Image)

Theorem 3.3.18 (Supremum of the Absolute-Value Image)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup |A| = \max\{|\sup A|, |\inf A|\}, \qquad |A| = \{|a| : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded.}$$
$$\sup\{|a| : a \in A\} = \max\{|\sup A|, |\inf A|\}.$$

Remark (Predicate reading).

$$\operatorname{Supremum}(s, A) \wedge \operatorname{Infimum}(t, A) \implies \operatorname{Supremum}(\max\{|s|, |t|\}, |A|).$$

Remark (Negated quantified statement).

$$\sup |A| \neq \max\{|\sup A|, |\inf A|\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\operatorname{Supremum}(s, A) \wedge \operatorname{Infimum}(t, A) \wedge \neg \operatorname{Supremum}(\max\{|s|, |t|\}, |A|)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if the farthest absolute-value size were not achieved at one of the two extremal thresholds. Since every $a \in A$ lies between $\inf A$ and $\sup A$, its distance from zero is bounded by the larger endpoint magnitude.

Remark (Failure mode decomposition).

$$\inf A \le a \le \sup A \Longrightarrow |a| \le \max\{|\sup A|, |\inf A|\}.$$

Remark (Interpretation). The absolute-value image measures distance from the origin. For a bounded subset of the real line, the largest possible distance is controlled by whichever extremal endpoint lies farther from zero.

Remark (Dependencies).

Supremum; Infimum; Maximum; Every Element Lies Below the Supremum.

Proposition (Bounds for Suprema and Infima Under Set Addition)

Proposition 3.3.19 (Bounds for Suprema and Infima Under Set Addition)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf A + \inf B \le \inf(A + B) \qquad \textit{and} \qquad \sup(A + B) \le \sup A + \sup B.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{array}{l}\text{Let } A, B \subseteq \mathbb{R} \text{ be nonempty and bounded.}\\ \inf A + \inf B \le \inf(A + B) \quad \wedge \quad \sup(A + B) \le \sup A + \sup B.\end{array}$$

Remark (Predicate reading).

$$\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \wedge \text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \Longrightarrow t_A + t_B \le \inf(A+B) \le \sup(A+$$

Remark (Negated quantified statement).

$$\inf(A + B) < \inf A + \inf B \quad \text{or} \quad \sup A + \sup B < \sup(A + B)$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\big[\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \wedge \inf(A+B) < t_A + t_B\big] \vee \big[\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \wedge s_A + s_B <$$

is a contradictory configuration.

Remark (Failure modes). The lower inequality could fail only if a sum fell below the sum of the two lower thresholds. The upper inequality could fail only if a sum rose above the sum of the two upper thresholds.

Remark (Failure mode decomposition).

$$\inf A \le a \le \sup A, \quad \inf B \le b \le \sup B \Longrightarrow \inf A + \inf B \le a + b \le \sup A + \sup B.$$

Remark (Interpretation). Every element of the sum set is built from one element of A and one element of B. Adding the individual lower and upper controls gives immediate lower and upper controls for every such sum.

Remark (Dependencies).

Infimum; Supremum; Supremum of a Sum Set.

Proposition (Upper Bounds Depend on the Ambient Order)

Proposition 3.3.20 (Upper Bounds Depend on the Ambient Order)**.** *Let $(S, \leq_S)$ and $(T, \leq_T)$ be ordered sets, and let $A \subseteq S \cap T$. Whether u is an upper bound of A must be interpreted relative to the chosen ambient order. Go to proof.*

Remark (Standard quantified statement).

$$u \text{ is an upper bound of } A \text{ in } S \iff u \in S \text{ and } \forall x \in A\ (x \leq_S u),$$
$$u \text{ is an upper bound of } A \text{ in } T \iff u \in T \text{ and } \forall x \in A\ (x \leq_T u).$$

Remark (Predicate reading).

$$\mathrm{UpperBound}_S(u, A) \iff u \in S \wedge \forall x \in A\ (x \leq_S u).$$

Remark (Negated quantified statement).

$$\text{The phrase "}u\text{ is an upper bound of }A\text{"}$$

is not well-determined until the ambient ordered set and its order relation are fixed.

Remark (Negation predicate reading).

$$\neg\, \mathrm{WellDetermined}(\mathrm{UpperBound}(u, A))$$

when the ambient order is omitted.

Remark (Failure modes). Ambiguity occurs when the same set A is considered inside different ordered ambient sets or under different order relations. The candidate bound may belong to one ambient set and not another, or the relevant comparison relation may change.

Remark (Failure mode decomposition).

$$\mathrm{UpperBound}_S(u, A) \text{ and } \mathrm{UpperBound}_T(u, A)$$

are separate assertions unless $S = T$ and $\leq_S = \leq_T$ on the elements being compared.

Remark (Interpretation). An upper bound is not only about the subset; it is also about where the subset lives. Changing the ambient order changes the universe of possible bounds and the comparison used to test them.

Remark (Dependencies).

Upper Bound.

Proposition (Suprema Depend on the Ambient Set)

Proposition 3.3.21 (Suprema Depend on the Ambient Set)**.** *Let $A \subseteq S \subseteq S'$ be subsets of an ordered set. The supremum of A relative to S, if it exists, need not agree with the supremum of A relative to S'.* *Go to proof.*

Remark (Standard quantified statement).

$$\sup_S A \text{ and } \sup_{S'} A$$

are ambient-relative objects and may differ in existence or value.

Remark (Predicate reading).

$$\mathrm{Supremum}_S(s, A) \quad \text{and} \quad \mathrm{Supremum}_{S'}(s, A)$$

are different assertions unless the ambient upper-bound sets agree.

Remark (Negated quantified statement).

$$\sup_S A = \sup_{S'} A$$

is not guaranteed by $A \subseteq S \subseteq S'$ alone.

Remark (Negation predicate reading).

$$\mathrm{Supremum}_{S'}(s, A) \wedge \neg\, \mathrm{Supremum}_S(s, A)$$

is possible when the larger ambient set contains the least upper bound but the smaller one does not.

Remark (Failure modes). The ambient-relative supremum can fail to transfer because the candidate least upper bound may not belong to the smaller ambient set, or because the smaller ambient set may have a different collection of upper bounds.

Remark (Failure mode decomposition).

$$s = \sup_{S'} A, \quad s \notin S \Longrightarrow s \text{ cannot be the supremum of } A \text{ relative to } S.$$

Remark (Interpretation). Suprema are least upper bounds inside a chosen universe. Enlarging or shrinking that universe can create or remove the point that should serve as the least ceiling.

Remark (Dependencies).

Upper Bound; Supremum.

Theorem (Ambient Existence of Supremum)

Theorem 3.3.22 (Ambient Existence of Supremum)**.** *There are ordered sets $S \subseteq S'$ and a subset $A \subseteq S$ such that A has a supremum relative to S' but has no supremum relative to S.* *Go to proof.*

Remark (Standard quantified statement).

$$\exists S \subseteq S' \; \exists A \subseteq S : \quad \sup_{S'} A \text{ exists} \quad \text{and} \quad \sup_{S} A \text{ does not exist.}$$

Remark (Predicate reading).

$$\exists S \subseteq S' \; \exists A \subseteq S \; \exists s \in S' : \mathrm{Supremum}_{S'}(s, A) \wedge \neg \exists t \in S \;\; \mathrm{Supremum}_{S}(t, A).$$

Remark (Negated quantified statement).

Every ambient enlargement preserves and reflects existence of suprema.

This assertion is false.

Remark (Negation predicate reading).

$$\forall S \subseteq S' \; \forall A \subseteq S : \left[\exists s \in S' \; \mathrm{Supremum}_{S'}(s, A) \Longrightarrow \exists t \in S \; \mathrm{Supremum}_{S}(t, A)\right]$$

is false.

Remark (Failure modes). The transfer from S' to S fails when the least upper bound exists in the larger ambient set but is missing from the smaller one. The smaller ambient set may contain upper bounds without containing a least upper bound.

Remark (Failure mode decomposition).

$$s = \sup_{S'} A, \quad s \notin S, \quad \text{and no } t \in S \text{ is least among the upper bounds of } A \text{ in } S.$$

Remark (Interpretation). Completeness is an ambient property. A larger ordered universe can fill a gap that remains absent in a smaller ordered universe.

Remark (Dependencies).
Upper Bound; Supremum; Suprema Depend on the Ambient Set.

Lemma (Rational Gap Example for Suprema)

Lemma 3.3.23 (Rational Gap Example for Suprema)**.** *Let*

$$A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Then A is nonempty and bounded above in $\mathbb{Q}$, but A has no supremum relative to $\mathbb{Q}$. *Go to proof.*

Remark (Standard quantified statement).

$$A = \{q \in \mathbb{Q} : q^2 < 2\} \quad \Longrightarrow \quad \sup_{\mathbb{Q}} A \text{ does not exist.}$$

Remark (Predicate reading).

$$\neg \exists s \in \mathbb{Q} \;\; \mathrm{Supremum}_{\mathbb{Q}}(s, A), \qquad A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Remark (Negated quantified statement).

$$\exists s \in \mathbb{Q}\ (s = \sup_{\mathbb{Q}} A)$$

is false for this set.

Remark (Negation predicate reading).

$$\exists s \in \mathbb{Q}\ \text{Supremum}_{\mathbb{Q}}(s, A)$$

is false.

Remark (Failure modes). A rational candidate below $\sqrt{2}$ fails to be an upper bound. A rational candidate above $\sqrt{2}$ is an upper bound but is not least, because another rational upper bound lies between it and $\sqrt{2}$.

Remark (Failure mode decomposition).

$$\begin{cases} s^2 < 2 & \Longrightarrow s \text{ is not an upper bound of } A, \\ s^2 > 2 & \Longrightarrow s \text{ is not the least upper bound of } A. \end{cases} \qquad (s \in \mathbb{Q})$$

Remark (Interpretation). The set wants $\sqrt{2}$ as its least ceiling, but that number is not rational. The rational ambient set therefore has a visible gap where the supremum should be.

Remark (Dependencies).

Upper Bound; Supremum; Ambient Existence of Supremum.

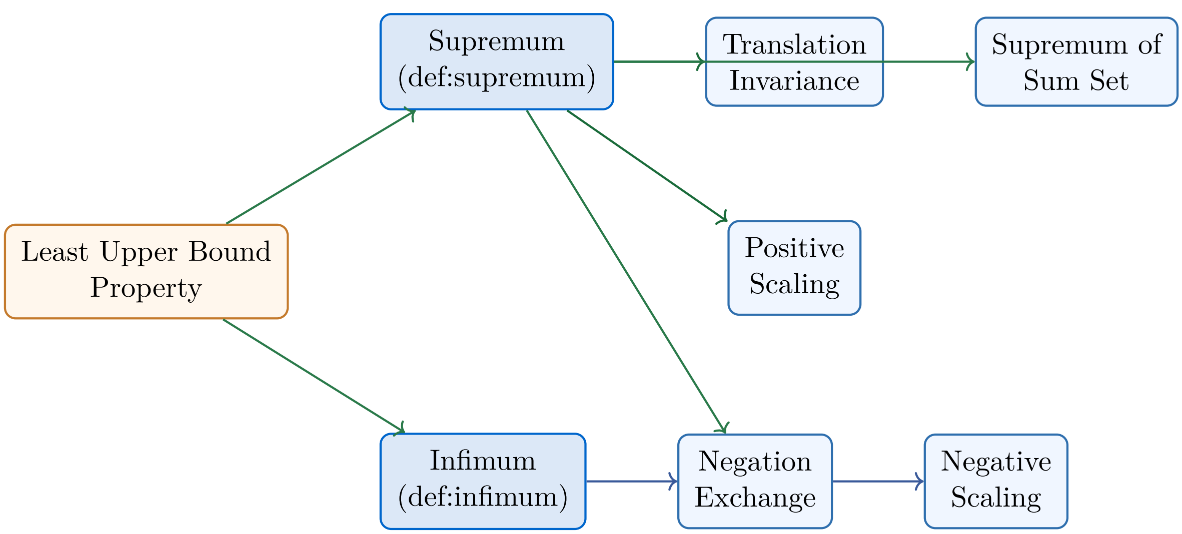
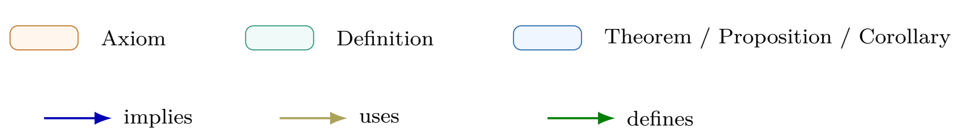


Figure 3.11: Bound Algebra Cluster Dependency Map. The map originates from the ‘Least Upper Bound Property‘ (Axiom) and branches into the dual properties of the continuum. The Negation Exchange serves as the structural bridge between supremum and infimum algebra.



3.3.1 Bound Algebra

Chapter 4

Functions

SERIES	VOLUME III	BOOK I	CHAPTER 4
From Cantor to Ito	Classical Analysis	Bounds, Sequences, and Series	**Functions**

4.1 Algebra of Functions

Algebra of Functions — Quick Reference

Statement	**Role**
Injective composition	Injectivity is preserved by composition
Surjective composition	Surjectivity is preserved by composition
Bijective composition	Bijections compose to bijections
Inverse bijection	Bijections have bijective inverses
Preimage algebra	Preimages preserve unions and intersections exactly

Algebra of Functions

Remark (Intervals on $\mathbb{R}$). Every open interval (a, b) consists entirely of cluster points; $[a, b]$ is closed and bounded. Natural domains for functions are typically intervals or unions of intervals.

Proposition

Proposition 4.1.1 (Composition of Injections Is Injective). *Let $f : A \to B$ and let $g : B \to C$. If f is injective and g is injective, then $g \circ f : A \to C$ is injective.*

Go to proof.

Remark (Standard quantified statement).

$$\Big(\forall a_1, a_2 \in A\ (f(a_1) = f(a_2) \Rightarrow a_1 = a_2)\Big) \land \Big(\forall b_1, b_2 \in B\ (g(b_1) = g(b_2) \Rightarrow b_1 = b_2)\Big)$$
$$\Rightarrow \forall a_1, a_2 \in A\ (g(f(a_1)) = g(f(a_2)) \Rightarrow a_1 = a_2).$$

Remark (Definition predicate reading).

$$\text{Injective}(f) \land \text{Injective}(g) \implies \text{Injective}(g \circ f).$$

Remark (Negated quantified statement).

$$\exists a_1, a_2 \in A\ \big(a_1 \neq a_2\ \land\ g(f(a_1)) = g(f(a_2))\big).$$

Remark (Negation predicate reading).

$$\neg\ \text{Injective}(g \circ f).$$

Remark (Failure modes). The composite fails to be injective exactly when two distinct inputs of A are sent to the same output in C after applying both functions.

Remark (Failure mode decomposition).

$$\exists a_1, a_2 \in A\ \big(a_1 \neq a_2\ \land\ g(f(a_1)) = g(f(a_2))\big).$$

Remark (Interpretation). Injectivity survives composition: equality after $g \circ f$ can be pulled back first through g and then through f.

Remark (Dependencies).

- Composition
- Function

Proposition

Proposition 4.1.2 (Composition of Surjections Is Surjective)**.** Surjective(f) $\land$ Surjective$(g) \Rightarrow$ Surjective$(g \circ f)$.

Go to proof.

Remark (Standard quantified statement).

$$\text{Surjective}(f) \land \text{Surjective}(g)\ \Rightarrow\ \forall c \in C,\ \exists a \in A : g(f(a)) = c.$$

Remark (Definition predicate reading).

$$\text{Surjective}(g \circ f, C)\ :\equiv\ \forall c \in C,\ \exists a \in A : (g \circ f)(a) = c.$$

Remark (Dependencies).

- Composition

- Function

Corollary

Corollary 4.1.3 (Composition of Bijections Is Bijective). *Let $f : A \to B$ and let $g : B \to C$. If f is bijective and g is bijective, then $g \circ f : A \to C$ is bijective.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\big(\forall a_1, a_2 \in A\ (f(a_1) = f(a_2) \Rightarrow a_1 = a_2)\big) \wedge \big(\forall b \in B\ \exists a \in A\ (f(a) = b)\big)\Big]$$
$$\wedge \Big[\big(\forall b_1, b_2 \in B\ (g(b_1) = g(b_2) \Rightarrow b_1 = b_2)\big) \wedge \big(\forall c \in C\ \exists b \in B\ (g(b) = c)\big)\Big]$$
$$\Rightarrow \Big[\big(\forall a_1, a_2 \in A\ (g(f(a_1)) = g(f(a_2)) \Rightarrow a_1 = a_2)\big) \wedge \big(\forall c \in C\ \exists a \in A\ (g(f(a)) = c)\big)\Big].$$

Remark (Definition predicate reading).

$$\text{Bijective}(f) \wedge \text{Bijective}(g) \implies \text{Bijective}(g \circ f).$$

Remark (Negated quantified statement).

$$\exists a_1, a_2 \in A\ \big(a_1 \neq a_2\ \wedge\ g(f(a_1)) = g(f(a_2))\big)$$
$$\vee\ \exists c \in C\ \forall a \in A\ \big(g(f(a)) \neq c\big).$$

Remark (Negation predicate reading).

$$\neg\ \text{Bijective}(g \circ f).$$

Remark (Failure modes). A composite bijection can fail only by losing injectivity or by missing some target value in C.

Remark (Failure mode decomposition).

$$\neg\ \text{Bijective}(g \circ f) \iff \neg\ \text{Injective}(g \circ f)\ \vee\ \neg\ \text{Surjective}(g \circ f).$$

Remark (Interpretation). A bijection is an injective and surjective function. Since both of those properties are preserved by composition, bijectivity is preserved too.

Remark (Dependencies).

- Composition
- Function

Proposition

Proposition 4.1.4 (Inverse of a Bijection Is a Bijection). Bijective(f) $\Rightarrow$ Bijective(f^{-1}).
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Bijective}(f) \Rightarrow \mathrm{InverseFunction}(f, f^{-1}),\ f^{-1} \circ f = \mathrm{id}_A,\ f \circ f^{-1} = \mathrm{id}_B.$$

Remark (Definition predicate reading).

$$\mathrm{InverseFunction}(f, f^{-1}) \ :\equiv\ f^{-1} \circ f = \mathrm{id}_A \ \wedge\ f \circ f^{-1} = \mathrm{id}_B.$$

Remark (Dependencies).

- Composition
- Function

Proposition

Proposition 4.1.5 (Preimage Distributes over Union and Intersection)**.** *Let* $f : A \to B$*,* $S, T \subseteq B$*. Then:*

(i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.

(ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.

(iii) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &(i) \quad x \in f^{-1}(S \cup T) \iff f(x) \in S \vee f(x) \in T \iff x \in f^{-1}(S) \cup f^{-1}(T). \\ &(ii) \quad x \in f^{-1}(S \cap T) \iff f(x) \in S \wedge f(x) \in T \iff x \in f^{-1}(S) \cap f^{-1}(T). \end{aligned}$$

Remark (Negated quantified statement).

$$x \notin f^{-1}(S) \iff f(x) \notin S \iff x \in f^{-1}(S^c).$$

Remark (Interpretation). Preimage is a complete lattice homomorphism: it preserves all set operations including complement. Contrast with image: $f(A \cap B) \subseteq f(A) \cap f(B)$ with strict containment possible when f is not injective. Preimage identities are load-bearing in topology (preimages of open sets are open) and measure theory (preimages of measurable sets are measurable).

Remark (Dependencies).

- Function

4.2 Real-Valued Functions

Real-Valued Functions — Quick Reference

Concept	Meaning
Pointwise sum	Algebraic addition of function values
Pointwise product	Product fg, distinct from composition
Pointwise quotient	Quotient where denominator is nonzero
Bounded function	A single bound controls all values on A
Supremum on a set	Least upper bound of $f(A)$
Maximum point	Point where the supremum is attained
Monotone function	Increasing or decreasing behavior on a set
Constant function	Same value at every input

Real-Valued Functions

Definition (Pointwise Sum of Functions)

Definition 4.2.1 (Pointwise Sum of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise sum $f + g : X \to \mathbb{R}$ is the function defined by

$$(f+g)(x) = f(x) + g(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((f+g)(x) = f(x) + g(x)\big).$$

Remark (Interpretation). The sum is computed output by output; at each input, add the two function values at that same input.

Remark (Dependencies).

- Function
- Subset
- Addition on $\mathbb{R}$

Definition (Pointwise Difference of Functions)

Definition 4.2.2 (Pointwise Difference of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise difference $f - g : X \to \mathbb{R}$ is the function defined by

$$(f-g)(x) = f(x) - g(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((f-g)(x) = f(x) - g(x)\big).$$

Remark (Interpretation). The difference subtracts the output of g from the output of f at each input separately.

Remark (Dependencies).

- Function
- Subset
- Subtraction on $\mathbb{R}$

Definition (Pointwise Product of Functions)

Definition 4.2.3 (Pointwise Product of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise product $fg : X \to \mathbb{R}$ is the function defined by

$$(fg)(x) = f(x)g(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((fg)(x) = f(x)g(x)\big).$$

Remark (Interpretation). The product is not composition: it multiplies the two real outputs at the same input.

Remark (Dependencies).

- Function
- Subset
- Multiplication on $\mathbb{R}$

Definition (Pointwise Scalar Multiple of a Function)

Definition 4.2.4 (Pointwise Scalar Multiple of a Function)**.** Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, and let $c \in \mathbb{R}$. The pointwise scalar multiple $cf : X \to \mathbb{R}$ is the function defined by

$$(cf)(x) = cf(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((cf)(x) = cf(x)\big).$$

Remark (Interpretation). Scalar multiplication stretches every output of the function by the same real scalar.

Remark (Dependencies).

- Function

- Subset
- Multiplication on $\mathbb{R}$

Definition (Pointwise Absolute Value of a Function)

Definition 4.2.5 (Pointwise Absolute Value of a Function)**.** Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$. The pointwise absolute value $|f| : X \to \mathbb{R}$ is the function defined by

$$|f|(x) = |f(x)| \qquad (x \in X).$$

Remark (Standard quantified statement)*.*

$$\forall x \in X \ \big(|f|(x) = |f(x)|\big).$$

Remark (Interpretation)*.* The absolute value operation applies the ordinary real absolute value to each output.

Remark (Dependencies)*.*

- Function
- Subset
- Absolute Value on $\mathbb{R}$

Definition (Pointwise Maximum of Two Functions)

Definition 4.2.6 (Pointwise Maximum of Two Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise maximum $\max(f, g) : X \to \mathbb{R}$ is the function defined by

$$\max(f, g)(x) = \max\{f(x), g(x)\} \qquad (x \in X).$$

Remark (Standard quantified statement)*.*

$$\forall x \in X \ \big(\max(f, g)(x) = \max\{f(x), g(x)\}\big).$$

Remark (Interpretation)*.* At each input, the new function keeps the larger of the two output values.

Remark (Dependencies)*.*

- Function
- Subset
- Maximum

Definition (Pointwise Quotient of Functions)

Definition 4.2.7 (Pointwise Quotient of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. Suppose $g(x) \neq 0$ for every $x \in X$. The pointwise quotient $f/g : X \to \mathbb{R}$ is the function defined by

$$(f/g)(x) = \frac{f(x)}{g(x)} \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\big(\forall x \in X \; (g(x) \neq 0)\big) \implies \forall x \in X \; \big((f/g)(x) = f(x)/g(x)\big).$$

Remark (Interpretation). The quotient is defined pointwise only when the denominator function never vanishes on the domain.

Remark (Dependencies).

- Function
- Subset
- Division on $\mathbb{R}$

Definition (Linear Combination of Real-Valued Functions)

Definition 4.2.8 (Linear Combination of Real-Valued Functions)**.** Let $X \subseteq \mathbb{R}$, let $f, g : X \to \mathbb{R}$, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha f + \beta g : X \to \mathbb{R}$ is the function defined by

$$(\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \; \big((\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x)\big).$$

Remark (Interpretation). A linear combination is built only from scalar multiples and sums. It is the shared algebraic pattern behind sum rules, constant-multiple rules, and finite linear-combination rules for functions.

Remark (Dependencies).

- Pointwise Scalar Multiple of a Function
- Pointwise Sum of Functions

Definition (Real-Linear Rule)

Definition 4.2.9 (Real-Linear Rule)**.** Let $\mathcal{C}$ be a class of real-valued functions closed under real linear combinations. A rule T on $\mathcal{C}$ is real-linear if, whenever $f, g \in \mathcal{C}$ and $\alpha, \beta \in \mathbb{R}$,

$$T(\alpha f + \beta g) = \alpha T(f) + \beta T(g)$$

whenever both sides are defined.

Remark (Standard quantified statement).

$$f, g \in \mathcal{C} \implies T(\alpha f + \beta g) = \alpha T(f) + \beta T(g).$$

Remark (Interpretation). Linearity records exactly the behavior of preserving addition and scalar multiplication. Nonlinear operations such as products, quotients, absolute values, and order comparisons require their own dependency information.

Remark (Dependencies).

- Function
- Linear Combination of Real-Valued Functions

Definition (Bounded Above Function)

Definition 4.2.10 (Bounded Above Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded above on A if there exists $M \in \mathbb{R}$ such that

$$f(x) \le M \qquad (x \in A).$$

Remark (Standard quantified statement).

$$\exists M \in \mathbb{R}\, \forall x \in A\, \big(f(x) \le M\big).$$

Remark (Negated quantified statement).

$$\forall M \in \mathbb{R}\, \exists x \in A\, \big(f(x) > M\big).$$

Remark (Failure modes). No proposed real ceiling works: whenever a candidate upper bound is chosen, some input has an output larger than that candidate.

Remark (Failure mode decomposition).

$$\forall M \in \mathbb{R}\, \exists x \in A\, \big(f(x) > M\big).$$

Remark (Interpretation). Bounded above means all output values lie below one common real ceiling.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$
- Bounded Above

Definition (Bounded Below Function)

Definition 4.2.11 (Bounded Below Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded below on A if there exists $m \in \mathbb{R}$ such that

$$m \leq f(x) \qquad (x \in A).$$

Remark (Standard quantified statement)*.*

$$\exists m \in \mathbb{R}\ \forall x \in A\ \big(m \leq f(x)\big).$$

Remark (Negated quantified statement)*.*

$$\forall m \in \mathbb{R}\ \exists x \in A\ \big(f(x) < m\big).$$

Remark (Failure modes)*.* No proposed real floor works: whenever a candidate lower bound is chosen, some input has an output smaller than that candidate.

Remark (Failure mode decomposition)*.*

$$\forall m \in \mathbb{R}\ \exists x \in A\ \big(f(x) < m\big).$$

Remark (Interpretation)*.* Bounded below means all output values lie above one common real floor.

Remark (Dependencies)*.*

- Function
- Subset
- Order on $\mathbb{R}$
- Bounded Below

Definition (Bounded Function)

Definition 4.2.12 (Bounded Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded on A if there exists $B > 0$ such that

$$|f(x)| \leq B \qquad (x \in A).$$

Remark (Standard quantified statement)*.*

$$\exists B > 0\ \forall x \in A\ \big(|f(x)| \leq B\big).$$

Remark (Definition predicate reading)*.*

$$\mathrm{BoundedOn}(f, A) \ :=\ \exists B > 0\ \forall x \in A\ \big(|f(x)| \leq B\big).$$

Remark (Negated quantified statement).

$$\forall B > 0\ \exists x \in A\ \big(|f(x)| > B\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BoundedOn}(f, A) \iff \forall B > 0\ \exists x \in A\ \big(|f(x)| > B\big).$$

Remark (Failure modes). Every symmetric interval around 0 fails to contain all output values of f on A.

Remark (Failure mode decomposition).

$$\forall B > 0\ \exists x \in A\ \big(f(x) > B\ \vee\ f(x) < -B\big).$$

Remark (Interpretation). A bounded function has all of its values trapped inside some symmetric interval $[-B, B]$. Being bounded above alone is not enough; the function $f(x) = -x$ on $\mathbb{R}$ is bounded above by 0 but is not bounded.

Remark (Dependencies).

- Bounded Above Function
- Bounded Below Function

Definition (Supremum of a Function on a Set)

Definition 4.2.13 (Supremum of a Function on a Set)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The supremum of f on A is

$$\sup_{x \in A} f(x) = \sup f(A),$$

provided the supremum of $f(A)$ exists.

Remark (Standard quantified statement).

$$s = \sup_{x \in A} f(x) \iff \big(\forall x \in A\ (f(x) \le s)\big)\ \wedge\ \big(\forall u \in \mathbb{R}\ ((\forall x \in A\ (f(x) \le u)) \Rightarrow s \le u)\big).$$

Remark (Interpretation). The supremum is the least real ceiling for the output set $f(A)$; it need not be an output value of f.

Remark (Dependencies).

- Function
- Subset
- Image Set
- Supremum

Definition (Infimum of a Function on a Set)

Definition 4.2.14 (Infimum of a Function on a Set)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The infimum of f on A is

$$\inf_{x \in A} f(x) = \inf f(A),$$

provided the infimum of $f(A)$ exists.

Remark (Standard quantified statement).

$$t = \inf_{x \in A} f(x) \iff \big(\forall x \in A \; (t \leq f(x))\big) \;\wedge\; \big(\forall \ell \in \mathbb{R} \; ((\forall x \in A \; (\ell \leq f(x))) \Rightarrow \ell \leq t)\big).$$

Remark (Interpretation). The infimum is the greatest real floor for the output set $f(A)$; it need not be an output value of f.

Remark (Dependencies).

- Function
- Subset
- Image Set
- Infimum

Definition (Maximum Point of a Function)

Definition 4.2.15 (Maximum Point of a Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x^* \in A$ is a maximum point of f on A if

$$f(x) \leq f(x^*) \qquad (x \in A).$$

Remark (Standard quantified statement).

$$x^* \in A \;\wedge\; \forall x \in A \; \big(f(x) \leq f(x^*)\big).$$

Remark (Negated quantified statement).

$$x^* \notin A \;\vee\; \exists x \in A \; \big(f(x) > f(x^*)\big).$$

Remark (Failure modes). A proposed maximum point fails either by not belonging to the domain or by being beaten by another output value.

Remark (Failure mode decomposition).

$$x^* \notin A \;\vee\; \exists x \in A \; \big(f(x) > f(x^*)\big).$$

Remark (Interpretation). A maximum point is an input where the largest output value is attained.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$
- Maximum

Definition (Minimum Point of a Function)

Definition 4.2.16 (Minimum Point of a Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x_* \in A$ is a minimum point of f on A if

$$f(x_*) \leq f(x) \qquad (x \in A).$$

Remark (Standard quantified statement)*.*

$$x_* \in A \;\wedge\; \forall x \in A \,\big(f(x_*) \leq f(x)\big).$$

Remark (Negated quantified statement)*.*

$$x_* \notin A \;\vee\; \exists x \in A \,\big(f(x) < f(x_*)\big).$$

Remark (Failure modes)*.* A proposed minimum point fails either by not belonging to the domain or by being undercut by another output value.

Remark (Failure mode decomposition)*.*

$$x_* \notin A \;\vee\; \exists x \in A \,\big(f(x) < f(x_*)\big).$$

Remark (Interpretation)*.* A minimum point is an input where the smallest output value is attained.

Remark (Dependencies)*.*

- Function
- Subset
- Order on $\mathbb{R}$
- Minimum

Definition (Increasing Function)

Definition 4.2.17 (Increasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is increasing on A if

$$x_1 < x_2 \implies f(x_1) \leq f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \ \big(x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)\big).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) > f(x_2)\big).$$

Remark (Failure modes). The function fails to be increasing if some later input has a smaller output than an earlier input.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) > f(x_2)\big).$$

Remark (Interpretation). As the input moves to the right in A, the output is not allowed to go down.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$

Definition (Strictly Increasing Function)

Definition 4.2.18 (Strictly Increasing Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly increasing on A if

$$x_1 < x_2 \implies f(x_1) < f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \ \big(x_1 < x_2 \Rightarrow f(x_1) < f(x_2)\big).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) \geq f(x_2)\big).$$

Remark (Failure modes). Strict increase fails if some later input has an output that is no larger than the earlier output.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) \geq f(x_2)\big).$$

Remark (Interpretation). Strict increase requires every genuine move to the right in the domain to produce a genuine increase in output.

Remark (Dependencies).

- Function
- Subset
- Strict Order on $\mathbb{R}$

Definition (Decreasing Function)

Definition 4.2.19 (Decreasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is decreasing on A if

$$x_1 < x_2 \implies f(x_1) \geq f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \left(x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)\right).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \left(x_1 < x_2 \ \wedge \ f(x_1) < f(x_2)\right).$$

Remark (Failure modes)*.* The function fails to be decreasing if some later input has a larger output than an earlier input.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \left(x_1 < x_2 \ \wedge \ f(x_1) < f(x_2)\right).$$

Remark (Interpretation)*.* As the input moves to the right in A, the output is not allowed to go up.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$

Definition (Strictly Decreasing Function)

Definition 4.2.20 (Strictly Decreasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly decreasing on A if

$$x_1 < x_2 \implies f(x_1) > f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \left(x_1 < x_2 \Rightarrow f(x_1) > f(x_2)\right).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \left(x_1 < x_2 \ \wedge \ f(x_1) \leq f(x_2)\right).$$

Remark (Failure modes). Strict decrease fails if some later input has an output that is no smaller than the earlier output.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) \leq f(x_2)\big).$$

Remark (Interpretation). Strict decrease requires every genuine move to the right in the domain to produce a genuine decrease in output.

Remark (Dependencies).

- Function
- Subset
- Strict Order on $\mathbb{R}$

Definition (Monotone Function)

Definition 4.2.21 (Monotone Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is monotone on A if f is increasing on A or decreasing on A.

Remark (Standard quantified statement).

$$\big(\forall x_1, x_2 \in A\ (x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2))\big) \ \vee \ \big(\forall x_1, x_2 \in A\ (x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2))\big).$$

Remark (Definition predicate reading).

$$\mathrm{MonotoneFunction}(f, A) \ :\Leftrightarrow \ \big(\forall x_1, x_2 \in A\ (x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2))\big) \vee \big(\forall x_1, x_2 \in A\ (x_1 < x_2 \Rightarrow f(x_1) \geq$$

Remark (Negated quantified statement).

$$\big(\exists x_1, x_2 \in A\ (x_1 < x_2 \wedge f(x_1) > f(x_2))\big) \ \wedge \ \big(\exists y_1, y_2 \in A\ (y_1 < y_2 \wedge f(y_1) < f(y_2))\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{MonotoneFunction}(f, A)$$

Remark (Failure modes). A nonmonotone function has evidence of both kinds of reversal: one pair where the outputs go down as inputs increase, and another pair where the outputs go up as inputs increase.

Remark (Failure mode decomposition).

$$\begin{aligned}&\exists x_1, x_2 \in A\ (x_1 < x_2 \wedge f(x_1) > f(x_2))\\ &\qquad \wedge \ \exists y_1, y_2 \in A\ (y_1 < y_2 \wedge f(y_1) < f(y_2)).\end{aligned}$$

Remark (Interpretation). A monotone function moves consistently in one direction: nondecreasing or nonincreasing.

Remark (Dependencies).

- Increasing Function
- Decreasing Function

Definition (Constant Function)

Definition 4.2.22 (Constant Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is constant on A if

$$f(x_1) = f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement)*.*

$$\forall x_1, x_2 \in A \ \big(f(x_1) = f(x_2)\big).$$

Remark (Negated quantified statement)*.*

$$\exists x_1, x_2 \in A \ \big(f(x_1) \neq f(x_2)\big).$$

Remark (Failure modes)*.* Constancy fails as soon as two inputs in the domain have different outputs.

Remark (Failure mode decomposition)*.*

$$\exists x_1, x_2 \in A \ \big(f(x_1) \neq f(x_2)\big).$$

Remark (Interpretation)*.* A constant function assigns the same real value to every input in its domain.

Remark (Dependencies)*.*

- Function
- Subset

4.3 Subsets of $\mathbb{R}$

Subsets of $\mathbb{R}$ — Quick Reference

Concept	**Meaning**
Epsilon neighbourhood	Unpunctured condition $\lvert x - c\rvert < r$
Deleted neighbourhood	Punctured condition $0 < \lvert x - c\rvert < \delta$
Cluster point	Every neighbourhood contains a distinct point of X
Adherent point	Every neighbourhood contains some point of X
Closure	Points adherent to X
Closed set	Set containing its closure or sequential limits
Bounded set	Set contained in a large interval
Heine–Borel	Closed and bounded iff sequentially compact in $\mathbb{R}$

Subsets of $\mathbb{R}$

Remark (Terminological note). Cluster point = limit point = accumulation point. Bartle and Lebl use *cluster point*; Rudin uses *limit point*; Tao uses both. Adherent point (Tao) is the weaker notion that includes isolated points.

Definition (ε-Neighbourhood)

Definition 4.3.1 (ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **ε-neighbourhood** of x is

$$V_\varepsilon(x) := (x - \varepsilon, x + \varepsilon) = \{y \in \mathbb{R} : |y - x| < \varepsilon\}.$$

Remark (Standard quantified statement).

$$y \in V_\varepsilon(x) \iff |y - x| < \varepsilon.$$

Remark (Interpretation). The ε-neighbourhood is the open interval of radius ε centered at x.

Remark (Dependencies).

- Open Ball on the Real Line

Definition (Deleted ε-Neighbourhood)

Definition 4.3.2 (Deleted ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **deleted ε-neighbourhood** of x is

$$V_\varepsilon^*(x) := V_\varepsilon(x) \setminus \{x\} = \{y \in \mathbb{R} : 0 < |y - x| < \varepsilon\}.$$

Remark (Standard quantified statement).

$$y \in V_\varepsilon^*(x) \iff 0 < |y - x| < \varepsilon.$$

Remark (Interpretation). The deleted neighbourhood removes the base point. It is the punctured input region used in limit definitions.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Cluster Point)

Definition 4.3.3 (Cluster Point)**.** Let $X \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. The point x is a **cluster point** of X if

$$\forall \varepsilon > 0,\ \exists\, y \in X \setminus \{x\} :\ |y - x| < \varepsilon.$$

Remark (Standard quantified statement).

$$x \text{ is a cluster point of } X \iff \forall \varepsilon > 0,\ \exists\, y \in X \setminus \{x\} :\ |y - x| < \varepsilon.$$

Remark (Definition predicate reading).

$$x \text{ is a cluster point of } X \;:\equiv\; \forall \varepsilon > 0,\; V_\varepsilon^*(x) \cap X \neq \emptyset.$$

Remark (Negated quantified statement).

$$\neg\, x \text{ is a cluster point of } X \iff \exists \varepsilon > 0,\; \forall\, y \in X \setminus \{x\} :\; \neg |y - x| < \varepsilon.$$

Remark (Negation predicate reading).

$$\neg\, x \text{ is a cluster point of } X \;:\equiv\; \exists \varepsilon > 0 :\; V_\varepsilon^*(x) \cap X = \emptyset.$$

Remark (Interpretation). A cluster point is approached by X via distinct witnesses arbitrarily close. A cluster point need not belong to X: 0 is a cluster point of $(0,1)$ but $0 \notin (0,1)$. The condition $\lim_{x \to c} f(x) = L$ requires $\mathrm{ClusterPoint}(c, \mathrm{dom}(f))$; without it the punctured ball condition is vacuous.

Remark (Dependencies).

- Deleted ε-Neighbourhood

Theorem

Theorem 4.3.4 (Sequential Characterization of Cluster Points)**.** *c is a cluster point of A if and only if there exists $(a_n) \subseteq A \setminus \{c\}$ with $a_n \to c$.*

Go to proof.

Remark (Standard quantified statement).

$$c \text{ is a cluster point of } A \iff \exists (a_n) \subseteq A \setminus \{c\} :\; a_n \to c.$$

Remark (Interpretation). The forward direction: use $\varepsilon = 1/n$ to build the witness sequence. The reverse direction is immediate since the sequence witnesses the ball condition for every ε. This bridges the ε-δ definition with the sequential language.

Remark (Dependencies).

- Cluster Point

Definition (Adherent Point)

Definition 4.3.5 (Adherent Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **adherent point** of X if

$$\forall \varepsilon > 0\; \exists y \in X\; (|y - x| < \varepsilon).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\; \big(V_\varepsilon(x) \cap X \neq \emptyset\big).$$

Remark (Definition predicate reading).

$$\operatorname{AdherentPoint}(x, X) \;:=\; \forall \varepsilon > 0\ \exists y \in X\ (|y - x| < \varepsilon).$$

Remark (Negated quantified statement).

$$\exists \varepsilon > 0\ \forall y \in X\ (|y - x| \geq \varepsilon).$$

Remark (Interpretation). An adherent point can be witnessed by points of X arbitrarily close to x, including possibly x itself.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Isolated Point)

Definition 4.3.6 (Isolated Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in X$. The point x is an **isolated point** of X if

$$\exists \varepsilon > 0\ \big(V_\varepsilon(x) \cap X = \{x\}\big).$$

Remark (Standard quantified statement).

$$\exists \varepsilon > 0\ \forall y \in X\ \big(|y - x| < \varepsilon \Rightarrow y = x\big).$$

Remark (Definition predicate reading).

$$\operatorname{IsolatedPoint}(x, X) \;:=\; x \in X \wedge \exists \varepsilon > 0\ \big(V_\varepsilon(x) \cap X = \{x\}\big).$$

Remark (Negated quantified statement).

$$\forall \varepsilon > 0\ \exists y \in X \setminus \{x\}\ (|y - x| < \varepsilon).$$

Remark (Interpretation). An isolated point belongs to the set but has a neighbourhood containing no other points of the set.

Remark (Dependencies).

- Adherent Point
- Cluster Point

Definition (Interior Point)

Definition 4.3.7 (Interior Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **interior point** of X if

$$\exists \varepsilon > 0\ \big(V_\varepsilon(x) \subseteq X\big).$$

Remark (Standard quantified statement).

$$\exists \varepsilon > 0\ \forall y \in \mathbb{R}\ \big(|y - x| < \varepsilon \Rightarrow y \in X\big).$$

Remark (Definition predicate reading).

$$\text{InteriorPoint}(x, X) \;:=\; \exists \varepsilon > 0 \; \big(V_\varepsilon(x) \subseteq X\big).$$

Remark (Interpretation). An interior point has a full open neighbourhood contained in the set.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Boundary Point)

Definition 4.3.8 (Boundary Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is a **boundary point** of X if

$$\forall \varepsilon > 0 \; \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \; \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Remark (Definition predicate reading).

$$\text{BoundaryPoint}(x, X) \;:=\; \forall \varepsilon > 0 \; \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Remark (Interpretation). A boundary point sees both the set and its complement in every neighbourhood.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Interior)

Definition 4.3.9 (Interior)**.** Let $X \subseteq \mathbb{R}$. The **interior** of X is

$$X^\circ := \{x \in \mathbb{R} : \exists \varepsilon > 0 \; V_\varepsilon(x) \subseteq X\}.$$

Remark (Standard quantified statement).

$$x \in X^\circ \iff \exists \varepsilon > 0 \; V_\varepsilon(x) \subseteq X.$$

Remark (Interpretation). The interior collects exactly the points around which the set contains a whole neighbourhood.

Remark (Dependencies).

- Interior Point

Definition (Boundary)

Definition 4.3.10 (Boundary)**.** Let $X \subseteq \mathbb{R}$. The **boundary** of X is

$$\partial X := \{x \in \mathbb{R} : \forall \varepsilon > 0\ (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset)\}.$$

Remark (Standard quantified statement).

$$x \in \partial X \iff \forall \varepsilon > 0\ (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset).$$

Remark (Interpretation). The boundary records the points where every neighbourhood touches both inside and outside of the set.

Remark (Dependencies).

- Boundary Point

Definition (Closure)

Definition 4.3.11 (Closure)**.**

$$\overline{X} := \{x \in \mathbb{R} : \forall \varepsilon > 0\ V_\varepsilon(x) \cap X \neq \emptyset\}.$$

Remark (Standard quantified statement).

$$x \in \overline{X} \iff x \text{ is an adherent point of } X \iff \forall \varepsilon > 0 : V_\varepsilon(x) \cap X \neq \emptyset.$$

Remark (Interpretation). The closure adds all adherent points of X, so it fills in the limiting points that can be reached by points of X.

Remark (Canonical examples). $\overline{(a,b)} = [a,b]$; $\overline{\mathbb{Q}} = \mathbb{R}$; $\overline{\mathbb{Z}} = \mathbb{Z}$; $\overline{\{1/n\}} = \{1/n\} \cup \{0\}$.

Remark (Dependencies).

- Adherent Point

Lemma

Lemma 4.3.12 (Elementary Properties of Closures)**.** *Let $X, Y \subseteq \mathbb{R}$. Then: (i) $X \subseteq \overline{X}$; (ii) $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$; (iii) $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$; (iv) $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$.*
Go to proof.

Remark (Standard quantified statement).

$$\overline{X \cup Y} = \overline{X} \cup \overline{Y}; \quad \overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}.$$

Remark (Failure of equality in (iii)). Take $X = (0,1)$, $Y = (1,2)$: $\overline{X \cap Y} = \emptyset$ but $\overline{X} \cap \overline{Y} = \{1\}$.

Remark (Interpretation). Closure preserves inclusions and finite unions, but intersection can lose limit points that are approached separately by the two sets.

Remark (Dependencies).

- Closure

Definition (Closed Subset of $\mathbb{R}$)

Definition 4.3.13 (Closed Subset of $\mathbb{R}$)**.** E is closed $\iff \overline{E} = E$.

Remark (Standard quantified statement).

$$E \text{ is closed} \iff \forall x \in \mathbb{R} : \ x \text{ is an adherent point of } E \Rightarrow x \in E.$$

Remark (Definition predicate reading).

$$E \text{ is closed} :\equiv \forall x : \ x \text{ is an adherent point of } E \Rightarrow x \in E.$$

Remark (Negated quantified statement).

$$\neg\, E \text{ is closed} \iff \exists x \notin E : \ x \text{ is an adherent point of } E.$$

Remark (Negation predicate reading).

$$\neg\, E \text{ is closed} :\equiv \exists x \notin E : \ \forall \varepsilon > 0 : V_\varepsilon(x) \cap E \neq \emptyset.$$

Remark (Interpretation)*.* E is closed iff it contains its own boundary. Examples: $[a, b]$ closed; (a, b) not closed; $\mathbb{R}$ and $\emptyset$ both closed; $\mathbb{Q}$ not closed.

Remark (Dependencies).

- Closure

Corollary

Corollary 4.3.14 (Closed iff Sequentially Closed)**.** *X is closed* $\iff \forall (a_n) \subseteq X,\ a_n \to x \Rightarrow x \in X$*.*

Go to proof.

Remark (Standard quantified statement).

$$X \text{ is closed} \iff \forall (a_n) \subseteq X,\ a_n \to x \Rightarrow x \in X.$$

Remark (Interpretation)*.* A closed set cannot be escaped by convergent sequences. Every adherent point has a witness sequence; closedness forces the limit into the set.

Remark (Dependencies).

- Closed Subset of $\mathbb{R}$
- Sequential Characterization of Cluster Points

Corollary

Corollary 4.3.15 (Every Point of an Interval Is a Cluster Point)**.** *If I is an interval, then every $x \in I$ is a cluster point of I.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall x \in I \, \forall \varepsilon > 0 \, \exists y \in I \setminus \{x\} \, (|y - x| < \varepsilon).$$

Remark (Interpretation)*.* This justifies intervals as natural domains for function limits. $\lim_{x \to c} f(x) = L$ requires $\mathrm{ClusterPoint}(c, \mathrm{dom}(f))$; on any interval this is free.

Remark (Dependencies)*.*

- Cluster Point

Definition (Bounded Subset of $\mathbb{R}$)

Definition 4.3.16 (Bounded Subset of $\mathbb{R}$)**.** X is bounded $\iff \exists M > 0, \ \forall x \in X : |x| \leq M$.

Remark (Standard quantified statement)*.*

$$\exists M > 0 \, \forall x \in X \, (|x| \leq M).$$

Remark (Negated quantified statement)*.*

$$\forall M > 0 \, \exists x \in X \, (|x| > M).$$

Remark (Interpretation)*.* A bounded subset of $\mathbb{R}$ fits inside some symmetric interval $[-M, M]$.

Remark (Dependencies)*.*

- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

Theorem

Theorem 4.3.17 (Heine–Borel Theorem for $\mathbb{R}$)**.** *X is closed$\wedge X$ is bounded $\iff$ every $(a_n) \subseteq X$ has a subsequence converging to a limit in X.*

Go to proof.

Remark (Standard quantified statement)*.*

$$X \text{ is closed} \wedge X \text{ is bounded} \iff \forall (a_n) \subseteq X, \ \exists (a_{n_k}), \ \exists L \in X : a_{n_k} \to L.$$

Remark (Definition predicate reading).

$$\text{SequentiallyCompact}(X) \;:\equiv\; X \text{ is closed} \wedge X \text{ is bounded}.$$

Remark (Negated quantified statement).

$$\neg\big(X \text{ is closed} \wedge X \text{ is bounded}\big) \iff \exists (a_n) \subseteq X : \text{ no subsequence converges to a limit in } X.$$

Remark (Failure modes). The compactness conclusion fails when either closedness or boundedness fails.

Remark (Failure mode decomposition). Either X is bounded fails — a sequence escapes to $\pm\infty$ — or X is closed fails — a convergent subsequence exists but its limit lies outside X.

Remark (Interpretation). Closed prevents limits from escaping through boundary gaps; bounded prevents escape to infinity. Both are required. Forward: bounded gives Bolzano–Weierstrass; closed keeps the limit in X. Reverse: negate either hypothesis to construct a sequence with no convergent subsequence in X.

Remark (Dependencies).

- Closed Subset of $\mathbb{R}$
- Bounded Subset of $\mathbb{R}$

Definition (True Near a Point)

Definition 4.3.18 (True Near a Point)**.** Property $P(f(x))$ is **true near** x_0 if $\exists \delta > 0,\ \forall x : 0 < |x - x_0| < \delta) \Rightarrow P(f(x))$.

Remark (Standard quantified statement).

$$P(f(x)) \text{ true near } x_0 \iff \exists \delta > 0,\ \forall x : 0 < |x - x_0| < \delta \Rightarrow P(f(x)).$$

Remark (Interpretation). $\lim_{x \to x_0} f(x) = L$ states: for every $\varepsilon > 0$, the property $|f(x) - L| < \varepsilon$ is true near x_0.

Remark (Dependencies).

- ε-Neighbourhood

Chapter 5

Discrete Calculus

5.1 Motivation

Motivation

Classical calculus studies functions defined on continuous domains, typically functions $f : \mathbb{R} \to \mathbb{R}$. Its central tools are the derivative and the integral:

- The derivative measures *instantaneous change*.
- The integral accumulates *total change*.

However, many important mathematical objects are inherently *discrete*. Examples include

- sequences a_n,
- recurrence relations,
- sums of integers or combinatorial quantities,
- discrete probability distributions,
- algorithms and computational processes.

In these contexts the domain is typically the natural numbers

$$f : \mathbb{Z} \to \mathbb{R}$$

rather than the real line.

To study change in such settings, calculus is replaced by its discrete counterpart.

Discrete Analogs of Calculus

The central idea of *discrete calculus* is that the roles of derivative and integral are played by two simple operations.

Continuous vs. Discrete Calculus

Continuous Calculus	Discrete Calculus
Derivative $\dfrac{d}{dx}f(x)$	Forward difference $\Delta f(n) = f(n+1) - f(n)$
Integral $\displaystyle\int f(x)\,dx$	Summation $\sum f(n)$

The forward difference operator measures change between consecutive values of a function.

$$\Delta f(n) = f(n+1) - f(n).$$

This quantity plays the same role for sequences that the derivative plays for functions of a real variable.

Likewise, summation serves as the discrete analogue of integration.

$$\sum_{k=a}^{b} f(k)$$

accumulates the values of a function across a finite discrete interval.

Telescoping Phenomena

One of the most important identities in discrete mathematics arises when differences are summed.

$$\sum_{k=a}^{b-1} \big(f(k+1) - f(k)\big) = f(b) - f(a).$$

The intermediate terms cancel, leaving only the endpoints. Such sums are called *telescoping sums*.

This identity is the discrete counterpart of the *Fundamental Theorem of Calculus*.

Why Discrete Calculus Matters

Discrete calculus appears throughout mathematics:

- in combinatorics and enumerative formulas,
- in recurrence relations and generating functions,
- in probability theory and stochastic processes,

- in numerical analysis and finite difference methods,
- in computer science and algorithm analysis.

For students of analysis, discrete calculus is particularly valuable because it reveals the structural skeleton behind many identities involving sums and sequences.

Many techniques used to estimate limits of sequences or series ultimately rely on discrete analogues of differential identities.

A Structural View

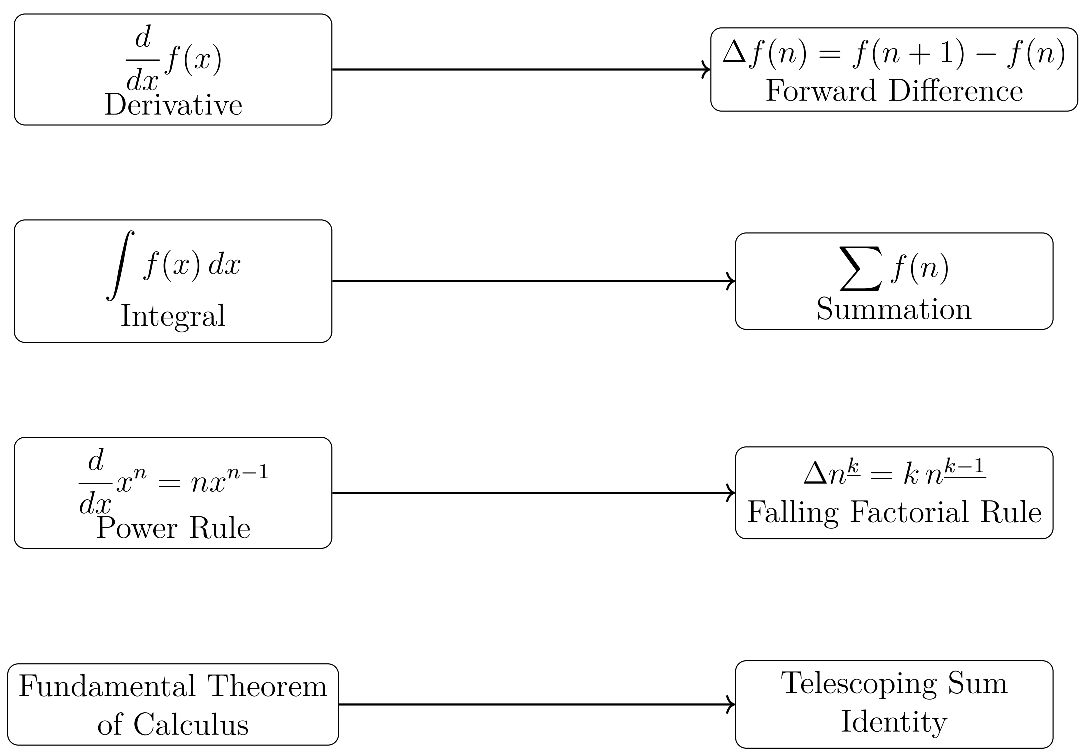
Continuous calculus studies how functions change along the real line.

Discrete calculus studies how functions change along the integers.

Despite the difference in setting, the two theories mirror each other remarkably closely. Understanding this parallel provides powerful intuition and useful techniques for manipulating sums, sequences, and recurrence relations.

In the sections that follow we develop the basic tools of discrete calculus, beginning with the forward difference operator and the algebra of finite differences.

Structural Correspondence Between Continuous and Discrete Calculus



5.2 Foundations

5.2.1 Foundations

Definition (Discrete Function)

A *discrete function* is a function

$$f : \mathbb{Z} \to \mathbb{R}$$

(or more generally $f : \mathbb{N} \to \mathbb{R}$). It assigns a real value to each integer input, producing a sequence $f(0)$, $f(1)$, $f(2)$, ...

Remark (Fully quantified form). $\forall n \in \mathbb{Z},\ f(n) \in \mathbb{R}$. The domain is the integers (or natural numbers); the codomain is the real line.

Remark (Discrete vs. continuous). In discrete calculus the independent variable moves in *integer steps* rather than continuously. Change is therefore measured between successive integer inputs rather than through infinitesimal limits. The fundamental operator of discrete calculus is the *difference operator*

$$\Delta f(n) = f(n+1) - f(n),$$

which plays a role analogous to the derivative in continuous calculus.

Remark (Sequences as discrete functions). A discrete function is simply another way of viewing a *sequence*. A sequence $a_0, a_1, a_2, \ldots$ is identified with

$$f : \mathbb{N} \to \mathbb{R}, \qquad f(n) = a_n.$$

Several equivalent notations appear in the literature: $f(n)$, a_n, $(a_n)_{n\in\mathbb{N}}$, $\{a_n\}_{n=0}^{\infty}$. All describe the same object. The functional notation $f(n)$ is preferred in discrete calculus because the difference operator acts on it much as a derivative acts on functions in continuous calculus.

5.2.2 Summation Notation

Definition (Summation)

Let f be a function defined on the integers. For integers $a \leq b$, the expression

$$\sum_{k=a}^{b} f(k)$$

denotes the sum $f(a) + f(a+1) + f(a+2) + \cdots + f(b)$. The symbol $\sum$ (Greek capital sigma) represents summation; k is the *index of summation*; a and b are the *lower* and *upper limits*; $f(k)$ is the *summand*.

Remark (Fully quantified form). $\displaystyle\sum_{k=a}^{b} f(k) \ := \ f(a) + f(a+1) + \cdots + f(b)$, where $a, b \in \mathbb{Z}$, $a \leq b$, and $f : \mathbb{Z} \to \mathbb{R}$.

Remark (Dummy variable). The index of summation is a *dummy variable*: $\sum_{k=a}^{b} f(k) = \sum_{i=a}^{b} f(i)$. Its name carries no information about the value of the sum.

Basic Properties of Summation

Remark (Basic properties of summation). For functions f and g and constants c,

$$\sum_{k=m}^{n}(f(k)+g(k)) = \sum_{k=m}^{n} f(k) + \sum_{k=m}^{n} g(k),$$

$$\sum_{k=m}^{n} c\, f(k) = c \sum_{k=m}^{n} f(k).$$

These express the *linearity* of summation.

Splitting Sums

Remark (Splitting sums). A sum over an interval may be divided into smaller pieces:

$$\sum_{k=a}^{c} f(k) = \sum_{k=a}^{b} f(k) + \sum_{k=b+1}^{c} f(k) \qquad \text{whenever } a \le b < c.$$

Changing the Index

Remark (Changing the index). It is often useful to shift the summation index. For example,

$$\sum_{k=0}^{n} f(k) = \sum_{j=1}^{n+1} f(j-1).$$

Such substitutions frequently simplify expressions or align sums with known formulas.

Empty Sums

Remark (Empty sums). If the upper bound is smaller than the lower bound, the sum contains no terms. By convention,

$$\sum_{k=a}^{b} f(k) = 0 \qquad \text{when } a > b.$$

This convention simplifies many algebraic identities.

Nested Summations

Remark (Nested summations). Summations may be nested:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} = \sum_{i=1}^{n} \big(a_{i1} + a_{i2} + \cdots + a_{im} \big).$$

The inner sum is evaluated first. When the limits are independent, the order of summation may be exchanged:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} = \sum_{j=1}^{m} \sum_{i=1}^{n} a_{ij}.$$

Remark (Analogy with integration). Nested sums behave like iterated integrals: exchanging summation order corresponds to Fubini's theorem in the discrete setting.

Common Examples

Remark (Common examples).

$$\sum_{k=1}^{n} 1 = n, \qquad \sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

These formulas appear throughout discrete mathematics and analysis.

5.2.3 The Binomial Theorem

Theorem 5.2.1 (Binomial Theorem). *For any integer $n \geq 0$ and real numbers x, y,*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k,$$

where the binomial coefficients *are*

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Remark (Standard quantified statement). $\forall n \in \mathbb{N},\ \forall x, y \in \mathbb{R} : (x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$

Remark (Combinatorial meaning). The binomial coefficients count the number of ways to choose k factors of y from the n factors in the product $(x+y)(x+y)\cdots(x+y)$. Thus $\binom{n}{k}$ is the number of ways to select k objects from a set of n, and the theorem describes how a power of a sum expands into a sum of products.

Basic Properties of Binomial Coefficients

Remark (Basic properties of binomial coefficients)*.*

$$\binom{n}{0} = 1, \qquad \binom{n}{n} = 1, \qquad \binom{n}{k} = \binom{n}{n-k}.$$

They also satisfy the *Pascal recursion*:

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}.$$

This recursion produces Pascal's triangle, in which each entry is the sum of the two entries directly above it.

$$\begin{array}{ccccccccccc}
 & & & & & \binom{0}{0} & & & & & \\
 & & & & \binom{1}{0} & & \binom{1}{1} & & & & \\
 & & & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & & & \\
 & & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & & \\
 & \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4} & \\
\binom{5}{0} & & \binom{5}{1} & & \binom{5}{2} & & \binom{5}{3} & & \binom{5}{4} & & \binom{5}{5}
\end{array}$$

Remark (Pascal's triangle)*.* Each entry $\binom{n}{k}$ is obtained from the sum of the two entries above it: $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. These coefficients appear throughout discrete calculus, especially in formulas for higher finite differences and in Newton's forward difference expansion.

Connection with Discrete Calculus

Remark (Connection with discrete calculus)*.* The binomial theorem plays a central role in discrete calculus because difference operators naturally produce binomial coefficients.

Proposition 5.2.2 (Higher difference formula)**.** *For any function f,*

$$\Delta^n f(x) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x+k).$$

Go to proof.

Remark (Dependencies)*.*

- Binomial Theorem.
- Forward Difference Operator.

- Higher Differences.

Remark (Standard quantified statement). $\forall n \in \mathbb{N},\ \forall f : \mathbb{Z} \to \mathbb{R},\ \forall x \in \mathbb{Z} :\ \Delta^n f(x) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x+k)$.

Remark (Proof idea). Applying the difference operator $\Delta = E - I$ (where E is the shift operator) repeatedly gives $\Delta^n = (E-I)^n$. Expanding by the binomial theorem yields the coefficients $(-1)^{n-k}\binom{n}{k}$, and $E^k f(x) = f(x+k)$. This derivation is carried out in full in the Difference Operators section.

Explicit Examples

Remark (Explicit examples). Applying the difference operator repeatedly to a function f yields

$$\begin{aligned} \Delta f(x) &= f(x+1) - f(x), \\ \Delta^2 f(x) &= f(x+2) - 2f(x+1) + f(x), \\ \Delta^3 f(x) &= f(x+3) - 3f(x+2) + 3f(x+1) - f(x). \end{aligned}$$

The coefficients $1, 2, 1$ and $1, 3, 3, 1$ are precisely the rows of Pascal's triangle, confirming that binomial coefficients determine the pattern of coefficients in higher-order finite differences.

5.3 Algebra of Summations

Summation Algebra — Quick Reference

Linearity	$\sum(af + bg) = a\sum f + b\sum g$
Splitting	$\sum_a^c = \sum_a^b + \sum_{b+1}^c$
Index shift	$\sum_{k=a}^b f(k) = \sum_{k=a+1}^{b+1} f(k-1)$
Constant sum	$\sum_{k=a}^b c = (b-a+1)c$
Empty sum	$\sum_{k=a}^b f(k) = 0$ when $a > b$
Nested sums	$\sum_i \sum_j a_{ij} = \sum_j \sum_i a_{ij}$ (independent limits)

Remark (Orientation). The difference operator and telescoping identity provide the background for the basic algebraic rules used to manipulate summations. These identities allow sums to be combined, split, and reindexed while preserving their value.

Linearity

Remark (Linearity). Summation is a linear operator. For functions f and g and constants a and b,

$$\sum_{k=m}^{n} (a\,f(k) + b\,g(k)) = a\sum_{k=m}^{n} f(k) + b\sum_{k=m}^{n} g(k).$$

This allows constants to be factored out of a sum and sums of expressions to be separated.

Example. $\sum_{k=1}^{n}(2k+3) = 2\sum_{k=1}^{n} k + 3\sum_{k=1}^{n} 1.$

Splitting a Sum

Remark (Splitting a sum)*.* A sum over an interval may be divided into pieces:

$$\sum_{k=a}^{c} f(k) = \sum_{k=a}^{b} f(k) + \sum_{k=b+1}^{c} f(k) \qquad \text{whenever } a \le b < c.$$

Example. $\sum_{k=1}^{10} f(k) = \sum_{k=1}^{5} f(k) + \sum_{k=6}^{10} f(k).$

Index Renaming

Remark (Index renaming)*.* The index of summation is a *dummy variable*:

$$\sum_{k=a}^{b} f(k) = \sum_{i=a}^{b} f(i).$$

This is useful when manipulating expressions containing several sums, since the index variable can be renamed to avoid conflicts.

Index Shifting

Remark (Index shifting)*.* It is often convenient to shift the summation index:

$$\sum_{k=a}^{b} f(k) = \sum_{k=a+1}^{b+1} f(k-1).$$

Index shifting allows sums to be aligned so that terms cancel or combine properly. It appears frequently in proofs involving recurrence relations and discrete calculus identities.

Example. $\sum_{k=0}^{n} f(k+1) = \sum_{k=1}^{n+1} f(k).$

Constant Sum

Remark (Constant sum)*.* If the summand does not depend on the index, the sum counts terms:

$$\sum_{k=a}^{b} c = (b-a+1)\,c.$$

The quantity $b-a+1$ is the number of integers between a and b, inclusive.

Example. $\sum_{k=3}^{7} 5 = 5 \cdot 5 = 25.$

Empty Sum

Remark (Empty sum). If the upper limit is smaller than the lower limit, the sum contains no terms. By convention,

$$\sum_{k=a}^{b} f(k) = 0 \qquad \text{whenever } a > b.$$

This convention simplifies many algebraic identities and allows formulas to remain valid without special cases.

Nested Sums

Remark (Nested sums). Summations may be nested:

$$\sum_{i=1}^{n}\sum_{j=1}^{m} a_{ij} = \sum_{i=1}^{n}\Big(a_{i1} + a_{i2} + \cdots + a_{im}\Big).$$

When the limits are independent, the order may be exchanged:

$$\sum_{i=1}^{n}\sum_{j=1}^{m} a_{ij} = \sum_{j=1}^{m}\sum_{i=1}^{n} a_{ij}.$$

Remark (Analogy with double integrals). Exchanging summation order is the discrete analogue of Fubini's theorem for iterated integrals.

Common Summation Identities

Remark (Common summation identities). Many sums that appear in discrete mathematics and discrete calculus have closed-form expressions. The following identities are among the most frequently encountered.

Common Summation Identities

$$\sum_{k=1}^{n} 1 = n \qquad \sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$$

$$\sum_{k=0}^{n} r^k = \frac{1-r^{n+1}}{1-r} \quad (r \neq 1) \qquad \sum_{k=0}^{n} \binom{n}{k} = 2^n$$

Remark (Role in discrete calculus). Polynomial summation formulas play an important role when working with finite differences and Newton series, while geometric sums arise naturally in recurrence relations and generating functions. Many of these identities can be derived systematically using the falling factorial basis and the Fundamental Theorem of Finite Calculus introduced later in this chapter.

5.4 Difference Operators

5.4.1 Difference Operators

Discrete calculus studies functions defined on the integers and the way their values change from one integer to the next. The central tool for measuring this change is the *difference operator*, which plays a role analogous to the derivative in continuous calculus.

Motivation

In ordinary calculus the derivative measures the rate of change of a function:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

In the discrete setting the independent variable changes in steps of size 1. Instead of taking a limit, we measure the difference between successive values of the function directly.

Forward Difference

Definition 5.4.1 (Forward Difference Operator)**.** Let $f : \mathbb{Z} \to \mathbb{R}$. The *forward difference operator* Δ is defined by

$$\Delta f(n) = f(n+1) - f(n).$$

The function Δf measures the change in f when the input increases by one unit.

Remark (Fully quantified form). $\forall f : \mathbb{Z} \to \mathbb{R},\ \forall n \in \mathbb{Z} :\ \Delta f(n) := f(n+1) - f(n).$

Remark (Analogy with differentiation). The operator Δ is the discrete analogue of the derivative. While the derivative measures infinitesimal change through a limit, the forward difference measures change between successive integer inputs exactly, with no limit required.

Example

Consider the function $f(n) = n^2$. Applying the difference operator gives

$$\begin{aligned}
\Delta f(n) &= f(n+1) - f(n) \\
&= (n+1)^2 - n^2 \\
&= 2n + 1.
\end{aligned}$$

Thus the forward difference of the quadratic function n^2 is the linear function $2n+1$.

Remark (Degree reduction). This mirrors a familiar fact from ordinary calculus: $\frac{d}{dx}x^2 = 2x$. Just as derivatives reduce the degree of a polynomial by one, finite differences also lower the degree of polynomial sequences. Note, however, that $\Delta(n^2) = 2n + 1$ rather than $2n$; ordinary powers do not behave as cleanly under Δ as they do under differentiation. This motivates introducing the *falling factorial* basis later in this chapter.

5.4.2 Higher Differences

Definition 5.4.2 (Higher Differences)**.** The *second difference* of f is defined by

$$\Delta^2 f(n) = \Delta(\Delta f(n)).$$

More generally, the k-th difference $\Delta^k f(n)$ denotes the k-fold application of the difference operator, defined recursively by

$$\Delta^0 f = f, \qquad \Delta^k f = \Delta(\Delta^{k-1} f).$$

Remark (Dependencies).

- Forward Difference Operator.

Remark (Fully quantified form). $\forall k \in \mathbb{N},\ \forall f : \mathbb{Z} \to \mathbb{R},\ \forall n \in \mathbb{Z}:\ \Delta^k f(n) := (\underbrace{\Delta \circ \cdots \circ \Delta}_{k})\, f(n)$

Example

Let $f(n) = n^2$. Then

$$\begin{aligned} \Delta f(n) &= 2n + 1, \\ \Delta^2 f(n) &= \Delta(2n+1) = 2(n+1) + 1 - (2n+1) = 2, \\ \Delta^3 f(n) &= \Delta(2) = 0. \end{aligned}$$

Remark (Analogy with higher derivatives). Higher differences behave analogously to higher derivatives. For polynomial functions of degree d, the d-th difference is constant and all higher differences are zero — just as the d-th derivative of a degree-d polynomial is constant and the next derivative vanishes.

5.4.3 Properties of the Difference Operator

The forward difference operator satisfies algebraic properties that closely mirror the familiar rules of differentiation.

Linearity

Proposition 5.4.3 (Linearity of Δ)**.** *For functions $f, g : \mathbb{Z} \to \mathbb{R}$ and constants $a, b \in \mathbb{R}$,*

$$\Delta(af + bg) = a\,\Delta f + b\,\Delta g.$$

Remark (Dependencies).

- Forward Difference Operator.

Remark (Fully quantified form). $\forall f, g : \mathbb{Z} \to \mathbb{R},\ \forall a, b \in \mathbb{R},\ \forall n \in \mathbb{Z}:\ \Delta(af + bg)(n) = a\,\Delta f(n) + b\,\Delta g(n)$.

Proof. By definition of the difference operator,

$$\begin{aligned}\Delta(af+bg)(n) &= (af+bg)(n+1)-(af+bg)(n)\\ &= af(n+1)+bg(n+1)-af(n)-bg(n)\\ &= a(f(n+1)-f(n))+b(g(n+1)-g(n))\\ &= a\,\Delta f(n)+b\,\Delta g(n). \quad \square\end{aligned}$$

Constant Rule

Proposition 5.4.4 (Constant rule)**.** *If c is a constant function, then $\Delta c = 0$.*

Remark (Dependencies).

- Forward Difference Operator.

Remark (Fully quantified form). $\forall c \in \mathbb{R},\ \forall n \in \mathbb{Z}:\ \Delta c(n) = 0.$

Proof. $\Delta c(n) = c - c = 0.$ $\square$

Remark (Analogy). This mirrors $\frac{d}{dx}c = 0$ in continuous calculus.

The Shift Operator and Operator Algebra

Definition 5.4.5 (Shift Operator)**.** The *shift operator* E and the *identity operator* I are defined by

$$Ef(n) = f(n+1), \qquad If(n) = f(n).$$

Remark (Fully quantified form). $\forall f:\mathbb{Z}\to\mathbb{R},\ \forall n\in\mathbb{Z}:\ Ef(n) = f(n+1)$ and $If(n) = f(n)$.

Theorem 5.4.6 ($\Delta = E - I$)**.**

$$\Delta = E - I.$$

Remark (Dependencies).

- Forward Difference Operator.
- Shift Operator.

Remark (Fully quantified form). $\forall f:\mathbb{Z}\to\mathbb{R},\ \forall n\in\mathbb{Z}:\ \Delta f(n) = (E-I)f(n) = f(n+1) - f(n)$.

Proof. Applying the operator $E - I$ to a function f gives

$$(E-I)f(n) = Ef(n) - If(n) = f(n+1) - f(n) = \Delta f(n). \quad \square$$

Remark (Higher differences via the binomial theorem). Expressing $\Delta = E - I$ reveals that higher differences can be computed using ordinary algebra. Applying the binomial theorem to $\Delta^n = (E - I)^n$ gives

$$\Delta^n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} E^k.$$

Since $E^k f(n) = f(n+k)$, this yields

$$\Delta^n f(n) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(n+k).$$

This explains why binomial coefficients and Pascal's triangle appear naturally throughout discrete calculus: they are a direct consequence of the algebraic identity $\Delta = E - I$.

5.4.4 Difference Tables and Polynomial Detection

One of the most useful tools in discrete calculus is the *difference table*. By repeatedly applying the difference operator to a sequence, one can detect whether the sequence arises from a polynomial function and determine its degree.

Definition 5.4.7 (Difference Table)**.** Given a sequence $f(0), f(1), f(2), \ldots$, the *difference table* is constructed by listing the successive values of the function and then computing their forward differences $\Delta f(n) = f(n+1) - f(n)$, then $\Delta^2 f$, $\Delta^3 f$, and so on.

Remark (Dependencies).

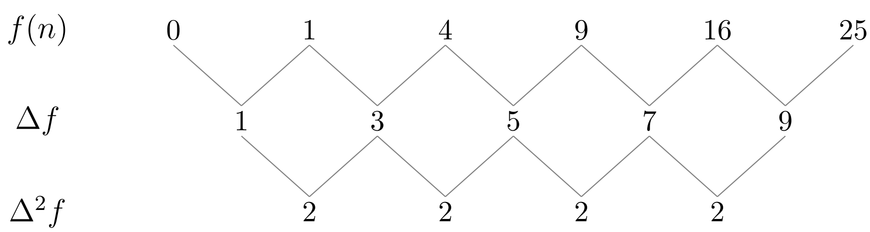
- Forward Difference Operator.
- Higher Differences.

Example

For $f(n) = n^2$:

n	$f(n)$	$\Delta f(n)$	$\Delta^2 f(n)$
0	0	1	
1	1	3	2
2	4	5	2
3	9	7	2
4	16	9	2
5	25		

The second differences are constant (all equal to 2), and the third differences are zero.



Theorem 5.4.8 (Polynomial detection)**.** *Let $f(n)$ be a sequence generated by a polynomial of degree d. Then $\Delta^d f(n)$ is constant, and $\Delta^{d+1} f(n) = 0$.*

Remark (Dependencies).

- Forward Difference Operator.
- Higher Differences.
- Difference Table.
- Binomial Theorem.

Proof. Write $f(n) = a_d n^d + a_{d-1} n^{d-1} + \cdots + a_0$ with $a_d \neq 0$. We show by induction on d that Δf is a polynomial of degree $d-1$ with leading coefficient $d\,a_d$.

Base case ($d = 0$): a constant polynomial has $\Delta f = 0$. ✓

Inductive step: Assume the result holds for all polynomials of degree less than d. Then

$$\Delta f(n) = f(n+1) - f(n) = a_d\big((n+1)^d - n^d\big) + (\text{lower-degree terms}).$$

Expanding $(n+1)^d - n^d = d\,n^{d-1} + \binom{d}{2} n^{d-2} + \cdots$, the leading term is $d\,a_d\,n^{d-1}$. Thus Δf is a polynomial of degree $d-1$ with leading coefficient $d\,a_d \neq 0$.

By induction, applying Δ a further $d-1$ times yields the constant $\Delta^d f(n) = d!\,a_d$, and one further application gives 0. □

Remark (Fully quantified form). $\forall d \in \mathbb{N},\ \forall f : \mathbb{Z} \to \mathbb{R}$ polynomial of degree d:
$\exists c \in \mathbb{R} : \forall n \in \mathbb{Z},\ \Delta^d f(n) = c$, and $\forall n \in \mathbb{Z},\ \Delta^{d+1} f(n) = 0$.

Remark (Consequence). Difference tables provide a practical method for identifying polynomial sequences and determining their degree. This observation leads naturally to the *falling factorial basis*: a family of polynomials that interact particularly cleanly with the difference operator, forming the foundation for Newton's discrete analogue of the Taylor expansion.

5.5 Telescoping Sums

5.5.1 Telescoping Sums

One of the most useful patterns in summation is the phenomenon of *telescoping*. In a telescoping sum, consecutive terms cancel with one another, leaving only a small number of terms from the beginning and end of the expression.

A Simple Example

Consider the sum

$$(2-1) + (3-2) + (4-3) + (5-4).$$

Expanding gives

$$2 - 1 + 3 - 2 + 4 - 3 + 5 - 4.$$

Most terms cancel in adjacent pairs:

$$\cancel{2} - 1 + \cancel{3} - \cancel{2} + \cancel{4} - \cancel{3} + 5 - \cancel{4}.$$

After cancellation, only the first and last terms remain: $5 - 1$. Thus

$$(2-1) + (3-2) + (4-3) + (5-4) = 5 - 1.$$

Definition (Telescoping Sum)

A sum is called a *telescoping sum* if consecutive terms cancel when the sum is expanded, leaving only a few remaining terms — typically the first and last.

General Form

The general telescoping structure is

$$\sum_{k=a}^{b-1} (f(k+1) - f(k)).$$

Writing out the terms explicitly:

$$\begin{aligned} &(f(a+1) - f(a)) \\ &+ (f(a+2) - f(a+1)) \\ &+ (f(a+3) - f(a+2)) \\ &\ \ \vdots \\ &+ (f(b) - f(b-1)). \end{aligned}$$

All intermediate terms cancel in pairs, leaving only $f(b) - f(a)$.

Remark (Why "telescoping"). The name comes from an old collapsible telescope: each section slides into the next. The intermediate terms in the sum "slide together" and vanish, collapsing the entire sum down to its endpoints.

The Telescoping Identity

Theorem (Telescoping Identity)

For any function $f : \mathbb{Z} \to \mathbb{R}$ and integers $a \leq b$,

$$\sum_{k=a}^{b-1} (f(k+1) - f(k)) = f(b) - f(a).$$

Remark (Fully quantified form). $\forall f : \mathbb{Z} \to \mathbb{R},\ \forall a, b \in \mathbb{Z},\ a \leq b : \displaystyle\sum_{k=a}^{b-1} (f(k+1) - f(k)) = f(b) - f(a)$.

Proof. Expanding the sum gives

$$(f(a+1) - f(a)) + (f(a+2) - f(a+1)) \\ + \cdots + (f(b) - f(b-1)).$$

Every intermediate term $f(a+1), f(a+2), \ldots, f(b-1)$ appears once with a positive sign and once with a negative sign, so they all cancel. The remaining terms are $f(b)$ and $-f(a)$, giving $f(b) - f(a)$. ☐

Cancellation Pattern

The cancellation can be seen explicitly:

$$\begin{array}{cc}
f(a+1) - f(a) & -f(a) \\
f(a+2) - f(a+1) & \cancel{f(a+1)} - \cancel{f(a+1)} \\
f(a+3) - f(a+2) & \cancel{f(a+2)} - \cancel{f(a+2)} \\
\vdots & \vdots \\
f(b) - f(b-1) & f(b)
\end{array}$$

Remark (Intuition)*.* Each intermediate term appears twice in the expanded sum: once with a positive sign and once with a negative sign. These pairs cancel, leaving only the first and last terms.

Operator Form

Using the forward difference operator $\Delta f(n) = f(n+1) - f(n)$, the telescoping identity takes the compact form

$$\sum_{k=a}^{b-1} \Delta f(k) = f(b) - f(a).$$

Remark (Discrete Fundamental Theorem of Calculus)*.* This identity is the discrete analogue of the Fundamental Theorem of Calculus:

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

It reveals a deep relationship between the difference operator and summation: summation is the inverse of differencing, just as integration is the inverse of differentiation. This connection is made precise in the next section through the concept of the *discrete antiderivative*.

5.6 Discrete Antiderivatives

5.6.1 Discrete Antiderivatives

The telescoping identity established in the previous section,

$$\sum_{k=a}^{b-1} \Delta f(k) = f(b) - f(a),$$

reveals that summation acts as an inverse operation to the difference operator, much like integration acts as an inverse to differentiation in ordinary calculus. This motivates the following definition.

Definition 5.6.1 (Discrete Antiderivative)**.** Let $f : \mathbb{Z} \to \mathbb{R}$ be a discrete function. A function $F : \mathbb{Z} \to \mathbb{R}$ is called a *discrete antiderivative* (or *discrete indefinite sum*) of f if

$$\Delta F(n) = f(n),$$

that is, $F(n+1) - F(n) = f(n)$ for all $n \in \mathbb{Z}$.

Remark (Dependencies).

- Forward Difference Operator.

Remark (Fully quantified form). $\forall f, F : \mathbb{Z} \to \mathbb{R} :\ F$ is a discrete antiderivative of $f \iff \forall n \in \mathbb{Z},\ F(n+1) - F(n) = f(n)$.

Remark (Analogy). A discrete antiderivative plays the same role in discrete calculus that an antiderivative plays in ordinary calculus: it reverses the fundamental differentiation-like operator of the theory. Instead of reversing differentiation, it reverses the forward difference Δ.

Examples

Example 1. Let $f(n) = 1$. Take $F(n) = n$. Then

$$\Delta F(n) = (n+1) - n = 1 = f(n). \checkmark$$

Example 2. Let $f(n) = n$. Take $F(n) = \frac{n(n-1)}{2}$. Then

$$\Delta F(n) = \frac{(n+1)n}{2} - \frac{n(n-1)}{2} = \frac{n(n+1-n+1)}{2} = n = f(n). \checkmark$$

Remark (Connection to falling factorials). Both examples above are instances of a general pattern: the discrete antiderivative of the falling factorial $n^{\underline{k}}$ is $\frac{1}{k+1} n^{\underline{k+1}}$, exactly mirroring the antiderivative rule for ordinary powers. This is developed in the Polynomial Basis section.

Constant of Integration

Discrete antiderivatives are not unique.

Proposition 5.6.2 (Constant of integration)**.** *If F is a discrete antiderivative of f, then so is $F + C$ for any constant $C \in \mathbb{R}$.*

Remark (Dependencies).

- Discrete Antiderivative.
- Forward Difference Operator.

Remark (Fully quantified form). $\forall C \in \mathbb{R} : \ \Delta F = f \Rightarrow \Delta(F + C) = f$.

Proof.

$$\Delta(F + C)(n) = (F + C)(n + 1) - (F + C)(n) = F(n + 1) - F(n) = \Delta F(n) = f(n). \quad \square$$

Fundamental Theorem of Finite Calculus

Theorem (Fundamental Theorem of Finite Calculus)

Let F be a discrete antiderivative of f, so that $\Delta F(n) = f(n)$ for all n. Then

$$\sum_{k=a}^{b-1} f(k) = F(b) - F(a).$$

Remark (Fully quantified form). $\forall f, F : \mathbb{Z} \to \mathbb{R}, \ \forall a, b \in \mathbb{Z}, \ a \leq b : \ (\forall n, \ \Delta F(n) = f(n)) \Rightarrow \sum_{k=a}^{b-1} f(k) = F(b) - F(a)$.

Proof. Since $f(k) = \Delta F(k)$ for all k, substituting gives

$$\sum_{k=a}^{b-1} f(k) = \sum_{k=a}^{b-1} \Delta F(k).$$

By the Telescoping Identity (Theorem 5.5.1),

$$\sum_{k=a}^{b-1} \Delta F(k) = F(b) - F(a). \quad \square$$

Remark (Analogy with continuous calculus). This theorem is the discrete analogue of the Fundamental Theorem of Calculus,

$$\int_a^b f'(x)\, dx = f(b) - f(a).$$

In continuous calculus, differentiation and integration are inverse operations. In discrete calculus, the analogous inverse pair is the difference operator Δ and summation. Any sum can be evaluated by finding a discrete antiderivative of the summand — just as any definite integral can be evaluated by finding an antiderivative.

Remark (Practical consequence). The theorem reduces the problem of computing $\sum_{k=a}^{b-1} f(k)$ to the algebraic problem of finding F with $\Delta F = f$. Combined with the falling factorial rules developed in the next section, this gives a systematic method for evaluating a wide class of sums in closed form.

5.7 Polynomial Basis

5.7.1 Polynomial Basis: Falling Factorials

In ordinary calculus, polynomial functions are naturally expressed using the monomial powers $1, x, x^2, x^3, \ldots$ because the derivative acts simply on them: $\frac{d}{dx}x^n = nx^{n-1}$.

In discrete calculus, however, the difference operator does not behave as cleanly on ordinary powers. For example,

$$\Delta(n^2) = (n+1)^2 - n^2 = 2n + 1.$$

Although the degree decreases by one, the result contains an extra constant term. For this reason it is more natural to work with a different family of polynomials — the *falling factorials* — which interact with Δ exactly as monomials interact with $\frac{d}{dx}$.

Definition (Falling Factorial)

For a nonnegative integer k, the *falling factorial* (or *falling power*) is defined by

$$x^{\underline{k}} = x(x-1)(x-2)\cdots(x-k+1) \qquad (k \text{ factors}).$$

By convention, $x^{\underline{0}} = 1$.

Remark (Fully quantified form). $\forall k \in \mathbb{N},\ \forall x \in \mathbb{R}:\ x^{\underline{k}} := \prod_{j=0}^{k-1}(x-j)$, with $x^{\underline{0}} := 1$.

Remark (Relation to factorials). When $x = n \in \mathbb{N}$ and $k \le n$, the falling factorial equals $\frac{n!}{(n-k)!}$. In particular, $n^{\underline{n}} = n!$. Thus falling factorials generalise factorials to arbitrary inputs.

The first few falling factorials are:

$$\begin{aligned} x^{\underline{0}} &= 1, \\ x^{\underline{1}} &= x, \\ x^{\underline{2}} &= x(x-1), \\ x^{\underline{3}} &= x(x-1)(x-2). \end{aligned}$$

The Difference Rule for Falling Factorials

Theorem (Difference Rule for Falling Factorials)

For $k \geq 1$,
$$\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}.$$

Remark (Fully quantified form). $\forall k \in \mathbb{N},\ k \geq 1,\ \forall x \in \mathbb{R}:\ (x+1)^{\underline{k}} - x^{\underline{k}} = k\, x^{\underline{k-1}}$.

Proof. Observe that
$$(x+1)^{\underline{k}} = (x+1)\, x\, (x-1) \cdots (x-k+2).$$

Therefore
$$\Delta\, x^{\underline{k}} = (x+1)^{\underline{k}} - x^{\underline{k}} = x(x-1)\cdots(x-k+2)\big[(x+1)-(x-k+1)\big].$$

The bracket simplifies:
$$(x+1)-(x-k+1) = k.$$

The remaining product $x(x-1)\cdots(x-k+2)$ is exactly $x^{\underline{k-1}}$. Thus
$$\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}.$$
□

Remark (Mirror of the power rule). This formula exactly mirrors the continuous power rule $\frac{d}{dx}x^k = kx^{k-1}$. Because of this clean behaviour under Δ, falling factorials form the natural polynomial basis for discrete calculus.

Remark (Antiderivative rule). The theorem also yields the discrete antiderivative rule: since $\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}$, we have
$$\Delta\left(\frac{x^{\underline{k+1}}}{k+1}\right) = x^{\underline{k}}.$$

Thus $\frac{x^{\underline{k+1}}}{k+1}$ is a discrete antiderivative of $x^{\underline{k}}$, mirroring the continuous rule $\int x^k\, dx = \frac{x^{k+1}}{k+1}$.

Summary Table

$f(x)$	$\Delta f(x)$
$x^{\underline{1}} = x$	1
$x^{\underline{2}} = x(x-1)$	$2x$
$x^{\underline{3}} = x(x-1)(x-2)$	$3x(x-1)$
$x^{\underline{4}} = x(x-1)(x-2)(x-3)$	$4x(x-1)(x-2)$

Polynomial Representation in the Falling Factorial Basis

Every polynomial can be written as a linear combination of falling factorials:

$$f(x) = c_0 + c_1\, x + c_2\, x^{\underline{2}} + c_3\, x^{\underline{3}} + \cdots + c_n\, x^{\underline{n}}.$$

Remark (Role in Newton's expansion). This representation plays the same role in discrete calculus that the Taylor expansion plays in continuous calculus. The coefficients c_k are determined by the successive finite differences of f evaluated at 0: $c_k = \Delta^k f(0)/k!$. This leads directly to Newton's forward difference expansion, developed in the Expansions section.

Consequences of the Difference Rule

The difference rule $\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}$ (Theorem 5.7.1) has several important consequences.

Higher Differences of Falling Factorials

Applying the rule repeatedly gives:

$$\Delta^m\, x^{\underline{k}} = \begin{cases} \dfrac{k!}{(k-m)!}\, x^{\underline{k-m}} & \text{if } m \le k, \\ 0 & \text{if } m > k. \end{cases}$$

In particular, evaluating at $x = 0$:

$$\Delta^m\, x^{\underline{k}}\Big|_{x=0} = \begin{cases} k! & \text{if } m = k, \\ 0 & \text{if } m \ne k. \end{cases}$$

Remark (Basis orthogonality). This vanishing property means the falling factorials behave like a "triangular" basis: Δ^m kills all $x^{\underline{k}}$ with $k < m$, and returns a scalar multiple for $k = m$. This is what makes the coefficient extraction in Newton's expansion work cleanly.

Connection to Binomial Coefficients

The higher difference formula from the Binomial Theorem section (Proposition 5.2.2),

$$\Delta^n f(x) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x+k),$$

together with the difference rule, gives an explicit formula for the coefficients of the falling factorial expansion:

$$c_k = \frac{\Delta^k f(0)}{k!}.$$

Remark (Pascal's triangle revisited). The coefficients appearing in higher finite differences follow exactly the pattern of Pascal's triangle — a direct consequence of the algebraic identity $\Delta = E - I$ and the binomial theorem.

Worked Example: Expanding n^2

We find the falling factorial expansion of $f(n) = n^2$.

Computing the difference table at $n = 0$:

$$f(0) = 0, \quad \Delta f(0) = 1, \quad \Delta^2 f(0) = 2, \quad \Delta^3 f(0) = 0.$$

The coefficients are $c_k = \Delta^k f(0)/k!$:

$$c_0 = 0, \quad c_1 = 1, \quad c_2 = \frac{2}{2!} = 1, \quad c_3 = 0.$$

Therefore:

$$n^2 = 0 + n^{\underline{1}} + n^{\underline{2}} = n + n(n-1) = n^2 - n + n = n^2.\checkmark$$

Remark (Practical value). Once a function is expressed in the falling factorial basis, both differencing and antidifferencing become purely mechanical, following the rule $\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}$ and its inverse. This makes the Fundamental Theorem of Finite Calculus immediately applicable to compute sums like $\sum_{k=0}^{n-1} k^2$ without clever tricks.

5.8 Calculus Rules

5.8.1 Calculus Rules

With the difference operator and the polynomial basis in place, we can now develop the full suite of differentiation-like rules for discrete calculus. The linearity of Δ was established in the Difference Operators section (Proposition 5.4.3). Here we add the *product rule* — the discrete counterpart of the Leibniz rule $(fg)' = f'g + fg'$.

Product Rule

Theorem (Discrete Product Rule)

For functions $f, g : \mathbb{Z} \to \mathbb{R}$,

$$\Delta(fg)(n) = f(n)\, \Delta g(n) + g(n+1)\, \Delta f(n).$$

Remark (Fully quantified form). $\forall f, g : \mathbb{Z} \to \mathbb{R},\ \forall n \in \mathbb{Z} :\ (f(n+1)g(n+1)) - (f(n)g(n)) = f(n)(g(n+1) - g(n)) + g(n+1)(f(n+1) - f(n))$.

Proof. By definition of Δ,

$$\Delta(fg)(n) = f(n+1)g(n+1) - f(n)g(n).$$

Add and subtract $f(n)g(n+1)$:

$$= f(n+1)g(n+1) - f(n)g(n+1) + f(n)g(n+1) - f(n)g(n).$$

Grouping the first two and last two terms:

$$= g(n+1)\big(f(n+1) - f(n)\big) + f(n)\big(g(n+1) - g(n)\big) = g(n+1)\, \Delta f(n) + f(n)\, \Delta g(n). \quad \square$$

Remark (Comparison with the continuous product rule). In continuous calculus, $(fg)' = f'g + fg'$ is symmetric in f and g. In discrete calculus the rule is slightly asymmetric: one factor is evaluated at n and the other at $n+1$. This asymmetry is unavoidable and is a characteristic feature of the discrete setting.

Remark (Consequence — summation by parts). The product rule leads directly to the summation by parts formula: summing both sides of $\Delta(fg) = f\,\Delta g + g(E)\,\Delta f$ over $k = a$ to $b-1$ and applying the Fundamental Theorem gives the formula in the next section.

Summation by Parts

Summation by parts is the discrete analogue of integration by parts. It is derived from the product rule exactly as the continuous version is derived from the Leibniz rule.

Theorem (Summation by Parts)

For functions $f, g : \mathbb{Z} \to \mathbb{R}$ and integers $a \le b$,

$$\sum_{k=a}^{b-1} f(k)\,\Delta g(k) = \Big[f(k)g(k)\Big]_{k=a}^{k=b} - \sum_{k=a}^{b-1} g(k+1)\,\Delta f(k).$$

That is,

$$\sum_{k=a}^{b-1} f(k)\,\Delta g(k) = f(b)g(b) - f(a)g(a) - \sum_{k=a}^{b-1} g(k+1)\,\Delta f(k).$$

Remark (Fully quantified form). $\forall f, g : \mathbb{Z} \to \mathbb{R},\ \forall a, b \in \mathbb{Z},\ a \le b : \ \sum_{k=a}^{b-1} f(k)\Delta g(k) = f(b)g(b) - f(a)g(a) - \sum_{k=a}^{b-1} g(k+1)\Delta f(k)$.

Proof. By the Discrete Product Rule (Theorem 5.8.1),

$$\Delta(fg)(k) = f(k)\,\Delta g(k) + g(k+1)\,\Delta f(k).$$

Rearranging,

$$f(k)\,\Delta g(k) = \Delta(fg)(k) - g(k+1)\,\Delta f(k).$$

Sum both sides from $k = a$ to $k = b-1$:

$$\sum_{k=a}^{b-1} f(k)\,\Delta g(k) = \sum_{k=a}^{b-1} \Delta(fg)(k) - \sum_{k=a}^{b-1} g(k+1)\,\Delta f(k).$$

Applying the Fundamental Theorem of Finite Calculus (Theorem 5.6.1) to the first sum on the right gives

$$\sum_{k=a}^{b-1} \Delta(fg)(k) = f(b)g(b) - f(a)g(a),$$

which completes the proof. ∎

Remark (Analogy with integration by parts). The continuous integration by parts formula,

$$\int_a^b f\,dg = \left[fg\right]_a^b - \int_a^b g\,df,$$

is recovered from summation by parts by replacing $\sum$ with $\int$ and Δ with d. The asymmetry in the discrete formula — $g(k+1)$ rather than $g(k)$ in the remaining sum — reflects the shift inherent in the product rule.

Remark (Applications). Summation by parts is useful for evaluating sums involving products, particularly when one factor is easy to sum (as a falling factorial) and the other is easy to difference. Classic applications include evaluating $\sum kr^k$ (where $f(k)=k$, $\Delta g(k)=r^k$ leads to a geometric-type sum) and establishing Abel's summation formula.

5.9 Newton Forward Difference Expansion

5.9.1 Newton's Forward Difference Expansion

In continuous calculus, smooth functions can be expanded as power series called Taylor series:

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}\,x^k = f(0) + f'(0)\,x + \frac{f''(0)}{2!}\,x^2 + \cdots .$$

The coefficients are determined by successive derivatives evaluated at 0, and the basis functions are ordinary powers x^k.

Discrete calculus has a precise analogue: the *Newton forward difference expansion*, in which derivatives are replaced by finite differences and ordinary powers are replaced by falling factorials.

Theorem (Newton Forward Difference Expansion)

Let $f : \mathbb{Z} \to \mathbb{R}$. Then f can be expressed as

$$f(x) = \sum_{k=0}^{n} \frac{\Delta^k f(0)}{k!}\,x^{\underline{k}} = f(0) + \Delta f(0)\,x + \frac{\Delta^2 f(0)}{2!}\,x^{\underline{2}} + \frac{\Delta^3 f(0)}{3!}\,x^{\underline{3}} + \cdots ,$$

where $x^{\underline{k}} = x(x-1)\cdots(x-k+1)$ is the falling factorial. For a polynomial of degree n, the expansion is exact (the series terminates at $k=n$).

Remark (Fully quantified form). $\forall f : \mathbb{Z} \to \mathbb{R}$ polynomial of degree $\le n$, $\forall x \in \mathbb{Z}:\ f(x) = \sum_{k=0}^{n} \binom{x}{k} \Delta^k f(0)$, where $\binom{x}{k} := x^{\underline{k}}/k!$.

Structural Parallel with the Taylor Series

The Newton expansion mirrors the Taylor series at every point. The correspondence is exact:

Taylor Series (continuous)	Newton Expansion (discrete)
Coefficients $f^{(k)}(0)/k!$	Coefficients $\Delta^k f(0)/k!$
Basis functions x^k	Basis functions $x^{\underline{k}}$
Derivative $f^{(k)}$	Difference Δ^k
Convergence required in general	Exact for polynomials

Remark (Why falling factorials play the role of powers). In the Taylor series, powers x^k are the natural basis because the derivative satisfies $\frac{d}{dx}x^k = kx^{k-1}$ and $\frac{d^m}{dx^m}x^k\big|_{x=0} = k!$ when $m = k$, and 0 otherwise. The falling factorials satisfy the same two properties under Δ: $\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}$ and $\Delta^m\, x^{\underline{k}}\big|_{x=0} = k!$ when $m = k$, else 0. This "triangular" property is exactly what allows the coefficients $c_k = \Delta^k f(0)/k!$ to be read off independently, level by level.

Proof Sketch

We construct the expansion term by term, matching the values of f at successive integers.

The constant term $f(0)$ matches f at $x = 0$.

Adding the linear term $\Delta f(0) \cdot x$ matches f at $x = 0$ and $x = 1$.

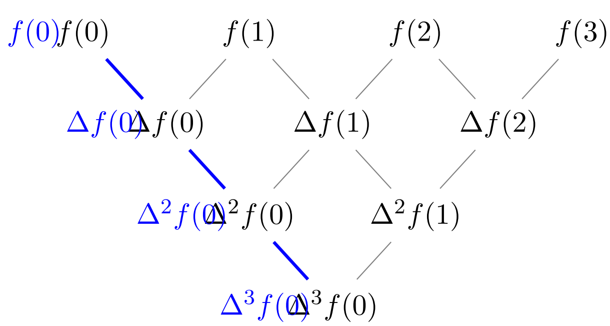
Adding the quadratic term $\frac{\Delta^2 f(0)}{2!}\, x^{\underline{2}}$ corrects the value at $x = 2$ while leaving earlier matches intact, because $x^{\underline{2}} = x(x-1)$ vanishes at $x = 0$ and $x = 1$.

At each stage, $x^{\underline{k}}$ vanishes at $x = 0, 1, \ldots, k-1$, so the k-th term corrects only the value at $x = k$. The coefficient $\Delta^k f(0)/k!$ achieves this correction by the triangular property noted above.

Reading Coefficients from the Difference Table

The coefficients in the Newton expansion are read from the *left edge* of the difference table — the "Newton diagonal":

$$f(0), \quad \Delta f(0), \quad \Delta^2 f(0), \quad \Delta^3 f(0), \ldots$$



Each level of the difference table contributes exactly one term to the falling-factorial expansion.

Worked Example: Expanding $f(n) = n^3$

We compute the difference table at $n = 0$:

n	$f(n)$	$\Delta f(n)$	$\Delta^2 f(n)$	$\Delta^3 f(n)$
0	0	1	6	6
1	1	7	12	
2	8	19		
3	27			

The coefficients are $c_k = \Delta^k f(0)/k!$:

$$c_0 = 0, \quad c_1 = 1, \quad c_2 = \frac{6}{2} = 3, \quad c_3 = \frac{6}{6} = 1.$$

Therefore the Newton expansion is:

$$n^3 = n + 3\, n^{\underline{2}} + n^{\underline{3}} = n + 3n(n-1) + n(n-1)(n-2).$$

Verification: $n + 3n^2 - 3n + n^3 - 3n^2 + 2n = n^3$. ✓

Application: Computing $\sum_{k=0}^{n-1} k^2$

Using the expansion $k^2 = k^{\underline{1}} + k^{\underline{2}}$ and the Fundamental Theorem of Finite Calculus (Theorem 5.6.1),

$$\sum_{k=0}^{n-1} k^2 = \sum_{k=0}^{n-1} k^{\underline{1}} + \sum_{k=0}^{n-1} k^{\underline{2}}.$$

Since $\Delta\left(\frac{k^{\underline{2}}}{2}\right) = k^{\underline{1}}$ and $\Delta\left(\frac{k^{\underline{3}}}{3}\right) = k^{\underline{2}}$, the Fundamental Theorem gives:

$$\sum_{k=0}^{n-1} k^{\underline{1}} = \frac{n^{\underline{2}}}{2} - \frac{0^{\underline{2}}}{2} = \frac{n(n-1)}{2},$$

$$\sum_{k=0}^{n-1} k^{\underline{2}} = \frac{n^{\underline{3}}}{3} - \frac{0^{\underline{3}}}{3} = \frac{n(n-1)(n-2)}{3}.$$

Adding:

$$\sum_{k=0}^{n-1} k^2 = \frac{n(n-1)}{2} + \frac{n(n-1)(n-2)}{3} = \frac{n(n-1)}{6}\big[3 + 2(n-2)\big] = \frac{n(n-1)(2n-1)}{6}.$$

Remark (Check against the standard formula). The standard formula is $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

Our result $\sum_{k=0}^{n-1} k^2 = \frac{(n-1)n(2n-1)}{6}$ matches: substitute $n \mapsto n-1$ in the standard formula to get $\frac{(n-1)n(2n-1)}{6}$. ✓

This example illustrates the power of the method: the sum was evaluated by pure algebraic antidifferentiation, with no need for induction or clever manipulation.

5.10 Special Functions

5.10.1 Special Functions in Discrete Calculus

Just as continuous calculus has its distinguished special functions — the exponential e^{ax}, the trigonometric functions, and the logarithm — discrete calculus has analogous functions that behave simply under the difference operator.

The Discrete Exponential

In continuous calculus, the exponential function e^{ax} is its own derivative (up to a constant):

$$\frac{d}{dx}e^{ax} = a\,e^{ax}.$$

The analogous function in discrete calculus is the *discrete exponential* $(1+a)^n$.

Definition 5.10.1 (Discrete Exponential)**.** For a constant $a \neq -1$ and $n \in \mathbb{Z}$, the *discrete exponential* with base $(1+a)$ is the function

$$E_a(n) = (1+a)^n.$$

Remark (Fully quantified form). $\forall a \in \mathbb{R} \setminus \{-1\},\ \forall n \in \mathbb{Z} :\ E_a(n) := (1+a)^n.$

Proposition 5.10.2 (Discrete exponential rule)**.**

$$\Delta\,(1+a)^n = a\,(1+a)^n.$$

Remark (Dependencies).

- Discrete Exponential.
- Forward Difference Operator.

Remark (Fully quantified form). $\forall a \in \mathbb{R} \setminus \{-1\},\ \forall n \in \mathbb{Z} :\ (1+a)^{n+1} - (1+a)^n = a(1+a)^n.$

Proof.

$$\Delta\,(1+a)^n = (1+a)^{n+1} - (1+a)^n = (1+a)^n\big[(1+a) - 1\big] = a(1+a)^n. \qquad \square$$

Remark (Analogy). The continuous exponential satisfies $\frac{d}{dx}e^{ax} = ae^{ax}$. The discrete exponential satisfies $\Delta(1+a)^n = a(1+a)^n$. The base e in continuous calculus is replaced by $(1+a)$ in discrete calculus. When a is small, $(1+a) \approx e^a$, recovering the continuous limit.

Remark (Antiderivative). From the proposition, a discrete antiderivative of $a(1+a)^n$ is $(1+a)^n$ itself. Equivalently, a discrete antiderivative of $(1+a)^n$ is $\frac{(1+a)^n}{a}$. Combined with the Fundamental Theorem of Finite Calculus, this gives the geometric series formula:

$$\sum_{k=0}^{n-1}(1+a)^k = \frac{(1+a)^n - 1}{a} \qquad (a \neq 0).$$

5.10.2 The Function 2^n in Discrete Calculus

Among all discrete exponentials, the case $a = 1$ — giving $(1+1)^n = 2^n$ — is perhaps the most important. It appears in combinatorics as the count of subsets, in probability as the sample space of fair coin sequences, in computer science as the natural measure of binary information, and in discrete calculus as a canonical test function. This section develops its properties systematically.

Differencing 2^n

Setting $a = 1$ in the discrete exponential rule (Proposition 5.10.2) gives immediately:

Proposition ($\Delta 2^n = 2^n$)

$$\Delta\, 2^n = 2^n.$$

Remark (Standard quantified statement). $\forall n \in \mathbb{Z} :\ 2^{n+1} - 2^n = 2^n$.

Proof.
$$\Delta\, 2^n = 2^{n+1} - 2^n = 2 \cdot 2^n - 2^n = (2-1) \cdot 2^n = 2^n. \qquad \square$$

Remark (Self-similarity). This says that 2^n is its own forward difference. In continuous calculus, the analogous property is $\frac{d}{dx}e^x = e^x$: the exponential is its own derivative. The function 2^n is the discrete analogue of e^x in the sense that $\Delta(2^n) = 2^n$. More precisely, $2^n = e^{n \ln 2}$, and the relationship $\Delta(2^n) = 2^n$ corresponds to the special case $a = 1$ of $\Delta(1+a)^n = a(1+a)^n$.

Antiderivative and Geometric Sum

Since $\Delta(2^n) = 2^n$, a discrete antiderivative of 2^n is 2^n itself (up to a constant).

Proposition (Antiderivative of 2^n)

A discrete antiderivative of 2^n is 2^n itself:

$$\Delta(2^n) = 2^n \qquad \text{so} \qquad \sum_{k=0}^{n-1} 2^k = 2^n - 1.$$

Remark (Standard quantified statement). $\forall n \in \mathbb{N} : \displaystyle\sum_{k=0}^{n-1} 2^k = 2^n - 1$.

Proof. Applying the Fundamental Theorem of Finite Calculus (Theorem 5.6.1) with $F(k) = 2^k$ and $f(k) = \Delta F(k) = 2^k$:

$$\sum_{k=0}^{n-1} 2^k = F(n) - F(0) = 2^n - 2^0 = 2^n - 1. \qquad \square$$

Remark (Verification by expansion). Expanding: $1 + 2 + 4 + 8 + \cdots + 2^{n-1}$. Multiply the partial sum $S = \sum_{k=0}^{n-1} 2^k$ by 2 to get $2S = \sum_{k=1}^{n} 2^k$. Subtracting: $2S - S = 2^n - 1$, so $S = 2^n - 1$. This is the classical derivation of the geometric series formula; the discrete calculus approach via antidifferentiation gives the same result mechanically.

Higher Differences of 2^n

Proposition 5.10.3 (Higher differences of 2^n)**.** *For all $k \geq 0$,*

$$\Delta^k\, 2^n = 2^n.$$

Remark (Dependencies).

- Higher Differences.
- Discrete Exponential Rule.

Remark (Standard quantified statement). $\forall k \in \mathbb{N},\ \forall n \in \mathbb{Z}:\ \Delta^k(2^n) = 2^n$.

Proof. By induction. The base case $k = 0$ is $\Delta^0(2^n) = 2^n$ by definition. For the inductive step, assume $\Delta^k(2^n) = 2^n$. Then

$$\Delta^{k+1}(2^n) = \Delta(\Delta^k(2^n)) = \Delta(2^n) = 2^n. \qquad \square$$

Remark (Consequence for difference tables). Since every finite difference of 2^n is 2^n again, the difference table of 2^n consists entirely of the sequence 2^n at every level. This is the discrete analogue of the fact that the Taylor series of e^x has the same coefficient pattern at every derivative level. In particular, 2^n is *not* a polynomial — a polynomial of degree d would reach 0 at the $(d+1)$-th difference row. The non-termination of the difference table is the signature of a genuinely transcendental discrete function.

2^n in Combinatorics and the Newton Expansion

The Binomial Theorem (Theorem 5.2.1) gives a direct connection: setting $x = y = 1$ yields

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

This counts the total number of subsets of an n-element set: there are $\binom{n}{k}$ subsets of size k, and the sum over all k gives 2^n.

More deeply, the Newton expansion (Theorem 5.9.1) of 2^n in the falling factorial basis is:

$$2^n = \sum_{k=0}^{n} \binom{n}{k} = \sum_{k=0}^{n} \frac{n^{\underline{k}}}{k!},$$

since $\Delta^k(2^n)\big|_{n=0} = 2^0 = 1$ for every k. This expresses 2^n as the sum of all normalised falling factorials up to degree n, and reveals why binomial coefficients count subsets: they are the coefficients of 2^n in its own Newton expansion.

Remark (Analogy with e^x). In continuous calculus, $e^x = \sum_{k=0}^{\infty} x^k/k!$. The Newton expansion gives $2^n = \sum_{k=0}^{n} n^{\underline{k}}/k!$. Both express a canonical exponential as a sum of normalised basis functions, with all coefficients equal to 1. The analogy is exact: ordinary powers replace falling factorials, and the infinite series terminates at degree n in the discrete case because 2^n has degree exactly n in the falling factorial basis.

Discrete Differential Equation

In continuous calculus, the equation $f' = f$ (with $f(0) = 1$) has the unique solution $f(x) = e^x$. In discrete calculus, the analogous *difference equation* is:

$$\Delta F(n) = F(n), \qquad F(0) = 1.$$

That is, $F(n+1) - F(n) = F(n)$, so $F(n+1) = 2F(n)$. The unique solution to this initial value problem is $F(n) = 2^n$.

Proposition 5.10.4 (Discrete IVP)**.** *The unique function $F : \mathbb{N} \to \mathbb{R}$ satisfying*

$$F(n+1) = 2\,F(n), \qquad F(0) = 1$$

is $F(n) = 2^n$.

Remark (Dependencies).

- Discrete Exponential.

Remark (Standard quantified statement). $\forall n \in \mathbb{N} : \ (F(n+1) = 2F(n) \text{ and } F(0) = 1) \Rightarrow F(n) = 2^n$.

Proof. Uniqueness: if G is another solution, then $G(0) = 1 = F(0)$, and inductively $G(n+1) = 2G(n) = 2F(n) = F(n+1)$, so $G = F$. Existence: verify directly that $2^{n+1} = 2 \cdot 2^n$ and $2^0 = 1$. □

Remark (Significance). This characterises 2^n as the unique fixed point of "multiply by 2" starting from 1. The analogous characterisation of e^x is: unique solution to $f' = f$, $f(0) = 1$. Both are initial-value problems in their respective calculi, and both have closed-form solutions that serve as the canonical exponential of that calculus.

Difference Formulas for Special Functions

The table below collects the difference formulas for the principal special functions of discrete calculus.

Remark (Difference formulas).

Function	Forward Difference
$x^{\underline{k}}$ (falling factorial)	$k\,x^{\underline{k-1}}$
$(1+a)^n$ (discrete exponential)	$a\,(1+a)^n$
$\binom{x}{k} = x^{\underline{k}}/k!$	$\binom{x}{k-1}$
c (constant)	0
n	1
$n^{\underline{2}} = n(n-1)$	$2n$

Remark (Binomial coefficient as a special case). The formula $\Delta\binom{x}{k} = \binom{x}{k-1}$ follows immediately from the falling factorial rule $\Delta\,x^{\underline{k}} = k\,x^{\underline{k-1}}$ by dividing through by $k!$. It says that the binomial coefficient $\binom{x}{k}$ is its own natural "unit" in discrete calculus, analogous to the monomial $x^k/k!$ in the Taylor expansion.

Antiderivative Table

By reversing the difference formulas, we obtain the corresponding discrete antiderivatives:

Remark (Discrete antiderivatives).

Function $f(n)$	**Antiderivative** $F(n)$ **with** $\Delta F = f$
$x^{\underline{k}}$	$\dfrac{x^{\underline{k+1}}}{k+1}$
$(1+a)^n$ $(a \neq 0)$	$\dfrac{(1+a)^n}{a}$
$\binom{x}{k}$	$\binom{x}{k+1}$
c (constant)	cn

Remark (Using the Fundamental Theorem). Each antiderivative in the table, combined with the Fundamental Theorem of Finite Calculus (Theorem 5.6.1), yields an immediate summation formula. For instance, from $\Delta\left(\frac{x^{\underline{k+1}}}{k+1}\right) = x^{\underline{k}}$:

$$\sum_{j=0}^{n-1} j^{\underline{k}} = \frac{n^{\underline{k+1}}}{k+1}.$$

This is the discrete analogue of $\int_0^n x^k\,dx = \frac{n^{k+1}}{k+1}$.

Chapter 6

Sequences

SERIES
From Cantor to Ito

VOLUME III
Classical Analysis

BOOK I
Bounds, Sequences, and Series

CHAPTER 6
Sequences

6.1 Sequence Definitions

6.1.1 Definitions

Definition (Sequence)

Definition 6.1.1 (Sequence)**.** Let X be a nonempty set. A **sequence** in X is a function

$$x : \mathbb{N} \to X.$$

The value $x(n)$ is written x_n and called the ***n*-th term** of the sequence. The sequence itself is denoted $(x_n)_{n\in\mathbb{N}}$, or simply (x_n) when the index set is clear.

Remark (Standard quantified statement).

$$\forall X \neq \varnothing\ \forall x : \mathbb{N} \to X\ \big(\mathrm{Sequence}(x, X) \iff x : \mathbb{N} \to X\big).$$

Remark (Definition predicate reading).

$$\mathrm{Sequence}(x_n, X)\ :=\ x : \mathbb{N} \to X.$$

Remark (Interpretation). A sequence is not a set of terms but a function: the index carries independent weight. Two sequences (x_n) and (y_n) in X are equal precisely when $x_n = y_n$ for every $n \in \mathbb{N}$, so order and repetition are intrinsic to the object. The notational collapse from $x(n)$ to x_n is conventional suppression of function-application syntax; the underlying object remains a function $\mathbb{N} \to X$. When $X = \mathbb{R}$, the sequence is called a **real sequence**, which is the default ambient context for this chapter.

Remark (Dependencies).

- Function
- Natural Numbers

Definition (Real Sequence)

Definition 6.1.2 (Real Sequence)**.** A **real sequence** is a sequence whose image is contained in $\mathbb{R}$. Equivalently, a real sequence is a function

$$x : \mathbb{N} \to \mathbb{R}.$$

Remark (Standard quantified statement).

$$\text{RealSequence}(x_n) \iff x : \mathbb{N} \to \mathbb{R}.$$

Remark (Definition predicate reading).

$$\text{RealSequence}(x_n) \ := \ \text{Sequence}(x_n, \mathbb{R}).$$

Remark (Interpretation). A real sequence is not a new kind of indexing object; it is the specialization of a sequence to the ambient space $\mathbb{R}$. Thus the generic function structure is inherited from sequences, while order, absolute value, and arithmetic enter through the real codomain.

Remark (Dependencies).

- Sequence
- Real Numbers

Example (Constant Sequence). Let $c \in \mathbb{R}$. The **constant sequence** is defined by

$$x_n := c \quad \forall n \in \mathbb{N}.$$

The constant sequence converges to c: given any $\varepsilon > 0$, take $N = 1$, and observe $|x_n - c| = 0 < \varepsilon$ for all $n \geq 1$. It is the degenerate case of convergence in which the tail is not merely eventually close to L but identically equal to it from the first term.

Example (The Sequence $(1/n)$). The sequence (x_n) defined by

$$x_n := \frac{1}{n} \quad \forall n \in \mathbb{N}$$

converges to 0. Given $\varepsilon > 0$, the Archimedean property supplies $N \in \mathbb{N}$ with $N > 1/\varepsilon$, and then for all $n \geq N$,

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

This is the canonical example of a sequence that approaches its limit monotonically from above, never reaching it. It also serves as the standard witness in squeeze arguments and in the proof that $\inf\{1/n : n \in \mathbb{N}\} = 0$.

Example (The Sequence (n)). The sequence (x_n) defined by

$$x_n := n \quad \forall n \in \mathbb{N}$$

is divergent. For any proposed limit $L \in \mathbb{R}$, take $\varepsilon = 1$; the Archimedean property supplies $n \in \mathbb{N}$ with $n > L + 1$, so $|x_n - L| = |n - L| > 1 = \varepsilon$ for infinitely many n. The sequence is unbounded, and every convergent sequence is bounded, so divergence also follows from that theorem once it is established. This is the canonical example of divergence to $+\infty$: the terms are not oscillating outside a neighborhood of L but growing without bound in a single direction.

6.1.2 Null and Constant Sequences

Null and Constant Sequences

Definition (Constant Sequence)

Definition 6.1.3 (Constant Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **constant** exactly when there exists $c \in \mathbb{R}$ such that $x_n = c$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &(x_n) \text{ is constant} \iff \exists c \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n = c).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{ConstantSequence}(x_n, c) :\equiv \forall n \in \mathbb{N}\ (x_n = c).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &(x_n) \text{ is not constant} \iff \forall c \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x_n \neq c).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\exists c \in \mathbb{R}\ \text{ConstantSequence}(x_n, c).$$

Remark (Failure modes). Every proposed constant value fails at some index.

Remark (Failure mode decomposition).

$$\forall c \in \mathbb{R} \qquad \underbrace{\exists n \in \mathbb{N}\ (x_n \neq c)}_{\text{a term differs from the proposed constant value}}.$$

Remark (Interpretation). A constant sequence has no genuine dependence on the index. The indexing is still part of the object, but every index is assigned the same real value.

Remark (Dependencies).

- Real Sequence

Definition (Null Sequence)

Definition 6.1.4 (Null Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is a **null sequence** exactly when for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n| < \varepsilon$ for every $n \geq N$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is a null sequence} \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{NullSequence}(x_n) :\equiv \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is not a null sequence} \iff \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ (|x_n| \geq \varepsilon).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{NullSequence}(x_n).$$

Remark (Failure modes). Some positive tolerance is missed infinitely far out in the sequence.

Remark (Failure mode decomposition).

$$\exists \varepsilon > 0\ \forall N \in \mathbb{N} \qquad \underbrace{\exists n \geq N\ (|x_n| \geq \varepsilon)}_{\text{the tail is not trapped inside the } \varepsilon\text{-band}} \qquad .$$

Remark (Interpretation). A null sequence is a sequence whose tail is eventually contained in every symmetric neighborhood of zero. The first finitely many terms are irrelevant; only the eventual behavior matters.

Remark (Dependencies).

- Real Sequence

Theorem (Constant Sequence Convergence)

Theorem 6.1.5 (Constant Sequence Convergence)**.** *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) converges to c.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence and } c \in \mathbb{R}.\\&\big(\forall n \in \mathbb{N}\ (x_n = c)\big) \implies \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - c| < \varepsilon).\end{aligned}$$

Remark (Predicate reading).

$$\text{ConstantSequence}(x_n, c) \implies \text{ConvergentSequence}(x_n, c).$$

Remark (Interpretation). A constant sequence is already equal to its proposed limit at every index. Every positive tolerance is therefore satisfied by any tail of the sequence.

Remark (Dependencies).

- Constant Sequence
- Convergent Sequence

Theorem (Zero Sequence is Null)

Theorem 6.1.6 (Zero Sequence is Null)**.** *Let (x_n) be a real sequence. If $x_n = 0$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\big(\forall n \in \mathbb{N}\ (x_n = 0)\big) \implies \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon).\end{aligned}$$

Remark (Predicate reading).

$$\text{ConstantSequence}(x_n, 0) \implies \text{NullSequence}(x_n).$$

Remark (Interpretation). The zero sequence satisfies every null-sequence tolerance immediately because the distance from each term to zero is exactly zero.

Remark (Dependencies).

- Null Sequence
- Constant Sequence

Theorem (Constant Null Sequence)

Theorem 6.1.7 (Constant Null Sequence)**.** *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence and } c \in \mathbb{R}.\\ &\big(\forall n \in \mathbb{N}\ (x_n = c)\big) \implies \Big(\big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon)\big] \iff c = 0\Big).\end{aligned}$$

Remark (Predicate reading).

$$\text{ConstantSequence}(x_n, c) \implies \big(\text{NullSequence}(x_n) \iff c = 0\big).$$

Remark (Interpretation). A sequence that never changes can approach zero only when its fixed value is already zero. Conversely, the fixed value zero gives the zero sequence.

Remark (Dependencies).

- Constant Sequence
- Null Sequence

Definition (Ultimately Constant Sequence)

Definition 6.1.8 (Ultimately Constant Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **ultimately constant** exactly when there exist $N \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is ultimately constant} \iff \exists N \in \mathbb{N}\, \exists c \in \mathbb{R}\, \forall n \geq N\ (x_n = c).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, c) :\equiv \exists N \in \mathbb{N}\, \forall n \geq N\ (x_n = c).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is not ultimately constant} \iff \forall N \in \mathbb{N}\, \forall c \in \mathbb{R}\, \exists n \geq N\ (x_n \neq c).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\exists c \in \mathbb{R}\ \ \text{UltimatelyConstantSequence}(x_n, c).$$

Remark (Failure modes). No tail of the sequence becomes permanently equal to a single real value.

Remark (Failure mode decomposition).

$$\forall N \in \mathbb{N}\, \forall c \in \mathbb{R} \qquad \underbrace{\exists n \geq N\ (x_n \neq c)}_{\text{the proposed tail value fails later in the sequence}}.$$

Remark (Interpretation). An ultimately constant sequence may vary on an initial finite segment, but its tail becomes fixed. The concept isolates eventual behavior from finite initial behavior.

Remark (Dependencies).

- Real Sequence

Theorem (Difference from the Limit is Null)

Theorem 6.1.9 (Difference from the Limit is Null)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The sequence (x_n) converges to L if and only if the sequence $(x_n - L)$ is a null sequence.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence and } L \in \mathbb{R}.\\&\big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon)\big]\\&\qquad \iff \qquad \big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon)\big].\end{aligned}$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \iff \text{NullSequence}(x_n - L).$$

Remark (Interpretation). Subtracting the proposed limit translates the convergence problem to the special case of convergence to zero. Null sequences are therefore the local normal form for sequence convergence.

Remark (Dependencies).

- Convergent Sequence
- Null Sequence
- Pointwise Difference of Sequences

Theorem (Ultimately Constant Sequence Convergence)

Theorem 6.1.10 (Ultimately Constant Sequence Convergence)**.** *Let (x_n) be a real sequence. If there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$, then (x_n) converges to c.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&\forall c \in \mathbb{R}\ \Big(\big[\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = c)\big]\\&\qquad \Longrightarrow \qquad \big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - c| < \varepsilon)\big]\Big).\end{aligned}$$

Remark (Predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, c) \Longrightarrow \text{ConvergentSequence}(x_n, c).$$

Remark (Interpretation). Once the tail is equal to a fixed value, all sufficiently late terms are already exactly at the proposed limit.

Remark (Dependencies).

- Ultimately Constant Sequence
- Convergent Sequence

Theorem (Constant Implies Ultimately Constant)

Theorem 6.1.11 (Constant Implies Ultimately Constant)**.** *Every constant real sequence is ultimately constant.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&\big[\exists c \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n = c)\big]\\&\qquad\implies\qquad \big[\exists N \in \mathbb{N}\ \exists c \in \mathbb{R}\ \forall n \geq N\ (x_n = c)\big].\end{aligned}$$

Remark (Predicate reading).

$$\text{ConstantSequence}(x_n, c) \implies \text{UltimatelyConstantSequence}(x_n, c).$$

Remark (Interpretation). A constant sequence has already stabilized from the first index, so it is automatically stabilized on some tail.

Remark (Dependencies).

- Constant Sequence
- Ultimately Constant Sequence

Theorem (Ultimately Zero Sequence is Null)

Theorem 6.1.12 (Ultimately Zero Sequence is Null)**.** *Let (x_n) be a real sequence. If there exists $N_0 \in \mathbb{N}$ such that $x_n = 0$ for every $n \geq N_0$, then (x_n) is a null sequence.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&\big[\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = 0)\big]\\&\qquad\implies\qquad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon).\end{aligned}$$

Remark (Predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, 0) \implies \text{NullSequence}(x_n).$$

Remark (Interpretation). If the tail is identically zero, then every sufficiently late term lies inside every positive tolerance around zero.

Remark (Dependencies).

- Ultimately Constant Sequence
- Null Sequence

Theorem (Ultimately Constant Null Sequence)

Theorem 6.1.13 (Ultimately Constant Null Sequence)**.** *Let (x_n) be a real sequence. Suppose there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$. Then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &\forall c \in \mathbb{R} \Big(\big[\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = c)\big] \\ &\qquad \implies \qquad \Big(\big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon)\big] \iff c = 0 \Big)\Big). \end{aligned}$$

Remark (Predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, c) \implies \big(\text{NullSequence}(x_n) \iff c = 0\big).$$

Remark (Interpretation). Only the eventual value controls whether an ultimately constant sequence is null. A finite initial segment cannot prevent or create null behavior.

Remark (Dependencies).

- Ultimately Constant Sequence
- Null Sequence

Theorem (Tail Equality Preserves Convergence)

Theorem 6.1.14 (Tail Equality Preserves Convergence)**.** *Let (x_n) and (y_n) be real sequences. If there exists $N_0 \in \mathbb{N}$ such that $x_n = y_n$ for every $n \geq N_0$, then for every $L \in \mathbb{R}$, (x_n) converges to L if and only if (y_n) converges to L.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ and } (y_n) \text{ be real sequences.} \\ &\big[\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = y_n)\big] \\ &\qquad \implies \qquad \forall L \in \mathbb{R} \Big(\big[\forall \varepsilon > 0\ \exists N_x \in \mathbb{N}\ \forall n \geq N_x\ (|x_n - L| < \varepsilon)\big] \\ &\qquad\qquad \iff \qquad \big[\forall \varepsilon > 0\ \exists N_y \in \mathbb{N}\ \forall n \geq N_y\ (|y_n - L| < \varepsilon)\big]\Big). \end{aligned}$$

Remark (Predicate reading).

$$\text{EventuallyEqual}(x_n, y_n) \implies \forall L \in \mathbb{R}\ \big(\text{ConvergentSequence}(x_n, L) \iff \text{ConvergentSequence}(y_n, L)\big). \blacksquare$$

Remark (Interpretation). Convergence is a tail property. Changing finitely many initial terms cannot change whether a sequence converges to a specified real number.

Remark (Dependencies).

- Convergent Sequence

Definition (Sequence Bounded Above)

Definition 6.1.15 (Sequence Bounded Above)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded above** exactly when there exists $M \in \mathbb{R}$ such that $x_n \le M$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &(x_n) \text{ is bounded above} \iff \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{BoundedAboveSequence}(x_n) :\Longleftrightarrow \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &(x_n) \text{ is not bounded above} \iff \forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ (M < x_n).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{BoundedAboveSequence}(x_n).$$

Remark (Failure modes). Every proposed upper bound is exceeded by some term of the sequence.

Remark (Failure mode decomposition).

$$\forall M \in \mathbb{R} \qquad \underbrace{\exists n \in \mathbb{N}\ (M < x_n)}_{\text{a term rises above the proposed ceiling}} \qquad .$$

Remark (Interpretation). A bounded-above sequence has a finite ceiling. Failure means no real number serves as a ceiling for all terms.

Remark (Dependencies).

- Real Sequence

Definition (Sequence Bounded Below)

Definition 6.1.16 (Sequence Bounded Below)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded below** exactly when there exists $m \in \mathbb{R}$ such that $m \le x_n$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is bounded below} \iff \exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n).\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\text{BoundedBelowSequence}(x_n) :\equiv \exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n).$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is not bounded below} \iff \forall m \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x_n < m).\end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg\,\text{BoundedBelowSequence}(x_n).$$

Remark (Failure modes)*.* Every proposed lower bound is undercut by some term of the sequence.

Remark (Failure mode decomposition)*.*

$$\forall m \in \mathbb{R} \quad \underbrace{\exists n \in \mathbb{N}\ (x_n < m)}_{\text{a term falls below the proposed floor}} .$$

Remark (Interpretation)*.* A bounded-below sequence has a finite floor. Failure means no real number serves as a floor for all terms.

Remark (Dependencies)*.*

- Real Sequence

Definition (Bounded Sequence)

Definition 6.1.17 (Bounded Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded** exactly when there exists $M > 0$ such that $|x_n| \le M$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is bounded} \iff \exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \le M).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{BoundedSequence}(x_n) :\equiv \exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \leq M).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &(x_n) \text{ is not bounded} \iff \forall M > 0\ \exists n \in \mathbb{N}\ (|x_n| > M).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BoundedSequence}(x_n).$$

Remark (Failure modes). Every symmetric bound around zero is escaped by some term of the sequence.

Remark (Failure mode decomposition).

$$\forall M > 0 \qquad \underbrace{\exists n \in \mathbb{N}\ (|x_n| > M)}_{\text{a term escapes the symmetric band of radius } M} \qquad .$$

Remark (Interpretation). A bounded sequence is trapped in a finite symmetric band around zero. This is equivalent to having both a finite ceiling and a finite floor.

Remark (Dependencies).

- Real Sequence

Theorem (Eventually Bounded Above Tail Formulation)

Theorem 6.1.18 (Eventually Bounded Above Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded above if and only if there exist $N_0 \in \mathbb{N}$ and $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \geq N_0$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\big[\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \leq M)\big]\\ &\qquad \iff \qquad \big[\exists N_0 \in \mathbb{N}\ \exists M \in \mathbb{R}\ \forall n \geq N_0\ (x_n \leq M)\big].\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{BoundedAboveSequence}(x_n) \iff \exists M \in \mathbb{R}\ \ \mathrm{Eventually}\big(x_n,\ x_n \leq M\big).$$

Remark (Interpretation). One-sided boundedness is unchanged by ignoring finitely many initial terms, because finitely many exceptional values can be absorbed into a larger upper bound.

Remark (Dependencies).

- Sequence Bounded Above

Theorem (Eventually Bounded Below Tail Formulation)

Theorem 6.1.19 (Eventually Bounded Below Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded below if and only if there exist $N_0 \in \mathbb{N}$ and $m \in \mathbb{R}$ such that $m \le x_n$ for every $n \ge N_0$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\big[\exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\big]\\ &\qquad \iff \qquad \big[\exists N_0 \in \mathbb{N}\ \exists m \in \mathbb{R}\ \forall n \ge N_0\ (m \le x_n)\big].\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{BoundedBelowSequence}(x_n) \iff \exists m \in \mathbb{R}\ \ \mathrm{Eventually}\big(x_n,\ m \le x_n\big).$$

Remark (Interpretation)*.* One-sided lower boundedness is unchanged by ignoring finitely many initial terms, because finitely many exceptional values can be absorbed into a lower bound.

Remark (Dependencies).

- Sequence Bounded Below

Theorem (Eventually Bounded Tail Formulation)

Theorem 6.1.20 (Eventually Bounded Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if there exist $N_0 \in \mathbb{N}$ and $M > 0$ such that $|x_n| \le M$ for every $n \ge N_0$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\big[\exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \le M)\big]\\ &\qquad \iff \qquad \big[\exists N_0 \in \mathbb{N}\ \exists M > 0\ \forall n \ge N_0\ (|x_n| \le M)\big].\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{BoundedSequence}(x_n) \iff \exists M > 0\ \ \mathrm{Eventually}\big(x_n,\ |x_n| \le M\big).$$

Remark (Interpretation)*.* Boundedness is a tail property up to finitely many exceptions. A finite initial segment can be enclosed by increasing the bound.

Remark (Dependencies).

- Bounded Sequence

Theorem (Bounded Sequence If and Only If Bounded Above and Below)

Theorem 6.1.21 (Bounded Sequence If and Only If Bounded Above and Below)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if it is both bounded above and bounded below.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&\big[\exists K > 0\ \forall n \in \mathbb{N}\ (|x_n| \le K)\big]\\&\qquad\iff\qquad \big[\exists m \in \mathbb{R}\ \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n \wedge x_n \le M)\big].\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{BoundedSequence}(x_n) \iff \big(\mathrm{BoundedBelowSequence}(x_n) \wedge \mathrm{BoundedAboveSequence}(x_n)\big).$$

Remark (Interpretation). An absolute-value bound gives a symmetric interval around zero. Having both a floor and a ceiling gives an interval that may not be symmetric, but it can be enlarged to a symmetric one.

Remark (Dependencies).

- Bounded Sequence
- Sequence Bounded Above
- Sequence Bounded Below

Theorem (Absolute Bound Gives Upper and Lower Bounds)

Theorem 6.1.22 (Absolute Bound Gives Upper and Lower Bounds)**.** *Let (x_n) be a real sequence, and let $K > 0$. If $|x_n| \le K$ for every $n \in \mathbb{N}$, then $-K \le x_n \le K$ for every $n \in \mathbb{N}$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&\forall K > 0\ \big[\forall n \in \mathbb{N}\ (|x_n| \le K)\big]\\&\qquad\implies\qquad \forall n \in \mathbb{N}\ (-K \le x_n \wedge x_n \le K).\end{aligned}$$

Remark (Interpretation). An absolute-value inequality simultaneously controls the positive and negative directions. The upper barrier is K, and the lower barrier is $-K$.

Remark (Dependencies).

- Bounded Sequence
- Sequence Bounded Above
- Sequence Bounded Below

Theorem (Upper and Lower Bounds Give an Absolute Bound)

Theorem 6.1.23 (Upper and Lower Bounds Give an Absolute Bound)**.** *Let (x_n) be a real sequence. If there exist $m, M \in \mathbb{R}$ such that $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then there exists $K > 0$ such that $|x_n| \leq K$ for every $n \in \mathbb{N}$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } (x_n) \text{ be a real sequence.}\\
&\big[\exists m \in \mathbb{R}\ \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \leq x_n \wedge x_n \leq M)\big]\\
&\qquad \Longrightarrow \qquad \big[\exists K > 0\ \forall n \in \mathbb{N}\ (|x_n| \leq K)\big].
\end{aligned}$$

Remark (Interpretation). Two one-sided barriers can always be replaced by one symmetric barrier around zero. Any positive K at least as large as both $|m|$ and $|M|$ works.

Remark (Dependencies).

- Sequence Bounded Above
- Sequence Bounded Below
- Bounded Sequence

6.2 Examples and Counterexamples

Sequence Examples — Quick Reference

Example	**Role**
Constant sequence	Basic convergent model
Reciprocal sequence	Canonical null sequence
Alternating null sequence	Sign oscillation with convergence to 0
Oscillating sequence	Bounded nonconvergent model
Geometric sequence	Ratio-driven convergence and divergence
Bounded not convergent	Boundedness alone is insufficient
Small differences not Cauchy	Vanishing increments do not ensure Cauchy

Examples and Counterexamples

Examples provide test objects for the definitions. A good sequence example usually isolates one behavior: settling to a limit, oscillating between several values, escaping to infinity, or satisfying a tempting condition that is still too weak to force convergence.

Example (Constant Sequence). Let $c \in \mathbb{R}$ and define $x_n = c$ for every $n \in \mathbb{N}$. Then (x_n) converges to c.

Remark (Interpretation). A constant sequence is the baseline example of convergence: every term is already equal to the proposed limit.

Example (Reciprocal Sequence). Define $x_n = 1/n$ for every $n \in \mathbb{N}$. Then (x_n) is decreasing, bounded below by 0, and converges to 0.

Remark (Interpretation). The reciprocal sequence is the standard positive null sequence. It is also the model for estimates of the form "eventually smaller than any fixed positive tolerance."

Example (Alternating Null Sequence). Define

$$x_n = \frac{(-1)^n}{n} \qquad \text{for every } n \in \mathbb{N}.$$

Then (x_n) is not monotone, but it converges to 0.

Remark (Interpretation). Oscillation in sign does not by itself prevent convergence. The amplitudes must be watched: here the oscillation collapses to zero.

Example (Oscillating Sequence). Define

$$x_n = (-1)^n \qquad \text{for every } n \in \mathbb{N}.$$

Then (x_n) is bounded and divergent. Its even-indexed subsequence converges to 1, and its odd-indexed subsequence converges to -1.

Remark (Interpretation). This is the canonical oscillator. The sequence never settles because two subsequences settle to different limits.

Example (Geometric Sequence). Let $r \in \mathbb{R}$ and define $x_n = r^n$ for every $n \in \mathbb{N}$. Then:

- if $|r| < 1$, then $x_n \to 0$;
- if $r = 1$, then $x_n \to 1$;
- if $r = -1$, then (x_n) oscillates and diverges;
- if $|r| > 1$, then (x_n) is unbounded.

Remark (Interpretation). The geometric sequence is the standard laboratory for rate, sign, and magnitude. The threshold $|r| = 1$ separates decay from growth.

Example (Bounded Does Not Imply Convergent). The sequence $x_n = (-1)^n$ is bounded, since $|x_n| \leq 1$ for every $n \in \mathbb{N}$, but it does not converge.

Remark (Interpretation). Boundedness prevents escape to infinity, but it does not prevent persistent oscillation.

Example (Vanishing Successive Differences Do Not Imply Cauchy). Let

$$x_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} \qquad \text{for every } n \in \mathbb{N}.$$

Then

$$x_{n+1} - x_n = \frac{1}{n+1} \to 0,$$

but (x_n) is not Cauchy.

Remark (Interpretation). Cauchy control requires all sufficiently late terms to be mutually close. Controlling only adjacent jumps is weaker.

6.3 Convergence

Definition (Convergence of a Sequence)

Definition 6.3.1 (Convergence of a Sequence)**.** Let $(x_n) \subseteq \mathbb{R}$. The sequence (x_n) **converges** to $L \in \mathbb{R}$ if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|x_n - L| < \varepsilon\big).$$

The value L is called the **limit** of (x_n), written $x_n \to L$ or $\lim\limits_{n\to\infty} x_n = L$. A sequence that converges to some $L \in \mathbb{R}$ is called **convergent**; otherwise it is called **divergent**.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\\ &\quad\Big(x_n \to L \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|x_n - L| < \varepsilon\big)\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \ :=\ \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \ \mathrm{OutputTolerance}(x_n, L, \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}x_n \not\to L \iff &\exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\\ &\big(|x_n - L| \geq \varepsilon\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\mathrm{NotConvergentTo}(x_n, L) \ :=\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| \geq \varepsilon\big).$$

Remark (Failure modes). Failure of convergence to a fixed L means that some positive tolerance cannot be controlled on any tail. The escaped terms may oscillate, drift monotonically away, or grow without bound, but all such behavior is captured by the same persistent-tolerance obstruction.

Remark (Failure mode decomposition). The negation decomposes into a single structural obstruction:

$$\underbrace{\exists \varepsilon > 0}_{\text{a fixed tolerance}} \quad \underbrace{\forall N \in \mathbb{N}}_{\text{no tail suffices}} \quad \underbrace{\exists n \geq N}_{\text{terms escape infinitely often}} \quad \underbrace{|x_n - L| \geq \varepsilon}_{\text{outside the target interval}} \quad .$$

The sequence fails to converge to L not by drifting away once, but by returning outside the ε-interval infinitely often regardless of how far along the tail one looks.

Remark (Interpretation). The quantifier order $\forall \varepsilon \, \exists N \, \forall n \geq N$ encodes a strict dependency: N is permitted to depend on ε, but not on n. This asymmetry is load-bearing. Reversing the outer quantifiers to $\exists N \, \forall \varepsilon$ would assert that a single index N controls all tolerances simultaneously, which is a strictly stronger and generally false condition. The condition $|x_n - L| < \varepsilon$ is the membership assertion $x_n \in (L - \varepsilon, \, L + \varepsilon)$; convergence is therefore the statement that the tail of the sequence is eventually contained in every ε-neighborhood of L.

Remark (Dependencies).

- Real Sequence

Definition (Convergence — Neighborhood Formulation)

Definition 6.3.2 (Convergence — Neighborhood Formulation)**.** Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The sequence (x_n) **converges** to L if for every $\varepsilon > 0$ the ε-neighborhood

$$V_\varepsilon(L) \ := \ (L - \varepsilon, \, L + \varepsilon)$$

contains all but finitely many terms of (x_n); that is,

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(x_n \in V_\varepsilon(L)\big).$$

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \, \forall L \in \mathbb{R} \\ \Big(x_n \to L \iff \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(x_n \in V_\varepsilon(L)\big)\Big).$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(x_n, L) \ := \ \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(x_n \in V_\varepsilon(L)\big).$$

Remark (Negated quantified statement).

$$x_n \not\to L \iff \exists \varepsilon > 0 \, \forall N \in \mathbb{N} \, \exists n \geq N \\ \big(x_n \notin V_\varepsilon(L)\big).$$

Remark (Negation predicate reading).

$$\text{NotConvergentTo}(x_n, L) \ := \ \exists \varepsilon > 0 \, \forall N \in \mathbb{N} \, \exists n \geq N \, \big(x_n \notin V_\varepsilon(L)\big).$$

Remark (Failure modes). Failure of the neighborhood formulation is exactly failure to trap a tail in every neighborhood of L. Different examples may escape by oscillation, by monotone drift, or by unbounded growth, but the formal failure is always the existence of one neighborhood that every tail misses at least once.

Remark (Failure mode decomposition).

$$\underbrace{\exists \varepsilon > 0}_{\text{a fixed neighborhood}} \quad \underbrace{\forall N \in \mathbb{N}}_{\text{no tail suffices}} \quad \underbrace{\exists n \geq N}_{\text{terms escape infinitely often}} \quad \underbrace{x_n \notin V_\varepsilon(L)}_{\text{outside every candidate neighborhood}} .$$

The sequence fails to converge to L precisely when some neighborhood of L contains only finitely many terms: infinitely many terms lie permanently outside it.

Remark (Interpretation). The neighborhood formulation reframes convergence as a set-containment condition rather than an inequality condition. The assertion $x_n \in V_\varepsilon(L)$ is identical to $|x_n - L| < \varepsilon$; the two definitions are logically equivalent and the choice between them is one of language, not content. The neighborhood language is the natural bridge to the topological formulation: in a general topological space, open sets replace ε-neighborhoods and the definition reads identically with $V_\varepsilon(L)$ replaced by any open set U containing L. The phrase *all but finitely many terms* is the canonical prose compression of the quantifier block $\exists N \in \mathbb{N}\ \forall n \geq N$, and is standard in the literature.

Remark (Dependencies).

- Convergence of a Sequence
- ε-neighbourhood

Theorem (Equivalence of Convergence Formulations)

Theorem 6.3.3 (Equivalence of Convergence Formulations). *Let* $(x_n) \subseteq \mathbb{R}$ *and let* $L \in \mathbb{R}$. *The following statements are equivalent.*

(a) $x_n \to L$.

(b) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(|x_n - L| < \varepsilon\big)$.

(c) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(L - \varepsilon < x_n < L + \varepsilon\big)$.

(d) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(x_n \in V_\varepsilon(L)\big)$.

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}$$
$$\Big((a) \iff (b) \iff (c) \iff (d)\Big).$$

Remark (Theorem predicate reading).

$$\text{EquivalentConvergenceForms}(x_n, L) \ :=\ \text{ConvergentSequence}(x_n, L) \iff \text{NeighborhoodConvergentSeque}$$

Remark (Interpretation). The four conditions are the same geometric fact written in four notational registers. Condition (a) is the informal assertion. Condition (b) is the canonical ε-K definition: the absolute value formulation. Condition (c) makes the interval structure explicit by expanding $|x_n - L| < \varepsilon$ into its equivalent double inequality $L - \varepsilon < x_n < L + \varepsilon$; this is useful in proofs where upper and lower bounds are handled separately. Condition (d) is the neighborhood formulation: $x_n \in V_\varepsilon(L)$ is identical to conditions (b) and (c) but frames convergence as set membership, which is the natural language for the topological generalization. The index K plays the same role as N in the base definition Definition 6.3.1; the letter changes here to signal that K is the tail offset in the sense of Definition 6.4.1.

Remark (Dependencies).

- Convergence of a Sequence
- Convergence — Neighborhood Formulation

6.3.1 Limits

Theorem (Uniqueness of Limits)

Theorem 6.3.4 (Uniqueness of Limits). *Let $(x_n) \subseteq \mathbb{R}$. If $x_n \to L$ and $x_n \to K$, then $L = K$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \forall K \in \mathbb{R}$$
$$\Big(\big(x_n \to L \ \wedge\ x_n \to K \big) \implies L = K \Big).$$

Remark (Theorem predicate reading).

$$\mathrm{UniqueSequentialLimit}(x_n, L, K) \ :=\ \big(\mathrm{ConvergentSequence}(x_n, L) \wedge \mathrm{ConvergentSequence}(x_n, K)\big) \implies L =$$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq \mathbb{R}\ \exists L \in \mathbb{R}\ \exists K \in \mathbb{R}$$
$$\Big(\big(x_n \to L \ \wedge\ x_n \to K \big) \ \wedge\ L \neq K \Big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{UniqueSequentialLimit}(x_n, L, K) \ :=\ \mathrm{ConvergentSequence}(x_n, L) \wedge \mathrm{ConvergentSequence}(x_n, K) \wedge L \neq K.$$

Remark (Failure modes). The only failure mode is the existence of one real sequence with two distinct real limits. In $\mathbb{R}$ this is impossible because distinct real numbers can be separated by disjoint neighborhoods.

Remark (Failure mode decomposition).

$$\underbrace{\exists (x_n) \subseteq \mathbb{R}}_{\text{some sequence}} \ \underbrace{x_n \to L \ \wedge \ x_n \to K}_{\text{converges to two values}} \ \underbrace{L \neq K}_{\text{distinct limits}} .$$

The theorem asserts this configuration is impossible in $\mathbb{R}$. The failure mode is not merely exotic: it occurs in non-Hausdorff topological spaces where distinct points cannot be separated by disjoint open sets. The proof that it cannot occur in $\mathbb{R}$ is precisely an argument that $\mathbb{R}$ has enough room to separate any two distinct points by disjoint ε-neighborhoods.

Remark (Interpretation). The theorem licenses the notation $\lim\limits_{n\to\infty} x_n = L$: the definite article is justified precisely because the limit, when it exists, is unique. Without uniqueness the limit would be a relation rather than a function, and the notation would be ambiguous. The proof strategy is a separation argument: if $L \neq K$ then $|L - K| > 0$, and choosing $\varepsilon = |L - K|/2$ produces two disjoint neighborhoods that the tail of (x_n) cannot simultaneously occupy. The Hausdorff remark in the failure mode block is the forward pointer: when sequences are studied in general topological spaces, uniqueness of limits is equivalent to the Hausdorff separation axiom.

Remark (Dependencies).

- Convergence of a Sequence
- Triangle inequality

Theorem (Limit Preserves Eventual Order)

Theorem 6.3.5 (Limit Preserves Eventual Order)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. Suppose that $x_n \to L$ and $y_n \to M$. If there exists $N_0 \in \mathbb{N}$ such that $x_n \le y_n$ for every $n \ge N_0$, then $L \le M$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall (x_n), (y_n) \subseteq \mathbb{R}\ \forall L, M \in \mathbb{R} \\ &\quad \Big(x_n \to L \ \wedge \ y_n \to M \ \wedge \ \exists N_0 \in \mathbb{N}\ \forall n \ge N_0 \ (x_n \le y_n) \Big) \\ &\quad \implies \quad L \le M. \end{aligned}$$

Remark (Theorem predicate reading).

$$\big(\mathrm{ConvergentSequence}(x_n, L) \wedge \mathrm{ConvergentSequence}(y_n, M) \wedge \mathrm{Eventually}(x_n,\ x_n \le y_n)\big) \implies L \le M.$$

Remark (Interpretation). An inequality that holds on a tail survives passage to the limit, but only as a weak inequality. Strict inequalities on the terms may collapse at the limit, as in $0 < x_n$ with $x_n \to 0$.

Remark (Dependencies).

- Convergence of a Sequence
- Uniqueness of Limits

Theorem (Strict Limit Separation Gives Eventual Order)

Theorem 6.3.6 (Strict Limit Separation Gives Eventual Order)**.** *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$. If $A < B$, then there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n), (y_n) \subseteq \mathbb{R}\ \forall A, B \in \mathbb{R} \\
&\quad \Big(x_n \to A \ \wedge\ y_n \to B \ \wedge\ A < B \Big) \\
&\quad \implies \quad \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n < y_n).
\end{aligned}$$

Remark (Theorem predicate reading).

$$\big(\mathrm{ConvergentSequence}(x_n, A) \wedge \mathrm{ConvergentSequence}(y_n, B) \wedge A < B\big) \implies \mathrm{Eventually}(x_n,\ x_n < y_n).$$

Remark (Interpretation). If the limiting values have a positive gap, then sufficiently late terms inherit the same order. The proof uses the open interval between A and B: eventually (x_n) lies near A and (y_n) lies near B, with room left between the two tails.

Remark (Dependencies).

- Convergence of a Sequence
- Uniqueness of Limits

Theorem (Eventual Strict Comparison Preserves Weak Limit Order)

Theorem 6.3.7 (Eventual Strict Comparison Preserves Weak Limit Order)**.** *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$, then $A \leq B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n > y_n$ for every $n \geq N$, then $A \geq B$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n), (y_n) \subseteq \mathbb{R}\ \forall A, B \in \mathbb{R} \\
&\quad \Big(x_n \to A \ \wedge\ y_n \to B \ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n < y_n) \Big) \implies A \leq B, \\
&\quad \Big(x_n \to A \ \wedge\ y_n \to B \ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n > y_n) \Big) \implies A \geq B.
\end{aligned}$$

Remark (Theorem predicate reading).

$$\big(\text{ConvergentSequence}(x_n, A) \wedge \text{ConvergentSequence}(y_n, B) \wedge \text{Eventually}(x_n,\ x_n < y_n)\big) \implies A \leq B,$$
$$\big(\text{ConvergentSequence}(x_n, A) \wedge \text{ConvergentSequence}(y_n, B) \wedge \text{Eventually}(x_n,\ x_n > y_n)\big) \implies A \geq B.$$

Remark (Interpretation). Strict comparison of terms does not force strict comparison of limits. The strict inequality can collapse in the limit, so the conclusion is weak: for example $1/n > 0$ for every $n \in \mathbb{N}$, but both sides have limit 0.

Remark (Dependencies).

- Limit Preserves Eventual Order

Theorem (Constant Comparison for Sequence Limits)

Theorem 6.3.8 (Constant Comparison for Sequence Limits)**.** *Let (x_n) be a real sequence, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n \leq B$ for every $n \geq N$, then $A \leq B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n < B$ for every $n \geq N$, then $A \leq B$.

(c) If there exists $N \in \mathbb{N}$ such that $x_n \geq B$ for every $n \geq N$, then $A \geq B$.

(d) If there exists $N \in \mathbb{N}$ such that $x_n > B$ for every $n \geq N$, then $A \geq B$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R}\ \forall A, B \in \mathbb{R} \\
&\quad \Big(x_n \to A\ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \leq B)\Big) \implies A \leq B, \\
&\quad \Big(x_n \to A\ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n < B)\Big) \implies A \leq B, \\
&\quad \Big(x_n \to A\ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \geq B)\Big) \implies A \geq B, \\
&\quad \Big(x_n \to A\ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n > B)\Big) \implies A \geq B.
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n,\ x_n \leq B) \implies A \leq B, \\
&\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n,\ x_n < B) \implies A \leq B, \\
&\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n,\ x_n \geq B) \implies A \geq B, \\
&\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n,\ x_n > B) \implies A \geq B.
\end{aligned}$$

Remark (Interpretation). This is the constant version of the comparison theorem. A fixed real number B may be treated as the constant sequence with value B, so each conclusion is a direct application of eventual order preservation. Again, strict term inequalities produce only weak inequalities between limits.

Remark (Dependencies).

- Constant Sequence
- Limit Preserves Eventual Order
- Eventual Strict Comparison Preserves Weak Limit Order

Theorem (Constant Squeeze Theorem)

Theorem 6.3.9 (Constant Squeeze Theorem)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If there exists $N_0 \in \mathbb{N}$ such that $L \le x_n \le L$ for every $n \ge N_0$, then $x_n \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\\ &\quad \Big(\exists N_0 \in \mathbb{N}\ \forall n \ge N_0\ (L \le x_n \le L)\Big)\\ &\quad \implies \quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \ge N\ \big(|x_n - L| < \varepsilon\big).\end{aligned}$$

Remark (Theorem predicate reading).

$$\text{Eventually}(x_n,\ L \le x_n \le L) \implies \text{ConvergentSequence}(x_n, L).$$

Remark (Interpretation)*.* This is the degenerate squeeze theorem: the lower and upper barriers are the same constant sequence. It says that eventual equality to a real number is enough for convergence to that real number.

Remark (Dependencies).

- Convergence of a Sequence
- Constant Sequence

Theorem (Sequence Squeeze Theorem)

Theorem 6.3.10 (Sequence Squeeze Theorem)**.** *Let (a_n), (x_n), and (b_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $a_n \to L$ and $b_n \to L$. If there exists $N_0 \in \mathbb{N}$ such that $a_n \le x_n \le b_n$ for every $n \ge N_0$, then $x_n \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (a_n), (x_n), (b_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\\ &\quad \Big(a_n \to L\ \wedge\ b_n \to L\ \wedge\ \exists N_0 \in \mathbb{N}\ \forall n \ge N_0\ (a_n \le x_n \le b_n)\Big)\\ &\quad \implies \quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \ge N\ \big(|x_n - L| < \varepsilon\big).\end{aligned}$$

Remark (Theorem predicate reading).

$$\big(\mathrm{ConvergentSequence}(a_n, L) \wedge \mathrm{ConvergentSequence}(b_n, L) \wedge \mathrm{Eventually}(x_n,\ a_n \le x_n \le b_n)\big) \implies \mathrm{Convergent}$$

Remark (Interpretation). The middle sequence is trapped between two tails that approach the same number. Once both barriers lie inside an ε-neighborhood of L, the trapped sequence must also lie inside that neighborhood.

Remark (Dependencies).

- Convergence of a Sequence
- Limit Preserves Eventual Order

Theorem (Absolute-Value Squeeze Theorem)

Theorem 6.3.11 (Absolute-Value Squeeze Theorem)**.** *Let (x_n) and (u_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $u_n \to 0$. If there exists $N_0 \in \mathbb{N}$ such that $|x_n - L| \le u_n$ for every $n \ge N_0$, then $x_n \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall (x_n), (u_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R} \\ &\quad \Big(u_n \to 0\ \wedge\ \exists N_0 \in \mathbb{N}\ \forall n \ge N_0\ (|x_n - L| \le u_n)\Big) \\ &\quad \implies \quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \ge N\ \big(|x_n - L| < \varepsilon\big). \end{aligned}$$

Remark (Theorem predicate reading).

$$\big(\mathrm{ConvergentSequence}(u_n, 0) \wedge \mathrm{Eventually}(x_n,\ |x_n - L| \le u_n)\big) \implies \mathrm{ConvergentSequence}(x_n, L).$$

Remark (Interpretation). This is the estimate form of the squeeze theorem. Instead of constructing two explicit bounding sequences, it bounds the error $|x_n - L|$ by a null sequence.

Remark (Dependencies).

- Convergence of a Sequence
- Null Sequence
- Sequence Squeeze Theorem

6.3.2 Algebra of Limits

Algebra of Convergent Sequences

The algebra of convergent sequences records that limits are compatible with the ordinary algebraic operations on real numbers. The pointwise operations below turn real sequences into new real sequences, and the theorem blocks that follow state how convergence passes through those operations. The reciprocal and quotient laws require nonzero denominator hypotheses, while the square-root law requires nonnegative terms.

Definition (Pointwise Sum of Sequences)

Definition 6.3.12 (Pointwise Sum of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise sum** of (x_n) and (y_n) is the real sequence $(x_n + y_n)$ defined by

$$(x+y)_n := x_n + y_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement)*.*

Let (x_n) and (y_n) be real sequences.

$$\forall n \in \mathbb{N} \ \big((x+y)_n = x_n + y_n\big).$$

Remark (Interpretation)*.* The sum sequence is formed term by term. No limiting process is involved in the definition; convergence enters only when the behavior of the resulting sequence is studied.

Remark (Dependencies)*.*

- Real Sequence

Definition (Pointwise Difference of Sequences)

Definition 6.3.13 (Pointwise Difference of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise difference** of (x_n) and (y_n) is the real sequence $(x_n - y_n)$ defined by

$$(x-y)_n := x_n - y_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement)*.*

Let (x_n) and (y_n) be real sequences.

$$\forall n \in \mathbb{N} \ \big((x-y)_n = x_n - y_n\big).$$

Remark (Interpretation)*.* The difference sequence subtracts corresponding terms. It is equivalently the sum of (x_n) with the pointwise negation of (y_n).

Remark (Dependencies)*.*

- Real Sequence
- Pointwise Sum of Sequences

Definition (Scalar Multiple of a Sequence)

Definition 6.3.14 (Scalar Multiple of a Sequence)**.** Let (x_n) be a real sequence, and let $\alpha \in \mathbb{R}$. The **scalar multiple** $\alpha(x_n)$ is the real sequence (αx_n) defined by

$$(\alpha x)_n := \alpha x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and } \alpha \in \mathbb{R}.$$
$$\forall n \in \mathbb{N} \ \big((\alpha x)_n = \alpha x_n\big).$$

Remark (Interpretation). Scalar multiplication rescales every term by the same real number. It changes the vertical scale of the sequence but not its indexing.

Remark (Dependencies).

- Real Sequence

Definition (Linear Combination of Real Sequences)

Definition 6.3.15 (Linear Combination of Real Sequences)**.** Let (x_n) and (y_n) be real sequences, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha(x_n) + \beta(y_n)$ is the real sequence $(\alpha x_n + \beta y_n)$ defined by

$$(\alpha x + \beta y)_n := \alpha x_n + \beta y_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences and } \alpha, \beta \in \mathbb{R}.$$
$$\forall n \in \mathbb{N} \ \big((\alpha x + \beta y)_n = \alpha x_n + \beta y_n\big).$$

Remark (Interpretation). A linear combination of sequences is built term by term from scalar multiples and sums. It is the common algebraic shape behind sequence limit rules and Cauchy closure under addition and scalar multiplication.

Remark (Dependencies).

- Scalar Multiple of a Sequence
- Pointwise Sum of Sequences

Definition (Pointwise Negation of a Sequence)

Definition 6.3.16 (Pointwise Negation of a Sequence)**.** Let (x_n) be a real sequence. The **pointwise negation** of (x_n) is the real sequence $(-x_n)$ defined by

$$(-x)_n := -x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\forall n \in \mathbb{N} \ \big((-x)_n = -x_n\big).$$

Remark (Interpretation). Pointwise negation reflects the sequence across zero. It is the special scalar multiple corresponding to the scalar -1.

Remark (Dependencies).

- Scalar Multiple of a Sequence

Definition (Pointwise Product of Sequences)

Definition 6.3.17 (Pointwise Product of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise product** of (x_n) and (y_n) is the real sequence $(x_n y_n)$ defined by

$$(xy)_n := x_n y_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ and } (y_n) \text{ be real sequences.}$$
$$\forall n \in \mathbb{N} \, \big((xy)_n = x_n y_n\big).$$

Remark (Interpretation). The product sequence multiplies corresponding terms. The product law for limits is subtler than the sum law because one factor must be controlled while the other is approximated.

Remark (Dependencies).

- Real Sequence

Definition (Reciprocal Sequence)

Definition 6.3.18 (Reciprocal Sequence)**.** Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$. The **reciprocal sequence** of (x_n) is the real sequence $(1/x_n)$ defined by

$$\left(\frac{1}{x}\right)_n := \frac{1}{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N} \, (x_n \neq 0).$$
$$\forall n \in \mathbb{N} \left(\left(\frac{1}{x}\right)_n = \frac{1}{x_n}\right).$$

Remark (Interpretation). The reciprocal sequence is defined only when division by each term is legal. For limit theorems, a nonzero limit guarantees eventual nonzero behavior, but the sequence-level construction itself requires nonzero terms wherever it is defined.

Remark (Dependencies).

- Real Sequence

Definition (Pointwise Quotient of Sequences)

Definition 6.3.19 (Pointwise Quotient of Sequences)**.** Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$. The **pointwise quotient** of (x_n) by (y_n) is the real sequence (x_n/y_n) defined by

$$\left(\frac{x}{y}\right)_n := \frac{x_n}{y_n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n), (y_n) \text{ be real sequences with } \forall n \in \mathbb{N}\, (y_n \neq 0).\\ &\forall n \in \mathbb{N} \left(\left(\frac{x}{y}\right)_n = \frac{x_n}{y_n}\right).\end{aligned}$$

Remark (Interpretation). The quotient sequence is the product of (x_n) with the reciprocal of (y_n). Its convergence law is therefore a product law plus a reciprocal law.

Remark (Dependencies).

- Pointwise Product of Sequences
- Reciprocal Sequence

Definition (Square Sequence)

Definition 6.3.20 (Square Sequence)**.** Let (x_n) be a real sequence. The **square sequence** of (x_n) is the real sequence (x_n^2) defined by

$$(x^2)_n := x_n^2 \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\forall n \in \mathbb{N}\, \big((x^2)_n = x_n^2\big).\end{aligned}$$

Remark (Interpretation). The square sequence is the product of a sequence with itself. Its limit law is therefore a direct specialization of the product law.

Remark (Dependencies).

- Pointwise Product of Sequences

Definition (Absolute Value Sequence)

Definition 6.3.21 (Absolute Value Sequence)**.** Let (x_n) be a real sequence. The **absolute value sequence** of (x_n) is the real sequence $(|x_n|)$ defined by

$$(|x|)_n := |x_n| \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\forall n \in \mathbb{N} \; \big((|x|)_n = |x_n|\big).$$

Remark (Interpretation). The absolute value sequence records the magnitude of each term. It removes sign information while preserving distance from zero.

Remark (Dependencies).

- Real Sequence

Definition (Square Root Sequence)

Definition 6.3.22 (Square Root Sequence)**.** Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$. The **square root sequence** of (x_n) is the real sequence $(\sqrt{x_n})$ defined by

$$(\sqrt{x})_n := \sqrt{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N}\ (0 \leq x_n).$$
$$\forall n \in \mathbb{N} \; \big((\sqrt{x})_n = \sqrt{x_n}\big).$$

Remark (Interpretation). The square root sequence is defined term by term on the nonnegative real terms. The nonnegativity hypothesis is part of the construction, not a proof artifact.

Remark (Dependencies).

- Real Sequence

Theorem (Limit of a Scalar Multiple)

Theorem 6.3.23 (Limit of a Scalar Multiple)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If $x_n \to L$, then $\alpha x_n \to \alpha L$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L, \alpha \in \mathbb{R}.$$
$$\big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon)\big]$$
$$\Longrightarrow \qquad \big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|\alpha x_n - \alpha L| < \varepsilon)\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \Longrightarrow \text{ConvergentSequence}(\alpha x_n, \alpha L).$$

Remark (Interpretation). Multiplying every term by a fixed scalar multiplies the limiting value by the same scalar. The case $\alpha = 0$ collapses the sequence to the constant zero sequence.

Remark (Dependencies).

- Convergence of a Sequence
- Scalar Multiple of a Sequence

Theorem (Limit of a Sum)

Theorem 6.3.24 (Limit of a Sum). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n + y_n \to L + M$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}.$$
$$\big[x_n \to L \ \wedge \ y_n \to M\big] \implies \big[x_n + y_n \to L + M\big].$$

Remark (Predicate reading).

$$\big(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)\big) \implies \text{ConvergentSequence}(x_n + y_n, L + M).$$

Remark (Interpretation). The sum law says that convergent sequences may be added before or after taking limits. The proof splits a target tolerance into two smaller tolerances, one for each summand.

Remark (Dependencies).

- Convergence of a Sequence
- Pointwise Sum of Sequences
- Triangle Inequality

Theorem (Limit of a Negation)

Theorem 6.3.25 (Limit of a Negation). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $-x_n \to -L$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}.$$
$$\big[x_n \to L\big] \implies \big[-x_n \to -L\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(-x_n, -L).$$

Remark (Interpretation). Negation is scalar multiplication by -1, so this theorem is the scalar multiple law in its most common special case.

Remark (Dependencies).

- Limit of a Scalar Multiple
- Pointwise Negation of a Sequence

Theorem (Limit of a Difference)

Theorem 6.3.26 (Limit of a Difference)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n - y_n \to L - M$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{gathered}\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}. \\ \big[x_n \to L \ \wedge \ y_n \to M\big] \implies \big[x_n - y_n \to L - M\big].\end{gathered}$$

Remark (Predicate reading).

$$\big(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)\big) \implies \text{ConvergentSequence}(x_n - y_n, L - M).$$

Remark (Interpretation). The difference law follows by adding (x_n) to the negation of (y_n). It is therefore not a new analytic mechanism, but a useful named rule.

Remark (Dependencies).

- Limit of a Sum
- Limit of a Negation
- Pointwise Difference of Sequences

Theorem (Limit of a Product)

Theorem 6.3.27 (Limit of a Product)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n y_n \to LM$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{gathered}\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}. \\ \big[x_n \to L \ \wedge \ y_n \to M\big] \implies \big[x_n y_n \to LM\big].\end{gathered}$$

Remark (Predicate reading).

$$\big(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)\big) \Longrightarrow \text{ConvergentSequence}(x_n y_n, LM). \blacksquare$$

Remark (Interpretation). The product law uses both approximation and boundedness. A convergent sequence is eventually bounded, so one factor can be controlled while the other is made small.

Remark (Dependencies).

- Convergence of a Sequence
- Pointwise Product of Sequences
- Bounded Sequence

Theorem (Nonzero Limit is Eventually Nonzero)

Theorem 6.3.28 (Nonzero Limit is Eventually Nonzero)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $x_n \neq 0$ for every $n \geq N$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}.$$
$$\big[L \neq 0 \ \wedge \ x_n \to L\big] \Longrightarrow \exists N \in \mathbb{N}\, \forall n \geq N\ (x_n \neq 0).$$

Remark (Interpretation). A sequence converging to a nonzero number eventually stays away from zero. This is the denominator-control result needed for reciprocal and quotient rules.

Remark (Dependencies).

- Convergence of a Sequence

Theorem (Limit of a Reciprocal)

Theorem 6.3.29 (Limit of a Reciprocal)**.** *Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then $1/x_n \to 1/L$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N}\ (x_n \neq 0), \text{ and let } L \neq 0.$$
$$\big[x_n \to L\big] \Longrightarrow \big[\frac{1}{x_n} \to \frac{1}{L}\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \Longrightarrow \text{ConvergentSequence}\left(\frac{1}{x_n}, \frac{1}{L}\right).$$

Remark (Interpretation). Reciprocation is continuous away from zero. The nonzero limit supplies a tail on which the denominators stay uniformly separated from zero.

Remark (Dependencies).

- Reciprocal Sequence
- Nonzero Limit is Eventually Nonzero

Theorem (Limit of a Quotient)

Theorem 6.3.30 (Limit of a Quotient)**.** *Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$, and let $L, M \in \mathbb{R}$ with $M \neq 0$. If $x_n \to L$ and $y_n \to M$, then $x_n/y_n \to L/M$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences with } \forall n \in \mathbb{N}\ (y_n \neq 0), \text{ and let } M \neq 0.$$
$$\big[x_n \to L \ \wedge \ y_n \to M\big] \implies \big[\frac{x_n}{y_n} \to \frac{L}{M}\big].$$

Remark (Predicate reading).

$$\big(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)\big) \implies \text{ConvergentSequence}\left(\frac{x_n}{y_n}, \frac{L}{M}\right).$$

Remark (Interpretation). The quotient law is the product law applied to (x_n) and the reciprocal of (y_n). The condition $M \neq 0$ prevents the limiting denominator from collapsing.

Remark (Dependencies).

- Pointwise Quotient of Sequences
- Limit of a Product
- Limit of a Reciprocal

Theorem (Limit of a Square)

Theorem 6.3.31 (Limit of a Square)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $x_n^2 \to L^2$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}.$$
$$\big[x_n \to L\big] \implies \big[x_n^2 \to L^2\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(x_n^2, L^2).$$

Remark (Interpretation). Squaring preserves limits because it is multiplication of a sequence by itself.

Remark (Dependencies).

- Square Sequence
- Limit of a Product

Theorem (Limit of an Absolute Value)

Theorem 6.3.32 (Limit of an Absolute Value)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $|x_n| \to |L|$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}.\\ &\big[x_n \to L\big] \implies \big[|x_n| \to |L|\big].\end{aligned}$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(|x_n|, |L|).$$

Remark (Interpretation). Taking absolute values cannot increase distance by more than the original distance: the magnitude function is continuous on $\mathbb{R}$.

Remark (Dependencies).

- Absolute Value Sequence
- Triangle Inequality

Theorem (Positive Limit is Eventually Positive)

Theorem 6.3.33 (Positive Limit is Eventually Positive)**.** *Let (x_n) be a real sequence, and let $L > 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $0 < x_n$ for every $n \geq N$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence and let } L > 0.\\ &\big[x_n \to L\big] \implies \exists N \in \mathbb{N}\ \forall n \geq N\ (0 < x_n).\end{aligned}$$

Remark (Interpretation). Convergence to a positive limit eventually traps the sequence inside the positive half-line. This result is the sign-control counterpart of the eventual nonzero theorem.

Remark (Dependencies).

- Convergence of a Sequence

Theorem (Limit of a Square Root)

Theorem 6.3.34 (Limit of a Square Root)**.** *Let (x_n) be a real sequence such that $0 \le x_n$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$. If $x_n \to L$, then $0 \le L$ and $\sqrt{x_n} \to \sqrt{L}$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N}\ (0 \le x_n), \text{ and let } L \in \mathbb{R}.$$
$$\big[x_n \to L\big] \implies \big[0 \le L \ \wedge\ \sqrt{x_n} \to \sqrt{L}\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(\sqrt{x_n}, \sqrt{L}).$$

Remark (Interpretation). The square-root function is continuous on the nonnegative real line. Near zero the proof uses the order relation; away from zero it can be rationalized by multiplying by the conjugate.

Remark (Dependencies).

- Square Root Sequence
- Convergence of a Sequence

Theorem (Polynomial Sequence Limit)

Theorem 6.3.35 (Polynomial Sequence Limit)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p \in \mathbb{R}[t]$ be a polynomial. If $x_n \to L$, then $p(x_n) \to p(L)$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence, } L \in \mathbb{R}, \text{ and } p \in \mathbb{R}[t].$$
$$\big[x_n \to L\big] \implies \big[p(x_n) \to p(L)\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(p(x_n), p(L)).$$

Remark (Interpretation). Polynomial limits are built from finitely many additions, scalar multiples, and products. The theorem packages those algebra laws into one reusable composition principle.

Remark (Dependencies).

- Limit of a Sum
- Limit of a Scalar Multiple
- Limit of a Product
- Limit of a Square

Theorem (Rational Sequence Limit)

Theorem 6.3.36 (Rational Sequence Limit)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p, q \in \mathbb{R}[t]$ be polynomials. Suppose $q(L) \neq 0$ and $q(x_n) \neq 0$ for every $n \in \mathbb{N}$. If $x_n \to L$, then*

$$\frac{p(x_n)}{q(x_n)} \to \frac{p(L)}{q(L)}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } (x_n) \text{ be a real sequence, } L \in \mathbb{R}, \text{ and } p, q \in \mathbb{R}[t].\\
&\big[q(L) \neq 0 \wedge \forall n \in \mathbb{N}\ (q(x_n) \neq 0) \wedge x_n \to L\big]\\
&\qquad \Longrightarrow \qquad \left[\frac{p(x_n)}{q(x_n)} \to \frac{p(L)}{q(L)}\right].
\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \Longrightarrow \mathrm{ConvergentSequence}\left(\frac{p(x_n)}{q(x_n)}, \frac{p(L)}{q(L)}\right).$$

Remark (Interpretation). A rational expression is a quotient of two polynomial expressions. The condition $q(L) \neq 0$ keeps the limiting denominator legitimate, and the pointwise condition $q(x_n) \neq 0$ keeps every term of the quotient sequence defined.

Remark (Dependencies).

- Polynomial Sequence Limit
- Limit of a Quotient
- Nonzero Limit is Eventually Nonzero

6.4 Tails

Definition (M-Tail of a Sequence)

Definition 6.4.1 (M-Tail of a Sequence)**.** Let $(x_n) \subseteq \mathbb{R}$ and let $M \in \mathbb{N}$. The M**-tail** of (x_n) is the sequence

$$(x_n)_{n \geq M} \;:=\; \big(x_M,\, x_{M+1},\, x_{M+2},\, \ldots\big).$$

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall M \in \mathbb{N}\ \Big((x_n)_{n \geq M} = (x_M, x_{M+1}, x_{M+2}, \ldots) \Big).$$

Remark (Definition predicate reading).

$$\text{TailSequence}(x_n, M, y_k)\ :=\ \forall k \in \mathbb{N}\ \big(y_k = x_{M+k-1} \big).$$

Remark (Interpretation). The M-tail discards the first $M-1$ terms and retains everything from index M onward. It is the formal object underlying the phrase *all but finitely many terms*: to say that all but finitely many terms of (x_n) satisfy a property P is precisely to say that some M-tail of (x_n) satisfies P termwise. The index M is an offset into the sequence, not a bound on the terms themselves; two sequences with identical M-tails share all asymptotic properties, differing only in finitely many initial values which convergence arguments discard entirely.

Remark (Dependencies).

- Real Sequence

Theorem (Convergence of a Tail)

Theorem 6.4.2 (Convergence of a Tail)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $m \in \mathbb{N}$. The m-tail $(x_{m+n})_{n\in\mathbb{N}}$ converges if and only if (x_n) converges. Moreover, in this case,*

$$\lim_{n\to\infty} x_{m+n}\ =\ \lim_{n\to\infty} x_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall (x_n) \subseteq \mathbb{R}\ \forall m \in \mathbb{N} \\ &\quad \Big(\big(x_{m+n}\big)_{n\in\mathbb{N}} \text{ converges} \iff (x_n) \text{ converges} \Big) \\ &\quad \wedge\ \Big((x_n) \text{ converges} \implies \lim_{n\to\infty} x_{m+n} = \lim_{n\to\infty} x_n \Big). \end{aligned}$$

Remark (Theorem predicate reading).

$$\text{TailConvergenceInvariant}(x_n, m)\ :=\ \Big(\exists L \in \mathbb{R}\ \text{ConvergentSequence}(x_n, L)\Big) \iff \Big(\exists L \in \mathbb{R}\ \text{ConvergentSeq}$$

Remark (Contrapositive quantified statement).

$$\begin{aligned} &\forall (x_n) \subseteq \mathbb{R}\ \forall m \in \mathbb{N} \\ &\quad \Big(\big(x_{m+n}\big)_{n\in\mathbb{N}} \text{ diverges} \iff (x_n) \text{ diverges} \Big). \end{aligned}$$

Remark (Contrapositive predicate reading).

$$\neg\,\text{TailConvergenceInvariant}(x_n, m)\ :=\ \big((x_n) \text{ diverges} \iff (x_{m+n})_{n\in\mathbb{N}} \text{ diverges} \big).$$

Remark (Interpretation). Convergence is a *tail property*: it depends only on the eventual behavior of the sequence, not on any finite initial segment. Discarding the first m terms neither creates nor destroys convergence, and when the limit exists it is preserved exactly. This theorem formalizes the informal principle that finitely many terms are irrelevant to asymptotic behavior. It is the rigorous justification for the common proof move of assuming without loss of generality that n is large enough for some auxiliary condition to hold: one simply passes to a suitable m-tail, applies the argument there, and concludes for the full sequence by this theorem. The contrapositive is equally useful: to show (x_n) diverges it suffices to show that some m-tail diverges.

Remark (Dependencies).

- Convergence of a Sequence
- M-Tail of a Sequence
- Uniqueness of Limits

Theorem (Convergence by Domination)

Theorem 6.4.3 (Convergence by Domination)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. Let (a_n) be a sequence of positive real numbers with $\lim_{n\to\infty} a_n = 0$. If there exist a constant $c > 0$ and an index $m \in \mathbb{N}$ such that*

$$|x_n - L| \ \le \ c\, a_n \qquad \forall n \ge m,$$

then $\lim_{n\to\infty} x_n = L$.
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \forall (a_n) \subseteq (0,\infty)$$
$$\Big(\Big(\lim_{n\to\infty} a_n = 0 \ \wedge\ \exists c > 0\ \exists m \in \mathbb{N}\ \forall n \ge m\ \big(|x_n - L| \le c\, a_n\big)\Big) \implies \lim_{n\to\infty} x_n = L\Big).$$

Remark (Contrapositive quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \forall (a_n) \subseteq (0,\infty)$$
$$\Big(\lim_{n\to\infty} x_n \ne L \implies \lim_{n\to\infty} a_n \ne 0\ \vee\ \forall c > 0\ \forall m \in \mathbb{N}\ \exists n \ge m\ \big(|x_n - L| > c\, a_n\big)\Big).$$

Remark (Contrapositive predicate reading).

$$\mathrm{NotDominatedByNullError}(x_n, L, a_n) \ :=\ \lim_{n\to\infty} a_n \ne 0 \vee \forall c > 0\, \forall m \in \mathbb{N}\, \exists n \ge m \big(|x_n - L| > c\, a_n\big).$$

Remark (Interpretation). The theorem is a *domination principle*: it is not necessary to verify the ε-N condition directly for (x_n). It suffices to find a null sequence (a_n) – one converging to zero – that dominates the error $|x_n - L|$ on some tail, up to a fixed scalar c. The scalar c

is irrelevant to the conclusion: since $a_n \to 0$, the product $c\,a_n \to 0$ regardless of the size of c, and any $\varepsilon > 0$ is eventually beaten by $c\,a_n$. This theorem is the rigorous engine behind the common proof pattern of bounding $|x_n - L|$ by a simpler expression and observing that the simpler expression tends to zero. The canonical instance is $a_n = 1/n$, where domination by c/n is sufficient to conclude convergence to L.

Remark (Dependencies).

- Convergence of a Sequence
- M-Tail of a Sequence

Theorem (Ratio Limit Less Than One Implies Null)

Theorem 6.4.4 (Ratio Limit Less Than One Implies Null)**.** *Let (x_n) be a sequence of positive real numbers. Suppose that the limit*

$$L := \lim_{n\to\infty} \frac{x_{n+1}}{x_n}$$

exists. If $L < 1$, then (x_n) converges and

$$\lim_{n\to\infty} x_n = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq (0,\infty)\ \forall L \in \mathbb{R}$$
$$\Big(\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = L \ \wedge\ L < 1\Big) \implies \lim_{n\to\infty} x_n = 0.$$

Remark (Theorem predicate reading).

$$\big(\text{ConvergentSequence}\big(x_{n+1}/x_n, L\big) \wedge L < 1\big) \implies \text{NullSequence}(x_n).$$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq (0,\infty)\ \exists L \in \mathbb{R}$$
$$\Big(\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = L \ \wedge\ L < 1 \ \wedge\ \lim_{n\to\infty} x_n \neq 0\Big).$$

Remark (Negation predicate reading).

$$\text{ConvergentSequence}\big(x_{n+1}/x_n, L\big) \wedge L < 1 \wedge \text{NotConvergentTo}(x_n, 0).$$

Remark (Failure modes). A failure would be a positive sequence whose consecutive ratios settle below one in the limit but whose terms do not tend to zero. The theorem says this cannot happen: once the ratio limit is less than one, some tail has ratios bounded by a fixed number $r < 1$, forcing geometric decay.

Remark (Failure mode decomposition).

$$\underbrace{(x_n) \subseteq (0,\infty)}_{\text{positive terms}} \quad \underbrace{\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = L < 1}_{\text{eventual ratio gap}} \quad \underbrace{\lim_{n\to\infty} x_n \neq 0}_{\text{fails to be null}}.$$

The positive-term hypothesis makes each ratio meaningful. The strict gap $L < 1$ gives a tail on which $x_{n+1} \leq r x_n$ for some $r < 1$, and repeated iteration then gives domination by a null geometric sequence.

Remark (Interpretation). This is the sequence form of the ratio test. The theorem does not merely say that the terms decrease eventually; it says they decrease at an eventually geometric rate. The strict inequality $L < 1$ is essential. When the ratio limit equals 1, the conclusion may hold, as with $1/n$, or fail, as with the constant sequence 1.

Remark (Dependencies).

- Convergence of a Sequence
- Null Sequence
- M-Tail of a Sequence
- Convergence by Domination

6.5 Monotonicity

Monotonicity of Sequences

Monotonicity is an order condition on successive terms of a sequence. It is independent of convergence by itself, but when combined with boundedness it becomes one of the fundamental convergence mechanisms in real analysis.

Definition (Increasing Sequence)

Definition 6.5.1 (Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **increasing** if

$$x_n \leq x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is increasing} \iff \forall n \in \mathbb{N}\ (x_n \leq x_{n+1}).$$

Remark (Definition predicate reading).

$$\text{MonotoneIncreasingSequence}(x_n) :\equiv \forall n \in \mathbb{N}\ (x_n \leq x_{n+1}).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not increasing} \iff \exists n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Negation predicate reading).

$$\neg \, \text{MonotoneIncreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Interpretation). An increasing sequence may stay flat. The term "increasing" is therefore used here in the non-strict sense: each term is at least as large as the one before it.

Remark (Dependencies).

- Real Sequence

Definition (Decreasing Sequence)

Definition 6.5.2 (Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **decreasing** if

$$x_{n+1} \le x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is decreasing} \iff \forall n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Definition predicate reading).

$$\text{MonotoneDecreasingSequence}(x_n) \coloneqq \forall n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not decreasing} \iff \exists n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Negation predicate reading).

$$\neg \, \text{MonotoneDecreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Interpretation). A decreasing sequence may stay flat. The order condition says the sequence never rises as the index advances.

Remark (Dependencies).

- Real Sequence

Definition (Strictly Increasing Sequence)

Definition 6.5.3 (Strictly Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **strictly increasing** if

$$x_n < x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is strictly increasing} \iff \forall n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Definition predicate reading).

$$\text{StrictlyIncreasingSequence}(x_n) \coloneqq \forall n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not strictly increasing} \iff \exists n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Negation predicate reading).

$$\neg\,\text{StrictlyIncreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Interpretation). Strict increase excludes equality between consecutive terms. Every strictly increasing sequence is increasing, but the converse need not hold.

Remark (Dependencies).

- Real Sequence
- Increasing Sequence

Definition (Strictly Decreasing Sequence)

Definition 6.5.4 (Strictly Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **strictly decreasing** if

$$x_{n+1} < x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is strictly decreasing} \iff \forall n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Definition predicate reading).

$$\text{StrictlyDecreasingSequence}(x_n) \coloneqq \forall n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not strictly decreasing} \iff \exists n \in \mathbb{N}\ (x_n \le x_{n+1}).$$

Remark (Negation predicate reading).

$$\neg\,\text{StrictlyDecreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_n \le x_{n+1}).$$

Remark (Interpretation). Strict decrease excludes equality between consecutive terms. Every strictly decreasing sequence is decreasing, but the converse need not hold.

Remark (Dependencies).

- Real Sequence

- Decreasing Sequence

Definition (Monotone Sequence)

Definition 6.5.5 (Monotone Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **monotone** if it is increasing or decreasing.

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is monotone}\\ &\quad\iff \quad \big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \lor \big[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\big].\end{aligned}$$

Remark (Definition predicate reading).

$$\text{MonotoneSequence}(x_n) :\!= \text{MonotoneIncreasingSequence}(x_n) \lor \text{MonotoneDecreasingSequence}(x_n).$$

Remark (Negated quantified statement).

$$\begin{aligned}&(x_n) \text{ is not monotone}\\ &\quad\iff \quad \big[\exists n \in \mathbb{N}\ (x_{n+1} < x_n)\big] \land \big[\exists m \in \mathbb{N}\ (x_m < x_{m+1})\big].\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{MonotoneSequence}(x_n) \iff \neg\,\text{MonotoneIncreasingSequence}(x_n) \land \neg\,\text{MonotoneDecreasingSequence}(x_n).$$

Remark (Interpretation). Monotonicity means the sequence has one global direction. It may move upward, move downward, or stay constant, but it cannot reverse direction.

Remark (Dependencies).

- Increasing Sequence
- Decreasing Sequence

Definition (Eventually Increasing Sequence)

Definition 6.5.6 (Eventually Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually increasing** if there exists $N \in \mathbb{N}$ such that

$$x_n \le x_{n+1} \qquad \text{for every } n \ge N.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is eventually increasing} \iff \exists N \in \mathbb{N}\ \forall n \ge N\ (x_n \le x_{n+1}).$$

Remark (Definition predicate reading).

$$\text{EventuallyIncreasingSequence}(x_n) :\!= \exists N \in \mathbb{N}\ \forall n \ge N\ (x_n \le x_{n+1}).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not eventually increasing} \iff \forall N \in \mathbb{N}\ \exists n \geq N\ (x_{n+1} < x_n).$$

Remark (Negation predicate reading).

$$\neg\,\text{EventuallyIncreasingSequence}(x_n) \iff \forall N \in \mathbb{N}\ \exists n \geq N\ (x_{n+1} < x_n).$$

Remark (Interpretation). Eventual increase allows finitely many early reversals. Only the tail of the sequence must move in a nondecreasing direction.

Remark (Dependencies).

- M-Tail of a Sequence
- Increasing Sequence

Definition (Eventually Decreasing Sequence)

Definition 6.5.7 (Eventually Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually decreasing** if there exists $N \in \mathbb{N}$ such that

$$x_{n+1} \leq x_n \qquad \text{for every } n \geq N.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is eventually decreasing} \iff \exists N \in \mathbb{N}\ \forall n \geq N\ (x_{n+1} \leq x_n).$$

Remark (Definition predicate reading).

$$\text{EventuallyDecreasingSequence}(x_n) :\equiv \exists N \in \mathbb{N}\ \forall n \geq N\ (x_{n+1} \leq x_n).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not eventually decreasing} \iff \forall N \in \mathbb{N}\ \exists n \geq N\ (x_n < x_{n+1}).$$

Remark (Negation predicate reading).

$$\neg\,\text{EventuallyDecreasingSequence}(x_n) \iff \forall N \in \mathbb{N}\ \exists n \geq N\ (x_n < x_{n+1}).$$

Remark (Interpretation). Eventual decrease allows the initial terms to behave irregularly. The order condition begins only after some finite threshold.

Remark (Dependencies).

- M-Tail of a Sequence
- Decreasing Sequence

Definition (Eventually Monotone Sequence)

Definition 6.5.8 (Eventually Monotone Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually monotone** if it is eventually increasing or eventually decreasing.

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is eventually monotone}\\ &\quad\iff \quad \big[\exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \leq x_{n+1})\big] \vee \big[\exists N \in \mathbb{N}\ \forall n \geq N\ (x_{n+1} \leq x_n)\big].\end{aligned}$$

Remark (Definition predicate reading).

$$\text{EventuallyMonotoneSequence}(x_n) :\!= \text{EventuallyIncreasingSequence}(x_n) \vee \text{EventuallyDecreasingSequence}(x$$

Remark (Negated quantified statement).

$$\begin{aligned}&(x_n) \text{ is not eventually monotone}\\ &\quad\iff \quad \big[\forall N \in \mathbb{N}\ \exists n \geq N\ (x_{n+1} < x_n)\big] \wedge \big[\forall N \in \mathbb{N}\ \exists n \geq N\ (x_n < x_{n+1})\big].\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{EventuallyMonotoneSequence}(x_n) \iff \neg\,\text{EventuallyIncreasingSequence}(x_n) \wedge \neg\,\text{EventuallyDecreasingSe}$$

Remark (Interpretation). Eventual monotonicity is the tail version of monotonicity. It is often the right hypothesis in applications, because finitely many early terms do not affect convergence.

Remark (Dependencies).

- Eventually Increasing Sequence
- Eventually Decreasing Sequence
- Convergence of a Tail

Theorem (Monotone Convergence Theorem)

Theorem 6.5.9 (Monotone Convergence Theorem)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and bounded above, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

(b) If (x_n) is decreasing and bounded below, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \wedge \big[\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\big] \Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ \Big(\lim_{n\to\infty} x_n = L \wedge L = \sup\{x_n : n \in \mathbb{N}\} \Big), \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\big] \wedge \big[\exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\big] \Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ \Big(\lim_{n\to\infty} x_n = L \wedge L = \inf\{x_n : n \in \mathbb{N}\} \Big).
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\text{MonotoneIncreasingSequence}(x_n) \wedge \text{BoundedAboveSequence}(x_n) \implies \text{ConvergentSequence}\big(x_n, \sup\{x_n : n \\
&\text{MonotoneDecreasingSequence}(x_n) \wedge \text{BoundedBelowSequence}(x_n) \implies \text{ConvergentSequence}\big(x_n, \inf\{x_n : n
\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\exists (x_n) \subseteq \mathbb{R}\ \Big(\big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \wedge \big[\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\big] \\
&\qquad\qquad \wedge (x_n) \text{ does not converge} \Big), \\
&\exists (x_n) \subseteq \mathbb{R}\ \Big(\big[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\big] \wedge \big[\exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\big] \\
&\qquad\qquad \wedge (x_n) \text{ does not converge} \Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned}
&\text{MonotoneIncreasingSequence}(x_n) \wedge \text{BoundedAboveSequence}(x_n) \wedge \neg\exists L \in \mathbb{R}\ \ \text{ConvergentSequence}(x_n, L), \\
&\text{MonotoneDecreasingSequence}(x_n) \wedge \text{BoundedBelowSequence}(x_n) \wedge \neg\exists L \in \mathbb{R}\ \ \text{ConvergentSequence}(x_n, L).
\end{aligned}$$

Remark (Failure modes). The theorem would fail only if a one-directional bounded sequence could avoid settling to a real limit. Completeness of $\mathbb{R}$ rules this out: an increasing bounded-above sequence is forced toward the least upper bound of its range, while a decreasing bounded-below sequence is forced toward the greatest lower bound of its range.

Remark (Failure mode decomposition).

$$\underbrace{\forall n \in \mathbb{N}\ (x_n \le x_{n+1})}_{\text{one direction}} \quad \underbrace{\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)}_{\text{ceiling}} \quad \underbrace{(x_n) \text{ does not converge}}_{\text{forbidden by completeness}}.$$

The decreasing case is dual: reverse the inequalities and replace the ceiling by a floor.

Remark (Interpretation). Boundedness prevents escape, and monotonicity prevents oscillation. Together they force convergence. The theorem is one of the standard forms in which the completeness of the real numbers becomes a sequence theorem.

Remark (Dependencies).

- Convergence of a Sequence
- Increasing Sequence
- Decreasing Sequence
- Sequence Bounded Above
- Sequence Bounded Below
- Supremum
- Infimum

Theorem (Strict Monotonicity Implies Monotonicity)

Theorem 6.5.10 (Strict Monotonicity Implies Monotonicity)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is strictly increasing, then (x_n) is increasing.

(b) If (x_n) is strictly decreasing, then (x_n) is decreasing.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_n < x_{n+1})\big] \implies \big[\forall n \in \mathbb{N}\ (x_n \leq x_{n+1})\big] \Big), \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_{n+1} < x_n)\big] \implies \big[\forall n \in \mathbb{N}\ (x_{n+1} \leq x_n)\big] \Big).
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\text{StrictlyIncreasingSequence}(x_n) \implies \text{MonotoneIncreasingSequence}(x_n), \\
&\text{StrictlyDecreasingSequence}(x_n) \implies \text{MonotoneDecreasingSequence}(x_n).
\end{aligned}$$

Remark (Interpretation). Strict monotonicity is stronger than non-strict monotonicity because every strict inequality is also a weak inequality.

Remark (Dependencies).

- Strictly Increasing Sequence
- Strictly Decreasing Sequence
- Increasing Sequence
- Decreasing Sequence

Theorem (Bounded Monotone Sequence Equivalences)

Theorem 6.5.11 (Bounded Monotone Sequence Equivalences)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing, then (x_n) converges if and only if (x_n) is bounded above.

(b) If (x_n) is decreasing, then (x_n) converges if and only if (x_n) is bounded below.

(c) If (x_n) is monotone, then (x_n) converges if and only if (x_n) is bounded.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\\ &\quad \big[\forall n \in \mathbb{N}\ (x_n \leq x_{n+1})\big] \implies \Big(\exists L \in \mathbb{R}\ (x_n \to L) \iff \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \leq M)\Big),\\ &\quad \big[\forall n \in \mathbb{N}\ (x_{n+1} \leq x_n)\big] \implies \Big(\exists L \in \mathbb{R}\ (x_n \to L) \iff \exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \leq x_n)\Big).\end{aligned}$$

Remark (Theorem predicate reading).

$$\mathrm{MonotoneIncreasingSequence}(x_n) \implies \Big(\exists L \in \mathbb{R}\ \ \mathrm{ConvergentSequence}(x_n, L) \iff \mathrm{BoundedAboveSequence}$$

$$\mathrm{MonotoneDecreasingSequence}(x_n) \implies \Big(\exists L \in \mathbb{R}\ \ \mathrm{ConvergentSequence}(x_n, L) \iff \mathrm{BoundedBelowSequence}$$

Remark (Interpretation). For monotone sequences, the only obstruction to real convergence is escape in the monotone direction. Boundedness removes that obstruction.

Remark (Dependencies).

- Monotone Convergence Theorem
- Bounded Sequence
- Convergence of a Sequence

Theorem (Eventually Monotone Convergence Theorem)

Theorem 6.5.12 (Eventually Monotone Convergence Theorem)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is eventually increasing and bounded above, then (x_n) converges.

(b) If (x_n) is eventually decreasing and bounded below, then (x_n) converges.

(c) If (x_n) is eventually monotone and bounded, then (x_n) converges.

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \Big(\exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \leq x_{n+1}) \wedge \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \leq M)\Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ (x_n \to L), \\
&\quad \Big(\exists N \in \mathbb{N}\ \forall n \geq N\ (x_{n+1} \leq x_n) \wedge \exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \leq x_n)\Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ (x_n \to L).
\end{aligned}
$$

Remark (Theorem predicate reading).

$$
\begin{aligned}
&\mathrm{EventuallyIncreasingSequence}(x_n) \wedge \mathrm{BoundedAboveSequence}(x_n) \implies \exists L \in \mathbb{R}\ \mathrm{ConvergentSequence}(x_n, L \\
&\mathrm{EventuallyDecreasingSequence}(x_n) \wedge \mathrm{BoundedBelowSequence}(x_n) \implies \exists L \in \mathbb{R}\ \mathrm{ConvergentSequence}(x_n, L
\end{aligned}
$$

Remark (Interpretation). This is the tail version of the Monotone Convergence Theorem. Once the tail is monotone and bounded in the right direction, finite initial behavior is irrelevant.

Remark (Dependencies).

- Eventually Increasing Sequence
- Eventually Decreasing Sequence
- Convergence of a Tail
- Monotone Convergence Theorem

Theorem (Unbounded Monotone Divergence)

Theorem 6.5.13 (Unbounded Monotone Divergence)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and is not bounded above, then $x_n \to +\infty$.

(b) If (x_n) is decreasing and is not bounded below, then $x_n \to -\infty$.

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_n \leq x_{n+1})\big] \wedge \big[\forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ (M < x_n)\big]\Big) \\
&\quad \implies \quad \forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ (M < x_n), \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_{n+1} \leq x_n)\big] \wedge \big[\forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x_n < M)\big]\Big) \\
&\quad \implies \quad \forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n < M).
\end{aligned}
$$

Remark (Theorem predicate reading).

$$\text{MonotoneIncreasingSequence}(x_n) \wedge \neg\,\text{BoundedAboveSequence}(x_n) \implies \text{DivergesToPositiveInfinity}(x_n),$$
$$\text{MonotoneDecreasingSequence}(x_n) \wedge \neg\,\text{BoundedBelowSequence}(x_n) \implies \text{DivergesToNegativeInfinity}(x_n).$$

Remark (Interpretation). For a monotone sequence, unboundedness has a direction. An increasing sequence that has no upper bound eventually exceeds every real number; the decreasing case is the order-dual statement.

Remark (Dependencies).

- Divergence to Positive Infinity
- Divergence to Negative Infinity
- Increasing Sequence
- Decreasing Sequence
- Sequence Bounded Above
- Sequence Bounded Below

Theorem (Algebraic Transformations Preserve Monotonicity)

Theorem 6.5.14 (Algebraic Transformations Preserve Monotonicity)**.** *Let (x_n) be a real sequence and let $c, \alpha \in \mathbb{R}$.*

(a) If (x_n) is increasing, then $(x_n + c)$ is increasing.

(b) If (x_n) is decreasing, then $(x_n + c)$ is decreasing.

(c) If (x_n) is increasing and $\alpha > 0$, then (αx_n) is increasing.

(d) If (x_n) is increasing and $\alpha < 0$, then (αx_n) is decreasing.

(e) If (x_n) is increasing, then $(-x_n)$ is decreasing.

(f) If (x_n) is decreasing, then $(-x_n)$ is increasing.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R}\ \forall c, \alpha \in \mathbb{R} \\
&\quad \big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \implies \big[\forall n \in \mathbb{N}\ (x_n + c \le x_{n+1} + c)\big], \\
&\quad \big(\alpha > 0 \wedge \forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big) \implies \big[\forall n \in \mathbb{N}\ (\alpha x_n \le \alpha x_{n+1})\big], \\
&\quad \big(\alpha < 0 \wedge \forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big) \implies \big[\forall n \in \mathbb{N}\ (\alpha x_{n+1} \le \alpha x_n)\big].
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\text{MonotoneIncreasingSequence}(x_n) \Longrightarrow \text{MonotoneIncreasingSequence}(x_n + c),\\
&\text{MonotoneIncreasingSequence}(x_n) \wedge \alpha > 0 \Longrightarrow \text{MonotoneIncreasingSequence}(\alpha x_n),\\
&\text{MonotoneIncreasingSequence}(x_n) \wedge \alpha < 0 \Longrightarrow \text{MonotoneDecreasingSequence}(\alpha x_n).
\end{aligned}$$

Remark (Interpretation). Translation preserves order, multiplication by a positive scalar preserves order, and multiplication by a negative scalar reverses order. Negation is the special case $\alpha = -1$.

Remark (Dependencies).

- Pointwise Sum of Sequences
- Scalar Multiple of a Sequence
- Pointwise Negation of a Sequence
- Increasing Sequence
- Decreasing Sequence

6.6 Subsequences

Subsequences

Subsequences record what remains after passing to an infinite ordered selection of indices. They are the basic tool for extracting structure from a sequence: limits, oscillation, boundedness, compactness, and failure of convergence are all detected by suitable subsequences.

Definition (Strictly Increasing Index Map)

Definition 6.6.1 (Strictly Increasing Index Map)**.** Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a function. The function σ is a **strictly increasing index map** if

$$k < \ell \Longrightarrow \sigma(k) < \sigma(\ell) \qquad \text{for all } k, \ell \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\sigma : \mathbb{N} \to \mathbb{N} \text{ is strictly increasing} \Longleftrightarrow \forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)).$$

Remark (Definition predicate reading).

$$\text{StrictlyIncreasing}(\sigma) :\!= \forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)).$$

Remark (Negated quantified statement).

$$\sigma \text{ is not strictly increasing} \Longleftrightarrow \exists k, \ell \in \mathbb{N}\ (k < \ell \wedge \sigma(\ell) \leq \sigma(k)).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{StrictlyIncreasing}(\sigma) \iff \exists k, \ell \in \mathbb{N}\ (k < \ell \wedge \sigma(\ell) \le \sigma(k)).$$

Remark (Interpretation). The index map selects terms without repetition and without changing their order. This is the structural difference between a subsequence and an arbitrary rearrangement.

Remark (Dependencies).

- Function
- Natural Numbers
- Strict Order Successor Characterization

Definition (Subsequence)

Definition 6.6.2 (Subsequence)**.** Let (x_n) be a real sequence. A **subsequence** of (x_n) is a sequence $(x_{\sigma(k)})_{k\in\mathbb{N}}$, where $\sigma : \mathbb{N} \to \mathbb{N}$ is a strictly increasing index map.

Remark (Standard quantified statement).

$$\begin{aligned} &(y_k) \text{ is a subsequence of } (x_n) \\ &\iff \quad \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \wedge \forall k \in \mathbb{N}\ (y_k = x_{\sigma(k)})\Big). \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Subsequence}(y_k, x_n) :\!= \exists \sigma\ \big(\mathrm{StrictlyIncreasing}(\sigma) \wedge \forall k \in \mathbb{N}\ (y_k = x_{\sigma(k)})\big).$$

Remark (Negated quantified statement).

$$\begin{aligned} &(y_k) \text{ is not a subsequence of } (x_n) \\ &\iff \quad \forall \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\exists k, \ell \in \mathbb{N}\ (k < \ell \wedge \sigma(\ell) \le \sigma(k)) \vee \exists k \in \mathbb{N}\ (y_k \ne x_{\sigma(k)})\Big). \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Subsequence}(y_k, x_n).$$

Remark (Interpretation). A subsequence keeps the original chronological order but may skip terms. The new index k counts the selected terms, while $\sigma(k)$ records where each selected term came from in the original sequence.

Remark (Dependencies).

- Real Sequence
- Strictly Increasing Index Map

Definition (Subsequential Limit)

Definition 6.6.3 (Subsequential Limit)**.** Let (x_n) be a real sequence and let $L \in \mathbb{R}$. The real number L is a **subsequential limit** of (x_n) if there exists a subsequence $(x_{\sigma(k)})$ such that

$$\lim_{k\to\infty} x_{\sigma(k)} = L.$$

Remark (Standard quantified statement)*.*

$$\begin{aligned}&L \text{ is a subsequential limit of } (x_n)\\ \iff \quad &\exists \sigma : \mathbb{N} \to \mathbb{N} \left(\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \land \lim_{k\to\infty} x_{\sigma(k)} = L\right).\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{SubsequentialLimit}(x_n, L) :\equiv \exists \sigma\ \big(\mathrm{StrictlyIncreasing}(\sigma) \land \mathrm{ConvergentSequence}(x_{\sigma(k)}, L)\big).$$

Remark (Interpretation)*.* Subsequential limits detect limiting behavior along an infinite selection of terms. A sequence may have no ordinary limit but still have one or more subsequential limits.

Remark (Dependencies)*.*

- Subsequence
- Convergence of a Sequence

Definition (Has Convergent Subsequence)

Definition 6.6.4 (Has Convergent Subsequence)**.** Let (x_n) be a real sequence. The sequence (x_n) **has a convergent subsequence** if there exist a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ and a real number L such that

$$\lim_{k\to\infty} x_{\sigma(k)} = L.$$

Remark (Standard quantified statement)*.*

$$\begin{aligned}&(x_n) \text{ has a convergent subsequence}\\ \iff \quad &\exists \sigma : \mathbb{N} \to \mathbb{N}\ \exists L \in \mathbb{R} \left(\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \land \lim_{k\to\infty} x_{\sigma(k)} = L\right).\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{HasConvergentSubsequence}(x_n) :\equiv \exists L \in \mathbb{R}\ \ \mathrm{SubsequentialLimit}(x_n, L).$$

Remark (Interpretation)*.* This property asks for at least one convergent infinite selection. It does not assert convergence of the whole sequence and does not require uniqueness of the subsequential limit.

Remark (Dependencies).

- Subsequential Limit

Theorem (Subsequence Indices Dominate the Identity)

Theorem 6.6.5 (Subsequence Indices Dominate the Identity)**.** *Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a strictly increasing index map. Then*

$$k \leq \sigma(k) \qquad \textit{for every } k \in \mathbb{N}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \sigma : \mathbb{N} \to \mathbb{N} \left(\left[\forall k, \ell \in \mathbb{N} \, (k < \ell \implies \sigma(k) < \sigma(\ell)) \right] \implies \left[\forall k \in \mathbb{N} \, (k \leq \sigma(k)) \right] \right).$$

Remark (Theorem predicate reading).

$$\text{StrictlyIncreasing}(\sigma) \implies \forall k \in \mathbb{N} \, (k \leq \sigma(k)).$$

Remark (Interpretation). A subsequence cannot outrun the original indexing backwards. By the time the subsequence has selected its kth term, it must have reached at least the kth original position.

Remark (Dependencies).

- Strictly Increasing Index Map
- Well-Ordering Principle

Theorem (Subsequences Preserve Limits)

Theorem 6.6.6 (Subsequences Preserve Limits)**.** *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then every subsequence of (x_n) converges to L.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall (x_n) \subseteq \mathbb{R} \, \forall L \in \mathbb{R} \, \forall \sigma : \mathbb{N} \to \mathbb{N} \\ &\quad \Big(x_n \to L \land \forall k, \ell \in \mathbb{N} \, (k < \ell \implies \sigma(k) < \sigma(\ell)) \Big) \implies \lim_{k \to \infty} x_{\sigma(k)} = L. \end{aligned}$$

Remark (Theorem predicate reading).

$$\text{ConvergentSequence}(x_n, L) \land \text{StrictlyIncreasing}(\sigma) \implies \text{ConvergentSequence}(x_{\sigma(k)}, L).$$

Remark (Interpretation). Subsequences cannot escape a limit already possessed by the whole sequence. The domination of subsequence indices by the identity transfers every tail estimate for (x_n) to a tail estimate for $(x_{\sigma(k)})$.

Remark (Dependencies).

- Subsequence
- Convergence of a Sequence
- Subsequence Indices Dominate the Identity

Theorem (Subsequential Limit of a Convergent Sequence)

Theorem 6.6.7 (Subsequential Limit of a Convergent Sequence)**.** *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then L is the only possible subsequential limit of (x_n).*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L, K \in \mathbb{R}$$
$$\Big(x_n \to L \wedge K \text{ is a subsequential limit of } (x_n)\Big) \Longrightarrow K = L.$$

Remark (Theorem predicate reading).

$$\text{ConvergentSequence}(x_n, L) \wedge \text{SubsequentialLimit}(x_n, K) \Longrightarrow K = L.$$

Remark (Interpretation). A convergent sequence has no hidden alternative limiting behavior. Every convergent subsequence inherits the same limit, and uniqueness of limits then forces equality.

Remark (Dependencies).

- Subsequences Preserve Limits
- Uniqueness of Limits

Theorem (Divergence by Two Subsequential Limits)

Theorem 6.6.8 (Divergence by Two Subsequential Limits)**.** *Let (x_n) be a real sequence. If (x_n) has two distinct subsequential limits, then (x_n) does not converge.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L, K \in \mathbb{R}$$
$$\Big(L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n) \wedge L \neq K\Big)$$
$$\Longrightarrow \quad (x_n) \text{ does not converge.}$$

Remark (Theorem predicate reading).

$$\text{SubsequentialLimit}(x_n, L) \wedge \text{SubsequentialLimit}(x_n, K) \wedge L \neq K \Longrightarrow \neg \exists A \in \mathbb{R}\ \text{ConvergentSequence}(x_n, A).$$

Remark (Interpretation). Two different subsequential limits reveal persistent oscillation or splitting. If the whole sequence had a limit, every subsequence would have to inherit that same limit.

Remark (Dependencies).

- Subsequential Limit
- Subsequential Limit of a Convergent Sequence

Theorem (Boundedness Passes to Subsequences)

Theorem 6.6.9 (Boundedness Passes to Subsequences)**.** *Every subsequence of a bounded real sequence is bounded.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall \sigma : \mathbb{N} \to \mathbb{N}\\ &\quad \Big(\exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \le M) \land \forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell))\Big)\\ &\quad \Longrightarrow \quad \exists M > 0\ \forall k \in \mathbb{N}\ (|x_{\sigma(k)}| \le M).\end{aligned}$$

Remark (Theorem predicate reading).

$$\mathrm{BoundedSequence}(x_n) \land \mathrm{StrictlyIncreasing}(\sigma) \Longrightarrow \mathrm{BoundedSequence}(x_{\sigma(k)}).$$

Remark (Interpretation). A subsequence cannot create new values; it only selects from the original sequence. Any global bound for the original sequence is automatically a bound for the selection.

Remark (Dependencies).

- Bounded Sequence
- Subsequence

Theorem (Monotonicity Passes to Subsequences)

Theorem 6.6.10 (Monotonicity Passes to Subsequences)**.** *Every subsequence of an increasing real sequence is increasing, and every subsequence of a decreasing real sequence is decreasing.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall \sigma : \mathbb{N} \to \mathbb{N}\\ &\quad \Big(\big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \land \big[\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell))\big]\Big)\\ &\quad \Longrightarrow \quad \forall k \in \mathbb{N}\ (x_{\sigma(k)} \le x_{\sigma(k+1)}).\end{aligned}$$

Remark (Theorem predicate reading).

$$\text{MonotoneIncreasingSequence}(x_n) \wedge \text{StrictlyIncreasing}(\sigma) \Longrightarrow \text{MonotoneIncreasingSequence}(x_{\sigma(k)}).$$ ∎

Remark (Interpretation). Selecting terms in the original order preserves the order pattern already present in the sequence. This is why the index map must be strictly increasing.

Remark (Dependencies).

- Increasing Sequence
- Decreasing Sequence
- Subsequence

Theorem (Subsequence of a Subsequence)

Theorem 6.6.11 (Subsequence of a Subsequence)**.** *Every subsequence of a subsequence of* (x_n) *is a subsequence of* (x_n).
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall \sigma, \tau : \mathbb{N} \to \mathbb{N} \\ &\quad \Big(\big[\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell))\big] \wedge \big[\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \tau(k) < \tau(\ell))\big] \Big) \\ &\quad \Longrightarrow \quad (x_{\sigma(\tau(k))})_{k \in \mathbb{N}} \text{ is a subsequence of } (x_n).\end{aligned}$$

Remark (Theorem predicate reading).

$$\text{StrictlyIncreasing}(\sigma) \wedge \text{StrictlyIncreasing}(\tau) \Longrightarrow \text{Subsequence}(x_{\sigma(\tau(k))}, x_n).$$

Remark (Interpretation). Passing to subsequences is closed under iteration. The composed index map $\sigma \circ \tau$ is still strictly increasing.

Remark (Dependencies).

- Subsequence
- Composition

Theorem (Eventual Properties Pass to Subsequences)

Theorem 6.6.12 (Eventual Properties Pass to Subsequences)**.** *Let* $P(n)$ *be a property of indices. If there exists* $N \in \mathbb{N}$ *such that* $P(n)$ *holds for every* $n \geq N$*, then for every strictly increasing index map* σ *there exists* $K \in \mathbb{N}$ *such that* $P(\sigma(k))$ *holds for every* $k \geq K$.
Go to proof.

Remark (Standard quantified statement).

$$\Big(\exists N \in \mathbb{N}\ \forall n \geq N\ P(n) \wedge \forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell))\Big) \\ \Longrightarrow \quad \exists K \in \mathbb{N}\ \forall k \geq K\ P(\sigma(k)).$$

Remark (Theorem predicate reading).

$$\text{Eventually}(x_n, P) \wedge \text{StrictlyIncreasing}(\sigma) \Longrightarrow \text{Eventually}(x_{\sigma(k)}, P).$$

Remark (Interpretation). Subsequences eventually enter every tail of the original sequence. Any property that holds on a tail of the original therefore holds on a tail of the subsequence.

Remark (Dependencies).

- Subsequence Indices Dominate the Identity
- Subsequence

Theorem (Frequent Properties Yield Subsequences)

Theorem 6.6.13 (Frequent Properties Yield Subsequences)**.** *Let $P(n)$ be a property of indices. If for every $N \in \mathbb{N}$ there exists $n \geq N$ such that $P(n)$ holds, then there exists a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \in \mathbb{N}$.*

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall N \in \mathbb{N}\ \exists n \geq N\ P(n)\Big] \Longrightarrow \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \wedge \forall k \in \mathbb{N}\ P(\sigma(k))\Big).$$

Remark (Theorem predicate reading).

$$\text{Frequently}(x_n, P) \Longrightarrow \exists \sigma\ \big(\text{StrictlyIncreasing}(\sigma) \wedge \forall k \in \mathbb{N}\ P(\sigma(k))\big).$$

Remark (Interpretation). An infinitely recurring property can be threaded into a subsequence. The construction chooses one later witness at each step.

Remark (Dependencies).

- Well-Ordering Principle
- Subsequence

Theorem (Subsequential Limits Respect Bounds)

Theorem 6.6.14 (Subsequential Limits Respect Bounds)**.** *Let (x_n) be a real sequence and let L be a subsequential limit of (x_n). If $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then $m \leq L \leq M$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L, m, M \in \mathbb{R}$$
$$\Big(L \text{ is a subsequential limit of } (x_n) \wedge \forall n \in \mathbb{N}\ (m \le x_n \le M)\Big) \Longrightarrow m \le L \le M.$$

Remark (Theorem predicate reading).

$$\mathrm{SubsequentialLimit}(x_n, L) \wedge \forall n \in \mathbb{N}\ (m \le x_n \le M) \Longrightarrow m \le L \le M.$$

Remark (Interpretation). Bounds pass to subsequences, and limits preserve eventual inequalities. Hence subsequential limits remain inside any closed interval that contains all terms.

Remark (Dependencies).

- Boundedness Passes to Subsequences
- Constant Comparison for Sequence Limits

Theorem (Squeeze Passes to Subsequences)

Theorem 6.6.15 (Squeeze Passes to Subsequences)**.** *Let (a_n), (x_n), and (b_n) be real sequences. If $a_n \le x_n \le b_n$ eventually, then for every subsequence $(x_{\sigma(k)})$ there exists $K \in \mathbb{N}$ such that*

$$a_{\sigma(k)} \le x_{\sigma(k)} \le b_{\sigma(k)} \qquad \text{for every } k \ge K.$$

Go to proof.

Remark (Standard quantified statement).

$$\Big(\exists N \in \mathbb{N}\ \forall n \ge N\ (a_n \le x_n \le b_n) \wedge \forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell))\Big)$$
$$\Longrightarrow \quad \exists K \in \mathbb{N}\ \forall k \ge K\ (a_{\sigma(k)} \le x_{\sigma(k)} \le b_{\sigma(k)}).$$

Remark (Theorem predicate reading).

$$\mathrm{Eventually}(x_n,\ a_n \le x_n \le b_n) \wedge \mathrm{StrictlyIncreasing}(\sigma) \Longrightarrow \mathrm{Eventually}(x_{\sigma(k)},\ a_{\sigma(k)} \le x_{\sigma(k)} \le b_{\sigma(k)}).$$

Remark (Interpretation). Squeeze hypotheses are tail hypotheses, so they survive passage to subsequences.

Remark (Dependencies).

- Eventual Properties Pass to Subsequences
- Sequence Squeeze Theorem

Theorem (Monotone Subsequence Theorem)

Theorem 6.6.16 (Monotone Subsequence Theorem)**.** *Every real sequence has a monotone subsequence.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \Big[\forall k \in \mathbb{N}\ (x_{\sigma(k)} \leq x_{\sigma(k+1)}) \vee \forall k \in \mathbb{N}\ (x_{\sigma(k+1)} \leq x$$

Remark (Theorem predicate reading).

$$\exists \sigma\ \big(\mathrm{StrictlyIncreasing}(\sigma) \wedge \mathrm{MonotoneSequence}(x_{\sigma(k)})\big).$$

Remark (Interpretation). Every sequence contains an infinite ordered pattern. The proof uses the peak dichotomy: either there are infinitely many terms larger than all later terms, or there is an infinite rising chain.

Remark (Dependencies).

- Monotone Sequence
- Subsequence
- Well-Ordering Principle

Theorem (Bolzano-Weierstrass Theorem for Sequences)

Theorem 6.6.17 (Bolzano-Weierstrass Theorem for Sequences)**.** *Every bounded real sequence has a convergent subsequence.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \Big(\exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \leq M) \implies \exists \sigma : \mathbb{N} \to \mathbb{N}\ \exists L \in \mathbb{R}\ \Big[\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \lim_{k \to \infty}$$

Remark (Theorem predicate reading).

$$\mathrm{BoundedSequence}(x_n) \implies \mathrm{HasConvergentSubsequence}(x_n).$$

Remark (Interpretation). Bolzano-Weierstrass is the compactness principle for bounded real sequences. Boundedness prevents escape, and the monotone subsequence theorem supplies an ordered subsequence to which the Monotone Convergence Theorem applies.

Remark (Dependencies).

- Bounded Sequence
- Monotone Subsequence Theorem
- Boundedness Passes to Subsequences
- Bounded Monotone Sequence Equivalences

Theorem (Sequential Compactness of Closed Bounded Intervals)

Theorem 6.6.18 (Sequential Compactness of Closed Bounded Intervals)**.** *Let $a, b \in \mathbb{R}$ with $a \leq b$. Every sequence in $[a, b]$ has a convergent subsequence whose limit belongs to $[a, b]$.*
Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b \in \mathbb{R}\ \forall (x_n) \subseteq \mathbb{R}$$
$$\Big(a \leq b \wedge \forall n \in \mathbb{N}\ (a \leq x_n \leq b)\Big)$$
$$\implies \quad \exists \sigma : \mathbb{N} \to \mathbb{N}\ \exists L \in \mathbb{R}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \lim_{k\to\infty} x_{\sigma(k)} = L \wedge a \leq L \leq b\Big).$$

Remark (Theorem predicate reading)*.*

$$\big[\forall n \in \mathbb{N}\ (a \leq x_n \leq b)\big] \implies \exists L \in [a, b]\ \text{SubsequentialLimit}(x_n, L).$$

Remark (Interpretation)*.* Closed bounded intervals are sequentially compact: a sequence trapped in the interval has a convergent subsequence, and closedness keeps the subsequential limit inside the interval.

Remark (Dependencies)*.*

- Bolzano-Weierstrass Theorem for Sequences
- Subsequential Limits Respect Bounds

6.7 Divergence

Divergence — Quick Reference

Concept	Meaning
Divergent sequence	Sequence that does not converge in $\mathbb{R}$
Diverges to $+\infty$	Eventually exceeds every real bound
Diverges to $-\infty$	Eventually lies below every real bound
Oscillatory sequence	Divergence by persistent oscillation
Infinite divergence	Divergence to infinity implies real divergence
Two subsequential limits	Distinct subsequential limits force divergence
Bounded divergence	Bounded divergence yields multiple subsequential limits

Divergence

Divergence is not a single behavior. A sequence can fail to converge because it escapes upward, escapes downward, oscillates between incompatible limiting patterns, or remains bounded while refusing to settle. The purpose of this section is to separate these failure modes and connect them to subsequences.

Definition (Divergent Sequence)

Definition 6.7.1 (Divergent Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **divergent** if there is no $L \in \mathbb{R}$ such that $x_n \to L$.

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is divergent}\\ &\quad\iff \quad \forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| \geq \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{DivergentSequence}(x_n) :\equiv \forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| \geq \varepsilon\big).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not divergent} \iff \exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|x_n - L| < \varepsilon\big).$$

Remark (Negation predicate reading).

$$\neg\,\text{DivergentSequence}(x_n) \iff \exists L \in \mathbb{R}\ \ \text{ConvergentSequence}(x_n, L).$$

Remark (Interpretation). Divergence is the formal negation of real convergence. It says that every candidate limit fails at some fixed tolerance on every tail.

Remark (Dependencies).

- Real Sequence
- Convergence of a Sequence

Definition (Divergence to Positive Infinity)

Definition 6.7.2 (Divergence to Positive Infinity)**.** Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $+\infty$ if

$$\forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(M < x_n\big).$$

Remark (Standard quantified statement).

$$x_n \to +\infty \iff \forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(M < x_n\big).$$

Remark (Definition predicate reading).

$$\text{DivergesToPositiveInfinity}(x_n) :\equiv \forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(M < x_n\big).$$

Remark (Negated quantified statement).

$$x_n \not\to +\infty \iff \exists M \in \mathbb{R}\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(x_n \leq M\big).$$

Remark (Interpretation). Divergence to $+\infty$ means every horizontal threshold is eventually left below the sequence. It is not convergence to a real number; it is escape from the real line in the positive direction.

Remark (Dependencies).

- Real Sequence

Definition (Divergence to Negative Infinity)

Definition 6.7.3 (Divergence to Negative Infinity)**.** Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $-\infty$ if

$$\forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(x_n < M\big).$$

Remark (Standard quantified statement).

$$x_n \to -\infty \iff \forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(x_n < M\big).$$

Remark (Definition predicate reading).

$$\mathrm{DivergesToNegativeInfinity}(x_n) \coloneqq \forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(x_n < M\big).$$

Remark (Negated quantified statement).

$$x_n \not\to -\infty \iff \exists M \in \mathbb{R}\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(M \leq x_n\big).$$

Remark (Interpretation). Divergence to $-\infty$ is escape past every lower threshold. It is the order-dual of divergence to $+\infty$.

Remark (Dependencies).

- Real Sequence

Definition (Oscillatory Sequence)

Definition 6.7.4 (Oscillatory Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **oscillatory** if it is divergent, does not diverge to $+\infty$, and does not diverge to $-\infty$.

Remark (Standard quantified statement).

$$\begin{aligned}
&(x_n) \text{ is oscillatory} \\
\iff \quad &\Big[\forall L \in \mathbb{R}\, \exists \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(|x_n - L| \geq \varepsilon\big)\Big] \\
&\wedge \Big[\exists M \in \mathbb{R}\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(x_n \leq M\big)\Big] \\
&\wedge \Big[\exists M \in \mathbb{R}\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(M \leq x_n\big)\Big].
\end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned}\mathrm{OscillatorySequence}(x_n) \coloneqq\ & \mathrm{DivergentSequence}(x_n)\\ & \land\ \neg\,\mathrm{DivergesToPositiveInfinity}(x_n)\\ & \land\ \neg\,\mathrm{DivergesToNegativeInfinity}(x_n).\end{aligned}$$

Remark (Interpretation). Oscillation means failure to settle without one-directional escape. The sequence may remain bounded, as $(-1)^n$ does, or it may be unbounded while still repeatedly returning in incompatible directions.

Remark (Dependencies).

- Divergent Sequence
- Divergence to Positive Infinity
- Divergence to Negative Infinity

Theorem (Divergence to Infinity Implies Real Divergence)

Theorem 6.7.5 (Divergence to Infinity Implies Real Divergence)**.** *Let (x_n) be a real sequence. If $x_n \to +\infty$ or $x_n \to -\infty$, then (x_n) is divergent.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\Big[\forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(M < x_n\big)\Big]\\ &\quad \lor \Big[\forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(x_n < M\big)\Big]\\ &\qquad \Longrightarrow \forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| \geq \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{DivergesToPositiveInfinity}(x_n) \lor \mathrm{DivergesToNegativeInfinity}(x_n) \Longrightarrow \mathrm{DivergentSequence}(x_n).$$

Remark (Interpretation). Infinity divergence is a stronger, directional failure of real convergence. A sequence escaping past every real threshold cannot approach a finite real limit.

Remark (Dependencies).

- Divergent Sequence
- Divergence to Positive Infinity
- Divergence to Negative Infinity

Theorem (Two Subsequential Limits Force Divergence)

Theorem 6.7.6 (Two Subsequential Limits Force Divergence)**.** *Let (x_n) be a real sequence. If (x_n) has two subsequential limits $L, K \in \mathbb{R}$ with $L \neq K$, then (x_n) is divergent.*

Go to proof.

Remark (Standard quantified statement).

$$\exists L, K \in \mathbb{R} \left(L \neq K \wedge L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n)\right) \\ \implies \forall A \in \mathbb{R}\, \exists \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists n \geq N \, \big(|x_n - A| \geq \varepsilon\big).$$

Remark (Definition predicate reading).

$$\exists L, K \in \mathbb{R} \, \big(L \neq K \wedge \text{SubsequentialLimit}(x_n, L) \wedge \text{SubsequentialLimit}(x_n, K)\big) \\ \implies \text{DivergentSequence}(x_n).$$

Remark (Interpretation). Convergence forces all subsequences to inherit the same limit. Two different subsequential limits therefore witness an irreparable conflict in the tail behavior of the original sequence.

Remark (Dependencies).

- Subsequential Limit
- Subsequences Preserve Limits
- Uniqueness of Limits
- Divergence by Two Subsequential Limits

Theorem (Unbounded Above Sequence Has a Positive-Infinity Subsequence)

Theorem 6.7.7 (Unbounded Above Sequence Has a Positive-Infinity Subsequence)**.** *Let (x_n) be a real sequence. If (x_n) is not bounded above, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to +\infty$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall M \in \mathbb{R}\, \exists n \in \mathbb{N} \, \big(M < x_n\big)\Big] \\ \implies \exists \sigma : \mathbb{N} \to \mathbb{N} \left(\forall k, \ell \in \mathbb{N} \, (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall M \in \mathbb{R}\, \exists K \in \mathbb{N}\, \forall k \geq K \, \big(M < x_{\sigma(k)}\big)\right).$$

Remark (Definition predicate reading).

$$\neg\, \text{BoundedAboveSequence}(x_n) \implies \exists \sigma \, \big(\text{StrictlyIncreasing}(\sigma) \wedge \text{DivergesToPositiveInfinity}(x_{\sigma(k)})\big).$$

Remark (Interpretation). Unboundedness above may be sparse. This theorem says the high values can still be selected in increasing index order to form a subsequence that escapes to $+\infty$.

Remark (Dependencies).

- Sequence Bounded Above

- Subsequence
- Divergence to Positive Infinity

Theorem (Unbounded Below Sequence Has a Negative-Infinity Subsequence)

Theorem 6.7.8 (Unbounded Below Sequence Has a Negative-Infinity Subsequence)**.** *Let (x_n) be a real sequence. If (x_n) is not bounded below, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to -\infty$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ \big(x_n < M\big)\Big]$$
$$\implies \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall M \in \mathbb{R}\ \exists K \in \mathbb{N}\ \forall k \geq K\ \big(x_{\sigma(k)} < M\big)\Big).$$

Remark (Definition predicate reading).

$$\neg\,\mathrm{BoundedBelowSequence}(x_n) \implies \exists \sigma\ \big(\mathrm{StrictlyIncreasing}(\sigma) \wedge \mathrm{DivergesToNegativeInfinity}(x_{\sigma(k)})\big).$$

Remark (Interpretation). Unboundedness below can be converted into a selected tail that escapes past every lower threshold. The proof is the order-reversed version of the positive-infinity subsequence construction.

Remark (Dependencies).

- Sequence Bounded Below
- Subsequence
- Divergence to Negative Infinity

Theorem (Bounded Divergence Produces Two Subsequential Limits)

Theorem 6.7.9 (Bounded Divergence Produces Two Subsequential Limits)**.** *Let (x_n) be a bounded real sequence. If (x_n) is divergent, then there exist $L, K \in \mathbb{R}$ with $L \neq K$ such that L and K are subsequential limits of (x_n).*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\exists M > 0\ \forall n \in \mathbb{N}\ \big(|x_n| \leq M\big)\Big] \wedge \Big[\forall A \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - A| \geq \varepsilon\big)\Big]$$
$$\implies \exists L, K \in \mathbb{R}\ \Big(L \neq K \wedge L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n)\Big).$$

Remark (Definition predicate reading).

$$\mathrm{BoundedSequence}(x_n) \wedge \mathrm{DivergentSequence}(x_n) \implies \exists L, K \in \mathbb{R}\ \big(L \neq K \wedge \mathrm{SubsequentialLimit}(x_n, L) \wedge \mathrm{Subse}$$

Remark (Interpretation). Bounded divergence cannot be explained by escape to infinity. In the real line, compactness forces convergent subsequences, and divergence forces at least two incompatible subsequential limits.

Remark (Dependencies).

- Bounded Sequence
- Divergent Sequence
- Subsequential Limit
- Bolzano-Weierstrass Theorem for Sequences
- Subsequences Preserve Limits

6.8 Liminf and Limsup

Liminf–Limsup — Quick Reference

Concept	**Meaning**
Tail supremum sequence	Supremum of each tail
Tail infimum sequence	Infimum of each tail
Limit superior	Limit of tail suprema
Limit inferior	Limit of tail infima
Liminf below limsup	Lower envelope never exceeds upper envelope
Convergence criterion	Convergence iff liminf equals limsup
Largest subsequential limit	Limsup is the top cluster value
Smallest subsequential limit	Liminf is the bottom cluster value
Oscillation criterion	Gap detects nonconvergence
Eventual-order squeeze	Squeeze estimates transfer to limsup and liminf

Liminf and Limsup

The limit superior and limit inferior measure the eventual upper and lower envelopes of a bounded sequence. Instead of asking whether all late terms settle to one number, they ask how high and how low the tails continue to reach. This makes them natural tools for detecting convergence, oscillation, and subsequential limits.

Definition (Tail Supremum Sequence)

Definition 6.8.1 (Tail Supremum Sequence)**.** Let (x_n) be a bounded real sequence. The **tail supremum sequence** of (x_n) is the real sequence (s_n) defined by

$$s_n := \sup\{x_k : k \geq n\} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(s_n = \sup\{x_k : k \geq n\}\right).$$

Remark (Interpretation). The term s_n records the least ceiling of the n-th tail. As n grows, earlier terms are discarded, so the tail has fewer opportunities to reach large values.

Remark (Dependencies).

- Bounded Sequence
- Supremum
- M-Tail of a Sequence

Definition (Tail Infimum Sequence)

Definition 6.8.2 (Tail Infimum Sequence)**.** Let (x_n) be a bounded real sequence. The **tail infimum sequence** of (x_n) is the real sequence (i_n) defined by

$$i_n := \inf\{x_k : k \geq n\} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(i_n = \inf\{x_k : k \geq n\}\right).$$

Remark (Interpretation). The term i_n records the greatest floor of the n-th tail. As earlier terms are discarded, the tail loses opportunities to reach small values.

Remark (Dependencies).

- Bounded Sequence
- Infimum
- M-Tail of a Sequence

Definition (Limit Superior of a Sequence)

Definition 6.8.3 (Limit Superior of a Sequence)**.** Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. The **limit superior** of (x_n) is

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} s_n.$$

Remark (Standard quantified statement).

$$\limsup_{n\to\infty} x_n = L \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N \left(|s_n - L| < \varepsilon\right).$$

Remark (Definition predicate reading).

$$\text{LimsupSequence}(x_n, L) :\equiv \text{ConvergentSequence}(s_n, L).$$

Remark (Interpretation). The limit superior is the eventual upper envelope of the sequence. It is the number to which the ceilings of the tails descend.

Remark (Dependencies).

- Tail Supremum Sequence
- Convergence of a Sequence

Definition (Limit Inferior of a Sequence)

Definition 6.8.4 (Limit Inferior of a Sequence)**.** Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. The **limit inferior** of (x_n) is

$$\liminf_{n\to\infty} x_n := \lim_{n\to\infty} i_n.$$

Remark (Standard quantified statement).

$$\liminf_{n\to\infty} x_n = L \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|i_n - L| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\text{LiminfSequence}(x_n, L) :\equiv \text{ConvergentSequence}(i_n, L).$$

Remark (Interpretation). The limit inferior is the eventual lower envelope of the sequence. It is the number to which the floors of the tails ascend.

Remark (Dependencies).

- Tail Infimum Sequence
- Convergence of a Sequence

Theorem (Tail Suprema Are Decreasing)

Theorem 6.8.5 (Tail Suprema Are Decreasing)**.** *Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. Then (s_n) is decreasing.*

Go to proof.

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N}\ \big(s_{n+1} \leq s_n\big).$$

Remark (Definition predicate reading).

$$\text{MonotoneDecreasingSequence}(s_n).$$

Remark (Interpretation). The $(n+1)$-st tail is contained in the n-th tail. Passing from a set to a smaller subset cannot raise its supremum.

Remark (Dependencies).

- Tail Supremum Sequence
- Supremum Is Monotone Under Inclusion

Theorem (Tail Infima Are Increasing)

Theorem 6.8.6 (Tail Infima Are Increasing)**.** *Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. Then (i_n) is increasing.*
Go to proof.

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(i_n \leq i_{n+1}\right).$$

Remark (Definition predicate reading).

$$\text{MonotoneIncreasingSequence}(i_n).$$

Remark (Interpretation). The $(n+1)$-st tail is contained in the n-th tail. Passing from a set to a smaller subset cannot lower its infimum.

Remark (Dependencies).

- Tail Infimum Sequence
- Infimum

Theorem (Liminf Is Below Limsup)

Theorem 6.8.7 (Liminf Is Below Limsup)**.** *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n.$$

Remark (Interpretation). Every tail infimum lies below every corresponding tail supremum. Passing to the limits preserves this order.

Remark (Dependencies).

- Limit Inferior of a Sequence
- Limit Superior of a Sequence

- Constant Comparison for Sequence Limits

Theorem (Convergence iff Liminf Equals Limsup)

Theorem 6.8.8 (Convergence iff Liminf Equals Limsup)**.** *Let (x_n) be a bounded real sequence, and let $L \in \mathbb{R}$. Then $x_n \to L$ if and only if*

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = L.$$

Go to proof.

Remark (Standard quantified statement).

$$x_n \to L \quad \Longleftrightarrow \quad \liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = L.$$

Remark (Definition predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \Longleftrightarrow \big(\mathrm{LiminfSequence}(x_n, L) \wedge \mathrm{LimsupSequence}(x_n, L)\big).$$

Remark (Interpretation)*.* Convergence occurs exactly when the eventual lower envelope and eventual upper envelope collapse to the same number.

Remark (Dependencies).

- Limit Inferior of a Sequence
- Limit Superior of a Sequence
- Convergence of a Sequence
- Sequence Squeeze Theorem

Theorem (Limsup Is the Largest Subsequential Limit)

Theorem 6.8.9 (Limsup Is the Largest Subsequential Limit)**.** *Let (x_n) be a bounded real sequence, and let $S = \limsup_{n\to\infty} x_n$. Then S is a subsequential limit of (x_n), and every subsequential limit of (x_n) is less than or equal to S.*
Go to proof.

Remark (Standard quantified statement).

$$S = \limsup_{n\to\infty} x_n \implies \Big[S \text{ is a subsequential limit of } (x_n) \wedge \forall L \in \mathbb{R} \,\big(L \text{ is a subsequential limit of } (x_n) \implies L \le S\big)\Big].$$

Remark (Definition predicate reading).

$$\mathrm{LimsupSequence}(x_n, S) \implies \big(\mathrm{SubsequentialLimit}(x_n, S) \wedge \forall L \in \mathbb{R}\ (\mathrm{SubsequentialLimit}(x_n, L) \implies L \le S)\big).$$

Remark (Interpretation). The limit superior is not merely an upper bound on subsequential behavior. It is actually attained as a subsequential limit, and no subsequential limit lies above it.

Remark (Dependencies).

- Limit Superior of a Sequence
- Subsequential Limit
- Bolzano-Weierstrass Theorem for Sequences

Theorem (Liminf Is the Smallest Subsequential Limit)

Theorem 6.8.10 (Liminf Is the Smallest Subsequential Limit)**.** *Let (x_n) be a bounded real sequence, and let $I = \liminf_{n\to\infty} x_n$. Then I is a subsequential limit of (x_n), and every subsequential limit of (x_n) is greater than or equal to I.*

Go to proof.

Remark (Standard quantified statement).

$$I = \liminf_{n\to\infty} x_n \implies \Big[I \text{ is a subsequential limit of } (x_n) \land \forall L \in \mathbb{R}\ \big(L \text{ is a subsequential limit of } (x_n) \implies I \le L\big)\Big].$$

Remark (Definition predicate reading).

$$\text{LiminfSequence}(x_n, I) \implies \big(\text{SubsequentialLimit}(x_n, I) \land \forall L \in \mathbb{R}\ (\text{SubsequentialLimit}(x_n, L) \implies I \le L)\big).$$

Remark (Interpretation). The limit inferior is attained as a subsequential limit and no subsequential limit lies below it.

Remark (Dependencies).

- Limit Inferior of a Sequence
- Subsequential Limit
- Bolzano-Weierstrass Theorem for Sequences

Theorem (Oscillation Criterion via Liminf and Limsup)

Theorem 6.8.11 (Oscillation Criterion via Liminf and Limsup)**.** *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n < \limsup_{n\to\infty} x_n$$

if and only if (x_n) has two distinct subsequential limits.

Go to proof.

Remark (Standard quantified statement).

$$\liminf_{n\to\infty} x_n < \limsup_{n\to\infty} x_n$$
$$\iff \quad \exists L, K \in \mathbb{R}\ \big(L \neq K \wedge L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n)\big).$$

Remark (Definition predicate reading).

$$\exists I, S \in \mathbb{R}\ \big(\mathrm{LiminfSequence}(x_n, I) \wedge \mathrm{LimsupSequence}(x_n, S) \wedge I < S\big)$$
$$\iff \quad \exists L, K \in \mathbb{R}\ \big(L \neq K \wedge \mathrm{SubsequentialLimit}(x_n, L) \wedge \mathrm{SubsequentialLimit}(x_n, K)\big).$$

Remark (Interpretation). A strict gap between the lower and upper envelopes is exactly persistent oscillation in bounded sequences. The sequence keeps returning near at least two incompatible limiting levels.

Remark (Dependencies).

- Limsup Is the Largest Subsequential Limit
- Liminf Is the Smallest Subsequential Limit
- Two Subsequential Limits Force Divergence

Theorem (Limsup Comparison Under Eventual Order)

Theorem 6.8.12 (Limsup Comparison Under Eventual Order). *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N$, then*

$$\limsup_{n\to\infty} x_n \leq \limsup_{n\to\infty} y_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \leq y_n)$$
$$\implies \quad \limsup_{n\to\infty} x_n \leq \limsup_{n\to\infty} y_n.$$

Remark (Interpretation). Eventual order of sequences passes to eventual order of their upper tail envelopes. Taking the limiting ceiling preserves that comparison.

Remark (Dependencies).

- Limit Superior of a Sequence
- Limit Preserves Eventual Order

Theorem (Liminf Comparison Under Eventual Order)

Theorem 6.8.13 (Liminf Comparison Under Eventual Order)**.** *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \le y_n$ for every $n \ge N$, then*

$$\liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} y_n.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned} &\exists N \in \mathbb{N}\ \forall n \ge N\ (x_n \le y_n) \\ &\implies \quad \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} y_n. \end{aligned}$$

Remark (Interpretation)*.* Eventual order also passes to the lower tail envelopes. If one sequence is eventually below another, its eventual floor cannot sit above the other's eventual floor.

Remark (Dependencies)*.*

- Limit Inferior of a Sequence
- Limit Preserves Eventual Order

Theorem (Limsup Squeeze Under Eventual Order)

Theorem 6.8.14 (Limsup Squeeze Under Eventual Order)**.** *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \text{for every } n \ge N,$$

then

$$\limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} b_n.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned} &\exists N \in \mathbb{N}\ \forall n \ge N\ (a_n \le x_n \le b_n) \\ &\implies \quad \limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} b_n. \end{aligned}$$

Remark (Interpretation)*.* If one sequence is eventually trapped between two others, then its eventual upper envelope is trapped between their eventual upper envelopes.

Remark (Dependencies)*.*

- Limsup Comparison Under Eventual Order

Theorem (Liminf Squeeze Under Eventual Order)

Theorem 6.8.15 (Liminf Squeeze Under Eventual Order)**.** *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \text{for every } n \ge N,$$

then

$$\liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} b_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\exists N \in \mathbb{N}\, \forall n \ge N\ (a_n \le x_n \le b_n) \\ &\quad \implies \quad \liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} b_n. \end{aligned}$$

Remark (Interpretation). The lower-envelope version of the squeeze principle says that eventual two-sided order traps the eventual floors as well as the eventual ceilings.

Remark (Dependencies).

- Liminf Comparison Under Eventual Order

Remark (Illustrative cases). For the alternating sequence $x_n = (-1)^n$, the tail suprema are constantly 1 and the tail infima are constantly -1, so

$$\limsup_{n\to\infty}(-1)^n = 1, \qquad \liminf_{n\to\infty}(-1)^n = -1.$$

For any convergent sequence $x_n \to L$, both envelopes collapse to L.

6.9 Order-Theoretic Limit Constructions

Order-Theoretic Limits — Quick Reference

Statement	**Role**
Increasing sequence limit	Limit realized as a supremum
Decreasing sequence limit	Limit realized as an infimum
Tail extremals converge	Tail suprema and infima are monotone
Bounded sequence limsup/liminf	Bounded sequences have asymptotic envelopes

Order-Theoretic Limit Constructions

Completeness lets order data produce limits. An increasing sequence bounded above converges to the least ceiling of its range; a decreasing sequence bounded below converges to the greatest floor of its range. Applying these same ideas to tail suprema and tail infima produces limsup and liminf for every bounded sequence.

Theorem (Increasing Sequence Limit as Supremum)

Theorem 6.9.1 (Increasing Sequence Limit as Supremum)**.** *Let (x_n) be an increasing real sequence that is bounded above. Then*

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall (x_n) \subseteq \mathbb{R} \left(\left[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\right] \land \left[\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\right] \implies \lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}\right).$$

Remark (Theorem predicate reading)*.*

MonotoneIncreasingSequence$(x_n)\land$BoundedAboveSequence$(x_n) \implies$ ConvergentSequence$\big(x_n, \sup\{x_n : n \in$

Remark (Interpretation)*.* The supremum is the only possible destination: all terms lie below it, and the defining approximation property of the supremum forces the sequence eventually above every lower tolerance.

Remark (Dependencies)*.*

- Increasing Sequence
- Sequence Bounded Above
- Supremum
- Monotone Convergence Theorem

Theorem (Decreasing Sequence Limit as Infimum)

Theorem 6.9.2 (Decreasing Sequence Limit as Infimum)**.** *Let (x_n) be a decreasing real sequence that is bounded below. Then*

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall (x_n) \subseteq \mathbb{R} \left(\left[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\right] \land \left[\exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\right] \implies \lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}\right).$$

Remark (Theorem predicate reading).

$$\text{MonotoneDecreasingSequence}(x_n) \wedge \text{BoundedBelowSequence}(x_n) \implies \text{ConvergentSequence}\big(x_n, \inf\{x_n : n \in$$

Remark (Interpretation). The infimum is the decreasing analogue of the supremum target: the sequence falls toward its greatest lower barrier.

Remark (Dependencies).

- Decreasing Sequence
- Sequence Bounded Below
- Infimum
- Monotone Convergence Theorem

Theorem (Tail Suprema and Infima Converge)

Theorem 6.9.3 (Tail Suprema and Infima Converge)**.** *Let* (x_n) *be a bounded real sequence. Define*

$$s_n = \sup\{x_k : k \geq n\}, \qquad i_n = \inf\{x_k : k \geq n\}.$$

Then (s_n) *and* (i_n) *converge.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left((x_n) \text{ is bounded} \implies \exists S, I \in \mathbb{R} \left[\lim_{n \to \infty} \sup\{x_k : k \geq n\} = S \wedge \lim_{n \to \infty} \inf\{x_k : k \geq n\} = I \right] \right).$$

Remark (Theorem predicate reading).

$$\text{BoundedSequence}(x_n) \implies \exists S, I \in \mathbb{R} \; \big(\text{ConvergentSequence}(\sup\{x_k : k \geq n\}, S) \wedge \text{ConvergentSequence}(\inf\{$$

Remark (Interpretation). Tail suprema decrease because later tails contain fewer terms; tail infima increase for the same reason. Boundedness supplies the opposite bound needed for monotone convergence.

Remark (Dependencies).

- Tail Supremum Sequence
- Tail Infimum Sequence
- Tail Suprema Are Decreasing
- Tail Infima Are Increasing
- Monotone Convergence Theorem

Theorem (Bounded Sequence Has Limsup and Liminf)

Theorem 6.9.4 (Bounded Sequence Has Limsup and Liminf)**.** *Every bounded real sequence has a limit superior and a limit inferior.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left((x_n) \text{ is bounded} \implies \exists S, I \in \mathbb{R} \left(S = \limsup_{n\to\infty} x_n \wedge I = \liminf_{n\to\infty} x_n \right) \right).$$

Remark (Theorem predicate reading).

$$\mathrm{BoundedSequence}(x_n) \implies \exists S, I \in \mathbb{R} \left(\mathrm{LimsupSequence}(x_n, S) \wedge \mathrm{LiminfSequence}(x_n, I) \right).$$

Remark (Interpretation). For bounded sequences, limsup and liminf are not extra assumptions. They are guaranteed order-theoretic limits of the tail envelopes.

Remark (Dependencies).

- Limit Superior of a Sequence
- Limit Inferior of a Sequence
- Tail Suprema and Infima Converge

6.10 Cluster Values of Sequences

Cluster Values — Quick Reference

Concept	Meaning
Cluster value	Value approached along a subsequence
Cluster equals subsequential limit	Converts frequent approach to subsequences
Bounded sequences have cluster values	Compactness produces accumulation
Extremal cluster values	Limsup and liminf are the outer cluster values

Cluster Values of Sequences

A convergent sequence has only one limiting value. A bounded divergent sequence may still have values that it approaches along selected subsequences. Cluster values name this repeated limiting behavior without requiring the entire sequence to converge.

Definition (Cluster Value of a Sequence)

Definition 6.10.1 (Cluster Value of a Sequence)**.** Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The number L is a **cluster value** of (x_n) if for every $\varepsilon > 0$ and every $N \in \mathbb{N}$ there exists $n \geq N$ such that

$$|x_n - L| < \varepsilon.$$

Remark (Standard quantified statement).

$$L \text{ is a cluster value of } (x_n) \iff \forall \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\mathrm{SequenceClusterValue}(x_n, L) \coloneqq \forall \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$L \text{ is not a cluster value of } (x_n) \iff \exists \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|x_n - L| \geq \varepsilon\big).$$

Remark (Interpretation). No matter how far out in the sequence one looks, some later term returns inside every neighborhood of L.

Remark (Dependencies).

- Real Sequence

Theorem (Cluster Values Are Subsequential Limits)

Theorem 6.10.2 (Cluster Values Are Subsequential Limits). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. Then L is a cluster value of (x_n) if and only if L is a subsequential limit of (x_n).*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \Big(L \text{ is a cluster value of } (x_n) \iff L \text{ is a subsequential limit of } (x_n)\Big).$$

Remark (Theorem predicate reading).

$$\mathrm{SequenceClusterValue}(x_n, L) \iff \mathrm{SubsequentialLimit}(x_n, L).$$

Remark (Interpretation). The neighborhood formulation and the subsequence formulation describe the same phenomenon: infinitely many opportunities to get arbitrarily close to L.

Remark (Dependencies).

- Cluster Value of a Sequence
- Subsequential Limit
- Frequent Properties Yield Subsequences

Theorem (Bounded Sequences Have Cluster Values)

Theorem 6.10.3 (Bounded Sequences Have Cluster Values). *Every bounded real sequence has at least one cluster value.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left(\left[\exists M > 0 \; \forall n \in \mathbb{N} \; (|x_n| \le M) \right] \implies \exists L \in \mathbb{R} \; \big(L \text{ is a cluster value of } (x_n) \big) \right).$$

Remark (Theorem predicate reading).

$$\text{BoundedSequence}(x_n) \implies \exists L \in \mathbb{R} \; \text{SequenceClusterValue}(x_n, L).$$

Remark (Interpretation). Boundedness traps the sequence in a finite interval, and Bolzano-Weierstrass extracts a convergent subsequence from that trap.

Remark (Dependencies).

- Bounded Sequence
- Bolzano-Weierstrass Theorem for Sequences
- Cluster Values Are Subsequential Limits

Theorem (Limsup and Liminf Are Extremal Cluster Values)

Theorem 6.10.4 (Limsup and Liminf Are Extremal Cluster Values)**.** *Let* (x_n) *be a bounded real sequence. Then* $\limsup_{n\to\infty} x_n$ *is the largest cluster value of* (x_n)*, and* $\liminf_{n\to\infty} x_n$ *is the smallest cluster value of* (x_n)*.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \Bigg((x_n) \text{ is bounded} \implies \\
&\quad \Big[\limsup_{n\to\infty} x_n \text{ is a cluster value of } (x_n) \wedge \forall L \in \mathbb{R} \; \big(L \text{ is a cluster value of } (x_n) \implies L \le \limsup_{n\to\infty} x_n \big) \Big] \\
&\quad \wedge \Big[\liminf_{n\to\infty} x_n \text{ is a cluster value of } (x_n) \wedge \forall L \in \mathbb{R} \; \big(L \text{ is a cluster value of } (x_n) \implies \liminf_{n\to\infty} x_n \le L \big) \Big] \Bigg).
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\text{BoundedSequence}(x_n) \implies \\
&\quad \text{SequenceClusterValue}\big(x_n, \limsup_{n\to\infty} x_n\big) \wedge \forall L \in \mathbb{R} \; \big(\text{SequenceClusterValue}(x_n, L) \implies L \le \limsup_{n\to\infty} x_n\big), \\
&\text{BoundedSequence}(x_n) \implies \\
&\quad \text{SequenceClusterValue}\big(x_n, \liminf_{n\to\infty} x_n\big) \wedge \forall L \in \mathbb{R} \; \big(\text{SequenceClusterValue}(x_n, L) \implies \liminf_{n\to\infty} x_n \le L\big).
\end{aligned}$$

Remark (Interpretation). Limsup and liminf describe the top and bottom edges of the sequence's subsequential behavior.

Remark (Dependencies).

- Limit Superior of a Sequence

- Limit Inferior of a Sequence
- Limsup Is the Largest Subsequential Limit
- Liminf Is the Smallest Subsequential Limit
- Cluster Values Are Subsequential Limits

6.11 Cauchy

Cauchy Sequences

Cauchy sequences measure internal stabilization. Instead of comparing the terms to an already named limit, the Cauchy condition compares sufficiently late terms to each other. Over the real numbers this internal criterion is equivalent to convergence, and that equivalence is one of the central sequence forms of completeness.

Definition (Cauchy Sequence)

Definition 6.11.1 (Cauchy Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is a **Cauchy sequence** if

$$\forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall m, n \geq N\, \big(|x_m - x_n| < \varepsilon\big).$$

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is Cauchy}\\ &\quad\iff\quad \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall m, n \geq N\, \big(|x_m - x_n| < \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\operatorname{CauchySequence}(x_n) :\!= \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall m, n \geq N\, \big(|x_m - x_n| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&(x_n) \text{ is not Cauchy}\\ &\quad\iff\quad \exists \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists m, n \geq N\, \big(|x_m - x_n| \geq \varepsilon\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg \operatorname{CauchySequence}(x_n) \iff \exists \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists m, n \geq N\, \big(|x_m - x_n| \geq \varepsilon\big).$$

Remark (Interpretation). The Cauchy condition says that the tails of the sequence eventually have arbitrarily small internal spread. It does not name a candidate limit; it only asserts that late terms become mutually close.

Remark (Dependencies).

- Real Sequence

Theorem (Convergent Sequences Are Cauchy)

Theorem 6.11.2 (Convergent Sequences Are Cauchy)**.** *Let (x_n) be a real sequence. If (x_n) converges, then (x_n) is Cauchy.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\\ &\quad \Big(\big(\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon) \big)\\ &\qquad \Longrightarrow \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon) \Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R}\ \Big(\exists L \in \mathbb{R}\ \ \mathrm{ConvergentSequence}(x_n, L) \Longrightarrow \mathrm{CauchySequence}(x_n) \Big).$$

Remark (Interpretation)*.* Once every sufficiently late term is close to the same limit, any two sufficiently late terms are close to each other. This theorem is the easy half of the Cauchy criterion.

Remark (Dependencies).

- Convergence of a Sequence
- Cauchy Sequence

Theorem (Cauchy Sequences Are Bounded)

Theorem 6.11.3 (Cauchy Sequences Are Bounded)**.** *Every Cauchy real sequence is bounded.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\\ &\quad \Big(\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon)\\ &\qquad \Longrightarrow \exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \leq M) \Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R}\ \big(\mathrm{CauchySequence}(x_n) \Longrightarrow \mathrm{BoundedSequence}(x_n) \big).$$

Remark (Interpretation). A Cauchy sequence has one tightly controlled tail. The finitely many terms before that tail can be absorbed into a single larger bound.

Remark (Dependencies).

- Cauchy Sequence
- Bounded Sequence

Theorem (Cauchy Sequence with a Convergent Subsequence)

Theorem 6.11.4 (Cauchy Sequence with a Convergent Subsequence)**.** *Let (x_n) be a Cauchy real sequence. If a subsequence of (x_n) converges to $L \in \mathbb{R}$, then (x_n) converges to L.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}$$
$$\Big(\big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon)\big]$$
$$\wedge \big[\exists \sigma : \mathbb{N} \to \mathbb{N}\ \big(\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \wedge \forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall k \geq K\ (|x_{\sigma(k)} - L| < \varepsilon)\big)\big]$$
$$\Longrightarrow \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon)\Big).$$

Remark (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}$$
$$\Big(\mathrm{CauchySequence}(x_n) \wedge \mathrm{SubsequentialLimit}(x_n, L) \Longrightarrow \mathrm{ConvergentSequence}(x_n, L)\Big).$$

Remark (Interpretation). The Cauchy condition prevents the rest of the tail from wandering away from any convergent subsequence. A single subsequential limit therefore determines the limit of the whole sequence.

Remark (Dependencies).

- Cauchy Sequence
- Subsequence
- Subsequential Limit
- Convergence of a Sequence

Theorem (Cauchy Criterion for Real Sequences)

Theorem 6.11.5 (Cauchy Criterion for Real Sequences)**.** *Let (x_n) be a real sequence. Then (x_n) converges if and only if (x_n) is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\\&\quad\Big(\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon)\\&\qquad\iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon)\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R}\ \Big(\exists L \in \mathbb{R}\ \text{ConvergentSequence}(x_n, L) \iff \text{CauchySequence}(x_n)\Big).$$

Remark (Interpretation). This theorem is the sequence form of completeness of $\mathbb{R}$. In an incomplete ambient space, a sequence may be internally stable without converging inside that space.

Remark (Dependencies).

- Convergent Sequences Are Cauchy
- Cauchy Sequences Are Bounded
- Bolzano-Weierstrass Theorem for Sequences
- Cauchy Sequence with a Convergent Subsequence

Theorem (Cauchy Criterion via Tails)

Theorem 6.11.6 (Cauchy Criterion via Tails)**.** *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that the N-tail has pairwise term distances less than ε.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is Cauchy}\\&\quad\iff\quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall p, q \in \mathbb{N}\ \big(|x_{N+p} - x_{N+q}| < \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall p, q \in \mathbb{N}\ \big(|x_{N+p} - x_{N+q}| < \varepsilon\big).$$

Remark (Interpretation). The Cauchy condition is a tail property. Reindexing a sufficiently late tail does not change the substance of the condition; it only changes the names of the indices.

Remark (Dependencies).

- Cauchy Sequence

- M-Tail of a Sequence

Theorem (Cauchy Tail Diameter Criterion)

Theorem 6.11.7 (Cauchy Tail Diameter Criterion)**.** *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if every sufficiently late tail has arbitrarily small pairwise diameter.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &(x_n) \text{ is Cauchy} \\ &\iff \quad \forall \varepsilon > 0\ \exists N_0 \in \mathbb{N}\ \forall N \geq N_0 \\ &\qquad \forall m, n \geq N\ \big(|x_m - x_n| < \varepsilon\big). \end{aligned}$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \iff \forall \varepsilon > 0\ \exists N_0 \in \mathbb{N}\ \forall N \geq N_0\ \forall m, n \geq N\ \big(|x_m - x_n| < \varepsilon\big).$$

Remark (Interpretation). The tail-diameter form says that once one tail is small, every later tail is small as well. It records the monotone nature of tail control without introducing additional notation for diameters.

Remark (Dependencies).

- Cauchy Sequence
- M-Tail of a Sequence

Theorem (Cauchy Sequences Have Vanishing Successive Differences)

Theorem 6.11.8 (Cauchy Sequences Have Vanishing Successive Differences)**.** *If (x_n) is a Cauchy real sequence, then*

$$|x_{n+1} - x_n| \to 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall (x_n) \subseteq \mathbb{R} \\ &\quad \Big(\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon) \\ &\qquad \implies \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_{n+1} - x_n| < \varepsilon)\Big). \end{aligned}$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \implies \text{NullSequence}\big(|x_{n+1} - x_n|\big).$$

Remark (Interpretation)*.* Cauchy control applies to every pair of late indices, so it applies in particular to consecutive late indices. The converse is false in general: small successive steps do not prevent accumulated drift.

Remark (Dependencies)*.*

- Cauchy Sequence
- Null Sequence

Theorem (Scalar Multiples of Cauchy Sequences)

Theorem 6.11.9 (Scalar Multiples of Cauchy Sequences)**.** *Let (x_n) be a Cauchy real sequence, and let $\alpha \in \mathbb{R}$. Then (αx_n) is Cauchy.*
Go to proof.

Remark (Standard quantified statement)*.*

$$\forall (x_n) \subseteq \mathbb{R}\ \forall \alpha \in \mathbb{R} \\ \Big((x_n) \text{ is Cauchy} \implies (\alpha x_n) \text{ is Cauchy} \Big).$$

Remark (Definition predicate reading)*.*

$$\text{CauchySequence}(x_n) \implies \text{CauchySequence}(\alpha x_n).$$

Remark (Interpretation)*.* Multiplying all terms by one fixed real number rescales all pairwise tail distances by the same fixed factor.

Remark (Dependencies)*.*

- Cauchy Sequence
- Scalar Multiple of a Sequence

Theorem (Sums of Cauchy Sequences)

Theorem 6.11.10 (Sums of Cauchy Sequences)**.** *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n + y_n)$ is Cauchy.*
Go to proof.

Remark (Standard quantified statement)*.*

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R} \\ \Big((x_n) \text{ is Cauchy} \land (y_n) \text{ is Cauchy} \implies (x_n + y_n) \text{ is Cauchy} \Big).$$

Remark (Definition predicate reading)*.*

$$\text{CauchySequence}(x_n) \land \text{CauchySequence}(y_n) \implies \text{CauchySequence}(x_n + y_n).$$

Remark (Interpretation). The triangle inequality splits the tail distance of the sum into the two tail distances already controlled by the original sequences.

Remark (Dependencies).

- Cauchy Sequence
- Pointwise Sum of Sequences

Theorem (Differences of Cauchy Sequences)

Theorem 6.11.11 (Differences of Cauchy Sequences)**.** *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n - y_n)$ is Cauchy.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \implies (x_n - y_n) \text{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \mathrm{CauchySequence}(y_n) \implies \mathrm{CauchySequence}(x_n - y_n).$$

Remark (Interpretation). The difference theorem is the sum theorem applied after negating one sequence. It is often the form used to compare two approximating sequences.

Remark (Dependencies).

- Sums of Cauchy Sequences
- Scalar Multiples of Cauchy Sequences
- Pointwise Difference of Sequences

Theorem (Linear Combinations of Cauchy Sequences)

Theorem 6.11.12 (Linear Combinations of Cauchy Sequences)**.** *Let (x_n) and (y_n) be Cauchy real sequences, and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha x_n + \beta y_n)$ is Cauchy.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}\ \forall \alpha \in \mathbb{R}\ \forall \beta \in \mathbb{R}$$
$$\Big((x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \implies (\alpha x_n + \beta y_n) \text{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \mathrm{CauchySequence}(y_n) \implies \mathrm{CauchySequence}(\alpha x_n + \beta y_n).$$

Remark (Interpretation). The Cauchy real sequences are closed under finite real linear combinations. This is the additive vector-space part of Cauchy algebra.

Remark (Dependencies).

- Linear Combination of Real Sequences
- Scalar Multiples of Cauchy Sequences
- Sums of Cauchy Sequences

Theorem (Products of Cauchy Sequences)

Theorem 6.11.13 (Products of Cauchy Sequences)**.** *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n y_n)$ is Cauchy.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \implies (x_n y_n) \text{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \wedge \text{CauchySequence}(y_n) \implies \text{CauchySequence}(x_n y_n).$$

Remark (Interpretation). Products require boundedness in addition to tail closeness. The boundedness is supplied automatically because every Cauchy real sequence is bounded.

Remark (Dependencies).

- Cauchy Sequences Are Bounded
- Pointwise Product of Sequences

Theorem (Reciprocals of Cauchy Sequences)

Theorem 6.11.14 (Reciprocals of Cauchy Sequences)**.** *Let (x_n) be a Cauchy real sequence. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|x_n| \geq c$ for every $n \geq N_0$, then $(1/x_n)$ is Cauchy.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \text{ is Cauchy} \wedge \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|x_n| \geq c)$$
$$\implies (1/x_n) \text{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|x_n| \geq c) \Longrightarrow \mathrm{CauchySequence}(1/x_n).$$

Remark (Interpretation). The lower bound away from zero prevents division from amplifying small tail differences without control. Without it, reciprocals need not preserve Cauchy behavior.

Remark (Dependencies).

- Cauchy Sequence
- Reciprocal Sequence

Theorem (Quotients of Cauchy Sequences)

Theorem 6.11.15 (Quotients of Cauchy Sequences)**.** *Let (x_n) and (y_n) be Cauchy real sequences. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|y_n| \geq c$ for every $n \geq N_0$, then (x_n/y_n) is Cauchy.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \wedge \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|y_n| \geq c)$$
$$\Longrightarrow (x_n/y_n) \text{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \mathrm{CauchySequence}(y_n) \wedge \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|y_n| \geq c) \Longrightarrow \mathrm{CauchySequence}(x_n/$$

Remark (Interpretation). The quotient theorem is the product theorem combined with the reciprocal theorem for a denominator eventually bounded away from zero.

Remark (Dependencies).

- Products of Cauchy Sequences
- Reciprocals of Cauchy Sequences
- Pointwise Quotient of Sequences

Theorem (Absolute Values of Cauchy Sequences)

Theorem 6.11.16 (Absolute Values of Cauchy Sequences)**.** *Let (x_n) be a Cauchy real sequence. Then $(|x_n|)$ is Cauchy.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left((x_n) \text{ is Cauchy} \implies (|x_n|) \text{ is Cauchy} \right).$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \implies \text{CauchySequence}(|x_n|).$$

Remark (Interpretation). The reverse triangle inequality bounds the distance between absolute values by the original distance between terms.

Remark (Dependencies).

- Cauchy Sequence
- Absolute Value of a Sequence

6.12 Applications of Sequences

Sequence Applications — Quick Reference

Construction	Role
Newton sequence for $\sqrt{2}$	Iterative approximation model
Newton approximation	Converges to $\sqrt{2}$
Factorial partial sums	Series approximants for e
Approximation of e	Factorial sums converge to e
Compound-interest sequence	Classical exponential approximants
Compound-interest approximation	Compound interest converges to e
Decimal truncation sequence	Decimal approximants from below
Decimal truncations	Truncations converge to the target real

Applications of Sequences

The convergence theorems for sequences become numerical tools once the sequence is computable. Newton iteration turns an equation into a convergent recursive approximation; factorial partial sums and compound-interest terms approximate the Euler number; decimal truncations approximate arbitrary real numbers by finite decimals.

Definition (Newton Sequence for $\sqrt{2}$)

Definition 6.12.1 (Newton Sequence for $\sqrt{2}$)**.** The **Newton sequence for** $\sqrt{2}$ is the real sequence (x_n) defined by

$$x_1 := \frac{3}{2}, \qquad x_{n+1} := \frac{1}{2}\left(x_n + \frac{2}{x_n}\right) \quad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$x_1 = \frac{3}{2} \quad \wedge \quad \forall n \in \mathbb{N} \left(x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right) \right).$$

Remark (Interpretation). The recursion is Newton's method applied to the equation $x^2 - 2 = 0$. Each new term averages the current estimate with the correction $2/x_n$.

Remark (Dependencies).

- Real Sequence

Theorem (Newton Approximation of $\sqrt{2}$)

Theorem 6.12.2 (Newton Approximation of $\sqrt{2}$)**.** *Let (x_n) be the Newton sequence for $\sqrt{2}$. Then (x_n) converges and*

$$\lim_{n\to\infty} x_n = \sqrt{2}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \left(|x_n - \sqrt{2}| < \varepsilon \right).$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(x_n, \sqrt{2}).$$

Remark (Interpretation). The Newton sequence is decreasing after the first term and bounded below by $\sqrt{2}$, so monotone convergence supplies a limit. Passing the recursion to the limit identifies the limit as the positive solution of $L^2 = 2$.

Remark (Dependencies).

- Newton Sequence for $\sqrt{2}$
- Monotone Convergence Theorem
- Rational Sequence Limit
- Square Root Sequence

Definition (Factorial Partial Sums)

Definition 6.12.3 (Factorial Partial Sums)**.** The **factorial partial-sum sequence** is the real sequence (s_n) defined by

$$s_n := \sum_{k=0}^{n} \frac{1}{k!} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(s_n = \sum_{k=0}^{n} \frac{1}{k!} \right).$$

Remark (Interpretation). The sequence adds the reciprocal factorial terms one at a time. The terms decrease very rapidly, so the partial sums stabilize quickly.

Remark (Dependencies).

- Real Sequence

Theorem (Factorial Partial Sums Approximate e)

Theorem 6.12.4 (Factorial Partial Sums Approximate e)**.** *Let (s_n) be the factorial partial-sum sequence. Then (s_n) converges to the Euler number e:*

$$\lim_{n\to\infty} s_n = e.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N \; \big(|s_n - e| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\mathrm{ConvergentSequence}(s_n, e).$$

Remark (Interpretation). This is the sequence form of the factorial-series definition of e. In practice the approximation is efficient because the error after n terms is controlled by a rapidly shrinking factorial tail.

Remark (Dependencies).

- Factorial Partial Sums
- The Number e
- Cauchy Criterion for Real Sequences

Definition (Compound-Interest Sequence)

Definition 6.12.5 (Compound-Interest Sequence)**.** The **compound-interest sequence** is the real sequence (c_n) defined by

$$c_n := \left(1 + \frac{1}{n}\right)^n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N}\ \left(c_n = \left(1 + \frac{1}{n}\right)^n\right).$$

Remark (Interpretation). The term c_n is the value of one unit of principal after one unit of time when interest is compounded n times at nominal rate 1.

Remark (Dependencies).

- Real Sequence

Theorem (Compound-Interest Approximation of e)

Theorem 6.12.6 (Compound-Interest Approximation of e)**.** *Let (c_n) be the compound-interest sequence. Then*

$$\lim_{n\to\infty} c_n = e.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|c_n - e| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(c_n, e).$$

Remark (Interpretation). The compound-interest approximation gives a second natural sequence converging to the same constant as the factorial partial sums. It is slower numerically, but it explains why e appears in continuous growth.

Remark (Dependencies).

- Compound-Interest Sequence
- The Number e
- Sequence Squeeze Theorem

Definition (Decimal Truncation Sequence)

Definition 6.12.7 (Decimal Truncation Sequence)**.** Let $\alpha \in \mathbb{R}$. The **decimal truncation sequence** of α is the real sequence (d_n) defined by

$$d_n := \frac{\lfloor 10^n \alpha \rfloor}{10^n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N}\ \left(d_n = \frac{\lfloor 10^n \alpha \rfloor}{10^n}\right).$$

Remark (Interpretation). The number d_n is obtained by cutting α off after n decimal places. Truncation is one-sided: it does not round to the nearest decimal.

Remark (Dependencies).

- Real Sequence

Theorem (Decimal Truncations Converge)

Theorem 6.12.8 (Decimal Truncations Converge)**.** *Let $\alpha \in \mathbb{R}$, and let (d_n) be the decimal truncation sequence of α. Then*

$$\lim_{n\to\infty} d_n = \alpha.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|d_n - \alpha| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(d_n, \alpha).$$

Remark (Interpretation). Decimal truncations converge because the truncation error is always less than 10^{-n}. This is one of the most concrete examples of convergence: finite decimal data can approximate a real number to arbitrary accuracy.

Remark (Dependencies).

- Decimal Truncation Sequence
- Sequence Squeeze Theorem

Chapter 7

Series

SERIES
From Cantor to Ito

VOLUME III
Classical Analysis

BOOK I
Bounds, Sequences, and Series

CHAPTER 7
Series

7.1 Finite Series

Remark (Exposition). Tao introduces finite series by first fixing a finite real sequence indexed by consecutive integers and then defining its sum recursively Tao [2006]. The source bundles these conventions into one definition; here they are split into atomic definitions.

Definition (Finite Real Sequence on an Integer Interval)

Definition 7.1.1 (Finite Real Sequence on an Integer Interval)**.** Let $m, n \in \mathbb{Z}$ with $m \leq n$. A finite real sequence indexed from m to n is an assignment of a real number a_i to each integer i satisfying $m \leq i \leq n$. We denote such a sequence by $(a_i)_{i=m}^n$.

Remark (Interpretation). The indices form the integer interval $\{m, m+1, \ldots, n\}$. The notation $(a_i)_{i=m}^n$ records both the terms and the finite range of indices on which the terms are defined.

Remark (Dependencies).

- Real Sequence

Definition (Empty Finite Sum)

Definition 7.1.2 (Empty Finite Sum)**.** Let $m, n \in \mathbb{Z}$. If $n < m$, the finite sum from m to n is defined by

$$\sum_{i=m}^{n} a_i := 0.$$

Remark (Interpretation). When the upper index lies below the lower index, there are no terms to add. The sum is therefore assigned the additive identity 0.

Remark (Dependencies).

- Finite Real Sequence on an Integer Interval

Definition (Finite Sum)

Definition 7.1.3 (Finite Sum)**.** Let $m, n \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant integer indices. The notation

$$\sum_{i=m}^{n} a_i$$

denotes the finite sum, or finite series, of the terms indexed from m to n.

Remark (Standard quantified statement).

$$\sum_{i=m}^{n} a_i \quad \text{denotes the finite sum of the terms } a_i \text{ indexed by } m \leq i \leq n.$$

Remark (Interpretation). This definition fixes the notation and name of the object being defined. The value of the object is then determined by the empty-sum convention and the recursive rule below.

Remark (Dependencies).

- Finite Real Sequence on an Integer Interval

Definition (Recursive Finite Sum)

Definition 7.1.4 (Recursive Finite Sum)**.** Let $m \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant indices. The finite sum, also called a finite series, is defined recursively from the empty finite sum by the rule

$$\sum_{i=m}^{n+1} a_i := \left(\sum_{i=m}^{n} a_i\right) + a_{n+1} \qquad \text{whenever } n \geq m - 1.$$

Remark (Standard quantified statement).

$$\forall m, n \in \mathbb{Z}, \quad n \geq m - 1 \implies \sum_{i=m}^{n+1} a_i := \left(\sum_{i=m}^{n} a_i\right) + a_{n+1}.$$

Remark (Interpretation). The recursive rule says that extending the upper endpoint by one index extends the sum by adding exactly the new last term. Together with the empty-sum convention, this determines every finite sum.

Remark (Examples). For a sequence indexed from m, the first few cases are

$$\sum_{i=m}^{m-2} a_i = 0, \qquad \sum_{i=m}^{m-1} a_i = 0, \qquad \sum_{i=m}^{m} a_i = a_m,$$

and

$$\sum_{i=m}^{m+1} a_i = a_m + a_{m+1}, \qquad \sum_{i=m}^{m+2} a_i = a_m + a_{m+1} + a_{m+2}.$$

Remark (Dependencies).

- Empty Finite Sum
- Finite Sum
- Finite Real Sequence on an Integer Interval

Lemma (Splitting a Finite Sum)

Lemma 7.1.5 (Splitting a Finite Sum)**.** *Let $m \le n < p$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \le i \le p$. Then*

$$\sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p} a_i = \sum_{i=m}^{p} a_i.$$

Go to proof.

Remark (Interpretation). A finite sum over one integer interval may be split into two adjacent finite sums without changing its value.

Remark (Dependencies).

- Recursive Finite Sum

Definition (Summation over a Finite Set)

Definition 7.1.6 (Summation over a Finite Set)**.** Let X be a finite set with n elements, where $n \in \mathbb{N}$, and let $f : X \to \mathbb{R}$. Choose a bijection

$$g : \{i \in \mathbb{N} : 1 \le i \le n\} \to X.$$

The finite sum of f over X is defined by

$$\sum_{x \in X} f(x) := \sum_{i=1}^{n} f(g(i)).$$

Remark (Standard quantified statement).

$$X = \{g(1), \dots, g(n)\} \implies \sum_{x \in X} f(x) := \sum_{i=1}^{n} f(g(i)).$$

Remark (Interpretation). The definition computes a sum over an unordered finite set by first listing its elements through a bijection from $\{1, \ldots, n\}$. The notation $\sum_{x \in X} f(x)$ records the set being summed over; the chosen listing is auxiliary.

Remark (Dependencies).

- Finite Sum

Proposition (Finite Summations Are Well-Defined)

Proposition 7.1.7 (Finite Summations Are Well-Defined). *Let X be a finite set with n elements, where $n \in \mathbb{N}$, let $f : X \to \mathbb{R}$, and let*

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X \qquad \textit{and} \qquad h : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X$$

be bijections. Then

$$\sum_{i=1}^{n} f(g(i)) = \sum_{i=1}^{n} f(h(i)).$$

Go to proof.

Remark (Interpretation). The value of a finite summation over a finite set does not depend on which bijection is used to list the elements of that set. Tao notes that the corresponding issue for summations over infinite sets is more complicated.

Remark (Dependencies).

- Summation over a Finite Set

Proposition (Empty Finite Set Summation)

Proposition 7.1.8 (Empty Finite Set Summation). *If X is empty, and $f : X \to \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = 0.$$

Go to proof.

Remark (Interpretation). The finite summation over the empty set has value 0.

Remark (Dependencies).

- Summation over a Finite Set

Proposition (Singleton Finite Set Summation)

Proposition 7.1.9 (Singleton Finite Set Summation). *If X consists of a single element, $X = \{x_0\}$, and $f : X \to \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = f(x_0).$$

Go to proof.

Remark (Interpretation). Summing a function over a one-element set gives the value of the function at that element.

Remark (Dependencies).

- Summation over a Finite Set

Proposition (Substitution for Finite Set Summation)

Proposition 7.1.10 (Substitution for Finite Set Summation)**.** *If X is a finite set, $f : X \to \mathbb{R}$ is a function, and $g : Y \to X$ is a bijection, then*

$$\sum_{x \in X} f(x) = \sum_{y \in Y} f(g(y)).$$

Go to proof.

Remark (Interpretation). A finite summation over X may be computed by summing over any set bijective with X, after composing the summand with the bijection.

Remark (Dependencies).

- Summation over a Finite Set
- Finite Summations Are Well-Defined

Proposition (Integer Interval as a Finite Set)

Proposition 7.1.11 (Integer Interval as a Finite Set)**.** *Let $n \leq m$ be integers, and let*

$$X := \{i \in \mathbb{Z} : n \leq i \leq m\}.$$

If a_i is a real number assigned to each integer $i \in X$, then

$$\sum_{i=n}^{m} a_i = \sum_{i \in X} a_i.$$

Go to proof.

Remark (Interpretation). Summing over a consecutive integer interval agrees with summing over that same interval regarded as a finite set.

Remark (Dependencies).

- Finite Sum
- Summation over a Finite Set

Proposition (Disjoint Union Finite Set Summation)

Proposition 7.1.12 (Disjoint Union Finite Set Summation)**.** *Let X and Y be disjoint finite sets, so $X \cap Y = \emptyset$, and let $f : X \cup Y \to \mathbb{R}$ be a function. Then*

$$\sum_{z \in X \cup Y} f(z) = \left(\sum_{x \in X} f(x) \right) + \left(\sum_{y \in Y} f(y) \right).$$

Go to proof.

Remark (Interpretation)*.* A sum over a disjoint union splits into the sum over each component set.

Remark (Dependencies)*.*

- Summation over a Finite Set

Proposition (Addition Rule for Finite Set Summation)

Proposition 7.1.13 (Addition Rule for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions. Then*

$$\sum_{x \in X} (f(x) + g(x)) = \sum_{x \in X} f(x) + \sum_{x \in X} g(x).$$

Go to proof.

Remark (Interpretation)*.* Finite set summation distributes over pointwise addition.

Remark (Dependencies)*.*

- Summation over a Finite Set

Proposition (Scalar Multiplication Rule for Finite Set Summation)

Proposition 7.1.14 (Scalar Multiplication Rule for Finite Set Summation)**.** *Let X be a finite set, let $f : X \to \mathbb{R}$ be a function, and let c be a real number. Then*

$$\sum_{x \in X} c f(x) = c \sum_{x \in X} f(x).$$

Go to proof.

Remark (Interpretation)*.* A constant scalar may be pulled outside a finite set summation.

Remark (Dependencies)*.*

- Summation over a Finite Set

Proposition (Monotonicity for Finite Set Summation)

Proposition 7.1.15 (Monotonicity for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions such that $f(x) \le g(x)$ for all $x \in X$. Then*

$$\sum_{x\in X} f(x) \le \sum_{x\in X} g(x).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\big(\forall x \in X,\ f(x) \le g(x)\big) \implies \sum_{x\in X} f(x) \le \sum_{x\in X} g(x).$$

Remark (Interpretation)*.* Pointwise order between two functions on a finite set gives the same order between their finite sums.

Remark (Dependencies)*.*

- Summation over a Finite Set

Proposition (Triangle Inequality for Finite Set Summation)

Proposition 7.1.16 (Triangle Inequality for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ be a function. Then*

$$\left|\sum_{x\in X} f(x)\right| \le \sum_{x\in X} |f(x)|.$$

Go to proof.

Remark (Interpretation)*.* The absolute value of a finite set summation is bounded by the corresponding finite set summation of absolute values.

Remark (Dependencies)*.*

- Summation over a Finite Set

Lemma (Iterated Summation over a Product)

Lemma 7.1.17 (Iterated Summation over a Product)**.** *Let X and Y be finite sets, and let $f : X \times Y \to \mathbb{R}$ be a function. Then*

$$\sum_{x\in X}\left(\sum_{y\in Y} f(x,y)\right) = \sum_{(x,y)\in X\times Y} f(x,y).$$

Go to proof.

Remark (Interpretation). Summing first over Y for each $x \in X$, and then summing those values over X, gives the same value as summing over the product set $X \times Y$.

Remark (Dependencies).

- Summation over a Finite Set

Corollary (Fubini's Theorem for Finite Series)

Corollary 7.1.18 (Fubini's Theorem for Finite Series). *Let X and Y be finite sets, and let $f : X \times Y \to \mathbb{R}$ be a function. Then*

$$\sum_{x\in X}\left(\sum_{y\in Y} f(x,y)\right) = \sum_{(x,y)\in X\times Y} f(x,y) = \sum_{(y,x)\in Y\times X} f(x,y) = \sum_{y\in Y}\left(\sum_{x\in X} f(x,y)\right).$$

Go to proof.

Remark (Standard quantified statement).

$$\sum_{x\in X}\left(\sum_{y\in Y} f(x,y)\right) = \sum_{y\in Y}\left(\sum_{x\in X} f(x,y)\right).$$

Remark (Interpretation). For a function on a finite product set, the order of summation over the two finite factors may be interchanged.

Remark (Dependencies).

- Iterated Summation over a Product
- Summation over a Finite Set

Lemma (Reindexing a Finite Sum)

Lemma 7.1.19 (Reindexing a Finite Sum). *Let $m \le n$ be integers, let k be an integer, and let a_i be a real number assigned to each integer i satisfying $m \le i \le n$. Then*

$$\sum_{i=m}^{n} a_i = \sum_{j=m+k}^{n+k} a_{j-k}.$$

Go to proof.

Remark (Interpretation). Shifting every index by the same integer changes the labels of the summation but not the ordered list of terms being summed.

Remark (Dependencies).

- Recursive Finite Sum

Lemma (Finite Sum of Sums)

Lemma 7.1.20 (Finite Sum of Sums)**.** *Let $m \le n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \le i \le n$. Then*

$$\sum_{i=m}^{n}(a_i + b_i) = \left(\sum_{i=m}^{n} a_i\right) + \left(\sum_{i=m}^{n} b_i\right).$$

Go to proof.

Remark (Interpretation)*.* Finite summation distributes over termwise addition.

Remark (Dependencies)*.*

- Recursive Finite Sum

Lemma (Finite Sum of Scalar Multiples)

Lemma 7.1.21 (Finite Sum of Scalar Multiples)**.** *Let $m \le n$ be integers, let a_i be a real number assigned to each integer i satisfying $m \le i \le n$, and let $c \in \mathbb{R}$. Then*

$$\sum_{i=m}^{n}(ca_i) = c\left(\sum_{i=m}^{n} a_i\right).$$

Go to proof.

Remark (Interpretation)*.* A common scalar factor may be pulled outside a finite sum.

Remark (Dependencies)*.*

- Recursive Finite Sum

Lemma (Triangle Inequality for Finite Series)

Lemma 7.1.22 (Triangle Inequality for Finite Series)**.** *Let $m \le n$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \le i \le n$. Then*

$$\left|\sum_{i=m}^{n} a_i\right| \le \sum_{i=m}^{n} |a_i|.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall m, n \in \mathbb{Z}, \quad m \le n \Longrightarrow \left|\sum_{i=m}^{n} a_i\right| \le \sum_{i=m}^{n} |a_i|.$$

Remark (Interpretation)*.* The absolute value of a finite series is bounded by the corresponding finite series of absolute values.

Remark (Dependencies).

- Recursive Finite Sum

Lemma (Comparison Test for Finite Series)

Lemma 7.1.23 (Comparison Test for Finite Series)**.** *Let $m \le n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \le i \le n$. Suppose that $a_i \le b_i$ for all i satisfying $m \le i \le n$. Then*

$$\sum_{i=m}^{n} a_i \le \sum_{i=m}^{n} b_i.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i\, (m \le i \le n \Longrightarrow a_i \le b_i) \Longrightarrow \sum_{i=m}^{n} a_i \le \sum_{i=m}^{n} b_i.$$

Remark (Interpretation). Termwise comparison of two finite series gives comparison of their finite sums.

Remark (Dependencies).

- Recursive Finite Sum

7.2 Infinite Series

Definition (Formal Infinite Series)

Definition 7.2.1 (Formal Infinite Series)**.** A formal infinite series is an expression of the form

$$\sum_{n=m}^{\infty} a_n,$$

where m is an integer, and a_n is a real number for every integer $n \ge m$. This series is sometimes written as

$$a_m + a_{m+1} + a_{m+2} + \cdots .$$

Remark (Standard quantified statement).

$$\sum_{n=m}^{\infty} a_n \quad \text{denotes a formal infinite series with } m \in \mathbb{Z} \text{ and } a_n \in \mathbb{R} \text{ for every } n \ge m.$$

Remark (Interpretation). The notation records an infinite list of real terms beginning at the integer index m. At this stage it is a formal expression rather than a claim that a value has already been assigned to the series.

Remark (Dependencies).

- Real Sequence

Chapter 8

Function Sequences

SERIES	VOLUME III	BOOK I	CHAPTER 8
From Cantor to Ito	Classical Analysis	Bounds, Sequences, and Series	**Function Sequences**

Chapter 9

Elementary Functions

9.1 Hyperbolic Functions

Hyperbolic Functions — Quick Reference

Concept	Meaning
Hyperbolic functions	sinh and cosh from even/odd exponential parts
Hyperbolic Pythagorean identity	$\cosh^2 x - \sinh^2 x = 1$
Addition formulas	Hyperbolic angle-addition identities
Tanh limits	$\tanh x \to \pm 1$ as $x \to \pm\infty$
Inverse hyperbolic functions	Logarithmic inverse formulas

Hyperbolic Functions

Definition (Hyperbolic Functions)

Definition 9.1.1 (Hyperbolic Functions)**.** For all $x \in \mathbb{R}$:

$$\cosh x := \frac{e^x + e^{-x}}{2}, \qquad \sinh x := \frac{e^x - e^{-x}}{2}, \qquad \tanh x := \frac{\sinh x}{\cosh x}.$$

Additional: $\operatorname{sech} x := 1/\cosh x$, $\operatorname{csch} x := 1/\sinh x$, $\coth x := \cosh x/\sinh x$.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \ \cosh x = \frac{e^x + e^{-x}}{2} \geq 1, \ \sinh x = \frac{e^x - e^{-x}}{2}, \ e^x = \cosh x + \sinh x.$$

Remark (Definition predicate reading).

$$\mathrm{HyperbolicCosh}(x,y) \;:\equiv\; y = \frac{e^x + e^{-x}}{2}. \qquad \mathrm{HyperbolicSinh}(x,y) \;:\equiv\; y = \frac{e^x - e^{-x}}{2}.$$

$$\mathrm{HyperbolicTanh}(x,y) \;:\equiv\; \cosh x \neq 0 \;\wedge\; y = \frac{\sinh x}{\cosh x}.$$

Remark (Definition predicate reading).

$\mathrm{HyperbolicPythagorean}(x) \;:\equiv\; \mathrm{HyperbolicCosh}(x,c) \wedge \mathrm{HyperbolicSinh}(x,s) \wedge c^2 - s^2 = 1.$

Remark (Negated quantified statement).

$$\neg\, \mathrm{HyperbolicTanh}(x,y) \iff y \neq \frac{\sinh x}{\cosh x}.$$

Note $\cosh x > 0$ for all x, so HyperbolicTanh is defined on all of $\mathbb{R}$.

Remark (Contrast with trigonometric functions).

Trigonometric	**Hyperbolic**
$\cos^2 x + \sin^2 x = 1$	$\cosh^2 x - \sinh^2 x = 1$
$(\cos x)' = -\sin x$	$(\cosh x)' = \sinh x$
$(\sin x)' = \cos x$	$(\sinh x)' = \cosh x$
$e^{ix} = \cos x + i \sin x$	$e^x = \cosh x + \sinh x$
Parametrises unit circle	Parametrises unit hyperbola

The minus sign in $\cosh^2 - \sinh^2 = 1$ versus $\cos^2 + \sin^2 = 1$ reflects the substitution $x \mapsto ix$ in Euler's formula.

Proposition (Hyperbolic Pythagorean Identity)

Remark (Dependencies).

- Exponential Function
- $\mathbb{R}$ Is a Field

Proposition 9.1.2 (Hyperbolic Pythagorean Identity)**.** *For all $x \in \mathbb{R}$:*

$$\cosh^2 x - \sinh^2 x = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} :\; \cosh^2 x - \sinh^2 x = 1.$$

Remark (Negated quantified statement).

$$\neg\,(\forall x : \cosh^2 x - \sinh^2 x = 1) \iff \exists x \in \mathbb{R} : \cosh^2 x - \sinh^2 x \neq 1.$$

The negation has no witness — the identity holds universally.

Remark (Proof). Direct computation:

$$\cosh^2 x - \sinh^2 x = \frac{(e^x + e^{-x})^2}{4} - \frac{(e^x - e^{-x})^2}{4} = \frac{4}{4} = 1.$$

Proposition (Addition Formulas for Hyperbolic Functions)

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- $\mathbb{R}$ Is a Field

Proposition 9.1.3 (Addition Formulas for Hyperbolic Functions)**.** *For all $x, y \in \mathbb{R}$:*

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y, \qquad \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R} : \; \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \wedge \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y. \blacksquare$$

Remark (Definition predicate reading).

$$\mathrm{HyperbolicAddition}(x, y) \; :\equiv \; \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \wedge \cosh(x+y) = \cosh x \cosh y + \sinh x$$

Remark (Negated quantified statement).

$$\neg\,(\text{addition formula for } \sinh) \iff \exists x, y \in \mathbb{R} : \; \sinh(x+y) \neq \sinh x \cosh y + \cosh x \sinh y.$$

The negation has no witness.

Remark (Proof strategy). Substitute the definitions and expand, or note the formulas follow from $e^{x+y} = e^x e^y$ by taking even and odd parts.

Remark (Comparison with trigonometric addition formulas). The sinh formula matches $\sin(x+y)$ exactly. The cosh formula differs from $\cos(x+y)$ only in the sign of the last term: $+\sinh x \sinh y$ versus $-\sin x \sin y$. The sign difference is a direct consequence of $\cosh^2 - \sinh^2 = 1$ versus $\cos^2 + \sin^2 = 1$.

Proposition (Limits of tanh)

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Proposition 9.1.4 (Limits of tanh)**.**

$$\lim_{x\to+\infty} \tanh x = 1, \qquad \lim_{x\to-\infty} \tanh x = -1.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{LimitAtInfinity}(\tanh x,\ +\infty,\ 1)\ \wedge\ \mathrm{LimitAtInfinity}(\tanh x,\ -\infty,\ -1).$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(\tanh x,\ +\infty,\ 1)\ :\equiv\ \forall\varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x:\ x > M \Rightarrow \mathrm{OutputTolerance}(\tanh x, 1, \varepsilon).$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to+\infty} \tanh x = 1\right) \iff \exists\varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M:\ |\tanh x - 1| \geq \varepsilon_0.$$

Remark (Proof strategy). Write $\tanh x = (1 - e^{-2x})/(1 + e^{-2x})$. As $x \to +\infty$, $e^{-2x} \to 0$, giving $\tanh x \to 1$. The limit at $-\infty$ follows by oddness: $\tanh(-x) = -\tanh(x)$.

Definition

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- Limit at Infinity
- Limit of a Quotient

Definition 9.1.5 (Inverse Hyperbolic Functions)**.** Since sinh is strictly increasing and bijective $\mathbb{R} \to \mathbb{R}$, its inverse arcsinh is defined on all of $\mathbb{R}$. Since cosh is strictly increasing on $[0, \infty)$ with range $[1, \infty)$, its inverse arccosh is defined on $[1, \infty)$. Since tanh is strictly increasing with range $(-1, 1)$, arctanh is defined on $(-1, 1)$.
Explicit logarithmic formulas:

$$\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right), \quad x \in \mathbb{R}.$$
$$\operatorname{arccosh} x = \ln\left(x + \sqrt{x^2 - 1}\right), \quad x \geq 1.$$
$$\operatorname{arctanh} x = \tfrac{1}{2} \ln\left(\frac{1 + x}{1 - x}\right), \quad |x| < 1.$$

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \ \sinh(\operatorname{arcsinh} x) = x \ \wedge \ \operatorname{arcsinh}(\sinh x) = x.$$

$$\forall x \geq 1 : \ \cosh(\operatorname{arccosh} x) = x. \qquad \forall |x| < 1 : \ \tanh(\operatorname{arctanh} x) = x.$$

Remark (Definition predicate reading).

$$\operatorname{Arcsinh}(y, x) \ :\equiv \ x \in \mathbb{R} \ \wedge \ \sinh x = y \ :\equiv \ x = \ln\left(y + \sqrt{y^2 + 1}\right).$$

$$\operatorname{Arctanh}(y, x) \ :\equiv \ |y| < 1 \ \wedge \ \tanh x = y \ :\equiv \ x = \tfrac{1}{2} \ln\left(\frac{1 + y}{1 - y}\right).$$

Remark (Negated quantified statement).

$$\neg \operatorname{Arctanh}(y, x) \iff |y| \geq 1 \ \vee \ \tanh x \neq y.$$

Remark (Derivation of arcsinh). Set $y = \sinh x = (e^x - e^{-x})/2$ and let $u = e^x > 0$. Then $y = (u - 1/u)/2$, so $u^2 - 2yu - 1 = 0$. The positive root is $u = y + \sqrt{y^2 + 1}$, giving $x = \ln(y + \sqrt{y^2 + 1})$.

Remark (Derivatives). $(\operatorname{arcsinh} x)' = 1/\sqrt{x^2 + 1}$; $(\operatorname{arccosh} x)' = 1/\sqrt{x^2 - 1}$ for $x > 1$; $(\operatorname{arctanh} x)' = 1/(1 - x^2)$ for $|x| < 1$. These follow from the inverse function theorem applied to sinh, cosh, and tanh respectively.

Remark (Dependencies).

- Hyperbolic Functions
- Natural Logarithm
- Continuous Inverse Theorem
- Absolute Value on $\mathbb{R}$

9.2 The Logarithm

Logarithmic Functions — Quick Reference

Concept	Meaning
Natural logarithm	Inverse of the exponential on $(0, \infty)$
Fundamental logarithmic limit	$\ln(1 + x)/x \to 1$
Power dominates log	Logarithmic growth is below every positive power

The Logarithm

Definition (Natural Logarithm)

Definition 9.2.1 (Natural Logarithm)**.** The **natural logarithm** $\ln : (0, \infty) \to \mathbb{R}$ is the inverse of the exponential function: for $x > 0$,

$$\ln x \;:=\; \text{the unique } y \in \mathbb{R} \text{ such that } e^y = x.$$

Equivalently, $\ln x$ is defined by the integral

$$\ln x \;:=\; \int_1^x \frac{1}{t}\, dt.$$

Remark (Standard quantified statement).

$$\forall x > 0 : \; e^{\ln x} = x. \qquad \forall y \in \mathbb{R} : \; \ln(e^y) = y.$$

Remark (Definition predicate reading).

$$\mathrm{NaturalLog}(x, y) \;:\equiv\; x > 0 \;\wedge\; e^y = x \;:\equiv\; x > 0 \;\wedge\; y = \int_1^x \frac{1}{t}\, dt.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{\ln(1+x)}{x},\, 0,\, 1\right) \;:\equiv\; \forall \varepsilon > 0,\; \exists \delta > 0,\; \forall x :\; \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}\Bigl(\frac{\ln}{}$$

Remark (Negated quantified statement).

$$\neg\, \mathrm{NaturalLog}(x, y) \iff x \le 0 \;\vee\; e^y \ne x.$$

Remark (Two definitions, same function). The inverse-of-exponential definition requires e^x to be established first. The integral definition is self-contained and is preferred when building analysis from scratch. Both yield the same function.

Remark (Key properties).

(i) $\ln 1 = 0$; $\ln e = 1$.

(ii) **Functional equation:** $\ln(xy) = \ln x + \ln y$ for all $x, y > 0$.

(iii) $\ln(x/y) = \ln x - \ln y$; $\ln(x^r) = r \ln x$ for $r \in \mathbb{R}$.

(iv) ln is strictly increasing on $(0, \infty)$.

(v) $\ln x \to +\infty$ as $x \to \infty$; $\ln x \to -\infty$ as $x \to 0^+$.

(vi) $(\ln x)' = 1/x$ for all $x > 0$.

Remark (General logarithm). For $a > 0$, $a \neq 1$, the **logarithm base** a is $\log_a x := \ln x / \ln a$. The common logarithm is $\log_{10}$; the binary logarithm is $\log_2$.

Proposition (Fundamental Logarithmic Limit)

Remark (Dependencies).

- Exponential Function
- Inverse Bijection
- Continuous Inverse Theorem

Proposition 9.2.2 (Fundamental Logarithmic Limit)**.**

$$\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{\ln(1+x)}{x} - 1 \right| < \varepsilon.$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1 \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta \ \wedge \ \left| \frac{\ln(1+x)}{x} - 1 \right| \geq \varepsilon_0.$$

Remark (Equivalence with exponential limit). The fundamental exponential limit $\lim(e^x - 1)/x = 1$ and the fundamental logarithmic limit $\lim \ln(1+x)/x = 1$ are equivalent: each follows from the other via the substitution $x \leftrightarrow e^x - 1$. Both express the first-order Taylor approximation of the respective function at 0.

Remark (Proof strategy). From the series: $\ln(1+x) = x - x^2/2 + x^3/3 - \cdots$ (valid for $|x| < 1$). Dividing by x gives $1 - x/2 + x^2/3 - \cdots \to 1$ as $x \to 0$ by continuity of power series.

Proposition (Every Power Dominates the Logarithm)

Remark (Dependencies).

- Natural Logarithm
- Fundamental Exponential Limit
- Function Limit

Proposition 9.2.3 (Every Power Dominates the Logarithm)**.** *For every $r > 0$:*

$$\lim_{x\to\infty} \frac{\ln x}{x^r} = 0 \qquad \text{and} \qquad \lim_{x\to 0^+} x^r \ln x = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r > 0 : \lim_{x\to\infty} x^{-r} \ln x = 0 \;\wedge\; \lim_{x\to 0^+} x^r \ln x = 0.$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(x^{-r}\ln x,\, +\infty,\, 0) \;:\equiv\; \forall \varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x :\ x > M \Rightarrow \mathrm{OutputTolerance}(x^{-r}\ln x, 0, \varepsilon).$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to\infty} \frac{\ln x}{x^r} = 0\right) \iff \exists \varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M :\ x^{-r}|\ln x| \geq \varepsilon_0.$$

Remark (Proof strategy). For the first: substitute $x = e^t$ to get $t/e^{rt} \to 0$ by exponential dominance. For the second: substitute $x = 1/t$ and reduce to the first: $x^r \ln x = (-\ln t)/t^r \to 0$.

Remark (Significance). These limits establish the position of $\ln x$ in the growth hierarchy: $\ln x$ is dominated by every positive power. Combined with the exponential dominance result, the full ordering is $\ln x \ll x^r \ll e^x$ as $x \to \infty$ for any $r > 0$. The second limit says that near 0^+, the positive power x^r wins over the logarithmic blow-up $|\ln x| \to \infty$. Used in integrability arguments near 0.

Remark (Dependencies).

- Natural Logarithm
- Power Function
- Exponential Dominates Every Power
- Limit at Infinity

9.3 Power Functions and the Exponential

Power and Exponential Functions — Quick Reference

Concept	Meaning
Power function	Real powers x^r on positive inputs
The number e	Series and compound-interest limit definitions
Exponential function	Power series with $e^{x+y} = e^x e^y$
Fundamental exponential limit	$(e^x - 1)/x \to 1$
Compound-interest limit	$(1 + 1/n)^n \to e$
Growth hierarchy	e^x dominates powers at infinity

Power Functions and the Exponential

Definition (Power Function)

Definition 9.3.1 (Power Function)**.** Let $n \in \mathbb{N}$. The **integer power function** is $f : \mathbb{R} \to \mathbb{R}$, $f(x) := x^n$. For $r \in \mathbb{R}$ and $x > 0$, the **real power** is defined by

$$x^r \;:=\; e^{r \ln x}.$$

Remark (Standard quantified statement)*.*

$$\forall r \in \mathbb{R},\ \forall x > 0 :\ x^r = e^{r \ln x}.$$

For $r \in \mathbb{N}$ this reduces to the r-fold product $x \cdot x \cdots x$.

Remark (Definition predicate reading)*.*

$$\mathrm{RealPower}(x, r, y) \;:\equiv\; x > 0 \;\wedge\; y = e^{r \ln x}.$$

Remark (Laws of exponents)*.* For $x, y > 0$ and $r, s \in \mathbb{R}$:

$$x^r \cdot x^s = x^{r+s}, \quad (x^r)^s = x^{rs}, \quad (xy)^r = x^r y^r, \quad x^0 = 1, \quad x^1 = x.$$

Each follows from the corresponding property of the exponential via the definition $x^r = e^{r \ln x}$.

Remark (Interpretation)*.* Defining $x^r := e^{r \ln x}$ unifies all three cases — integer, rational, irrational exponents — into a single formula and makes differentiability of x^r immediate from that of e^x and $\ln x$ via the chain rule: $(x^r)' = r x^{r-1}$.

Definition (The Number e)

Remark (Dependencies).

- Integer Powers in a Field
- Exponential Function
- Natural Logarithm
- Multiplication on $\mathbb{R}$

Definition 9.3.2 (The Number e)**.** The **Euler number** is

$$e \;:=\; \sum_{n=0}^{\infty} \frac{1}{n!} \;=\; \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^{n} \;\approx\; 2.71828\ldots$$

Both expressions define the same real number.

Remark (Standard quantified statement).

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} \;\wedge\; e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^{n}.$$

Remark (Interpretation). The series $\sum 1/n!$ converges rapidly by comparison with a geometric series. The limit $(1 + 1/n)^n$ arises from continuous compound interest and is proved equal to the series sum via the fundamental logarithmic limit. Both representations are in common use.

Remark (Dependencies).

- Sequence
- Convergent Sequence

Definition (Exponential Function)

Definition 9.3.3 (Exponential Function)**.** The **exponential function** $\exp : \mathbb{R} \to (0, \infty)$ is defined by the absolutely convergent power series

$$e^{x} \;:=\; \exp(x) \;:=\; \sum_{n=0}^{\infty} \frac{x^{n}}{n!}, \qquad x \in \mathbb{R}.$$

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} :\; e^{x} = \sum_{n=0}^{\infty} \frac{x^{n}}{n!}, \quad \text{convergent absolutely since } \sum \frac{|x|^{n}}{n!} < \infty.$$

Remark (Definition predicate reading).

$$\text{Exponential}(x, y) \;:\equiv\; y = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Remark (Key properties). (i) $e^0 = 1$; $e^1 = e$.

(ii) **Functional equation:** $e^{x+y} = e^x \cdot e^y$ for all $x, y \in \mathbb{R}$.

(iii) $e^x > 0$ for all x; $e^{-x} = 1/e^x$.

(iv) e^x is strictly increasing and bijective $\mathbb{R} \to (0, \infty)$.

(v) $(e^x)' = e^x$ — the exponential is its own derivative.

Remark (General exponential). For $a > 0$, $a \neq 1$: $a^x := e^{x \ln a}$. Strictly increasing if $a > 1$, strictly decreasing if $0 < a < 1$. $(a^x)' = a^x \ln a$. Only e^x is its own derivative.

Proposition (Fundamental Exponential Limit)

Remark (Dependencies).

- The Number e
- Integer Powers in a Field
- Sequence
- Convergent Sequence

Proposition 9.3.4 (Fundamental Exponential Limit)**.**

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{e^x - 1}{x} - 1 \right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}\left(\frac{e^x - 1}{x},\ 0,\ 1\right) \;:\equiv\; \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \text{InputTolerance}(x, 0, \delta) \Rightarrow \text{OutputTolerance}\Big(\frac{e^x -}{x}$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to 0} \frac{e^x - 1}{x} = 1 \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta \ \wedge\ \left| \frac{e^x - 1}{x} - 1 \right| \geq \varepsilon_0.$$

Remark (Interpretation). This is the statement $(e^x)'\big|_{x=0} = 1$. The series gives $(e^x - 1)/x = 1 + x/2! + x^2/3! + \cdots \to 1$ as $x \to 0$ by continuity of power series at their center. Combined with the functional equation, this implies $(e^x)' = e^x$ everywhere.

Proposition (Compound Interest Limit)

Remark (Dependencies).

- Exponential Function
- Function Limit
- Derivative at a Point

Proposition 9.3.5 (Compound Interest Limit)**.**

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists M > 0:\ x > M \Rightarrow \left|\left(1+\frac{1}{x}\right)^x - e\right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}\big(\left(1+\tfrac{1}{x}\right)^x, +\infty, e\big) \ :\equiv\ \forall \varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x:\ x > M \Rightarrow \mathrm{OutputTolerance}\big(\left(1+\tfrac{1}{x}\right)^x, e, \varepsilon\big)$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to\infty}\left(1+\tfrac{1}{x}\right)^x = e\right) \iff \exists \varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M:\ \left|\left(1+\frac{1}{x}\right)^x - e\right| \geq \varepsilon_0.$$

Remark (Proof strategy). Write $(1 + 1/x)^x = e^{x \ln(1+1/x)}$. Set $t = 1/x \to 0^+$ and use the fundamental logarithmic limit $\ln(1 + t)/t \to 1$. Then $x \ln(1 + 1/x) = \ln(1 + t)/t \to 1$, so $(1 + 1/x)^x = e^{x \ln(1+1/x)} \to e^1 = e$ by continuity of the exponential.

Proposition (Exponential Dominates Every Power)

Remark (Dependencies).

- The Number e
- Exponential Function
- Natural Logarithm
- Fundamental Logarithmic Limit
- Limit at Infinity

Proposition 9.3.6 (Exponential Dominates Every Power)**.** *For every $r > 0$:*

$$\lim_{x\to\infty} \frac{x^r}{e^x} = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r > 0 : \lim_{x\to\infty} x^r e^{-x} = 0.$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(x^r e^{-x}, +\infty, 0) \;:\equiv\; \forall\varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x :\ x > M \Rightarrow \mathrm{OutputTolerance}(x^r e^{-x}, 0, \varepsilon).$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to\infty} \frac{x^r}{e^x} = 0\right) \iff \exists\varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M :\ x^r e^{-x} \geq \varepsilon_0.$$

Remark (Proof strategy). Pick integer $n > r$. The series bound $e^x \geq x^n/n!$ gives $x^r/e^x \leq n!\, x^{r-n} \to 0$ since $r - n < 0$.

Remark (Growth hierarchy). For any $r > 0$:

$$\ln x \;\ll\; x^r \;\ll\; e^x \qquad \text{as } x \to \infty.$$

The exponential eventually dominates every polynomial and every power function. This ordering is used constantly in estimates, dominated convergence arguments, and comparison tests.

Remark (Dependencies).

- Power Function
- Exponential Function
- Limit at Infinity
- Squeeze Theorem for Function Limits

9.4 Trigonometric Functions

Trigonometric Functions — Quick Reference

Concept	Meaning
Sine and cosine	Defined analytically by power series
Pythagorean identity	$\sin^2 x + \cos^2 x = 1$
Addition formulas	Angle addition identities from exponentials
Other trig functions	Tangent, cotangent, secant, cosecant
Inverse trig functions	Inverses on monotone restricted domains
Fundamental trig limit	$\sin x/x \to 1$ as $x \to 0$
Second trig limit	$(1 - \cos x)/x^2 \to 1/2$

Trigonometric Functions

Definition (Sine and Cosine)

Definition 9.4.1 (Sine and Cosine via Power Series)**.** The **sine** and **cosine** functions are defined for all $x \in \mathbb{R}$ by the absolutely convergent power series

$$\sin x \;:=\; \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \;=\; x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x \;:=\; \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \;=\; 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \; \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

both absolutely convergent by the ratio test.

Remark (Definition predicate reading).

$$\mathrm{Sine}(x, y) \;:\equiv\; y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}. \qquad \mathrm{Cosine}(x, y) \;:\equiv\; y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.$$

Remark (Key properties). (i) $\sin 0 = 0$; $\cos 0 = 1$.

(ii) $\sin(-x) = -\sin x$ (odd); $\cos(-x) = \cos x$ (even).

(iii) $(\sin x)' = \cos x$; $(\cos x)' = -\sin x$.

(iv) $|\sin x| \le 1$; $|\cos x| \le 1$ for all x.

(v) $\sin^2 x + \cos^2 x = 1$ (Pythagorean identity).

(vi) Periodic: $\sin(x+2\pi) = \sin x$; $\cos(x+2\pi) = \cos x$.

Remark (Relationship to e^{ix}). Euler's formula: $e^{ix} = \cos x + i \sin x$, where e^{ix} is defined by the complex exponential series. This gives $\cos x = \mathrm{Re}(e^{ix})$ and $\sin x = \mathrm{Im}(e^{ix})$. Euler's identity $e^{i\pi} + 1 = 0$ follows immediately.

Proposition (Pythagorean Identity)

Remark (Dependencies).

- Real Numbers
- Integer Powers in a Field
- Sequence
- Convergent Sequence

Proposition 9.4.2 (Pythagorean Identity)**.** *For all $x \in \mathbb{R}$:*

$$\sin^2 x + \cos^2 x = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \ \sin^2 x + \cos^2 x = 1.$$

Remark (Definition predicate reading).

$$\mathrm{PythagoreanIdentity}(x) \ :\equiv \ \mathrm{Sine}(x, s) \ \wedge \ \mathrm{Cosine}(x, c) \ \wedge \ s^2 + c^2 = 1.$$

Remark (Negated quantified statement).

$$\neg \left(\forall x : \sin^2 x + \cos^2 x = 1\right) \iff \exists x \in \mathbb{R} : \sin^2 x + \cos^2 x \neq 1.$$

The negation has no witness — the identity holds universally.

Remark (Proof strategy). Define $f(x) := \sin^2 x + \cos^2 x$. Then $f'(x) = 2 \sin x \cos x + 2 \cos x(-\sin x) = 0$, so f is constant. Since $f(0) = 0 + 1 = 1$, the identity follows.

Proposition (Addition Formulas)

Remark (Dependencies).

- Sine and Cosine
- Derivative at a Point
- Zero Derivative Implies Constant

Proposition 9.4.3 (Addition Formulas)**.** *For all $x, y \in \mathbb{R}$:*

$$\sin(x+y) = \sin x \cos y + \cos x \sin y, \qquad \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}: \ \sin(x+y) = \sin x \cos y + \cos x \sin y \ \wedge \ \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

Remark (Definition predicate reading).

$$\mathrm{TrigAddition}(x, y) \ :\equiv \ \sin(x+y) = \sin x \cos y + \cos x \sin y \wedge \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

Remark (Negated quantified statement).

$$\neg\,(\text{addition formula for } \sin) \iff \exists x, y \in \mathbb{R}: \ \sin(x+y) \neq \sin x \cos y + \cos x \sin y.$$

The negation has no witness.

Remark (Proof strategy). From the Cauchy product: $e^{i(x+y)} = e^{ix}e^{iy}$. Taking real and imaginary parts yields both addition formulas simultaneously.

Remark (Derived identities).

$$\begin{aligned}
\sin(2x) &= 2 \sin x \cos x \\
\cos(2x) &= \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1 \\
\sin^2 x &= \tfrac{1}{2}(1 - \cos 2x) \\
\cos^2 x &= \tfrac{1}{2}(1 + \cos 2x)
\end{aligned}$$

Definition

Remark (Dependencies).

- Sine and Cosine
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition 9.4.4 (Tangent, Cotangent, Secant, Cosecant)**.**

$$\tan x := \frac{\sin x}{\cos x}, \quad \cot x := \frac{\cos x}{\sin x}, \quad \sec x := \frac{1}{\cos x}, \quad \csc x := \frac{1}{\sin x}.$$

$\tan x$ and $\sec x$ are defined where $\cos x \neq 0$, i.e., $x \neq \pi/2 + k\pi$ for $k \in \mathbb{Z}$.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R},\ \cos x \neq 0 :\ \tan x = \frac{\sin x}{\cos x}.$$

Remark (Definition predicate reading).

$$\mathrm{Tangent}(x, y)\ :\equiv\ \cos x \neq 0\ \wedge\ y = \frac{\sin x}{\cos x}.$$

Remark (Pythagorean identities for tan). Dividing $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$: $1 + \tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$: $\cot^2 x + 1 = \csc^2 x$.

Definition

Remark (Dependencies).

- Sine and Cosine
- $\mathbb{R}$ Is a Field

Definition 9.4.5 (Inverse Trigonometric Functions)**.**

- $\arcsin : [-1, 1] \to [-\pi/2, \pi/2]$ is the inverse of $\sin\big|_{[-\pi/2,\pi/2]}$.
- $\arccos : [-1, 1] \to [0, \pi]$ is the inverse of $\cos\big|_{[0,\pi]}$.
- $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ is the inverse of $\tan\big|_{(-\pi/2,\pi/2)}$.

Remark (Standard quantified statement).

$$\forall y \in [-1, 1] :\ \sin(\arcsin y) = y. \qquad \forall x \in [-\pi/2, \pi/2] :\ \arcsin(\sin x) = x.$$

Remark (Definition predicate reading).

$$\mathrm{Arctan}(y, x)\ :\equiv\ x \in (-\pi/2, \pi/2)\ \wedge\ \tan x = y.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}\left(\frac{\sin x}{x}, 0, 1\right) :\equiv \forall\varepsilon > 0, \exists\delta > 0, \forall x : \text{InputTolerance}(x, 0, \delta) \Rightarrow \text{OutputTolerance}\left(\frac{\sin x}{x},\right.$$

Remark (Negated quantified statement).

$$\neg\ \text{Arctan}(y, x) \iff x \notin (-\pi/2, \pi/2) \lor \tan x \neq y.$$

Remark (Key limits of arctan).

$$\lim_{x\to+\infty} \arctan x = \frac{\pi}{2}, \qquad \lim_{x\to-\infty} \arctan x = -\frac{\pi}{2}.$$

arctan is bounded, strictly increasing, and continuous on $\mathbb{R}$; the Monotone Limit Theorem gives the limits as the supremum and infimum of its range.

Theorem (Fundamental Trigonometric Limit)

Remark (Dependencies).

- Sine and Cosine
- Tangent, Cotangent, Secant, Cosecant
- Continuous Inverse Theorem
- Closed Interval

Theorem 9.4.6 (Fundamental Trigonometric Limit)**.**

$$\lim_{x\to 0} \frac{\sin x}{x} = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall\varepsilon > 0, \exists\delta > 0 : 0 < |x| < \delta \Rightarrow \left|\frac{\sin x}{x} - 1\right| < \varepsilon.$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to 0} \frac{\sin x}{x} = 1\right) \iff \exists\varepsilon_0 > 0, \forall\delta > 0, \exists x : 0 < |x| < \delta \land \left|\frac{\sin x}{x} - 1\right| \geq \varepsilon_0.$$

Remark (Proof strategy — series). From the series: $\sin x/x = 1 - x^2/3! + x^4/5! - \cdots \to 1$ as $x \to 0$ by continuity of power series at their center.

Remark (Proof strategy — geometric squeeze). For $0 < x < \pi/2$, area comparison of two triangles and a circular sector gives $\cos x \leq \sin x/x \leq 1$. Since $\cos x \to 1$ as $x \to 0^+$, the Squeeze Theorem gives $\sin x/x \to 1$. Even symmetry extends this to $x \to 0^-$.

Remark (Significance). This is the statement $(\sin x)'\big|_{x=0} = 1$, which implies $(\sin x)' = \cos x$ everywhere. It underlies all trigonometric differentiation.

Proposition (Second Fundamental Trigonometric Limit)

Remark (Dependencies).

- Sine and Cosine
- Function Limit
- Squeeze Theorem for Function Limits

Proposition 9.4.7 (Second Fundamental Trigonometric Limit)**.**

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{1-\cos x}{x^2} - \frac{1}{2} \right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{1-\cos x}{x^2},\ 0,\ \tfrac{1}{2}\right) \ :\equiv\ \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}\Bigg($$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}\right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta\ \wedge\ \left| \frac{1-\cos x}{x^2} - \frac{1}{2} \right| \geq \varepsilon_0.$$

Remark (Proof strategy). Use the half-angle identity $1 - \cos x = 2\sin^2(x/2)$:

$$\frac{1-\cos x}{x^2} = \frac{2\sin^2(x/2)}{x^2} = \frac{1}{2}\left(\frac{\sin(x/2)}{x/2}\right)^2 \to \frac{1}{2}\cdot 1^2 = \frac{1}{2}.$$

Remark (Dependencies).

- Sine and Cosine
- Trigonometric Addition Formulas
- Fundamental Trigonometric Limit
- Function Limit

Chapter 10

Continuity

SERIES	VOLUME III	BOOK II	CHAPTER 9
From Cantor to Ito	Classical Analysis	Functions, Continuity, and Differentiation	Continuity

10.1 Limits

One-Sided Limits — Quick Reference

Concept/Statement	Meaning/Role
Right-hand limit	Approach the point from inputs larger than c.
Left-hand limit	Approach the point from inputs smaller than c.
Right sequential criterion	Test right limits by right-approaching sequences.
Left sequential criterion	Test left limits by left-approaching sequences.
Matching criterion	Two-sided limits exist exactly when side limits agree.

Limits of Functions — Quick Reference

Concept/Statement	Meaning/Role
Function limit	Punctured input control near a cluster point.
Sequential limit	All approaching sequences have the same output limit.
Neighbourhood limit	Neighbourhood formulation of the same condition.
Uniqueness	A function has at most one limit at a point.
Sequential criterion	Sequence convergence is the main working test.
Local boundedness	Finite limits bound the function near the point.
Squeeze theorem	Ordered comparison transfers limits.
Locality	Only behaviour near the approach point matters.

Algebra of Limits — Quick Reference

Concept/Statement	Meaning/Role
Sum rule	Limits add pointwise.
Difference rule	Limits subtract pointwise.
Scalar multiple rule	Scalars pass through limits.
Product rule	Products of finite limits multiply.
Quotient rule	Division is valid when the denominator limit is nonzero.

Limits of Functions

Definition (ε-Closeness of a Function to a Value)

Definition 10.1.1 (ε-Closeness of a Function to a Value)**.** Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, and let $\varepsilon > 0$. The function f is **ε-close to L** if $|f(x) - L| \leq \varepsilon$ for all $x \in X$.

Remark (Standard quantified statement).

$$f \ \varepsilon\text{-close to } L \iff \forall x \in X \ \big(|f(x) - L| \leq \varepsilon\big).$$

Remark (Interpretation). A global condition: every output lies within ε of L, regardless of where in X the input lies. This is Tao's scaffolding toward the limit definition, which localises the same output condition to a punctured neighbourhood of c.

Definition (Local δ-Closeness Near a Point)

Definition 10.1.2 (Local δ-Closeness Near a Point)**.** Let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, let x_0 be adherent to X, and let $\varepsilon, \delta > 0$. The function f is **ε-close to L near x_0 within δ** if

$$\forall x \in X \ \big(|x - x_0| < \delta \Rightarrow |f(x) - L| \leq \varepsilon\big).$$

Remark (Standard quantified statement).

$$\forall x \in X : \ |x - x_0| < \delta \ \Rightarrow \ |f(x) - L| \leq \varepsilon.$$

Remark (Interpretation). A local condition: only outputs from the δ-ball around x_0 need lie within ε of L. The input condition here is Within — unpunctured. The limit definition replaces this with InputTolerance (punctured) and asserts that for every ε such a δ exists.

Remark (Dependencies).

- ε-Closeness of a Function to a Value

Definition (Limit of a Function)

Definition 10.1.3 (Limit of a Function)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. A real number L is the **limit of f at** c, written $\lim_{x \to c} f(x) = L$, if

$$\forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x \in A \; 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Go to uniqueness proof.

Remark (Standard quantified statement)*.*

$$\forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x \in A \; 0 < |x - c| < \delta \;\Rightarrow\; |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading)*.* Define

$$\begin{aligned} \mathrm{OutputTol}(\varepsilon) &\iff \varepsilon > 0, \\ \mathrm{InputTol}(\delta) &\iff \delta > 0, \\ \mathrm{PuncturedNbhd}(x, c, \delta) &\iff 0 < |x - c| < \delta, \\ \mathrm{OutputNbhd}(y, L, \varepsilon) &\iff |y - L| < \varepsilon. \end{aligned}$$

Then

$$\mathrm{FunctionLimit}(f, A, c, L) \iff \forall \varepsilon \Big(\mathrm{OutputTol}(\varepsilon) \Rightarrow \exists \delta \left(\mathrm{InputTol}(\delta) \wedge \forall x \in A \left[\mathrm{PuncturedNbhd}(x, c, \delta) \Rightarrow \mathrm{O}\right.\right.\right.$$

Remark (Negated quantified statement)*.*

$$\exists \varepsilon_0 > 0 \, \forall \delta > 0 \, \exists x \in A \; 0 < |x - c| < \delta \;\wedge\; |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading)*.*

$$\begin{aligned} \neg\, \mathrm{FunctionLimit}(f, A, c, L) &\iff \exists \varepsilon_0 \Big(\mathrm{OutputTol}(\varepsilon_0) \wedge \forall \delta \left(\mathrm{InputTol}(\delta) \Rightarrow \exists x \in A \left[\mathrm{PuncturedNbhd}(x, c, \delta)\right.\right.\right. \\ &\iff \exists \varepsilon_0 > 0 \, \forall \delta > 0 \, \exists x \in A \, \big(0 < |x - c| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big). \end{aligned}$$

Remark (Failure modes)*.*

- The proposed value L is the wrong target: the function approaches some value $M \neq L$ near c.
- The function does not settle near any single value as x approaches c through A.
- The outputs repeatedly stay at least a fixed distance from L no matter how tightly the inputs are restricted around c.

Remark (Failure mode decomposition)*.*

$$\neg\, \mathrm{FunctionLimit}(f, A, c, L) \iff \exists \varepsilon_0 > 0 \, \forall \delta > 0 \, \exists x \in A \, \Big(\underbrace{0 < |x - c| < \delta}_{\text{arbitrarily close to } c} \wedge \underbrace{|f(x) - L| \geq \varepsilon_0}_{\text{fixed output gap from } L}\Big).$$

Remark (Interpretation). The quantifier structure is asymmetric by design. The output tolerance ε is a *demand*: the caller specifies how precise the output must be. The input tolerance δ is a *response*: the definition asserts one always exists. The input condition is punctured (InputTolerance, not Within) because f may not be defined at c, and because continuity — not limits — is the concept that includes c itself. This is the deepest structural difference between the two definitions. Four critical features: (i) f need not be defined at c; (ii) c must be a cluster point of A else the condition is vacuously satisfied by every L; (iii) δ may depend on ε; (iv) the condition is about all x near c simultaneously, not just one.

Remark (Dependencies).

- Cluster Point

Definition (Sequential Definition of Limit)

Definition 10.1.4 (Sequential Definition of Limit)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The limit $\lim_{x\to c} f(x) = L$ in the **sequential sense** holds if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, the sequence $(f(x_n))$ converges to L.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq A \setminus \{c\} \ \big(x_n \to c \Rightarrow f(x_n) \to L\big).$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}_{\text{seq}}(f, c, L) \ :\equiv \ \forall (x_n) \subseteq A\setminus\{c\} : \ \text{ConvergentSequence}(x_n, c) \ \Rightarrow \ \text{ConvergentSequence}(f(x_n),$$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq A \setminus \{c\} \ \big(x_n \to c \ \wedge \ f(x_n) \not\to L\big).$$

Remark (Interpretation). The sequence (x_n) must lie in $A \setminus \{c\}$: the term $x_n = c$ is excluded, mirroring the $0 <$ in the ε-δ form. The sequential form is the primary tool for both proving limits (via the bridge principle) and disproving them (exhibit one bad sequence). The ε-δ form is better for constructing explicit δ functions.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Definition (Neighbourhood Form of Limit)

Definition 10.1.5 (Neighbourhood Form of Limit)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The limit $\lim_{x\to c} f(x) = L$ in the **neighbourhood sense** holds if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a deleted δ-neighbourhood $V_\delta^*(c)$ such that

$$x \in V_\delta^*(c) \cap A \quad \Longrightarrow \quad f(x) \in V_\varepsilon(L).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in V^*_\delta(c) \cap A\ f(x) \in V_\varepsilon(L).$$

Remark (Interpretation). The neighbourhood form packages the same local limit condition in terms of sets around the input point and output value. The domain is still restricted to A, and the neighbourhood of c is deleted so the value of f at c itself is irrelevant.

Remark (Dependencies).

- Limit of a Function
- ε-Neighbourhood
- Deleted ε-Neighbourhood

Proposition

Proposition 10.1.6 (Neighbourhood Form of the Limit)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ if and only if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a δ-neighbourhood $V_\delta(c)$ such that $x \in V^*_\delta(c) \cap A$ implies $f(x) \in V_\varepsilon(L)$.*

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in V^*_\delta(c) \cap A\ f(x) \in V_\varepsilon(L)\Big].$$

Remark (Interpretation). The equivalence is purely notational: $V^*_\delta(c) \cap A$ is exactly the set of $x \in A$ satisfying $0 < |x - c| < \delta$, and $f(x) \in V_\varepsilon(L)$ is exactly $\mathrm{OutputTolerance}(f(x), L, \varepsilon)$. The neighbourhood form makes the geometric content explicit: the image of every punctured δ-slice of A must fit inside the ε-ball around L. This is the form of continuity used in metric space topology.

Remark (Dependencies).

- Limit of a Function
- Neighbourhood Form of Limit
- ε-Neighbourhood
- Deleted ε-Neighbourhood

Theorem

Theorem 10.1.7 (Uniqueness of Limits of Functions)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then f has at most one limit at c.*

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c} f(x) = L \ \wedge \ \lim_{x\to c} f(x) = L' \ \Rightarrow \ L = L'.$$

Remark (Failure modes). Failure of uniqueness would mean that two distinct real numbers both satisfy the limit definition at the same cluster point.

Remark (Failure mode decomposition). Non-uniqueness would require two values $L \neq L'$ each satisfying the limit definition. The $\varepsilon = |L-L'|/2$ argument shows any x within $\min(\delta_1, \delta_2)$ of c satisfies both $|f(x)-L| < \varepsilon$ and $|f(x)-L'| < \varepsilon$, but then $|L-L'| \leq |L-f(x)|+|f(x)-L'| < 2\varepsilon = |L-L'|$, a contradiction. The cluster point condition is essential: it guarantees such an x exists.

Remark (Interpretation). The limit, when it exists, is uniquely determined by the local behavior of f near c. Uniqueness is what licenses writing $\lim_{x\to c} f(x) = L$ as an equation rather than an assertion.

Remark (Dependencies).

- Limit of a Function

Proposition

Proposition 10.1.8 (Sequential Criterion for Function Limits). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, one has $f(x_n) \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall\varepsilon > 0 \ \exists\delta > 0 \ \forall x \in A \ 0 < |x-c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \setminus \{c\} \ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Negated quantified statement).

$$\neg \lim_{x\to c} f(x) = L \iff \exists (x_n) \subseteq A \setminus \{c\} : \ x_n \to c \ \wedge \ f(x_n) \not\to L.$$

Remark (Interpretation). The Sequential Criterion is the engine of the algebra of limits. Once it is in hand:

$$\text{algebra of sequence limits} \xrightarrow{\text{Sequential Criterion}} \text{algebra of function limits.}$$

Every sequence result — sum, product, quotient, squeeze — transfers to function limits by feeding a sequence into the criterion and reading the result back. The negation is equally powerful: to disprove $\lim_{x\to c} f(x) = L$, exhibit one sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$ and $f(x_n) \not\to L$.

Remark (Dependencies).

- Limit of a Function
- Sequential Definition of Limit

Corollary

Corollary 10.1.9 (Three Equivalent Forms of the Function Limit)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The ε-δ form of $\lim_{x\to c} f(x) = L$, the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\Big[\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]\\ \iff &\Big[\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in V_\delta^*(c) \cap A\ f(x) \in V_\varepsilon(L)\Big]\\ \iff &\Big[\forall(x_n) \subseteq A \setminus \{c\}\ x_n \to c \Rightarrow f(x_n) \to L\Big].\end{aligned}$$

Remark (Interpretation). All three encodings capture the same mathematical condition. The ε-δ form is best for proving explicit bounds. The neighbourhood form is best for geometric reasoning and metric space generalisation. The sequential form is best for importing sequence results and constructing counterexamples.

Remark (Dependencies).

- Limit of a Function
- Neighbourhood Form of Limit
- Sequential Definition of Limit
- Neighbourhood Form of the Limit
- Sequential Criterion for Function Limits

Theorem

Theorem 10.1.10 (Limit Implies Local Boundedness)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ for some $L \in \mathbb{R}$, then there exist $\delta, M > 0$ such that $|f(x)| \le M$ for all $x \in A$ with $0 < |x - c| < \delta$.*
Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c} f(x) = L\ \Rightarrow\ \exists\delta > 0\ \exists M > 0\ \forall x \in A\ \big(0 < |x - c| < \delta \Rightarrow |f(x)| \le M\big).$$

Remark (Proof sketch). Apply the definition with $\varepsilon = 1$: get $\delta > 0$ such that $|f(x) - L| < 1$ holds for all x satisfying $0 < |x - c| < \delta$. The triangle inequality then gives $|f(x)| \le |f(x) - L| + |L| < 1 + |L|$. Set $M = 1 + |L|$.

Remark (Interpretation). A finite limit does more than control the output near L: it prevents the function from growing without bound near c. Local boundedness is a coarser statement than limit existence, but it is what is needed for the product and quotient rules.

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 10.1.11 (Bounded Limits). *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $a, b \in \mathbb{R}$. If $a \leq f(x) \leq b$ for all $x \in A$ with $x \neq c$, and $\lim_{x\to c} f(x) = L$, then $a \leq L \leq b$.*

Go to proof.

Remark (Standard quantified statement).

$$\big(\forall x \in A \setminus \{c\} : \; a \leq f(x) \leq b\big) \;\wedge\; \lim_{x\to c} f(x) = L \;\Rightarrow\; a \leq L \leq b.$$

Remark (Interpretation). If every nearby value of f is trapped in $[a, b]$, the limit cannot escape outside $[a, b]$. This is the function-limit analogue of the sequence result that limits preserve non-strict inequalities. The proof applies the Sequential Criterion and that sequence result directly.

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 10.1.12 (Squeeze Theorem for Function Limits). *Let $f, g, h : A \to \mathbb{R}$ and let c be a cluster point of A. Suppose $\lim_{x\to c} f(x) = L = \lim_{x\to c} g(x) = L$, and suppose there exists $\delta_0 > 0$ such that*

$$\forall x \in A, \; 0 < |x - c| < \delta_0 \;\Rightarrow\; f(x) \leq h(x) \leq g(x).$$

Then $\lim_{x\to c} h(x) = L$.

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c} f(x) = L = \lim_{x\to c} g(x) = L \;\wedge\; f \leq h \leq g \text{ near } c \;\Rightarrow\; \lim_{x\to c} h(x) = L.$$

Remark (Interpretation). The proof routes entirely through the Sequential Criterion: convert to sequences, apply the sequential squeeze theorem, convert back. No new ε-δ argument is needed. The condition $f \leq h \leq g$ is only required near c, not globally.

Remark (Dependencies).

- Limit of a Function

- Bounded Limits

Theorem

Theorem 10.1.13 (Limits Are Local)**.** *Let $E \subseteq X \subseteq \mathbb{R}$, let x_0 be a cluster point of E, let $f : X \to \mathbb{R}$, and let $\delta > 0$. Then*

$$\lim_{x\to x_0;\, x\in E} f(x) = L \iff \lim_{x\to x_0;\, x\in E\cap(x_0-\delta,x_0+\delta)} f(x) = L.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\lim_{x\to x_0;\, x\in E} f(x) = L \iff \lim_{x\to x_0;\, x\in E\cap(x_0-\delta,\, x_0+\delta)} f(x) = L.$$

Remark (Interpretation)*.* The existence and value of FunctionLimit(f, x_0, L) depend only on the behavior of f within any fixed neighbourhood of x_0. Points outside that neighbourhood are irrelevant. This justifies focusing all limit arguments on small balls around c without any loss of generality.

Remark (Dependencies)*.*

- Limit of a Function

Proposition

Proposition 10.1.14 (Limit of Absolute Value)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, then $\lim_{x\to c} |f(x)| = |L|$.*
Go to proof.

Remark (Standard quantified statement)*.*

$$\lim_{x\to c} f(x) = L \Rightarrow \lim_{x\to c} |f(x)| = |L|.$$

Remark (Negated quantified statement)*.*

$$\lim_{x\to c} |f(x)| = |L| \not\Rightarrow \lim_{x\to c} f(x) = L.$$

Remark (Failure modes)*.* The absolute values may converge while the signs of $f(x)$ oscillate or approach the opposite sign from the proposed limiting value.

Remark (Proof strategy)*.* The reverse triangle inequality $\big||f(x)| - |L|\big| \le |f(x) - L|$ feeds the ε-δ definition directly: the same δ that delivers $|f(x) - L| < \varepsilon$ also delivers $\big||f(x)| - |L|\big| < \varepsilon$.

Remark (Interpretation)*.* Taking absolute values can collapse sign information, so convergence of f forces convergence of $|f|$, but convergence of $|f|$ alone does not recover the original limiting sign.

Remark (Dependencies)*.*

- Limit of a Function

Theorem

Theorem 10.1.15 (Composition of Limits)**.** *Let* $f : A \to B$*, let* $g : B \to \mathbb{R}$*, let* c_1 *be a cluster point of* A*, and let* $c_2 \in \mathbb{R}$ *be a cluster point of* B*. Suppose* $\lim_{x\to c_1} f(x) = c_2$ *and* $\lim_{y\to c_2} g(y) = L$*. If* $c_2 \in B$*, assume also that* $g(c_2) = L$*. Then* $\lim_{x\to c_1} g(f(x)) = L$*.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\lim_{x\to c_1} f(x) = c_2 \;\wedge\; \lim_{y\to c_2} g(y) = L \;\wedge\; \big(c_2 \in B \Rightarrow g(c_2) = L\big) \;\Rightarrow\; \lim_{x\to c_1} g(f(x)) = L.$$

Remark (Failure modes)*.* The composition conclusion can fail if the outer function is not controlled at the intermediate value reached by the inner limit.

Remark (Failure mode decomposition)*.* The hypothesis $g(c_2) = L$ is essential. Without it the composition can fail: take $f \equiv c_2$ (constant) and $g(c_2) \neq L$. Then $g(f(x)) = g(c_2) \neq L$ for all x, so $\lim_{x\to c_1} g(f(x)) = L$ fails even though both component limits exist. The failure is that $f(x) = c_2$ is excluded from the punctured condition on g, but f hits c_2 constantly.

Remark (Interpretation)*.* Composition of limits requires care precisely because the input condition is punctured. The extra hypothesis $g(c_2) = L$ bridges the gap: when $f(x) = c_2$, the output condition $g(f(x)) = g(c_2) = L$ is satisfied directly. This hypothesis is equivalent to requiring g to be continuous at c_2 (within B).

Remark (Dependencies)*.*

- Limit of a Function

Theorem

Theorem 10.1.16 (Monotone Limit Theorem)**.** *Let* $a < b$ *and let* $f : (a, b) \to \mathbb{R}$ *be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing* f *and every* $x_0 \in (a, b)$*:*

$$f(x_0^-) = \sup\{\, f(x) : a < x < x_0 \,\}, \qquad f(x_0^+) = \inf\{\, f(x) : x_0 < x < b \,\}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$f \text{ is monotone on } (a,b) \wedge \exists B > 0\ \forall x \in (a,b)\ (|f(x)| \le B) \;\Rightarrow\; \forall x_0 \in (a,b) : \lim_{x\to x_0^-} f(x) = f(x_0^-) \wedge \lim_{x\to x_0^+} f($$

Remark (Interpretation)*.* Boundedness guarantees the supremum and infimum used as candidate limits exist. Monotonicity forces convergence toward them from each side: for increasing f, the values $f(x)$ for $x < x_0$ form a bounded increasing net approaching their supremum. No oscillation is possible. The theorem establishes that monotone functions can only have jump discontinuities — never oscillatory ones.

Remark (Dependencies).

- Monotone Function
- Bounded Function
- Left-Hand Limit
- Right-Hand Limit

Proposition

Proposition 10.1.17 (Cauchy Criterion for Function Limits)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ exists for some $L \in \mathbb{R}$ if and only if*

$$\forall \ \mathrm{OutputTolerance}(\varepsilon),\ \exists \ \mathrm{InputTolerance}(\delta),\ \forall x, y \in A:\ 0 < |x-c| < \delta \wedge 0 < |y-c| < \delta \ \Rightarrow \ |f(x)-f(y)$$

Go to proof.

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R}\ \Big(\lim_{x\to c} f(x) = L\Big) \iff \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y \in A:\ 0 < |x-c| < \delta \wedge 0 < |y-c| < \delta \Rightarrow |f(x)-f(y)|$$

Remark (Interpretation). The Cauchy criterion tests limit existence without knowing L. Two outputs from any sufficiently small punctured ball must lie within ε of each other — the outputs cluster even if their common limit is unknown. This is the function-limit analogue of the Cauchy completeness criterion for sequences, and relies on the completeness of $\mathbb{R}$ in exactly the same way.

Remark (Dependencies).

- Limit of a Function

Algebra of Limits

Theorem

Theorem 10.1.18 (Sum Rule for Function Limits)**.** *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f+g)(x) = L + M.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\Big[\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x \in A\ 0 < |x-c| < \delta_f \Rightarrow |f(x) - L| < \varepsilon\Big] \\ &\quad \wedge \Big[\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x \in A\ 0 < |x-c| < \delta_g \Rightarrow |g(x) - M| < \varepsilon\Big] \\ &\quad \Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow |(f+g)(x) - (L+M)| < \varepsilon. \end{aligned}$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(f + g, c, L + M).$$

Remark (Interpretation). Limits respect addition: sufficiently close values of f and g add to values of $f + g$ close to the sum of the two limits.

Remark (Dependencies).

- Limit of a Function
- Pointwise Sum of Functions

Theorem

Theorem 10.1.19 (Difference Rule for Function Limits)**.** *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f - g)(x) = L - M.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right]$$
$$\Rightarrow \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |(f - g)(x) - (L - M)| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(f - g, c, L - M).$$

Remark (Interpretation). The difference rule is the sum rule applied to $f + (-g)$.

Remark (Dependencies).

- Limit of a Function
- Pointwise Difference of Functions

Theorem

Theorem 10.1.20 (Scalar Multiple Rule for Function Limits)**.** *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $\alpha \in \mathbb{R}$. If $\lim_{x\to c} f(x) = L$, then*

$$\lim_{x\to c}(\alpha f)(x) = \alpha L.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \Rightarrow \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |\alpha f(x) - \alpha L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \implies \text{FunctionLimit}(\alpha f, c, \alpha L).$$

Remark (Interpretation)*.* Multiplying a function by a fixed scalar multiplies its limiting value by that same scalar.

Remark (Dependencies).

- Limit of a Function
- Pointwise Scalar Multiple of a Function

Theorem

Theorem 10.1.21 (Product Rule for Function Limits)**.** *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c} (fg)(x) = LM.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right]$$
$$\Rightarrow \forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in A \; 0 < |x - c| < \delta \Rightarrow |f(x)g(x) - LM| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(fg, c, LM).$$

Remark (Interpretation)*.* The product rule combines closeness of f to L with local boundedness of g near c.

Remark (Dependencies).

- Limit of a Function
- Pointwise Product of Functions

Theorem

Theorem 10.1.22 (Quotient Rule for Function Limits)**.** *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, $\lim_{x\to c} g(x) = M$, and $M \neq 0$, then there exists $\delta_0 > 0$ such that $g(x) \neq 0$ whenever $x \in A$ and $0 < |x - c| < \delta_0$, and*

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right] \wedge M \neq 0$$
$$\Rightarrow \exists \delta_0 > 0\ \forall x \in A\ 0 < |x - c| < \delta_0 \Rightarrow g(x) \neq 0,$$
$$\text{and} \quad \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow \left|\frac{f(x)}{g(x)} - \frac{L}{M}\right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \wedge \mathrm{FunctionLimit}(g, c, M) \wedge M \neq 0 \implies \mathrm{FunctionLimit}(f/g, c, L/M).$$

Remark (Interpretation). A nonzero limiting denominator stays away from zero near the limit point, so division becomes locally legitimate and the quotient limit follows.

Remark (Dependencies).

- Limit of a Function
- Pointwise Quotient of Functions

One-Sided Limits

Definition (Right-Hand Limit)

Definition 10.1.23 (Right-Hand Limit)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. A real number L is the **right-hand limit** of f at c if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ c < x < c + \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{RightLimit}(f, c, L) \ :=\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Negated quantified statement).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{RightLimit}(f, c, L).$$

Remark (Failure modes). A proposed right-hand limit fails when some positive output tolerance is violated arbitrarily close to c from the right.

Remark (Failure mode decomposition).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Interpretation). Only inputs to the right of c are tested. The value of f at c, if it is defined, is irrelevant to the right-hand limit.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Definition (Left-Hand Limit)

Definition 10.1.24 (Left-Hand Limit)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. A real number L is the **left-hand limit** of f at c if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ c - \delta < x < c \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{LeftLimit}(f, c, L)\ :=\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Negated quantified statement).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading).

$$\neg\ \mathrm{LeftLimit}(f, c, L).$$

Remark (Failure modes). A proposed left-hand limit fails when some positive output tolerance is violated arbitrarily close to c from the left.

Remark (Failure mode decomposition).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Interpretation). Only inputs to the left of c are tested. This is the mirror image of the right-hand limit definition.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Theorem

Theorem 10.1.25 (Sequential Criterion for Right-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. Then $\lim_{x \to c^+} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (c, \infty)$ with $x_n \to c$, one has $f(x_n) \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \cap (c, \infty)\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Definition predicate reading).

$$\text{RightLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (c, \infty)\ x_n \to c \Rightarrow f(x_n) \to L.$$

Remark (Interpretation). Right-hand limits can be tested by sequences approaching c from above. Every such sequence must force the corresponding output sequence toward L.

Remark (Dependencies).

- Right-Hand Limit
- Sequential Definition of Limit

Corollary

Corollary 10.1.26 (Sequential Criterion for Left-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. Then $\lim_{x \to c^-} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (-\infty, c)$ with $x_n \to c$, one has $f(x_n) \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \cap (-\infty, c)\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Definition predicate reading).

$$\text{LeftLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (-\infty, c)\ x_n \to c \Rightarrow f(x_n) \to L.$$

Remark (Interpretation). Left-hand limits can be tested by sequences approaching c from below. The criterion is the left-sided analogue of the right-hand sequential criterion.

Remark (Dependencies).

- Left-Hand Limit

- Sequential Definition of Limit

Theorem

Theorem 10.1.27 (Two-Sided Limit via One-Sided Limits)**.** *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If c is a cluster point of both $A \cap (c, \infty)$ and $A \cap (-\infty, c)$, then $\lim_{x \to c} f(x) = L$ if and only if both $\lim_{x \to c^+} f(x) = L$ and $\lim_{x \to c^-} f(x) = L$.*
Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned} &\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big] \\ &\iff \Big[\forall \varepsilon > 0\ \exists \delta_+ > 0\ \forall x \in A\ 0 < x - c < \delta_+ \Rightarrow |f(x) - L| < \varepsilon\Big] \\ &\quad \wedge \Big[\forall \varepsilon > 0\ \exists \delta_- > 0\ \forall x \in A\ 0 < c - x < \delta_- \Rightarrow |f(x) - L| < \varepsilon\Big]. \end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{FunctionLimit}(f, c, L) \iff \mathrm{RightLimit}(f, c, L) \wedge \mathrm{LeftLimit}(f, c, L).$$

Remark (Negated quantified statement)*.*

$$\begin{aligned} &\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < |x - c| < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0 \\ &\iff \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\Big] \\ &\quad \vee \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\Big]. \end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg\, \mathrm{FunctionLimit}(f, c, L) \iff \neg\, \mathrm{RightLimit}(f, c, L) \vee \neg\, \mathrm{LeftLimit}(f, c, L).$$

Remark (Failure modes)*.* A two-sided limit fails precisely when the right-hand limit fails, the left-hand limit fails, or the two one-sided behaviours do not meet at the same value.

Remark (Failure mode decomposition)*.*

$$\neg\, \mathrm{FunctionLimit}(f, c, L) \iff \neg\, \mathrm{RightLimit}(f, c, L) \vee \neg\, \mathrm{LeftLimit}(f, c, L).$$

Remark (Interpretation)*.* A two-sided limit is exactly the agreement of the two one-sided tests. Both sides must approach the same value.

Remark (Dependencies)*.*

- Limit of a Function
- Right-Hand Limit
- Left-Hand Limit

10.2 Point Continuity

Oscillation — Quick Reference

Concept/Statement	Meaning/Role
Oscillation on a set	Diameter of the image over a set.
Oscillation at a point	Limiting local oscillation near the point.
Zero oscillation criterion	Continuity is exactly vanishing point oscillation.
Discontinuity set	Discontinuities are detected by positive oscillation.

Discontinuity — Quick Reference

Concept/Statement	Meaning/Role
Point of discontinuity	Negation of continuity at a point.
Sequential discontinuity	A bad sequence witnesses failure.
Neighbourhood discontinuity	Neighbourhood form of the same failure.
Equivalence lemma	The discontinuity formulations agree.
Types of discontinuity	Removable, jump, and essential behavior.

Continuity — Quick Reference

Concept/Statement	Meaning/Role
Relative neighbourhood	Neighbourhoods are read inside the domain.
Point continuity	The unpunctured ε-δ condition.
Neighbourhood continuity	Images of nearby inputs stay near the value.
Sequential continuity	Convergent input sequences preserve limits.
Oscillation criterion	Continuity is equivalent to zero local oscillation.

Continuity

Definition (Relative Neighborhood)

Definition 10.2.1 (Relative Neighborhood)**.** Let $E \subseteq \mathbb{R}$ and $a \in E$. A set $U \subseteq \mathbb{R}$ is a neighborhood of a in E if and only if there exists $\delta > 0$ such that $U = E \cap (a-\delta, a+\delta)$. Equivalently, every neighborhood of a in E is a set of the form $U_E(a) := E \cap (a-\delta, a+\delta)$ for some $\delta > 0$.

Remark (Standard quantified statement).

$$\text{Let } E \subseteq \mathbb{R},\ a \in E,\ \text{and } U \subseteq \mathbb{R}.$$
$$U \text{ is a neighborhood of } a \text{ in } E \iff \exists \delta > 0 \ \big(U = E \cap (a-\delta, a+\delta)\big).$$

Remark (Definition predicate reading).

$$\text{RelativeNeighborhood}(U, E, a) := \exists \delta > 0 \ \big(U = E \cap (a-\delta, a+\delta)\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } E \subseteq \mathbb{R},\ a \in E,\ \text{and } U \subseteq \mathbb{R}.\\ &U \text{ fails to be a neighborhood of } a \text{ in } E \iff \forall \delta > 0\ \big(U \neq E \cap (a-\delta, a+\delta)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{RelativeNeighborhood}(U, E, a) \iff \forall \delta > 0\ \big(U \neq E \cap (a-\delta, a+\delta)\big).$$

Remark (Failure modes). Failure occurs precisely when no radius δ makes U coincide with the relative interval slice: for every $\delta > 0$ either U contains a point outside $E \cap (a-\delta, a+\delta)$ or the slice contains a point missing from U.

Remark (Failure mode decomposition).

$$\forall \delta > 0\ \underbrace{\big(\exists x \in U :\ x \notin E \cap (a-\delta, a+\delta)\big)}_{\text{extraneous point}} \vee \underbrace{\big(\exists y \in E \cap (a-\delta, a+\delta) :\ y \notin U\big)}_{\text{missing point}}.$$

Remark (Interpretation). Relative neighborhoods capture the local behavior of E around a by intersecting E with an ordinary open interval centered at a. The construction restricts the ambient notion of openness to the geometry of E, so it is suited to relative topologies. Failure simply means that U never matches the precise slice of E at any scale, typically because U omits some nearby points of E or includes points that fall outside E near a.

Remark (Dependencies).

- Subset
- Epsilon Neighbourhood
- Order on $\mathbb{R}$

Definition (Continuity at a Point)

Definition 10.2.2 (Continuity at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is **continuous at** c if and only if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A : \big(|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &f \text{ is continuous at } c \iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{ContinuousAt}(f, A, c) :\equiv \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.$$
$$f \text{ is not continuous at } c \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f, A, c) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0\big).$$

Remark (Failure modes). Continuity at c fails precisely when there exists a persistent output gap $\varepsilon_0 > 0$ such that no matter how small the chosen neighborhood of c is, a point of A within that neighborhood keeps the function values at least ε_0 away from $f(c)$. The single failure branch records this inescapable oscillation near c.

Remark (Failure mode decomposition).

$$\underbrace{\exists \varepsilon_0 > 0}_{\text{persistent gap}}\quad \underbrace{\forall \delta > 0}_{\text{every neighborhood}}\quad \underbrace{\exists x \in A}_{\text{nearby witness}}\ :\ |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0.$$

Remark (Interpretation). Continuity at c means that the output $f(x)$ can be kept arbitrarily close to $f(c)$ by taking x sufficiently close to c while staying in the domain A. The negated form highlights that a discontinuity arises only when there is a fixed positive separation in output that persists no matter how tightly one zooms in on c, so nearby inputs cannot force the outputs to settle near $f(c)$. This captures the geometric intuition that the graph of a continuous function has no abrupt jumps or breaks at c.

Remark (Dependencies).

- Definition: Limit of a Function

Definition (Continuity at a Point — Neighbourhood Form)

Definition 10.2.3 (Continuity at a Point — Neighbourhood Form)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is continuous at c if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$f\big(A \cap (c - \delta, c + \delta)\big) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &f \text{ is continuous at } c \iff \forall \varepsilon > 0\ \exists \delta > 0 :\\ &\qquad f\big(A \cap (c - \delta, c + \delta)\big) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{ContinuousAt}(f, c) :\!= \forall \varepsilon > 0\ \exists \delta > 0 :\ f\big(A \cap (c - \delta, c + \delta)\big) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &\neg\big(f \text{ is continuous at } c\big) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0 :\\ &\qquad f\big(A \cap (c-\delta, c+\delta)\big) \nsubseteq (f(c)-\varepsilon_0, f(c)+\varepsilon_0).\end{aligned}$$

Remark (Negation predicate reading).

$$\text{DiscontinuousAt}(f, c) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0 :\ f\big(A \cap (c-\delta, c+\delta)\big) \nsubseteq (f(c)-\varepsilon_0, f(c)+\varepsilon_0).$$

Remark (Failure modes). Failure occurs precisely when some fixed output tolerance cannot be matched by any neighborhood about c. This manifests by finding points of A arbitrarily close to c whose images either exceed the upper endpoint $f(c)+\varepsilon_0$ or fall below the lower endpoint $f(c)-\varepsilon_0$.

Remark (Failure mode decomposition).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \underbrace{\Big(\exists x \in A \cap (c-\delta, c+\delta) : f(x) \geq f(c)+\varepsilon_0\Big)}_{\text{overshoot}} \vee \underbrace{\Big(\exists x \in A \cap (c-\delta, c+\delta) : f(x) \leq f(c)-\varepsilon_0}_{\text{undershoot}}$$

Remark (Interpretation). Continuity at c demands that every output tolerance around $f(c)$ has a matching neighborhood around c whose image remains confined to the tolerance band. The negation says that one tolerance band resists every neighborhood choice, so some sequence of points from A approaches c while their images either spike above or dip below $f(c)$ by at least a fixed amount. Geometrically, arbitrarily tight vertical strips around the graph above c must be achievable by shrinking the horizontal window; failure means the graph keeps escaping that strip no matter how closely the input variable hugs c.

Definition (Continuity at a Point — Sequential Form)

Definition 10.2.4 (Continuity at a Point — Sequential Form)**.** Let $A \subseteq \mathbb{R}$, let $c \in A$, and let $f\colon A \to \mathbb{R}$. The function f is continuous at c if and only if for every sequence $(x_n)_{n\in\mathbb{N}}$ in A, the convergence $\lim_{n\to\infty} x_n = c$ implies $\lim_{n\to\infty} f(x_n) = f(c)$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ c \in A,\ f\colon A \to \mathbb{R}.\\ &f \text{ is continuous at } c \iff \forall (x_n)_{n\in\mathbb{N}} \subseteq A\ \Big(\lim_{n\to\infty} x_n = c \Rightarrow \lim_{n\to\infty} f(x_n) = f(c)\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{ContinuousAt}(f, c) \coloneqq \forall x_n\ \Big(\text{Sequence}(x_n, A) \wedge \text{ConvergentSequence}(x_n, c) \Rightarrow \text{ConvergentSequence}(f(x_n), f$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ c \in A,\ f\colon A \to \mathbb{R}.\\ &f \text{ is not continuous at } c \iff \exists (x_n)_{n\in\mathbb{N}} \subseteq A\ \Big(\lim_{n\to\infty} x_n = c \wedge \lim_{n\to\infty} f(x_n) \neq f(c)\Big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f,c) \iff \exists x_n \Big(\mathrm{Sequence}(x_n, A) \wedge \mathrm{ConvergentSequence}(x_n, c) \wedge \neg\,\mathrm{ConvergentSequence}(f(x_n$$

Remark (Failure modes). Continuity at c fails sequentially when there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in A with $x_n \to c$ but $f(x_n)$ does not converge to $f(c)$. This manifests in two distinct ways: either $f(x_n)$ converges to some limit different from $f(c)$, or $f(x_n)$ fails to converge entirely (oscillating without settling or diverging to infinity).

Remark (Failure mode decomposition).

$$\neg(f \text{ continuous at } c) \iff \exists (x_n)_{n\in\mathbb{N}} \subseteq A : \underbrace{\lim_{n\to\infty} x_n = c}_{\text{valid approach}} \wedge \underbrace{\lim_{n\to\infty} f(x_n) \neq f(c)}_{\text{image sequence fails}}.$$

Remark (Interpretation). The sequential characterization asserts that information about f near c is completely encoded by its action on limits of sequences taken within the domain. Any approach to c through admissible points must transport convergent input sequences to convergent output sequences with the expected limit; conversely, a single pathological sequence suffices to expose a discontinuity. Geometrically, this captures the idea that f cannot introduce jumps or oscillations that persist along every infinitesimal path toward c.

Remark (Dependencies).

Definition 10.2.2.

Characterization of Discontinuity

Definition (Point of Discontinuity)

Definition 10.2.5 (Point of Discontinuity). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $a \in A$. The function f is discontinuous at a if and only if f is not continuous at a.

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ a \in A.\\ &f \text{ is discontinuous at } a \iff \neg\Big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon\big)\Big). \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{DiscontinuousAt}(f, a) := \neg\,\mathrm{ContinuousAt}(f, a).$$

Remark (Negated quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ a \in A.\\ &f \text{ is discontinuous at } a \iff \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x-a| < \delta \wedge |f(x) - f(a)| \geq \varepsilon\big). \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DiscontinuousAt}(f, a) \iff \mathrm{ContinuousAt}(f, a) \iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-a| < \delta \Rightarrow |f(x)-f(a)| <$$

Remark (Failure modes). Discontinuity arises either because the nearby values of f do not approach any single real number as $x \to a$, or because a well-defined limit exists but disagrees with the actual value $f(a)$.

Remark (Failure mode decomposition).

$$\underbrace{\neg\exists L \in \mathbb{R}\ \mathrm{FunctionLimit}(f, A, a, L)}_{\text{no limit exists}} \vee \underbrace{\big(\exists L \in \mathbb{R}\ \mathrm{FunctionLimit}(f, A, a, L) \wedge L \neq f(a)\big)}_{\text{limit-value mismatch}}.$$

Remark (Interpretation). A point of discontinuity is detected by the breakdown of the usual epsilon-delta control. Understanding removable, jump, and essential cases guides how to remedy or classify singular behavior in proofs.

Remark (Dependencies).

Definition 10.2.2.

Definition (Sequential Characterization of Discontinuity)

Definition 10.2.6 (Sequential Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}} \subseteq A$ such that $x_n \to c$ while $f(x_n) \not\to f(c)$.

Remark (Standard quantified statement).

$$f \text{ is discontinuous at } c \iff \exists (x_n) \subseteq A\ \big[x_n \to c\ \wedge\ f(x_n) \not\to f(c)\big].$$

Remark (Definition predicate reading).

$$\mathrm{DiscontinuousAt}(f, c) := \exists (x_n) \subseteq A :\ x_n \to c\ \wedge\ \neg\big(f(x_n) \to f(c)\big).$$

Remark (Negated quantified statement).

$$f \text{ is continuous at } c \iff \forall (x_n) \subseteq A :\ x_n \to c\ \Rightarrow\ f(x_n) \to f(c).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DiscontinuousAt}(f, c) \iff \mathrm{ContinuousAt}(f, c) \iff \forall (x_n) \subseteq A :\ x_n \to c\ \Rightarrow\ f(x_n) \to f(c).$$ ∎

Remark (Failure modes). The characterization fails — meaning f is NOT sequentially discontinuous — when every sequence (x_n) in A converging to c has $f(x_n) \to f(c)$. No sequential witness exists.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{DiscontinuousAt}(f, c) \iff \underbrace{\forall (x_n) \subseteq A :\ x_n \to c\ \Rightarrow\ f(x_n) \to f(c)}_{\text{every approach preserves the limit}}.$$

Remark (Interpretation). A discontinuity at c can be certified by a single sequence approaching c whose images refuse to settle at $f(c)$.

Remark (Dependencies).

Definition 10.2.4; Definition 10.2.5.

Definition (Neighbourhood Characterization of Discontinuity)

Definition 10.2.7 (Neighbourhood Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a neighborhood V of $f(c)$ such that for every neighborhood U of c relative to A there exists $x \in U \cap A$ with $f(x) \notin V$.

Remark (Standard quantified statement).

$$f \text{ is discontinuous at } c \iff \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A\ \big(f(x) \notin V\big).$$

Remark (Definition predicate reading).

$$\mathrm{DiscontinuousAt}(f, c) := \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A :\ f(x) \notin V.$$

Remark (Negated quantified statement).

$$f \text{ is continuous at } c \iff \forall V \text{ nbhd of } f(c)\ \exists U \text{ nbhd of } c\ \forall x \in U \cap A\ \big(f(x) \in V\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DiscontinuousAt}(f, c) \iff \mathrm{ContinuousAt}(f, c) \iff \forall V \text{ nbhd of } f(c)\ \exists U \text{ nbhd of } c\ \forall x \in U \cap A\ \big(f(x) \in V$$

Remark (Failure modes)*.* The neighbourhood discontinuity characterization fails — meaning f is actually continuous — when every output neighbourhood V of $f(c)$ can be matched by some input neighbourhood U such that f maps all of $U \cap A$ into V. No perpetually-missed output neighbourhood exists.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{DiscontinuousAt}(f, c) \iff \forall V \text{ nbhd of } f(c)\ \underbrace{\exists U \text{ nbhd of } c}_{\text{some input window}} :\ \underbrace{\forall x \in U \cap A :\ f(x) \in V}_{\text{all outputs inside tolerance}}.$$

Remark (Interpretation). Discontinuity means a fixed output neighborhood V is perpetually missed no matter how tightly the input is restricted around c.

Remark (Dependencies).

Definition 10.2.1.

Lemma (Equivalent Formulations of Discontinuity at a Point)

Lemma 10.2.8 (Equivalent Formulations of Discontinuity at a Point)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The following are equivalent:*

1. *c is a point of discontinuity in the ε-δ sense.*
2. *c is a sequential point of discontinuity.*
3. *c is a neighbourhood point of discontinuity.*

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&(1)\ \exists\varepsilon_0>0\ \forall\delta>0\ \exists x\in A:\ 0<|x-c|<\delta\wedge|f(x)-f(c)|\geq\varepsilon_0\\
\iff\ &(2)\ \exists(x_n)\subseteq A:\ x_n\to c\wedge\neg\big(f(x_n)\to f(c)\big)\\
\iff\ &(3)\ \exists V\text{ nbhd of }f(c)\ \forall U\text{ nbhd of }c\ \exists x\in U\cap A:\ f(x)\notin V.
\end{aligned}
$$

Remark (Predicate reading).

$$
\begin{aligned}
\mathrm{DiscontinuousAt}(f,c)&\iff\exists\varepsilon_0>0\ \forall\delta>0\ \exists x\in A:\ 0<|x-c|<\delta\wedge|f(x)-f(c)|\geq\varepsilon_0\\
&\iff\exists(x_n)\subseteq A:\ x_n\to c\wedge\neg\big(f(x_n)\to f(c)\big)\\
&\iff\exists V\text{ nbhd of }f(c)\ \forall U\text{ nbhd of }c\ \exists x\in U\cap A:\ f(x)\notin V.
\end{aligned}
$$

Remark (Failure modes). The equivalence cannot fail for functions at limit points of A: the proof that all three characterizations are equivalent is constructive in both directions. In the degenerate case where c is not a limit point of A, the punctured-neighbourhood conditions are vacuously satisfied by every L, and the equivalence becomes trivial rather than informative.

Remark (Interpretation). The three characterizations of discontinuity are mutually convertible. Analysts select whichever viewpoint simplifies a particular argument.

Remark (Dependencies).

- Point of Discontinuity
- Sequential Characterization of Discontinuity
- Neighbourhood Characterization of Discontinuity

Definition (Types of Discontinuity)

Definition 10.2.9 (Types of Discontinuity)**.** Let $A\subseteq\mathbb{R}$, let $f:A\to\mathbb{R}$, and let $x_0\in A$ with f discontinuous at x_0.

(i) *Removable* iff $\exists L\in\mathbb{R}:\ \lim_{x\to x_0}f(x)=L$ and $L\neq f(x_0)$.

(ii) *Jump* iff $\exists L_-,L_+\in\mathbb{R}:\ \lim_{x\to x_0^-}f(x)=L_-$, $\lim_{x\to x_0^+}f(x)=L_+$, and $L_-\neq L_+$.

(iii) *Essential* iff it is not removable.

Remark (Standard quantified statement).

$$
\begin{aligned}
&\text{Removable }\iff\exists L\in\mathbb{R}\ \Big(\lim_{x\to x_0}f(x)=L\wedge L\neq f(x_0)\Big).\\
&\text{Jump }\iff\exists L_-,L_+\in\mathbb{R}\ \Big(\lim_{x\to x_0^-}f(x)=L_-\wedge\lim_{x\to x_0^+}f(x)=L_+\wedge L_-\neq L_+\Big).\\
&\text{Essential }\iff\neg\exists L\in\mathbb{R}\ \Big(\lim_{x\to x_0}f(x)=L\wedge L\neq f(x_0)\Big).
\end{aligned}
$$

Remark (Definition predicate reading).

$$
\begin{aligned}
\mathrm{RemovableDiscontinuity}(f,x_0)&:=\exists L\in\mathbb{R}\ \big(\mathrm{FunctionLimit}(f,A,x_0,L)\wedge L\neq f(x_0)\big).\\
\mathrm{JumpDiscontinuity}(f,x_0)&:=\exists L_-,L_+\in\mathbb{R}\ \big(\mathrm{LeftLimit}(f,A,x_0,L_-)\wedge\mathrm{RightLimit}(f,A,x_0,L_+)\wedge L_-\neq\\
\mathrm{EssentialDiscontinuity}(f,x_0)&:=\neg\,\mathrm{RemovableDiscontinuity}(f,x_0).
\end{aligned}
$$

Remark (Negated quantified statement).

$$\begin{aligned}
\neg\text{Removable} &\iff \forall L \in \mathbb{R}\ \big(\lim_{x\to x_0} f(x) \neq L \vee L = f(x_0)\big),\\
\neg\text{Jump} &\iff \forall L_-, L_+ \in \mathbb{R}\ \big(\lim_{x\to x_0^-} f(x) \neq L_- \vee \lim_{x\to x_0^+} f(x) \neq L_+ \vee L_- = L_+\big),\\
\neg\text{Essential} &\iff \text{Removable}.
\end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned}
\neg\,\text{RemovableDiscontinuity}(f, x_0) &\iff \forall L \in \mathbb{R}\ \big(\neg\,\text{FunctionLimit}(f, A, x_0, L) \vee L = f(x_0)\big).\\
\neg\,\text{JumpDiscontinuity}(f, x_0) &\iff \forall L_-, L_+ \in \mathbb{R}\ \big(\neg\,\text{LeftLimit}(f, A, x_0, L_-) \vee \neg\,\text{RightLimit}(f, A, x_0, L_+)\\
\neg\,\text{EssentialDiscontinuity}(f, x_0) &\iff \text{RemovableDiscontinuity}(f, x_0).
\end{aligned}$$

Remark (Failure modes).

- Removable discontinuity fails when no bilateral limit exists at x_0, or when the bilateral limit exists but equals $f(x_0)$ (making the function actually continuous).
- Jump discontinuity fails when at least one of the one-sided limits does not exist, or when both one-sided limits exist but are equal (making it potentially removable or continuous).
- Essential discontinuity fails when the discontinuity is removable (the bilateral limit exists and differs from $f(x_0)$).

Remark (Failure mode decomposition).

$$\begin{aligned}
\neg\,\text{RemovableDiscontinuity}(f, x_0) &\iff \underbrace{\neg\exists L \in \mathbb{R}\ \lim_{x\to x_0} f(x) = L}_{\text{no bilateral limit exists}} \vee \underbrace{\lim_{x\to x_0} f(x) = f(x_0)}_{\text{limit equals function value}}.\\
\neg\,\text{JumpDiscontinuity}(f, x_0) &\iff \underbrace{\neg\exists L_- \in \mathbb{R}\ \lim_{x\to x_0^-} f(x) = L_-}_{\text{no left limit}} \vee \underbrace{\neg\exists L_+ \in \mathbb{R}\ \lim_{x\to x_0^+} f(x) = L_+}_{\text{no right limit}}\\
&\qquad \vee \underbrace{\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x)}_{\text{one-sided limits agree}}.
\end{aligned}$$

Remark (Interpretation). Removable discontinuities can be healed by redefining $f(x_0)$. Jump discontinuities provide well-defined one-sided limits useful in integration theory. Essential discontinuities are pathological and require more global tools.

Remark (Dependencies).

- Point of Discontinuity
- Limit of a Function
- Left-Hand Limit
- Right-Hand Limit

Oscillation

Definition (Oscillation on a Set)

Definition 10.2.10 (Oscillation on a Set)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $E \subseteq A$. The oscillation of f on E is the extended real number

$$\omega(f;E) \coloneqq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\} = \operatorname{diam}(f(E)).$$

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ E \subseteq A,\ \omega \in [0,\infty].\\ &\omega(f;E) = \omega \iff \omega = \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{OscillationOn}(f,E,\omega) \coloneqq \omega = \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ E \subseteq A,\ \omega \in [0,\infty].\\ &\omega(f;E) \neq \omega \iff \omega \neq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.\end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{OscillationOn}(f,E,\omega) \iff \omega \neq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

Remark (Failure modes)*.* The proposed value ω fails either because it is not an upper bound for pairwise differences of function values on E, or because it is an upper bound but not the least such upper bound.

Remark (Failure mode decomposition)*.*

$$\underbrace{\exists x_1, x_2 \in E :\ |f(x_1) - f(x_2)| > \omega}_{\text{not an upper bound}} \vee \underbrace{\exists \eta < \omega\ \forall x_1, x_2 \in E :\ |f(x_1) - f(x_2)| \leq \eta}_{\text{not least}}.$$

Remark (Interpretation)*.* Oscillation on a set records the spread of the image — the diameter of $f(E)$. Small oscillation means all values of f on E lie close together. Zero oscillation means f is constant on E.

Remark (Dependencies)*.*

- Function
- Subset
- Supremum
- Absolute Value on $\mathbb{R}$

Definition (Oscillation at a Point)

Definition 10.2.11 (Oscillation at a Point)**.** Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $x_0 \in \mathbb{R}$ satisfy $U_\delta(x_0) \cap E \neq \varnothing$ for every $\delta > 0$, where $U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}$. The oscillation of f at x_0 is the extended real number

$$\omega(f; x_0) = \lim_{\delta \to 0^+} \omega\big(f;\, U_\delta(x_0) \cap E\big),$$

with $\omega(f; A)$ denoting the oscillation of f on the set A. Because the function $\delta \mapsto \omega\big(f;\, U_\delta(x_0) \cap E\big)$ is nonincreasing, this right-hand limit exists (possibly $+\infty$).

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } E \subseteq \mathbb{R},\ f : E \to \mathbb{R},\ x_0 \in \mathbb{R},\ \omega \in [0, \infty].\\
&\big[(\forall \delta > 0 :\ U_\delta(x_0) \cap E \neq \varnothing)\ \wedge\ \omega(f; x_0) = \omega\big]\\
&\iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall r\ (0 < r < \delta \Rightarrow |\omega(f; U_r(x_0) \cap E) - \omega| < \varepsilon).
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{OscillationAt}(f, x_0, \omega) :\equiv \lim_{\delta \to 0^+} \omega\big(f;\, U_\delta(x_0) \cap E\big) = \omega, \quad U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}.$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } E \subseteq \mathbb{R},\ f : E \to \mathbb{R},\ x_0 \in \mathbb{R},\ \omega \in [0, \infty].\\
&\neg\big[(\forall \delta > 0 :\ U_\delta(x_0) \cap E \neq \varnothing)\ \wedge\ \omega(f; x_0) = \omega\big]\\
&\iff \Big[\exists \delta_0 > 0 :\ U_{\delta_0}(x_0) \cap E = \varnothing\Big]\\
&\qquad \vee \Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E) - \omega| \geq \varepsilon_0)\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{OscillationAt}(f, x_0, \omega) \iff \Big[\exists \delta_0 > 0 :\ U_{\delta_0}(x_0) \cap E = \varnothing\Big] \vee \Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E$$

Remark (Failure modes). The definition fails in two distinct ways: either the point x_0 has a punctured neighborhood missing E, so there are no arbitrarily small samples to measure, or the local oscillation values near x_0 refuse to stabilize, preventing the limit from existing.

Remark (Failure mode decomposition).

$$\neg\, \mathrm{OscillationAt}(f, x_0, \omega) = \underbrace{\exists \delta_0 > 0 :\ U_{\delta_0}(x_0) \cap E = \varnothing}_{\text{no nearby sample points}} \vee \underbrace{\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E)}_{\text{oscillation limit does not exist}}$$

Remark (Interpretation). Oscillation at x_0 measures the largest possible jump in function values that survives after one zooms in arbitrarily tightly around x_0. The shrinking neighborhoods $U_r(x_0) \cap E$ supply finer and finer samples, and the oscillation on each such set records the spread of f there. If these spreads settle to a single number, that number captures the local variability; it is zero precisely when f becomes arbitrarily close to a single value, i.e., when f is continuous at x_0. Failure occurs either because x_0 has no nearby domain points or because the spreads fluctuate indefinitely, reflecting wild local behavior.

Remark (Dependencies).

Definition 10.2.10

Theorem (Continuity Characterised by Oscillation)

Theorem 10.2.12 (Continuity Characterised by Oscillation)**.** *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. Then f is continuous at c if and only if $\omega(f;c) = 0$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ c \in A. \\
&\big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\big) \\
&\iff \\
&\left(\inf_{\delta>0} \sup\big\{|f(x)-f(y)| : x, y \in A,\ |x-c| < \delta,\ |y-c| < \delta\big\} = 0\right).
\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ContinuousAt}(f,c) \iff \mathrm{OscillationAt}(f,c,0).$$

Remark (Negated quantified statement).

Let $A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ c \in A.$

$\Big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ (|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big) \wedge \Big(\inf_{\delta>0} \sup\{|f(x)-f(y)| : x, y \in A,\ |x-c| < \delta,$

$\vee$

$\Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ (|x-c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0)\Big) \wedge \Big(\inf_{\delta>0} \sup\{|f(x)-f(y)| : x, y \in A,\ |x-c| <$

Remark (Negation predicate reading).

$$\neg\big(\mathrm{ContinuousAt}(f,c) \iff \mathrm{OscillationAt}(f,c,0)\big).$$

Remark (Failure modes). The equivalence fails precisely in two mutually exclusive ways: either f remains continuous at c while the oscillation retains a positive lower bound, or $\omega(f;c)$ vanishes even though f does not satisfy the ε-δ continuity condition at c.

Remark (Failure mode decomposition).

$$\underbrace{\mathrm{ContinuousAt}(f,c) \wedge \neg\,\mathrm{OscillationAt}(f,c,0)}_{\text{Positive oscillation despite continuity}} \vee \underbrace{\neg\,\mathrm{ContinuousAt}(f,c) \wedge \mathrm{OscillationAt}(f,c,0)}_{\text{Zero oscillation without continuity}}.$$

Remark (Interpretation). The theorem states that measuring the oscillation of f near c captures exactly the same local regularity as the classical ε-δ definition of continuity. If oscillation collapses to zero, every pair of nearby function values becomes arbitrarily close, forcing $f(x)$ to approach $f(c)$. Conversely, any discontinuity manifests as a nonzero oscillation, because nearby points can produce separated outputs. The standard failure therefore arises from detecting a residual jump or oscillatory behavior in arbitrarily small neighborhoods of c.

Remark (Dependencies).

Definition 10.2.2, Definition 10.2.11.

Proposition

Proposition 10.2.13 (Discontinuity Set via Oscillation)**.** *Let $f : E \to \mathbb{R}$. Then*

$$D(f) := \{x \in E : f \textit{ is discontinuous at } x\} = \{x \in E : \omega(f;x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f;x) \geq 1/n\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } f : E \to \mathbb{R}.\\
&\forall x \in E : \; x \in D(f) \iff \omega(f;x) > 0 \iff \exists n \in \mathbb{N} : \; \omega(f;x) \geq \tfrac{1}{n}.
\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}
x \in D(f) &\iff \mathrm{DiscontinuousAt}(f,x)\\
&\iff \exists \omega \; (\mathrm{OscillationAt}(f,x,\omega) \wedge \omega > 0)\\
&\iff \exists n \in \mathbb{N} \; \exists \omega \; (\mathrm{OscillationAt}(f,x,\omega) \wedge \omega \geq \tfrac{1}{n}).
\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } f : E \to \mathbb{R}.\\
&\neg\Big[\forall x \in E \; \big(x \in D(f) \iff \omega(f;x) > 0 \iff \exists n \in \mathbb{N}\; \omega(f;x) \geq \tfrac{1}{n}\big)\Big]\\
&\iff \exists x \in E \; \Big((x \in D(f) \wedge \omega(f;x) = 0) \vee (x \notin D(f) \wedge \omega(f;x) > 0)\\
&\qquad\qquad \vee (\omega(f;x) > 0 \wedge \forall n \in \mathbb{N}\; \omega(f;x) < \tfrac{1}{n})\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned}
\exists x \in E \; \Big(&\exists \omega \; (\mathrm{DiscontinuousAt}(f,x) \wedge \mathrm{OscillationAt}(f,x,\omega) \wedge \omega = 0) \vee\\
&\exists \omega \; (\neg \, \mathrm{DiscontinuousAt}(f,x) \wedge \mathrm{OscillationAt}(f,x,\omega) \wedge \omega > 0) \vee\\
&\exists \omega \; (\mathrm{OscillationAt}(f,x,\omega) \wedge \omega > 0 \wedge \forall n \in \mathbb{N}\; \omega < \tfrac{1}{n})\Big).
\end{aligned}$$

Remark (Failure modes). The proposition fails precisely when some $x \in E$ is marked as discontinuous although its oscillation vanishes, or when x is declared continuous despite a positive oscillation, or when a positive oscillation never exceeds any reciprocal integer, contradicting the Archimedean step that converts $\omega(f;x) > 0$ into the countable union description.

Remark (Failure mode decomposition).

$$\exists x \in E \begin{cases} x \in D(f) \wedge \omega(f;x) = 0 & \text{(Type I)}, \\ x \notin D(f) \wedge \omega(f;x) > 0 & \text{(Type II)}, \\ \omega(f;x) > 0 \wedge \forall n \in \mathbb{N}\ \omega(f;x) < \frac{1}{n} & \text{(Type III)}. \end{cases}$$

Remark (Interpretation). Oscillation detects discontinuity quantitatively: every discontinuity forces $\omega(f;x)$ to stay above some positive threshold, and conversely the vanishing of oscillation characterises continuity. Because each set $\{x \in E : \omega(f;x) \geq 1/n\}$ is closed, the discontinuity set $D(f)$ is an F_σ, a countable union of closed sets. This structural description is the gateway to measure-theoretic criteria such as the Lebesgue characterisation of Riemann integrability, where the size of $D(f)$ controls integrability.

Remark (Dependencies).

Definition 10.2.5; Definition 10.2.11.

10.3 Global Theorems

Global Theorems — Quick Reference

Concept/Statement	Meaning/Role
Bounded on a set	The output set is bounded in $\mathbb{R}$.
Boundedness theorem	Continuous functions on $[a, b]$ are bounded.
Absolute extrema	Global maximum and minimum values on a set.
Extreme value theorem	Continuous functions on $[a, b]$ attain extrema.
Intermediate value theorem	Continuous functions do not skip values.
Image interval theorem	$f([a, b])$ is a closed bounded interval.
Darboux property	Interval values are preserved without gaps.

Intermediate Value Theorem

Definition (Bounded on a Set)

Definition 10.3.1 (Bounded on a Set)**.** Let $A \subseteq \mathbb{R}$ and $f\colon A \to \mathbb{R}$. The function f is bounded on A if and only if there exists a constant $M > 0$ such that $|f(x)| \leq M$ for every $x \in A$.

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R} \text{ and } f\colon A \to \mathbb{R}. \\ &f \text{ is bounded on } A \iff \exists M > 0\ \forall x \in A\ (|f(x)| \leq M). \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{BoundedOn}(f, A) :\!= \exists M > 0\ \forall x \in A\ (|f(x)| \leq M).$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \wedge \forall x \in [a,b]\ (f(x) \neq c) \implies \Big[\big(\forall x \in [a,b]\ f(x) > c\big) \vee \big(\forall x \in [a,b]\ f(x) < c\big)\Big].$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } f\colon A \to \mathbb{R}.\\ &\neg(f \text{ bounded on } A) \iff \forall M > 0\ \exists x \in A\ (|f(x)| > M).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BoundedOn}(f,A) \iff \forall M > 0\ \exists x \in A\ (|f(x)| > M).$$

Remark (Failure modes). Failure occurs precisely when every proposed positive bound M is exceeded by some value of $|f(x)|$ with $x \in A$.

Remark (Failure mode decomposition).

$$\underbrace{\forall M > 0}_{\text{no admissible bound}}\ \underbrace{\exists x \in A\ (|f(x)| > M)}_{\text{value exceeding } M}.$$

Remark (Interpretation). The definition packages the informal idea that the values of f stay within a fixed vertical strip over A. The witness M serves as a uniform ceiling for the magnitudes $|f(x)|$, so once such an M is known the graph of f never escapes the band $[-M, M]$. Failure means that the outputs of f grow without limit along some sequence of points inside A, so no finite strip can contain the entire image. Boundedness on a set is the foundational regularity hypothesis that underlies the boundedness and extreme value theorems for continuous functions on compact domains.

Remark (Dependencies).

- Function
- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

Theorem (Boundedness Theorem)

Theorem 10.3.2 (Boundedness Theorem)**.** *Let $a < b$ and let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. Then there exists $M > 0$ such that $|f(x)| \leq M$ for every $x \in [a,b]$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a < b \text{ and } f : [a,b] \to \mathbb{R}.\\ &\Big(\forall c \in [a,b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a,b]\big(|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big)\\ &\qquad \implies \exists M > 0\ \forall x \in [a,b]\ (|f(x)| \leq M).\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \Longrightarrow \mathrm{BoundedOn}(f,[a,b]).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a<b \text{ and } f:[a,b]\to\mathbb{R}.\\
&\Big(\forall c\in[a,b]\ \forall\varepsilon>0\ \exists\delta>0\ \forall x\in[a,b]\big(|x-c|<\delta\Rightarrow|f(x)-f(c)|<\varepsilon\big)\Big)\\
&\qquad\wedge\Big(\forall M>0\ \exists x\in[a,b]\ (|f(x)|>M)\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \wedge \neg\,\mathrm{BoundedOn}(f,[a,b]).$$

Remark (Failure modes). Each component hypothesis is indispensable: without closedness (for instance $f(x)=1/x$ on $(0,1]$) continuity does not prevent divergence near the missing endpoint; without bounded domain (for example $f(x)=x$ on $[0,\infty)$) linear growth forces unbounded values; without continuity (take $f(0)=0$ and $f(x)=1/x$ for $x\in(0,1]$) the discontinuity at 0 again creates arbitrarily large magnitudes.

Remark (Failure mode decomposition).

$$\neg\big[\mathrm{ClosedInterval}(a,b)\wedge\mathrm{BoundedSet}([a,b])\wedge\mathrm{ContinuousOn}(f,[a,b])\big]=\underbrace{\neg\,\mathrm{ClosedInterval}(a,b)}_{\text{missing endpoint}}\vee\underbrace{\neg\,\mathrm{Bounde}}_{\text{unbounde}}$$

Remark (Interpretation). Continuity ties nearby inputs to nearby outputs, and the compact interval $[a,b]$ can be covered by finitely many neighborhoods in which the oscillation of f is controlled; taking the maximal oscillation among this finite cover supplies a global bound. Failure typically arises when the domain lacks compactness: an omitted endpoint allows the function to escape control near the hole, an unbounded interval lets values diverge along the ray, and a point of discontinuity can spike the function without upsetting nearby behavior elsewhere.

Remark (Dependencies).

Definition 10.2.2; Definition 10.3.1.

Definition (Absolute Maximum and Absolute Minimum)

Definition 10.3.3 (Absolute Maximum and Absolute Minimum). Let $A\subseteq\mathbb{R}$ and let $f\colon A\to\mathbb{R}$. A point $x^*\in A$ is an absolute maximum of f on A if and only if $f(x^*)\ge f(x)$ for every $x\in A$. A point $x_*\in A$ is an absolute minimum of f on A if and only if $f(x_*)\le f(x)$ for every $x\in A$.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A\subseteq\mathbb{R},\ f\colon A\to\mathbb{R},\ x^*,x_*\in A.\\
&x^* \text{ is an absolute maximum of } f \text{ on } A \Longleftrightarrow \forall x\in A\ (f(x)\le f(x^*)).\\
&x_* \text{ is an absolute minimum of } f \text{ on } A \Longleftrightarrow \forall x\in A\ (f(x_*)\le f(x)).
\end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned}(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A)) &\iff (x^* \in A) \wedge \forall x \in A\ (f(x) \leq f(x^*)),\\(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A)) &\iff (x_* \in A) \wedge \forall x \in A\ (f(x_*) \leq f(x)).\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x^*, x_* \in \mathbb{R}.\\&x^* \text{ fails to be an absolute maximum of } f \text{ on } A\\&\iff (x^* \notin A) \vee \big(\exists x \in A\ (f(x) > f(x^*))\big).\\&x_* \text{ fails to be an absolute minimum of } f \text{ on } A\\&\iff (x_* \notin A) \vee \big(\exists x \in A\ (f(x_*) > f(x))\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned}\neg\big[(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A))\big] &\iff (x^* \notin A) \vee \big(\exists x \in A\ (f(x) > f(x^*))\big),\\\neg\big[(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A))\big] &\iff (x_* \notin A) \vee \big(\exists x \in A\ (f(x_*) > f(x))\big).\end{aligned}$$

Remark (Failure modes). An alleged absolute maximum can fail because the proposed point lies outside the domain A, or because some $x \in A$ produces a strictly larger function value. An alleged absolute minimum fails for the analogous reasons, either by leaving the domain or by admitting a point in A with a strictly smaller value.

Remark (Failure mode decomposition).

$$\begin{aligned}\neg\big[(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A))\big] &= \underbrace{(x^* \notin A)}_{\text{external point}} \vee \underbrace{\big(\exists x \in A\ (f(x) > f(x^*))\big)}_{\text{better value}},\\\neg\big[(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A))\big] &= \underbrace{(x_* \notin A)}_{\text{external point}} \vee \underbrace{\big(\exists x \in A\ (f(x_*) > f(x))\big)}_{\text{better value}}.\end{aligned}$$

Remark (Interpretation). An absolute maximum identifies the highest value that f attains on its domain A, together with a point of attainment, and the absolute minimum identifies the lowest attained value. These extrema describe the global ceiling and floor of the graph restricted to A, so they are the endpoints of the vertical range. Failure occurs either when the candidate point is not part of the domain or when another domain point produces a superior value, reflecting that global optimization is sensitive to both membership and comparative size. Geometrically the absolute maximum sits at the tallest point of the graph above A, while the absolute minimum sits at the lowest point.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$

- Maximum
- Minimum

Theorem (Extreme Value Theorem)

Theorem 10.3.4 (Extreme Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$, and let $f : [a, b] \to \mathbb{R}$ be continuous. There exist points $x_m, x_M \in [a, b]$ such that $f(x_m) \leq f(x) \leq f(x_M)$ for every $x \in [a, b]$. Equivalently, f attains both an absolute maximum and an absolute minimum on $[a, b]$. Go to proof.*

Remark (Standard quantified statement)*.*

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } f : [a, b] \to \mathbb{R} \text{ be continuous.}$$
$$\exists x_m, x_M \in [a, b]\ \forall x \in [a, b]\ (f(x_m) \leq f(x) \leq f(x_M)).$$

Remark (Predicate reading)*.*

$$\text{ContinuousOn}(f, [a, b]) \implies \exists x_m, x_M \in [a, b]\ \big(\text{Maximum}(f(x_M), f([a, b])) \wedge \text{Minimum}(f(x_m), f([a, b]))\big).$$

Remark (Failure modes)*.* The theorem can fail if the compact interval hypothesis is lost, or if the function is not continuous on the whole interval. Without compactness an extremal value may exist only as a limiting value, and without continuity the graph may jump over its supremum or infimum.

Remark (Failure mode decomposition)*.*

$$\underbrace{\neg(a < b)}_{\text{invalid interval}} \vee \underbrace{\neg\big(f : [a, b] \to \mathbb{R}\big)}_{\text{wrong function type}} \vee \underbrace{\exists c \in [a, b]\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in [a, b]\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon\big)}_{\text{loss of continuity}}.$$

Remark (Interpretation)*.* Continuity on a closed and bounded interval forces the image of the interval to inherit the compact features of the domain. The function cannot escape to infinity or oscillate without settling, so it must reach both its highest and lowest values somewhere in the interval. When the hypotheses fail, the typical obstruction is the loss of compactness: on an open interval the function might drift off without achieving its supremum, and without continuity it may jump over candidate extrema.

Remark (Dependencies)*.*

Uses Definition 10.3.3 and Theorem 10.3.2.

Theorem (Location of Roots)

Theorem 10.3.5 (Location of Roots)**.** *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous. If either $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$, then there exists $c \in (a, b)$ such that $f(c) = 0$. Go to proof.*

Remark (Standard quantified statement)*.*

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } f : [a, b] \to \mathbb{R} \text{ continuous.}$$
$$\big(f(a) < 0 < f(b) \vee f(a) > 0 > f(b)\big) \Rightarrow \exists c \in (a, b)\ \big(f(c) = 0\big).$$

Remark (Failure modes). The root conclusion can fail if f is not continuous on $[a,b]$, or if the endpoint values do not have opposite signs.

Remark (Failure mode decomposition).

$$\underbrace{\exists c \in [a,b]\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in [a,b]\ \big(|x-c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon\big)}_{\text{loss of continuity}} \vee \underbrace{\big(f(a)f(b) \geq 0\big)}_{\text{no sign change}}.$$

Remark (Interpretation). Continuity forces the image of the interval $[a,b]$ to contain every intermediate value between $f(a)$ and $f(b)$. A sign change therefore compels the graph to cross the horizontal axis, producing a root strictly inside the interval. Failure occurs when the endpoint values share the same sign or when f is not continuous, because then the image need not sweep across zero. Geometrically, the theorem asserts that a continuous curve connecting opposite sides of the x-axis must intersect the axis somewhere in between.

Remark (Dependencies).

Bolzano's Intermediate Value Theorem.

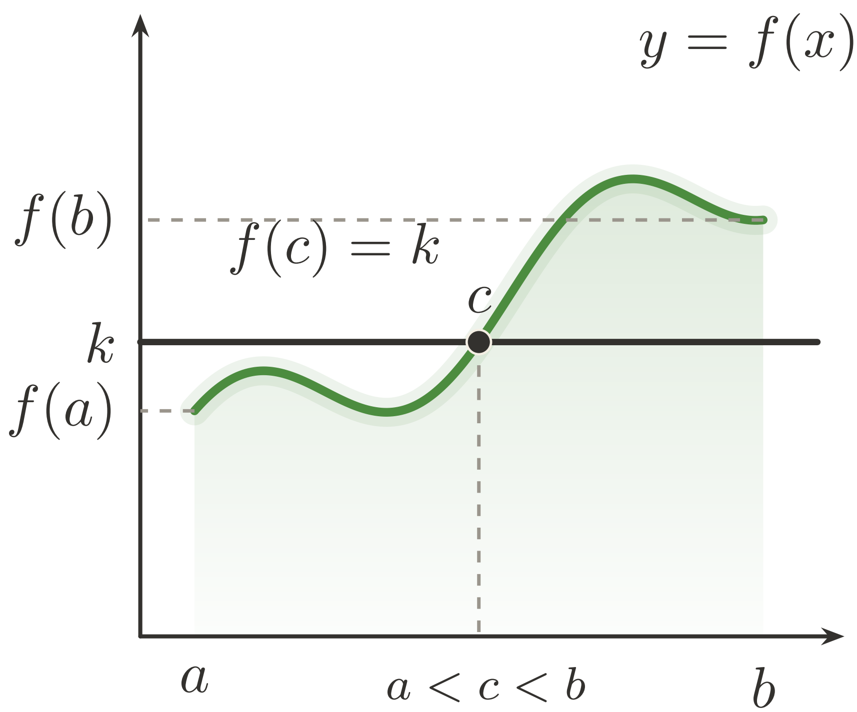


Figure 10.1: Intermediate Value Theorem: a continuous path hits every intermediate height.

Theorem (Bolzano's Intermediate Value Theorem)

Theorem 10.3.6 (Bolzano's Intermediate Value Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f\colon I \to \mathbb{R}$ be continuous on I, and choose $a, b \in I$ with $f(a) < k < f(b)$. Then there exists $c \in I$ such that $\min\{a,b\} < c < \max\{a,b\}$ and $f(c) = k$.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall a, b \in I\ \forall k \in \mathbb{R}\ \Big[\big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in$$

Remark (Failure modes). The conclusion can fail only if the continuity-on-an-interval hypothesis breaks down, or if the proposed level k is not strictly between the two attained endpoint values.

Remark (Failure mode decomposition).

$$\begin{aligned} &\underbrace{\exists x, y \in I\ \exists z \in \mathbb{R}\ \big(x < z < y \wedge z \notin I\big)}_{\text{not an interval}} \\ \vee\ &\underbrace{\exists c \in I\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in I\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon\big)}_{\text{loss of continuity}} \\ \vee\ &\underbrace{\neg\big(f(a) < k < f(b)\big)}_{\text{level not between endpoint values}}. \end{aligned}$$

Remark (Interpretation). A continuous real-valued function on an interval cannot skip any intermediate scalar between two attained values, so the graph must cross every horizontal level that lies strictly between $f(a)$ and $f(b)$. The conclusion provides an interior point c whose image equals the prescribed level k, ensuring root-finding for $f - k$. If the hypothesis fails, either f is not continuous or the level k does not lie between the endpoint values, and no such interior point need exist.

Remark (Dependencies).

- Function
- Subset
- Continuity at a Point
- Order on $\mathbb{R}$
- Maximum
- Minimum

Theorem (Image of Closed Bounded Interval Is Closed Bounded Interval)

Theorem 10.3.7 (Image of Closed Bounded Interval Is Closed Bounded Interval). *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then $f([a, b]) = [\min_{[a,b]} f, \max_{[a,b]} f]$. Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ \forall f : [a, b] \to \mathbb{R} :$$
$$\Big(a < b \wedge \forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big) \Longrightarrow f([a, b]) = [\min_{[a,b]} f$$

Remark (Predicate reading).

$$\text{ContinuousOn}(f, [a,b]) \Longrightarrow f([a,b]) = [\min_{[a,b]} f, \max_{[a,b]} f].$$

Remark (Interpretation). The statement combines the extreme value theorem, which supplies both a minimum and a maximum on $[a,b]$, with the intermediate value theorem, which fills in every intermediate value between them. Consequently the image $f([a,b])$ is a closed interval with endpoints given by the absolute extremal values of f on $[a,b]$, showing that continuous functions map compact intervals onto compact intervals without introducing gaps.

Remark (Dependencies).

Extreme Value Theorem, Bolzano's Intermediate Value Theorem

Theorem (Preservation of Intervals)

Theorem 10.3.8 (Preservation of Intervals). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be continuous. Then $f(I)$ is an interval. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R} : \\
&\quad \Big[\big(\forall x_1, x_2 \in I\ \forall z \in \mathbb{R}\ (x_1 < z < x_2 \Rightarrow z \in I)\big) \\
&\quad \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\big)\Big] \Longrightarrow \forall y_1, y_2 \in f(I)\ \forall y \in \mathbb{R}\ \big(y_1 <
\end{aligned}$$

Remark (Predicate reading).

$$\text{ContinuousOn}(f, I) \Longrightarrow \text{Interval}(f(I)).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\exists I \subseteq \mathbb{R}\ \exists f : I \to \mathbb{R}\ \exists y_1, y_2 \in f(I)\ \exists y \in \mathbb{R} : \\
&\quad \big(\forall x_1, x_2 \in I\ \forall z \in \mathbb{R}\ (x_1 < z < x_2 \Rightarrow z \in I)\big) \\
&\quad \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\big) \\
&\quad \wedge y_1 < y < y_2 \wedge y \notin f(I).
\end{aligned}$$

Remark (Negation predicate reading).

$$\text{Interval}(I) \wedge \text{ContinuousOn}(f, I) \wedge \neg\,\text{Interval}(f(I)).$$

Remark (Interpretation). Continuous functions cannot tear gaps open inside their images of intervals. Once the function achieves two output values, continuity and the Intermediate Value Theorem force every intermediate value to occur as well, so the image remains an interval even when I is unbounded or half-open. Failure occurs precisely when continuity breaks, because discontinuities can skip intermediate heights and leave disconnected image sets.

Remark (Dependencies).

thm:bolzano-intermediate-value

Theorem (Darboux Property)

Theorem 10.3.9 (Darboux Property)**.** *Let $a, b \in \mathbb{R}$ satisfy $a < b$, let $f : [a, b] \to \mathbb{R}$ be continuous, and fix $c \in \mathbb{R}$. If $f(x) \neq c$ for every $x \in [a, b]$, then either $f(x) > c$ for all $x \in [a, b]$ or $f(x) < c$ for all $x \in [a, b]$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\forall x \in [a, b]\ f(x) \neq c\Big) \Rightarrow \Big[\big(\forall x \in [a, b]\ f(x) > c\big) \vee \big(\forall x \in [a, b]\ f(x) < c\big)\Big].$$

Remark (Negated quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\forall x \in [a, b]\ f(x) \neq c\Big) \wedge \Big(\exists u \in [a, b]\ f(u) \leq c\Big) \wedge \Big(\exists v \in [a, b]\ f(v) \geq c\Big).$$

Remark (Negation predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \wedge \Big(\forall x \in [a, b]\ f(x) \neq c\Big) \wedge \neg\Big[\big(\forall x \in [a, b]\ f(x) > c\big) \vee \big(\forall x \in [a, b]\ f(x) < c\big)\Big].$$

Remark (Contrapositive quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\exists x \in [a, b]\ f(x) \leq c\Big) \wedge \Big(\exists y \in [a, b]\ f(y) \geq c\Big) \Rightarrow \exists z \in [a, b]\ f(z) = c.$$

Remark (Contrapositive predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \wedge \mathrm{MeetsBothSides}(f, [a, b], c) \Longrightarrow \exists z \in [a, b]\ (f(z) = c).$$

Remark (Interpretation). A continuous function on a compact interval cannot oscillate across a level c without actually attaining that level. Stated contrapositive fashion, if c is never taken then the entire graph lies strictly above or strictly below the horizontal line y=c. The only way this guarantee can fail is when the function produces values on both sides of c, in which case continuity forces a crossing point. Geometrically the conclusion describes the image of a connected set under a continuous map: it is an interval, so omitting c forces the image to stay in one of the open half-lines determined by c.

Remark (Dependencies).

Bolzano's Intermediate Value Theorem.

10.4 Uniform Continuity

Uniform Continuity — Quick Reference

Concept/Statement	Meaning/Role
Uniform continuity	One $\delta(\varepsilon)$ works for all pairs in the domain.
Sequential test	Close inputs with separated outputs witness failure.
Heine–Cantor	Continuity on $[a, b]$ upgrades to uniform continuity.
Cauchy preservation	Uniform continuity preserves Cauchy sequences.
Lipschitz	A global slope bound gives a uniform modulus.
Lipschitz $\Rightarrow$ uniform	Lipschitz control implies uniform continuity.
Bi-Lipschitz maps	Distances are controlled above and below.

Uniform Continuity

Definition (Uniform Continuity)

Definition 10.4.1 (Uniform Continuity)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is uniformly continuous on A if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, u \in A$ the implication $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon$ holds.

Remark (Standard quantified statement).

$$\begin{aligned}&(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R})\\ &\qquad \Longrightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, u \in A\ (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{UniformlyContinuous}(f, A) :\equiv \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, u \in A\ (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R})\\ &\qquad \vee \exists \varepsilon > 0\ \forall \delta > 0\ \exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{UniformlyContinuous}(f, A) \iff \exists \varepsilon > 0\ \forall \delta > 0\ \exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).$$

Remark (Failure modes). The definition fails exactly when a single positive tolerance ε_0 exposes pairs of inputs arbitrarily close together whose function values remain at least ε_0 apart, so no universal δ works for that ε_0.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{UniformlyContinuous}(f, A) = \underbrace{\exists \varepsilon_0 > 0}_{\text{frozen tolerance}} \wedge \underbrace{\forall \delta > 0}_{\text{no global control}} \wedge \underbrace{\exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon_0)}_{\text{bad witness pair}}.$$

Remark (Interpretation). Uniform continuity demands that a single proximity scale δ answers any prescribed accuracy ε simultaneously across the whole domain. The negation shows what goes wrong: a fixed output tolerance resists all choices of δ because pairs of nearby inputs can always be found that stretch the function values too far apart. Geometrically, the graph of a uniformly continuous function can be traversed with a ruler of length δ horizontally and ε vertically without ever overstepping the vertical tolerance, whereas a nonuniformly continuous function exhibits spikes that defeat every fixed ruler length.

Remark (Dependencies).

- Function
- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

Theorem (Algebra of Uniform Continuity on a General Domain)

Theorem 10.4.2 (Algebra of Uniform Continuity on a General Domain). *(a) For every set $A \subseteq \mathbb{R}$, every pair of functions $f, g : A \to \mathbb{R}$ that are uniformly continuous on A, and every scalar $c \in \mathbb{R}$, the functions $f + g$, $f - g$, and cf are uniformly continuous on A.*

(b) There exist a set $A \subseteq \mathbb{R}$ and uniformly continuous functions $f, g : A \to \mathbb{R}$ such that the product fg is not uniformly continuous on A.

Go to proof.

Remark (Standard quantified statement).

$$\forall A \subseteq \mathbb{R}\ \forall f, g : A \to \mathbb{R}\ \forall c \in \mathbb{R} :$$
$$\Big[\big(\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x, y \in A\ (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon)\big)$$
$$\wedge \big(\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x, y \in A\ (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon)\big)\Big] \Rightarrow \Big[\forall \varepsilon > 0\ \exists \delta_1 > 0\ \forall x, y \in A\ (|x -$$
$$\wedge \forall \varepsilon > 0\ \exists \delta_2 > 0\ \forall x, y \in A\ (|x - y| < \delta_2 \Rightarrow |(f - g)(x) - (f - g)(y)| < \varepsilon)$$
$$\wedge \forall \varepsilon > 0\ \exists \delta_3 > 0\ \forall x, y \in A\ (|x - y| < \delta_3 \Rightarrow |cf(x) - cf(y)| < \varepsilon)\Big],$$
$$\exists A \subseteq \mathbb{R}\ \exists f, g : A \to \mathbb{R} :\ \Big[\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x, y \in A\ (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon)\Big]$$
$$\wedge \Big[\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x, y \in A\ (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon)\Big]$$
$$\wedge \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in A\ \big(|x - y| < \delta \wedge |f(x)g(x) - f(y)g(y)| \geq \varepsilon_0\big).$$

Remark (Predicate reading).

$$\forall A \subseteq \mathbb{R}\ \forall f, g : A \to \mathbb{R}\ \forall c \in \mathbb{R} :$$
$$\big(\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A)\big) \Longrightarrow \big(\mathrm{UniformlyContinuous}(f + g, A) \wedge \mathrm{U}$$
$$\exists A \subseteq \mathbb{R}\ \exists f, g : A \to \mathbb{R} :\quad \mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A) \wedge \neg\, \mathrm{UniformlyContin}$$

Remark (Interpretation). Uniform continuity is stable under linear combinations on any common domain, so sums, differences, and scalar multiples inherit the same global control on oscillation from the original functions. The proof decomposes the linear expressions into epsilon-delta bounds that add and subtract without upsetting the uniform response. In contrast, products can amplify local variation when one factor lacks a bounded range, so even uniformly continuous multiplicands can produce a product that fails the global tolerance requirement. The theorem explains both the algebraic closure that analysts routinely exploit and the standard obstruction, guiding expectations when building uniformly continuous functions from simpler pieces.

Remark (Dependencies).

- Uniform Continuity
- Linear Combination of Real-Valued Functions
- Pointwise Difference of Functions

Theorem (Algebra of Uniform Continuity on a Bounded Domain)

Theorem 10.4.3 (Algebra of Uniform Continuity on a Bounded Domain)**.** *Let $A \subseteq \mathbb{R}$ be bounded, and let $f, g : A \to \mathbb{R}$ be uniformly continuous on A.*

(a) The product $x \mapsto f(x)g(x)$ is uniformly continuous on A.

(b) If there exists $k > 0$ such that $|g(x)| \geq k$ for all $x \in A$, then the quotient $x \mapsto f(x)/g(x)$ is uniformly continuous on A.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R} \text{ be bounded and } f, g : A \to \mathbb{R}.\\
&\big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big]\\
&\wedge \big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)\big]\\
&\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x)g(x) - f(y)g(y)| < \varepsilon),\\
&\big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big]\\
&\wedge \big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)\big]\\
&\wedge \big[\exists k > 0\ \forall x \in A\ (|g(x)| \geq k)\big]\\
&\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x)/g(x) - f(y)/g(y)| < \varepsilon).
\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}
&\text{Assume } A \subseteq \mathbb{R} \text{ is bounded and } f, g : A \to \mathbb{R}.\\
&\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A)\\
&\qquad \Longrightarrow \mathrm{UniformlyContinuous}(fg, A),\\
&\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A) \wedge \exists k > 0\ \forall x \in A\ (|g(x)| \geq k)\\
&\qquad \Longrightarrow \mathrm{UniformlyContinuous}(f/g, A).
\end{aligned}$$

Remark (Failure modes). The product conclusion can fail only when one of the hypotheses used to control the product is missing: the domain is not bounded, f is not uniformly continuous on A, or g is not uniformly continuous on A. The quotient conclusion can also fail if the denominator is not bounded away from zero on A.

Remark (Failure mode decomposition).

$$\underbrace{\neg\,\mathrm{BoundedSet}(A)}_{\text{unbounded domain}} \vee \underbrace{\neg\,\mathrm{UniformlyContinuous}(f,A)}_{\text{first factor not uniformly continuous}} \vee \underbrace{\neg\,\mathrm{UniformlyContinuous}(g,A)}_{\text{second factor not uniformly continuous}}$$
$$\vee \underbrace{\neg\exists k>0\ \forall x\in A:\ |g(x)|\geq k}_{\text{denominator not bounded away from zero}}.$$

Remark (Interpretation). Bounded subsets of $\mathbb{R}$ prevent the product of two uniformly continuous functions from developing large oscillations, so the shared modulus of continuity for f and g can be combined to control fg. The quotient is uniformly continuous whenever g never approaches zero, because the bounded-away-from-zero hypothesis allows inversion to act as another uniformly continuous operation that preserves the common modulus after scaling. Failure of boundedness or a loss of a positive lower bound for $|g|$ are the only ways these closure properties can break, and in either case the oscillation of the algebraic combination can no longer be controlled uniformly across the entire set.

Remark (Dependencies).

Definition 10.4.1.

Proposition (Sequential Characterization of Uniform Continuity)

Proposition 10.4.4 (Sequential Characterization of Uniform Continuity)**.** *Let $A\subseteq\mathbb{R}$ and let $f:A\to\mathbb{R}$. Then f is uniformly continuous on A if and only if for every pair of sequences (x_n) and (y_n) in A with $\lim_{n\to\infty}(x_n-y_n)=0$ we have $\lim_{n\to\infty}(f(x_n)-f(y_n))=0$.* *Go to proof.*

Remark (Standard quantified statement).

$$\Big[\forall\varepsilon>0\ \exists\delta>0\ \forall x,y\in A\ \big(|x-y|<\delta\Rightarrow|f(x)-f(y)|<\varepsilon\big)\Big]\iff\forall x_n,y_n:\mathbb{N}\to A\ \big(|x_n-y_n|\to0\Rightarrow|f(x_n$$

Remark (Predicate reading).

$$\mathrm{UniformlyContinuous}(f,A)\iff\forall x_n,y_n:\mathbb{N}\to A\ \big(|x_n-y_n|\to0\Rightarrow|f(x_n)-f(y_n)|\to0\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\exists A\subseteq\mathbb{R}\ \exists f:A\to\mathbb{R}:\\&\Big([\forall\varepsilon>0\ \exists\delta>0\ \forall x,y\in A\ (|x-y|<\delta\Rightarrow|f(x)-f(y)|<\varepsilon)]\\&\qquad\wedge\exists x_n,y_n:\mathbb{N}\to A\ \big(|x_n-y_n|\to0\wedge\neg(|f(x_n)-f(y_n)|\to0)\big)\Big)\\&\vee\Big([\exists\varepsilon_0>0\ \forall\delta>0\ \exists x,y\in A\ (|x-y|<\delta\wedge|f(x)-f(y)|\geq\varepsilon_0)]\wedge\forall x_n,y_n:\mathbb{N}\to A\ \big(|x_n-y_n|\to0\Rightarrow|\end{aligned}$$

Remark (Negation predicate reading).

$$\exists A \subseteq \mathbb{R}\ \exists f : A \to \mathbb{R} : \\ \Big(\text{UniformlyContinuous}(f, A) \wedge \exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)\Big) \\ \vee \Big(\neg\,\text{UniformlyContinuous}(f, A) \wedge \forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big)\Big).$$

Remark (Failure modes). The equivalence can fail in exactly two ways: either f is uniformly continuous while some pair of asymptotically coincident sequences in A yields outputs that do not converge together, or every such pair of sequences does converge together while f nonetheless fails to be uniformly continuous on A.

Remark (Failure mode decomposition).

$$\underbrace{\text{UniformlyContinuous}(f, A)}_{\text{uniform continuity holds}} \wedge \underbrace{\exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)}_{\text{sequential test fails}} \\ \vee\ \underbrace{\neg\,\text{UniformlyContinuous}(f, A)}_{\text{uniform continuity fails}} \wedge \underbrace{\forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big)}_{\text{sequential test holds}}.$$

Remark (Interpretation). Uniform continuity prevents the function from pulling together inputs that are asymptotically identical while letting their images drift apart. The sequential formulation packages this requirement into a tail test that ignores explicit epsilon-delta bookkeeping: whenever two points of A are eventually arbitrarily close, their function values must also be eventually arbitrarily close. Proving uniform continuity via sequences is often simpler, and conversely, producing two sequences that collide in A but whose images do not collide gives a concrete witness that uniform continuity fails. The two failure modes correspond to breaking one direction or the other of the claimed equivalence.

Remark (Dependencies).

Definition 10.4.1

Theorem (Heine–Cantor Theorem)

Theorem 10.4.5 (Heine–Cantor Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f is uniformly continuous on $[a, b]$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } f : [a, b] \to \mathbb{R}. \\ \Big[\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big] \\ \Longrightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in [a, b]\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon).$$

Remark (Predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \Rightarrow \text{UniformlyContinuous}(f, [a, b]).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } f : [a, b] \to \mathbb{R}.\\&\Big[\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big]\\&\wedge \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in [a, b]\ (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0)\Big].\end{aligned}$$

Remark (Negation predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \wedge \neg\, \mathrm{UniformlyContinuous}(f, [a, b]).$$

Remark (Interpretation). Continuity on the compact interval $[a, b]$ forces a single modulus of continuity that works simultaneously at every point, so oscillations of f cannot sharpen when inspecting smaller neighborhoods. Proofs usually argue by contradiction: assuming the negated form produces sequences of points whose distances shrink while their images stay apart, contradicting compactness-driven convergence. The theorem shows that closed bounded domains prevent the pathological behavior that obstructs uniform control on open or unbounded sets.

Remark (Dependencies).

Definition 10.2.2; Definition 10.4.1.

Proposition (Uniform Continuity Preserves Cauchy Sequences)

Proposition 10.4.6 (Uniform Continuity Preserves Cauchy Sequences)**.** *Let $S \subseteq \mathbb{R}$, let $f : S \to \mathbb{R}$, and let (x_n) be a sequence in S. If f is uniformly continuous on S and (x_n) is a Cauchy sequence in S, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } S \subseteq \mathbb{R},\ f : S \to \mathbb{R},\ (x_n)_{n\in\mathbb{N}} \subseteq S.\\&\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in S\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\Big]\\&\wedge \Big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_n - x_m| < \varepsilon)\Big]\\&\Rightarrow \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|f(x_n) - f(x_m)| < \varepsilon).\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{UniformlyContinuous}(f, S) \wedge \mathrm{CauchySequence}(x_n) \ \Rightarrow\ \mathrm{CauchySequence}(f(x_n)).$$

Remark (Interpretation). Uniform continuity controls the change in $f(x)$ solely through the distance between inputs, so any Cauchy sequence in the domain is sent to a sequence whose tails remain uniformly close. Ordinary pointwise continuity cannot guarantee this because the necessary input tolerance could shrink along the sequence, so uniformity is exactly the additional strength needed to transport the Cauchy property. The standard failure mode occurs when f is merely continuous and the sequence approaches a boundary point where continuity loses uniform control, producing large oscillations in the image sequence. Geometrically, uniform continuity supplies a single modulus of continuity that works for the entire path traced by (x_n), ensuring that eventually all image terms lie in an arbitrarily small common interval.

Remark (Dependencies).

Definition 10.4.1; Definition 6.11.1.

Definition (Lipschitz Condition)

Definition 10.4.7 (Lipschitz Condition)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $K > 0$. The function f satisfies the Lipschitz condition on A with constant K if and only if $|f(x) - f(u)| \le K|x - u|$ for every $x, u \in A$.

Remark (Standard quantified statement).

$$\begin{aligned}&(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R}) \wedge (K > 0)\\ &\qquad \Longrightarrow \forall x, u \in A \ \big(|f(x) - f(u)| \le K|x - u|\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Lipschitz}(f, A, K) \coloneqq (K > 0) \wedge \forall x, u \in A \ \big(|f(x) - f(u)| \le K|x - u|\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \vee (K \le 0)\\ &\qquad\qquad\qquad\qquad \vee \exists x, u \in A \ \big(|f(x) - f(u)| > K|x - u|\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Lipschitz}(f, A, K) \iff \neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \vee (K \le 0) \vee \exists x, u \in A \ \big(|f(x) - f(u)| > K|x - u|\big).$$

Remark (Failure modes). The Lipschitz condition fails when the ambient data are malformed, when the proposed constant is not positive, or when there exists a pair of points in A whose images separate faster than K times the separation of the points themselves.

Remark (Failure mode decomposition).

$$\underbrace{\neg(A \subseteq \mathbb{R})}_{\text{bad domain}} \vee \underbrace{\neg(f : A \to \mathbb{R})}_{\text{bad function type}} \vee \underbrace{K \le 0}_{\text{bad constant}} \vee \underbrace{\exists x, u \in A : \ |f(x) - f(u)| > K|x - u|}_{\text{excessive separation of outputs}}.$$

Remark (Interpretation). The Lipschitz condition requires a single constant K that simultaneously bounds the rate at which f can stretch distances across the entire set A. It encapsulates a global slope cap: every secant line through the graph has absolute slope at most K. Failure occurs once a single pair of points forces a steeper secant, signaling that no uniform control constant works on the whole domain. Geometrically, the graph of a Lipschitz function lies inside a cone of opening determined by K, so nearby points cannot be blown apart faster than K times their original spacing.

Remark (Dependencies).

- Function

Proposition (Lipschitz Implies Uniform Continuity)

Proposition 10.4.8 (Lipschitz Implies Uniform Continuity)**.** *Let $A \subseteq \mathbb{R}$ and $f : A \to \mathbb{R}$. If there exists $K \geq 0$ such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in A$, then f is uniformly continuous on A.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ \text{and suppose } \exists K \geq 0\, \forall x, y \in A\, (|f(x) - f(y)| \leq K|x - y|).$$
$$\forall \varepsilon > 0\, \exists \delta > 0\, \forall x, y \in A\, (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon).$$

Remark (Predicate reading).

$$\big[\exists K \geq 0 \ \text{Lipschitz}(f, A, K)\big] \Rightarrow \text{UniformlyContinuous}(f, A).$$

Remark (Contrapositive quantified statement). If f fails to be uniformly continuous on A, then f is not Lipschitz on A with any constant.

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R}.\\ &\Big[\exists \varepsilon_0 > 0\, \forall \delta > 0\, \exists x, y \in A\, (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0)\Big]\\ &\Longrightarrow \forall K \geq 0\, \exists x, y \in A\, (|f(x) - f(y)| > K|x - y|).\end{aligned}$$

Remark (Contrapositive predicate reading).

$$\neg\,\text{UniformlyContinuous}(f, A) \implies \neg \exists K \geq 0 :\ \text{Lipschitz}(f, A, K).$$

Remark (Interpretation). A Lipschitz bound controls the variation of f by the distance between inputs with a single constant K, so the same K yields a usable $\delta = \varepsilon/K$ for every pair of points at once, proving uniform continuity. When $K = 0$ the function is constant and trivially uniform. Failure of uniform continuity would force two points arbitrarily close together whose images stay a fixed distance apart, contradicting the Lipschitz inequality.

Remark (Strict implication note). The implication Lipschitz $\Rightarrow$ uniform continuity is strict. The canonical counterexample is $f(x) = \sqrt{x}$ on $[0, 1]$, which is uniformly continuous by the Heine–Cantor theorem but satisfies $|f(x) - f(0)|/|x - 0| = 1/\sqrt{x} \to \infty$ as $x \to 0^+$, so no Lipschitz constant exists at the origin. This distinction is relevant to GPU Lipschitz-constant estimation: uniform continuity of a simulation function does not supply a computable K for the Lipschitz condition; only an explicit Lipschitz bound does.

Remark (Dependencies).

Definition 10.4.7, Definition 10.4.1.

Theorem (Algebra of Lipschitz Functions)

Theorem 10.4.9 (Algebra of Lipschitz Functions)**.** *Let $A \subseteq \mathbb{R}$ and let $f, g : A \to \mathbb{R}$ be Lipschitz with constants $K_f, K_g \geq 0$ respectively.*

(a) For every $c \in \mathbb{R}$, the function cf is Lipschitz on A with constant $|c|K_f$.

(b) The functions $f + g$ and $f - g$ are Lipschitz on A with constant $K_f + K_g$.

(c) If $B \subseteq \mathbb{R}$, $h : B \to \mathbb{R}$ is Lipschitz with constant $K_h \geq 0$, and $f(A) \subseteq B$, then $h \circ f$ is Lipschitz on A with constant $K_h K_f$.

(d) If $|f(x)| \leq M_f$ and $|g(x)| \leq M_g$ for all $x \in A$, then fg is Lipschitz on A with constant $M_g K_f + M_f K_g$.

(e) If $|f(x)| \leq M_f$ for all $x \in A$, if $|g(x)| \geq k > 0$ for all $x \in A$, and if g is Lipschitz on A with constant K_g, then f/g is Lipschitz on A with constant $K_f/k + M_f K_g/k^2$.

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f, g : A \to \mathbb{R},\ K_f, K_g \geq 0,\\
&\forall x, y \in A:\ |f(x) - f(y)| \leq K_f|x - y| \text{ and } |g(x) - g(y)| \leq K_g|x - y|.\\
&\text{(a) } \forall c \in \mathbb{R},\ \forall x, y \in A:\ |cf(x) - cf(y)| \leq |c|K_f|x - y|.\\
&\text{(b) } \forall x, y \in A:\ |(f \pm g)(x) - (f \pm g)(y)| \leq (K_f + K_g)|x - y|.\\
&\text{(c) If } B \subseteq \mathbb{R},\ f(A) \subseteq B,\ h : B \to \mathbb{R},\ \forall u, v \in B:\ |h(u) - h(v)| \leq K_h|u - v|,\\
&\qquad \text{then } \forall x, y \in A:\ |h(f(x)) - h(f(y))| \leq K_h K_f|x - y|.\\
&\text{(d) If } |f(x)| \leq M_f,\ |g(x)| \leq M_g\ \forall x \in A,\ \text{then } \forall x, y \in A:\\
&\qquad |f(x)g(x) - f(y)g(y)| \leq (M_g K_f + M_f K_g)|x - y|.\\
&\text{(e) If } |f(x)| \leq M_f,\ |g(x)| \geq k > 0\ \forall x \in A,\ \text{then } \forall x, y \in A:\\
&\qquad \left|\frac{f(x)}{g(x)} - \frac{f(y)}{g(y)}\right| \leq \left(\frac{K_f}{k} + \frac{M_f K_g}{k^2}\right)|x - y|.
\end{aligned}
$$

Remark (Predicate reading).

$$
\begin{aligned}
&\mathrm{Lipschitz}(f, A, K_f) \wedge \mathrm{Lipschitz}(g, A, K_g)\\
&\implies \Big[\forall c \in \mathbb{R} : \mathrm{Lipschitz}(cf, A, |c|K_f)\Big]\\
&\qquad \wedge \mathrm{Lipschitz}(f + g, A, K_f + K_g) \wedge \mathrm{Lipschitz}(f - g, A, K_f + K_g)\\
&\qquad \wedge \Big(\mathrm{Lipschitz}(h, B, K_h) \wedge f(A) \subseteq B \Rightarrow \mathrm{Lipschitz}(h \circ f, A, K_h K_f)\Big)\\
&\qquad \wedge \Big[(|f| \leq M_f) \wedge (|g| \leq M_g) \Rightarrow \mathrm{Lipschitz}(fg, A, M_g K_f + M_f K_g)\Big]\\
&\qquad \wedge \left[(|f| \leq M_f) \wedge (|g| \geq k > 0) \Rightarrow \mathrm{Lipschitz}\left(\frac{f}{g}, A, \frac{K_f}{k} + \frac{M_f K_g}{k^2}\right)\right].
\end{aligned}
$$

Remark (Interpretation). Lipschitz functions are stable under the standard algebraic operations: scaling magnifies the constant proportionally, addition and subtraction add the errors, and composition multiplies constants because distortions accumulate multiplicatively. Products and quotients require uniform control of magnitudes; bounds on the factors prevent growth in the coefficients that accompany the derivatives of the product rule heuristic, while a positive lower bound on the denominator guarantees the quotient does not amplify oscillations uncontrollably. Together, these clauses justify building rich classes of Lipschitz maps from simpler ones without losing quantitative control.

Remark (Dependencies).

Definition 10.4.7, Definition 10.3.1

Definition (Bi-Lipschitz Map)

Definition 10.4.10 (Bi-Lipschitz Map)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The map f is bi-Lipschitz on A if and only if there exist constants $0 < \alpha \leq K$ such that for every $x, u \in A$,

$$\alpha|x-u| \leq |f(x)-f(u)| \leq K|x-u|.$$

Any such pair (α, K) is called a bi-Lipschitz constant pair for f on A.

Remark (Standard quantified statement).

$$(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R}) \\ \implies \exists \alpha, K \in \mathbb{R} \left(0 < \alpha \leq K \wedge \forall x, u \in A :\ \alpha|x-u| \leq |f(x)-f(u)| \leq K|x-u|\right).$$

Remark (Definition predicate reading).

$$\operatorname{BiLipschitz}(f, A, \alpha, K) \coloneqq (0 < \alpha \leq K) \wedge \forall x, u \in A\ \left(\alpha|x-u| \leq |f(x)-f(u)| \leq K|x-u|\right).$$

Remark (Negated quantified statement).

$$\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \\ \vee \forall \alpha, K \in \mathbb{R} \Big(0 < \alpha \leq K \Rightarrow \exists x, u \in A :\ |f(x)-f(u)| < \alpha|x-u| \vee |f(x) -$$

Remark (Negation predicate reading).

$$\neg \operatorname{BiLipschitz}(f, A, \alpha, K) \iff \neg(0 < \alpha \leq K) \vee \exists x, u \in A\ \left(|f(x)-f(u)| < \alpha|x-u| \vee |f(x)-f(u)| > K|x -\right.$$

Remark (Failure modes). The property fails either because f collapses some pair of points faster than the allowed lower constant α, or because it stretches some pair more than the upper constant K permits. Both distortions prevent a uniform two-sided control of distances.

Remark (Failure mode decomposition).

$$\exists x, u \in A :\quad \underbrace{|f(x)-f(u)| < \alpha|x-u|}_{\text{lower-distortion failure}} \quad \text{or} \quad \underbrace{|f(x)-f(u)| > K|x-u|}_{\text{upper-distortion failure}}.$$

Remark (Interpretation). A bi-Lipschitz map distorts every distance in A by factors lying between α and K, so it is simultaneously Lipschitz and has a Lipschitz inverse defined on its image. This two-sided control guarantees injectivity and preserves geometric features such as relative spacing up to uniform scaling, while failures arise precisely from excessive collapsing or stretching of some pair of points.

Remark (Dependencies).

- Function

Theorem (Inverse of a Bi-Lipschitz Map Is Lipschitz)

Theorem 10.4.11 (Inverse of a Bi-Lipschitz Map Is Lipschitz)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose there exist constants $\alpha, K > 0$ such that $\alpha|x-y| \le |f(x) - f(y)| \le K|x-y|$ for all $x, y \in A$. Then f is injective, the inverse mapping $f^{-1} : f(A) \to A$ is well defined, and*

$$\frac{1}{K}|u-v| \le |f^{-1}(u) - f^{-1}(v)| \le \frac{1}{\alpha}|u-v| \quad \textit{for all } u, v \in f(A).$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ \alpha, K > 0.\\
&\Big[\forall x, y \in A\ \big(\alpha|x-y| \le |f(x) - f(y)| \le K|x-y|\big)\Big]\\
&\Longrightarrow \Big[\forall x, y \in A\ (f(x) = f(y) \Rightarrow x = y)\\
&\qquad \wedge \exists g : f(A) \to A\ \big(\forall u \in f(A)\ (f(g(u)) = u)\big)\\
&\qquad \wedge \forall u, v \in f(A)\ \Big(\frac{1}{K}|u-v| \le |g(u) - g(v)| \le \frac{1}{\alpha}|u-v|\Big)\Big].
\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{BiLipschitz}(f, A, \alpha, K) \Rightarrow \big(\mathrm{Injective}(f) \wedge \mathrm{BiLipschitz}(f^{-1}, f(A), 1/K, 1/\alpha)\big).$$

Remark (Interpretation). A bi-Lipschitz bound simultaneously controls the expansion and contraction of f, so no two points of A can share the same image and the inverse is automatically a well-defined function on $f(A)$. The inequalities for f^{-1} follow by applying the original bi-Lipschitz bounds to pairs of preimages, showing that reversing the mapping preserves the same type of two-sided Lipschitz control with the reciprocals of the original constants. Failure would require collapsing two distinct points or losing the two-sided bounds, either of which contradicts the defining hypothesis.

Remark (Dependencies).

Definition 10.4.7; Definition 10.4.10.

10.5 Monotone Functions

Monotone Functions — Quick Reference

Concept/Statement	Meaning/Role
One-sided limits	Monotone functions have finite or infinite side limits.
Continuity criterion	Continuity means both side limits equal the value.
Jump	The size of the monotone discontinuity.
First-kind gaps	Monotone discontinuities are jumps.
Countability	Only countably many jumps can occur.
Continuous inverse	Strict monotonicity gives a continuous inverse.

Limits Superior and Inferior — Quick Reference

Concept/Statement	Meaning/Role
Function $\limsup$ and $\liminf$	Tail suprema become punctured-neighbourhood suprema.
Order relation	Always $\limsup \geq \liminf$.
Limit criterion	A finite limit exists exactly when they agree.

Monotone Functions and Continuity

Theorem (One-Sided Limits of Monotone Functions)

Theorem 10.5.1 (One-Sided Limits of Monotone Functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let c be interior to I, and let $f : I \to \mathbb{R}$ be increasing. Define*

$$L_- := \sup\{f(x) : x \in I,\ x < c\}, \qquad L_+ := \inf\{f(x) : x \in I,\ c < x\}.$$

Then both one-sided limits exist and satisfy

$$\lim_{x \to c^-} f(x) = L_-, \qquad \lim_{x \to c^+} f(x) = L_+.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&I \subseteq \mathbb{R} \wedge \big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge f : I \to \mathbb{R} \\
&\wedge \big(\forall x, y \in I\ (x < y \Rightarrow f(x) \leq f(y))\big) \wedge c \in I \wedge \exists a, b \in I\ (a < c < b) \\
&\wedge L_- = \sup\{f(x) : x \in I,\ x < c\} \wedge L_+ = \inf\{f(x) : x \in I,\ c < x\} \\
&\implies \\
&\forall \varepsilon > 0\ \exists \delta_- > 0\ \forall x \in I\ \big((0 < c - x < \delta_- \wedge x < c) \Rightarrow |f(x) - L_-| < \varepsilon\big) \\
&\wedge \forall \varepsilon > 0\ \exists \delta_+ > 0\ \forall x \in I\ \big((0 < x - c < \delta_+ \wedge c < x) \Rightarrow |f(x) - L_+| < \varepsilon\big).
\end{aligned}$$

Remark (Predicate reading).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ increasing, and c interior to I.

$$\mathrm{LeftLimit}\Big(f, c, \sup\{f(x) : x \in I,\ x < c\}\Big) \wedge \mathrm{RightLimit}\Big(f, c, \inf\{f(x) : x \in I,\ x > c\}\Big).$$

Remark (Interpretation). For an increasing function, the values just to the left of c form a monotone increasing set bounded above by any right neighborhood, so their supremum matches the left-hand limit. Symmetrically, the values just to the right form a monotone increasing tail bounded below, making the infimum coincide with the right-hand limit. The theorem shows that monotonicity alone guarantees both one-sided limits exist and equals these extremal values. Failure occurs when c is not an interior point or f is not monotone, because the comparison sets may be empty or oscillatory, preventing the suprema or infima from governing the boundary behavior.

Remark (Dependencies).

- Completeness of $\mathbb{R}$
- Monotone Function
- Supremum
- Infimum

Corollary (Continuity Criterion for Monotone Functions)

Corollary 10.5.2 (Continuity Criterion for Monotone Functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone increasing, and let c be an interior point of I. Then f is continuous at c if and only if its one-sided limits satisfy $f(c^-) = f(c) = f(c^+)$, where $f(c^-) = \lim_{x\to c^-} f(x)$ and $f(c^+) = \lim_{x\to c^+} f(x)$. Go to proof.*

Remark (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone increasing, and c an interior point of I.

$$\big(f \text{ is continuous at } c\big) \iff \left(\lim_{x\to c^-} f(x) = f(c) = \lim_{x\to c^+} f(x)\right).$$

Remark (Predicate reading).

$$\mathrm{ContinuousAt}(f, c) \iff \big(\mathrm{LeftLimit}(f, c, f(c)) \wedge \mathrm{RightLimit}(f, c, f(c))\big).$$

Remark (Negated quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone increasing, and c an interior point of I.

$$\neg\big(f \text{ is continuous at } c\big) \iff \left(\lim_{x\to c^-} f(x) < f(c)\right) \vee \left(f(c) < \lim_{x\to c^+} f(x)\right).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f,c) \iff \big(\exists L^- \left[\mathrm{LeftLimit}(f,c,L^-) \wedge L^- < f(c)\right]\big) \vee \big(\exists L^+ \left[\mathrm{RightLimit}(f,c,L^+) \wedge f(c) < L\right.$$

Remark (Failure modes). Continuity can fail only through a jump: either the graph arrives at c from the left strictly below $f(c)$, or it departs to the right strictly above $f(c)$. Both jumps reflect the inability of the one-sided limits to match the function value at the interior point.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{ContinuousAt}(f,c) = \underbrace{\big(\exists L^- \left[\mathrm{LeftLimit}(f,c,L^-) \wedge L^- < f(c)\right]\big)}_{\text{left jump at } c} \vee \underbrace{\big(\exists L^+ \left[\mathrm{RightLimit}(f,c,L^+) \wedge f(c) < L^+\right]}_{\text{right jump at } c}$$

Remark (Interpretation). A monotone graph possesses finite one-sided limits at every interior point. Continuity at c demands that these approach values exactly meet the function value, so the graph neither overshoots nor undershoots when passing through c. Any failure manifests as a jump discontinuity: the graph either leaps upward at c or remains below the right-hand approach. This criterion is indispensable when classifying discontinuities of monotone functions, since it converts continuity into a simple equality check between the function value and the adjacent plateau levels.

Remark (Dependencies).

Definition 10.2.2; Theorem 10.5.1.

Definition (Jump of a Function)

Definition 10.5.3 (Jump of a Function)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be increasing, and let c be an interior point of I. If the one-sided limits $f(c^-)$ and $f(c^+)$ exist, the jump of f at c is

$$j_f(c) := f(c^+) - f(c^-).$$

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall c \in I\ \forall j \in \mathbb{R} \\
&\Big[\big(\forall x,y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge \exists a,b \in I\ (a < c < b) \\
&\qquad \wedge \big(\forall x,y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \wedge \exists L_-, L_+ \in \mathbb{R}\ \big(\lim_{x\to c^-} f(x) = L_- \wedge \lim_{x \to c^+} f(x) = L_+\big)\Big] \\
&\qquad \Rightarrow \big(j = j_f(c) \iff j = f(c^+) - f(c^-)\big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{JumpOfFunction}(f,I,c,j) := \mathrm{Interval}(I) \wedge \mathrm{MonotoneIncreasing}(f,I) \wedge \mathrm{InteriorPoint}(c,I) \wedge \exists L_-, L_+ \in \mathbb{R}\ \big(\mathrm{Le}$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall c \in I\ \forall j \in \mathbb{R} : \\
&\neg\big(j = j_f(c)\big) \iff \neg\big(\forall x,y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \\
&\qquad \vee \neg\big(\forall x,y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \vee \neg\big(\exists a,b \in I\ (a < c < b)\big) \\
&\qquad \vee \forall L_-, L_+ \in \mathbb{R}\ \big(\lim_{x \to c^-} f(x) \ne L_- \vee \lim_{x \to c^+} f(x) \ne L_+ \vee j \ne L_+ - L_-\big).
\end{aligned}$$

Remark (Negation predicate reading).

$\neg\,\text{JumpOfFunction}(f, I, c, j) \iff \neg\,\text{Interval}(I) \vee \neg\,\text{MonotoneIncreasing}(f, I) \vee \neg\,\text{InteriorPoint}(c, I) \vee \forall L_-, L$

Remark (Failure modes). A proposed jump value fails when a required one-sided limit does not exist, or when both one-sided limits exist but the proposed value is not their difference.

Remark (Failure mode decomposition).

$$\underbrace{\neg\big(f(c^-)\text{ exists}\big)}_{\text{missing left limit}} \vee \underbrace{\neg\big(f(c^+)\text{ exists}\big)}_{\text{missing right limit}} \vee \underbrace{j \neq f(c^+) - f(c^-)}_{\text{incorrect jump magnitude}}.$$

Remark (Interpretation). For a monotone function the one-sided limits exist at every interior point, and their difference measures the vertical gap in the graph at that point. A zero jump means the two limiting heights agree; a positive jump records a discontinuity of first kind.

Remark (Dependencies).

Theorem 10.5.1.

Proposition (Discontinuities of a Monotone Function Are of First Kind)

Proposition 10.5.4 (Discontinuities of a Monotone Function Are of First Kind). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone, and let $a \in I$ be such that there exist points of I strictly less and strictly greater than a. If f is discontinuous at a, then both one-sided limits $\lim_{x\uparrow a} f(x)$ and $\lim_{x\downarrow a} f(x)$ exist in $\mathbb{R}$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } I \subseteq \mathbb{R} \text{ be an interval, } f : I \to \mathbb{R} \text{ monotone, and } a \in I \text{ interior to } I.$$

$$\big[f \text{ is discontinuous at } a\big] \implies \exists L^-, L^+ \in \mathbb{R} : \begin{cases} \lim\limits_{x\uparrow a} f(x) = L^-, \\ \lim\limits_{x\downarrow a} f(x) = L^+. \end{cases}$$

Remark (Predicate reading).

$\text{MonotoneFunction}(f, I) \wedge \text{DiscontinuousAt}(f, a) \implies \exists L^-, L^+ \in \mathbb{R}\ \big(\text{LeftLimit}(f, a, L^-) \wedge \text{RightLimit}(f, a, L$

Remark (Interpretation). A monotone graph can jump only by settling on distinct finite one-sided limits; it cannot oscillate wildly or blow up to infinity at a point of discontinuity. Thus every discontinuity of a monotone function is a jump discontinuity of the first kind, which explains why such functions are easy to integrate and why their exceptional sets are well behaved. Geometrically, the graph approaches a finite height from the left and another finite height from the right before making the jump.

Remark (Dependencies).

Definition 10.2.5

Corollary (Jump Intervals Are Disjoint)

Corollary 10.5.5 (Jump Intervals Are Disjoint). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be nondecreasing. If $a, b \in I$ are distinct points of discontinuity of f, then the open intervals $(f(a^-), f(a^+))$ and $(f(b^-), f(b^+))$ are disjoint.Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in I \ \Big[a \neq b \wedge \big(\exists \varepsilon_a > 0 \ \forall \delta > 0 \ \exists x \in I \ (|x - a| < \delta \wedge |f(x) - f(a)| \geq \varepsilon_a)\big) \wedge \big(\exists \varepsilon_b > 0 \ \forall \delta > 0 \ \exists y \in I \ (|y - b| <$$

Remark (Predicate reading).

$$\text{MonotoneFunction}(f, I) \wedge \begin{cases} a, b \in I, \\ a \neq b, \\ \text{DiscontinuousAt}(f, a), \\ \text{DiscontinuousAt}(f, b) \end{cases} \implies (f(a^-), f(a^+)) \cap (f(b^-), f(b^+)) = \varnothing.$$

Remark (Interpretation). The corollary states that a nondecreasing function assigns disjoint open value intervals to distinct jump discontinuities. Because the function never reverses direction, the lower one-sided limit at the later point cannot drop below the upper one-sided limit at the earlier point, so the corresponding ranges cannot overlap. This distinction of jump intervals ensures that the cumulative vertical jumps of a monotone function stay separated, preventing the graph from revisiting the same band of values via different discontinuities.

Remark (Dependencies).

Definition 10.5.3; Proposition 10.5.4

Theorem (Discontinuities of a Monotone Function Are Countable)

Theorem 10.5.6 (Discontinuities of a Monotone Function Are Countable)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be monotone. Then the set $D_f := \{c \in I : f \text{ is discontinuous at } c\}$ is at most countable. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } I \subseteq \mathbb{R} \text{ be an interval and } f : I \to \mathbb{R} \text{ be monotone.} \\ &\#\{c \in I : f \text{ fails to be continuous at } c\} \leq \aleph_0. \end{aligned}$$

Remark (Predicate reading).

$$\text{MonotoneFunction}(f, I) \implies \#\{c \in I : \text{DiscontinuousAt}(f, c)\} \leq \aleph_0.$$

Remark (Interpretation). Every discontinuity of a monotone function is a jump discontinuity with a nondegenerate vertical gap $(f(c^-), f(c^+))$. Distinct discontinuities yield pairwise disjoint gaps, and each such interval contains a rational number. Associating to every discontinuity the rational number inside its gap produces an injection of the discontinuity set into $\mathbb{Q}$, forcing the set to be countable. The only way the conclusion could fail would be the existence of uncountably many disjoint jump intervals, which is impossible because the rationals are countable.

Remark (Dependencies).

Theorem 10.5.1 and Corollary 10.5.5.

Corollary (Monotone Function Is Continuous on a Dense Set)

Corollary 10.5.7 (Monotone Function Is Continuous on a Dense Set)**.** *Let $f : [a, b] \to \mathbb{R}$ be monotone. Then the set $\{x \in [a, b] : f$ is continuous at $x\}$ is dense in $[a, b]$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{gathered}\text{Let } f : [a, b] \to \mathbb{R} \text{ be monotone.}\\ \forall u, v \in [a, b] \ (u < v \Rightarrow \exists c \in (u, v) \text{ such that } f \text{ is continuous at } c).\end{gathered}$$

Remark (Predicate reading).

$$\text{MonotoneFunction}(f, [a, b]) \Longrightarrow \text{DenseSubset}\big(\{x \in [a, b] : \text{ContinuousAt}(f, x)\}, [a, b]\big).$$

Remark (Negated quantified statement).

$$\exists u, v \in [a, b] \ \Big(u < v \wedge \forall c \in (u, v) \ \exists \varepsilon_c > 0 \ \forall \delta > 0 \ \exists x \in [a, b] \ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_c\big)\Big).$$

Remark (Negation predicate reading).

$$\exists u, v \in [a, b] \ \Big(u < v \wedge \forall c \in (u, v) \ \neg\, \text{ContinuousAt}(f, c)\Big).$$

Remark (Interpretation). A monotone function on a closed interval can only jump at most countably many times, so the points of discontinuity leave room inside every subinterval for a continuity point. Thus, although the function may fail to be continuous at isolated or countably many locations, continuity still occurs throughout any interval one inspects. This property ensures that monotone functions retain many features of continuous functions when arguments depend only on dense sets of continuity points.

Remark (Dependencies).

Discontinuities of a Monotone Function Are Countable

Proposition (Continuous Injective Is Strictly Monotone)

Proposition 10.5.8 (Continuous Injective Is Strictly Monotone)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f is injective if and only if f is strictly monotone on $[a, b]$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{gathered}\text{Let } f : [a, b] \to \mathbb{R} \text{ be continuous on } [a, b].\\ \big[\forall x, y \in [a, b] \ (f(x) = f(y) \Rightarrow x = y)\big] \Longleftrightarrow \big[\forall x, y \in [a, b] \ (x < y \Rightarrow f(x) < f(y))\big]\\ \vee \big[\forall x, y \in [a, b] \ (x < y \Rightarrow f(y) < f(x))\big].\end{gathered}$$

Remark (Predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \Longrightarrow \big(\text{Injective}(f) \Longleftrightarrow \text{StrictlyMonotone}(f, [a, b])\big).$$

Remark (Interpretation). A continuous function on a closed interval cannot reverse direction without repeating a value, so injectivity enforces a single increasing or decreasing trend across the entire domain. Conversely any strictly monotone map is automatically injective. The result emphasizes that topological constraints supplied by continuity interact with order-theoretic injectivity to force global monotonicity; failure occurs precisely when the graph turns back on itself, producing repeated values. Geometrically the continuous injective graph must cross each horizontal line at most once, which is only possible when the graph rises everywhere or falls everywhere.

Remark (Dependencies).

Bolzano's Intermediate Value Theorem.

Theorem (Continuous Inverse Theorem)

Theorem 10.5.9 (Continuous Inverse Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be strictly monotone and continuous on I. Then the inverse function $f^{-1} : f(I) \to I$ exists, is strictly monotone, and is continuous on $f(I)$. Go to proof.*

Remark (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$.

$$\Big[(\forall x, y \in I)(x < y \Rightarrow f(x) < f(y)) \wedge \forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big]$$

$$\Longrightarrow \Big[\exists g : f(I) \to I \text{ such that } (\forall x \in I)\, g(f(x)) = x \wedge (\forall y \in f(I))\, f(g(y)) = y$$

$$\wedge\ (\forall u, v \in f(I))(u < v \Rightarrow g(u) < g(v)) \wedge \forall y \in f(I)\ \forall \varepsilon > 0\ \exists \eta > 0\ \forall u \in f(I)\big(|u - y| < \eta \Rightarrow |g(u) - g$$

Remark (Interpretation). A continuous strictly monotone function on an interval never reverses the order of points, so it is automatically bijective onto its image. This theorem records that the inverse map retains the same order orientation and inherits continuity, reflecting the fact that no sudden jumps can occur when the original graph crosses each horizontal line exactly once. Failure would require either loss of monotonicity or a discontinuity, either of which would create a flat spot or a jump preventing the existence of a continuous inverse. Geometrically, the graph of f^{-1} is the reflection of the graph of f across the diagonal line $y = x$, so the smoothness and strict increase or decrease are visibly preserved by symmetry.

Remark (Dependencies).

- Bolzano's Intermediate Value Theorem
- Strictly Increasing Function
- Strictly Decreasing Function
- Continuity at a Point

Limits Superior and Inferior of Functions

Definition (Limits Superior and Inferior of a Function)

Definition 10.5.10 (Limits Superior and Inferior of a Function)**.** Let $f : E \to \mathbb{R}$ and let x_0 be an accumulation point of E. For each $\delta > 0$ set

$$U(\delta) = \{\, f(x) : x \in E,\ 0 < |x - x_0| < \delta \,\}.$$

The extended real numbers $\limsup_{x\to x_0} f(x)$ and $\liminf_{x\to x_0} f(x)$ are defined by

$$\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta), \qquad \liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta),$$

where each inner supremum or infimum is taken in $\overline{\mathbb{R}}$ and the outer infimum or supremum ranges over all $\delta > 0$.

Remark (Standard quantified statement)*.*

Let $f : E \to \mathbb{R},\ x_0$ an accumulation point of $E,$ and $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}.$
$\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta),$
$\liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta).$

Remark (Definition predicate reading)*.*

$$\text{FunctionLimsup}(L, f, E, x_0) :\!= L = \inf_{\delta>0} \sup U(\delta),$$
$$\text{FunctionLiminf}(l, f, E, x_0) :\!= l = \sup_{\delta>0} \inf U(\delta),$$

where $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}$.

Remark (Negated quantified statement)*.*

Let $f : E \to \mathbb{R},\ x_0$ an accumulation point of $E,$ and $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}.$
$\neg\Big[\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta) \ \wedge\ \liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta)\Big]$
$\iff \limsup_{x\to x_0} f(x) \neq \inf_{\delta>0} \sup U(\delta) \vee \liminf_{x\to x_0} f(x) \neq \sup_{\delta>0} \inf U(\delta).$

Remark (Negation predicate reading)*.*

$$\neg\big(\text{FunctionLimsup}(L, f, E, x_0) \wedge \text{FunctionLiminf}(l, f, E, x_0)\big)$$
$$\iff L \neq \inf_{\delta>0} \sup U(\delta) \vee l \neq \sup_{\delta>0} \inf U(\delta),$$

where $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}$.

Remark (Failure modes)*.* The definition can fail only in structurally distinct ways: punctured neighborhoods might be empty despite the accumulation-point hypothesis, or the upper envelopes may blow up while a finite limsup is asserted, or the lower envelopes may dive to negative infinity while a finite liminf is asserted.

Remark (Failure mode decomposition).

$$\underbrace{\exists \delta > 0 : \; U(\delta) = \varnothing}_{\text{missing punctured samples}} \vee \underbrace{\left(\inf_{\delta>0} \sup U(\delta) = +\infty \wedge L < +\infty\right)}_{\text{upper envelope blow-up}} \vee \underbrace{\left(\sup_{\delta>0} \inf U(\delta) = -\infty \wedge l > -\infty\right)}_{\text{lower envelope collapse}}.$$

Remark (Interpretation). The limit superior records the smallest height that eventually dominates all nearby function values, while the limit inferior records the largest depth that eventually underestimates all nearby values. Taking suprema within punctured neighborhoods captures every oscillation that can be witnessed arbitrarily close to x_0, and the outer infimum or supremum squeezes these oscillations to the sharpest possible bounds. Failure occurs only if the neighborhood sampling breaks down or if the oscillations race off to infinite extremal heights, signaling that no finite envelope can control the function near the accumulation point.

Remark (Dependencies).

- Function
- Supremum
- Infimum
- Epsilon Neighbourhood

Proposition

Proposition 10.5.11 ($\limsup \geq \liminf$)**.** *Let $A \subseteq \mathbb{R}$, let x_0 be a limit point of A, and let $f : A \to \mathbb{R}$. Then*

$$\limsup_{x \to x_0} f(x) \geq \liminf_{x \to x_0} f(x).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall A \subseteq \mathbb{R} \; \forall f : A \to \mathbb{R} \; \forall x_0 \in \mathbb{R} :$$
$$\Big(\forall \eta > 0 \; \exists x \in A \setminus \{x_0\} \; (|x - x_0| < \eta)\Big) \implies \limsup_{x \to x_0} f(x) \geq \liminf_{x \to x_0} f(x).$$

Remark (Interpretation). The limsup records the largest cluster value that f attains near x_0, while the liminf records the smallest cluster value. Because every subsequential limit contributing to the liminf also contributes to the limsup, the supremum of all cluster values cannot lie below their infimum. The only way equality holds is when all cluster values coincide, i.e. when the ordinary limit exists.

Remark (Dependencies).

Definition (Limits Superior and Inferior of a Function).

Proposition

Proposition 10.5.12 (Limit Exists iff $\limsup = \liminf$)**.** *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, let x_0 be a limit point of A with $x_0 \in \mathbb{R}$, and let $L \in \mathbb{R}$. Then $\lim_{x\to x_0} f(x)$ exists and equals L if and only if $\limsup_{x\to x_0} f(x) = \liminf_{x\to x_0} f(x) = L$. Go to proof.*

Remark (Standard quantified statement)*.*

$$\begin{gathered}\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x_0 \text{ a limit point of } A,\ L \in \mathbb{R}.\\ (\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A \setminus \{x_0\}\ (0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon))\\ \iff \quad \left(\limsup_{x\to x_0} f(x) = L = \liminf_{x\to x_0} f(x)\right).\end{gathered}$$

Remark (Predicate reading)*.*

$$\mathrm{FunctionLimit}(f, A, x_0, L) \iff \big(\limsup_{x\to x_0} f(x) = L = \liminf_{x\to x_0} f(x)\big).$$

Remark (Negated quantified statement)*.*

$$\begin{gathered}\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x_0 \text{ a limit point of } A,\ L \in \mathbb{R}.\\ \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A \setminus \{x_0\}\ \big(0 < |x - x_0| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big)\Big]\\ \iff \quad \big(\limsup_{x\to x_0} f(x) \neq L \vee \liminf_{x\to x_0} f(x) \neq L\big).\end{gathered}$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{FunctionLimit}(f, A, x_0, L) \iff \big(\limsup_{x\to x_0} f(x) \neq L \vee \liminf_{x\to x_0} f(x) \neq L\big).$$

Remark (Failure modes)*.* The criterion fails only when at least one side of the equivalence breaks: either the function fails to approach L in the epsilon-delta sense, or one of the two extremal envelopes near x_0 does not equal L.

Remark (Failure mode decomposition)*.*

$$\underbrace{\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A \setminus \{x_0\}\ \big(0 < |x - x_0| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big)}_{\text{limit condition fails}} \vee \underbrace{\limsup_{x\to x_0} f(x) \neq L}_{\text{upper envelope mismatch}} \vee \underbrace{\liminf_{x\to x_0} f(x) \neq}_{\text{lower envelope mism}}$$

Remark (Interpretation)*.* The two-sided limit of a real-valued function at a finite limit point exists precisely when the largest subsequential accumulation value and the smallest subsequential accumulation value coincide, in which case they both equal the limit. If the limsup exceeds the liminf, as happens for $f(x) = \sin(1/x)$ near $x_0 = 0$, the function oscillates between distinct cluster heights and therefore has no finite limit. The theorem isolates the limit as the unique height compatible with every oscillatory envelope and shows that agreeing envelopes are the exact signature of convergence in this local setting.

Remark (Dependencies)*.*

Definition 10.5.10

10.6 Limits at Infinity

Limits at Infinity — Quick Reference

Concept/Statement	Meaning/Role
Infinite adherent points	Unboundedness makes $\pm\infty$ available as approach points.
Limit at infinity	Replace $0 < \lvert x - c\rvert < \delta$ by sufficiently large $\lvert x\rvert$.
Sequential criterion	Limits at infinity are tested along escaping sequences.
Limit laws	Algebraic limit rules carry over unchanged.

Limits at Infinity

Definition (Infinite Adherent Points)

Definition 10.6.1 (Infinite Adherent Points)**.** Let $X \subseteq \mathbb{R}$. The point $+\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x > M$. The point $-\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x < M$.

Remark (Standard quantified statement).

$$\begin{aligned} +\infty \text{ is adherent to } X &\iff \forall M \in \mathbb{R}\ \exists x \in X\ (x > M), \\ -\infty \text{ is adherent to } X &\iff \forall M \in \mathbb{R}\ \exists x \in X\ (x < M). \end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned} \mathrm{PlusInfAdherent}(X) &:= \forall M \in \mathbb{R}\ \exists x \in X\ (x > M), \\ \mathrm{MinusInfAdherent}(X) &:= \forall M \in \mathbb{R}\ \exists x \in X\ (x < M). \end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned} \neg(+\infty \text{ is adherent to } X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \le M), \\ \neg(-\infty \text{ is adherent to } X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \ge M). \end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned} \neg\,\mathrm{PlusInfAdherent}(X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \le M), \\ \neg\,\mathrm{MinusInfAdherent}(X) &\iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \ge M). \end{aligned}$$

Remark (Failure modes). Infinite adherence fails at $+\infty$ exactly when X is bounded above, and fails at $-\infty$ exactly when X is bounded below.

Remark (Failure mode decomposition).

$$\underbrace{\exists M \in \mathbb{R}\ \forall x \in X\ (x \le M)}_{\text{finite upper bound}} \qquad \underbrace{\exists M \in \mathbb{R}\ \forall x \in X\ (x \ge M)}_{\text{finite lower bound}}.$$

Remark (Interpretation). Both finite and infinite adherence are instances of adherence in $\overline{\mathbb{R}}$: finite adherence uses balls; infinite adherence uses rays.

Definition (Limit at Infinity)

Definition 10.6.2 (Limit at Infinity)**.** Let $X \subseteq \mathbb{R}$ be such that for every $M \in \mathbb{R}$ there exists $x \in X$ with $x > M$, let $f : X \to \mathbb{R}$, and let $L \in \mathbb{R}$. The limit of f at $+\infty$ equals L if and only if for every $\varepsilon > 0$ there exists $M \in \mathbb{R}$ such that for every $x \in X$, $x > M$ implies $|f(x) - L| < \varepsilon$.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } X \subseteq \mathbb{R},\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M),\ f : X \to \mathbb{R},\ L \in \mathbb{R}.\\ &\lim_{x\to+\infty} f(x) = L \iff \forall \varepsilon > 0\ \exists M \in \mathbb{R}\ \forall x \in X\\ &\qquad (x > M \Rightarrow |f(x) - L| < \varepsilon).\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{LimitAtInfinity}(f, +\infty, L) :\equiv \forall \varepsilon > 0\ \exists M \in \mathbb{R}\ \forall x \in X\ (x > M \Rightarrow |f(x) - L| < \varepsilon).$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}&\text{Let } X \subseteq \mathbb{R},\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M),\ f : X \to \mathbb{R},\ L \in \mathbb{R}.\\ &\neg\big(\lim_{x\to+\infty} f(x) = L\big) \iff \exists \varepsilon_0 > 0\ \forall M \in \mathbb{R}\ \exists x \in X\\ &\qquad (x > M \land |f(x) - L| \geq \varepsilon_0).\end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{LimitAtInfinity}(f, +\infty, L) \iff \exists \varepsilon_0 > 0\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M \land |f(x) - L| \geq \varepsilon_0).$$

Remark (Failure modes)*.* The definition fails only when there is a fixed tolerance ε_0 that the function cannot satisfy on any tail of X, meaning every attempt to choose M leaves some point $x > M$ with $|f(x) - L| \geq \varepsilon_0$.

Remark (Failure mode decomposition)*.*

$$\underbrace{\exists \varepsilon_0 > 0}_{\text{fixed tolerance}}\ \underbrace{\forall M \in \mathbb{R}}_{\text{no tail succeeds}}\ \underbrace{\exists x \in X\ (x > M \land |f(x) - L| \geq \varepsilon_0)}_{\text{far-point violation}}.$$

Remark (Interpretation)*.* The limit at $+\infty$ demands that eventually the function values stay within any prescribed accuracy of L. Intuitively, once x is large enough, the graph of f must lie in the horizontal band of height ε around L. Failure occurs when some positive tolerance is violated infinitely often by points escaping to $+\infty$. This definition allows us to treat unbounded domains using the same epsilon control that governs finite limits, capturing the long-range behavior of f with respect to the ideal point at infinity.

Remark (Dependencies)*.*

Definition 10.6.1.

10.7 Approximation

Approximation — Quick Reference

Concept/Statement	Meaning/Role
Step approximation	Uniform approximation by step functions.
Piecewise linear function	Linear pieces joined across a partition.
PL approximation	Uniform approximation by polygonal functions.
Weierstrass	Uniform polynomial approximation on compact intervals.
Bernstein polynomial	Constructive polynomial approximants on $[0, 1]$.
Bernstein convergence	Bernstein polynomials converge uniformly.

Approximation of Continuous Functions

Theorem (Step Function Approximation)

Theorem 10.7.1 (Step Function Approximation). *Let $f : [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a step function $s_\varepsilon : [a, b] \to \mathbb{R}$ such that*

$$\sup_{x\in[a,b]} |f(x) - s_\varepsilon(x)| < \varepsilon.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } f : [a, b] \to \mathbb{R} \text{ be continuous and let } \varepsilon > 0.\\ &\exists s_\varepsilon : [a, b] \to \mathbb{R} \text{ step function } \forall x \in [a, b]\ (|f(x) - s_\varepsilon(x)| < \varepsilon).\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{StepApproximation}(f, s_\varepsilon, [a, b], \varepsilon) := \mathrm{StepFunction}(s_\varepsilon, [a, b]) \wedge \forall x \in [a, b]\ (|f(x) - s_\varepsilon(x)| < \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } f : [a, b] \to \mathbb{R} \text{ be continuous and let } \varepsilon > 0.\\ &\forall s : [a, b] \to \mathbb{R} \text{ step function } \exists x \in [a, b]\ (|f(x) - s(x)| \geq \varepsilon).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\exists s_\varepsilon\ \mathrm{StepApproximation}(f, s_\varepsilon, [a, b], \varepsilon) \iff \forall s : [a, b] \to \mathbb{R}\ \Big(\mathrm{StepFunction}(s, [a, b]) \Rightarrow \exists x \in [a, b]\ (|f(x) - s$$

Remark (Failure modes). The statement fails precisely when a continuous f and tolerance ε are fixed but every step function leaves an error of at least ε at some point in $[a, b]$.

Remark (Failure mode decomposition).

$$\underbrace{\forall s : [a, b] \to \mathbb{R} \text{ step function}}_{\text{all approximants}}\ \underbrace{\exists x \in [a, b]\ (|f(x) - s(x)| \geq \varepsilon)}_{\text{witnessed large deviation}}.$$

Remark (Interpretation). Uniform continuity on $[a, b]$ guarantees that f does not oscillate rapidly at small scales, so subdividing the interval finely enough allows a step function to track f within any prescribed uniform error. The typical construction samples f at one point of each subinterval and keeps that value constant across the piece. Failure would mean that no matter how the interval is partitioned, some subinterval forces an error at least ε, contradicting uniform continuity. The result is a foundational approximation fact used to show that continuous functions on compact intervals are Riemann integrable and to connect continuous functions with simple combinatorial models.

Remark (Dependencies).

Heine–Cantor Theorem.

Definition (Piecewise Linear Function)

Definition 10.7.2 (Piecewise Linear Function)**.** Let $a, b \in \mathbb{R}$ with $a < b$ and let $g : [a, b] \to \mathbb{R}$. The function g is piecewise linear if and only if there exist $m \in \mathbb{N}$ with $m \geq 1$ and points $a = x_0 < x_1 < \cdots < x_m = b$ such that $g|_{[x_{k-1}, x_k]}$ is affine for every $k \in \{1, \ldots, m\}$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } g : [a, b] \to \mathbb{R}.\\ &g \text{ is piecewise linear}\\ &\iff \exists m \in \mathbb{N} \Bigl(m \geq 1 \wedge \exists x_0, \ldots, x_m \in [a, b] \bigl[a = x_0 < x_1 < \cdots < x_m = b\\ &\qquad\qquad \wedge \forall k \in \{1, \ldots, m\}\ g|_{[x_{k-1}, x_k]} \text{ is affine}\bigr]\Bigr).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{PiecewiseLinear}(g, [a, b]) :\equiv \exists m \in \mathbb{N} \Bigl(m \geq 1 \wedge \exists x_0, \ldots, x_m \in [a, b] \bigl[a = x_0 < \cdots < x_m = b \wedge \forall k \in \{1, \ldots, m\}$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } g : [a, b] \to \mathbb{R}.\\ &g \text{ is not piecewise linear}\\ &\iff \forall m \in \mathbb{N} \Bigl(m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a, b] \bigl[a = x_0 < \cdots < x_m = b\\ &\qquad\qquad \Rightarrow \exists k \in \{1, \ldots, m\}\ g|_{[x_{k-1}, x_k]} \text{ is not affine}\bigr]\Bigr).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{PiecewiseLinear}(g, [a, b]) \iff \forall m \in \mathbb{N} \Bigl(m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a, b] \bigl[a = x_0 < \cdots < x_m = b \Rightarrow \exists k \in \{1$$

Remark (Failure modes). A proposed piecewise-linear structure can fail because no finite ordered partition of $[a, b]$ is available, or because for every such partition at least one subinterval has a restriction of g that is not affine.

Remark (Failure mode decomposition).

$$\underbrace{\forall m \in \mathbb{N}\ (m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a,b]\ (a = x_0 < \cdots < x_m = b \Rightarrow \cdots))}_{\text{no valid finite affine subdivision}} \vee \underbrace{\exists k \in \{1, \ldots, m\}\ g|_{[x_{k-1},x_k]} \text{ is not}}_{\text{non-affine segment}}$$

Remark (Interpretation). A piecewise linear function on $[a, b]$ has a graph made from finitely many straight line segments. The definition permits different affine rules on different subintervals, but requires a finite ordered list of nodes that covers the whole interval.

Remark (Dependencies).

- Function
- Closed Interval
- Natural Numbers
- Order on $\mathbb{R}$

Theorem

Theorem 10.7.3 (Piecewise Linear Approximation). *Let $f\colon [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a continuous piecewise linear function $\ell_\varepsilon\colon [a, b] \to \mathbb{R}$ such that $\sup_{x\in[a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } f\colon [a,b] \to \mathbb{R},\ f \text{ continuous},\ \varepsilon > 0.\\ &\exists \text{ continuous piecewise linear } \ell_\varepsilon\colon [a,b] \to \mathbb{R} : \sup_{x\in[a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon.\end{aligned}$$

Remark (Interpretation). Every continuous function on a compact interval can be uniformly approximated by a polygonal curve matching finitely many sampled values of the original graph. The construction refines the interval into a sufficiently fine partition guaranteed by compactness, then connects the sampled values with straight segments, producing a uniformly small error over the entire domain. Failure would require a uniform oscillation exceeding ε on some subinterval, which cannot occur once the partition is fine enough. Geometrically, the theorem states that continuous graphs admit arbitrarily close polyline shadows, providing a bridge from rough step approximations to smooth polynomial approximations.

Remark (Dependencies).

- Heine–Cantor Theorem
- Piecewise Linear Function
- Continuity at a Point

Theorem (Weierstrass Approximation Theorem)

Theorem 10.7.4 (Weierstrass Approximation Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. For every $\varepsilon > 0$ there exists a polynomial $p_\varepsilon \in \mathbb{R}[x]$ such that $\|f - p_\varepsilon\|_{\infty,[a,b]} < \varepsilon$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a < b \text{ in } \mathbb{R}, \text{ and let } f : [a, b] \to \mathbb{R} \text{ be continuous.}\\ &\forall \varepsilon > 0\ \exists p \in \mathbb{R}[x]\ (\|f - p\|_{\infty,[a,b]} < \varepsilon).\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \Longrightarrow \forall \varepsilon > 0\ \exists p \in \mathbb{R}[x]\ \big(\|f - p\|_{\infty,[a,b]} < \varepsilon\big).$$

Remark (Interpretation). Every continuous function on a closed interval can be uniformly approximated by polynomials, so polynomials are dense in the space of continuous functions under the supremum norm. The result holds because convolutions with Bernstein polynomials preserve continuity and converge uniformly to the original function. Failure would mean there exists a uniform tolerance that no polynomial can match, contradicting the approximation scheme built from Bernstein polynomials. Geometrically, the theorem states that the graph of a continuous function can be matched arbitrarily well by the graph of a polynomial, so polynomial graphs form a flexible mesh that can shadow any continuous curve on a compact interval.

Remark (Dependencies).

Depends on thm:bernstein-approximation.

Definition (Bernstein Polynomial)

Definition 10.7.5 (Bernstein Polynomial)**.** Let $n \in \mathbb{N}$ and let $f : [0, 1] \to \mathbb{R}$. A function $B_n : [0, 1] \to \mathbb{R}$ is called the nth Bernstein polynomial of f if and only if, for every $x \in [0, 1]$,

$$B_n(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1 - x)^{n-k}.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } n \in \mathbb{N},\ f : [0, 1] \to \mathbb{R},\ B_n : [0, 1] \to \mathbb{R}.\\ &B_n \text{ is the } n\text{th Bernstein polynomial of } f\\ &\Longleftrightarrow \forall x \in [0, 1]\ \left(B_n(x) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} x^k (1 - x)^{n-k}\right).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{BernsteinPolynomial}(B_n, f, n) :\!= \Big(B_n : [0, 1] \to \mathbb{R}\Big) \wedge \forall x \in [0, 1]\ \left(B_n(x) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} x^k (1-x)^{n-k}\right).$$

Remark (Interpretation). The nth Bernstein polynomial samples f at the uniformly spaced nodes k/n, weights those samples by the binomial probabilities of order n, and produces for each x an average that depends polynomially on x. This construction smooths f into a polynomial that agrees with f at the endpoints and approximates f uniformly as n grows, so describing it explicitly is essential for subsequent approximation theorems. Failure arises only when the candidate polynomial disagrees with the binomial averaging recipe at some point of the unit interval, reflecting an incorrect coefficient or evaluation in the polynomial expression.

Remark (Dependencies).

- Function
- Natural Numbers
- Multiplication on $\mathbb{R}$
- Addition on $\mathbb{R}$

Theorem (Bernstein Approximation Theorem)

Theorem 10.7.6 (Bernstein Approximation Theorem)**.** *Let $f : [0, 1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let B_n denote the nth Bernstein polynomial of f. Then for every $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that $\|f - B_n\|_\infty < \varepsilon$ for all $n \geq n_\varepsilon$. Go to proof.*

Remark (Standard quantified statement).

Let $f : [0, 1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let B_n be the nth Bernstein polynomial of f.
$\forall \varepsilon > 0\ \exists n_\varepsilon \in \mathbb{N}\ \forall n \geq n_\varepsilon :\ \|f - B_n\|_\infty < \varepsilon.$

Remark (Interpretation). The theorem asserts that the Bernstein polynomials of a continuous function on $[0, 1]$ converge uniformly to the original function, so the entire graph of f can be approximated to any prescribed sup-norm accuracy by explicit degree-n polynomials once n is large enough. The proof works by showing that the Bernstein operators preserve positivity and linear growth while averaging values of f, which forces the uniform error to vanish as $n \to \infty$. This matters because it gives an elementary constructive route from continuous functions to polynomials, strengthening the Weierstrass approximation theorem with a concrete approximating sequence. The approximation can fail only when f is not continuous on $[0, 1]$, in which case the Bernstein polynomials may smooth out jumps and no longer capture the original function in the sup norm. Geometrically, each B_n is a convex combination of the samples $f(k/n)$, so the polygons formed by these samples become finer and eventually hug the graph of f everywhere on the interval.

Remark (Dependencies).

Definition (Continuity at a Point), Definition (Bernstein Polynomial).

10.8 Gauge

Gauges and Tagged Partitions — Quick Reference

Concept/Statement	Meaning/Role
Interval partition	Finite subdivision of a compact interval.
Tagged partition	Each subinterval carries a selected tag.
Gauge	A position-dependent positive tolerance.
δ-fine partition	Each subinterval lies within its tag's gauge window.
Cousin's theorem	Every gauge on $[a, b]$ admits a fine tagged partition.
Mesh	The largest subinterval length.
Common refinement	A partition refining two given partitions.

Gauges and Tagged Partitions

Definition (Partition)

Definition 10.8.1 (Partition)**.** Let $a < b$ and set $I = [a, b]$. A finite ordered set $P = \{x_0, \ldots, x_n\}$ with $n \in \mathbb{N}$ and $n \geq 1$ is a partition of I if and only if $x_0 = a$, $x_n = b$, and $x_0 < x_1 < \cdots < x_n$. In this case the closed subintervals $[x_{i-1}, x_i]$ for $i = 1, \ldots, n$ subdivide I.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a < b,\ I = [a, b],\ n \in \mathbb{N},\ n \geq 1, \text{ and } P = \{x_0, \ldots, x_n\} \subseteq I.\\
&P \text{ is a partition of } I\\
&\iff \big(x_0 = a \wedge x_n = b \wedge \forall i \in \{1, \ldots, n\}\ (x_{i-1} < x_i)\big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{PartitionOfInterval}(P, I) :\equiv \exists n \in \mathbb{N}\ \Big(n \geq 1 \wedge P = \{x_0, \ldots, x_n\} \subseteq I \wedge x_0 = \min I \wedge x_n = \max I \wedge \forall i \in \{1, \ldots,$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a < b,\ I = [a, b],\ n \in \mathbb{N},\ n \geq 1, \text{ and } P = \{x_0, \ldots, x_n\} \subseteq I.\\
&P \text{ is not a partition of } I\\
&\iff \big((x_0 \neq a) \vee (x_n \neq b) \vee \exists i \in \{1, \ldots, n\}\ (x_{i-1} \geq x_i)\big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{PartitionOfInterval}(P, I) \iff \forall n \in \mathbb{N}\ \Big(n \geq 1 \Rightarrow \big(P \neq \{x_0, \ldots, x_n\} \vee P \nsubseteq I \vee x_0 \neq \min I \vee x_n \neq \max I \vee \exists i$$

Remark (Failure modes). The definition fails precisely in three ways: the left endpoint is omitted, the right endpoint is omitted, or the sequence of nodes ceases to be strictly increasing, leading to repeated or out-of-order points.

Remark (Failure mode decomposition).

$$\underbrace{x_0 \neq a}_{\text{left endpoint missing}} \quad \vee \quad \underbrace{x_n \neq b}_{\text{right endpoint missing}} \quad \vee \ \underbrace{\exists i \in \{1, \ldots, n\} \ (x_{i-1} \geq x_i)}_{\text{nodes not strictly increasing}}.$$

Remark (Interpretation). A partition records the finite list of nodes that slice a closed interval into adjacent subintervals. The strict ordering ensures that every interior point appears exactly once in this linear traversal, so the induced subintervals $[x_{i-1}, x_i]$ tile $[a, b]$ without overlap or gaps. Computations of Riemann sums, Darboux sums, and gauge integrals use partitions as the combinatorial scaffolding that supports sampling and measurement. Failure arises when the endpoints are not captured or when the nodes regress or stagnate, which would leave gaps or redundancies in the subdivision.

Remark (Dependencies).

- Closed Interval
- Natural Numbers
- Order on $\mathbb{R}$

Definition (Tagged Partition)

Definition 10.8.2 (Tagged Partition)**.** Let $a, b \in \mathbb{R}$ with $a < b$. A list $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$ is a tagged partition of $[a, b]$ if and only if the subintervals determine a partition $P = \{[x_{i-1}, x_i]\}_{i=1}^{n}$ of $[a, b]$ with $a = x_0 < x_1 < \cdots < x_n = b$, and each tag satisfies $t_i \in [x_{i-1}, x_i]$ for $i = 1, \ldots, n$. Equivalently, a tagged partition is exactly a partition P of $[a, b]$ together with one tag chosen inside every subinterval of P.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}. \\
&\dot{\mathcal{P}} \text{ is a tagged partition of } [a, b] \\
&\iff \big(a = x_0 < \cdots < x_n = b \wedge \forall i \in \{1, \ldots, n\} \ (t_i \in [x_{i-1}, x_i])\big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\text{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) :\equiv \Big(\text{Partition}(P, [a, b]) \wedge \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n} \wedge \forall i \ (t_i \in [x_{i-1}, x_i])\Big),$$

where $\text{Partition}(P, [a, b])$ abbreviates $a = x_0 < \cdots < x_n = b$ for $P = \{[x_{i-1}, x_i]\}_{i=1}^{n}$.

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}. \\
&\dot{\mathcal{P}} \text{ is not a tagged partition of } [a, b] \\
&\iff \Big(x_0 \neq a \vee x_n \neq b \vee \exists j \in \{1, \ldots, n\} \ (x_{j-1} \geq x_j) \vee \exists i \in \{1, \ldots, n\} \ (t_i \notin [x_{i-1}, x_i])\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{TaggedPartition}(\dot{\mathcal{P}},[a,b]) \iff (x_0 \neq a) \vee (x_n \neq b) \vee \Big(\exists j\ (x_{j-1} \geq x_j)\Big) \vee \Big(\exists i\ (t_i \notin [x_{i-1},x_i])\Big).$$

Remark (Failure modes). A list fails to be a tagged partition either because the subintervals do not form a valid partition (the mesh is disordered or has gaps) or because at least one chosen tag lies outside its designated subinterval.

Remark (Failure mode decomposition).

$$\underbrace{x_0 \neq a}_{\text{left endpoint defect}} \quad\vee\quad \underbrace{x_n \neq b}_{\text{right endpoint defect}} \quad\vee\ \underbrace{\exists j\ (x_{j-1} \geq x_j)}_{\text{ordering defect}} \ \vee\ \underbrace{\exists i\ (t_i \notin [x_{i-1},x_i])}_{\text{tag misplaced}}.$$

Remark (Interpretation). A tagged partition augments the usual subdivision of a closed interval by selecting a representative sampling point inside each component subinterval. These tags serve as evaluation points for Riemann sums and gauge integrals. The construction encapsulates both the geometric slicing of the interval and the analytic data needed for sample values. Failure occurs when the slices no longer tile the interval in order or when a sample point is chosen outside its slice, breaking the correspondence between tags and subintervals.

Remark (Dependencies).

Definition 10.8.1.

Definition (Gauge)

Definition 10.8.3 (Gauge). Let $a,b \in \mathbb{R}$ with $a < b$ and set $I = [a,b]$. A gauge on I is a function $\delta : I \to (0,\infty)$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a,b \in \mathbb{R} \text{ with } a<b,\ \text{put } I=[a,b],\ \text{and let } \delta : I \to \mathbb{R}.\\ &\delta \text{ is a gauge on } I \iff \forall x \in I\ (\delta(x) > 0).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Gauge}(\delta, I) \coloneqq \big(\delta : I \to \mathbb{R}\big) \wedge \forall x \in I\ (\delta(x) > 0).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } a,b \in \mathbb{R} \text{ with } a<b,\ \text{put } I=[a,b],\ \text{and let } \delta : I \to \mathbb{R}.\\ &\delta \text{ fails to be a gauge on } I \iff \exists x \in I\ (\delta(x) \leq 0).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Gauge}(\delta, I) \iff \neg\big(\delta : I \to \mathbb{R}\big) \vee \exists x \in I\ (\delta(x) \leq 0).$$

Remark (Failure modes). A purported gauge fails either because it is not a function from I to $\mathbb{R}$, or because it takes on a nonpositive value somewhere in I.

Remark (Failure mode decomposition).

$$\underbrace{\neg\big(\delta : I \to \mathbb{R}\big)}_{\text{wrong function type}} \ \vee\ \underbrace{\exists x \in I\ (\delta(x) \le 0)}_{\text{nonpositive gauge value}}$$

Remark (Interpretation). A gauge assigns to each point of the closed interval a positive radius, encoding how tightly a tagged subinterval must cluster around that point in fine partition arguments. The sole obstruction is positivity: if any value is zero or negative, the construction cannot deliver legitimate local scales, so the gauge structure breaks down.

Remark (Dependencies).

- Function
- Closed Interval
- Order on $\mathbb{R}$

Definition (δ-Fine Partition)

Definition 10.8.4 (δ-Fine Partition)**.** Let $[a, b]$ be a closed interval, let $\delta : [a, b] \to (0, \infty)$ be a gauge, and let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$ be a tagged partition of $[a, b]$. The tagged partition $\dot{\mathcal{P}}$ is δ-fine if and only if for every $i \in \{1, \ldots, n\}$ and every $x \in [x_{i-1}, x_i]$ the inequality $|x - t_i| < \delta(t_i)$ holds.

Remark (Standard quantified statement).

Let $[a, b]$ be a closed interval, $\delta : [a, b] \to (0, \infty)$, and $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$ a tagged partition of $[a, b]$. $\dot{\mathcal{P}}$ is δ-fine $\iff \forall i \in \{1, \ldots, n\}\ \forall x \in [x_{i-1}, x_i]\ (|x - t_i| < \delta(t_i))$.

Remark (Definition predicate reading).

$$\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta) \coloneqq \forall i \in \{1, \ldots, n\}\ \forall x \in [x_{i-1}, x_i]\ (|x - t_i| < \delta(t_i)).$$

Remark (Negated quantified statement).

$$\exists i \in \{1, \ldots, n\}\ \exists x \in [x_{i-1}, x_i]\ \big(|x - t_i| \ge \delta(t_i)\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta) \iff \exists i \in \{1, \ldots, n\}\ \exists x \in [x_{i-1}, x_i]\ (|x - t_i| \ge \delta(t_i)).$$

Remark (Failure modes). A tagged partition fails to be δ-fine precisely when some subinterval contains a point whose distance from its tag reaches or exceeds the available radius $\delta(t_i)$, so the corresponding subinterval is not contained in the prescribed open neighborhood around its tag.

Remark (Failure mode decomposition).

$$\underbrace{\exists i \in \{1, \ldots, n\}\ \exists x \in [x_{i-1}, x_i]}_{\text{choose offending subinterval and point}}\ \underbrace{(|x - t_i| \ge \delta(t_i))}_{\text{distance not controlled by } \delta}\ .$$

Remark (Interpretation). A δ-fine tagged partition refines the interval so that each subinterval sits entirely inside the open ball centered at its tag with radius dictated by the gauge. This ensures that when summing local contributions, every evaluation point remains within a tolerance chosen for that location, a key requirement for gauge-based integral constructions. Failure occurs when a subinterval is too wide for the tolerance prescribed at its tag, so some point of the subinterval lies on or outside the allowed neighborhood.

Remark (Dependencies).

Definition 10.8.3; Definition 10.8.2

Lemma

Lemma 10.8.5 (Every Point Is Covered by a Tag)**.** *Let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$ that is δ-fine on $[a, b]$, and let $x \in [a, b]$. Then there exists $i \in \{1, \ldots, n\}$ such that $|x - t_i| < \delta(t_i)$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a < b,\ \delta : [a, b] \to (0, \infty),\ \text{and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n \text{ be } \delta\text{-fine on } [a, b].$$
$$\forall x \in [a, b]\ \exists i \in \{1, \ldots, n\}\ (|x - t_i| < \delta(t_i)).$$

Remark (Interpretation). A δ-fine tagged partition matches each subinterval to a tag that lies inside the interval and records the gauge radius allowed at that tag. The lemma asserts that every point of the underlying interval falls within the gauge ball centered at one of the tags, so the gauge control at that tag can be applied to the point. This coverage property is the mechanism that transfers local control encoded by the gauge to arbitrary points of the interval.

Remark (Dependencies).

Definition 10.8.2, Definition 10.8.4.

Theorem (Cousin's Theorem)

Theorem 10.8.6 (Cousin's Theorem)**.** *Every gauge $\delta : [a, b] \to (0, \infty)$ on a closed interval $[a, b]$ admits a δ-fine tagged partition of $[a, b]$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } \delta : [a, b] \to (0, \infty).\\
&\exists m \in \mathbb{N}\ \exists x_0, \ldots, x_m\ \exists t_1, \ldots, t_m\\
&\qquad \big(a = x_0 < x_1 < \cdots < x_m = b \wedge \forall j \in \{1, \ldots, m\}\ (t_j \in [x_{j-1}, x_j])\\
&\qquad \wedge \forall j \in \{1, \ldots, m\}\ [x_{j-1}, x_j] \subseteq (t_j - \delta(t_j), t_j + \delta(t_j))\big).
\end{aligned}$$

Remark (Predicate reading).

$$\forall \delta : [a, b] \to (0, \infty) \quad \exists \dot{\mathcal{P}} \Big(\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) \wedge \mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta) \Big).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } \delta : [a, b] \to (0, \infty).\\
&\neg\Big(\text{there exists a } \delta\text{-fine tagged partition of } [a, b]\Big)\\
&\iff \forall m \in \mathbb{N}\ \forall x_0, \ldots, x_m\ \forall t_1, \ldots, t_m\\
&\qquad \Big[\big(a = x_0 < x_1 < \cdots < x_m = b \wedge \forall j\ (t_j \in [x_{j-1}, x_j])\big)\\
&\qquad \Rightarrow \exists j \in \{1, \ldots, m\}\ [x_{j-1}, x_j] \nsubseteq (t_j - \delta(t_j), t_j + \delta(t_j))\Big].
\end{aligned}$$

Remark (Negation predicate reading).

$$\exists \delta : [a, b] \to (0, \infty)\ \forall \dot{\mathcal{P}}\ \Big(\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) \Rightarrow \neg\,\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta)\Big).$$

Remark (Failure modes). Failure would mean that a positive gauge δ defeats every tagged partition: no matter how the interval is subdivided and tagged, at least one subinterval is not contained in the gauge neighborhood around its tag.

Remark (Failure mode decomposition).

$$\underbrace{\forall \dot{\mathcal{P}}\ \ \mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a, b])}_{\text{all tagged partitions}} \quad \Rightarrow \quad \underbrace{\neg\,\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta)}_{\text{some interval escapes its gauge ball}} \quad .$$

Remark (Interpretation). The theorem is the gauge analogue of compactness for a closed interval. Although the allowed radius may vary from point to point, finitely many tagged subintervals can still be chosen so that each lies inside the neighborhood dictated by its own tag. The standard bisection proof uses the Nested Interval Property to rule out a gauge that defeats all finite tagged partitions.

Remark (Dependencies).

Definition 10.8.3, Definition 10.8.2, Definition 10.8.4.

Remark (Gauge proofs of the four classical theorems). Each proof follows the same pattern: define a gauge δ from the continuity hypothesis, invoke Cousin's theorem for a δ-fine partition, apply the Covering Lemma to reduce to finitely many tags, derive the global conclusion.

Boundedness. At each t, continuity gives $\delta(t)$ with $\mathrm{Within}(x, t, \delta(t)) \Rightarrow |f(x) - f(t)| \leq 1$. A δ-fine partition with tags $t_1, \ldots, t_n$ gives $K = \max_i |f(t_i)|$; the Covering Lemma yields $|f(x)| \leq 1 + K$ for all x.

EVT. Let $M = \sup f(I)$; suppose $f(x) < M$ everywhere. At each t, take $\delta(t)$ with $\mathrm{Within}(x, t, \delta(t)) \Rightarrow f(x) < \frac{1}{2}(M + f(t))$. A δ-fine partition produces $\widetilde{M} = \frac{1}{2}(M + \max_i f(t_i)) < M$ bounding f everywhere — contradiction.

Location of Roots. Assume $f(t) \neq 0$ for all t. At each t, take $\delta(t)$ preserving $\mathrm{sgn}(f(t))$. A δ-fine partition forces consistent sign from $f(a) < 0$ to $f(b) > 0$ — contradiction.

Heine–Cantor. Given $\varepsilon > 0$, at each t let $\delta(t)$ satisfy $\mathrm{Within}(x, t, 2\delta(t)) \Rightarrow |f(x) - f(t)| \leq \varepsilon/2$. A δ-fine partition yields $\delta_\varepsilon = \min_i \delta(t_i) > 0$. For $|x - u| < \delta_\varepsilon$: find t_i covering x; then $|u - t_i| \leq |u - x| + |x - t_i| < 2\delta(t_i)$; so $|f(x) - f(u)| \leq \varepsilon$.

Tagged Partitions and Mesh

Definition (Mesh of a Partition)

Definition 10.8.7 (Mesh of a Partition)**.** Let $a, b \in \mathbb{R}$ with $a < b$, and let $P = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. A real number m is the *mesh* (or *norm*) of P if and only if

$$m = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

The mesh of P is denoted by $\|P\|$ and is the unique m satisfying this equivalence.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } P = \{x_0, \ldots, x_n\} \text{ be a partition of } [a, b].\\ &m \text{ is the mesh of } P \iff m = \max_{1 \le i \le n} (x_i - x_{i-1}).\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\operatorname{NormOfPartition}(P, m) \coloneqq m = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } P = \{x_0, \ldots, x_n\} \text{ be a partition of } [a, b].\\ &\neg\big(m \text{ is the mesh of } P\big) \iff \big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\big) \lor \big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\big).\end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg \operatorname{NormOfPartition}(P, m) \iff \big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\big) \lor \big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\big).$$

Remark (Failure modes)*.* Failure occurs either when m underestimates the largest subinterval of P or when m overshoots that largest subinterval.

Remark (Failure mode decomposition)*.*

$$\neg \operatorname{NormOfPartition}(P, m) = \underbrace{\big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\big)}_{\text{Underestimate}} \lor \underbrace{\big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\big)}_{\text{Overestimate}}.$$

Remark (Interpretation)*.* The mesh of a finite partition records the length of the widest subinterval and thus measures the resolution of the partition. Refining a partition strictly decreases or preserves this value, so a small mesh signals uniformly short subintervals, a critical input for Riemann and gauge integral estimates. Failure arises when a proposed number does not coincide with the actual maximum subinterval length, either because some subinterval is longer than the proposal or because all subintervals are shorter.

Remark (Dependencies)*.*

Definition 10.8.1.

Definition (Refinement)

Definition 10.8.8 (Refinement)**.** Let $[a, b] \subseteq \mathbb{R}$ and let P and Q be partitions of $[a, b]$. Then Q is a refinement of P if and only if every point of P is also a point of Q; equivalently, $P \subseteq Q$.

Remark (Standard quantified statement)*.*

$$\forall x\ (x \in P \Rightarrow x \in Q).$$

Remark (Definition predicate reading)*.*

$$\mathrm{Refinement}(Q, P) \coloneqq P \subseteq Q.$$

Remark (Negated quantified statement)*.*

$$\exists x\ (x \in P \wedge x \notin Q).$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{Refinement}(Q, P) \iff \exists x \in P\ (x \notin Q).$$

Remark (Failure modes)*.* Failure occurs precisely when some point listed in P fails to appear in Q, meaning Q omits at least one of the partition points it was supposed to inherit from P.

Remark (Failure mode decomposition)*.*

$$\underbrace{\exists x \in P\ (x \notin Q)}_{\text{point of } P \text{ omitted by } Q}.$$

Remark (Interpretation)*.* Refinement formalizes the idea of subdividing an interval more finely without losing any previously chosen partition points. The condition $P \subseteq Q$ guarantees that every subinterval defined by P is still represented when one passes to Q, allowing comparisons of sums or integrals computed on different meshes. Failure means that Q discards at least one point of P, so information tied to that breakpoint would be lost when transitioning between partitions.

Remark (Dependencies)*.*

Definition 10.8.1.

Definition (Common Refinement)

Definition 10.8.9 (Common Refinement)**.** Let $a < b$ and let P and Q be partitions of $[a, b]$. Their common refinement is the partition R obtained by taking the union of their node sets, so $R = P \cup Q$ as sets of points in $[a, b]$.

Remark (Standard quantified statement)*.*

$$\text{Let } a < b \text{ in } \mathbb{R}, \text{ and let } P, Q \text{ be partitions of } [a, b].$$
$$\forall R \text{ partition of } [a, b] :\ R \text{ is the common refinement of } P \text{ and } Q \iff R = P \cup Q.$$

Remark (Definition predicate reading).

$$\mathrm{CommonRefinement}(R, P, Q) \coloneqq (R = P \cup Q).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&a < b \land P, Q \subseteq [a, b] \land R \subseteq [a, b] \\
&\implies \neg\big(\forall x \in \mathbb{R}\ (x \in R \iff x \in P \cup Q)\big) \\
&\iff \big(\exists x \in R\ (x \notin P \cup Q)\big) \lor \big(\exists x \in P \cup Q\ (x \notin R)\big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{CommonRefinement}(R, P, Q) \iff R \neq P \cup Q.$$

Remark (Failure modes). The claim that a partition R is the common refinement of P and Q fails exactly when R either contains a node not present in $P \cup Q$ or omits a node that $P \cup Q$ contributes. These are the only two structural ways in which R can differ from $P \cup Q$.

Remark (Failure mode decomposition).

$$R \neq P \cup Q \iff \underbrace{\exists x \in R \setminus (P \cup Q)}_{\text{extra node}} \lor \underbrace{\exists x \in (P \cup Q) \setminus R}_{\text{missing node}}.$$

Remark (Interpretation). Forming the common refinement of two partitions simply collects every partition point that either partition already uses, producing the unique partition whose nodes are exactly those existing ones. No new nodes are created beyond those already in P or Q, and none are discarded; the construction merely merges the lists so that later comparisons, such as evaluating tagged sums on both partitions simultaneously, can be performed without further adjustment.

Remark (Dependencies).

Definition 10.8.1

Chapter 11

Differentiation

SERIES	VOLUME III	BOOK II	CHAPTER 10
From Cantor to Ito	Classical Analysis	Functions, Continuity, and Differentiation	**Differentiation**

11.1 The Tangent Problem

Secant and Tangent — Quick Reference

Concept	Meaning
Secant line	Line through two points of a graph
Tangent line	Limiting local line determined by the derivative

Remark (Section guide). Before the derivative is a formula it is a question: what straight line best matches a curve at a single point? The route in is indirect: choose a second point, draw the computable secant line, and let that point slide toward the first. If the secants settle toward a limiting line, that line is the tangent and its slope is the derivative.

Definition (Secant Line)

Definition 11.1.1 (Secant Line). Let $f : A \to \mathbb{R}$, and let $x_1, x_2 \in A$ with $x_1 \neq x_2$. The *secant line* determined by the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ is the unique line passing through these two points. Its slope is

$$m_{\mathrm{sec}}(x_1, x_2) := \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \ (x_1 \neq x_2 \Rightarrow \mathrm{SecantLine}(f, x_1, x_2) \text{ has slope } \tfrac{f(x_2)-f(x_1)}{x_2-x_1}).$$

Remark (Interpretation). With increments $\Delta x := x_2 - x_1$ and $\Delta y := f(x_2) - f(x_1)$, the secant slope is $\Delta y / \Delta x$ — the average rate of change of f over the interval $[x_1, x_2]$. This is an exact algebraic identity; no limit is involved. As $x_2 \to x_1$, if the secant slope converges, its limit is the derivative $f'(x_1)$ — the instantaneous rate of change. Geometrically, the secant line rotates about the fixed point $(x_1, f(x_1))$ as x_2 approaches x_1, and the limit position is the tangent line.

Definition (Difference Quotient)

Definition 11.1.2 (Difference Quotient)**.** Let $f : A \to \mathbb{R}$, let $c \in A$, and let $x \in A$ with $x \neq c$. The *difference quotient* of f from c to x is

$$m_{\text{sec}}(c, x) := \frac{f(x) - f(c)}{x - c}.$$

Remark (Interpretation). The difference quotient is the slope of the secant line through $(c, f(c))$ and $(x, f(x))$. It is the quantity whose limiting value, when it exists as $x \to c$, becomes the derivative.

Remark (Dependencies).

- Secant Line

Definition (Tangent Line)

Definition 11.1.3 (Tangent Line)**.** Let $f : A \to \mathbb{R}$, let $c \in A$, and suppose that f is differentiable at c. The *tangent line* to the graph of f at the point $(c, f(c))$ is the line through $(c, f(c))$ with slope $f'(c)$. Its equation is

$$y = f(c) + f'(c)(x - c).$$

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \;\Rightarrow\; \text{LinearApproximationAt}(f, c) = f(c) + f'(c)(x - c).$$

Remark (Interpretation). The tangent line is the limit of the family of secant lines through $(c, f(c))$ and $(x, f(x))$ as $x \to c$. More precisely: the slope $m_{\text{sec}}(c, x) = (f(x) - f(c))/(x - c)$ converges to $f'(c)$, so the tangent line is the line whose slope is that limit. It is also the unique linear function $\ell(x) = f(c) + f'(c)(x - c)$ satisfying $\ell(c) = f(c)$ and $\ell'(c) = f'(c)$ — the best first-order approximation to f near c. The error $f(x) - \ell(x) = o(x - c)$ as $x \to c$, meaning it vanishes faster than the displacement.

Remark (Dependencies).

- Definition: Secant Line
- Definition: Derivative at a Point

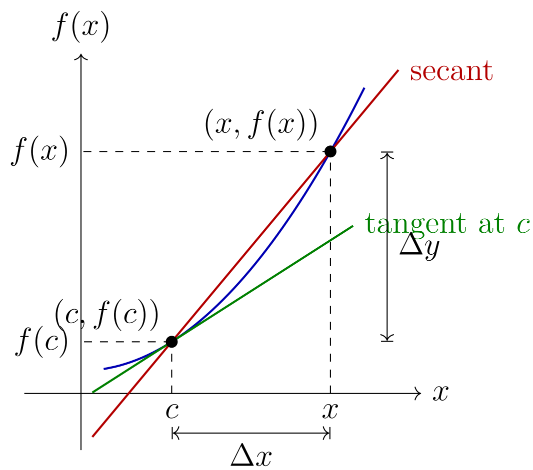


Figure 11.1: The secant line through $(c, f(c))$ and $(x, f(x))$ has slope $\Delta y/\Delta x$. As $x \to c$, the secant line rotates toward the tangent line at $(c, f(c))$, whose slope is $f'(c)$.

11.2 The Derivative

Derivative Definition — Quick Reference

Concept	**Meaning**
Derivative at a point	Epsilon–delta limiting secant slope
Example: derivative of x^2	The definition in a removable-singularity computation
Topological derivative	Neighbourhood formulation of the same limit
Sequential derivative	Sequence test for differentiability
Equivalent definitions	Epsilon–delta, topological, and sequential forms agree
h-form	Difference quotient written with $x = c + h$
Differentiability implies continuity	Differentiability is stronger than continuity
Example: one corner	Continuous but not differentiable at one point
Example: sawtooth	Continuous with infinitely many isolated corners
Example: Weierstrass function	Continuous and nowhere differentiable
Uniqueness	The derivative value is single-valued
Identity derivative	Base case $D(x) = 1$
Constant derivative	Base case $D(C) = 0$

Remark (Section guide). The derivative is the number the secant slopes approach, but the word “limit” has three useful faces. The epsilon–delta form is the contract; the topological form strips it to neighbourhoods; the sequential form is the disproving tool, since one bad approach sequence is enough.

Definition (Derivative at a Point)

Definition 11.2.1 (Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ be a real-valued function and $c \in \operatorname{int}(D)$. We say f is *differentiable at* c with *derivative* $L = f'(c)$ if and only if:

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D \Big(0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D \Big(0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

Remark (Definition predicate reading).

$$\operatorname{DerivativeAt}(f, c, L) \iff \operatorname{FunctionLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\, D,\, c,\, L \right).$$

The predicate $\operatorname{DerivativeAt}(f, c, L)$ asserts that L is the value of the limit of the difference quotient of f at c.

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in D \Big(0 < |x - c| < \delta\ \wedge\ \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \Big).$$

Remark (Negation predicate reading).

$$\neg \operatorname{DerivativeAt}(f, c, L) \iff \neg \operatorname{FunctionLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\, D,\, c,\, L \right).$$

Remark (Failure modes).

- The difference quotient has no finite limit at c (e.g. $|f(x) - f(c)|/|x - c| \to \infty$).
- The difference quotient oscillates without settling near any single value.
- The proposed value L is the wrong candidate.

Remark (Interpretation). The derivative is a limit of slopes of secant lines. The ε-δ form demands that the difference quotient can be made arbitrarily close to L by taking x sufficiently close to c. The input condition $0 < |x - c|$ is punctured: $x = c$ is excluded because $(f(c) - f(c))/(c - c)$ is $0/0$.

Remark (Dependencies).

- Limit of a Function

Example (Derivative of x^2 from the Definition)

Remark (Worked example). Let $f(x) = x^2$ and fix $c \in \mathbb{R}$. For $x \neq c$,

$$\frac{f(x) - f(c)}{x - c} = \frac{x^2 - c^2}{x - c} = x + c.$$

The singularity has been removed by exact algebra, so the limit as $x \to c$ is $2c$. Hence $f'(c) = 2c$. Read through Carathéodory, this is the factor $\varphi(x) = x + c$, continuous at c with $\varphi(c) = 2c$.

Definition (Topological Definition of Derivative at a Point)

Definition 11.2.2 (Topological Definition of Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ and $c \in \operatorname{int}(D)$. f is differentiable at c with derivative L if for every neighbourhood V of L, there exists a neighbourhood U of c such that for all $x \in (U \setminus \{c\}) \cap D$:

$$\frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Standard quantified statement).

$$\forall V \text{ neighbourhood of } L \; \exists U \text{ neighbourhood of } c \; \forall x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Definition predicate reading).

$$\mathrm{DerivativeAt}_{\mathrm{top}}(f, c, L) \iff \forall V \ni L \; \exists U \ni c \; \forall x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Negated quantified statement).

$$\exists V \text{ neighbourhood of } L \; \forall U \text{ neighbourhood of } c \; \exists x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \notin V.$$

Remark (Negation predicate reading).

$$\neg \, \mathrm{DerivativeAt}_{\mathrm{top}}(f, c, L) \iff \exists V \ni L \; \forall U \ni c \; \exists x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \notin V.$$

Remark (Failure modes). The topological derivative fails when a fixed output neighbourhood of L cannot be entered by the difference quotient no matter how tightly the input is restricted around c. Geometric reading: no punctured neighbourhood around c maps all secant slopes into any prescribed window around L.

Remark (Interpretation). The topological form replaces ε-balls with abstract neighbourhoods, making the definition coordinate-free and suited for generalisation to metric spaces and manifolds.

Remark (Dependencies).

- Function
- Relative Neighborhood
- Subtraction on $\mathbb{R}$
- Division on $\mathbb{R}$

Definition (Sequential Definition of Derivative at a Point)

Definition 11.2.3 (Sequential Definition of Derivative at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. We say that f is differentiable at c with derivative L if

$$\forall (x_n) \subseteq A \setminus \{c\} \ \big(x_n \to c \Rightarrow \tfrac{f(x_n)-f(c)}{x_n-c} \to L\big).$$

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq A \setminus \{c\} : \ x_n \to c \ \Rightarrow \ \frac{f(x_n) - f(c)}{x_n - c} \to L.$$

Remark (Definition predicate reading).

$$\mathrm{DerivativeAt}_{\mathrm{seq}}(f, c, L) \iff \forall (x_n) \subseteq A \setminus \{c\} : \ x_n \to c \ \Rightarrow \ \frac{f(x_n) - f(c)}{x_n - c} \to L.$$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq A \setminus \{c\} : \ x_n \to c \ \wedge \ \frac{f(x_n) - f(c)}{x_n - c} \not\to L.$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DerivativeAt}_{\mathrm{seq}}(f, c, L) \iff \exists (x_n) \subseteq A \setminus \{c\} : \ x_n \to c \ \wedge \ \frac{f(x_n) - f(c)}{x_n - c} \not\to L.$$

Remark (Failure modes). Failure occurs when a single approach sequence $(x_n) \to c$ exists along which the difference quotients either: (i) diverge to infinity (vertical tangent); (ii) oscillate without settling; (iii) converge but to a value other than L (wrong slope candidate). A single such sequence certifies non-differentiability.

Remark (Failure mode decomposition).

$$\underbrace{x_n \to c}_{\text{approach condition}} \quad \wedge \ \underbrace{\frac{f(x_n) - f(c)}{x_n - c} \not\to L}_{\text{quotient failure}}.$$

The two canonical witness types are: two subsequences with different limiting difference quotients (e.g., $f(x) = |x|$ at $c = 0$), or quotients growing without bound.

Remark (Interpretation). The sequential form is the primary tool for disproving differentiability: exhibit a single sequence $x_n \to c$ along which the difference quotient fails to converge to any single value.

Remark (Dependencies).

- Sequence
- Convergent Sequence
- Derivative at a Point
- Limit of a Function

Theorem (Equivalence of Definitions of the Derivative at a Point)

Theorem 11.2.4 (Equivalence of Definitions of the Derivative at a Point)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$ be a limit point of A. The following are equivalent:*

1. *f is differentiable at c in the ε-δ sense with derivative L.*
2. *f is differentiable at c in the topological sense with derivative L.*
3. *f is differentiable at c in the sequential sense with derivative L.*

Go to proof.

Remark (Standard quantified statement).

$$\text{(i)} \iff \text{(ii)} \iff \text{(iii)}.$$

Remark (Predicate reading).

$$\mathrm{DerivativeAt}(f, c, L) \iff \mathrm{DerivativeAt}_{\mathrm{top}}(f, c, L) \iff \mathrm{DerivativeAt}_{\mathrm{seq}}(f, c, L).$$

All three predicates name the same underlying condition on the difference quotient of f at c.

Remark (Negated quantified statement).

$$\begin{aligned} \neg\text{(i)} \iff \neg\text{(ii)} \iff \neg\text{(iii)} \\ \iff \exists (x_n) \subseteq A \setminus \{c\} : \ x_n \to c \ \wedge \ \frac{f(x_n) - f(c)}{x_n - c} \not\to L. \end{aligned}$$

Remark (Failure modes). The equivalence cannot produce contradictory verdicts: if c is a limit point of A, all three forms agree. Degenerate case: if c is not a limit point of A, the punctured-ball and sequence conditions are vacuously satisfied by every candidate L simultaneously, making the equivalence trivially true for all L rather than characterizing a unique derivative.

Remark (Interpretation). The three forms are notational restatements of the same underlying limit condition on the difference quotient. Analysts choose whichever form is most convenient for the proof at hand.

Remark (Dependencies).

- Derivative at a Point
- Topological Definition
- Sequential Definition

Proposition (Derivative: h-Form Equivalence)

Proposition 11.2.5 (Derivative: h-Form Equivalence)**.** *Let $f : A \to \mathbb{R}$ and $c \in \operatorname{int}(A)$. Then f is differentiable at c with derivative L if and only if*

$$\lim_{h\to 0} \frac{f(c+h)-f(c)}{h} = L.$$

Go to proof.

Remark (Standard quantified statement).

$$\operatorname{DerivativeAt}(f,c,L) \iff \lim_{h\to 0} \frac{f(c+h)-f(c)}{h} = L.$$

Remark (Negated quantified statement).

$$\neg \operatorname{DerivativeAt}(f,c,L) \iff \lim_{h\to 0} \frac{f(c+h)-f(c)}{h} \neq L.$$

Remark (Failure modes). The h-form fails in two branches: (i) The incremental ratio $(f(c+h)-f(c))/h$ has no finite limit as $h \to 0$ (corner, cusp, or vertical tangent at c). (ii) The ratio converges to a finite value different from L (proposed L is the wrong slope).

Remark (Interpretation). The h-form substitutes $x = c+h$ in the difference quotient. It is the standard notation in calculus and makes the incremental structure of the derivative explicit.

Remark (Dependencies).

- Derivative at a Point

Remark (Discrete and continuous change). The forward difference operator

$$\Delta f(n) = f(n+1) - f(n)$$

is the discrete analogue of the derivative. The h-form above is its continuous descendant: replace the fixed step 1 by a shrinking step h, divide by h, and pass to the limit.

The kinship matters, but so does the divergence. The discrete product rule has an unavoidable shift:

$$\Delta(fg)(n) = (\Delta f)(n)g(n) + f(n+1)(\Delta g)(n).$$

The continuous product rule loses this shift because $f(x+h) \to f(x)$ as $h \to 0$. Likewise, ordinary powers are not the clean basis for Δ; falling factorials are:

$$\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}.$$

Passing to the limit is therefore not just a sharpening of a quotient. It restores symmetries that the discrete operator must carry as shifts.

11.2.1 First Consequences

Theorem (Differentiable Implies Continuous)

Theorem 11.2.6 (Differentiable Implies Continuous)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$. If f is differentiable at c, then f is continuous at c.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\mathrm{DerivableAt}(f, c) \;\Rightarrow\; \mathrm{ContinuousAt}(f, c).$$

Remark (Predicate reading)*.*

$$\mathrm{DifferentiableAt}(f, c) \implies \mathrm{ContinuousAt}(f, c).$$

Remark (Failure modes)*.* The implication can only fail in the reverse direction: continuity at c does not force differentiability at c.

Remark (Failure mode decomposition)*.* The converse fails: $f(x) = |x|$ is continuous at 0 but not differentiable there.

Remark (Contrapositive quantified statement)*.*

$$\neg\mathrm{ContinuousAt}(f, c) \;\Rightarrow\; \neg\mathrm{DifferentiableAt}(f, c).$$

Remark (Contrapositive predicate reading)*.*

$$\neg\,\mathrm{ContinuousAt}(f, c) \implies \neg\,\mathrm{DifferentiableAt}(f, c).$$

Remark (Interpretation)*.* Differentiability is strictly stronger than continuity: the existence of a well-defined tangent slope forces the function values to coalesce at $f(c)$ as $x \to c$.

Remark (Dependencies)*.*

- Derivative at a Point
- Continuity at a Point

Remark (Converse failure)*.* The theorem runs one way only. Differentiability forces continuity, but continuity does not force differentiability. The negation of the converse is

$$\exists f\ \exists c:\ \mathrm{ContinuousAt}(f, c) \wedge \neg\,\mathrm{DifferentiableAt}(f, c).$$

The examples below escalate the failure: one corner, infinitely many isolated corners, and no differentiability anywhere.

Example (One Corner: $|x|$)

Remark (Worked example)*.* The function $f(x) = |x|$ is continuous on $\mathbb{R}$, but it is not differentiable at 0. Its one-sided derivatives are -1 and $+1$, so the two one-sided slopes do not agree. Thus a single corner is enough to separate continuity from differentiability.

Example (Finitely Many Corners: A Sawtooth)

Remark (Worked example). On a compact interval, a triangular sawtooth made from finitely many line segments can be continuous while failing to be differentiable at each corner. For example,

$$s(x) = \begin{cases} x, & 0 \le x \le 1, \\ 2 - x, & 1 \le x \le 2, \end{cases}$$

is continuous on $[0, 2]$, but the left derivative at 1 is 1 and the right derivative is -1. Adding more teeth gives any finite number of corner failures without breaking continuity.

Example (The Weierstrass Function)

Remark (Worked example). For $0 < a < 1$ and an odd integer b satisfying $ab > 1 + \frac{3}{2}\pi$, the Weierstrass function

$$W(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$$

is continuous on $\mathbb{R}$, because the series converges uniformly by the Weierstrass M-test. Classically, under the displayed condition, it is differentiable at no point. Thus continuity may hold everywhere while differentiability holds nowhere.

Remark (Proof status). The continuity of W follows from uniform convergence once the function-series machinery is available. The nowhere-differentiability proof is a classical result of Weierstrass and is deferred to the function-series development; it is used here only as a counterexample to the converse of Differentiable Implies Continuous.

Remark (Forward flag). Brownian motion has sample paths that are almost surely continuous everywhere and differentiable nowhere. This is the stochastic analogue of the Weierstrass phenomenon and one reason ordinary calculus must be replaced by the Itô integral in stochastic analysis.

Theorem (Uniqueness of the Derivative)

Theorem 11.2.7 (Uniqueness of the Derivative). *Let $f : A \to \mathbb{R}$ and $c \in A$. If f is differentiable at c, its derivative is unique.*

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DerivativeAt}(f, c, L) \ \wedge \ \mathrm{DerivativeAt}(f, c, L') \ \Rightarrow \ L = L'.$$

Remark (Predicate reading).

$$\mathrm{DerivativeAt}(f, c, L) \ \wedge \ \mathrm{DerivativeAt}(f, c, L') \ \Longrightarrow \ L = L'.$$

Remark (Failure modes). The uniqueness proof assumes $L \neq L'$ and takes $\varepsilon = |L - L'|/2 > 0$. The definition supplies δ making the difference quotient within ε of both L and L'

simultaneously, forcing $|L-L'| < 2\varepsilon = |L-L'|$, a contradiction. The cluster-point hypothesis on c is essential: without it, the punctured-ball condition is vacuous and the limit is trivially satisfied by every value.

Remark (Contrapositive quantified statement).

$$L \neq L' \;\Rightarrow\; \neg\big(\mathrm{DerivativeAt}(f,c,L) \;\wedge\; \mathrm{DerivativeAt}(f,c,L')\big).$$

Two distinct candidates cannot both be the derivative of f at c.

Remark (Contrapositive predicate reading).

$$L \neq L' \implies \neg\,\mathrm{DerivativeAt}(f,c,L) \;\vee\; \neg\,\mathrm{DerivativeAt}(f,c,L').$$

Remark (Interpretation). Uniqueness licenses writing $f'(c)$ as a definite value rather than a set of candidates.

Remark (Dependencies).

- Derivative at a Point

Proposition (Derivative of the Identity)

Proposition 11.2.8 (Derivative of the Identity)**.** *Let $f(x) = x$ for all $x \in \mathbb{R}$. Then $f'(c) = 1$ for every $c \in \mathbb{R}$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall c \in \mathbb{R} : \; \frac{d}{dx}[x]\Big|_{x=c} = 1.$$

Remark (Interpretation). The identity function has slope 1 everywhere. It is the base case for computing derivatives of polynomials via the algebra of derivatives.

Remark (Dependencies).

- Derivative at a Point
- Function

Proposition (Derivative of a Constant)

Proposition 11.2.9 (Derivative of a Constant)**.** *Let $k \in \mathbb{R}$ and let $f(x) = k$ for all $x \in \mathbb{R}$. Then $f'(c) = 0$ for every $c \in \mathbb{R}$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall k \in \mathbb{R}\, \forall c \in \mathbb{R} : \; \frac{d}{dx}[k]\Big|_{x=c} = 0.$$

Remark (Interpretation). A constant function has zero change across every secant line, so its instantaneous rate of change is zero everywhere. Together with the derivative of the identity, this is one of the base cases for the algebraic computation rules.

Remark (Dependencies).

- Derivative at a Point

11.3 One-Sided Derivatives

One-Sided Derivatives — Quick Reference

Concept	Meaning
Left-hand derivative	Limiting slope approached from the left
Right-hand derivative	Limiting slope approached from the right
Two-sided criterion	Differentiability iff both one-sided derivatives agree
Example: $\lvert x\rvert$ at 0	A continuous corner with unequal one-sided slopes

Remark (Section guide). A corner is a place where a curve has a left answer and a right answer and refuses to choose. One-sided derivatives interrogate each side alone, and their agreement criterion is the fastest way to certify many failures of differentiability.

Definition (Left-Hand Derivative)

Definition 11.3.1 (Left-Hand Derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a left-hand limit point of I. We say that f has a *left-hand derivative* at c if the limit

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_-(c)$:

$$f'_-(c) := \lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}.$$

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R}\, \forall \varepsilon > 0\, \exists \delta > 0\, \forall x \in I \Big(-(\delta) < x - c < 0 \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

In this case $f'_-(c) = L$.

Remark (Definition predicate reading).

$$\text{LeftDerivativeAt}(f, c, L) \iff \text{LeftLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\, c,\, L \right).$$

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R}\, \exists \varepsilon > 0\, \forall \delta > 0\, \exists x \in I \Big(-\delta < x - c < 0 \ \wedge\ \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \Big).$$

Remark (Negation predicate reading).

$$\forall L \in \mathbb{R}\ \neg\, \mathrm{LeftDerivativeAt}(f, c, L).$$

No finite L is the left-hand limit of the difference quotient.

Remark (Interpretation). The left-hand derivative $f'_-(c)$ is the limit of the difference quotient as x approaches c *strictly from the left*. It measures the instantaneous rate of change of f at c as seen from the left side. The right-hand derivative $f'_+(c)$ is the symmetric object from the right. At interior points of an interval, both one-sided derivatives are defined, and the full derivative $f'(c)$ exists if and only if they agree. At endpoints, only the inward one-sided derivative is accessible, and it serves as the boundary derivative. The canonical example: for $f(x) = |x|$ at $c = 0$, $f'_-(0) = -1$ and $f'_+(0) = +1$; since these disagree, $f'(0)$ does not exist.

Remark (Dependencies).

- Definition: Derivative at a Point

Definition (Right-Hand Derivative)

Definition 11.3.2 (Right-Hand Derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a right-hand limit point of I. We say that f has a *right-hand derivative* at c if the limit

$$\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_+(c)$:

$$f'_+(c) := \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}.$$

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I \left(0 < x - c < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

In this case $f'_+(c) = L$.

Remark (Definition predicate reading).

$$\mathrm{RightDerivativeAt}(f, c, L) \iff \mathrm{RightLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\ c,\ L \right).$$

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in I \left(0 < x - c < \delta\ \wedge\ \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \right).$$

Remark (Negation predicate reading).

$$\forall L \in \mathbb{R}\ \neg\, \mathrm{RightDerivativeAt}(f, c, L).$$

Remark (Interpretation). The right-hand derivative $f'_+(c)$ is the instantaneous rate of change seen from inputs larger than c. It is the endpoint derivative available at the left endpoint of a closed interval, and it is one half of the two-sided derivative test at interior points.

Remark (Dependencies).

- Definition: Derivative at a Point
- Definition: Left-Hand Derivative

Theorem

Theorem 11.3.3 (Differentiability and One-Sided Derivatives)**.** *Let $I \subseteq \mathbb{R}$ be an open interval and let $f : I \to \mathbb{R}$. Then f is differentiable at $c \in I$ if and only if both one-sided derivatives $f'_-(c)$ and $f'_+(c)$ exist and are equal. In that case,*

$$f'(c) = f'_-(c) = f'_+(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \iff \Big(\text{LeftDerivativeAt}(f, c, L) \wedge \text{RightDerivativeAt}(f, c, L)\Big) \text{ for some } L \in \mathbb{R}.$$

Remark (Negated quantified statement).

$$\neg\,\text{DifferentiableAt}(f, c) \iff \neg\Big(\text{LeftDerivativeAt}(f, c, L) \wedge \text{RightDerivativeAt}(f, c, L)\Big) \text{ for all } L \in \mathbb{R}.$$

That is: either a one-sided derivative fails to exist, or both exist but are unequal.

Remark (Interpretation). This theorem decomposes differentiability at an interior point into two one-sided conditions. It is the standard tool for proving non-differentiability: to show f is not differentiable at c, compute $f'_-(c)$ and $f'_+(c)$ and observe they disagree. For $f(x) = |x|$ at $c = 0$: $f'_-(0) = -1 \neq +1 = f'_+(0)$, so f is not differentiable at 0. The theorem also provides the correct definition of differentiability on a closed interval $[a, b]$: differentiable on the open interior (a, b) in the full sense, plus $f'_+(a)$ and $f'_-(b)$ both exist at the endpoints.

Remark (Dependencies).

- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Definition: Derivative at a Point

Example ($|x|$ Has No Derivative at 0)

Remark (Worked example). For $f(x) = |x|$ at $c = 0$, the one-sided difference quotients are

$$\frac{|x| - 0}{x - 0} = \frac{|x|}{x} = \begin{cases} -1, & x < 0, \\ 1, & x > 0. \end{cases}$$

Thus $f'_-(0) = -1$ and $f'_+(0) = 1$. The two values disagree, so the two-sided derivative does not exist at 0, even though f is continuous there.

11.4 Carathéodory and the Chain Rule

Carathéodory and Chain Rule — Quick Reference

Concept	Role
Carathéodory factorization	Continuous slope factor for differentiability
Carathéodory characterization	Differentiability iff the slope factor extends continuously
Chain Rule	Derivative of a composition
Leibniz Rule	n-th derivative of a product
Faa di Bruno	n-th derivative of a composition
Faa di Bruno at $n = 2$	Coefficient check for the second derivative
Faa di Bruno at $n = 3$	Coefficient check for the third derivative

11.4.1 Carathéodory Characterization

Remark (Section guide). Carathéodory replaces division by factorisation: $f(x) - f(c) = (x - c)\varphi(x)$, with φ continuous at c. Differentiability becomes continuity of the slope factor, and continuity composes. That is why the chain rule becomes clean rather than a fight with vanishing denominators.

Definition (Carathéodory Factorization)

Definition 11.4.1 (Carathéodory Factorization)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. We say that f admits a *Carathéodory factorization* at c if there exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \text{for all } x \in I.$$

In this case, f is differentiable at c and $f'(c) = \phi(c)$.

Remark (Standard quantified statement).

$$\exists \phi : I \to \mathbb{R}\ \big(\mathrm{ContinuousAt}(\phi, c)\ \wedge\ \forall x \in I\ [f(x) = f(c) + (x - c)\phi(x)]\big).$$

Remark (Definition predicate reading). The existence of a Carathéodory factorization at c is equivalent to $\mathrm{DifferentiableAt}(f, c)$ by the characterization theorem below. The factorizing function is explicit:

$$\phi(x) = \begin{cases} \dfrac{f(x) - f(c)}{x - c} & x \neq c, \\ f'(c) & x = c. \end{cases}$$

Differentiability of f at c is precisely the statement that this piecewise-defined ϕ is continuous at c.

Remark (Interpretation). The Carathéodory factorization replaces the difference quotient — a function with a removable discontinuity at c — with a function ϕ that is genuinely continuous at c. The factorization $f(x) - f(c) = (x - c)\phi(x)$ holds everywhere on I, including at

$x = c$ where both sides equal zero. The derivative is recovered as $\phi(c)$ without ever dividing by zero. This reframing converts analytical limit conditions into topological continuity conditions, which compose and factor more cleanly. It is the preferred tool for proving the product rule, quotient rule, and chain rule, because continuity of a product or composition is easier to establish than the existence of a limit of a quotient.

Carathéodory vs. Lipschitz. Both control the increment $f(x) - f(c)$ by the factor $(x - c)$, but differently. The Carathéodory factorization requires the slope function $\phi(x)$ to extend *continuously* to c; it controls the *behaviour* of the slope, not just its size. The Lipschitz condition $|f(x)-f(c)| \le L|x-c|$ only bounds the size of the increment. Carathéodory implies Lipschitz: if ϕ is continuous at c it is bounded near c, so $|f(x)-f(c)| = |\phi(x)||x-c| \le K|x-c|$ for some K near c. The converse fails: $f(x) = |x|$ is Lipschitz on $\mathbb{R}$ with constant 1 but the slope function takes values ± 1 on either side of 0 with no continuous extension.

Remark (Dependencies).

- Definition: Derivative at a Point
- Definition: Continuous at a Point

Theorem

Theorem 11.4.2 (Carathéodory Characterization of Differentiability). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. The following are equivalent:*

(i) f is differentiable at c.

(ii) There exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \text{for all } x \in I.$$

In this case, $\phi(c) = f'(c)$. *Go to proof. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}\text{DifferentiableAt}(f, c) \iff\ & \exists \phi : I \to \mathbb{R}\ \big(\text{ContinuousAt}(\phi, c) \\ & \wedge\ \forall x \in I\ [f(x) = f(c) + (x - c)\phi(x)]\big).\end{aligned}$$

In this case $\phi(c) = f'(c)$.

Remark (Interpretation). This theorem establishes that differentiability and the existence of a Carathéodory factorization are exactly the same condition. The proof is a two-direction construction: in the forward direction, define ϕ as the difference quotient extended by $f'(c)$ at c and verify continuity from the definition of the derivative; in the backward direction, recover the difference quotient from ϕ and read off its limit as $\phi(c)$. The theorem's power is that it converts every future differentiability argument into a continuity argument: to show $g \circ f$ is differentiable, find a continuous Carathéodory function for it; composing the Carathéodory functions of f and g yields one automatically.

Remark (Dependencies).

- Definition: Carathéodory Factorization
- Definition: Derivative at a Point
- Definition: Continuous at a Point

Theorem

Theorem 11.4.3 (Chain Rule). *Let $I, J \subseteq \mathbb{R}$ be intervals, let $f : I \to \mathbb{R}$ and $g : J \to \mathbb{R}$, and suppose that $f(I) \subseteq J$. Let $c \in I$. If f is differentiable at c and g is differentiable at $f(c)$, then the composition $g \circ f$ is differentiable at c, and*

$$(g \circ f)'(c) = g'(f(c))\, f'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, f(c)) \implies (g \circ f)'(c) = g'(f(c))f'(c).$$

Remark (Interpretation). The derivative of a composition is the derivative of the outside function evaluated at the inside value, multiplied by the derivative of the inside function.

Remark (Interpretation). Let ϕ_f and ϕ_g be the Carathéodory slope factors for f at c and g at $f(c)$. Then

$$g(f(x)) - g(f(c)) = \phi_g(f(x))(f(x) - f(c)) = \phi_g(f(x))\phi_f(x)(x - c).$$

The new slope factor is

$$\phi_{g\circ f}(x) = \phi_g(f(x))\phi_f(x),$$

which is continuous at c and has value $g'(f(c))f'(c)$. This proof never divides by $f(x) - f(c)$, so it avoids the failure in the informal $\Delta g/\Delta f \cdot \Delta f/\Delta x$ argument when $f(x) = f(c)$ near c.

Remark (Dependencies).

- Theorem: Carathéodory Characterization of Differentiability
- Theorem: Differentiable Implies Continuous
- Theorem: Composition of Limits
- Theorem: Product Rule for Function Limits

Theorem (Leibniz Rule)

Theorem 11.4.4 (Leibniz Rule). *Let $I \subseteq \mathbb{R}$ be an interval, let $x \in I$, and suppose that $f, g : I \to \mathbb{R}$ are n-times differentiable on a neighbourhood of x. Then fg is n-times differentiable at x, and*

$$(fg)^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Diff}^n(f,x) \wedge \mathrm{Diff}^n(g,x) \Longrightarrow (fg)^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

Remark (Interpretation). The binomial pattern is not an accident. Each of the n differentiations lands on either f or g, and $\binom{n}{k}$ counts the ways to send exactly k differentiations to f. At $n = 1$, this is the ordinary product rule.

Remark (Dependencies).

- Theorem: Product Rule
- Definition: Higher Derivatives

Theorem (Faa di Bruno's Formula)

Theorem 11.4.5 (Faa di Bruno's Formula). *Let $I, J \subseteq \mathbb{R}$ be intervals, let $x \in I$, suppose $g : I \to J$ is n-times differentiable on a neighbourhood of x, and suppose $f : J \to \mathbb{R}$ is n-times differentiable on a neighbourhood of $g(x)$. Then $f \circ g$ is n-times differentiable at x, and*

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\sum_{m=1}^{n} m k_m = n} \frac{n!}{\prod_{m=1}^{n} k_m!\,(m!)^{k_m}} f^{(k)}(g(x)) \prod_{m=1}^{n} \left(g^{(m)}(x)\right)^{k_m},$$

where $k_m \geq 0$ and $k = \sum_{m=1}^{n} k_m$. Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Diff}^n(g,x) \wedge \mathrm{Diff}^n(f,g(x)) \Longrightarrow (f \circ g)^{(n)}(x) = \sum_{\sum m k_m = n} \frac{n!}{\prod_m k_m!\,(m!)^{k_m}} f^{(k)}(g(x)) \prod_m \left(g^{(m)}(x)\right)^{k_m}.$$

Remark (Interpretation). The chain rule gives the first derivative of a composition. Repeated differentiation is deeper: product-rule terms and chain-rule terms interlace, and the bookkeeping is governed by integer partitions. Faa di Bruno's formula is the all-orders chain rule.

Remark (Hypotheses). Smoothness is more than this formula needs. The n-th order statement requires n derivatives of g near x and n derivatives of f near $g(x)$. At $n = 1$, the formula is exactly the chain rule.

Remark (Structural reading). The tuple $(k_1, \ldots, k_n)$ records an integer partition of n: k_m is the number of parts of size m. The coefficient

$$\frac{n!}{\prod_m k_m!\,(m!)^{k_m}}$$

counts set partitions with that block-size profile. The outer derivative $f^{(k)}$ is differentiated once per block, while each block of size m contributes one factor $g^{(m)}$.

Remark (Normalization). The formula uses the bare-product normalization: the coefficient contains $(m!)^{k_m}$, and the product contains $\big(g^{(m)}(x)\big)^{k_m}$. Equivalently, one may put $g^{(m)}(x)/m!$ inside the product and remove $(m!)^{k_m}$ from the denominator. Mixing these two conventions gives the wrong coefficients.

Remark (Cross-volume callback). The constraint $\sum_m mk_m = n$ is precisely the multiplicity form of an integer partition of n. Thus the number of summands is $p(n)$, the partition function from the discrete and number-system material, now appearing as bookkeeping for iterated differentiation.

Remark (Scope). This theorem is enrichment, not a critical dependency for the integration, measure, or manifold arcs. It is included because it is the complete all-orders chain rule and a useful callback to partitions.

Remark (Dependencies).

- Theorem: Chain Rule
- Theorem: Leibniz Rule
- Definition: Higher Derivatives
- Definition: Class C^k

Example (Faa di Bruno at $n = 2$)

Remark (Worked example). The partitions of 2 are represented by $(k_1, k_2) = (2, 0)$ and $(0, 1)$. The first gives $f''(g)(g')^2$, and the second gives $f'(g)g''$. Hence

$$(f \circ g)'' = f''(g)(g')^2 + f'(g)g''.$$

The coefficient on $f'(g)g''$ is 1, which is the quickest check that the normalization has not been mixed.

Example (Faa di Bruno at $n = 3$)

Remark (Worked example). The partitions of 3 produce the three terms

$$(f \circ g)''' = f'''(g)(g')^3 + 3f''(g)g'g'' + f'(g)g'''.$$

The middle coefficient 3 counts the three ways to choose which differentiation contributes the second-derivative block of g.

11.5 The Shape Questions

Shape Questions — Quick Reference

Concept	Meaning
Increasing at a point	Local right-up, left-down behavior
Decreasing at a point	Local right-down, left-up behavior
Relative minimum	Locally smallest function value
Relative maximum	Locally largest function value
Relative extremum	Relative minimum or maximum
Convex function	Graph lies below its chords
Concave function	Graph lies above its chords
Inflection point	Local change in bending behavior

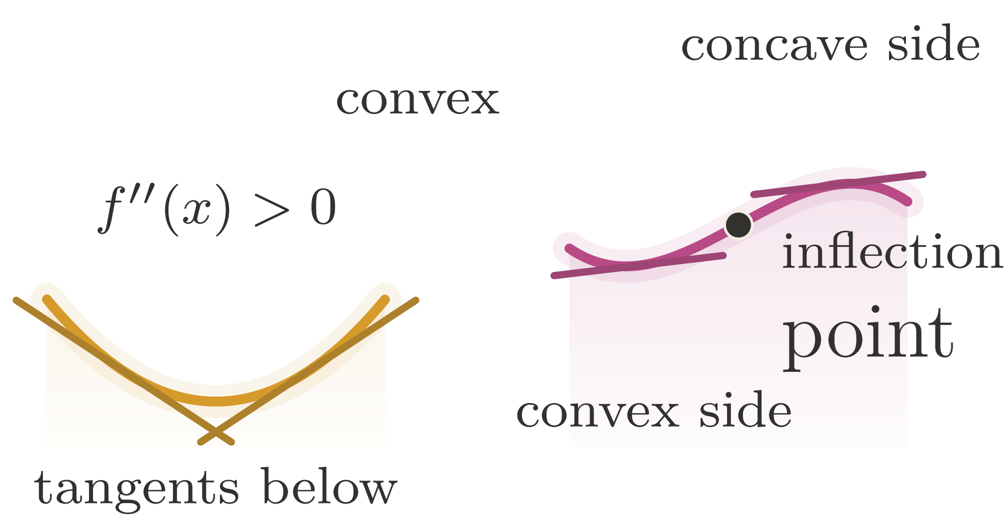


Figure 11.2: Convexity, concavity, and the change of bending at an inflection point.

Definition (Convex Function)

Definition 11.5.1 (Convex Function)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *convex* on I if for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Remark (Standard quantified statement)*.*

$$\forall x, y \in I \ \forall \lambda \in [0, 1] : \quad f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Remark (Interpretation)*.* Convexity is defined by chords, not derivatives. The graph lies below the line segment joining any two of its points. Derivative tests are later tools for detecting this geometric condition, not the definition of the condition.

Remark (Dependencies).

- Closed Interval

Definition (Concave Function)

Definition 11.5.2 (Concave Function)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *concave* on I if $-f$ is convex on I.

Remark (Standard quantified statement).

$$\mathrm{ConcaveOn}(f, I) \iff \mathrm{ConvexOn}(-f, I).$$

Remark (Interpretation)*.* Concavity is the downward-bending counterpart of convexity. Equivalently, the graph of a concave function lies above each chord joining two of its points.

Remark (Dependencies).

- Convex Function

Definition (Inflection Point)

Definition 11.5.3 (Inflection Point)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be an interior point. The point c is an *inflection point* of f if f changes convexity at c: there is a neighbourhood $(c - \delta, c + \delta) \subseteq I$ such that f is convex on one side of c and concave on the other side.

Remark (Standard quantified statement).

$$\mathrm{InflectionPoint}(f, c) \iff f \text{ changes convexity at } c.$$

Remark (Interpretation)*.* An inflection point is a change in the direction of bending. It is not merely a point where $f''(c) = 0$; that equation is only a necessary condition under additional hypotheses.

Remark (Dependencies).

- Convex Function
- Concave Function

11.5.1 Interior Extrema

Interior Extrema — Quick Reference

Statement	**Role**
Relative minimum	Local lower comparison point
Relative maximum	Local upper comparison point
Relative extremum	Umbrella term for local optima
Interior Extremum Theorem	Interior differentiable extremum has zero derivative
Necessary condition	Extrema occur at critical behavior
Critical-point corollary	Derivative zero or derivative undefined

Definition (Relative Minimum)

Definition 11.5.4 (Relative Minimum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative minimum* at c if there exists $\delta > 0$ such that

$$f(c) \leq f(x) \qquad \text{for all } x \in A \cap (c-\delta, c+\delta).$$

If, in addition,

$$f(c) < f(x) \qquad \text{for all } x \in \big(A \cap (c-\delta, c+\delta)\big) \setminus \{c\},$$

then f has a *strict relative minimum* at c.

Remark (Standard quantified statement).

$$\exists \delta > 0 \; \forall x \in A \; \big(|x-c| < \delta \implies f(c) \leq f(x)\big).$$

Remark (Definition predicate reading).

$$\mathrm{LocalMinimum}(f, c) \iff \exists \delta > 0 \; \forall x \in A \; \big(\mathrm{Within}(x, c, \delta) \implies f(c) \leq f(x)\big).$$

Remark (Interpretation). f has a relative minimum at c if $f(c)$ is the smallest value of f in some open neighbourhood of c within the domain. The word *relative* (or *local*) signals that we are comparing $f(c)$ only to nearby values, not to all values of f globally. The strict variant excludes the degenerate case where nearby points share the minimum value. The symmetric notion — relative maximum — reverses the inequality. Together they constitute a relative extremum. These are purely neighbourhood conditions; no derivative appears in the definitions.

Remark (Dependencies).

- Function
- Subset
- Open Interval
- Order on $\mathbb{R}$

Definition (Relative Maximum)

Definition 11.5.5 (Relative Maximum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative maximum* at c if there exists $\delta > 0$ such that

$$f(x) \leq f(c) \qquad \text{for all } x \in A \cap (c-\delta, c+\delta).$$

If, in addition,

$$f(x) < f(c) \qquad \text{for all } x \in \big(A \cap (c-\delta, c+\delta)\big) \setminus \{c\},$$

then f has a *strict relative maximum* at c.

Remark (Standard quantified statement).

$$\exists \delta > 0\ \forall x \in A\ \big(|x - c| < \delta \implies f(x) \le f(c)\big).$$

Remark (Definition predicate reading).

$$\mathrm{LocalMaximum}(f, c) \iff \exists \delta > 0\ \forall x \in A\ \big(\mathrm{Within}(x, c, \delta) \implies f(x) \le f(c)\big).$$

Remark (Interpretation). A relative maximum is a point whose function value is at least as large as all nearby function values in the domain.

Remark (Dependencies).

- Definition: Relative Minimum

Definition (Relative Extremum)

Definition 11.5.6 (Relative Extremum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative extremum* at c if f has either a relative minimum at c or a relative maximum at c.

Remark (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \iff \mathrm{LocalMinimum}(f, c) \lor \mathrm{LocalMaximum}(f, c).$$

Remark (Definition predicate reading).

$$\mathrm{LocalExtremum}(f, c) \iff \mathrm{LocalMinimum}(f, c) \lor \mathrm{LocalMaximum}(f, c).$$

Remark (Interpretation). Relative extrema are the local turning candidates of a graph: nearby values lie all on one side of $f(c)$.

Remark (Dependencies).

- Definition: Relative Minimum
- Definition: Relative Maximum

Theorem

Theorem 11.5.7 (Interior Extremum Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c and f is differentiable at c, then*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \land \mathrm{DifferentiableAt}(f, c) \Rightarrow \mathrm{DerivativeAt}(f, c, 0).$$

Remark (Contrapositive quantified statement).

$$\text{DerivativeAt}(f, c, L) \ \wedge \ L \neq 0 \ \Rightarrow \ \neg\,\text{LocalExtremum}(f, c).$$

If $f'(c)$ exists and is nonzero, then c is not a relative extremum.

Remark (Contrapositive predicate reading).

$$\text{DerivativeAt}(f, c, L) \ \wedge \ L \neq 0 \ \Rightarrow \ \neg\,\text{LocalMinimum}(f, c) \ \wedge \ \neg\,\text{LocalMaximum}(f, c).$$

Remark (Interpretation). The theorem gives a necessary condition for interior extrema under differentiability: the derivative must vanish. The geometric picture is immediate — at a local max or min, the tangent line must be horizontal, since the function is neither consistently increasing nor consistently decreasing. The proof is a sign argument: at a relative maximum, the difference quotient is non-positive for $x > c$ and non-negative for $x < c$; since the derivative is a single limit, it is squeezed to zero. The converse fails: $f'(c) = 0$ does not imply an extremum ($f(x) = x^3$ at $c = 0$). Critical points — where $f'(c) = 0$ or $f'(c)$ does not exist — are candidates for extrema, not guarantees.

Remark (Dependencies).

- Definition: Relative Extremum
- Definition: Derivative at a Point
- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Theorem: Differentiability and One-Sided Derivatives

Theorem

Theorem 11.5.8 (Necessary Condition for an Interior Extremum)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then either $f'(c)$ does not exist, or*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \ \Rightarrow \ \neg\,\text{DifferentiableAt}(f, c) \ \vee \ \text{DerivativeAt}(f, c, 0).$$

Remark (Interpretation). This is a weakening of the Interior Extremum Theorem that drops the differentiability hypothesis. Without assuming $f'(c)$ exists, the conclusion is a disjunction: either the derivative fails to exist at c, or it exists and equals zero. This is the correct form for identifying *critical points* — points that are candidates for extrema. The set of critical points is the union of points where f' is zero and points where f' does not exist.

Remark (Dependencies).

- Theorem: Interior Extremum Theorem

Corollary

Corollary 11.5.9 (Necessary Condition for an Extremum). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then*

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \;\Rightarrow\; \mathrm{DerivativeAt}(f, c, 0) \;\vee\; \neg\,\mathrm{DifferentiableAt}(f, c).$$

Remark (Interpretation). An extremum can occur only at a critical point: either the derivative vanishes or the derivative fails to exist.

Remark (Dependencies).

- Theorem: Necessary Condition for an Interior Extremum

11.6 The Mean Value Theorems

Mean Value Theorems — Quick Reference

Statement	**Role**
Rolle's Theorem	Equal endpoint values force a horizontal tangent
Mean Value Theorem	Some tangent slope equals the secant slope
Cauchy Mean Value Theorem	Compares two functions' rates of change
Derivative bound implies Lipschitz	Turns a slope bound into a global modulus

Remark (Section guide). Rolle's theorem says that a differentiable curve returning to the same height must level off somewhere between. The Mean Value Theorem tilts that picture: between any two points, some interior tangent runs parallel to the chord. Cauchy's version compares two functions at once and later powers L'Hôpital's rule.

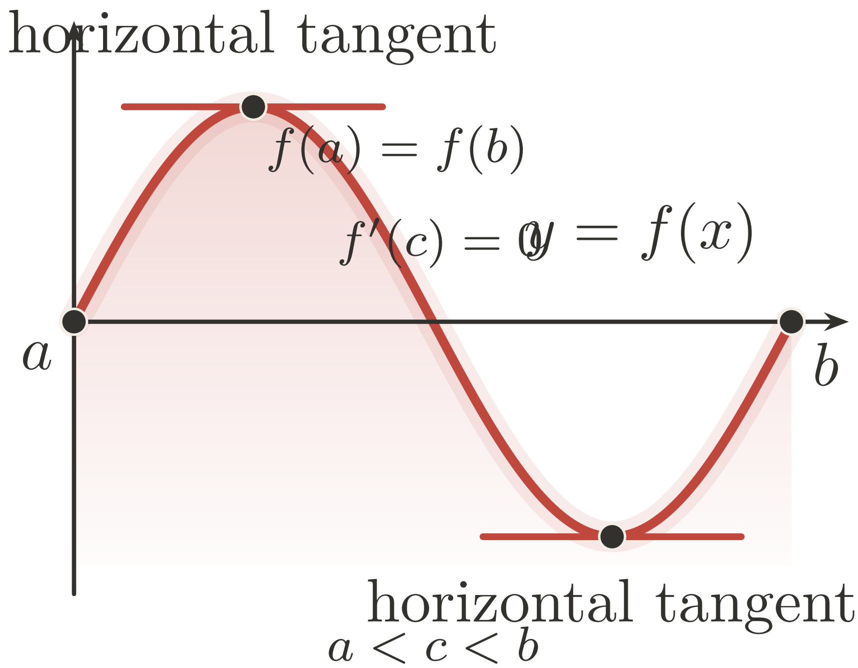


Figure 11.3: Rolle's Theorem: equal endpoint heights force a horizontal tangent.

Theorem (Rolle's Theorem)

Theorem 11.6.1 (Rolle's Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and satisfies $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\mathrm{ContinuousOn}(f, [a, b]) \;\wedge\; \forall x \in (a, b)\ \mathrm{DifferentiableAt}(f, x) \;\wedge\; f(a) = f(b)\\ &\quad \Rightarrow\ \exists c \in (a, b)\ \mathrm{DerivativeAt}(f, c, 0).\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}&\mathrm{ContinuousOn}(f, [a, b]) \;\wedge\; \mathrm{DifferentiableOn}(f, (a, b)) \;\wedge\; f(a) = f(b)\\ &\quad \Longrightarrow\ \exists c \in (a, b)\ :\ \mathrm{DerivativeAt}(f, c, 0).\end{aligned}$$

Remark (Failure modes). Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \mathrm{sgn}(x - 1/2)$ on $[0, 1]$. Then $f(0) = f(1) = -1$, but f has a jump discontinuity at $x = 1/2$. The derivative does not exist anywhere, so there is no horizontal tangent.

- **Differentiability fails:** Let $f(x) = |x - 1/2|$ on $[0, 1]$. Then $f(0) = f(1) = 1/2$, and f is continuous, but f has a corner at $x = 1/2$ where the derivative is undefined. For all $c \neq 1/2$, we have $f'(c) = \pm 1 \neq 0$.

- **Equal endpoints fail:** Let $f(x) = x$ on $[0, 1]$. Then f is continuous and differentiable with $f'(x) = 1 > 0$ everywhere, but $f(0) = 0 \neq 1 = f(1)$.

Remark (Contrapositive quantified statement).

$$\big(\forall c \in (a, b)\ :\ f'(c) \neq 0\big) \ \Rightarrow\ f \text{ not continuous on } [a, b] \vee f \text{ not differentiable on } (a, b) \vee f(a) \neq f(b).$$ ∎

Remark (Contrapositive predicate reading).

$$\big(\forall c \in (a, b)\ :\ \neg\,\mathrm{DerivativeAt}(f, c, 0)\big)$$
$$\implies \neg\,\mathrm{ContinuousOn}(f, [a, b])\ \vee\ \neg\,\mathrm{DifferentiableOn}(f, (a, b))\ \vee\ f(a) \neq f(b).$$

Remark (Interpretation). If a differentiable function returns to the same value at both endpoints of an interval, the derivative must vanish somewhere in between. Geometrically: if the curve starts and ends at the same height, at some interior point the tangent line is horizontal. The proof combines two ingredients — the Extreme Value Theorem guarantees that a continuous function on a closed bounded interval attains its maximum and minimum, and the Interior Extremum Theorem guarantees that at an interior extremum the derivative is zero. If both extrema occur at the endpoints then f is constant and $f' \equiv 0$ throughout. Rolle's Theorem is the special case of the MVT where the secant slope is zero.

Remark (Dependencies).

- Theorem: Interior Extremum Theorem
- Theorem: Extreme Value Theorem
- Definition: Derivative at a Point

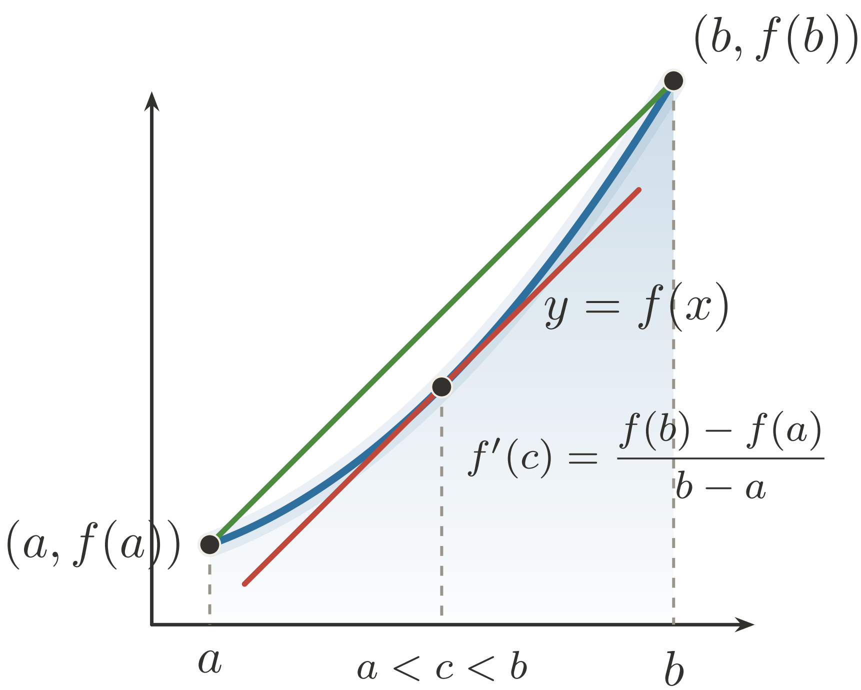


Figure 11.4: Mean Value Theorem: an interior tangent matches the secant slope.

Theorem (Mean Value Theorem)

Theorem 11.6.2 (Mean Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\mathrm{ContinuousOn}(f, [a, b]) \;\wedge\; \forall x \in (a, b)\; \mathrm{DifferentiableAt}(f, x)\\&\quad \Rightarrow\; \exists c \in (a, b)\; \mathrm{DerivativeAt}\Big(f,\, c,\, \tfrac{f(b)-f(a)}{b-a}\Big).\end{aligned}$$

Remark (Predicate reading)*.*

$$\begin{aligned}&\mathrm{ContinuousOn}(f, [a, b]) \;\wedge\; \mathrm{DifferentiableOn}(f, (a, b))\\&\quad \Longrightarrow\; \exists c \in (a, b) \;:\; \mathrm{DerivativeAt}\Big(f,\, c,\, \tfrac{f(b)-f(a)}{b-a}\Big).\end{aligned}$$

Remark (Failure modes)*.* Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \begin{cases} 0 & \text{if } x \in [0, 1/2) \\ 1 & \text{if } x \in [1/2, 1] \end{cases}$. Then f has a jump discontinuity at $x = 1/2$, and the secant slope is $\frac{f(1)-f(0)}{1-0} = 1$. But $f'(x) = 0$ wherever the derivative exists, so no interior point achieves the average rate of change.
- **Differentiability fails:** Let $f(x) = |x|$ on $[-1, 1]$. Then f is continuous and the secant slope is $\frac{f(1)-f(-1)}{1-(-1)} = \frac{1-1}{2} = 0$. But $f'(x) = \pm 1$ everywhere f is differentiable (i.e., for $x \neq 0$), so no interior point has derivative equal to the secant slope.

The MVT is an existence result. It guarantees some c exists but gives no formula for it.

Remark (Contrapositive quantified statement)*.*

$$\left(\forall c \in (a, b) \;:\; f'(c) \neq \frac{f(b) - f(a)}{b - a}\right) \;\Rightarrow\; f \text{ not continuous on } [a, b] \vee f \text{ not differentiable on } (a, b).$$

Remark (Contrapositive predicate reading)*.*

$$\begin{aligned}&\Big(\forall c \in (a, b) \;:\; \neg\,\mathrm{DerivativeAt}\Big(f,\, c,\, \tfrac{f(b)-f(a)}{b-a}\Big)\Big)\\&\quad \Longrightarrow\; \neg\,\mathrm{ContinuousOn}(f, [a, b]) \;\vee\; \neg\,\mathrm{DifferentiableOn}(f, (a, b)).\end{aligned}$$

Remark (Interpretation)*.* The MVT asserts that the instantaneous rate of change equals the average rate of change at some interior point. Geometrically: the tangent line at some point c is parallel to the secant line through $(a, f(a))$ and $(b, f(b))$. The proof reduces to Rolle's Theorem by subtracting the secant: define $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$, which satisfies $g(a) = g(b) = f(a)$, then apply Rolle's. The theorem is an existence result, not a construction: it guarantees some c exists but gives no formula for it. Its immediate corollaries — the constant function corollary, the monotonicity corollary, and the Lipschitz bound corollary — each flow from the MVT by a one-line argument and are the theorem's primary applications.

Remark (Dependencies).

- Theorem: Rolle's Theorem
- Definition: Derivative at a Point

Theorem (Cauchy Mean Value Theorem)

Theorem 11.6.3 (Cauchy Mean Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f, g : [a, b] \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that*

$$\big(f(b) - f(a)\big)g'(c) = \big(g(b) - g(a)\big)f'(c).$$

If, in addition, $g'(x) \neq 0$ for every $x \in (a, b)$, then $g(b) \neq g(a)$ and

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{ContinuousOn}(f, [a, b]) \wedge \text{ContinuousOn}(g, [a, b]) \wedge \text{DifferentiableOn}(f, (a, b)) \wedge \text{DifferentiableOn}(g, (a, b)) \\ &\quad\Longrightarrow \exists c \in (a, b) \ \big[(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)\big].\end{aligned}$$

Remark (Predicate reading). $\text{CauchyMVT}(f, g, [a, b])$ asserts the existence of an interior point where the velocity vector $(g'(c), f'(c))$ is parallel to the chord vector $(g(b)-g(a), f(b)-f(a))$. Taking $g(x) = x$ recovers the ordinary Mean Value Theorem.

Remark (Negated quantified statement).

$$\forall c \in (a, b) : \ \big(f(b) - f(a)\big)g'(c) \neq \big(g(b) - g(a)\big)f'(c).$$

Remark (Failure modes). The continuity and differentiability hypotheses fail for the same reasons they fail in the ordinary MVT: jumps break endpoint-to-interior control, and corners leave no derivative to match. The quotient form has the additional nonvanishing hypothesis $g' \neq 0$ on (a, b); without it the cross-multiplied conclusion may still hold, but the displayed quotient can be undefined.

Remark (Contrapositive quantified statement).

$$\begin{aligned}&\big(\forall c \in (a, b) : \ (f(b) - f(a))g'(c) \neq (g(b) - g(a))f'(c)\big) \\ &\quad\Longrightarrow \neg\,\text{ContinuousOn}(f, [a, b]) \vee \neg\,\text{ContinuousOn}(g, [a, b]) \vee \neg\,\text{DifferentiableOn}(f, (a, b)) \vee \neg\,\text{Differentia}\end{aligned}$$

Remark (Contrapositive predicate reading). If no interior point makes the derivative pair $(g'(c), f'(c))$ parallel to the chord pair $(g(b) - g(a), f(b) - f(a))$, then at least one of the four regularity hypotheses fails: continuity of f or g on $[a, b]$, or differentiability of f or g on (a, b).

Remark (Interpretation). This is the parametric Mean Value Theorem. For the planar curve $t \mapsto (g(t), f(t))$, some interior tangent is parallel to the chord joining its endpoints. It compares two rates of change rather than comparing one function to the horizontal axis. This is the form that later underlies L'Hopital-type ratio arguments and the Cauchy form of Taylor's remainder.

Remark (Dependencies).

- Theorem: Rolle's Theorem
- Definition: Derivative at a Point
- Theorem: Sum Rule
- Theorem: Constant Multiple Rule

Corollary (Derivative Bound Implies Lipschitz)

Corollary 11.6.4 (Derivative Bound Implies Lipschitz). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be differentiable on I. If there exists $M \geq 0$ such that*

$$|f'(x)| \leq M \qquad \text{for every } x \in I,$$

then f is Lipschitz on I with constant M:

$$|f(x) - f(y)| \leq M|x - y| \qquad \text{for every } x, y \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\big(\exists M \geq 0\ \forall x \in I : \ |f'(x)| \leq M\big) \Longrightarrow \forall x, y \in I : \ |f(x) - f(y)| \leq M|x - y|.$$

Remark (Predicate reading).

$$\text{BoundedDerivativeOn}(f, I, M) \Longrightarrow \text{Lipschitz}(f, I, M).$$

Remark (Failure modes). The derivative bound must hold throughout the interval. A function such as $f(x) = \sqrt{x}$ on $[0, 1]$ is continuous and uniformly continuous, but its derivative is unbounded near 0, so no global Lipschitz constant follows from this test. The interval hypothesis is also essential: across a disconnected gap, the MVT cannot compare the two components.

Remark (Interpretation). This corollary converts a pointwise slope bound into a global modulus of continuity. It is the precise bridge from derivative estimates to the Lipschitz condition: controlling the size of f' controls every secant slope on the interval.

Remark (Dependencies).

- Theorem: Mean Value Theorem
- Definition: Lipschitz Condition
- Definition: Derivative at a Point

11.7 Reading the Graph from f' and f''

11.7.1 First-Order Shape

First-Order Shape — Quick Reference

Statement	Role
Zero derivative	Vanishing derivative forces constant behavior
Equal derivatives	Functions differ by a constant
Nonnegative derivative	Detects nondecreasing behavior
Nonpositive derivative	Detects nonincreasing behavior
Positive derivative	Local increasing behavior
Negative derivative	Local decreasing behavior
First derivative test: max	Sign change detects a local maximum
First derivative test: min	Sign change detects a local minimum

Remark (Section guide). The Mean Value Theorem is now available, so the harvest can begin. Vanishing derivatives force constancy, derivative signs force monotonicity, and second derivatives detect bending when enough regularity is present. Darboux and the smoothness hierarchy show that derivatives carry hidden rigidity even when they are not continuous.

Definition (Increasing at a Point)

Definition 11.7.1 (Increasing at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *increasing at* c if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \leq f(c) \leq f(y)$.

Remark (Standard quantified statement).

$$\exists \delta > 0 \; \forall x \in A \cap (c - \delta, c) \; \forall y \in A \cap (c, c + \delta) \; \big(f(x) \leq f(c) \leq f(y)\big).$$

Remark (Interpretation). f is increasing at c if c lies between the values of f on either side in some punctured neighbourhood: f is smaller to the left and larger to the right (weakly). This is a strictly local notion — it says nothing about f outside the δ-neighbourhood of c. A function can be increasing at c without being increasing (globally or on any interval). The non-strict inequalities permit the function to be flat on one or both sides of c, distinguishing this from the notion of a strict local increase.

Remark (Dependencies).

- Definition: Strictly Increasing Function (functions chapter)

Definition (Decreasing at a Point)

Definition 11.7.2 (Decreasing at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *decreasing at* c if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \geq f(c) \geq f(y)$.

Remark (Standard quantified statement).

$$\exists \delta > 0 \, \forall x \in A \cap (c - \delta, c) \, \forall y \in A \cap (c, c + \delta) \, \big(f(x) \geq f(c) \geq f(y)\big).$$

Remark (Interpretation). The function is decreasing at c when nearby values to the left sit above $f(c)$ and nearby values to the right sit below it.

Remark (Dependencies).

- Definition: Increasing at a Point

Theorem (Zero Derivative Implies Constant)

Theorem 11.7.3 (Zero Derivative Implies Constant)**.** *Let $I := [a, b] \subseteq \mathbb{R}$ be a closed interval, and let $f : I \to \mathbb{R}$ be continuous on I and differentiable on (a, b). If*

$$f'(x) = 0 \qquad \textit{for all } x \in (a, b),$$

then f is constant on I. Go to proof.

Remark (Standard quantified statement).

$$\forall x \in (a, b) \; \text{DerivativeAt}(f, x, 0) \; \Rightarrow \; \exists C \in \mathbb{R} \, \forall x \in I \; (f(x) = C).$$

Remark (Contrapositive quantified statement).

$$\neg\big(\exists C \in \mathbb{R} \, \forall x \in I \; (f(x) = C)\big) \; \Rightarrow \; \exists x \in (a, b) \, \neg \, \text{DerivativeAt}(f, x, 0).$$

If f is not constant on I, then f' is not identically zero on (a, b).

Remark (Contrapositive predicate reading).

$$\neg \, \text{ConstantOn}(f, I) \; \Rightarrow \; \exists x \in (a, b) \; (f'(x) \neq 0).$$

Remark (Interpretation). The MVT is the engine here: if $f'(x) = 0$ everywhere on (a, b), then for any two points $x_1 < x_2$ in I, the MVT supplies a c with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$, so $f(x_1) = f(x_2)$. Since this holds for all pairs, f is constant. The theorem is the foundation of antiderivative uniqueness: two antiderivatives of the same function differ by a constant. The contrapositive is used in ODE uniqueness arguments — if a solution is not identically the zero solution, its derivative is not identically zero somewhere.

Remark (Dependencies).

- Theorem: Mean Value Theorem

Corollary

Corollary 11.7.4 (Equal Derivatives Imply Difference Is Constant)**.** *Let $I := [a, b] \subseteq \mathbb{R}$, and let $f, g : I \to \mathbb{R}$ be continuous on I and differentiable on (a, b). If*

$$f'(x) = g'(x) \qquad \text{for all } x \in (a, b),$$

then there exists $C \in \mathbb{R}$ such that $f(x) = g(x) + C$ for all $x \in I$. *Go to proof.*

Remark (Standard quantified statement).

$$\forall x \in (a, b)\ (f'(x) = g'(x)) \ \Rightarrow\ \exists C \in \mathbb{R}\ \forall x \in I\ (f(x) = g(x) + C).$$

Remark (Contrapositive quantified statement).

$$\forall C \in \mathbb{R}\ \exists x \in I\ (f(x) \neq g(x) + C) \ \Rightarrow\ \exists x \in (a, b)\ (f'(x) \neq g'(x)).$$

If $f - g$ is not a constant function, then f' and g' disagree somewhere.

Remark (Contrapositive predicate reading).

$$\neg\,\mathrm{ConstantOn}(f - g,\ I) \ \Rightarrow\ \exists x \in (a, b)\ (f'(x) \neq g'(x)).$$

Remark (Interpretation). This is the corollary that makes antiderivatives well-defined up to a constant: any two antiderivatives of the same function on an interval differ by a constant. The proof is one line — apply the previous theorem to $h = f - g$, whose derivative is $(f - g)' = f' - g' = 0$.

Remark (Dependencies).

- Theorem: Zero Derivative Implies Constant

Theorem (Nondecreasing Iff Nonnegative Derivative)

Theorem 11.7.5 (Nondecreasing Iff Nonnegative Derivative)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nondecreasing on I if and only if*

$$f'(x) \geq 0 \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{NondecreasingOn}(f, I) \iff \forall x \in I\ (f'(x) \geq 0).$$

Remark (Contrapositive quantified statement).

$$\exists x \in I\ (f'(x) < 0) \ \Rightarrow\ \neg\,\text{NondecreasingOn}(f, I).$$

If the derivative is negative at some point, the function is not nondecreasing.

Remark (Contrapositive predicate reading).

$$\exists x \in I\ (f'(x) < 0) \ \Rightarrow\ \neg\,\text{NondecreasingOn}(f, I).$$

Remark (Interpretation). The derivative is the instantaneous rate of change; requiring it to be nonnegative everywhere exactly characterises functions that never decrease. The forward direction (f nondecreasing $\Rightarrow f' \geq 0$) follows from the definition: the difference quotient is nonnegative when f is nondecreasing, so the limit is nonnegative. The reverse direction ($f' \geq 0 \Rightarrow$ nondecreasing) is a consequence of the MVT: for $x < y$, the MVT gives $f(y) - f(x) = f'(c)(y - x) \geq 0$.

Note carefully: $f' \geq 0$ characterises *nondecreasing* (weak), not *strictly increasing* (strict). Strict monotonicity requires $f' > 0$ at all but isolated points. The function $f(x) = x^3$ satisfies $f'(0) = 0$ but is strictly increasing; $f' \geq 0$ with equality only at isolated points is sufficient for strict increase, but $f' > 0$ everywhere is not necessary.

Remark (Dependencies).

- Theorem: Mean Value Theorem
- Definition: Derivative at a Point

Theorem

Theorem 11.7.6 (Nonincreasing Iff Nonpositive Derivative)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nonincreasing on I if and only if*

$$f'(x) \leq 0 \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{NonincreasingOn}(f, I) \iff \forall x \in I\ (f'(x) \leq 0).$$

Remark (Contrapositive quantified statement).

$$\exists x \in I\ (f'(x) > 0) \ \Rightarrow\ \neg\,\text{NonincreasingOn}(f, I).$$

Remark (Contrapositive predicate reading).

$$\exists x \in I\ (f'(x) > 0) \ \Rightarrow\ \neg\,\text{NonincreasingOn}(f, I).$$

Remark (Interpretation). Nonpositive derivative everywhere is exactly the infinitesimal signature of a function that never increases.

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative

Lemma

Lemma 11.7.7 (Positive Derivative Implies Locally Increasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) > 0$, then there exists $\delta > 0$ such that*

$$f(x) < f(c) \quad \textit{for all } x \in I \textit{ with } c - \delta < x < c,$$

and

$$f(x) > f(c) \quad \textit{for all } x \in I \textit{ with } c < x < c + \delta.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \wedge L > 0 \;\Rightarrow\; \exists \delta > 0 \, \forall x \in I \, \big(c-\delta < x < c \Rightarrow f(x) < f(c)\big) \wedge \big(c < x < c+\delta \Rightarrow f(x) >$$

Remark (Contrapositive quantified statement).

$$\forall \delta > 0 \, \exists x \in I \, \big((c-\delta < x < c \wedge f(x) \geq f(c)) \vee (c < x < c+\delta \wedge f(x) \leq f(c))\big) \;\Rightarrow\; f'(c) \leq 0.$$

Remark (Contrapositive predicate reading).

$$\neg\, \text{LocallyIncreasingAt}(f, c) \;\Rightarrow\; f'(c) \leq 0.$$

Remark (Interpretation). If the instantaneous rate of change is strictly positive at c, the function is genuinely climbing through c: it is strictly below $f(c)$ immediately to the left and strictly above immediately to the right. The proof is a direct ε-δ argument: take $\varepsilon = f'(c)/2 > 0$; the definition of the derivative supplies δ such that the difference quotient is within ε of $f'(c)$, forcing it to be positive, which then forces the correct sign on $f(x) - f(c)$. This lemma is the key ingredient in the proof of the First Derivative Test.

Remark (Dependencies).

- Definition: Derivative at a Point

Lemma

Lemma 11.7.8 (Negative Derivative Implies Locally Decreasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) < 0$, then there exists $\delta > 0$ such that*

$$f(x) > f(c) \quad \textit{for all } x \in I \textit{ with } c - \delta < x < c,$$

and

$$f(x) < f(c) \quad \textit{for all } x \in I \textit{ with } c < x < c + \delta.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DerivativeAt}(f,c,L) \wedge L < 0 \;\Rightarrow\; \exists\delta > 0\, \forall x \in I\, \big(c{-}\delta < x < c \Rightarrow f(x) > f(c)\big) \wedge \big(c < x < c{+}\delta \Rightarrow f(x) <$$

Remark (Contrapositive quantified statement).

$$\neg\,\text{LocallyDecreasingAt}(f,c) \;\Rightarrow\; f'(c) \geq 0.$$

Remark (Contrapositive predicate reading).

$$\neg\,\text{LocallyDecreasingAt}(f,c) \;\Rightarrow\; f'(c) \geq 0.$$

Remark (Interpretation). If the derivative at c is strictly negative, nearby values move downward as the input passes through c from left to right.

Remark (Dependencies).

- Lemma: Positive Derivative Implies Locally Increasing

Theorem (First Derivative Test: Relative Maximum)

Theorem 11.7.9 (First Derivative Test: Relative Maximum). *Let $I := [a,b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a,b)$. Suppose f is differentiable on (a,c) and (c,b). If there exists $\delta > 0$ such that $(c-\delta, c+\delta) \subseteq I$ and*

$$f'(x) \geq 0 \quad \text{for all } x \in (c-\delta, c), \qquad f'(x) \leq 0 \quad \text{for all } x \in (c, c+\delta),$$

then f has a relative maximum at c. *Go to proof.*

Remark (Standard quantified statement).

$$\exists\delta > 0 \left[(c-\delta, c+\delta) \subseteq I \;\wedge\; \forall x \in (c-\delta, c)\, (f'(x) \geq 0) \;\wedge\; \forall x \in (c, c+\delta)\, (f'(x) \leq 0)\right]$$
$$\Rightarrow\; \text{LocalMaximum}(f,c).$$

Remark (Interpretation). If the function is nondecreasing immediately to the left of c and nonincreasing immediately to the right, then c is a local maximum — the function climbs to c and then falls away. The proof integrates: by the nondecreasing characterisation applied on $(c-\delta, c)$, $f(x) \leq f(c)$ for $x < c$; by the nonincreasing characterisation on $(c, c+\delta)$, $f(x) \leq f(c)$ for $x > c$. The sign condition on f' replaces the Interior Extremum Theorem's converse, which does not hold in general; this is a sufficient condition, not a necessary one. The test is applicable even when $f'(c) = 0$ or $f'(c)$ does not exist.

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative
- Theorem: Nonincreasing Iff Nonpositive Derivative
- Definition: Relative Maximum

Theorem

Theorem 11.7.10 (First Derivative Test: Relative Minimum)**.** *Let $I := [a,b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a,b)$. Suppose f is differentiable on (a,c) and (c,b). If there exists $\delta > 0$ such that $(c-\delta, c+\delta) \subseteq I$ and*

$$f'(x) \le 0 \quad \text{for all } x \in (c-\delta, c), \qquad f'(x) \ge 0 \quad \text{for all } x \in (c, c+\delta),$$

then f has a relative minimum at c. Go to proof.

Remark (Standard quantified statement).

$$\exists \delta > 0 \left[(c-\delta, c+\delta) \subseteq I \ \wedge\ \forall x \in (c-\delta, c)\ (f'(x) \le 0)\ \wedge\ \forall x \in (c, c+\delta)\ (f'(x) \ge 0)\right] \\ \Rightarrow\ \mathrm{LocalMinimum}(f, c).$$

Remark (Interpretation)*.* If the derivative changes from nonpositive to nonnegative at c, the graph falls into c and then rises away from it, so c is a local minimum.

Remark (Dependencies).

- Theorem: First Derivative Test: Relative Maximum
- Definition: Relative Minimum

11.7.2 Second-Order Shape

Second-Order Shape — Quick Reference

Concept	Role
Second derivative	Derivative of the derivative
Higher derivatives	Iterated derivatives $f^{(n)}$
Convexity test	$f'' \ge 0$ detects convexity
Concavity test	$f'' \le 0$ detects concavity
Second derivative test	Classifies critical points
Inflection condition	Continuous f'' changes sign through zero

11.7.3 Higher-Order Derivatives

Definition (Second Derivative)

Definition 11.7.11 (Second Derivative)**.** Let f be differentiable on a neighbourhood of c. If f' is differentiable at c, the *second derivative* of f at c is

$$f''(c) := (f')'(c).$$

Remark (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f', c, f''(c)) \quad \Longleftrightarrow \quad f''(c) = (f')'(c).$$

Remark (Interpretation). The first derivative measures instantaneous slope. The second derivative measures how that slope itself changes.

Remark (Dependencies).

- Derivative at a Point

Definition (Higher Derivatives)

Definition 11.7.12 (Higher Derivatives)**.** Define $f^{(0)} := f$ and $f^{(1)} := f'$. For $n \geq 2$, if $f^{(n-1)}$ is differentiable at c, define

$$f^{(n)}(c) := (f^{(n-1)})'(c).$$

Remark (Standard quantified statement).

$$\forall n \geq 2: \quad f^{(n)}(c) = (f^{(n-1)})'(c)$$

whenever the derivative on the right exists.

Remark (Interpretation). Higher derivatives are obtained by iterating the derivative operator. Their existence is not automatic: each stage requires differentiability of the previous derivative.

Remark (Dependencies).

- Second Derivative

11.7.4 Second-Order Shape

Theorem (Second-Derivative Convexity Test)

Theorem 11.7.13 (Second-Derivative Convexity Test)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be twice differentiable on I. If*

$$f''(x) \geq 0 \qquad \text{for all } x \in I,$$

then f is convex on I. *Go to proof.*

Remark (Standard quantified statement).

$$\big(\forall x \in I,\ f''(x) \geq 0\big) \Longrightarrow \mathrm{ConvexOn}(f, I).$$

Remark (Interpretation). Nonnegative second derivative means the slopes do not decrease. This is the derivative-based signature of convex bending.

Remark (Dependencies).

- Second Derivative
- Convex Function

> **Theorem (Second-Derivative Concavity Test)**
>
> **Theorem 11.7.14** (Second-Derivative Concavity Test)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be twice differentiable on I. If*
>
> $$f''(x) \le 0 \qquad \text{for all } x \in I,$$
>
> *then f is concave on I. Go to proof.*

Remark (Standard quantified statement)*.*

$$\bigl(\forall x \in I,\ f''(x) \le 0\bigr) \Longrightarrow \mathrm{ConcaveOn}(f, I).$$

Remark (Interpretation)*.* Nonpositive second derivative means the slopes do not increase. This is the derivative-based signature of concave bending.

Remark (Interpretation)*.* The second-derivative tests are narrower than the chord definitions of convexity and concavity. The function $|x|$ is convex, but it is not twice differentiable at 0.

Remark (Dependencies)*.*

- Second Derivative
- Concave Function

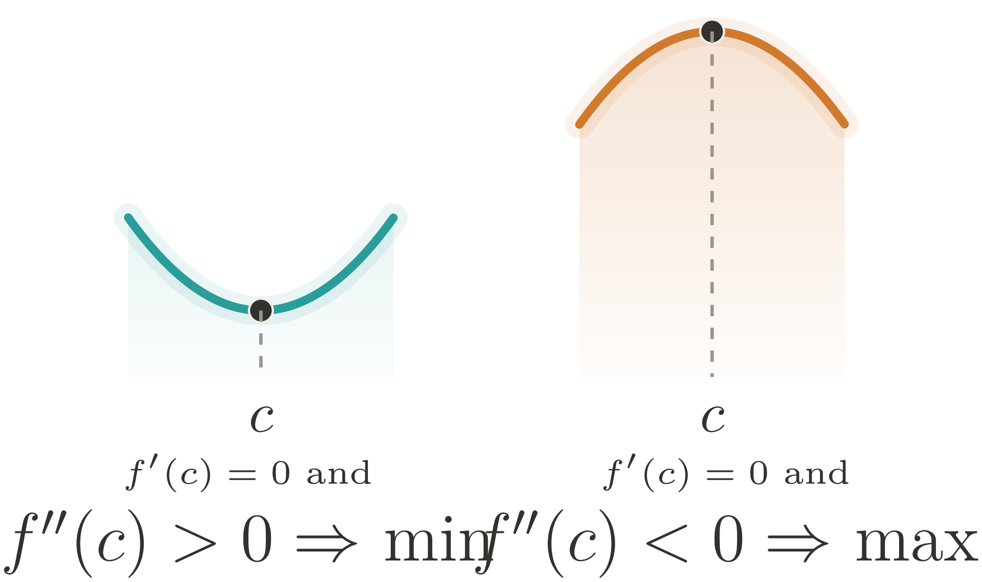


Figure 11.5: Second derivative test: the sign of $f''(c)$ decides the local shape at a critical point.

Theorem (Second Derivative Test)

Theorem 11.7.15 (Second Derivative Test)**.** *Let f be twice differentiable in a neighbourhood of c, and suppose $f'(c) = 0$.*

- *If $f''(c) > 0$, then f has a strict relative minimum at c.*
- *If $f''(c) < 0$, then f has a strict relative maximum at c.*

Go to proof.

Remark (Standard quantified statement).

$$f'(c) = 0 \wedge f''(c) > 0 \Rightarrow \text{StrictRelativeMinimum}(f, c),$$

and

$$f'(c) = 0 \wedge f''(c) < 0 \Rightarrow \text{StrictRelativeMaximum}(f, c).$$

Remark (Interpretation). At a critical point, positive second derivative means the graph bends upward; negative second derivative means it bends downward.

Remark (Dependencies).

- Second Derivative
- Relative Minimum
- Relative Maximum

Proposition (Inflection Point Necessary Condition)

Proposition 11.7.16 (Inflection Point Necessary Condition)**.** *Let f have an inflection point at c. If f'' exists on a neighbourhood of c and is continuous at c, then*

$$f''(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{InflectionPoint}(f, c) \wedge \text{ContinuousAt}(f'', c) \implies f''(c) = 0.$$

Remark (Interpretation). An inflection point is a change in bending. When the second derivative is continuous, such a change can pass through c only by passing through zero. The converse can fail: $f''(c) = 0$ does not by itself guarantee an inflection point.

Remark (Dependencies).

- Inflection Point
- Second Derivative
- Continuity at a Point

11.7.5 Darboux's Theorem

Darboux — Quick Reference

Statement	Role
Darboux's Theorem	Derivatives have the intermediate value property

Theorem (Darboux's Theorem)

Theorem 11.7.17 (Darboux's Theorem)**.** *Let $I := [a, b] \subseteq \mathbb{R}$ and let $f : I \to \mathbb{R}$ be differentiable on I. If k is a real number strictly between $f'(a)$ and $f'(b)$, then there exists $c \in (a, b)$ such that*

$$f'(c) = k.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall k \left(\min(f'(a), f'(b)) < k < \max(f'(a), f'(b)) \Rightarrow \exists c \in (a, b)\ (f'(c) = k)\right).$$

Remark (Interpretation). The Intermediate Value Theorem says continuous functions have the intermediate value property. Darboux's theorem says derivatives also have that property even when the derivative is not continuous. Thus a derivative cannot have a jump discontinuity. The proof uses differentiability of f to construct an auxiliary function $g(x) = f(x) - kx$ and then applies the Extreme Value Theorem and the Interior Extremum Theorem.

Remark (Dependencies).

- Theorem: Extreme Value Theorem
- Theorem: Interior Extremum Theorem
- Definition: Derivative at a Point

11.7.6 Smoothness Classes

Smoothness Classes — Quick Reference

Concept	Meaning
Class C^1	Differentiable with continuous first derivative
Class C^k	k derivatives with continuous top derivative
Class C^∞	Continuous derivatives of every order
Class C^ω	Equal to local Taylor series
Smoothness tower	Strict hierarchy $C^\omega \subsetneq C^\infty \subsetneq \cdots$
Class $C^{1,1}$	C^1 with Lipschitz first derivative
Placement of $C^{1,1}$	Refines the gap between C^1 and C^2
Bounded f''	Bounded second derivative gives $C^{1,1}$
Example: differentiable not C^1	Oscillatory derivative discontinuity at 0

Definition (Continuously Differentiable; Class C^1)

Definition 11.7.18 (Continuously Differentiable; Class C^1)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *continuously differentiable* on I, written $f \in C^1(I)$, if f is differentiable at every point of I and the derivative $f' : I \to \mathbb{R}$ is continuous on I.

Remark (Standard quantified statement)*.*

$$f \in C^1(I) \iff \big(\forall x \in I : \mathrm{DifferentiableAt}(f, x)\big) \wedge \mathrm{ContinuousOn}(f', I).$$

Remark (Interpretation)*.* The condition $f \in C^1(I)$ is strictly stronger than differentiability. A function may be differentiable everywhere while its derivative fails to be continuous; Darboux's Theorem still constrains that failure, because derivatives cannot jump. Thus the gap between differentiability and C^1 is oscillatory rather than jump-like.

Remark (Dependencies)*.*

- Definition: Derivative at a Point
- Definition: Continuity at a Point

Example (Differentiable Everywhere, but Not C^1)

Remark (Worked example)*.* Let

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

At the origin,

$$f'(0) = \lim_{x \to 0} \frac{x^2 \sin(1/x)}{x} = \lim_{x \to 0} x \sin(1/x) = 0$$

by squeezing. For $x \neq 0$,

$$f'(x) = 2x \sin(1/x) - \cos(1/x).$$

The first term tends to 0, but $\cos(1/x)$ oscillates, so f' is not continuous at 0. Thus f is differentiable on all of $\mathbb{R}$, but $f \notin C^1(\mathbb{R})$. The failure is oscillatory, not a jump, as Darboux's theorem requires.

Definition (Class C^k)

Definition 11.7.19 (Class C^k)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $k \in \mathbb{N}$, and let $f : I \to \mathbb{R}$. The function f is of *class* C^k on I, written $f \in C^k(I)$, if the derivatives $f^{(0)}, f^{(1)}, \ldots, f^{(k)}$ exist on I and $f^{(k)}$ is continuous on I. By convention, $C^0(I)$ is the class of continuous functions on I.

Remark (Standard quantified statement)*.*

$$f \in C^k(I) \iff \big(\forall j \in \{0, 1, \ldots, k\} : f^{(j)} \text{ exists on } I\big) \wedge \mathrm{ContinuousOn}\big(f^{(k)}, I\big).$$

Remark (Predicate reading).

$$\mathrm{ClassC}(f, I, k).$$

For a fixed interval I, the classes are nested:

$$f \in C^{k+1}(I) \Longrightarrow f \in C^k(I).$$

Remark (Negated quantified statement).

$$f \notin C^k(I) \iff \exists x \in I: \ \big[f^{(k)} \text{ fails to exist at } x\big] \vee \big[f^{(k)} \text{ exists on } I \wedge \neg\, \mathrm{ContinuousAt}(f^{(k)}, x)\big].$$

Remark (Negation predicate reading). $\neg\,\mathrm{ClassC}(f, I, k)$: somewhere the k-th derivative either does not exist or exists but is not continuous. A single point where the top derivative is discontinuous certifies $f \notin C^k(I)$.

Remark (Interpretation). The notation $C^k(I)$ records k layers of differentiability together with continuity at the top layer. This is the regularity scale used to state how many times a theorem may differentiate a function without leaving the controlled setting.

Remark (Dependencies).

- Definition: Higher Derivatives
- Definition: Continuously Differentiable; Class C^1

Definition (Smooth Function; Class C^∞)

Definition 11.7.20 (Smooth Function; Class C^∞)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *smooth*, or of *class* C^∞, written $f \in C^\infty(I)$, if $f \in C^k(I)$ for every $k \in \mathbb{N}$. Equivalently,

$$C^\infty(I) = \bigcap_{k=0}^{\infty} C^k(I).$$

Remark (Standard quantified statement).

$$f \in C^\infty(I) \iff \forall k \in \mathbb{N}: \ f \in C^k(I).$$

Remark (Negated quantified statement).

$$f \notin C^\infty(I) \iff \exists k \in \mathbb{N}: \ f \notin C^k(I).$$

A single order at which differentiability or continuity breaks demotes the function out of C^∞.

Remark (Interpretation). Smoothness is the regularity language imported by differential geometry. Smooth manifolds are assembled from C^∞ transition maps, and the tangent-map construction seeded by the differential uses this vocabulary.

Remark (Dependencies).

- Definition: Class C^k

Definition (Real-Analytic Function; Class C^ω)

Definition 11.7.21 (Real-Analytic Function; Class C^ω)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *real-analytic*, or of *class* C^ω, written $f \in C^\omega(I)$, if for every $a \in I$ there exists $r > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \qquad \text{for every } x \in (a-r, a+r) \cap I.$$

Remark (Standard quantified statement).

$$f \in C^\omega(I) \iff \forall a \in I\ \exists r > 0\ \forall x \in (a-r, a+r) \cap I :\ f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Remark (Interpretation). Analytic regularity means that the Taylor series does more than approximate the function: near each point, the series equals the function. The infinite Taylor series and its convergence are developed in Taylor's construction and error analysis, which supplies the series machinery behind this definition.

Remark (Dependencies).

- Definition: Smooth Function; Class C^∞
- Definition: Taylor Polynomial at a Point

Theorem (The Smoothness Tower)

Theorem 11.7.22 (The Smoothness Tower)**.** *For every nondegenerate interval I, the inclusions*

$$C^0(I) \supseteq C^1(I) \supseteq C^2(I) \supseteq \cdots \supseteq C^\infty(I) \supseteq C^\omega(I)$$

hold, and each displayed inclusion is strict. Go to proof.

Remark (Standard quantified statement).

$$\forall k \in \mathbb{N} :\ C^{k+1}(I) \subsetneq C^k(I), \qquad C^\omega(I) \subsetneq C^\infty(I).$$

Remark (Failure modes). The inclusions do not fail; only their strictness needs witnesses.

- $C^k \setminus C^{k+1}$: $\varphi_k(x) = x^k|x|$, whose k-th derivative is $(k+1)!\,|x|$, continuous but not differentiable at 0.
- $C^\infty \setminus C^\omega$: the flat function e^{-1/x^2}, smooth with all derivatives zero at 0, so its Maclaurin series is zero while the function is not.

These separators prove every displayed containment is a genuine drop.

Remark (Interpretation). The inclusions are definitional; strictness is witnessed by examples. On any nondegenerate interval, translated and rescaled copies of $x \mapsto x^k|x|$ separate C^k from C^{k+1}. Translated copies of the flat function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

separate C^∞ from C^ω: all derivatives at the flat point vanish, so the Taylor series is identically zero, while the function is positive on one side of that point.

Remark (Dependencies).

- Definition: Class C^k
- Definition: Smooth Function; Class C^∞
- Definition: Real-Analytic Function; Class C^ω
- Theorem: Darboux's Theorem

Remark (Section guide). Between C^1 and C^2 there is a regularity floor worth naming. A C^1 function has a continuous derivative; a C^2 function has a derivative that is itself differentiable. Many functions used in analysis live in the gap: their first derivative is continuous and controlled, but a classical second derivative may fail to exist at a few points. The control is a Lipschitz bound on f', and the class is written $C^{1,1}$.

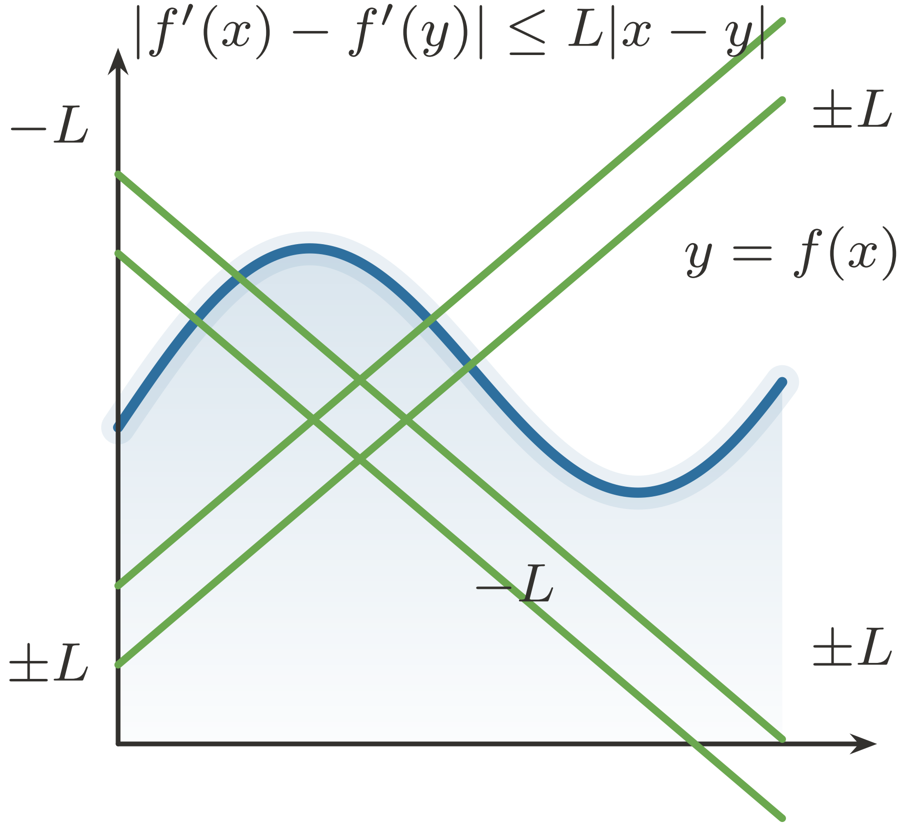


Figure 11.6: A Lipschitz derivative changes at a uniformly bounded rate.

Definition (Lipschitz-Derivative Class $C^{1,1}$)

Definition 11.7.23 (Lipschitz-Derivative Class $C^{1,1}$)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is of *class* $C^{1,1}$ on I, written $f \in C^{1,1}(I)$, if $f \in C^1(I)$ and f' is Lipschitz on I: there exists $L \geq 0$ such that

$$|f'(x) - f'(y)| \leq L|x - y| \qquad \text{for all } x, y \in I.$$

Remark (Standard quantified statement)*.*

$$f \in C^{1,1}(I) \iff f \in C^1(I) \wedge \exists L \geq 0 \; \forall x, y \in I : \; |f'(x) - f'(y)| \leq L|x - y|.$$

Remark (Definition predicate reading)*.*

$$\text{ClassC11}(f, I) \iff \text{ClassC}(f, I, 1) \wedge \exists L \geq 0 : \text{Lipschitz}(f', I, L).$$

The smallest admissible L is the Lipschitz constant of f':

$$\sup_{x \neq y} \frac{|f'(x) - f'(y)|}{|x - y|}.$$

Remark (Negated quantified statement)*.*

$$f \notin C^{1,1}(I) \iff f \notin C^1(I) \vee \big(\forall L \geq 0 \; \exists x, y \in I : \; |f'(x) - f'(y)| > L|x - y|\big).$$

Remark (Interpretation). $C^{1,1}$ says that f' behaves as if $|f''| \le L$, without requiring f'' to exist everywhere. It is the exact regularity tracked by many first-order estimates: the derivative is not merely continuous, but changes at a bounded rate.

Remark (Dependencies).

- Definition: Continuously Differentiable; Class C^1
- Corollary: Derivative Bound Implies Lipschitz

Theorem (Placement of $C^{1,1}$)

Theorem 11.7.24 (Placement of $C^{1,1}$). *For every nondegenerate interval I,*

$$C^{1,1}(I) \subsetneq C^1(I),$$

and $C^{1,1}(I)$ is not contained in $C^2(I)$. If K is a compact interval, then

$$C^2(K) \subseteq C^{1,1}(K) \subsetneq C^1(K),$$

and the first inclusion is strict. Go to proof.

Remark (Standard quantified statement).

$$\big(f \in C^{1,1}(I) \Rightarrow f \in C^1(I)\big), \qquad \big(K \text{ compact} \wedge f \in C^2(K) \Rightarrow f \in C^{1,1}(K)\big),$$

with the displayed containments strict.

Remark (Failure modes). The global inclusion $C^2(I) \subseteq C^{1,1}(I)$ is false on arbitrary intervals: $f(x) = x^3$ is $C^2(\mathbb{R})$, but $f'(x) = 3x^2$ is not Lipschitz on $\mathbb{R}$. Strictness is witnessed by two elementary functions.

- $C^{1,1} \setminus C^2$: $f(x) = x|x|$ has $f'(x) = 2|x|$, which is Lipschitz, but $f''(0)$ does not exist.
- $C^1 \setminus C^{1,1}$: $f(x) = |x|^{4/3}$ has continuous first derivative, but f' has unbounded difference quotients at 0.

Remark (Interpretation). $C^{1,1}$ is a genuine intermediate regularity condition, but it is not the same as C^2. A bounded second derivative forces the class; a Lipschitz first derivative may have corners; and a continuous first derivative may still vary too sharply to be Lipschitz.

Remark (Dependencies).

- Definition: Lipschitz-Derivative Class $C^{1,1}$
- Theorem: The Smoothness Tower
- Corollary: Derivative Bound Implies Lipschitz

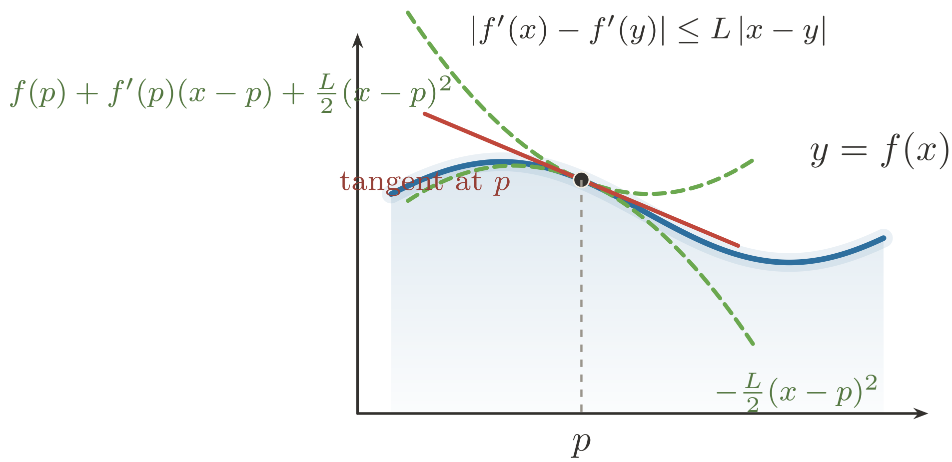


Figure 11.7: $C^{1,1}$ control gives quadratic upper and lower bounds around a tangent line.

Corollary (Bounded Second Derivative Implies $C^{1,1}$)

Corollary 11.7.25 (Bounded Second Derivative Implies $C^{1,1}$)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f \in C^2(I)$. If $|f''(x)| \le M$ for all $x \in I$, then $f \in C^{1,1}(I)$, and f' is Lipschitz with constant M. Go to proof.*

Remark (Standard quantified statement).

$$\big(f \in C^2(I) \wedge \forall x \in I:\ |f''(x)| \le M\big) \Rightarrow f \in C^{1,1}(I) \wedge \mathrm{Lipschitz}(f', I, M).$$

Remark (Interpretation). This is Derivative Bound Implies Lipschitz applied one level up, to f' in place of f. The converse requires almost-everywhere language and is therefore deferred.

Remark (Dependencies).

- Corollary: Derivative Bound Implies Lipschitz
- Definition: Class C^k
- Definition: Lipschitz-Derivative Class $C^{1,1}$

Remark (Forward flag). A Lipschitz derivative is differentiable at most points, but making that sentence precise requires measure zero. In one variable this passes through bounded variation; in several variables it becomes Rademacher's theorem. Those results belong to the integration and measure-theoretic development, not to this chapter.

11.8 The Algebra of Derivatives

Algebra of Derivatives — Quick Reference

Rule	Role
Constant multiple	Scalars pass through differentiation
Sum	Derivative distributes over addition
Product	Differentiates pointwise products
Quotient	Differentiates quotients where denominator is nonzero
Finite sum	Iterates the sum rule over finitely many terms
Extended product	Iterates the product rule over finitely many factors
Integer power	Differentiates powers built from products
Finite linear combination	Packages linearity in finite form
Inverse Function Theorem	Produces a C^1 local inverse
Inverse derivative	Gives the reciprocal derivative formula
L'Hôpital 0/0	Evaluates indeterminate zero ratios
L'Hôpital ∞/∞	Evaluates indeterminate divergent ratios

11.8.1 From Characterization to Computation

Remark (Section guide). With Carathéodory in hand, computing derivatives becomes bookkeeping with continuous slope factors. The basic rules record one structural fact: differentiation is a linear operator that also obeys a product law.

Remark (Interpretation). Carathéodory's characterization turns differentiability into continuity of a slope factor. If

$$f(x) - f(c) = (x - c)\phi_f(x), \qquad g(x) - g(c) = (x - c)\phi_g(x),$$

with ϕ_f and ϕ_g continuous at c, then the standard algebraic rules for derivatives follow from the algebra of continuous functions. The recurring move is no longer to manipulate a punctured quotient directly, but to construct a continuous factor whose value at c is the derivative.

Remark (Dependencies).

- Carathéodory Characterization of Differentiability
- Algebra of Limits

Theorem (Constant Multiple Rule)

Theorem 11.8.1 (Constant Multiple Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $\alpha \in \mathbb{R}$. If f is differentiable at c, then αf is differentiable at c, and*

$$(\alpha f)'(c) = \alpha f'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \;\Rightarrow\; \text{DerivativeAt}(\alpha f,\, c,\, \alpha f'(c)).$$

Remark (Interpretation). Multiplying a function by a constant scales its derivative by the same constant. The proof is one line: the scalar α factors out of the difference quotient algebraically, then the Constant Multiple Rule for Limits passes it through the limit. Differentiation is therefore homogeneous of degree one with respect to scalar multiplication.

Remark (Dependencies).

- Definition: Derivative at a Point
- Constant Multiple Rule for Limits

Theorem (Sum Rule)

Theorem 11.8.2 (Sum Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then $f + g$ is differentiable at c, and*

$$(f + g)'(c) = f'(c) + g'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \;\wedge\; \text{DifferentiableAt}(g, c) \;\Rightarrow\; \text{DerivativeAt}(f + g,\, c,\, f'(c) + g'(c)).$$

Remark (Interpretation). The derivative of a sum is the sum of the derivatives. The proof is immediate: the difference quotient of $f + g$ splits linearly into the sum of the individual difference quotients, and the Sum Rule for Limits distributes the limit across the sum. Together with the Constant Multiple Rule, this establishes that differentiation is a linear operation: $(\alpha f + \beta g)' = \alpha f' + \beta g'$.

Remark (Dependencies).

- Definition: Derivative at a Point
- Sum Rule for Limits

Theorem (Product Rule)

Theorem 11.8.3 (Product Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then fg is differentiable at c, and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \Rightarrow \text{DerivativeAt}(fg,\ c,\ f'(c)g(c)+f(c)g'(c)).$$

Remark (Interpretation). The derivative of a product is not simply the product of the derivatives. The correct formula adds two terms: the rate of change of f weighted by the current value of g, plus the rate of change of g weighted by the current value of f. The proof inserts the auxiliary term $f(x)g(c)$ to decouple the increments of f and g, isolating the individual difference quotients. Continuity of f at c (which follows from differentiability) is used to evaluate $\lim_{x\to c} f(x) = f(c)$ in one of the factors.

Remark (Dependencies).

- Definition: Derivative at a Point
- Theorem: Differentiable Implies Continuous
- Sum Rule for Limits
- Product Rule for Limits

Theorem (Quotient Rule)

Theorem 11.8.4 (Quotient Rule). *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. Suppose that f and g are differentiable at c and that $g(c) \neq 0$. Then f/g is differentiable at c, and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \wedge g(c) \neq 0 \Rightarrow \text{DerivativeAt}\left(\frac{f}{g},\ c,\ \frac{f'(c)g(c)-f(c)g'(c)}{(g(c))^2}\right).$$

Remark (Interpretation). The quotient rule is the product rule applied to $f \cdot g^{-1}$, but stated directly. The hypothesis $g(c) \neq 0$ ensures the quotient is defined near c — continuity of g (from its differentiability) and the quotient rule for limits guarantee $g(x) \neq 0$ in a deleted neighbourhood of c, so the difference quotient of f/g is well-defined throughout. The add-and-subtract technique with $f(c)g(c)$ decouples the increments of f and g in the numerator, exactly as in the product rule proof.

Remark (Dependencies).

- Definition: Derivative at a Point
- Theorem: Differentiable Implies Continuous
- Sum Rule for Limits
- Product Rule for Limits
- Quotient Rule for Limits

11.8.2 Closure

Corollary

Corollary 11.8.5 (Finite Sum Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\sum_{i=1}^n \alpha_i f_i$ is differentiable at c, and*

$$\left(\sum_{i=1}^n \alpha_i f_i\right)'(c) = \sum_{i=1}^n \alpha_i f_i'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \in \{1, \ldots, n\} \text{ DifferentiableAt}(f_i, c) \Rightarrow \text{DerivativeAt}\left(\textstyle\sum_i \alpha_i f_i,\ c,\ \sum_i \alpha_i f_i'(c)\right).$$

Remark (Interpretation). Finite sums and scalar weights do not introduce a new kind of differentiation rule; they package repeated use of the Sum Rule and Constant Multiple Rule.

Remark (Dependencies).

- Theorem: Sum Rule
- Theorem: Constant Multiple Rule

Corollary

Corollary 11.8.6 (Extended Product Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\prod_{i=1}^n f_i$ is differentiable at c, and*

$$\left(\prod_{i=1}^n f_i\right)'(c) = \sum_{k=1}^n f_k'(c) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \text{ DifferentiableAt}(f_i, c) \Rightarrow \text{DerivativeAt}\left(\textstyle\prod_i f_i,\ c,\ \sum_k f_k'(c) \prod_{i \neq k} f_i(c)\right).$$

Remark (Interpretation). In a finite product, each term takes one turn being differentiated while all other factors are held at their values.

Remark (Dependencies).

- Theorem: Product Rule

Corollary

Corollary 11.8.7 (Power Rule for Integer Powers of a Function)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $n \in \mathbb{N}$. If f is differentiable at c, then f^n is differentiable at c, and*

$$(f^n)'(c) = n(f(c))^{n-1} f'(c).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\text{DifferentiableAt}(f, c) \;\wedge\; n \in \mathbb{N} \;\Rightarrow\; \text{DerivativeAt}(f^n,\, c,\, n(f(c))^{n-1} f'(c)).$$

Remark (Interpretation)*.* The integer power rule is the extended product rule applied to n identical factors.

Remark (Dependencies)*.*

- Corollary: Extended Product Rule

Theorem

Theorem 11.8.8 (Finite Linear Combination Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i \right)'(c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall i \;\text{DifferentiableAt}(f_i, c) \;\Rightarrow\; \text{DerivativeAt}\left(\textstyle\sum_i \alpha_i f_i,\; c,\; \sum_i \alpha_i f_i'(c)\right).$$

Remark (Interpretation)*.* The Finite Linear Combination Rule is the full statement that differentiation is a *linear operator*: it commutes with scalar multiplication and addition simultaneously, for any finite collection of differentiable functions. This is not a separate theorem from the Constant Multiple and Sum Rules — it is their inductive closure. Stated in operator language: if $\mathcal{D}(I)$ denotes the vector space of functions differentiable at c, then $D_c : \mathcal{D}(I) \to \mathbb{R}$ defined by $D_c(f) = f'(c)$ is a linear functional. The Finite Linear Combination Rule says $D_c(\sum_i \alpha_i f_i) = \sum_i \alpha_i D_c(f_i)$.

Remark (Dependencies)*.*

- Real-Linear Rule
- Linear Combination of Real-Valued Functions
- Theorem: Constant Multiple Rule
- Theorem: Sum Rule

11.8.3 Global Forms on an Interval

Theorem

Theorem 11.8.9 (Constant Multiple Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If f is differentiable on I, then αf is differentiable on I, and*

$$(\alpha f)'(x) = \alpha f'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall x \in I\ \operatorname{DifferentiableAt}(f, x)\ \Rightarrow\ \forall x \in I\ \operatorname{DerivativeAt}(\alpha f,\, x,\, \alpha f'(x)).$$

Remark (Interpretation)*.* The pointwise constant-multiple rule becomes an interval statement by applying it at every point of the interval.

Remark (Dependencies)*.*

- Theorem: Constant Multiple Rule

Theorem

Theorem 11.8.10 (Sum Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then $f + g$ is differentiable on I, and*

$$(f + g)'(x) = f'(x) + g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall x \in I\ \big(\operatorname{DifferentiableAt}(f, x) \wedge \operatorname{DifferentiableAt}(g, x)\big)\ \Rightarrow\ \forall x \in I\ \operatorname{DerivativeAt}(f+g,\, x,\, f'(x)+g'(x)).$$

Remark (Interpretation)*.* The derivative of a sum is computed point by point across the whole interval.

Remark (Dependencies)*.*

- Theorem: Sum Rule

Theorem

Theorem 11.8.11 (Product Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then fg is differentiable on I, and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \ \big(\mathrm{DifferentiableAt}(f,x) \wedge \mathrm{DifferentiableAt}(g,x)\big) \ \Rightarrow \ \forall x \in I \ \ \mathrm{DerivativeAt}(fg,\ x,\ f'(x)g(x) + f(x)g'($$

Remark (Interpretation). The interval product rule records the product rule as an identity of derivative functions.

Remark (Dependencies).

- Theorem: Product Rule

Theorem

Theorem 11.8.12 (Quotient Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. Suppose that f and g are differentiable on I and that $g(x) \neq 0$ for all $x \in I$. Then f/g is differentiable on I, and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \ \big(\mathrm{DifferentiableAt}(f,x) \wedge \mathrm{DifferentiableAt}(g,x) \wedge g(x) \neq 0\big) \ \Rightarrow \ \forall x \in I \ \ \mathrm{DerivativeAt}\Big(\tfrac{f}{g},\ x,\ \tfrac{f'(x)g(x) - f}{(g(x))}$$

Remark (Interpretation). The quotient rule on an interval is the pointwise quotient rule with the nonvanishing denominator hypothesis required at every point.

Remark (Dependencies).

- Theorem: Quotient Rule

Corollary

Corollary 11.8.13 (Finite Sum Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \ldots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then $\sum_{i=1}^n \alpha_i f_i$ is differentiable on I, and*

$$\left(\sum_{i=1}^n \alpha_i f_i\right)'(x) = \sum_{i=1}^n \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \ \forall x \in I \ \ \mathrm{DifferentiableAt}(f_i, x) \ \Rightarrow \ \forall x \in I \ \ \mathrm{DerivativeAt}\big(\textstyle\sum_i \alpha_i f_i,\ x,\ \sum_i \alpha_i f_i'(x)\big).$$

Remark (Interpretation). Finite linear combinations commute with differentiation as functions on the whole interval.

Remark (Dependencies).

- Corollary: Finite Sum Rule

Corollary

Corollary 11.8.14 (Extended Product Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f_1, \ldots, f_n : I \to \mathbb{R}$. If each f_i is differentiable on I, then $\prod_{i=1}^n f_i$ is differentiable on I, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(x) = \sum_{k=1}^{n} f_k'(x) \prod_{\substack{i=1 \\ i \neq k}}^{n} f_i(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i\ \forall x \in I\ \ \text{DifferentiableAt}(f_i, x)\ \Rightarrow\ \forall x \in I\ \ \text{DerivativeAt}\Big(\textstyle\prod_i f_i,\ x,\ \sum_k f_k'(x) \prod_{i \neq k} f_i(x)\Big).$$

Remark (Interpretation). The product rule for many factors holds uniformly point by point on the interval.

Remark (Dependencies).

- Corollary: Extended Product Rule

Corollary

Corollary 11.8.15 (Power Rule for Integer Powers of a Function on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $n \in \mathbb{N}$. If f is differentiable on I, then f^n is differentiable on I, and*

$$(f^n)'(x) = n(f(x))^{n-1} f'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I\ \ \text{DifferentiableAt}(f, x)\ \wedge\ n \in \mathbb{N}\ \Rightarrow\ \forall x \in I\ \ \text{DerivativeAt}(f^n,\ x,\ n(f(x))^{n-1} f'(x)).$$

Remark (Interpretation). The interval power rule says that the pointwise integer-power formula assembles into an equality of derivative functions.

Remark (Dependencies).

- Corollary: Power Rule for Integer Powers of a Function

Theorem

Theorem 11.8.16 (Finite Linear Combination Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \ldots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(x) = \sum_{i=1}^{n} \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall i\ \forall x \in I\ \ \text{DifferentiableAt}(f_i, x)\ \Rightarrow\ \forall x \in I\ \ \text{DerivativeAt}\left(\textstyle\sum_i \alpha_i f_i,\ x,\ \sum_i \alpha_i f_i'(x)\right).$$

Remark (Interpretation)*.* The pointwise statements above — each proved at a single point c — extend to global statements when the hypotheses hold at every point of I. If f is differentiable on I, then $f' : I \to \mathbb{R}$ is itself a function and differentiation acts as a linear operator on the vector space $\mathcal{D}(I)$ of functions differentiable on I:

$$(\alpha f + \beta g)' = \alpha f' + \beta g', \qquad (fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

These equalities hold as functions on I. The Finite Linear Combination Rule in its global form is precisely the statement that $D : \mathcal{D}(I) \to \mathcal{F}(I)$ defined by $D(f) = f'$ is a linear map between function spaces, where $\mathcal{F}(I)$ denotes functions on I. The interval variants are not separate theorems — they are the pointwise results collected into function notation by applying the pointwise result at each $x \in I$.

Remark (Dependencies)*.*

- Real-Linear Rule
- Linear Combination of Real-Valued Functions
- Theorem: Finite Linear Combination Rule

When f is strictly monotone and differentiable on I with $f'(x) \neq 0$ for all $x \in I$, the inverse $g : J \to I$ is differentiable on $J = f(I)$ and

$$g' = \frac{1}{f' \circ g},$$

as functions on J. This is the global form of the Inverse Function Theorem for derivatives and is treated in the inverse function section.

11.8.4 Inverting the Operation

Theorem (Inverse Function Theorem, One Variable)

Theorem 11.8.17 (Inverse Function Theorem, One Variable)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$ be of class C^1, and let $c \in I$ satisfy $f'(c) \neq 0$. Then there exist open intervals $U \subseteq I$ and $V \subseteq \mathbb{R}$, with $c \in U$ and $f(c) \in V$, such that:*

1. *$f|_U : U \to V$ is a bijection;*
2. *the inverse $g = (f|_U)^{-1} : V \to U$ is of class C^1;*
3. *for every $y \in V$,*
$$g'(y) = \frac{1}{f'(g(y))}.$$

Go to proof.

Remark (Standard quantified statement).

$$f \in C^1(I) \wedge c \in I \wedge f'(c) \neq 0$$
$$\implies \exists U \ni c\ \exists V \ni f(c)\ \big[\text{Bijective}(f|_U, U, V) \wedge (f|_U)^{-1} \in C^1(V) \wedge \forall y \in V :\ ((f|_U)^{-1})'(y) = 1/f'((f|_U)^{-1}$$

Remark (Predicate reading). LocallyInvertible(f, c) holds with a C^1 local inverse. The condition $f'(c) \neq 0$ upgrades local set-theoretic invertibility to local analytic invertibility: the inverse exists as a differentiable function and has controlled derivative.

Remark (Failure modes). The hypotheses fail in different ways.

- If $f'(c) = 0$, the inverse may exist without being differentiable. The map $f(x) = x^3$ is globally bijective, but its inverse $x \mapsto x^{1/3}$ is not differentiable at 0.
- If f' is not continuous at c, the sign of f' need not persist near c. Mere differentiability with $f'(c) \neq 0$ is therefore not enough to force local monotonicity or local invertibility.

Remark (Contrapositive quantified statement).

$$\neg\,\text{LocallyInvertible}_{C^1}(f, c) \implies f \notin C^1(\text{near } c)\ \vee\ f'(c) = 0.$$

Remark (Contrapositive predicate reading). If f has no C^1 local inverse at c, then either f fails to be continuously differentiable near c, or its derivative vanishes there. The witnesses $x \mapsto x^3$ and $x \mapsto x + 2x^2 \sin(1/x)$ isolate the two failure mechanisms.

Remark (Interpretation). This theorem is the existence result behind the inverse-derivative formula. A continuous nonzero derivative keeps its sign on a small interval; constant sign forces strict monotonicity; strict monotonicity gives a local inverse; and the Carathéodory characterization upgrades that inverse to a differentiable map with reciprocal slope. In several variables, the condition $f'(c) \neq 0$ becomes invertibility of the differential.

Remark (Dependencies).

- Definition: Continuously Differentiable; Class C^1
- Theorem: Monotonicity from the Sign of the Derivative
- Theorem: Continuous Inverse Theorem
- Theorem: Carathéodory Characterization of Differentiability

Corollary (Derivative of the Inverse Function)

Corollary 11.8.18 (Derivative of the Inverse Function)**.** *Under the hypotheses of the one-variable Inverse Function Theorem, the local inverse g satisfies*

$$g' = \frac{1}{f' \circ g}$$

on V. More globally, if $f : I \to J$ is a C^1 bijection with $f'(x) \neq 0$ for every $x \in I$, then $f^{-1} : J \to I$ is C^1 and

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} \qquad \text{for every } y \in J.$$

Go to proof.

Remark (Standard quantified statement).

$$g = (f|_U)^{-1} \Longrightarrow \forall y \in V : \; g'(y) = \frac{1}{f'(g(y))}.$$

Remark (Interpretation). The formula says that undoing a differentiable change of variable reverses the local stretch factor. If f multiplies small increments near x by $f'(x)$, then its inverse multiplies corresponding increments near $f(x)$ by $1/f'(x)$.

Remark (Dependencies).

- Theorem: Inverse Function Theorem, One Variable
- Theorem: Chain Rule

11.8.5 Indeterminate Forms: L'Hôpital's Rule

Theorem (L'Hôpital's Rule, $0/0$ Form)

Theorem 11.8.19 (L'Hôpital's Rule, 0/0 Form). *Let $a < b$, and let $f, g : (a, b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0 \qquad \text{and} \qquad \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

The analogous statements at left-hand endpoints, at two-sided interior points, and at infinity follow by the same reductions. Go to proof.

Remark (Standard quantified statement).

$$\left(\lim_{x\to a^+} f(x) = 0 \wedge \lim_{x\to a^+} g(x) = 0 \wedge \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L\right) \Longrightarrow \lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

Remark (Failure modes). The rule is a one-way implication and is routinely misapplied.

- *Form not indeterminate.* If f/g is not genuinely $0/0$, differentiating numerator and denominator can answer a different problem.
- *$\lim f'/g'$ fails to exist.* With $f(x) = x + \sin x$ and $g(x) = x$ as $x \to +\infty$, the quotient $f/g \to 1$, but $f'/g' = 1 + \cos x$ oscillates. Absence of $\lim f'/g'$ says nothing by itself about $\lim f/g$.
- *g' vanishes near the target.* The denominator derivative must stay nonzero so the Cauchy Mean Value Theorem applies and the derivative ratio is defined.

Remark (Interpretation). The $0/0$ form is the Cauchy Mean Value Theorem read as a limit law. On a shrinking interval, a ratio of changes in f and g is represented by a ratio of derivatives at an interior point. If the derivative ratio settles, the original quotient is forced to settle to the same value.

Remark (Dependencies).

- Theorem: Cauchy Mean Value Theorem
- Definition: Limit of a Function
- Theorem: Squeeze Theorem for Function Limits

Theorem (L'Hôpital's Rule, ∞/∞ Form)

Theorem 11.8.20 (L'Hôpital's Rule, ∞/∞ Form)**.** *Let $a < b$, and let $f, g : (a, b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x\to a^+} |g(x)| = +\infty \qquad \textit{and} \qquad \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

No separate hypothesis on $\lim_{x\to a^+} f(x)$ is required. Go to proof.

Remark (Standard quantified statement).

$$\left(\lim_{x\to a^+} |g(x)| = +\infty \wedge \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L\right) \Longrightarrow \lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

Remark (Failure modes). L'Hôpital's Rule is a one-way implication and requires an indeterminate form. If the quotient is not of type $0/0$ or ∞/∞, differentiating numerator and denominator can change the problem. The derivative ratio must also have a limit: for $f(x) = x + \sin x$ and $g(x) = x$ as $x \to +\infty$, the quotient f/g tends to 1, but $f'/g' = 1 + \cos x$ has no limit.

Remark (Interpretation). The ∞/∞ form compares two functions whose magnitudes escape. The Cauchy Mean Value Theorem controls the quotient after subtracting fixed endpoint values, and the divergent denominator makes those fixed values negligible.

Remark (Dependencies).

- Theorem: Cauchy Mean Value Theorem
- Definition: Limit of a Function
- Theorem: Squeeze Theorem for Function Limits

11.9 Taylor: Construction and Its Error

Taylor Expansion — Quick Reference

Concept	**Meaning**
Taylor polynomial	Polynomial forced by derivative data at a point
Taylor remainder	Exact error after subtracting the Taylor polynomial
Maclaurin polynomial	Taylor polynomial centered at 0
Lagrange remainder	Error controlled by an intermediate derivative value
Peano remainder	Local asymptotic error smaller than the retained order
Example: $(\sin x - x)/x^3$	Taylor and L'Hôpital compared on one limit
Flat function	Smooth function whose Taylor series misses it

Taylor's theorem is the derivative's most ambitious claim: local behaviour to finite order is encoded by derivatives at one point, with an error term that can be estimated. The first-order case is the seed of the closing reframe, where the derivative stops being only a number $f'(c)$ and becomes the linear map $h \mapsto f'(c)h$.

Definition (Taylor Polynomial at a Point)

Definition 11.9.1 (Taylor Polynomial at a Point)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives $f^{(k)}(a)$ for $0 \leq k \leq n$. The **Taylor polynomial of order** n for f at a is

$$T_{n,a}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } I \subseteq \mathbb{R},\ a \in I,\ n \in \mathbb{N},\ f : I \to \mathbb{R}.\\ &\text{If } f^{(k)}(a) \text{ exists for every } 0 \leq k \leq n, \text{ then}\\ &\qquad \forall x \in I \left(T_{n,a}f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \right).\end{aligned}$$

Remark (Interpretation). The Taylor polynomial is the unique polynomial of degree at most n whose first n derivatives at a agree with those of f. Its coefficients are not chosen by fitting sample points; they are dictated by derivative data at one point.

Remark (Dependencies).

- Derivative at a Point
- Power Rule for Integer Powers of a Function
- Finite Sum Rule

Definition (Taylor Remainder)

Definition 11.9.2 (Taylor Remainder)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have a Taylor polynomial $T_{n,a}f$. The **Taylor remainder of order** n at a is the function $R_{n,a}f : I \to \mathbb{R}$ defined by

$$R_{n,a}f(x) := f(x) - T_{n,a}f(x).$$

Remark (Standard quantified statement).

$$\forall x \in I \ \big(R_{n,a}f(x) = f(x) - T_{n,a}f(x)\big).$$

Remark (Interpretation). The remainder is the exact error made by replacing f with its Taylor polynomial. Taylor's theorem is valuable because it gives usable formulas or estimates for this error.

Remark (Dependencies).

- Taylor Polynomial at a Point

Definition (Maclaurin Polynomial)

Definition 11.9.3 (Maclaurin Polynomial)**.** Let $n \in \mathbb{N}$, and let f be defined on an interval containing 0 with derivatives $f^{(k)}(0)$ for $0 \le k \le n$. The **Maclaurin polynomial of order** n for f is

$$T_{n,0}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k.$$

Remark (Standard quantified statement).

$$\forall x \in I \left(T_{n,0}f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k \right).$$

Remark (Interpretation). Maclaurin polynomials are Taylor polynomials centered at the origin. They are not a separate approximation theory; they are the centered-at-zero case of the same construction.

Remark (Dependencies).

- Taylor Polynomial at a Point

Theorem (Taylor's Theorem with Lagrange Remainder)

Theorem 11.9.4 (Taylor's Theorem with Lagrange Remainder)**.** *Let $a, b \in \mathbb{R}$ with $a < b$, let $n \in \mathbb{N}$, and let $f : [a, b] \to \mathbb{R}$. Suppose that $f, f', \ldots, f^{(n)}$ are continuous on $[a, b]$ and that $f^{(n+1)}$ exists on (a, b). Then, for every $x \in (a, b)$, there exists c strictly between a and x such that*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in (a, b)\ \exists c \in (a, x) \left(f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \right).$$

Remark (Interpretation). Taylor's theorem says that the error after truncating at degree n has the same shape as the next Taylor term, but with the next derivative evaluated at some intermediate point rather than at the center. The point c is generally not known explicitly; the theorem is most often used to estimate the size of the remainder.

Remark (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder
- Mean Value Theorem
- Finite Sum Rule
- Power Rule for Integer Powers of a Function

Corollary (Taylor Expansion with Peano Remainder)

Corollary 11.9.5 (Taylor Expansion with Peano Remainder)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives up to order n in a neighbourhood of a. If $f^{(n)}$ is continuous at a, then*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + o\big((x-a)^n\big) \qquad \text{as } x \to a.$$

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to a} \frac{f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k}{(x-a)^n} = 0.$$

Remark (Interpretation). The Peano form is a local asymptotic statement. It says that the Taylor polynomial captures all terms through order n, and the remaining error is smaller than $(x-a)^n$ near the center.

Remark (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder
- Limit of a Function
- Derivative at a Point

Corollary (First-Order Peano Remainder)

Corollary 11.9.6 (First-Order Peano Remainder)**.** *Let f be differentiable at c. Then, as $h \to 0$,*

$$f(c+h) = f(c) + f'(c)h + o(h).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \;\Rightarrow\; \lim_{h\to 0} \frac{f(c+h) - f(c) - f'(c)h}{h} = 0.$$

Remark (Interpretation). The first-order Peano formula is the derivative rewritten as an approximation theorem. It says that the affine approximation $f(c) + f'(c)h$ captures all first-order change, and the remaining error is smaller than h itself.

Remark (Dependencies).

- Derivative at a Point

Example ($(\sin x - x)/x^3$ Two Ways)

Remark (Worked example). Taylor expansion gives

$$\sin x = x - \frac{x^3}{6} + o(x^3),$$

so

$$\frac{\sin x - x}{x^3} = \frac{-x^3/6 + o(x^3)}{x^3} \longrightarrow -\frac{1}{6}.$$

L'Hôpital's rule reaches the same value by three differentiations:

$$\lim_{x\to 0} \frac{\sin x - x}{x^3} = \lim_{x\to 0} \frac{\cos x - 1}{3x^2} = \lim_{x\to 0} \frac{-\sin x}{6x} = \lim_{x\to 0} \frac{-\cos x}{6} = -\frac{1}{6}.$$

Taylor exposes the reason for the answer: the limit is the cubic coefficient.

Proposition (The Flat Function: Smooth but Not Analytic)

Proposition 11.9.7 (The Flat Function: Smooth but Not Analytic)**.** *Define $f : \mathbb{R} \to \mathbb{R}$ by*

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then:

1. $f \in C^\infty(\mathbb{R})$*;*
2. $f^{(n)}(0) = 0$ *for every* $n \geq 0$*;*
3. *the Maclaurin series of* f *is identically* 0*, so* $f \notin C^\omega$ *on any neighbourhood of* 0*.*

In particular, $C^\infty(\mathbb{R}) \setminus C^\omega(\mathbb{R}) \neq \varnothing$*.* *Go to proof.*

Remark (Standard quantified statement).

$$\big(\forall n \in \mathbb{N} : f^{(n)}(0) = 0\big) \wedge (\forall r > 0 \; \exists x : \; 0 < |x| < r \wedge f(x) \neq 0) \Longrightarrow f \in C^\infty \setminus C^\omega.$$

Remark (Interpretation). The Taylor series sees only derivative data at the center, and all of that data is zero for the flat function at 0. The function is nevertheless positive away from 0. This separates smoothness from analyticity and supplies the strictness witness referenced by the smoothness tower.

Remark (Dependencies).

- Definition: Higher Derivatives
- Definition: Smooth Function; Class C^∞
- Definition: Real-Analytic Function; Class C^ω
- Definition: Maclaurin Polynomial
- Theorem: The Smoothness Tower

11.9.1 The Differential

Remark (Interpretation). The derivative value $f'(c) \in \mathbb{R}$ is a number. The differential df_c is a linear map from increments to first-order changes. In one dimension these two objects are represented by the same scalar, but they play different roles.

Definition (Differentiability by a Differential)

Definition 11.9.8 (Differentiability by a Differential)**.** Let f be defined on a neighbourhood of $c \in \mathbb{R}$. The function f is *differentiable at c in the differential sense* if there exists a linear map $L : \mathbb{R} \to \mathbb{R}$ such that

$$f(c+h) - f(c) = L(h) + o(h) \qquad \text{as } h \to 0.$$

In that case the *differential* of f at c is $df_c := L$.

Remark (Standard quantified statement).

$$\exists L : \mathbb{R} \to \mathbb{R} \text{ linear such that } f(c+h) - f(c) = L(h) + o(h).$$

Remark (Interpretation). The differential is the linear part of the increment. The function itself is approximated affinely by $f(c) + df_c(x - c)$, while df_c alone is the linear map acting on the displacement.

Remark (Dependencies).

- First-Order Peano Remainder

Theorem (Differential and Derivative Agree)

Theorem 11.9.9 (Differential and Derivative Agree)**.** *Let f be defined on a neighbourhood of $c \in \mathbb{R}$. Then f is differentiable at c in the ordinary derivative sense if and only if it is differentiable at c in the differential sense. When this holds,*

$$df_c(h) = f'(c)h.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \iff \exists df_c \text{ with } df_c(h) = f'(c)h.$$

Remark (Interpretation). This theorem is the bridge from the slope number to the linear map. The one-dimensional derivative is multiplication by the scalar $f'(c)$.

Remark (Dependencies).

- Derivative at a Point
- Differentiability by a Differential

Theorem (Uniqueness of the Differential)

Theorem 11.9.10 (Uniqueness of the Differential)**.** *If $L_1, L_2 : \mathbb{R} \to \mathbb{R}$ are linear maps and*

$$f(c+h) - f(c) = L_1(h) + o(h) = L_2(h) + o(h) \qquad \text{as } h \to 0,$$

then $L_1 = L_2$. *Go to proof.*

Remark (Standard quantified statement).

$$\big(L_1(h) - L_2(h) = o(h)\big) \implies L_1 = L_2.$$

Remark (Interpretation). The differential is not a choice of linear approximation among many. If a first-order linear part exists, it is forced.

Remark (Dependencies).

- Differentiability by a Differential

Theorem (Differential Continuity Criterion)

Theorem 11.9.11 (Differential Continuity Criterion)**.** *If f is differentiable at c in the differential sense, then f is continuous at c.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{DifferentiableByDifferentialAt}(f, c) \implies \text{ContinuousAt}(f, c).$$

Remark (Interpretation). From $f(c+h) - f(c) = df_c(h) + o(h)$, both terms on the right tend to zero as $h \to 0$. Continuity is therefore immediate in the differential language.

Remark (Dependencies).

- Differentiability by a Differential

Theorem (Chain Rule for Differentials)

Theorem 11.9.12 (Chain Rule for Differentials)**.** *If f is differentiable at c and g is differentiable at $f(c)$, then*

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

Go to proof.

Remark (Standard quantified statement).

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

Remark (Interpretation). In one dimension this says that composing two scalar multiplications multiplies their scalars. In several variables the same formula becomes matrix multiplication.

Remark (Dependencies).

- Chain Rule
- Differential and Derivative Agree

Theorem (Linearity of the Differential)

Theorem 11.9.13 (Linearity of the Differential)**.** *If f and g are differentiable at c and $\alpha, \beta \in \mathbb{R}$, then*

$$d(\alpha f + \beta g)_c = \alpha\, df_c + \beta\, dg_c.$$

Go to proof.

Remark (Standard quantified statement).

$$d(\alpha f + \beta g)_c = \alpha\, df_c + \beta\, dg_c.$$

Remark (Interpretation). The differential packages the linearity of differentiation as equality of linear maps.

Remark (Interpretation). On a manifold, the same construction becomes a tangent map

$$T_c f : T_c M \to T_{f(c)} N.$$

The one-dimensional differential is the prototype: replace real increments by tangent vectors and replace scalar multiplication by a linear map between tangent spaces.

Remark (Dependencies).

- Finite Linear Combination Rule
- Differential and Derivative Agree

Chapter 12

Integration

SERIES	VOLUME III	BOOK IV	CHAPTER 11
From Cantor to Ito	Classical Analysis	Integration	Integration

12.1 Partitions and Tags

Remark (Reframing note). This chapter is the narrative version of the reference list. The reference orders the definitions correctly, but the real subject is the sequence of defects and patches:

$$\begin{aligned}\text{geometric problem} &\to \text{Cauchy} \to \text{Riemann}\\ &\to \text{Darboux} \to \text{Stieltjes}\\ &\to \text{Henstock–Kurzweil and McShane}\\ &\to \text{Lebesgue.}\end{aligned}$$

Each integral answers a concrete complaint about the one before it. The point is not to memorize several integrals, but to understand why each one had to be invented.

The Geometric Problem

Forget integrals for a moment. The physical question is always the same:

> Given a pointwise rate f, how much total quantity accumulates from a to b?

A speedometer reads $f(t)$; the accumulated quantity is distance. A wire has linear density $f(x)$; the accumulated quantity is mass. A curve $y = f(x)$ over $[a, b]$ asks for signed area. The only available strategy is:

1. Chop $[a, b]$ into thin slabs.
2. Pretend f is constant on each slab.
3. Add the resulting rectangles.

4. Refine the chopping and ask whether the sums settle.

Every construction below makes two choices: which height to use in a slab, and what it means for all these rectangle sums to settle to one number.

Remark (Construction comparison). The constructions differ by the height rule, the ruler, or the fineness rule:

Construction	Height rule	Ruler / fineness
Cauchy	$f(x_{i-1})$, left endpoint	Δx_i
Riemann	$f(\xi_i)$, arbitrary tag	Δx_i
Darboux	$\inf_I f,\ \sup_I f$ on each slab	Δx_i
Riemann–Stieltjes	$f(\xi_i)$, tagged height	$\Delta \alpha_i$
Henstock–Kurzweil	$f(\xi_i),\ \xi_i \in I_i$	$\Delta x_i,\ \delta(\xi_i)$-fine, tag tethered
McShane	$f(t_i),\ t_i$ may float	$\Delta x_i,\ \delta(t_i)$-fine, tag untethered

Henstock–Kurzweil does not change the sampled height or the ordinary ruler; it changes the admissible chopping by replacing a global mesh bound with a local gauge condition. McShane keeps the same gauge idea but frees the tag from the slab, which pulls the theory back toward Lebesgue's absolute notion of size.

Definition (Partition)

Definition 12.1.1 (Partition of a Compact Interval)**.** A *partition* of $[a, b]$ is a finite increasing list

$$P = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

Equivalently,

$$\begin{aligned}\text{Partition}(P, a, b) \iff {} & \exists n \in \mathbb{N}\, \exists x_0, \ldots, x_n \in \mathbb{R} \\ & \big(P = \{x_0, \ldots, x_n\} \wedge x_0 = a \wedge x_n = b\big) \\ & \wedge \forall i\, \big(1 \le i \le n \Rightarrow x_{i-1} < x_i\big).\end{aligned}$$

Definition (Subinterval Width)

Definition 12.1.2 (Subinterval Width)**.** The i-th slab has width

$$\Delta x_i = x_i - x_{i-1}.$$

Definition (Mesh)

Definition 12.1.3 (Mesh of a Partition)**.** The *mesh* or *norm* of P is the width of its fattest slab:

$$\|P\| = \max_{1 \le i \le n} \Delta x_i.$$

Remark (Geometric reading). The mesh is the resolution dial. The phrase "as the partition is refined" will usually mean that $\|P\| \to 0$: no slab is allowed to remain fat.

Definition (Tagged Partition)

Definition 12.1.4 (Tagged Partition). A *tagged partition* is a partition $P = \{x_0, \dots, x_n\}$ together with points $\xi_i \in [x_{i-1}, x_i]$. The tag is the point where the height $f(\xi_i)$ is sampled.

Definition (Refinement)

Definition 12.1.5 (Refinement of a Partition). A partition P' *refines* P when $P \subseteq P'$. Refinement adds cut-points; it does not move or delete existing cut-points.

Lemma (Common Refinement)

Lemma 12.1.6 (Common Refinement of Two Partitions). *Any two partitions P and P' of $[a, b]$ have a common refinement: order the finite set $P \cup P'$ increasingly.*

Remark (Why the lemma is load-bearing). This small fact is the reason independently chosen partitions can be compared. Whenever a proof says "refine them together," it is using this lemma. It is the structural engine behind Cauchy's existence proof and Darboux's squeeze.

12.2 The Cauchy Integral

Geometric Intuition

Cauchy makes the simplest possible height choice: always sample the left edge of each slab. The result is a staircase of left-anchored rectangles. As the slabs thin out, the question is whether those staircase areas settle.

Remark (Engineering reading). Cauchy's construction is the seed. It answers the geometric problem for the case where the curve is tame enough that left-edge staircases forget their initial sampling bias as the mesh shrinks. The construction is deliberately primitive: it fixes a height rule first and asks for convergence second.

Definition (Cauchy Sum)

Definition 12.2.1 (Left-Endpoint Cauchy Sum). For $f : [a, b] \to \mathbb{R}$ and $P = \{a = x_0 < \cdots < x_n = b\}$, define

$$C(f, P) = \sum_{i=1}^{n} f(x_{i-1})\, \Delta x_i.$$

Definition (Cauchy Integral)

Definition 12.2.2 (Cauchy Integral). The function f is *Cauchy-integrable* with value L if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall P\ \big(\|P\| < \delta \Rightarrow |C(f, P) - L| < \varepsilon\big).$$

The value L is denoted $\int_a^b f(x)\, dx$.

Proposition (Integral of a Constant)

Proposition 12.2.3 (Cauchy Integral of a Constant). *If $f(x) = c$ on $[a, b]$, then $\int_a^b c\,dx = c(b - a)$.*

Theorem (Linearity)

Theorem 12.2.4 (Linearity of the Cauchy Integral). *If f and g are Cauchy-integrable on $[a, b]$, then*

$$\int_a^b (\alpha f + \beta g)\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx.$$

Theorem (Monotonicity)

Theorem 12.2.5 (Monotonicity of the Cauchy Integral). *If f and g are Cauchy-integrable on $[a, b]$ and $f(x) \le g(x)$ for every $x \in [a, b]$, then*

$$\int_a^b f\,dx \le \int_a^b g\,dx.$$

Corollary (Bounds)

Corollary 12.2.6 (Bounds for the Cauchy Integral). *If f is continuous on $[a, b]$, $m = \min_{[a,b]} f$, and $M = \max_{[a,b]} f$, then*

$$m(b - a) \le \int_a^b f\,dx \le M(b - a).$$

Theorem (Triangle Inequality)

Theorem 12.2.7 (Triangle Inequality for the Cauchy Integral). *If f and $|f|$ are Cauchy-integrable on $[a, b]$, then*

$$\left|\int_a^b f\,dx\right| \le \int_a^b |f|\,dx.$$

Theorem (Interval Additivity)

Theorem 12.2.8 (Interval Additivity for the Cauchy Integral). *If $c \in [a, b]$ and f is Cauchy-integrable on the relevant intervals, then*

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx.$$

Definition (Oscillation on an Interval)

Definition 12.2.9 (Oscillation on an Interval)**.** For bounded f on an interval I, define

$$\omega_f(I) = \sup_I f - \inf_I f.$$

Oscillation is the error controller: if $\omega_f(I)$ is small, every sample height on I is close to every other sample height.

Remark (Oscillation as the visible error)*.* Figure 1.1 shows the same quantity geometrically: the Darboux gap $U(f,P) - L(f,P)$ is the sum of local oscillations $\omega_f(I_i)$ multiplied by the slab widths Δx_i. The trapezoid and Simpson panels are numerical side branches; the load-bearing panel here is the oscillation identity.

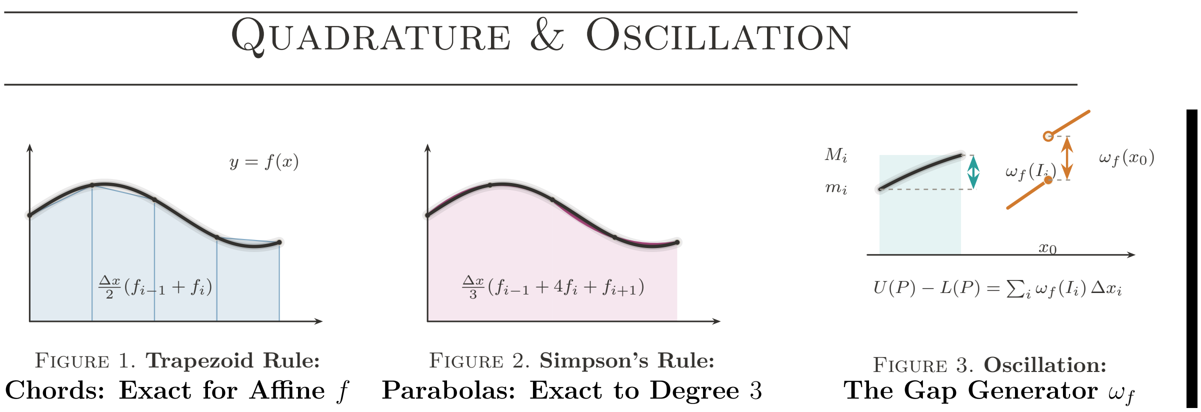


Figure 12.1: Quadrature and oscillation

Theorem (Continuous Implies Cauchy-Integrable)

Theorem 12.2.10 (Continuous Functions Are Cauchy-Integrable)**.** *Every continuous function $f : [a,b] \to \mathbb{R}$ is Cauchy-integrable.*

Remark (Proof skeleton)*.* Continuity on the compact interval gives uniform continuity. Hence short slabs have small oscillation. Given two fine partitions, pass to their common refinement and compare sums slab by slab; each re-sampling error is bounded by $\omega_f(I)$ times the slab width. The sums form a Cauchy net, and completeness of $\mathbb{R}$ supplies the limit.

Remark (What continuity buys)*.* Uniform continuity is the load-bearing ingredient. It converts pointwise control into one global scale: if the slabs are all shorter than that scale, then every slab has small oscillation. Without this global scale, Cauchy's left-endpoint definition has no mechanism for proving that unrelated partitions settle to the same number.

Remark (A first discontinuous edge case)*.* A single jump should still have an area. If

$$f(x) = \begin{cases} 0, & x < c, \\ 1, & x \geq c, \end{cases}$$

on $[a, b]$, the only dangerous slab is the one crossing c. Its total possible error is bounded by its width, and that width tends to zero as the mesh tends to zero. This example is not hard; the point is that Cauchy's definition gives no internal test explaining why it is harmless. The proof above proves continuity is sufficient, not that controlled discontinuities are allowed.

Theorem (Tag Independence)

Theorem 12.2.11 (Tag Independence for Continuous Functions)**.** *If f is continuous on $[a, b]$, then every choice of tags gives the same limit as the left-endpoint sums as $\|P\| \to 0$.*

Remark (Bug report). Cauchy has two defects. First, the left endpoint is privileged by convention; tag independence is a theorem only after continuity has already been assumed. Second, the existence proof gives no usable criterion for discontinuous functions. The patch request is to stop privileging the left endpoint and make tag independence part of the definition.

Remark (Patch request). The next definition should not say "sample on the left." It should say: sample anywhere, even adversarially, and require all fine enough choices to agree. Riemann is exactly Cauchy's tag-independence theorem promoted into the definition of integrability.

12.3 The Riemann Integral

Geometric Intuition

Riemann promotes Cauchy's tag-independence theorem into a definition. A function is integrable when every sufficiently fine staircase, even with adversarial tags, is forced near the same number.

Remark (Engineering reading). The sampling convention is no longer chosen by the author of the definition. It is chosen by a universal quantifier. A Riemann-integrable function is one whose accumulated area survives every possible choice of sample heights once the slabs are fine enough.

Definition (Riemann Sum)

Definition 12.3.1 (Riemann Sum)**.** For a tagged partition (P, ξ), define

$$S(f, P, \xi) = \sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

Definition (Riemann Integral)

Definition 12.3.2 (Riemann Integral)**.** A function $f : [a, b] \to \mathbb{R}$ is *Riemann-integrable* with value L if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall (P, \xi)\ \big(\|P\| < \delta \Rightarrow |S(f, P, \xi) - L| < \varepsilon\big).$$

Remark (The one symbol that changed everything). Cauchy quantifies over choppings. Riemann quantifies over choppings and every possible choice of heights:

$$\forall P \qquad \rightsquigarrow \qquad \forall (P, \xi).$$

That extra quantifier removes the arbitrary left endpoint and turns tag independence into the price of admission.

Remark (Payoff). The definition is no longer tied to continuity. It can admit bounded functions with discontinuities, because the question is not whether f is continuous but whether all fine tagged sums are forced into one narrow numerical corridor. The Cauchy criterion below makes that statement intrinsic: the value need not be known in advance.

Remark (The ruler function example). Thomae's function, also called the ruler function, is

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms,} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

It is discontinuous at every rational and continuous at every irrational, so its discontinuities are dense. Nevertheless its Riemann integral on any compact interval is 0. The reason is that the rational points where T can be large form only a finite set above any fixed height threshold. Fine partitions can isolate those finitely many tall spikes, while all remaining slabs have uniformly small height. This is exactly the kind of function Cauchy's continuity proof could not diagnose, but Riemann's tag-independent definition can admit.

Theorem (Continuous Implies Riemann-Integrable)

Theorem 12.3.3 (Continuous Functions Are Riemann-Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable.*

Theorem (Linearity)

Theorem 12.3.4 (Linearity of the Riemann Integral)**.** *If f and g are Riemann-integrable on $[a, b]$, then*

$$\int_a^b (\alpha f + \beta g)\, dx = \alpha \int_a^b f\, dx + \beta \int_a^b g\, dx.$$

Theorem (Monotonicity)

Theorem 12.3.5 (Monotonicity of the Riemann Integral)**.** *If f and g are Riemann-integrable on $[a, b]$ and $f(x) \le g(x)$ for every $x \in [a, b]$, then*

$$\int_a^b f\, dx \le \int_a^b g\, dx.$$

Theorem (Triangle Inequality)

Theorem 12.3.6 (Triangle Inequality for the Riemann Integral). *If f is Riemann-integrable on $[a, b]$, then $|f|$ is Riemann-integrable and*

$$\left|\int_a^b f\,dx\right| \leq \int_a^b |f|\,dx.$$

Theorem (Interval Additivity)

Theorem 12.3.7 (Interval Additivity for the Riemann Integral). *If $c \in [a, b]$, then f is Riemann-integrable on $[a, b]$ if and only if it is Riemann-integrable on $[a, c]$ and $[c, b]$, and in that case*

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx.$$

Theorem (Cauchy Criterion)

Theorem 12.3.8 (Cauchy Criterion for Riemann Integrability). *A bounded function is Riemann-integrable if and only if for every $\varepsilon > 0$ there is $\delta > 0$ such that any two tagged partitions (P, ξ) and (Q, η) with $\|P\|, \|Q\| < \delta$ have*

$$|S(f, P, \xi) - S(f, Q, \eta)| < \varepsilon.$$

Remark (Bug report). Riemann fixed the arbitrary tag, but the definition is operationally hostile. To verify integrability directly one must control all tagged partitions at once. The patch request is to remove tags from the verification problem: replace every possible sample height by the slab infimum and supremum, then squeeze.

Remark (Why this is hard to use). Riemann is logically clean but not mechanically friendly. Refining a partition does not make an arbitrary Riemann sum move monotonically, because every refinement lets the tags move again. There is no single upper or lower quantity improving under refinement. Darboux supplies exactly that missing monotone control.

12.4 The Darboux Integral

Geometric Intuition

Darboux stops chasing tags. On each slab, the adversary can do no better than the supremum and no worse than the infimum. So build the upper staircase from $\sup f$, the lower staircase from $\inf f$, and ask whether the gap can be squeezed to zero.

Remark (Engineering reading). Darboux is the usable face of Riemann integration. It does not change the integral; it changes the test. Instead of quantifying over every possible tag, it brackets all tags at once.

Definition (Darboux Sums)

Definition 12.4.1 (Lower and Upper Darboux Sums)**.** For bounded f on $[a, b]$, set

$$m_i = \inf_{[x_{i-1}, x_i]} f, \qquad M_i = \sup_{[x_{i-1}, x_i]} f,$$

and define

$$L(f, P) = \sum_i m_i \Delta x_i, \qquad U(f, P) = \sum_i M_i \Delta x_i.$$

Lemma (Monotonicity Under Refinement)

Lemma 12.4.2 (Darboux Sums Under Refinement)**.** *If P' refines P, then*

$$L(f, P) \le L(f, P') \le U(f, P') \le U(f, P).$$

Remark (Geometric reading). Thinner slabs give lower rectangles room to rise and upper rectangles room to fall. This monotonicity is exactly what Riemann sums lack when tags move adversarially.

Remark (One-point refinement proof idea). It is enough to add one cut-point c inside one slab and iterate. The infimum over the whole slab is less than or equal to the infimum over each piece, so the lower contribution rises. The supremum over the whole slab is greater than or equal to the supremum over each piece, so the upper contribution falls.

Definition (Darboux Integrability)

Definition 12.4.3 (Darboux Integrability)**.** The lower and upper integrals are

$$\underline{I} = \sup_P L(f, P), \qquad \overline{I} = \inf_P U(f, P).$$

The function f is *Darboux-integrable* if $\underline{I} = \overline{I}$.

Theorem (Darboux Criterion)

Theorem 12.4.4 (Darboux Criterion)**.** *A bounded function f is Darboux-integrable if and only if for every $\varepsilon > 0$ there exists a partition P such that*

$$U(f, P) - L(f, P) < \varepsilon.$$

Remark (The squeeze in one plate). Figure 1.2 places the Cauchy/Riemann tag choices, the lower–upper Darboux bracket, the refinement squeeze, and the gauge panel on the same page. The Darboux row is the chapter's hinge: replacing moving tags by inf and sup turns integrability into the checkable $U - L$ squeeze.

Probes of the Integral

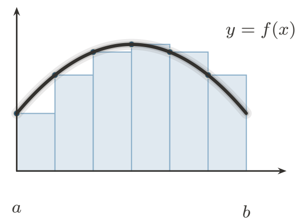


Figure 1. **Left Riemann Sum: Tag at the Left Endpoint**

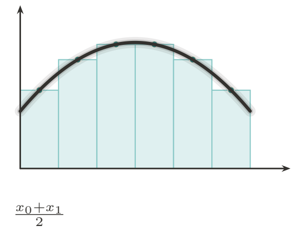


Figure 2. **Midpoint Rule: The Centered Tag**

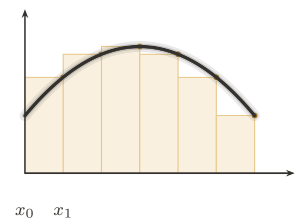


Figure 3. **Right Riemann Sum: Tag at the Right Endpoint**

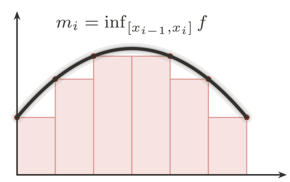


Figure 4. **Lower Darboux Sum: The Infimum Probe**

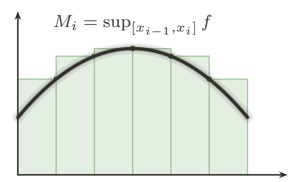


Figure 5. **Upper Darboux Sum: The Supremum Probe**

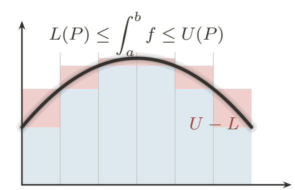


Figure 6. **The Squeeze: Integrability as a Closing Gap**

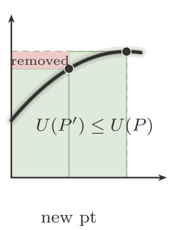


Figure 7. **Partition Refinement: Adding Points Tightens the Sums**

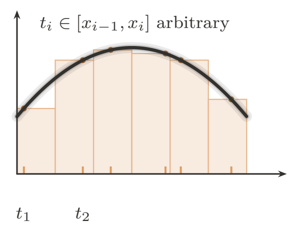


Figure 8. **Tagged Partition: The General Riemann Sum**

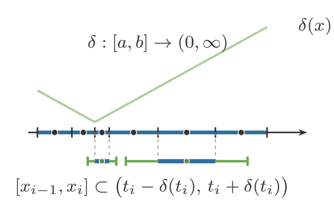


Figure 9. **The Gauge: δ-Fine Partitions (Henstock–Kurzweil)**

Figure 12.2: Probes of the integral

Theorem (Riemann Equals Darboux)

Theorem 12.4.5 (Equivalence of Riemann and Darboux Integrability)**.** *For bounded functions on $[a, b]$, Riemann integrability and Darboux integrability are equivalent, and the integral values agree.*

Theorem (Continuous Functions Are Integrable)

Theorem 12.4.6 (Continuous Functions Are Darboux-Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Theorem (Monotone Functions Are Integrable)

Theorem 12.4.7 (Monotone Functions Are Darboux-Integrable)**.** *Every monotone function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Remark (Why this is a harvest theorem). Monotone functions may have infinitely many discontinuities. Darboux still handles them because their total vertical movement is finite. On each slab the oscillation is just the endpoint jump across that slab, and these jumps telescope over the partition. Equal-width partitions therefore force $U(f, P) - L(f, P)$ to zero.

Theorem (Finitely Many Discontinuities)

Theorem 12.4.8 (Bounded Functions with Finitely Many Discontinuities Are Darboux-Integrable)**.** *If $f : [a, b] \to \mathbb{R}$ is bounded and has only finitely many discontinuities, then f is Darboux-integrable.*

Remark (Why finite bad sets are harmless). The Darboux gap is

$$U(f, P) - L(f, P) = \sum_i \omega_f([x_{i-1}, x_i])\Delta x_i.$$

A finite set of discontinuities can be trapped in intervals of arbitrarily small total width. On the complement, f is continuous on compact pieces, so its oscillation is uniformly small on fine slabs. Thus the bad slabs are short and the good slabs are flat.

Theorem (Algebra of Darboux-Integrable Functions)

Theorem 12.4.9 (Sums and Scalar Multiples of Darboux-Integrable Functions)**.** *If f and g are Darboux-integrable on $[a, b]$, then $\alpha f + \beta g$ is Darboux-integrable for all $\alpha, \beta \in \mathbb{R}$.*

Theorem (Products)

Theorem 12.4.10 (Products of Darboux-Integrable Functions)**.** *If f and g are Darboux-integrable on $[a, b]$, then fg is Darboux-integrable on $[a, b]$.*

Theorem (Absolute Values)

Theorem 12.4.11 (Absolute Values of Darboux-Integrable Functions)**.** *If f is Darboux-integrable on $[a, b]$, then $|f|$ is Darboux-integrable on $[a, b]$.*

Theorem (Continuous Composition)

Theorem 12.4.12 (Continuous Images of Darboux-Integrable Functions)**.** *If f is Darboux-integrable on $[a, b]$, φ is continuous on an interval containing $f([a, b])$, and f is bounded, then $\varphi \circ f$ is Darboux-integrable on $[a, b]$.*

Remark (Why the harvest is cheap). The Darboux criterion reduces closure questions to controlling the upper-lower gap. Boundedness controls coefficients and products, while continuity of φ converts small oscillation of f into small oscillation of $\varphi \circ f$. This is the point of switching from tags to a squeeze: theorems become estimates on $U(f, P) - L(f, P)$.

Remark (Bug report). Darboux is a better tool for the same class. It does not enlarge Riemann integrability. The remaining defects now fork. One can change the ruler Δx to $\Delta\alpha$, which leads to Riemann–Stieltjes. Or one can diagnose the size of the discontinuity set, which leads to Lebesgue and also to the gauge-integral cure.

Remark (Residual failures). The Dirichlet function $\mathbf{1}_{\mathbb{Q}}$ on $[0, 1]$ has infimum 0 and supremum 1 on every slab, since rationals and irrationals are dense. Thus $L(f, P) = 0$ and $U(f, P) = 1$ for every partition, so the Darboux gap never closes. This is not a Darboux-specific failure; by equivalence, it is the failure of the whole Riemann family. The failure modes in Figure 1.3 separate the common pathologies: Dirichlet breaks the continuity-almost-everywhere condition, Thomae survives it, unbounded examples violate the boundedness hypothesis, and pointwise limits show why Riemann integration has weak limit behavior.

Where Riemann Fails

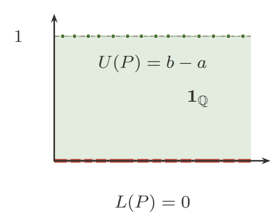


Figure 1. **Dirichlet $\mathbf{1}_{\mathbb{Q}}$: The Gap That Never Closes**

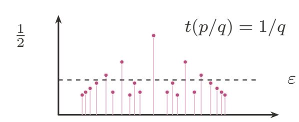


Figure 2. **Thomae's Function: Dense Jumps, Yet Integrable**

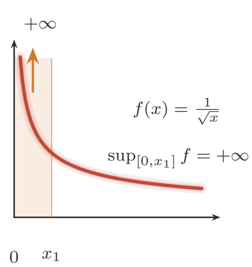


Figure 3. **Unbounded Integrand: Boundedness Is Not Optional**

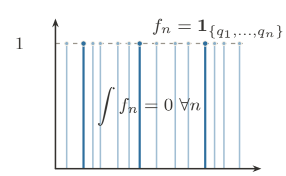


Figure 4. **Pointwise Limit:** $\lim \int \neq \int \lim$ **(Lebesgue's Cue)**

Figure 12.3: Failure modes of the Riemann family

Remark (What Darboux makes visible). For Thomae's function, Darboux sees the opposite picture. On every slab the infimum is 0, because every slab contains irrationals. The supremum can be large only near rationals with small denominator; above any fixed height threshold, there are only finitely many such rationals. A partition can isolate those finitely many tall rational spikes inside intervals of tiny total width, and the remaining upper rectangles are short. Thus the upper sum can be forced below any prescribed ε, while every lower sum is 0.

This is the practical gain: Riemann gives the correct definition, but Darboux gives the hand tool. Prove $U - L$ can be squeezed, and the Riemann value follows from the equivalence theorem.

Remark (The fork). Darboux exposes two separate defects:

Residual defect	Successor	Change
only ordinary length	Riemann–Stieltjes	$\Delta x_i \mapsto \Delta\alpha_i$
domain slabs see every bad set	Lebesgue / HK	measure bad sets or adapt fineness

Stieltjes is therefore a sideways generalization, not the final repair of Riemann's domain-partition limitation.

12.5 The Riemann–Stieltjes Integral

Geometric Intuition

Riemann and Darboux weigh each slab by ordinary length, Δx_i. Riemann–Stieltjes changes the ruler. A slab is weighed by how much an integrator α changes across it:

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}).$$

If $\alpha(x) = x$, ordinary Riemann integration returns. If α jumps, the integral sees a point mass.

Remark (Engineering reading). Stieltjes does not repair the discontinuity-set problem. It answers a different complaint: ordinary integration has only one ruler. By replacing Δx_i with $\Delta\alpha_i$, the integral can count regions with nonuniform weight and can encode discrete masses as jumps of α.

Definition (Total Variation)

Definition 12.5.1 (Total Variation)**.** The total variation of α on $[a, b]$ is

$$V_a^b(\alpha) = \sup_P \sum_i |\alpha(x_i) - \alpha(x_{i-1})|.$$

Definition (Bounded Variation)

Definition 12.5.2 (Bounded Variation)**.** The function α has *bounded variation* on $[a, b]$ if

$$V_a^b(\alpha) < \infty.$$

Geometrically, this says the ruler has finite total up-and-down travel.

Proposition (Monotone Implies BV)

Proposition 12.5.3 (Monotone Functions Have Bounded Variation)**.** *If α is monotone on $[a, b]$, then α has bounded variation.*

Remark (Why bounded variation is the honest ruler condition)*.* If α is increasing, the increments telescope:

$$\sum_i \big(\alpha(x_i) - \alpha(x_{i-1})\big) = \alpha(b) - \alpha(a).$$

Bounded variation is the signed analogue of that finite ruler budget. It is what prevents the weighted sums from accumulating uncontrolled oscillatory ruler error.

Definition (Riemann–Stieltjes Integral)

Definition 12.5.4 (Riemann–Stieltjes Integral)**.** The Riemann–Stieltjes sum is

$$\sum_i f(\xi_i)\big(\alpha(x_i) - \alpha(x_{i-1})\big).$$

When these sums have a tag-independent limit as $\|P\| \to 0$, the limit is denoted $\int_a^b f\,d\alpha$.

Theorem (Continuous Integrand, BV Integrator)

Theorem 12.5.5 (Continuous Integrand and BV Integrator)**.** *If f is continuous on $[a, b]$ and α has bounded variation, then $\int_a^b f\,d\alpha$ exists.*

Theorem (Bilinearity)

Theorem 12.5.6 (Bilinearity of the Riemann–Stieltjes Integral)**.** *Whenever the displayed integrals exist,*

$$\int_a^b (\lambda f + \mu g)\,d\alpha = \lambda \int_a^b f\,d\alpha + \mu \int_a^b g\,d\alpha$$

and

$$\int_a^b f\,d(\lambda\alpha + \mu\beta) = \lambda \int_a^b f\,d\alpha + \mu \int_a^b f\,d\beta.$$

Theorem (Interval Additivity)

Theorem 12.5.7 (Interval Additivity for the Riemann–Stieltjes Integral)**.** *If $c \in [a, b]$ and the relevant Riemann–Stieltjes integrals exist, then*

$$\int_a^b f\,d\alpha = \int_a^c f\,d\alpha + \int_c^b f\,d\alpha.$$

Theorem (Integration by Parts)

Theorem 12.5.8 (Riemann–Stieltjes Integration by Parts)**.** *If f and α are such that one of $\int_a^b f\,d\alpha$ and $\int_a^b \alpha\,df$ exists, then the other exists under the same Riemann–Stieltjes convention, and*

$$\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a).$$

Theorem (Smooth Integrator Reduction)

Theorem 12.5.9 (Reduction to Riemann Integration for a C^1 Integrator)**.** *If $\alpha \in C^1([a,b])$ and f is Riemann-integrable on $[a,b]$, then*

$$\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx.$$

Theorem (Step Integrator)

Theorem 12.5.10 (Step Integrators Give Finite Sums)**.** *If α is constant except for finitely many jumps at points c_k and f is continuous at each c_k, then*

$$\int_a^b f\,d\alpha = \sum_k f(c_k)\,\Delta\alpha(c_k).$$

Remark (Proof idea). The Cauchy common-refinement argument runs again. Uniform continuity makes $\omega_f(I)$ small on short slabs, and bounded variation controls $\sum_i |\Delta\alpha_i|$. The error has the form

$$\sum_i \omega_f(I_i)\,|\Delta\alpha_i|,$$

so small oscillation multiplied by finite total variation is small.

Remark (Sums as integrals). If α is a step function with jumps at points c_k, then the Stieltjes integral collapses to a weighted sum of the values $f(c_k)$ times the jumps. This is the conceptual doorway to probability: a cumulative distribution function is an integrator, and expectation is an integral against that integrator.

Remark (Step integrator calculation). Suppose α is constant except for finitely many jumps $\Delta\alpha(c_k) = \alpha(c_k+) - \alpha(c_k-)$. Any sufficiently fine partition has one tiny slab around each c_k carrying that jump, while all other slabs have $\Delta\alpha_i = 0$. If f is continuous at each c_k, then the tagged height on the k-th jump slab is forced toward $f(c_k)$, so

$$\int_a^b f\,d\alpha = \sum_k f(c_k)\,\Delta\alpha(c_k).$$

This is why a finite weighted sum is not a different species from an integral: it is a Stieltjes integral against a discrete ruler.

Remark (Shared-jump failure). The new pathology is real. On $[0,1]$, let $c = 1/2$, let

$$\alpha(x) = \mathbf{1}_{[c,1]}(x), \qquad f(x) = \mathbf{1}_{[c,1]}(x).$$

The only nonzero $\Delta\alpha_i$ occurs on the slab containing c. If the tag for that slab is chosen left of c, its contribution is 0; if the tag is chosen at or right of c, its contribution is 1. This discrepancy does not shrink with the slab width, because the ruler has a whole unit of mass concentrated at the jump. Tag independence fails, so the Riemann–Stieltjes integral does not exist in this convention.

Remark (Bug report). Stieltjes is a sideways generalization, not the Lebesgue fix. Shared discontinuities of f and α can destroy tag independence, and the standard existence theorem leans on the finite ruler budget $V_a^b(\alpha) < \infty$. Brownian paths have infinite variation almost surely, so stochastic integration cannot be ordinary pathwise Riemann–Stieltjes integration.

Remark (Patch request). The next stochastic patch is not "make Stieltjes a little better." Brownian motion forces a different limiting mode: an L^2 or probabilistic limit rather than a pathwise Riemann–Stieltjes limit. That is the structural reason Ito integration is not just Stieltjes integration with a rougher integrator.

12.6 The Henstock–Kurzweil Integral

Where This Sits

Darboux exposed the Riemann disease before measure theory names it precisely: one uniform fineness parameter must serve the whole interval. There are two ways forward. Lebesgue will change what is measured. The gauge integrals keep the tagged-sum picture and change exactly one symbol: the constant fineness $\delta > 0$ becomes a positive function $\delta(x)$.

Remark (Sibling cure). Henstock–Kurzweil belongs here, before McShane and the final measure-zero diagnosis, because it is still visibly a Riemann-style tagged-sum integral. It patches the uniform-mesh defect without abandoning slabs, tags, or sums. McShane will modify the tag rule one more time, producing a gauge-sum version of Lebesgue integration.

Geometric Intuition

A gauge is adaptive mesh refinement written as analysis. It is tiny where the function needs tight local control and generous where the function is calm. A tagged slab is allowed only if it fits inside the local radius prescribed by its own tag.

Remark (The one-symbol upgrade). Riemann uses a positive number:

$$\exists \delta > 0.$$

Henstock–Kurzweil uses a positive function:

$$\exists \delta : [a,b] \to (0,\infty).$$

The mesh condition $\|P\| < \delta$ becomes the local condition that each slab is δ-fine at its tag.

Definition (Gauge)

Definition 12.6.1 (Gauge on an Interval)**.** A *gauge* on $[a, b]$ is a function

$$\delta : [a, b] \to (0, \infty).$$

Definition (Delta-Fine Tagged Partition)

Definition 12.6.2 (δ-Fine Tagged Partition)**.** A tagged partition (P, ξ) is *δ-fine* if

$$[x_{i-1}, x_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Definition (Henstock–Kurzweil Integral)

Definition 12.6.3 (Henstock–Kurzweil Integral)**.** A function $f : [a, b] \to \mathbb{R}$ is *Henstock–Kurzweil integrable* with value A if

$$\forall \varepsilon > 0 \; \exists \delta : [a, b] \to (0, \infty) \; \forall (P, \xi) \; \big((P, \xi) \text{ is } \delta\text{-fine} \Rightarrow |S(f, P, \xi) - A| < \varepsilon\big).$$

The value is denoted (HK) $\int_a^b f$.

Lemma (Cousin)

Lemma 12.6.4 (Cousin's Lemma)**.** *For every gauge δ on $[a, b]$, there exists a δ-fine tagged partition of $[a, b]$.*

Remark (Non-vacuity). Cousin's Lemma prevents the definition from being empty. No matter how wildly the gauge varies, compactness of $[a, b]$ guarantees that at least one tagged partition respects all the local step-size demands.

Theorem (Riemann Sits Inside Henstock–Kurzweil)

Theorem 12.6.5 (Riemann Integrable Implies Henstock–Kurzweil Integrable)**.** *If f is Riemann-integrable on $[a, b]$, then f is Henstock–Kurzweil integrable on $[a, b]$, and the two integrals have the same value.*

Remark (Constant gauges). Riemann is the special case where the gauge is constant. If a Riemann proof hands over one mesh tolerance δ_0, the gauge proof uses $\delta(x) = \delta_0$ for every x.

Lemma (Straddle)

Lemma 12.6.6 (Straddle Lemma)**.** *If F is differentiable at ξ with $F'(\xi) = f(\xi)$, then for every $\varepsilon > 0$ there is $\delta(\xi) > 0$ such that whenever $u \le \xi \le v$, $u, v \in (\xi - \delta(\xi), \xi + \delta(\xi))$,*

$$|F(v) - F(u) - f(\xi)(v - u)| \le \varepsilon(v - u).$$

Theorem (FTC for the Gauge Integral)

Theorem 12.6.7 (Fundamental Theorem for the Henstock–Kurzweil Integral)**.** *If $F : [a,b] \to \mathbb{R}$ is differentiable at every point and $F' = f$, then f is Henstock–Kurzweil integrable and*

$$(\mathrm{HK}) \int_a^b f = F(b) - F(a).$$

Theorem (Continuous Functions Are Henstock–Kurzweil Integrable)

Theorem 12.6.8 (Continuous Functions Are Henstock–Kurzweil Integrable)**.** *Every continuous function $f : [a,b] \to \mathbb{R}$ is Henstock–Kurzweil integrable.*

Remark (What it buys)*.* On a compact interval,

$$\mathcal{R}[a,b] \subsetneq \mathcal{M}[a,b] = \mathcal{L}[a,b] \subsetneq \mathcal{HK}[a,b].$$

Figure 1.4 records this inclusion lattice and the standard witness at each frontier. The first strict inclusion is witnessed by functions like $\mathbf{1}_{\mathbb{Q}}$, which are Lebesgue-integrable but not Riemann integrable. The second is witnessed by derivatives such as that of $F(x) = x^2 \sin(1/x^2)$ with $F(0) = 0$: the derivative is HK-integrable by the gauge FTC even when its absolute value is not Lebesgue-integrable.

This inclusion statement anticipates the next volume's Lebesgue theory. The present chapter has not built Lebesgue integration yet; it has only isolated the Riemann-side machinery, the gauge machinery, and the reason a measure-theoretic successor is needed.

The Integration Lattice

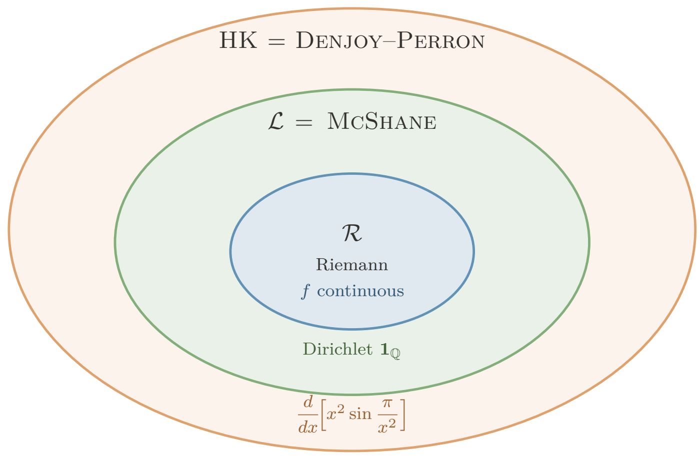


Figure. The Integration Lattice: each frontier is crossed by exactly one witness.

$\mathcal{R} \subsetneq \mathcal{L} = \text{McShane} \subsetneq \mathcal{HK}$, with the strict gaps witnessed by:

- $\mathcal{L} \setminus \mathcal{R}$: $\mathbf{1}_{\mathbb{Q}}$ — Lebesgue (and McShane) integrable, $\int = 0$; not Riemann.
- $\mathcal{HK} \setminus \mathcal{L}$: F' for $F(x) = x^2 \sin(\pi/x^2)$, $F(0) = 0$ — HK-integrable ($\int_0^1 F' = F(1) - F(0) = 0$) but $\int_0^1 |F'| = \infty$, so not Lebesgue.

Engine: $\mathcal{L}$ = McShane uses *free* tags (absolute integration); $\mathcal{HK}$ uses *tethered* tags, which preserve cancellation — the source of the outermost ring, and the reason HK integrates *every* everywhere-derivative.

Figure 12.4: The inclusion lattice for Riemann, Lebesgue–McShane, and Henstock–Kurzweil integration

Remark (Killer example). Let $F(x) = x^2 \sin(1/x^2)$ for $x \neq 0$, and set $F(0) = 0$. Then F is differentiable at 0, with $F'(0) = 0$, while for $x \neq 0$,

$$F'(x) = 2x \sin(1/x^2) - \frac{2}{x} \cos(1/x^2).$$

The oscillatory term has nonintegrable absolute size near 0, so this derivative is the standard witness that HK reaches beyond Lebesgue while the FTC remains exact.

Remark (HK's own bug report). Henstock–Kurzweil is not simply "the best integral." Lebesgue has stronger limit theorems, a complete L^1 normed-space home, and a clean abstract measure-space setting. HK is the jewel for the one-dimensional FTC and conditional accumulation; Lebesgue and Ito remain the engines for probability and stochastic integration.

Remark (Computational bridge). A gauge $\delta(x)$ is a spatially varying step-size field. Lipschitz control is the case where a constant gauge is enough; singular or non-Lipschitz regions demand a genuinely varying gauge. Cousin's Lemma is the compactness guarantee that a valid adaptive partition exists.

Remark (One-page summary).

Integral	Patch	Remaining defect
Cauchy	left rectangles settle for continuous f	privileged tag
Riemann	all tags must agree	hard to verify
Darboux	sup/inf squeeze	same class as Riemann
Stieltjes	change the ruler	needs BV; shared jumps
Henstock–Kurzweil	local gauge fineness	weaker limit machinery
McShane	free gauge tags	same class as Lebesgue
Lebesgue	measure level sets	abstract machinery required

The thread is the right-hand column: every integral is a direct response to a specific defect.

12.7 The McShane Integral

Where This Sits

McShane is the gauge integral that looks almost indistinguishable from Henstock–Kurzweil until one notices where the tags are allowed to live. HK keeps each tag inside its own slab. McShane lets a tag control a slab from outside it, as long as the slab lies inside the tag's gauge ball.

Remark (The tiny syntactic change). Henstock–Kurzweil says:

$$\xi_i \in [x_{i-1}, x_i] \quad \text{and} \quad [x_{i-1}, x_i] \subset (\xi_i - \delta(\xi_i), \xi_i + \delta(\xi_i)).$$

McShane drops only the first requirement. The interval still has to be locally small around its tag, but the tag need not lie in the interval it controls. That small freedom is decisive: it prevents the conditional cancellation that lets HK integrate every derivative.

Remark (HK versus McShane at a glance).

Integral	Gauge	Tag position	Class on $[a,b]$
Henstock–Kurzweil	$\delta(x)$	$\xi_i \in I_i$	$\mathcal{HK}$
McShane	$\delta(x)$	$\xi_i \in [a,b]$ and $I_i \subset B(\xi_i, \delta(\xi_i))$	$\mathcal{L}$

Both use a variable gauge. The difference is not the step-size field; it is whether the tag must sit inside the interval whose height it supplies. Figure 1.5 shows this tethered-versus-free distinction directly.

Two Ways to Tag

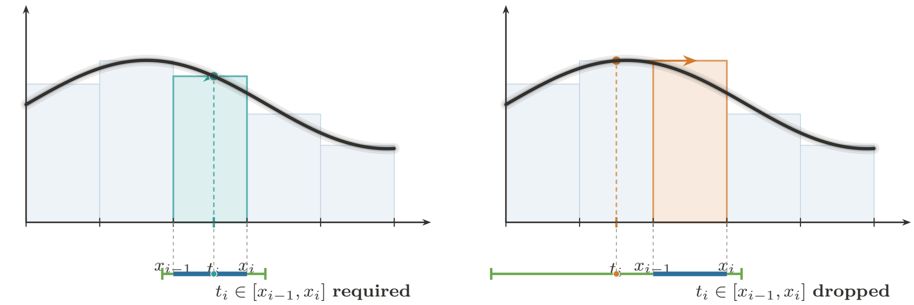


FIGURE 1. **Henstock–Kurzweil: Tag Tethered to Its Cell**

FIGURE 2. **McShane: The Tag Roams Free**

Same integrand, same gauge condition $I_i \subseteq (t_i - \delta(t_i),\, t_i + \delta(t_i))$. The *only* difference is whether the tag must sit in its own cell.

HK keeps the tether $\Rightarrow$ conditional convergence survives $\Rightarrow$ $\mathcal{HK}$ = Denjoy–Perron (nonabsolute). MCSHANE cuts it $\Rightarrow$ only absolute integrals remain $\Rightarrow$ McShane = $\mathcal{L}^1$ (Lebesgue).

Figure 12.5: Henstock–Kurzweil versus McShane fineness

Geometric Intuition

In HK, a tag is a sample point inside the slab. In McShane, a tag is more like a local address that authorizes nearby slabs. The gauge still says how far the tag's influence reaches, but the rectangle height may be chosen from a point near the slab rather than inside it. This makes the integral less forgiving about sign-changing singular behavior and much closer to Lebesgue's absolute notion of size.

Definition (McShane-Fine Tagged Partition)

Definition 12.7.1 (McShane-Fine Tagged Partition)**.** Let $\delta : [a,b] \to (0,\infty)$ be a gauge. A finite tagged partition $\{([u_i,v_i],\xi_i)\}_{i=1}^n$ is *McShane δ-fine* if the intervals $[u_i,v_i]$ are non-overlapping, cover $[a,b]$, each $\xi_i \in [a,b]$, and

$$[u_i,v_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Unlike an HK tagged partition, the tag ξ_i is not required to lie in $[u_i,v_i]$.

Remark (Untethered tag). Figure 1.6 isolates the McShane move: a tag may sit outside the cell whose height it supplies, provided the cell lies inside the tag's gauge ball. The ruler is still Δx_i; the change is entirely in the admissible tag geometry.

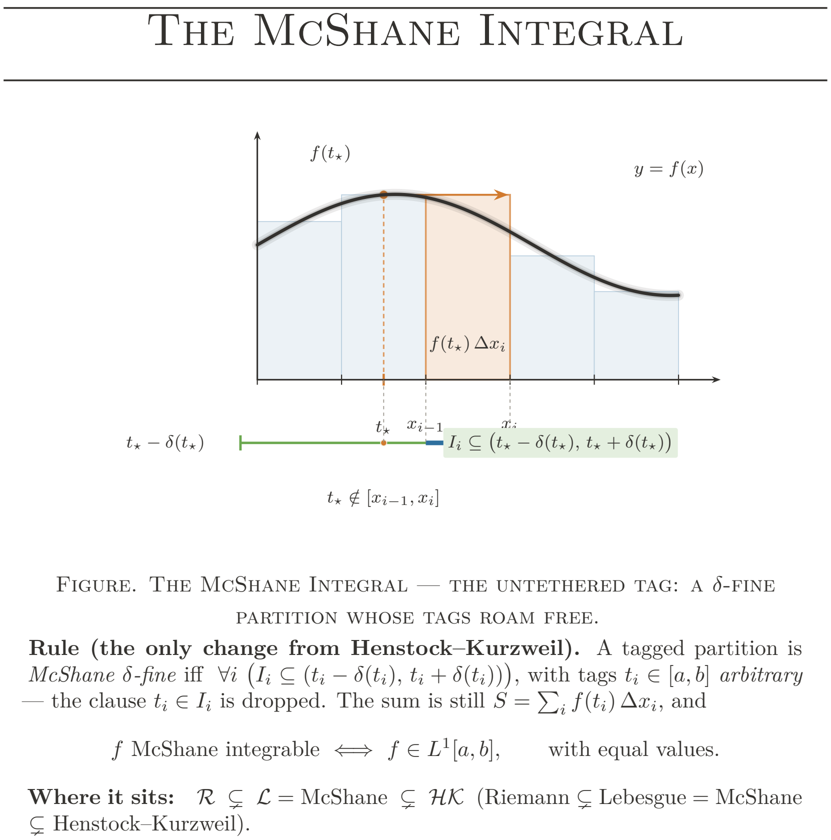


Figure 12.6: The McShane integral and the untethered tag

Definition (McShane Integral)

Definition 12.7.2 (McShane Integral)**.** A function $f : [a,b] \to \mathbb{R}$ is *McShane integrable* with value A if

$$\forall \varepsilon > 0\ \exists \delta : [a,b] \to (0,\infty)\ \forall (P,\xi)\ \big((P,\xi) \text{ is McShane } \delta\text{-fine} \Rightarrow |S(f,P,\xi) - A| < \varepsilon\big),$$

where

$$S(f,P,\xi) = \sum_i f(\xi_i)(v_i - u_i).$$

Remark (Relation to HK). Every McShane-fine partition is not necessarily HK-fine, because its tags may float outside their intervals. But every HK-fine partition is McShane-fine. Therefore McShane integrability is a stronger demand: the sums must survive all the usual gauge-fine tests and also the extra free-tag tests.

Theorem (McShane Sits Between Riemann and HK)

Theorem 12.7.3 (Riemann Integrable Implies McShane Integrable Implies Henstock–Kurzweil Integrable)**.** *On a compact interval,*

$$\mathcal{R}[a,b] \subseteq \mathcal{M}[a,b] \subseteq \mathcal{HK}[a,b],$$

and the integral values agree whenever a function belongs to the smaller class.

Theorem (McShane Equals Lebesgue)

Theorem 12.7.4 (McShane Integrability Equals Lebesgue Integrability)**.** *For real-valued functions on a compact interval,*

$$\mathcal{M}\rfloor\mathcal{S}\langle\dashv\backslash\rceil[a,b] = \mathcal{L}[a,b],$$

and the McShane and Lebesgue integral values agree.

Remark (The Lebesgue connection)*.* For real-valued functions on a compact interval, the McShane integral recovers the Lebesgue integral. In practical terms,

$$\mathcal{M}[a,b] = \mathcal{L}[a,b]$$

with equal values. Thus McShane is not a second route beyond Lebesgue in the way HK is. It is a gauge-sum presentation of Lebesgue integration.

Remark (Comparison chain)*.* Combining the Riemann inclusion, the McShane–Lebesgue identification, and the HK derivative witness gives

$$\mathcal{R}[a,b] \subsetneq \mathcal{M}\rfloor\mathcal{S}\langle\dashv\backslash\rceil[a,b] = \mathcal{L}[a,b] \subsetneq \mathcal{HK}[a,b].$$

The first strict inclusion is the old Riemann failure at dense null discontinuities; the second is the HK recovery of derivatives whose absolute size is not Lebesgue integrable.

Remark (Forward dependency)*.* The proof that every Riemann-integrable function is McShane-integrable uses the measure-zero diagnosis in the next section: Riemann integrability means the discontinuity set is null. The exposition places McShane first because it belongs syntactically with the gauge integrals; the proof ledger records that it imports the Lebesgue criterion.

Remark (Why free tags force Lebesgue behavior)*.* The HK derivative example

$$F'(x) = 2x\sin(1/x^2) - \frac{2}{x}\cos(1/x^2)$$

is conditionally integrable by HK because the tags remain inside the tiny slabs where the derivative is being locally linearized. McShane's free tags can sample nearby oscillatory peaks in a way that destroys this conditional balance. What survives that stronger test is exactly the Lebesgue-integrable, absolutely controlled part of the gauge world.

Remark (What it buys). McShane gives a tagged-sum construction of the Lebesgue integral without starting from simple functions or sigma-algebras. Conceptually, that is its job in this chapter: it shows that adaptive Riemann-style sums can reach the Lebesgue class, while HK shows that keeping tags inside slabs reaches beyond Lebesgue to conditionally integrable derivatives.

Remark (Bug report). McShane is elegant but not the terminal fix for this chapter's story. If one wants a tagged-sum definition of the Lebesgue integral, it is excellent. If one wants the perfect one-dimensional Fundamental Theorem of Calculus, HK is strictly stronger. And if one wants probability, stochastic processes, and abstract spaces, the next volume still needs measure theory explicitly.

Remark (Handoff to measure zero). McShane reveals that gauge machinery and Lebesgue size are compatible. The next section names the exact size condition already controlling the older Riemann–Darboux family: the discontinuity set must have measure zero.

12.8 Measure Zero and the Lebesgue Criterion

Geometric Intuition

Darboux integrability fails where upper and lower staircases refuse to meet. That refusal is local oscillation. Thus the exact Riemann question becomes: how large is the set where f is discontinuous?

Remark (Engineering reading). This final section is not another integral. It is the diagnostic scan of the entire Riemann–Darboux family, placed after Henstock–Kurzweil and McShane so the chapter ends by naming the exact obstruction that opens the Lebesgue volume. Once Darboux has made the gap visible, the remaining question is whether the bad part of the domain is small enough to cover by slabs of negligible total length.

Definition (Measure Zero)

Definition 12.8.1 (Measure Zero). A set $E \subseteq \mathbb{R}$ has *measure zero* if for every $\varepsilon > 0$ there are open intervals I_k such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \qquad \sum_{k=1}^{\infty} |I_k| < \varepsilon.$$

Definition (Point Oscillation)

Definition 12.8.2 (Point Oscillation). For bounded f on $[a, b]$, define

$$\omega_f(x) = \inf_{\delta>0} \omega_f\big((x - \delta, x + \delta) \cap [a, b]\big).$$

Then f is continuous at x exactly when $\omega_f(x) = 0$.

Remark (Oscillation levels). The sets

$$D_\eta = \{x : \omega_f(x) \geq \eta\}$$

separate the bad set by severity. The full discontinuity set is

$$D = \bigcup_{k=1}^{\infty} D_{1/k}.$$

The proof of the criterion works by showing that positive-size high oscillation forces a permanent Darboux gap, while null high-oscillation sets can be covered by intervals of tiny total length.

Theorem (Lebesgue Criterion)

Theorem 12.8.3 (Lebesgue Criterion for Riemann Integrability)**.** *A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable if and only if its discontinuity set*

$$D = \{x \in [a, b] : \omega_f(x) > 0\}$$

has measure zero.

Remark (Terminal diagnosis). Riemann integration works precisely when the set of bad points is invisible to length. The Dirichlet function $\mathbf{1}_{\mathbb{Q}}$ is discontinuous everywhere, so $D = [0, 1]$, not a null set. The Lebesgue patch is to stop partitioning the domain into slabs and instead measure the sets where the function takes given ranges of values. Henstock–Kurzweil keeps tagged sums, but replaces uniform fineness by local fineness; McShane shows that a slightly stricter gauge rule recovers the Lebesgue class itself. The pathology plate in Figure 1.3 is the visual companion to this diagnosis.

Remark (Lebesgue handoff). The slogan is precise: Riemann partitions the domain, so every slab that meets a dense bad set still sees the bad behavior. Lebesgue partitions the range: it groups points by height and measures the set carrying each height. For $\mathbf{1}_{\mathbb{Q}}$, the height 1 lives on a null set and therefore contributes nothing.

Appendix: Pathologies of the Riemann Integral

Remark (Source placement). This appendix is the converted support note from `integration-failures.tex`. Its standalone theorem machinery and document preamble have been removed; the material is kept as commentary so it does not duplicate the formal Lebesgue criterion in Section 1.8.3.

Remark (The one theorem behind the failures). The Riemann failures in Figure 1.3 are not a bag of examples. They are tests against the two hypotheses in the Lebesgue criterion: boundedness, and a discontinuity set of measure zero. Dirichlet fails because its discontinuity set is all of $[0, 1]$. Unbounded examples fail before the criterion even begins. Pointwise limits expose a different weakness: the Riemann class is not stable under the limit operations that Lebesgue integration was built to support.

Remark (Thomae as the diagnostic example). Thomae's function is the useful warning. Its discontinuities are dense, but they are countable and therefore null. Darboux can isolate the finitely many large spikes above any fixed threshold inside intervals of tiny total length; outside those intervals, the upper sums are uniformly small. Thus density of the bad set is not the obstruction. Size is.

Remark (Through-line). The atlas failure note should be read as the bridge from Darboux to Measure Zero: $U(f, P) - L(f, P)$ is local oscillation times length, so the only bad oscillation that matters is bad oscillation carried by a set of positive length. That observation closes the Riemann family and points directly to Lebesgue's range-partition construction.

Appendix: The Gauge Integrals

Remark (Source placement). This appendix is the converted support note from `gauge-integrals.tex`. The standalone preamble, local theorem declarations, and duplicate figure includes have been removed; the shared figures are the pinned figures already included in Sections 1.6 and 1.7.

Remark (The gauge framework). The gauge integrals replace Riemann's single global mesh tolerance by a local step-size controller $\delta : [a, b] \to (0, \infty)$. Henstock–Kurzweil and McShane then differ by one geometric clause. HK requires the tag to sit inside the interval it controls. McShane drops that tether and allows the tag to range over $[a, b]$, provided the interval lies inside the tag's gauge ball. The contrast is visualized in Figures 1.5 and 1.6.

Remark (Free tags and Lebesgue). McShane's free tags force absolute, Lebesgue-style behavior. Since a tag can authorize nearby intervals without lying inside them, the sums can test whether the mass assigned to height bands is genuinely controlled. On compact intervals this recovers the Lebesgue integral: $\mathcal{M}[a, b] = \mathcal{L}[a, b]$, with equal values.

Remark (Tethered tags and the FTC). HK keeps the tag tethered to its interval, which preserves local cancellation for derivatives. That is why the HK integral has the exact one-dimensional FTC: if F is differentiable everywhere, then F' is HK-integrable and (HK) $\int_a^b F' = F(b) - F(a)$, even when F' is not Lebesgue integrable.

Remark (Lattice). The resulting compact-interval lattice is

$$\mathcal{R}[a, b] \subsetneq \mathcal{L}[a, b] = \mathcal{M}[a, b] \subsetneq \mathcal{HK}[a, b],$$

as recorded in Figure 1.4. This appendix is kept after the main chapter because the book has not yet built the measure theory needed to prove the McShane–Lebesgue identification.

Bibliography

Terence Tao. *Analysis I*, volume 37 of *Texts and Readings in Mathematics*. Hindustan Book Agency, New Delhi, 2006. ISBN 978-8185931751.